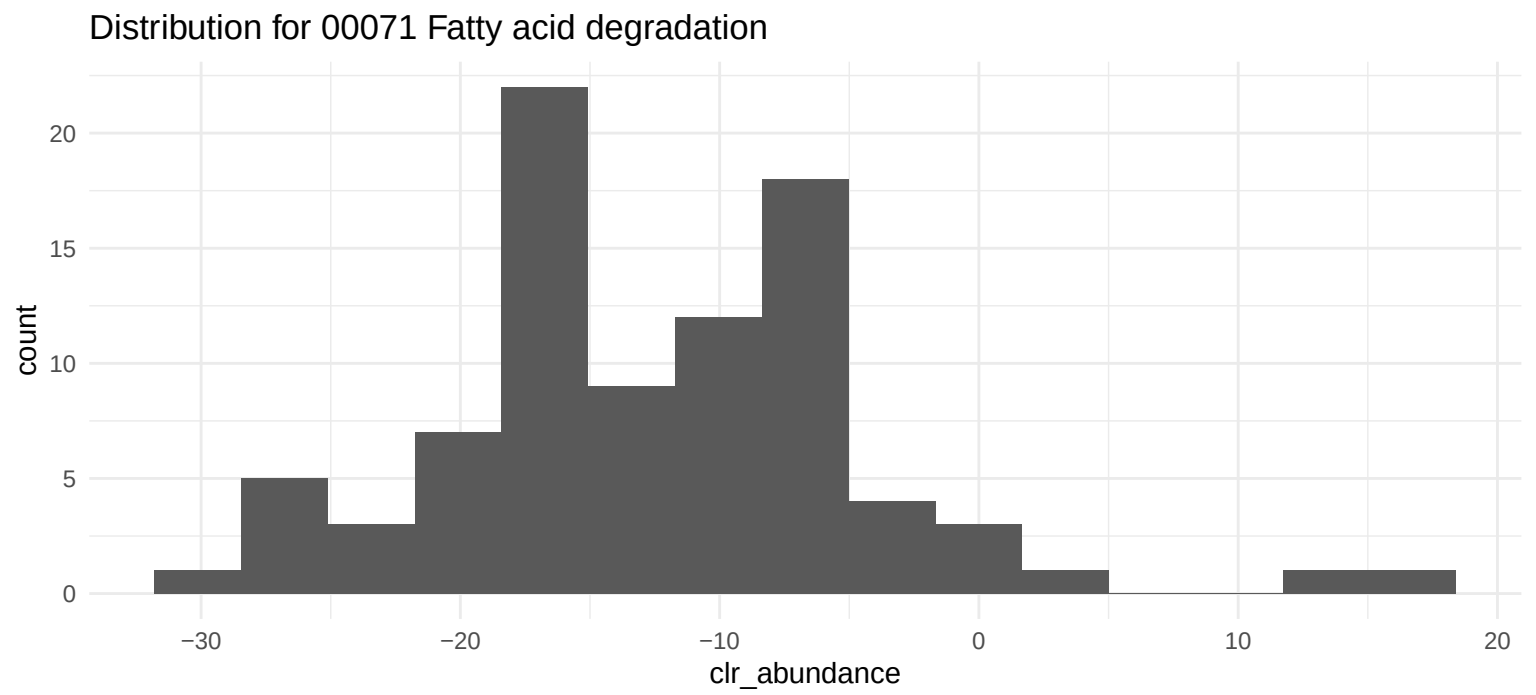
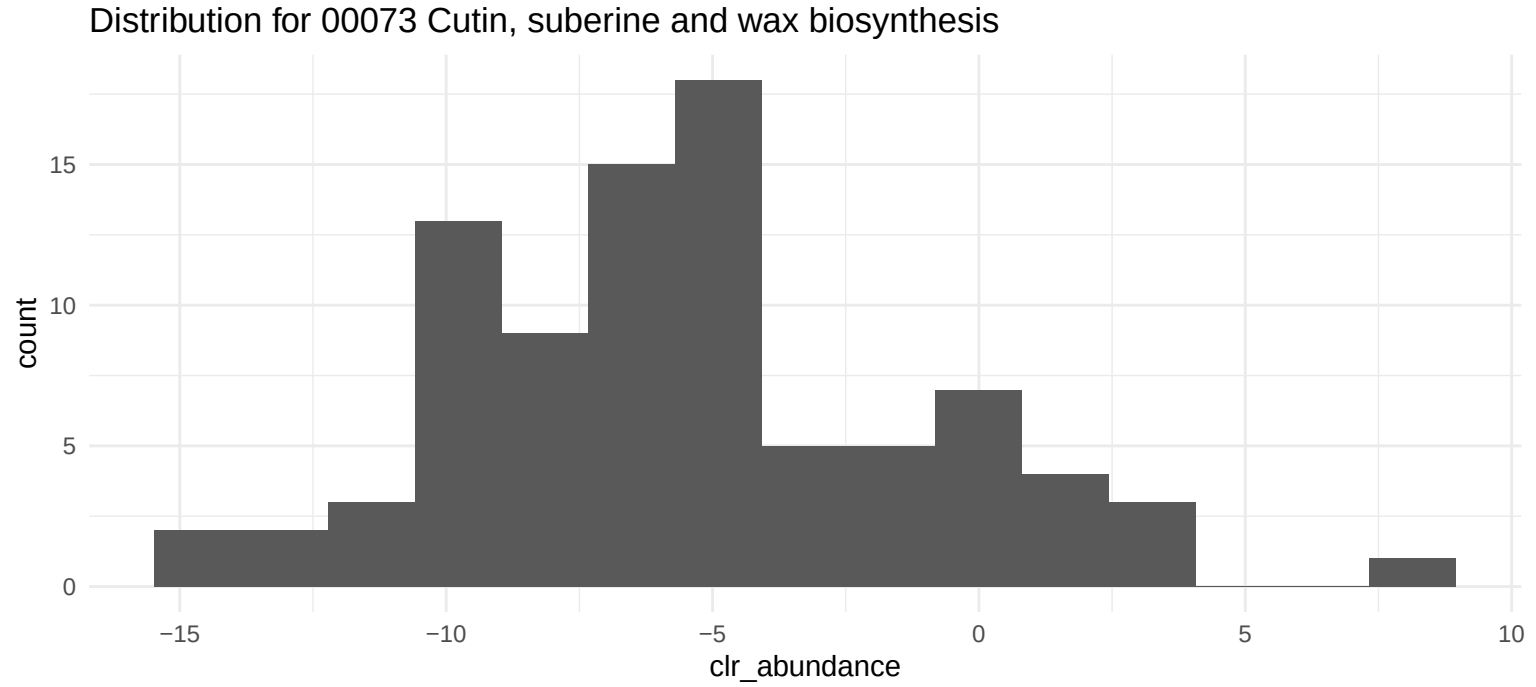
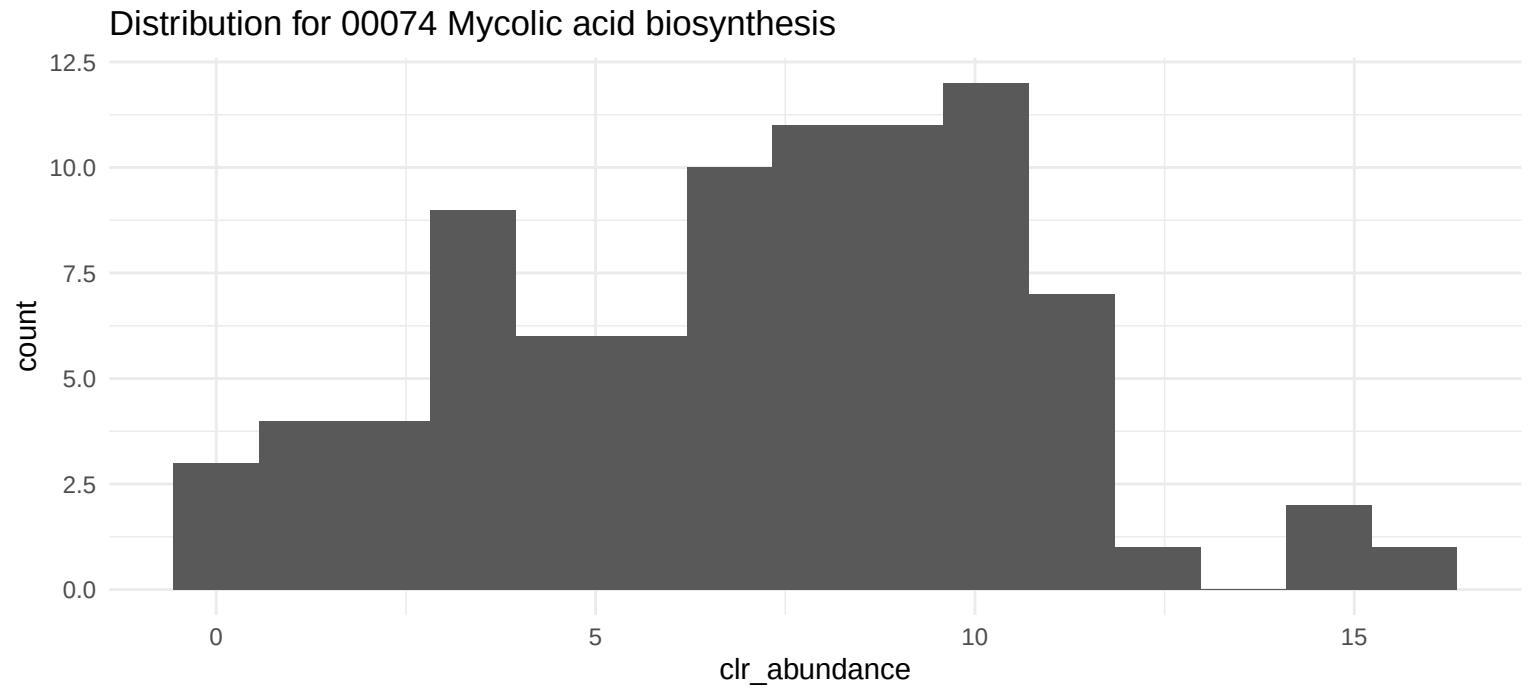
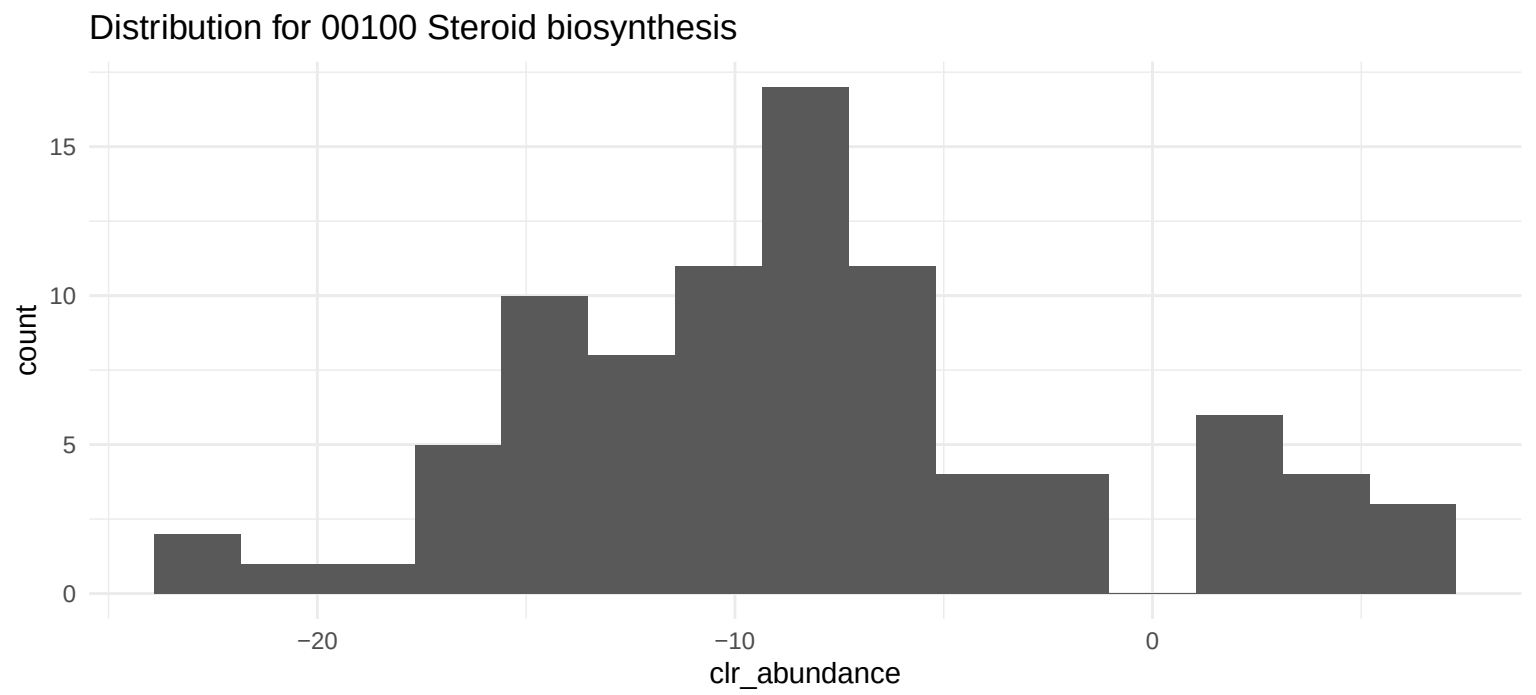
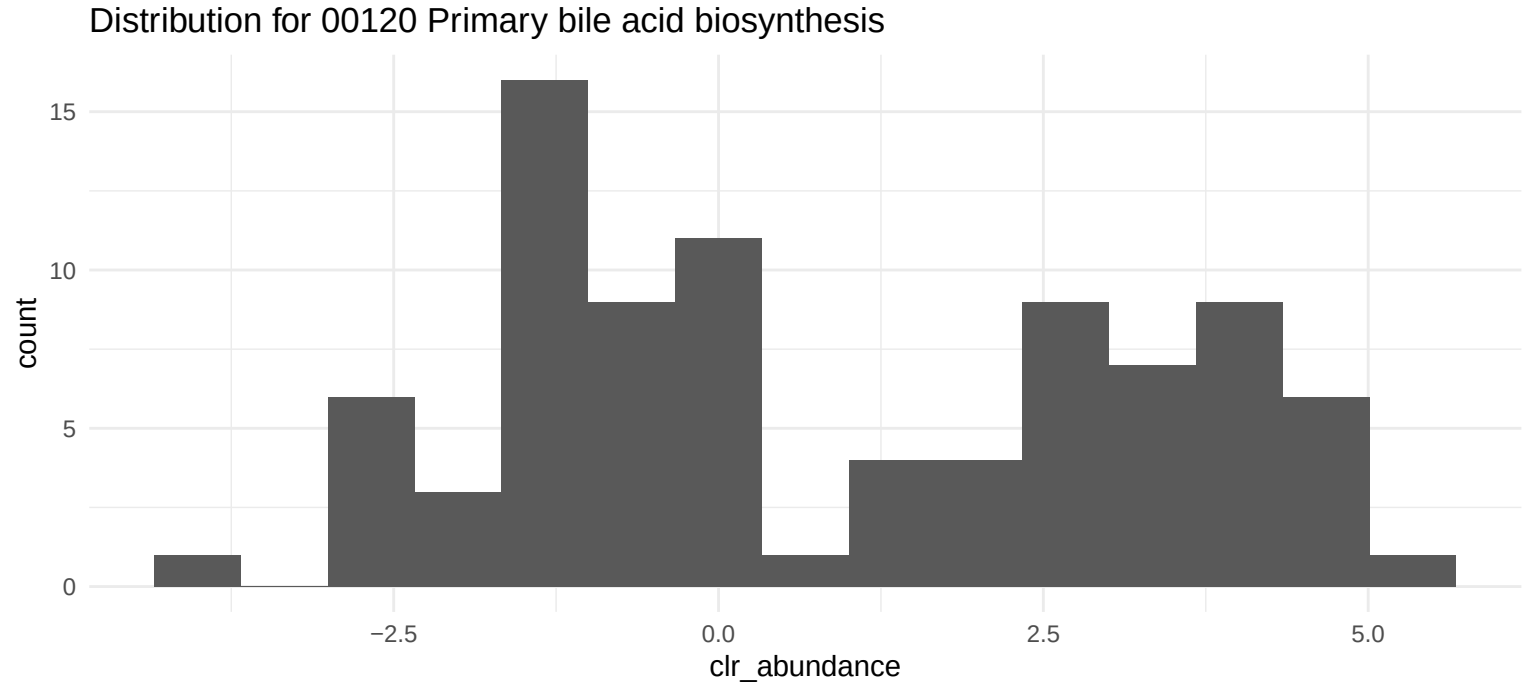
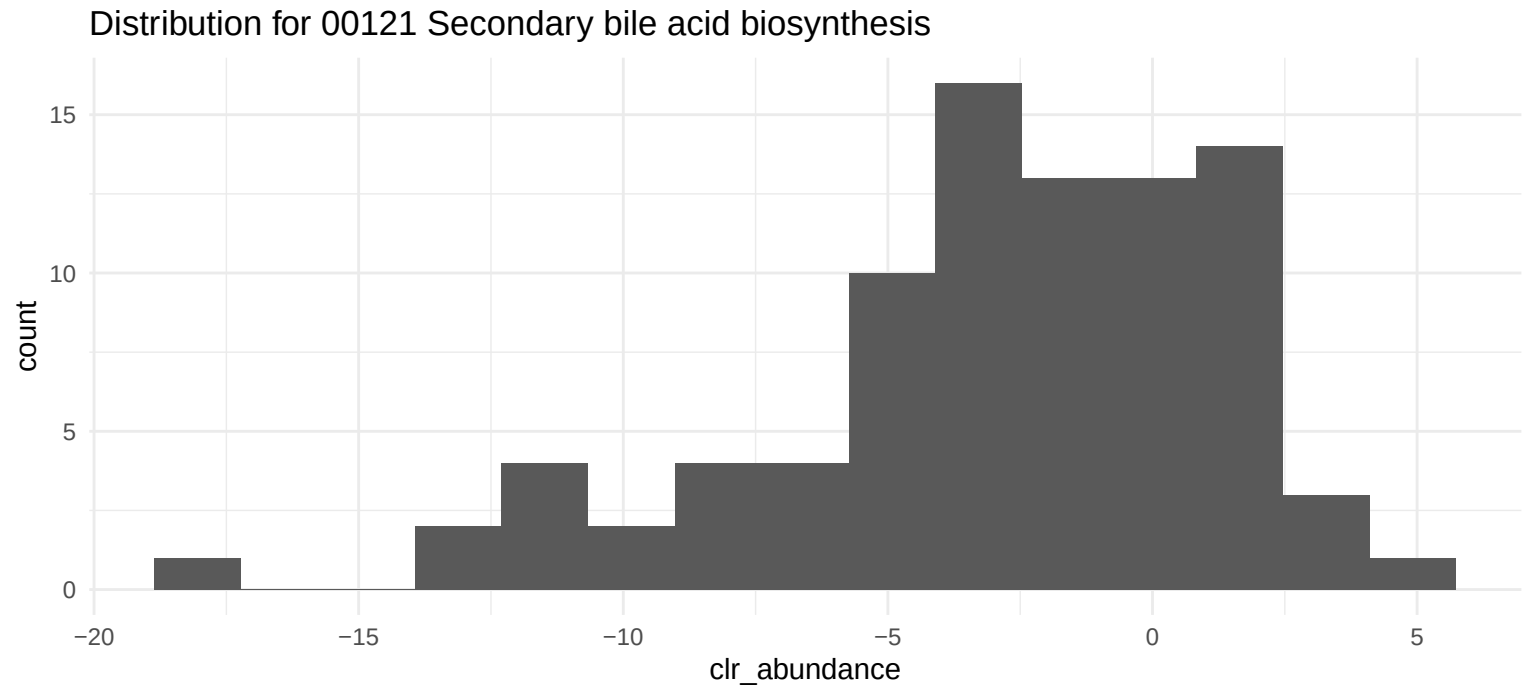

Distribution for 00071 Fatty acid degradation
count
clr_abundance

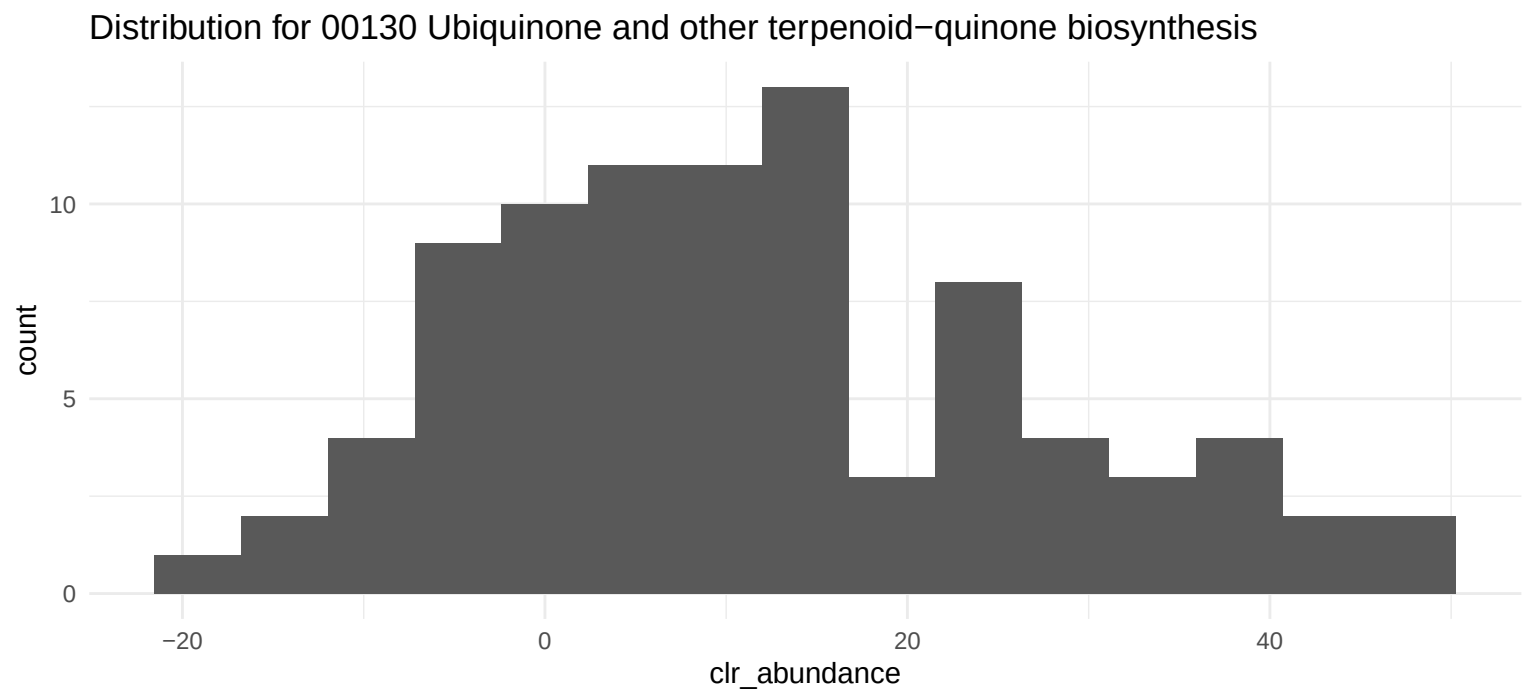

Distribution for 00073 Cutin, suberine and wax biosynthesis
count
clr_abundance

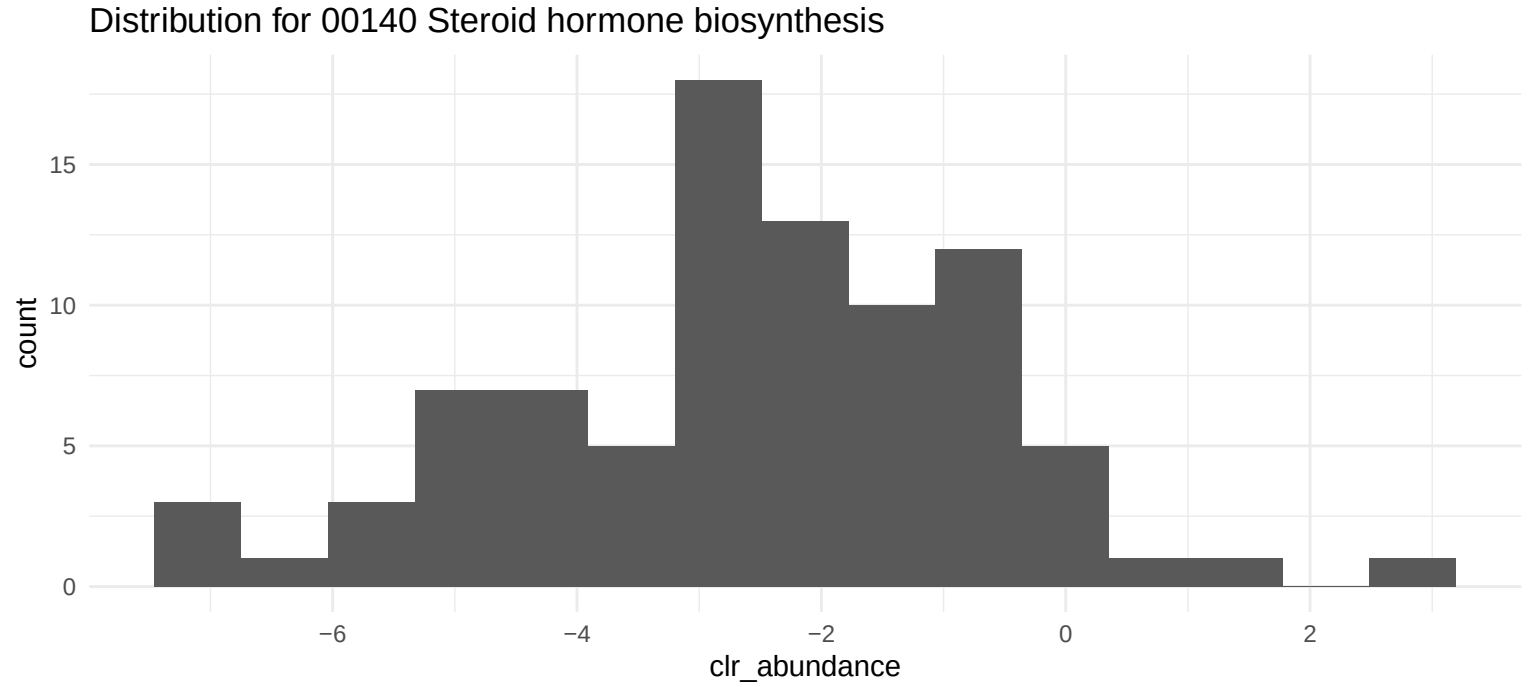

Distribution for 00074 Mycolic acid biosynthesis
count
clr_abundance

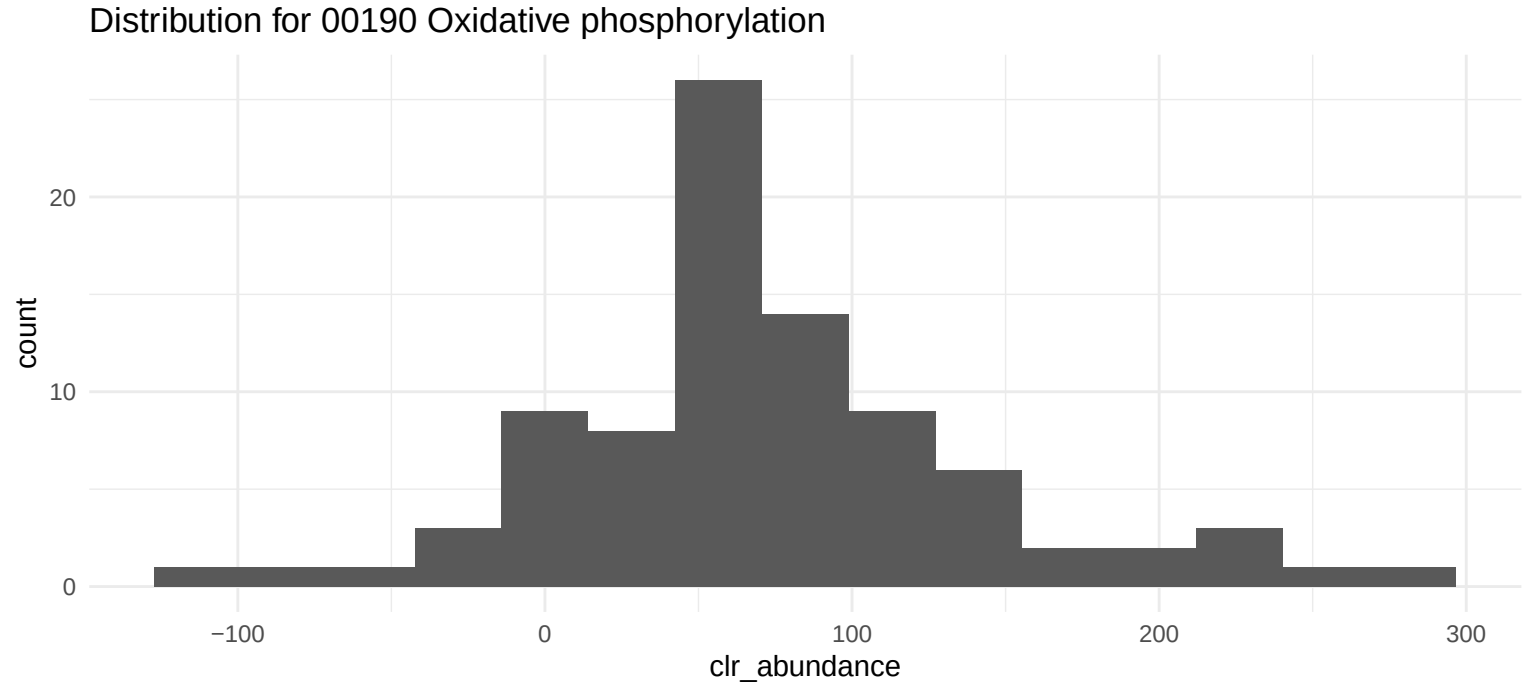

Distribution for 00100 Steroid biosynthesis
count
15
10
5
0
−20
−10
0
clr_abundance

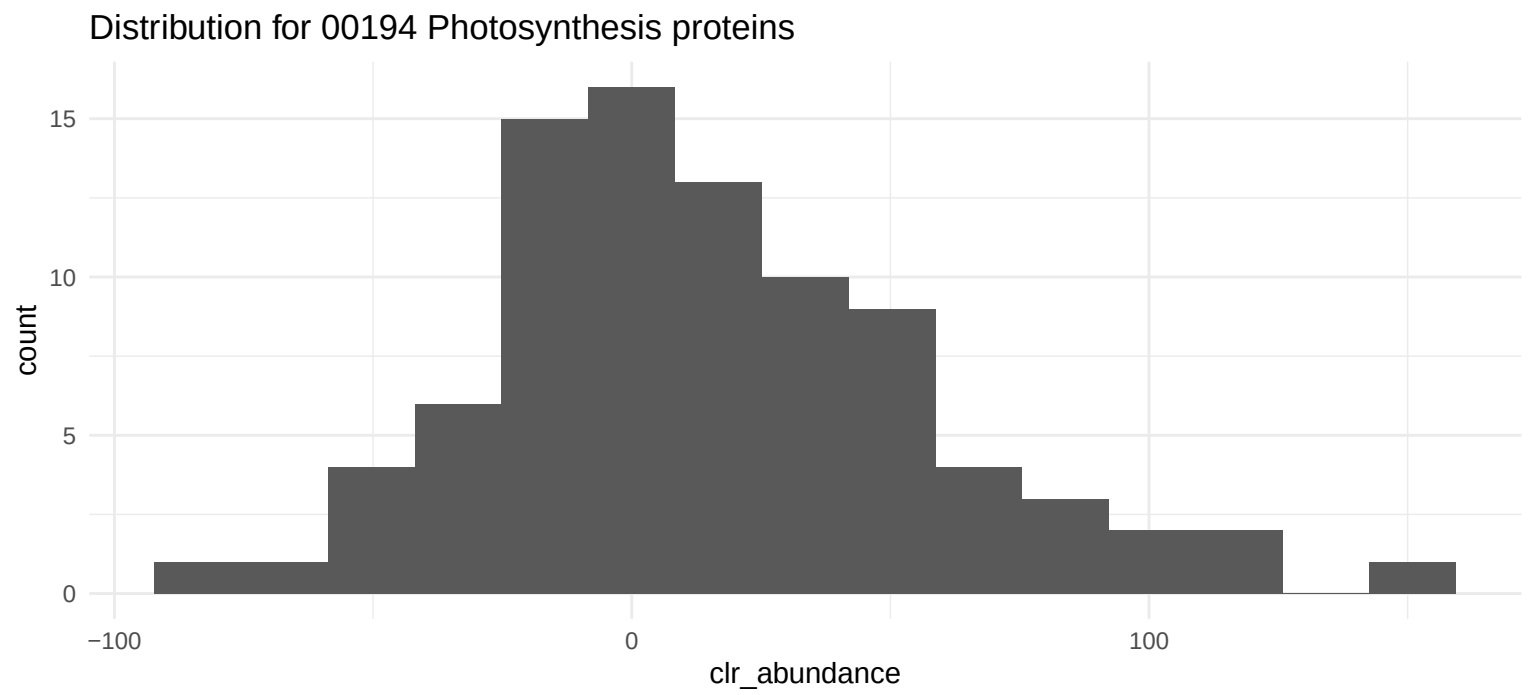

Distribution for 00120 Primary bile acid biosynthesis
count
15
10
5
0
−2.5
0.0
2.5
5.0
clr_abundance

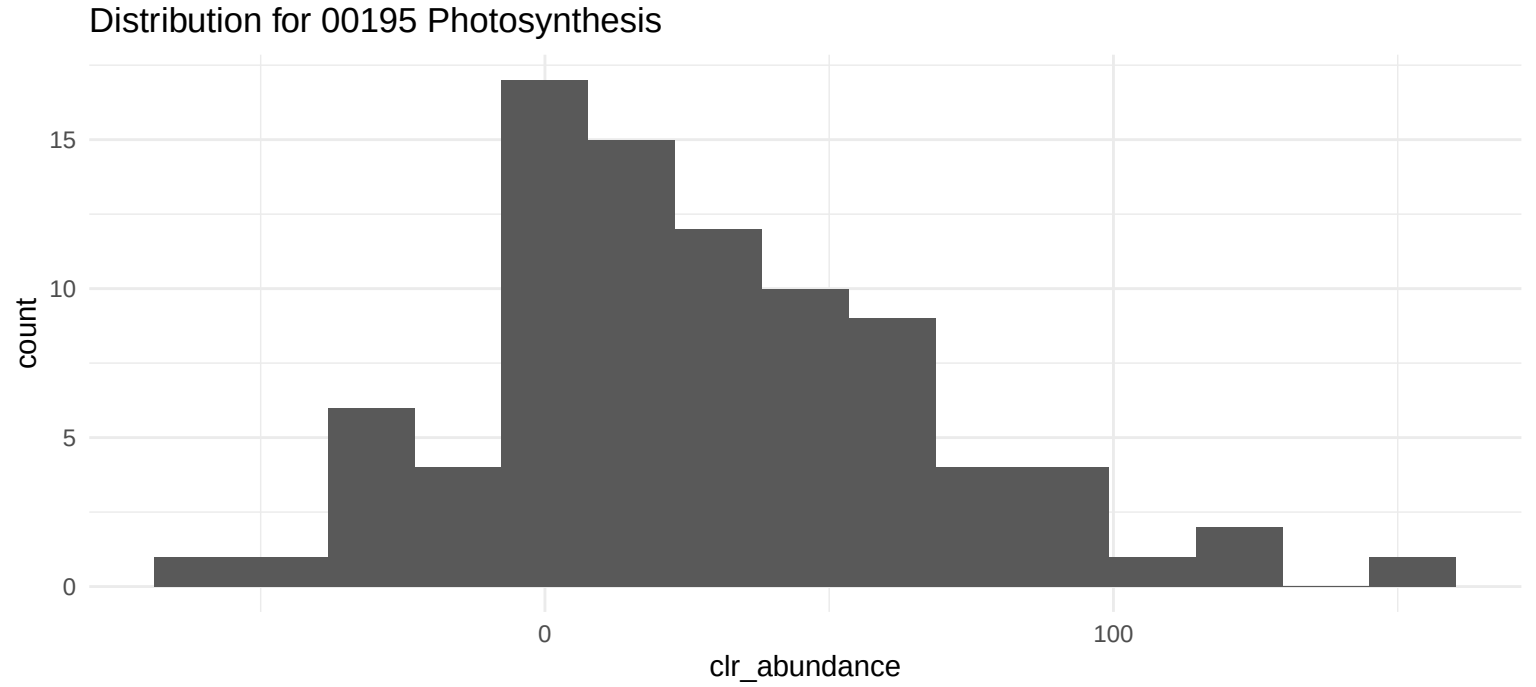

Distribution for 00121 Secondary bile acid biosynthesis
count
15
10
5
0
−20
−15
−10
−5
0
5
clr_abundance

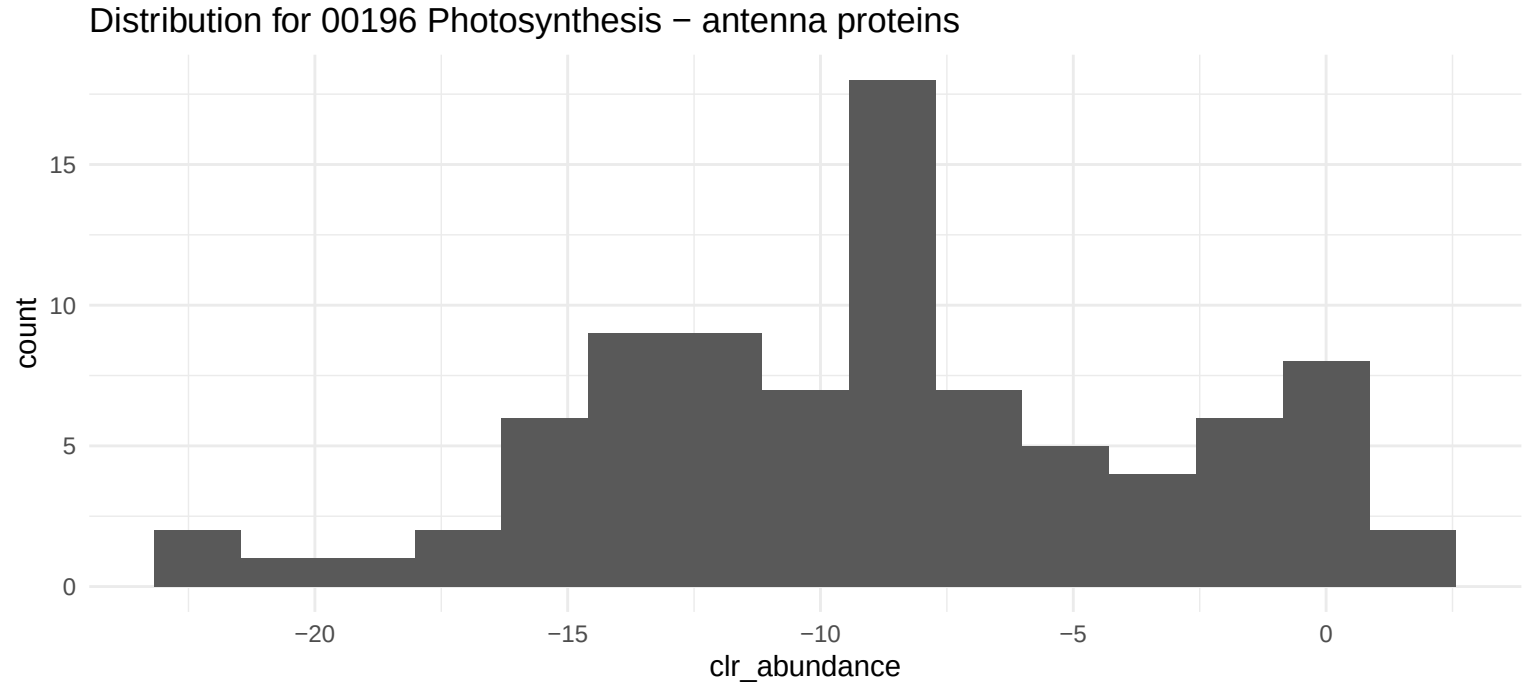

Distribution for 00130 Ubiquinone and other terpenoid−quinone biosynthesis
count
clr_abundance
0
5
10
−20
0
20
40

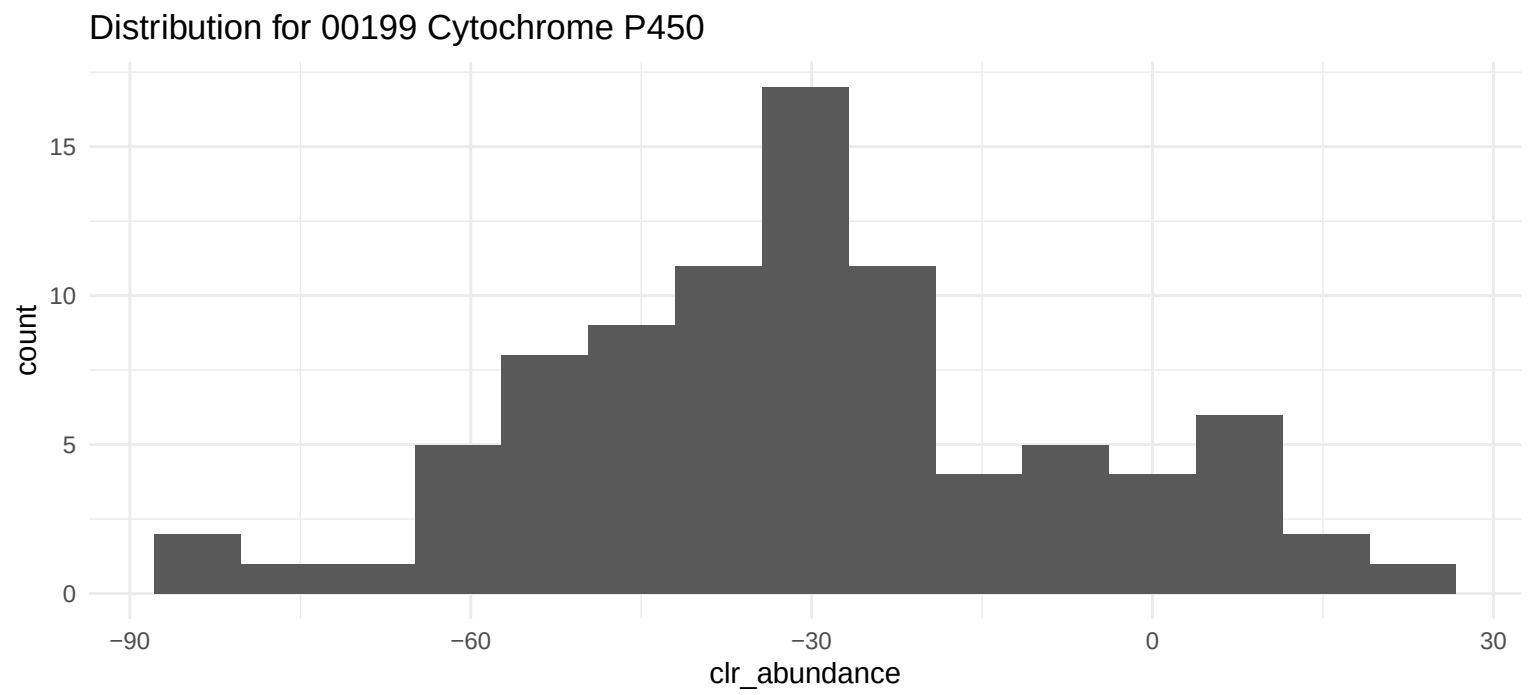

Distribution for 00140 Steroid hormone biosynthesis
count
clr_abundance
0
5
10
15
−6
−4
−2
0
2

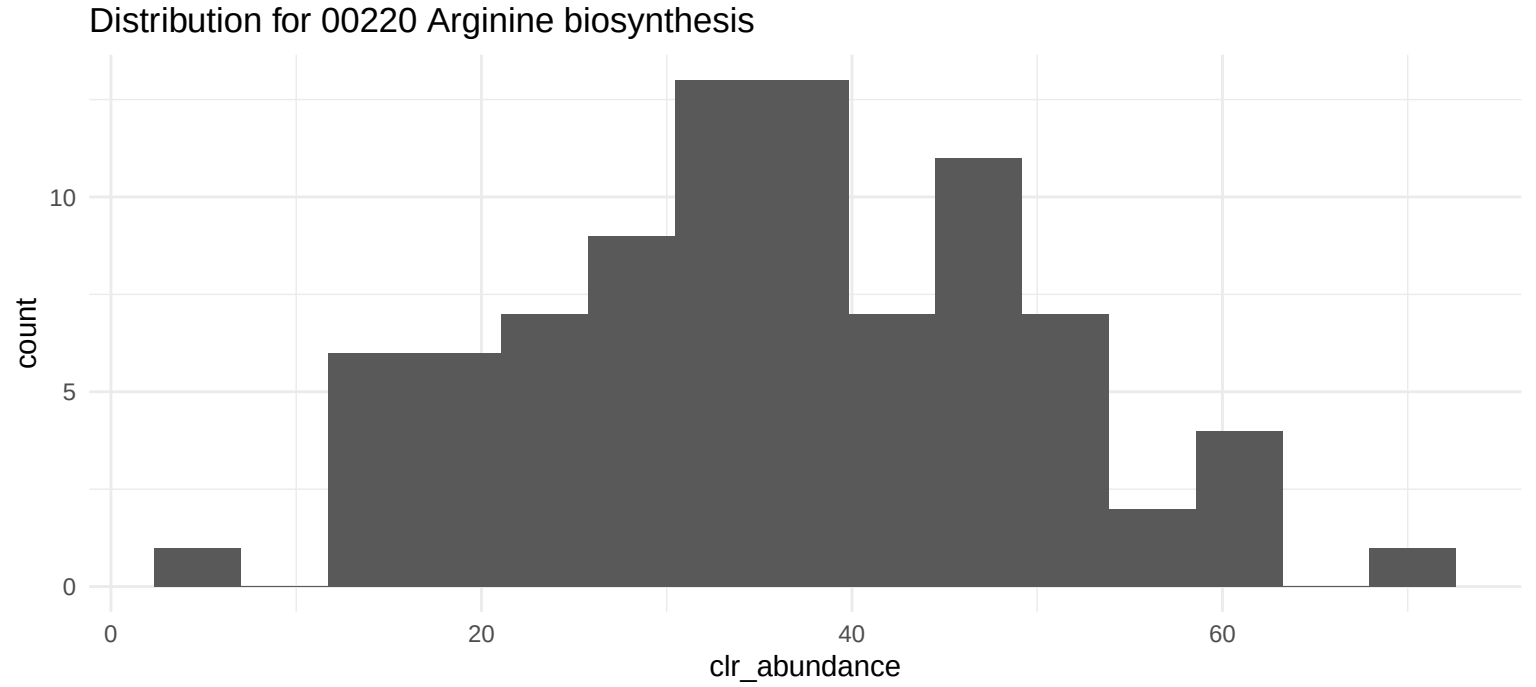

Distribution for 00190 Oxidative phosphorylation
count
clr_abundance
0
10
20
−100
0
100
200
300

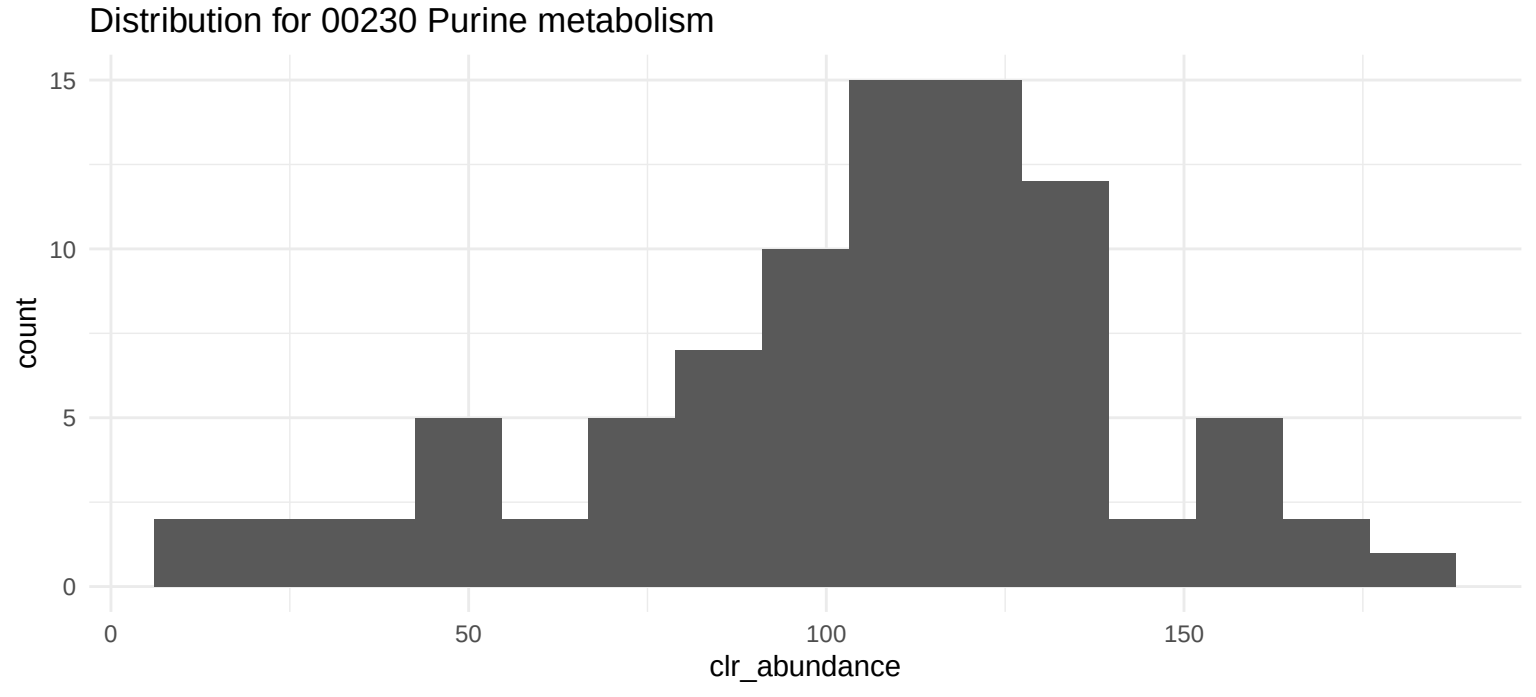

Distribution for 00194 Photosynthesis proteins
count
15
10
5
0
−100
0
100
clr_abundance


Distribution for 00195 Photosynthesis
count
15
10
5
0
0
100
clr_abundance

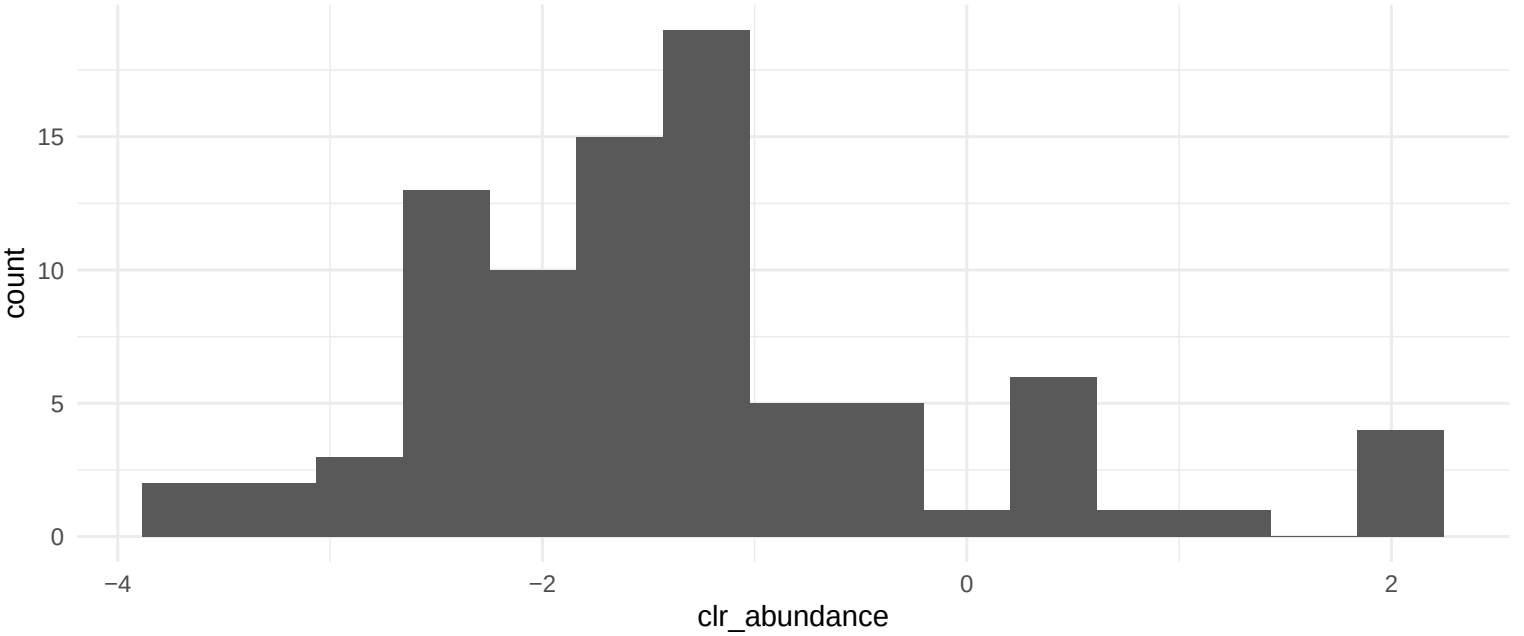

Distribution for 00196 Photosynthesis − antenna proteins
count
15
10
5
0
−20
−15
−10
−5
0
clr_abundance


Distribution for 00199 Cytochrome P450
count
15
10
5
0
−90
−60
−30
0
30
clr_abundance

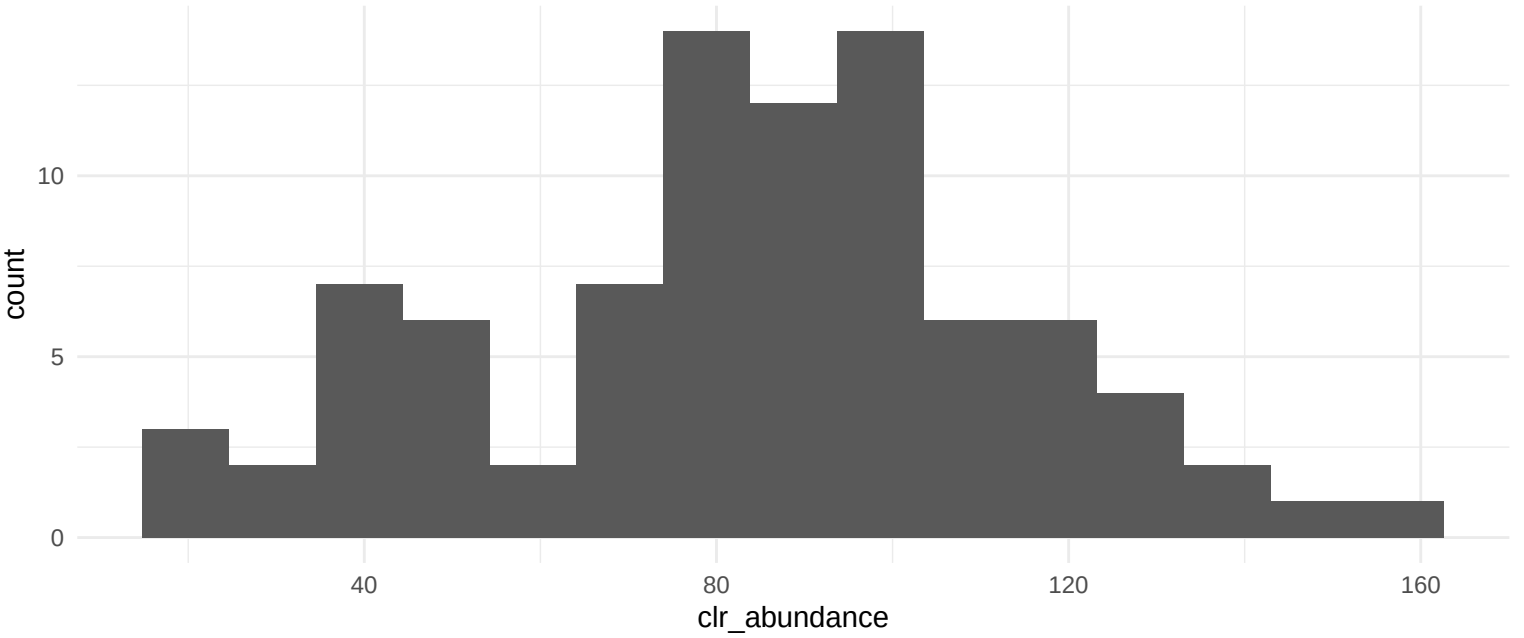

Distribution for 00220 Arginine biosynthesis
count
10
5
0
0
20
40
60
clr_abundance


Distribution for 00230 Purine metabolism
count
15
10
5
0
0
50
100
150
clr_abundance

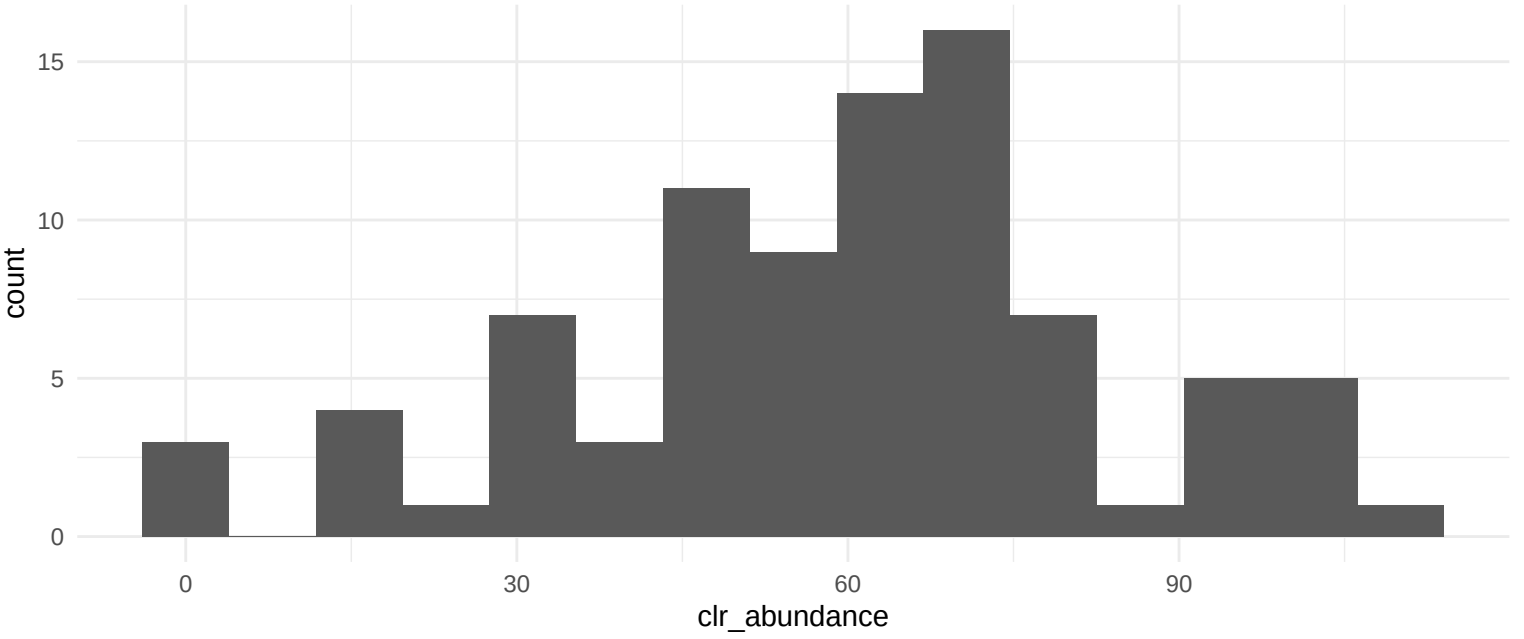
Distribution for 00232 Caffeine metabolism

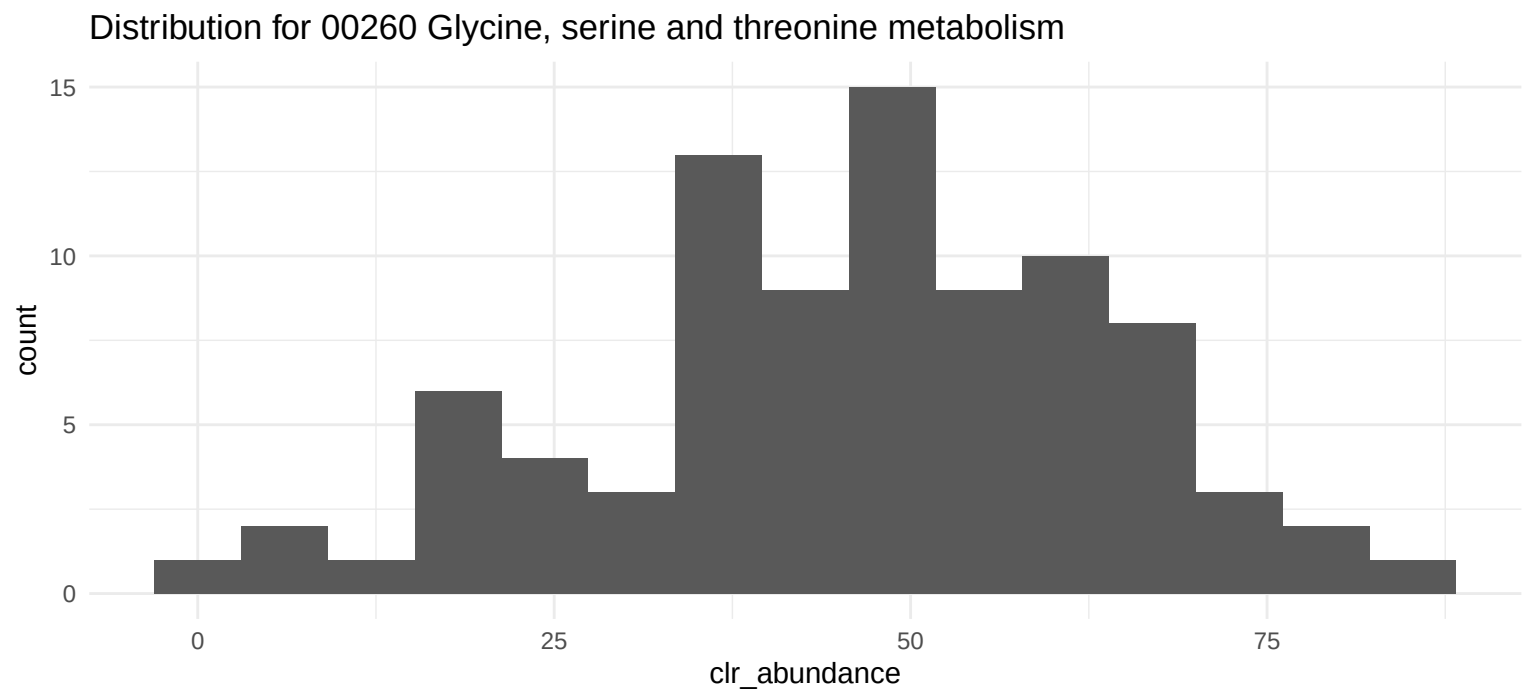


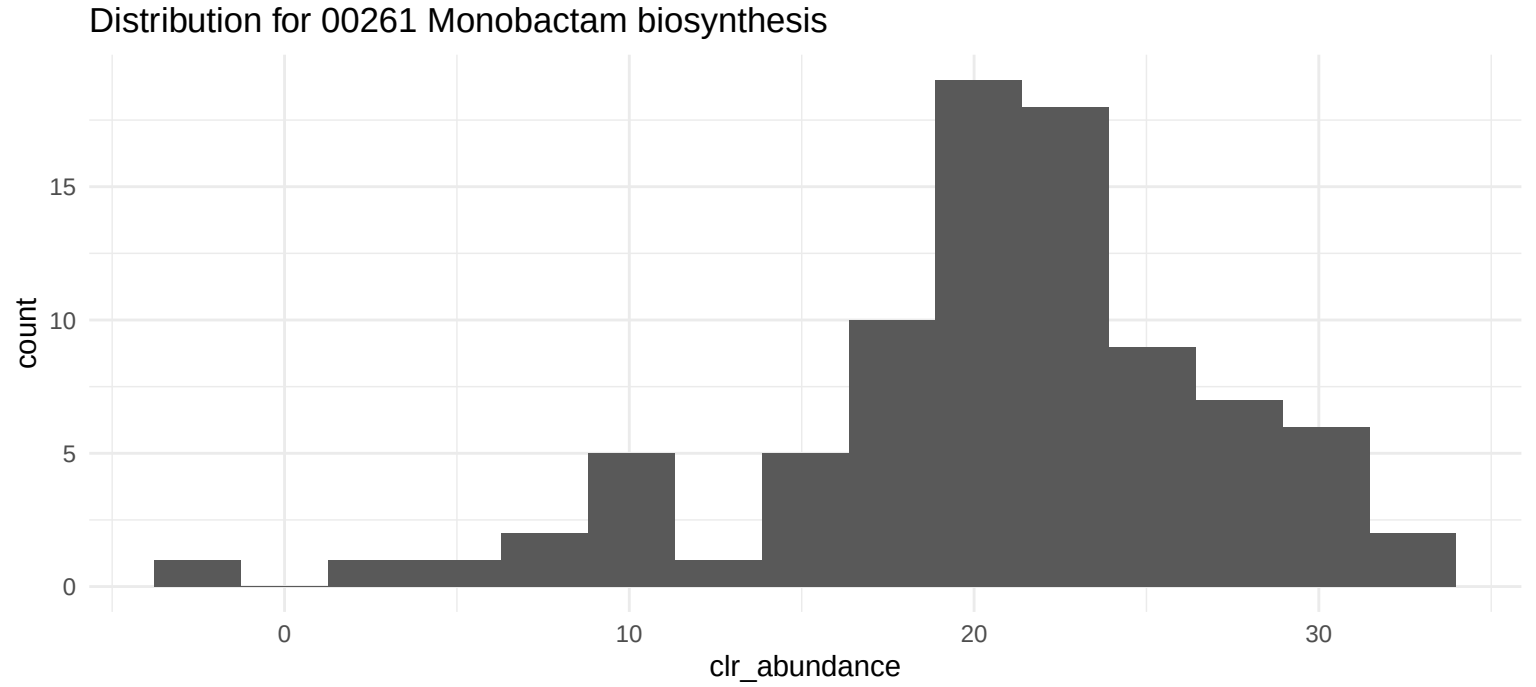
Distribution for 00240 Pyrimidine metabolism

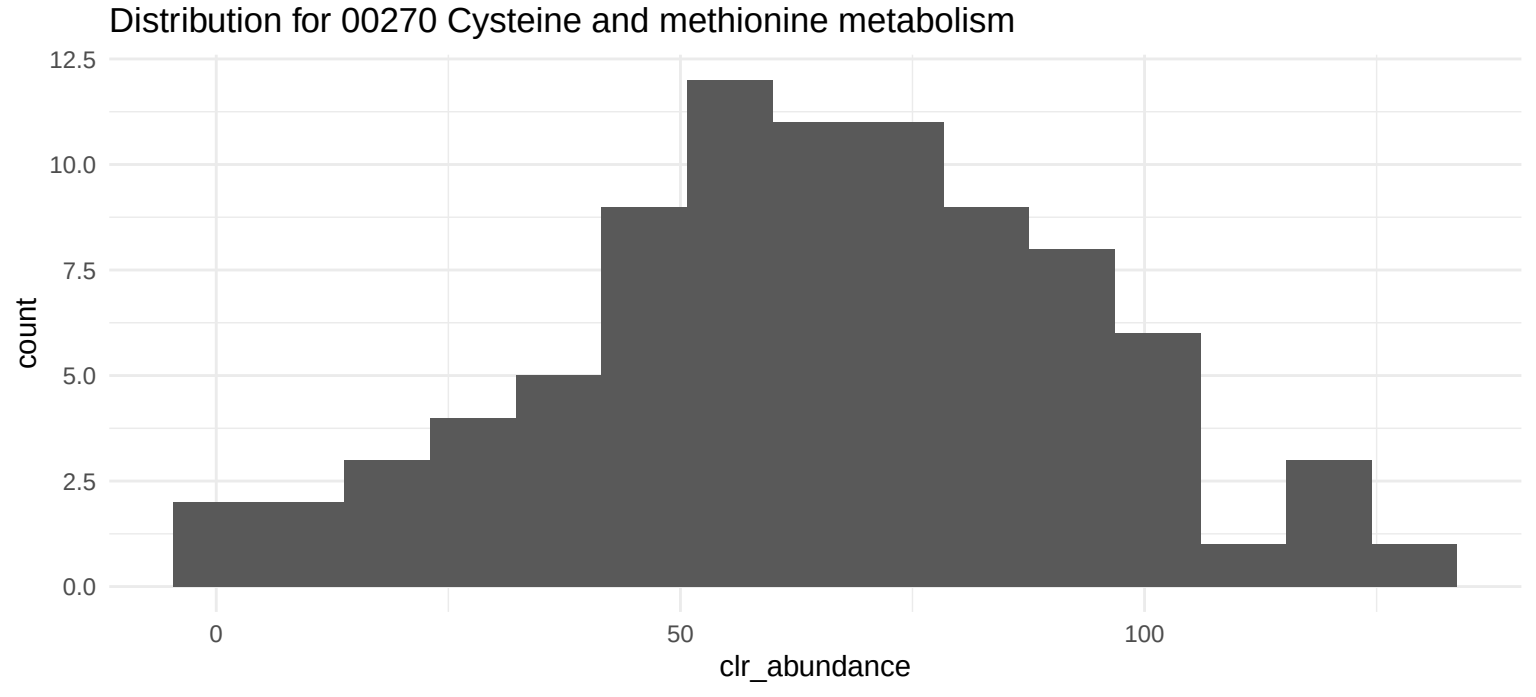


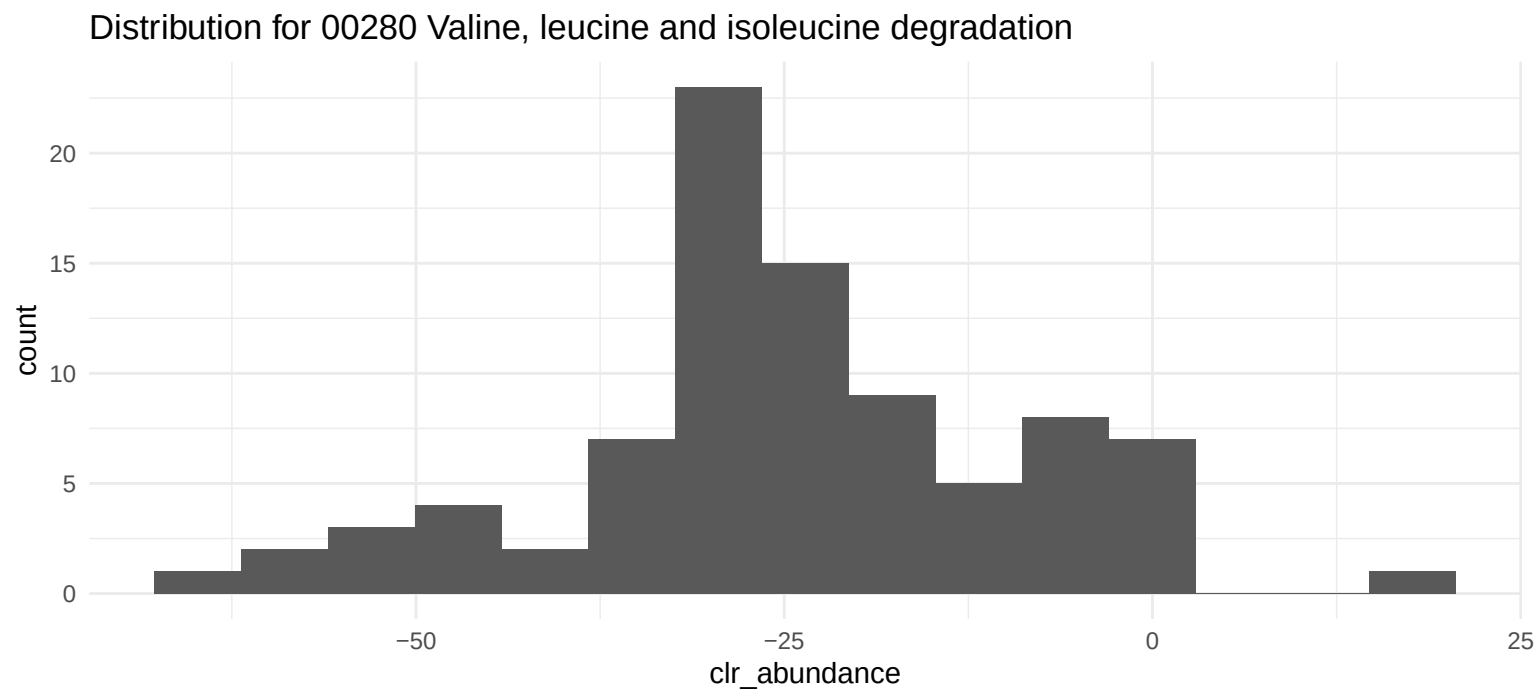
Distribution for 00250 Alanine, aspartate and glutamate metabolism

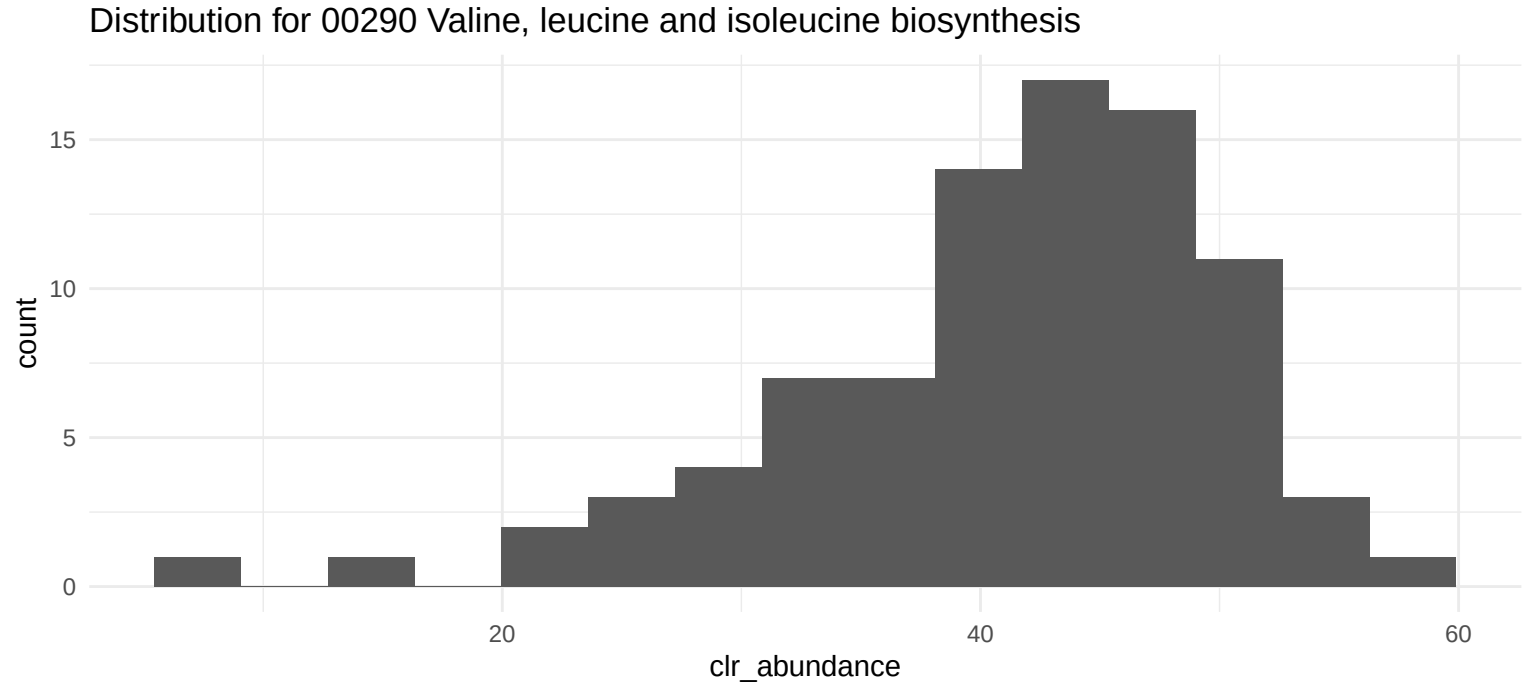


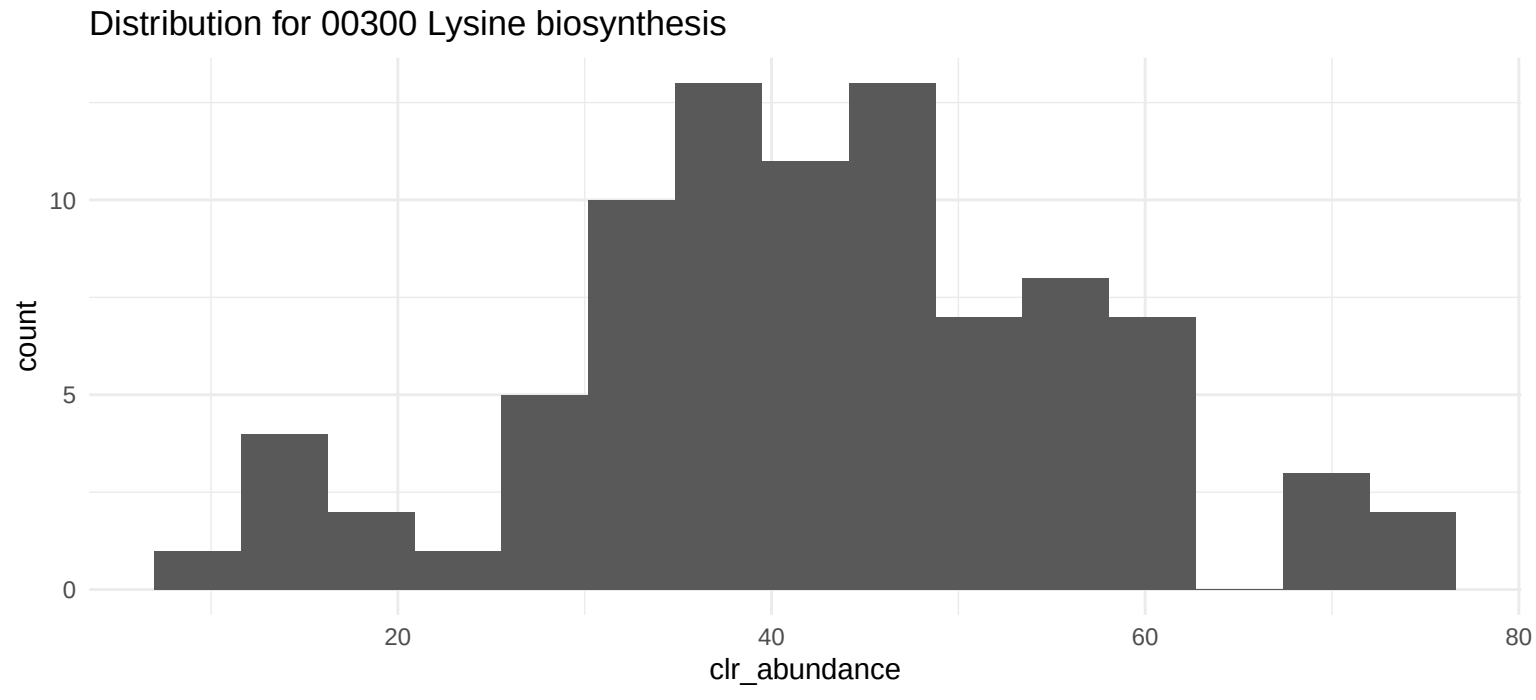

Distribution for 00260 Glycine, serine and threonine metabolism
count
clr_abundance
0
5
10
15
0
25
50
75

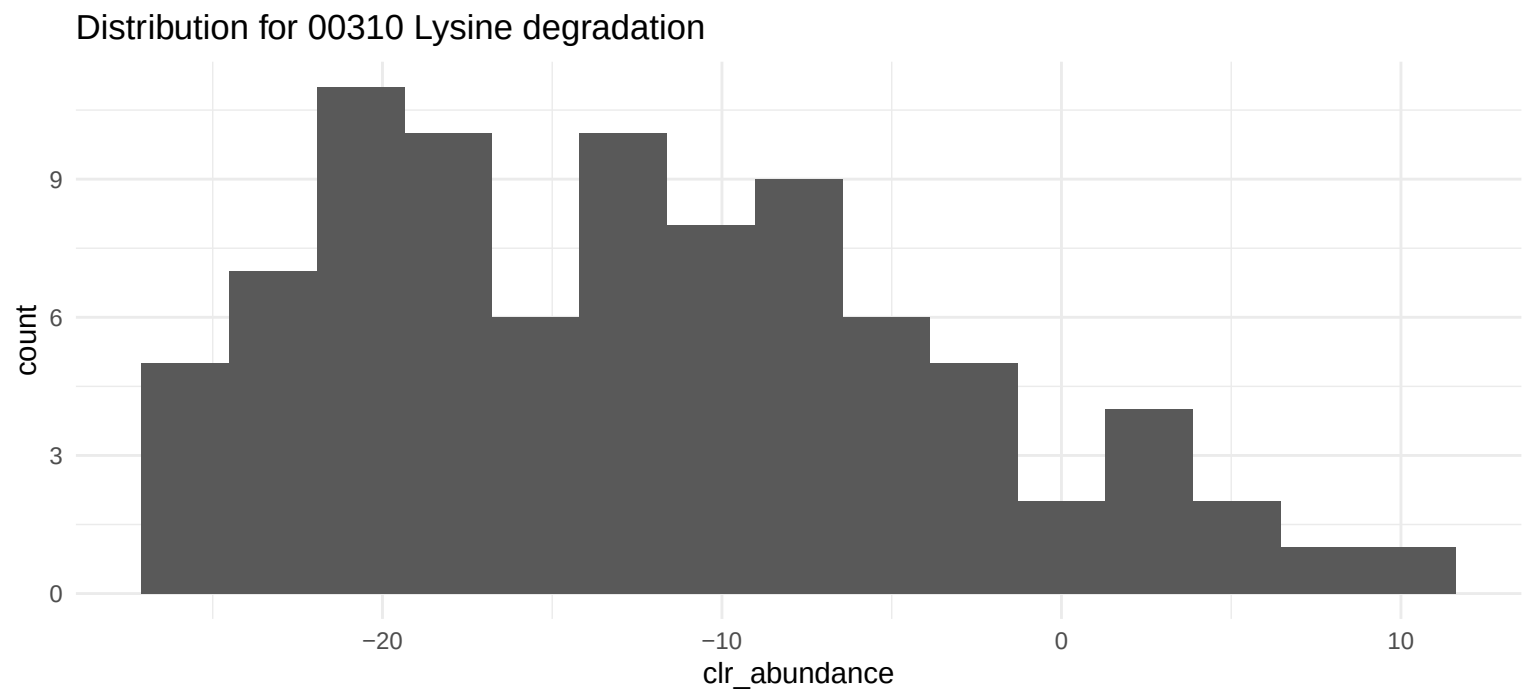

Distribution for 00261 Monobactam biosynthesis
count
clr_abundance
0
5
10
15
0
10
20
30

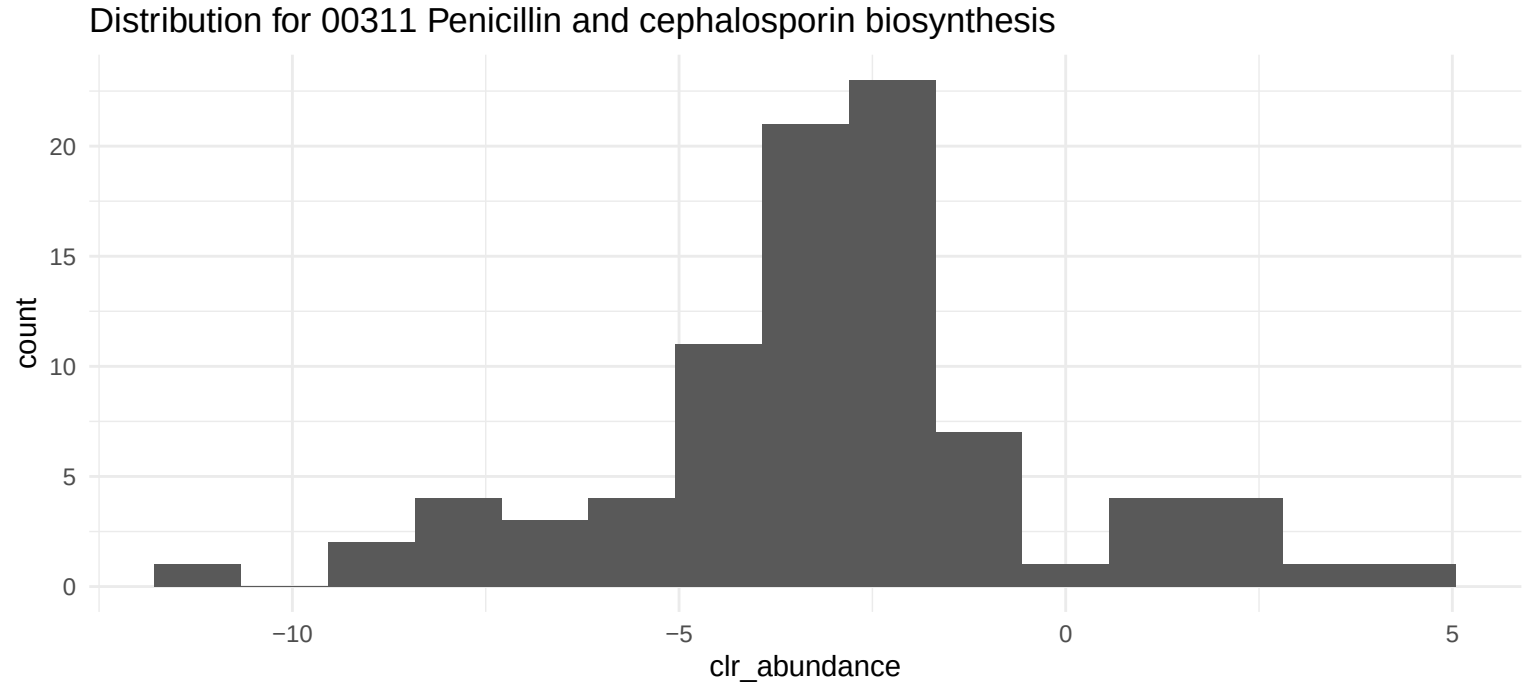

Distribution for 00270 Cysteine and methionine metabolism
count
clr_abundance
0.0
2.5
5.0
7.5
10.0
12.5
0
50
100

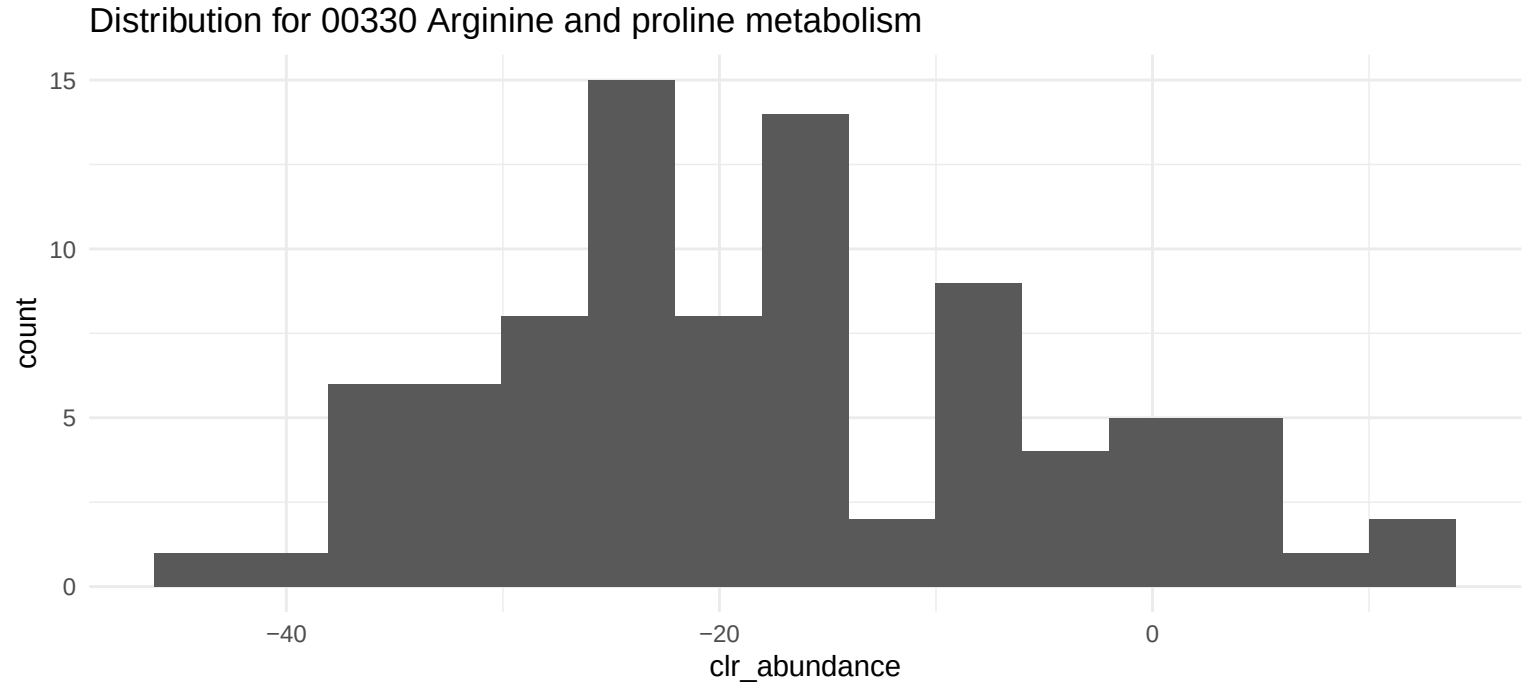

Distribution for 00280 Valine, leucine and isoleucine degradation
count
clr_abundance
0
5
10
15
20
−50
−25
0
25

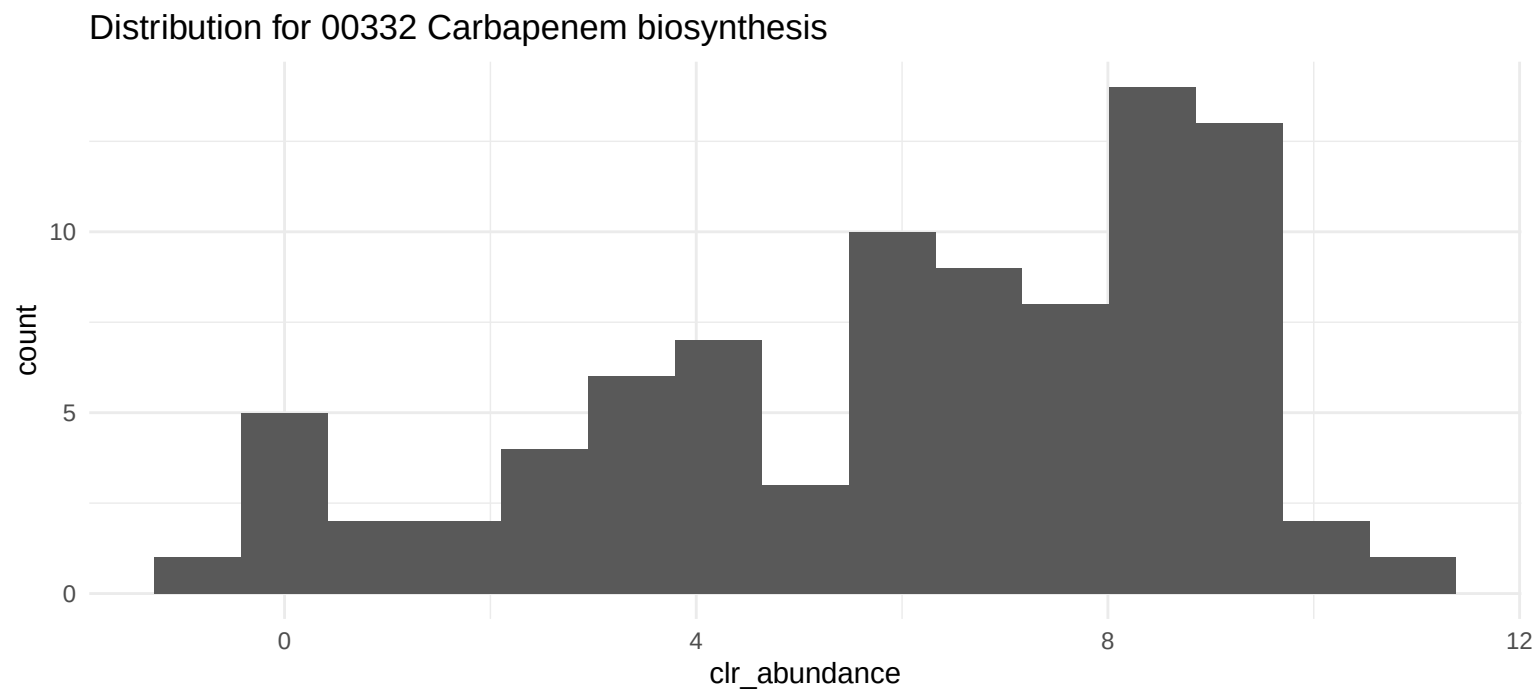

Distribution for 00290 Valine, leucine and isoleucine biosynthesis
count
clr_abundance
0
5
10
15
20
40
60

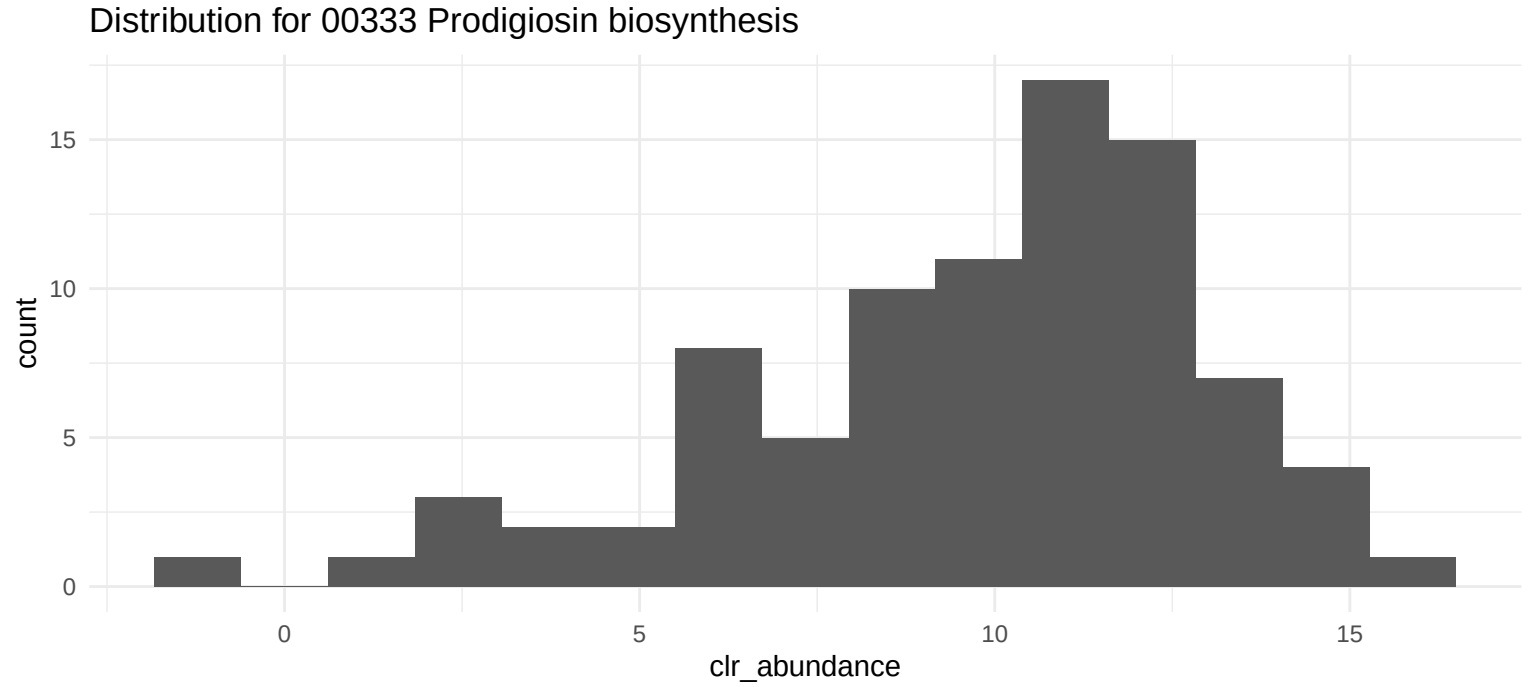

Distribution for 00300 Lysine biosynthesis
count
clr_abundance
0
5
10
20
40
60
80

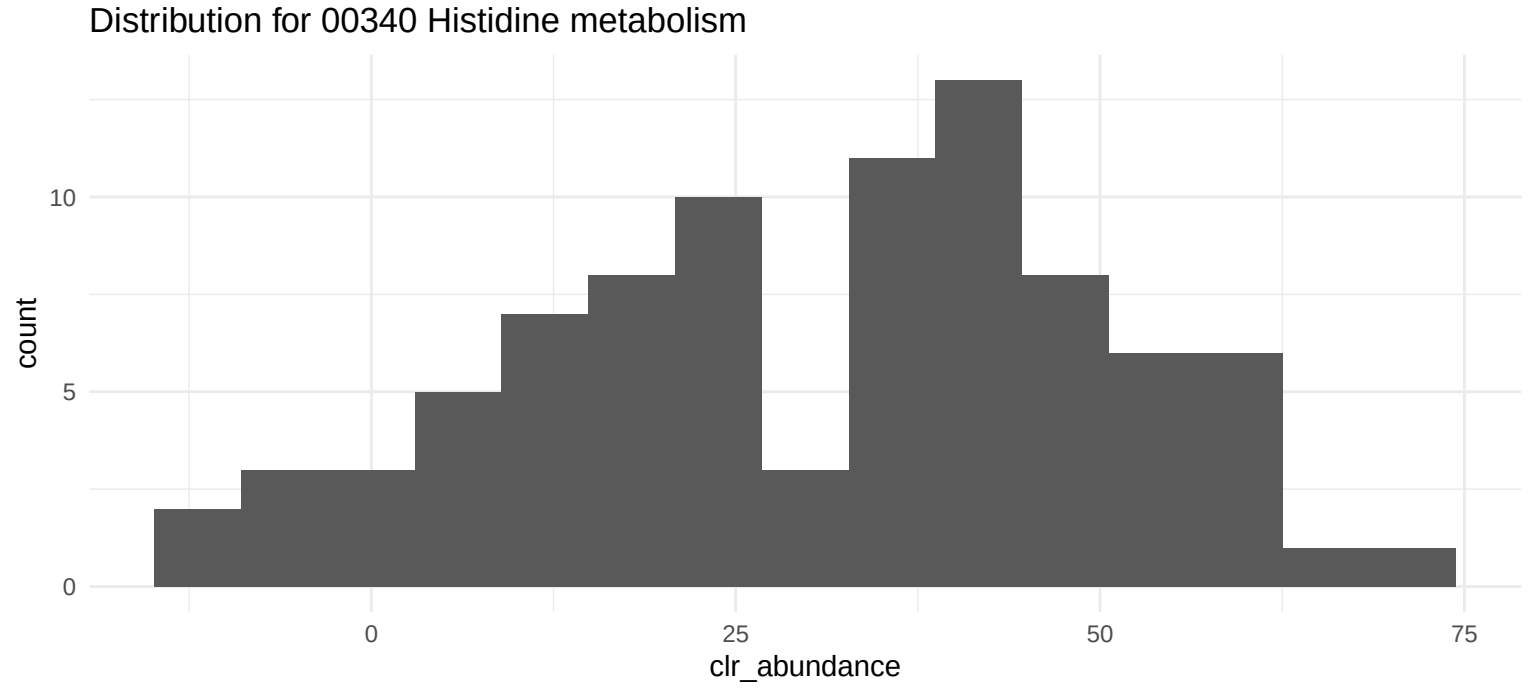

Distribution for 00310 Lysine degradation
count
clr_abundance
0
3
6
9
−20
−10
0
10


Distribution for 00311 Penicillin and cephalosporin biosynthesis
count
clr_abundance
0
5
10
15
20
−10
−5
0
5

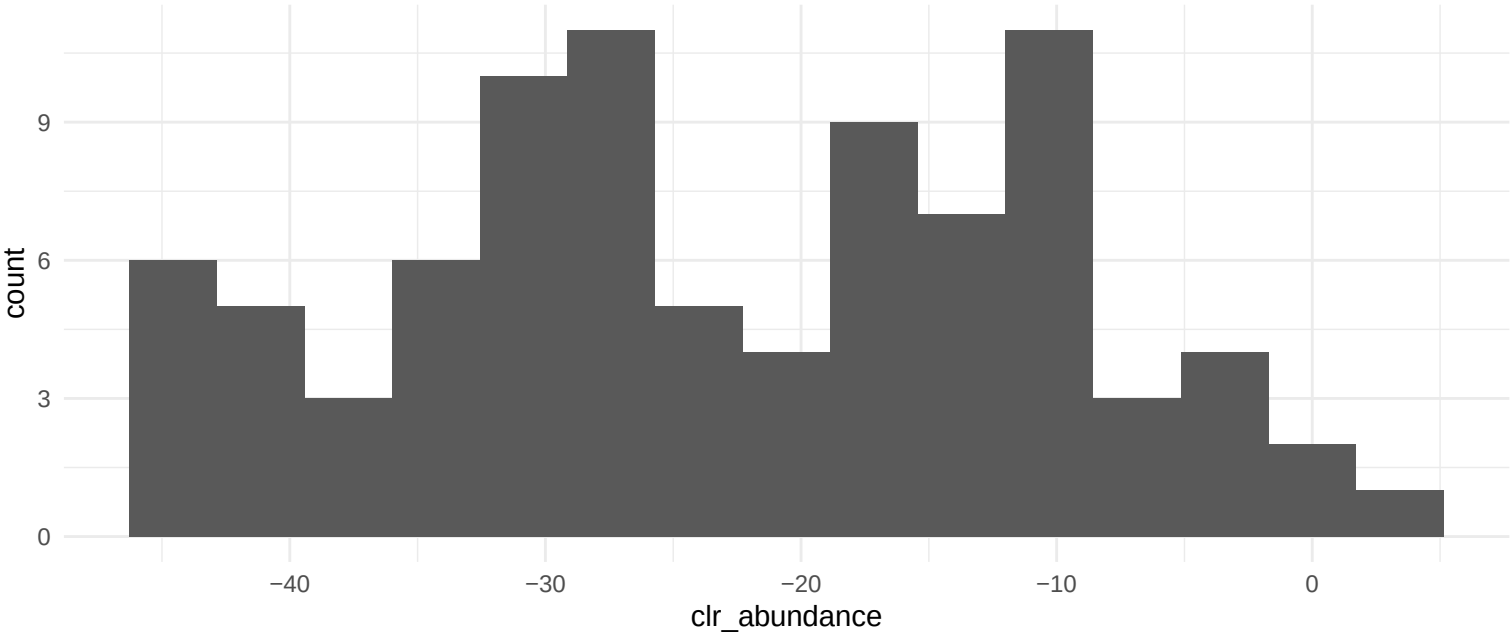

Distribution for 00330 Arginine and proline metabolism
count
clr_abundance
0
5
10
15
−40
−20
0


Distribution for 00332 Carbapenem biosynthesis
count
clr_abundance
0
5
10
0
4
8
12

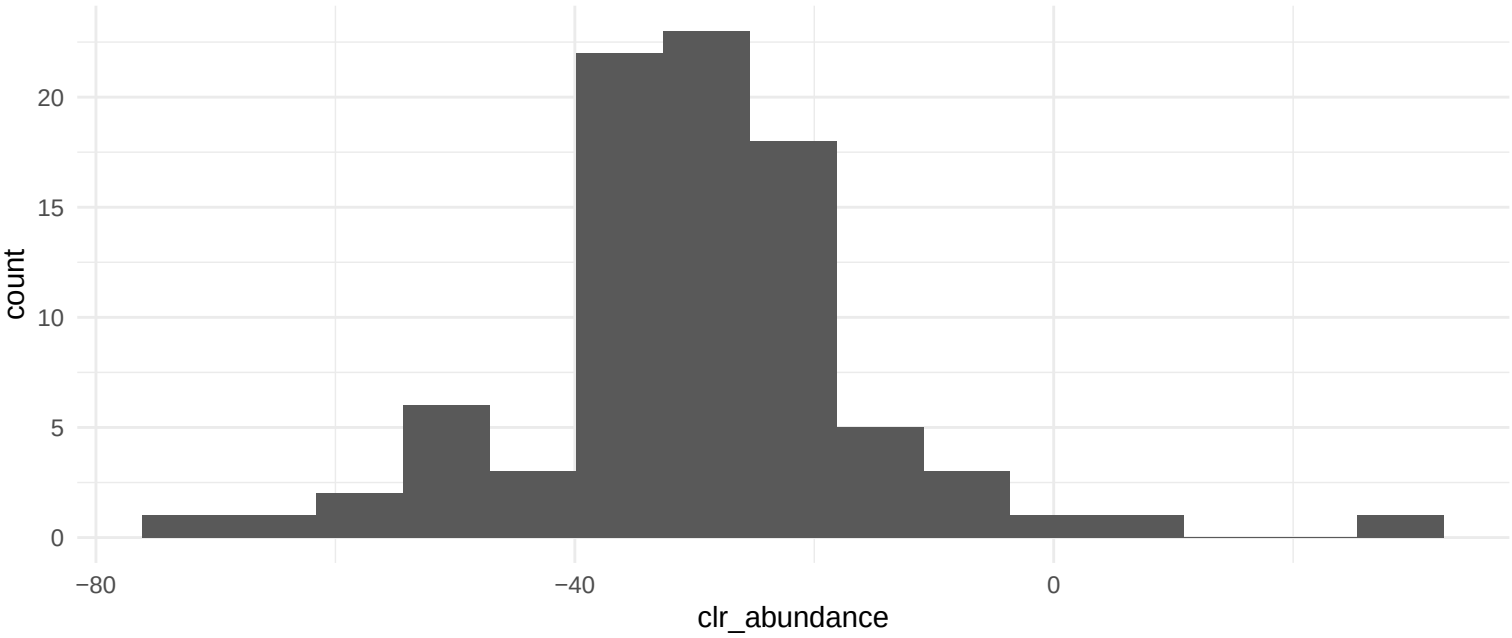

Distribution for 00333 Prodigiosin biosynthesis
count
clr_abundance
0
5
10
15
0
5
10
15


Distribution for 00340 Histidine metabolism
count
clr_abundance
0
5
10
0
25
50
75

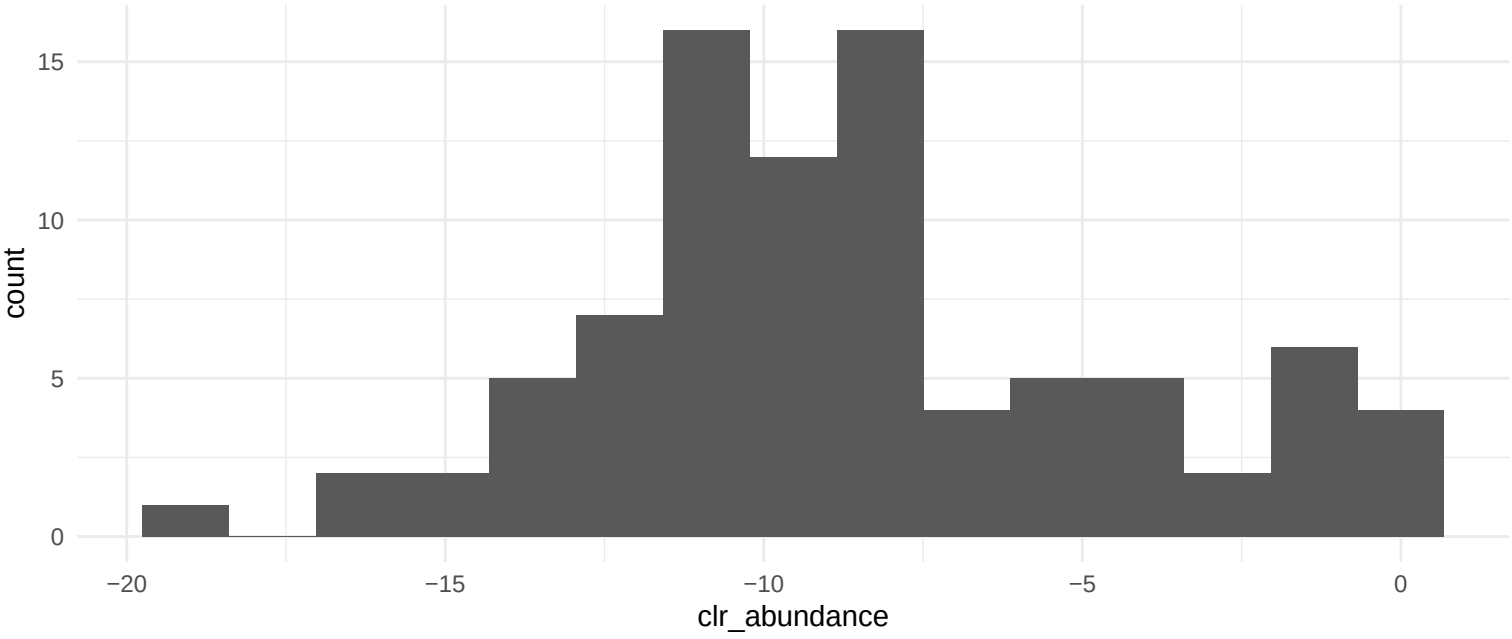
Distribution for 00350 Tyrosine metabolism

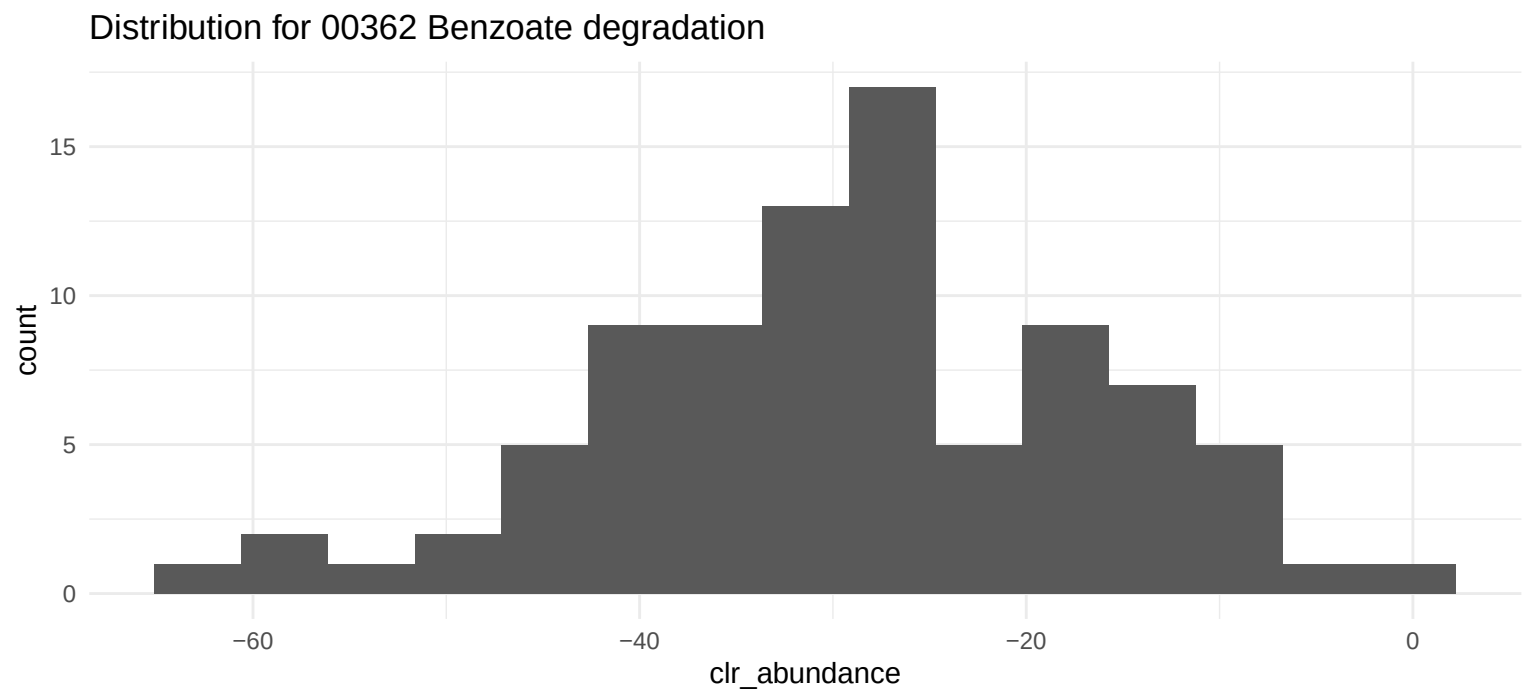


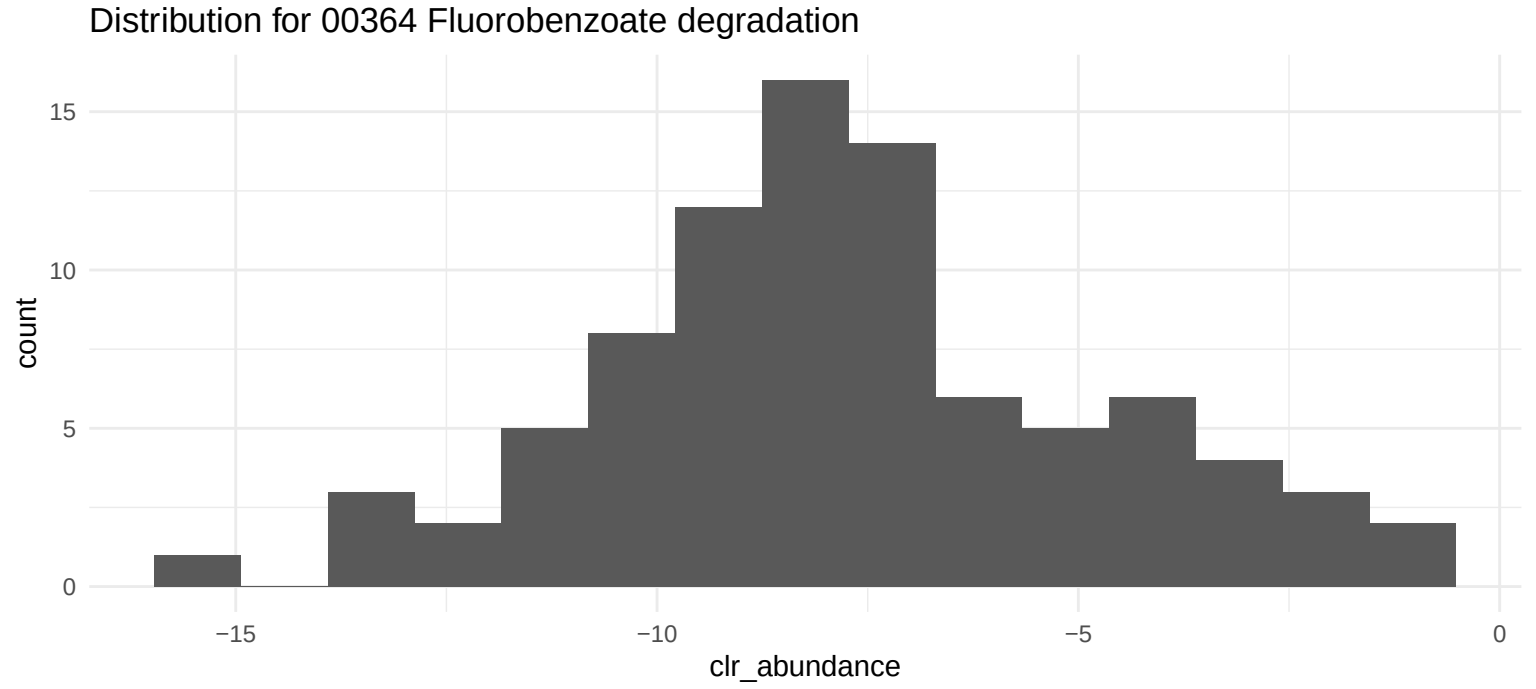
Distribution for 00360 Phenylalanine metabolism

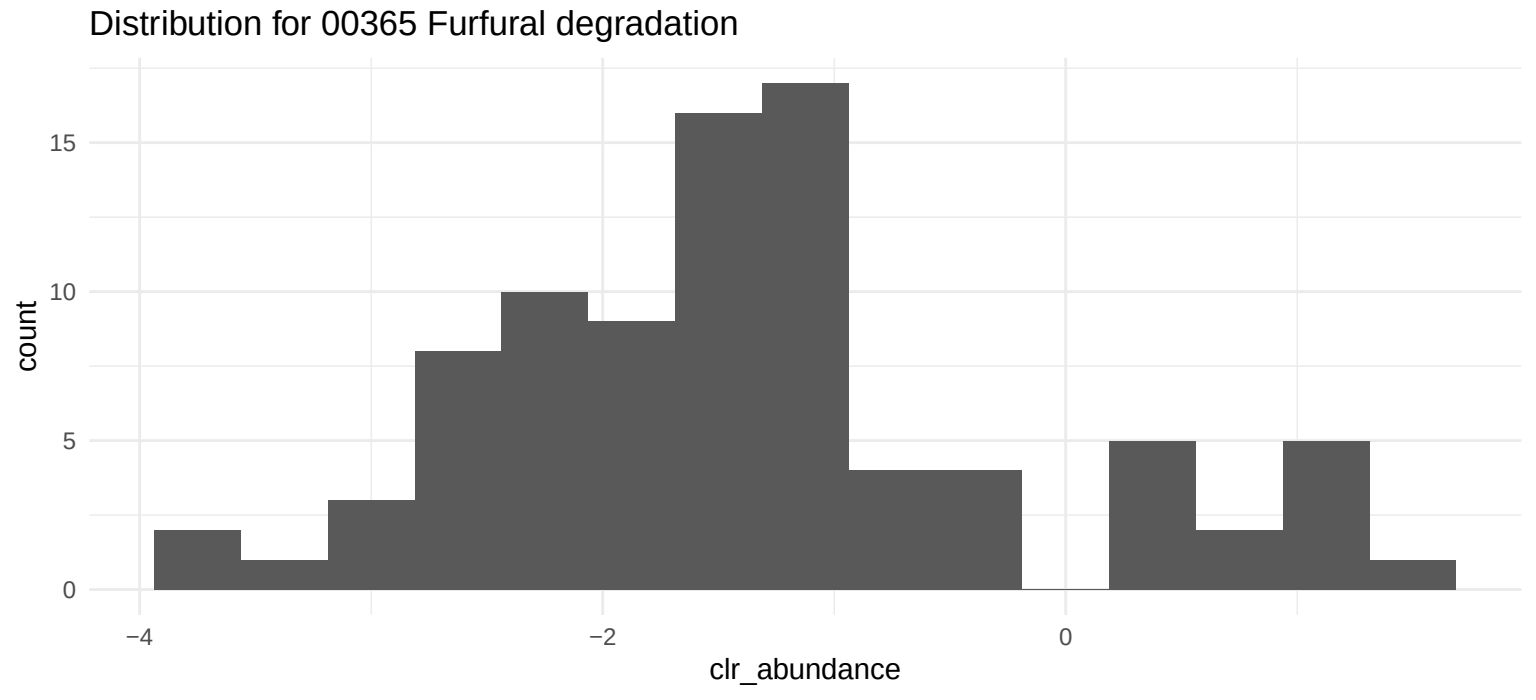


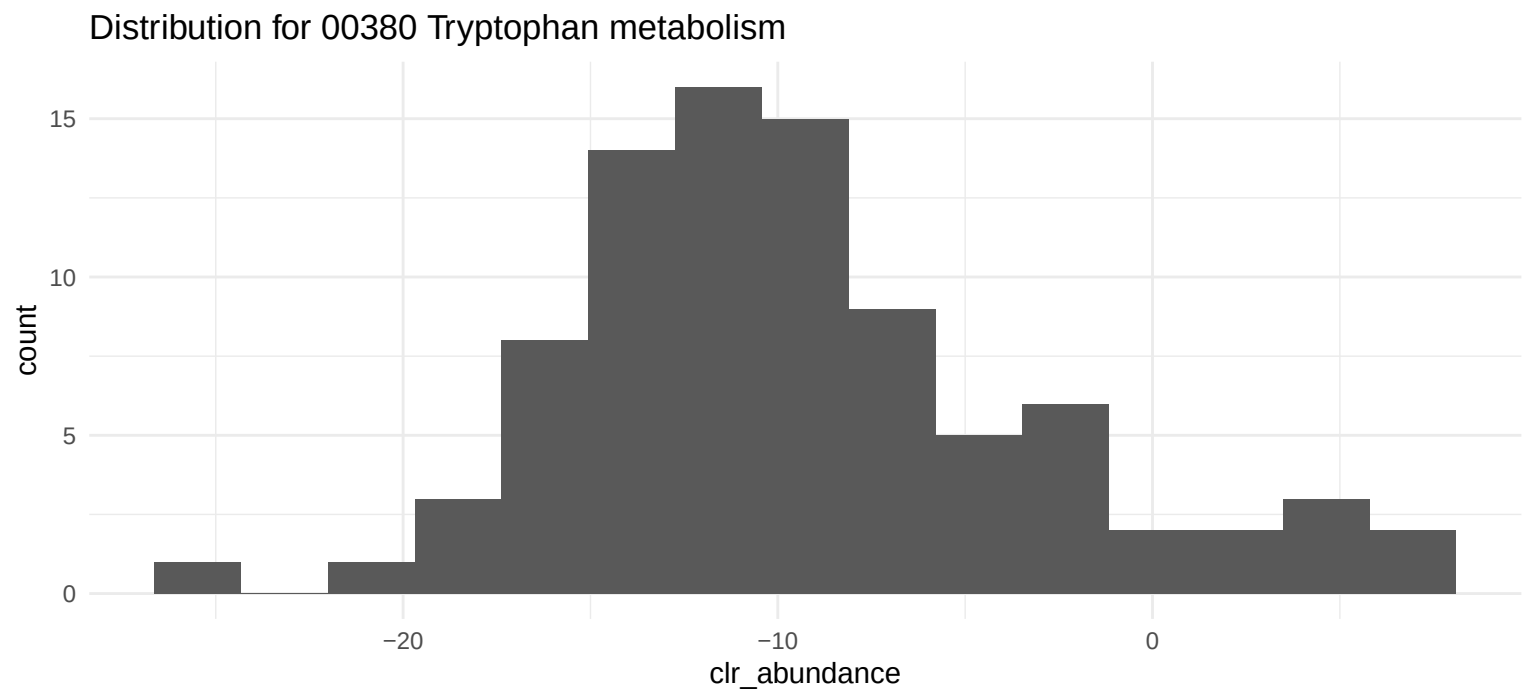
Distribution for 00361 Chlorocyclohexane and chlorobenzene degradation

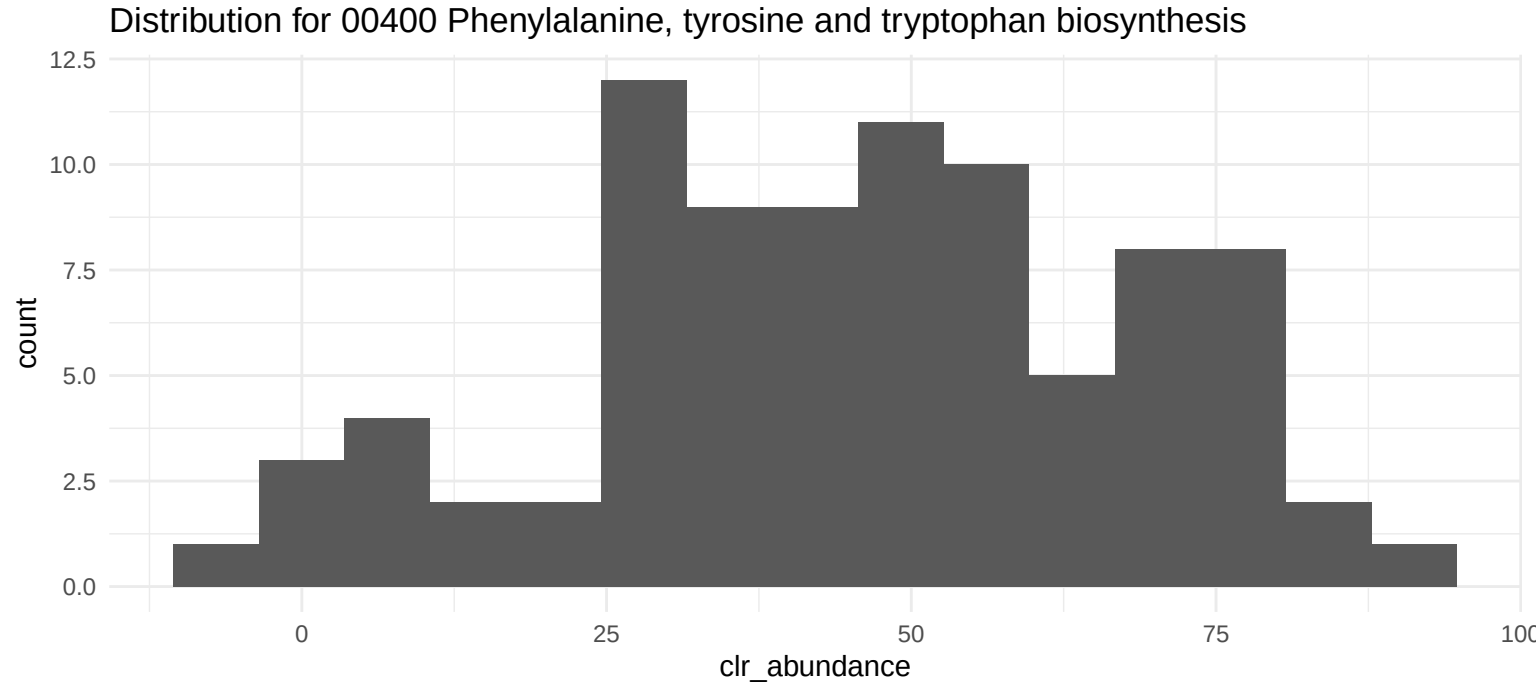


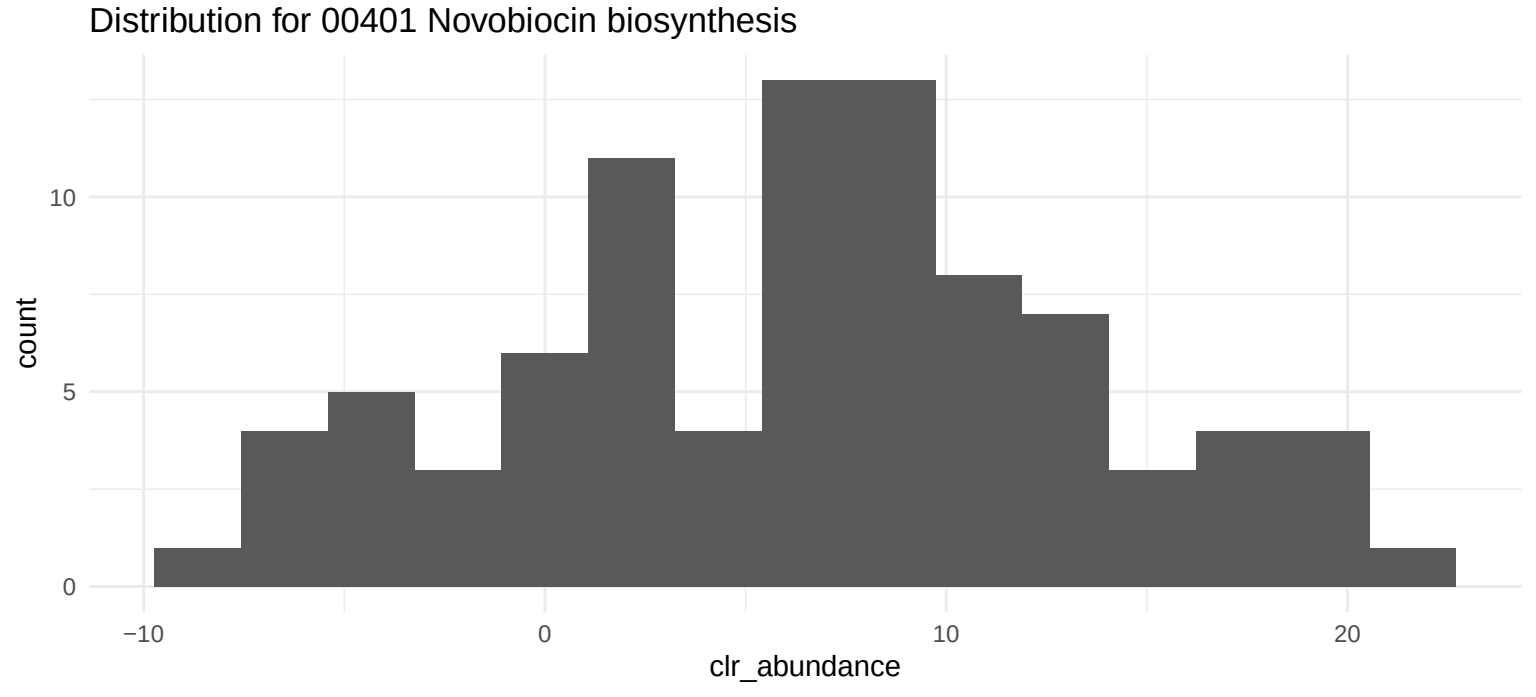

Distribution for 00362 Benzoate degradation
count
15
10
5
0
−60
−40
−20
0
clr_abundance


Distribution for 00364 Fluorobenzoate degradation
count
15
10
5
0
−15
−10
−5
0
clr_abundance

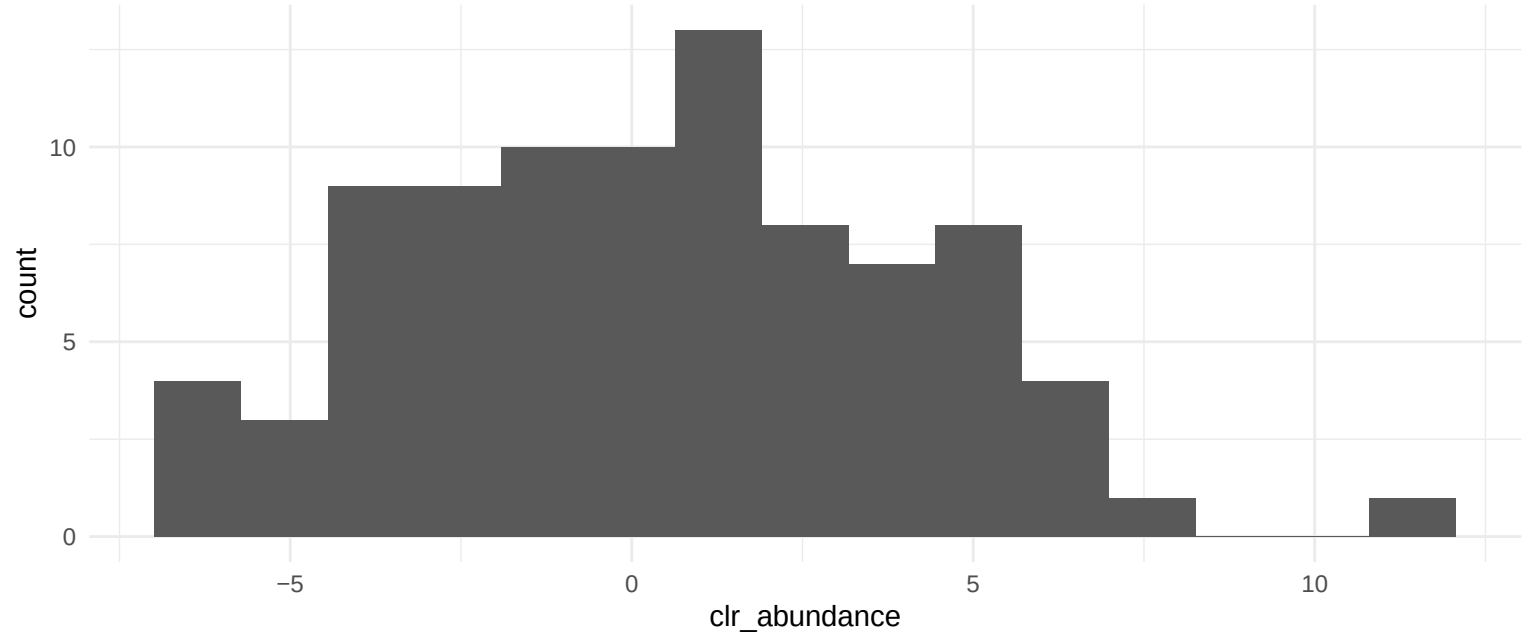

Distribution for 00365 Furfural degradation
count
15
10
5
0
−4
−2
0
clr_abundance


Distribution for 00380 Tryptophan metabolism
count
15
10
5
0
−20
−10
0
clr_abundance

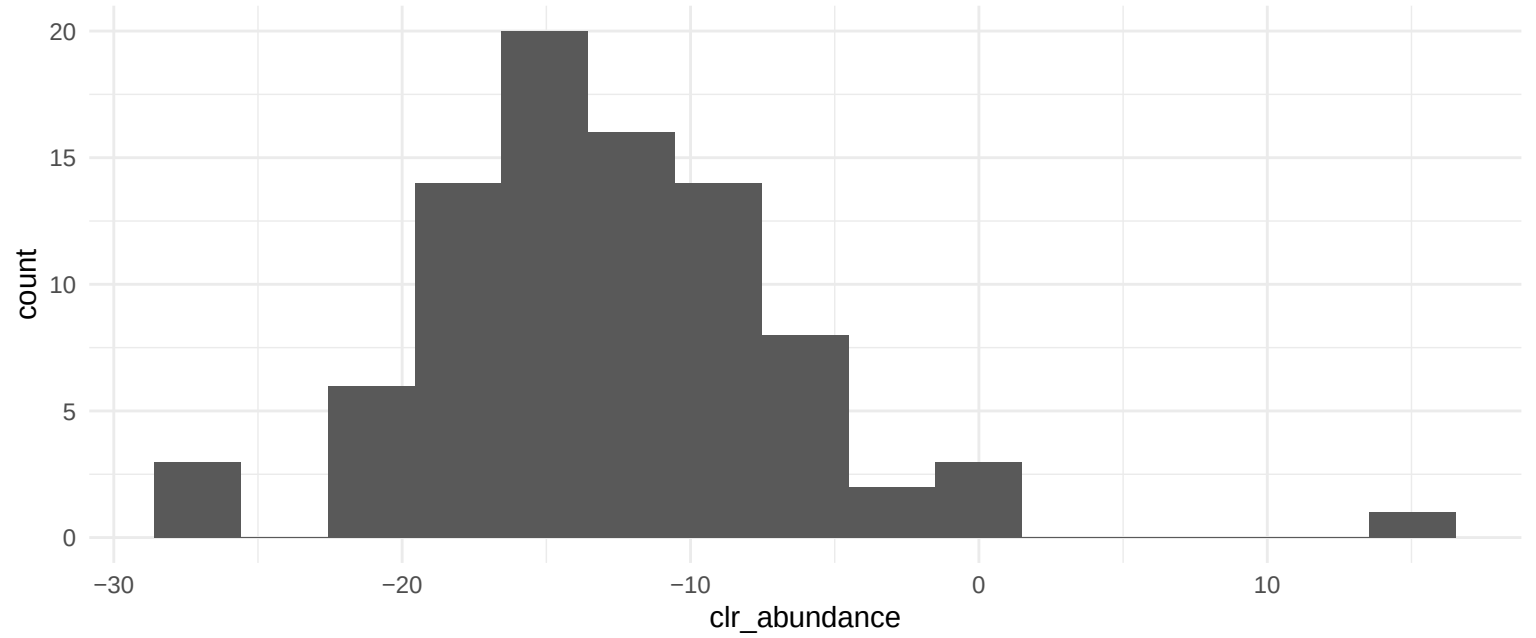

Distribution for 00400 Phenylalanine, tyrosine and tryptophan biosynthesis
count
12.5
10.0
7.5
5.0
2.5
0.0
0
25
50
75
100
clr_abundance


Distribution for 00401 Novobiocin biosynthesis
count
10
5
0
−10
0
10
20
clr_abundance

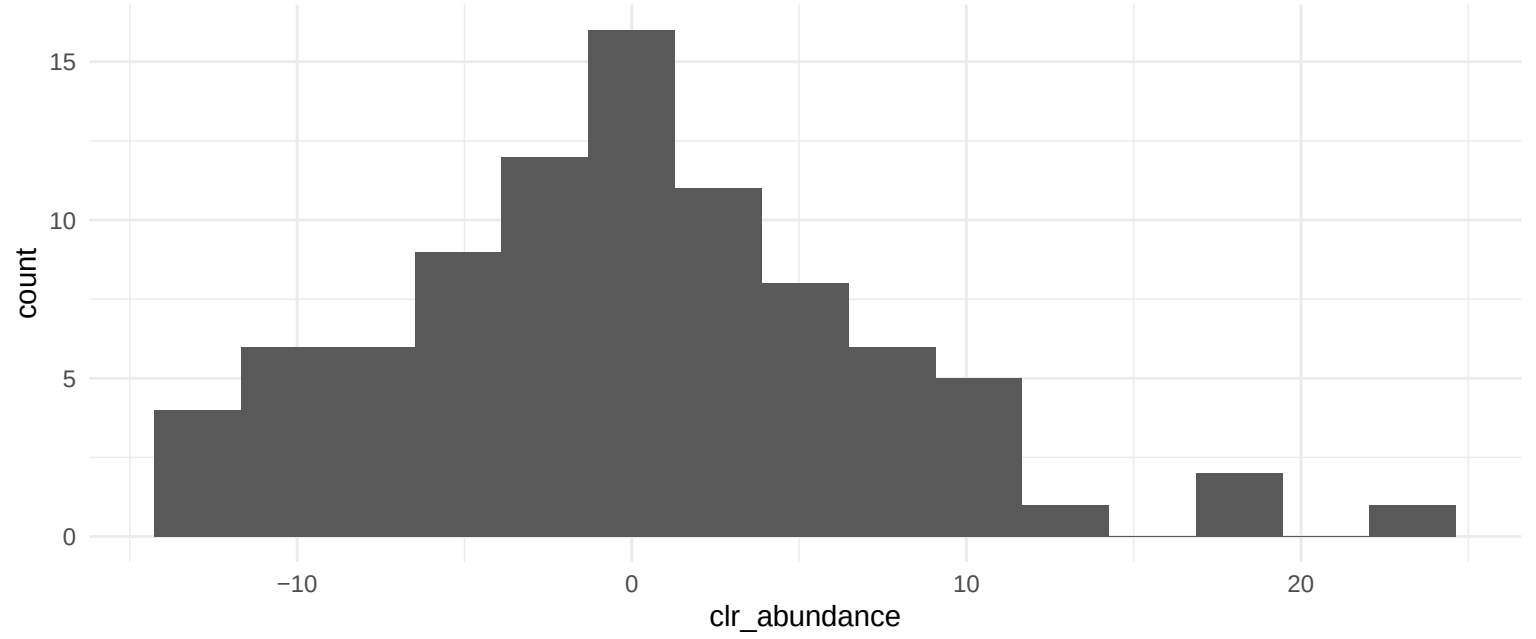
Distribution for 00405 Phenazine biosynthesis



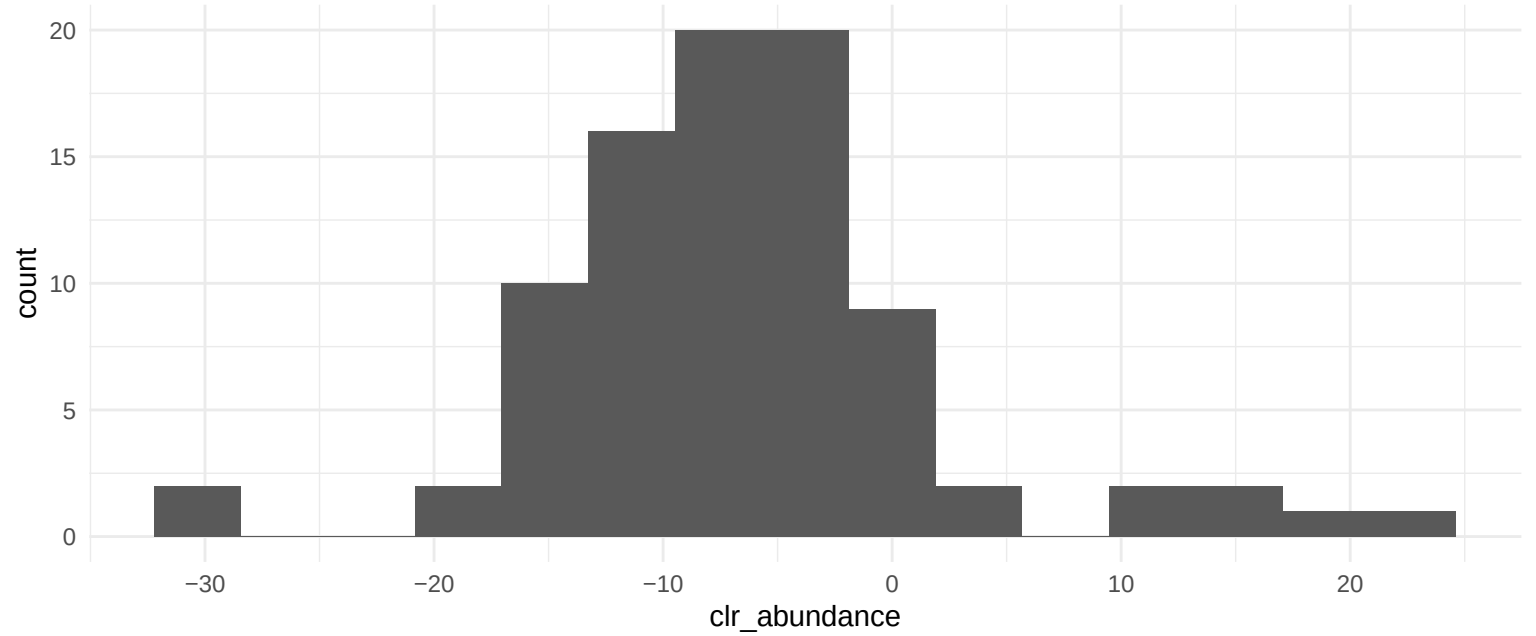
Distribution for 00410 beta−Alanine metabolism



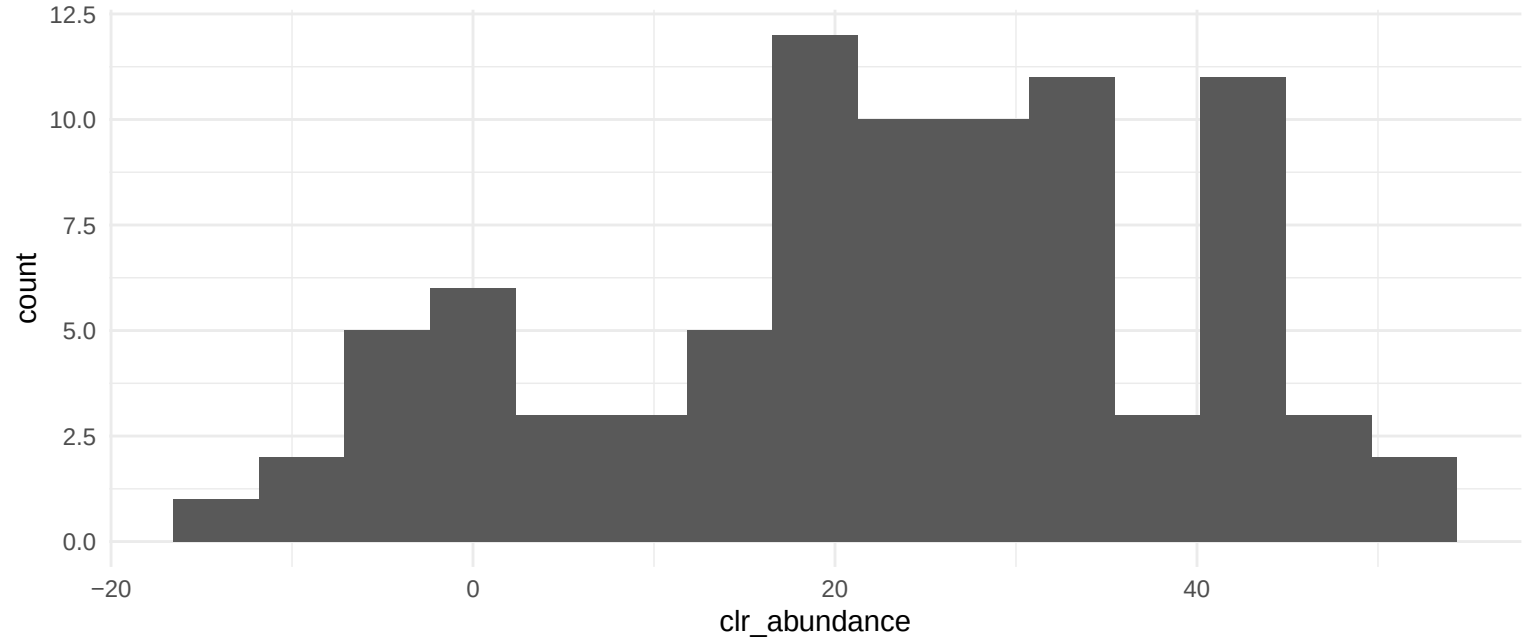
Distribution for 00430 Taurine and hypotaurine metabolism



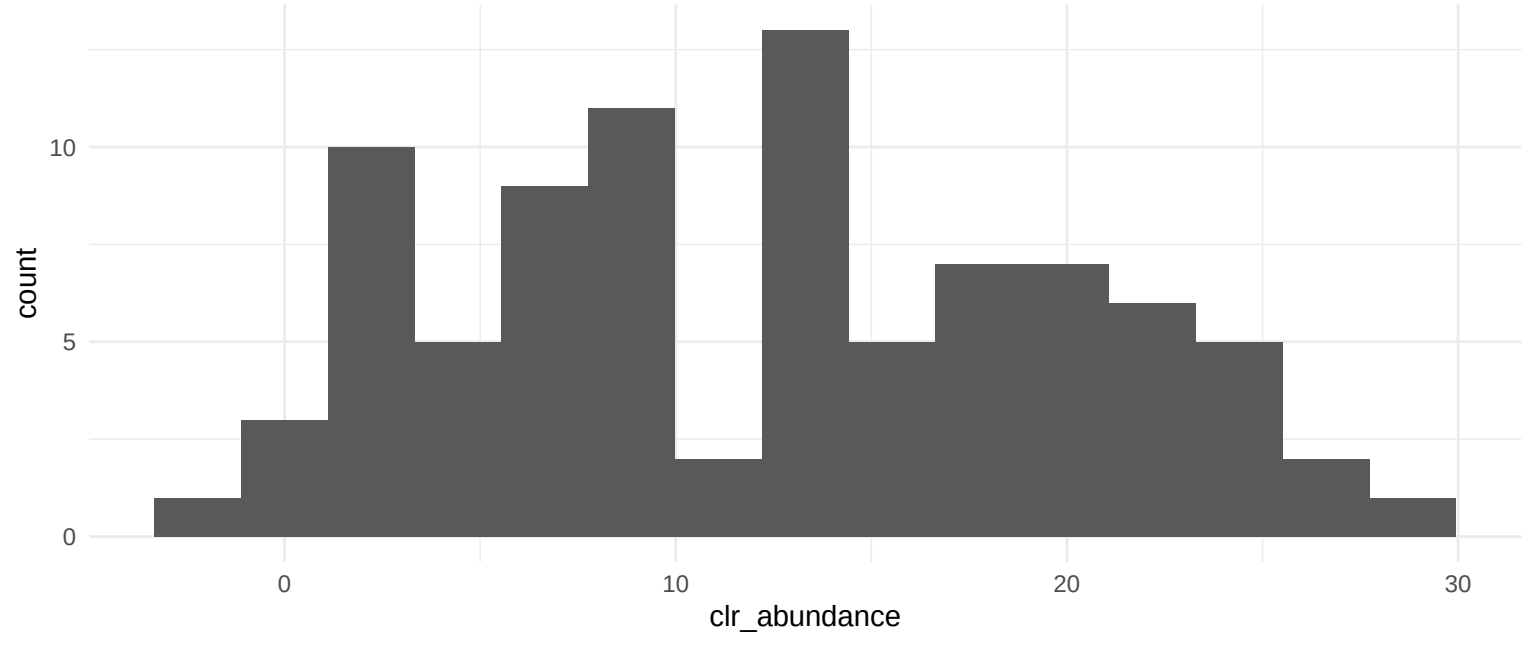
Distribution for 00440 Phosphonate and phosphinate metabolism

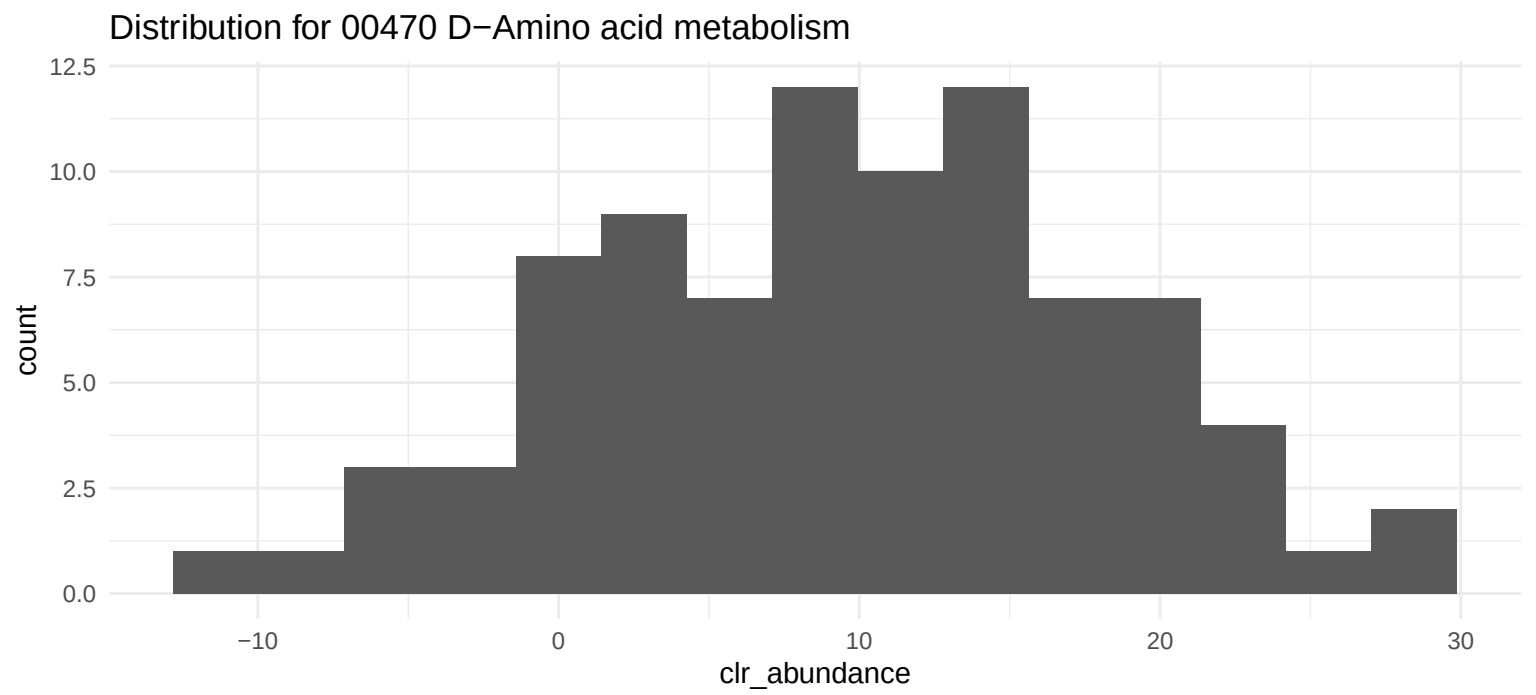


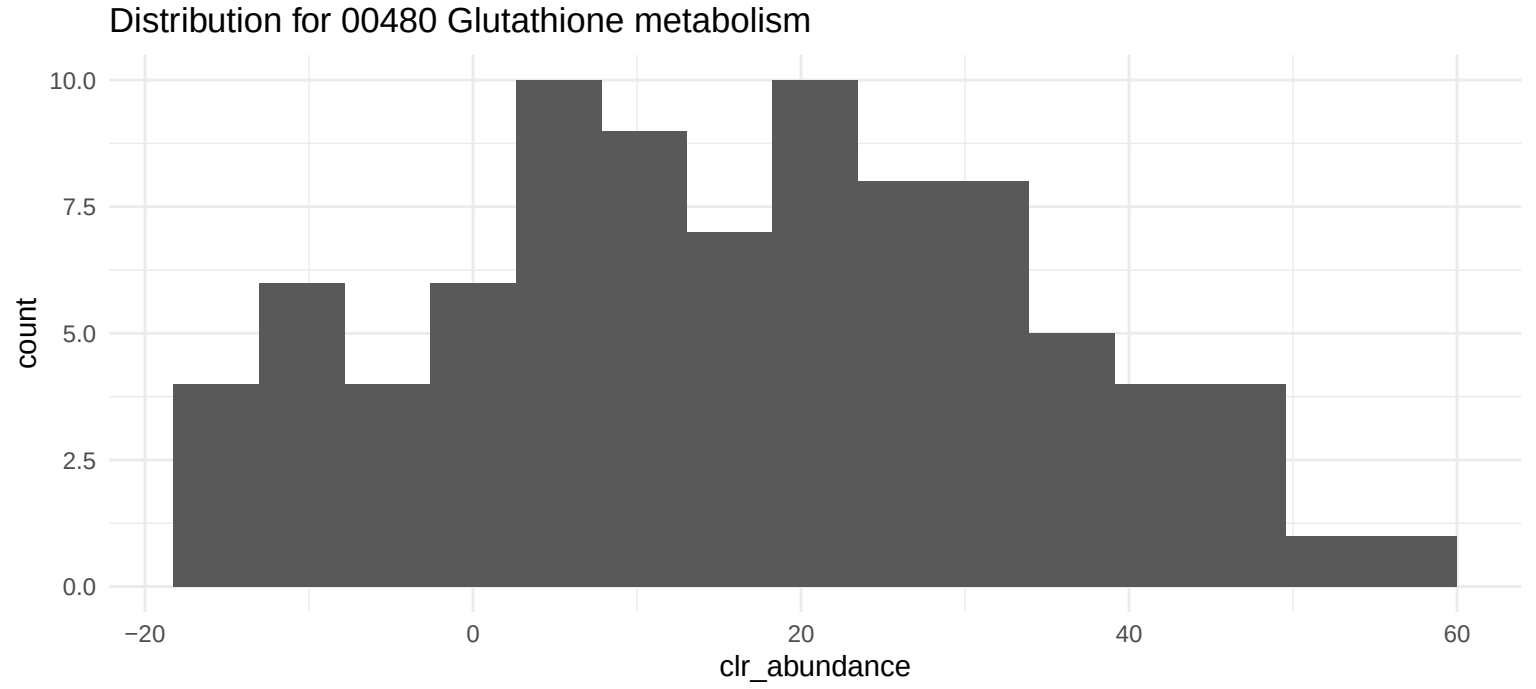
Distribution for 00450 Selenocompound metabolism

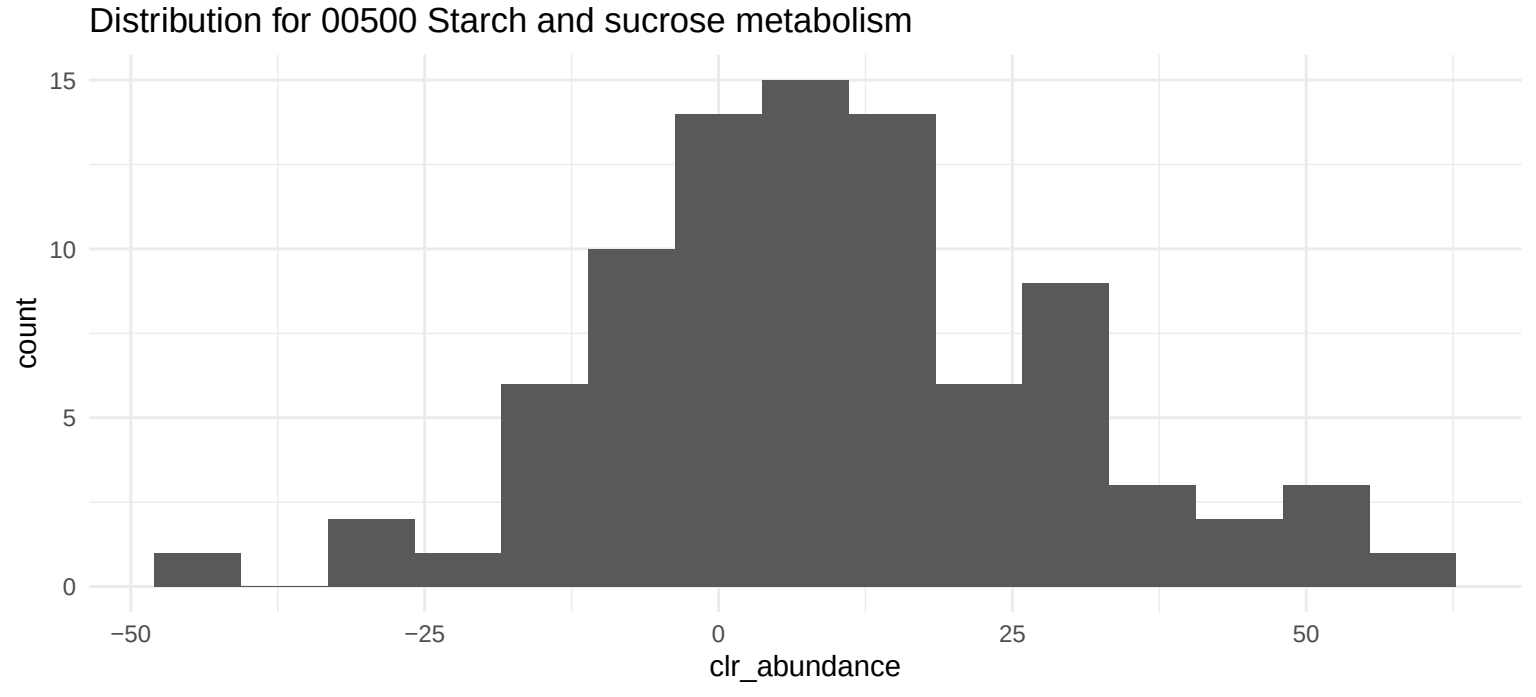


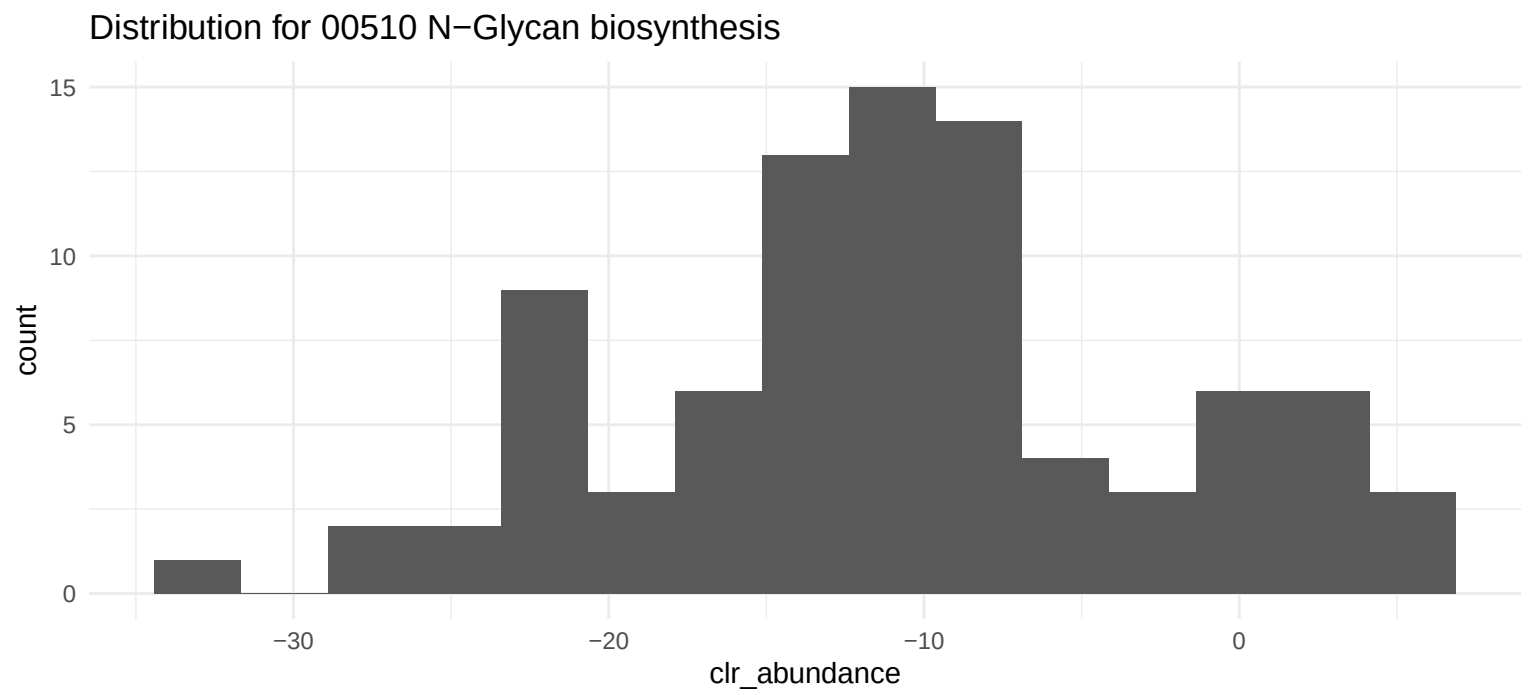
Distribution for 00460 Cyanoamino acid metabolism

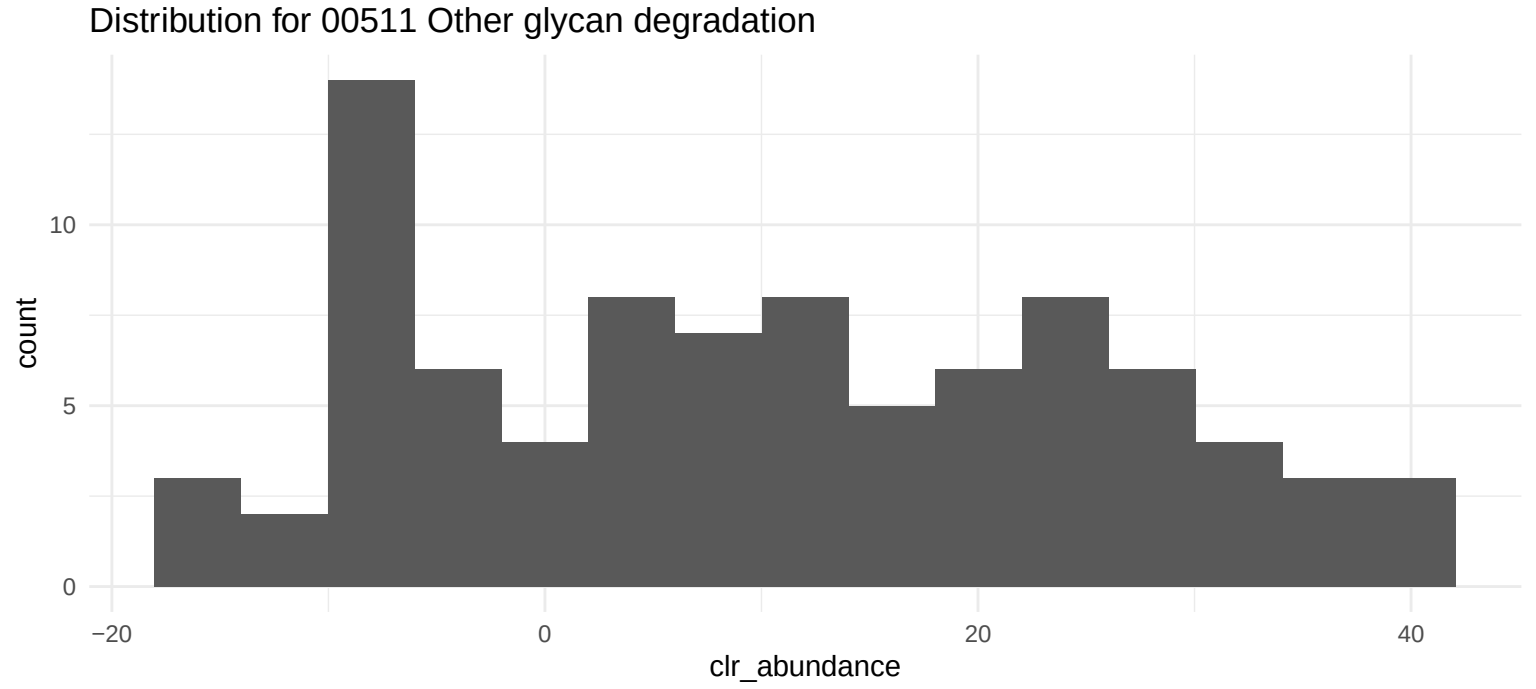


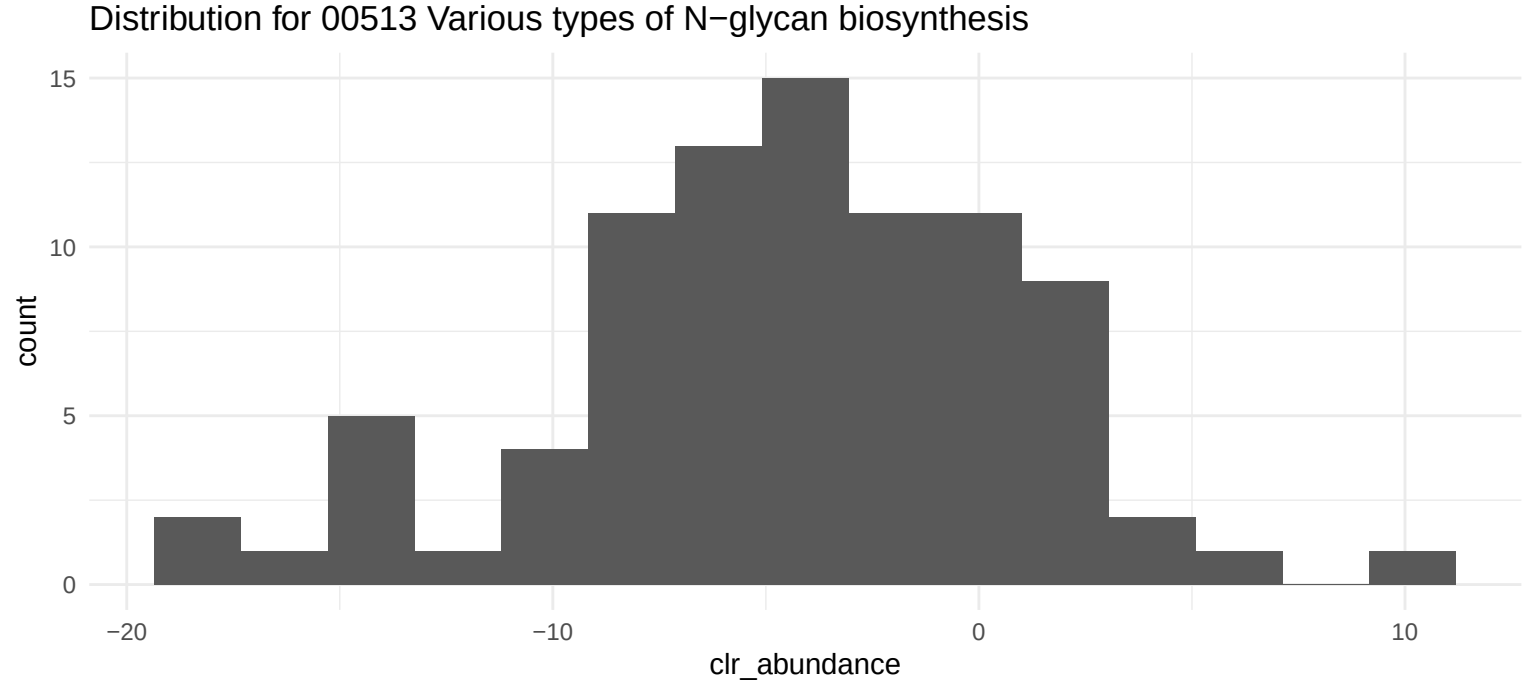

Distribution for 00470 D−Amino acid metabolism
count
clr_abundance
12.5
10.0
7.5
5.0
2.5
0.0
−10
0
10
20
30

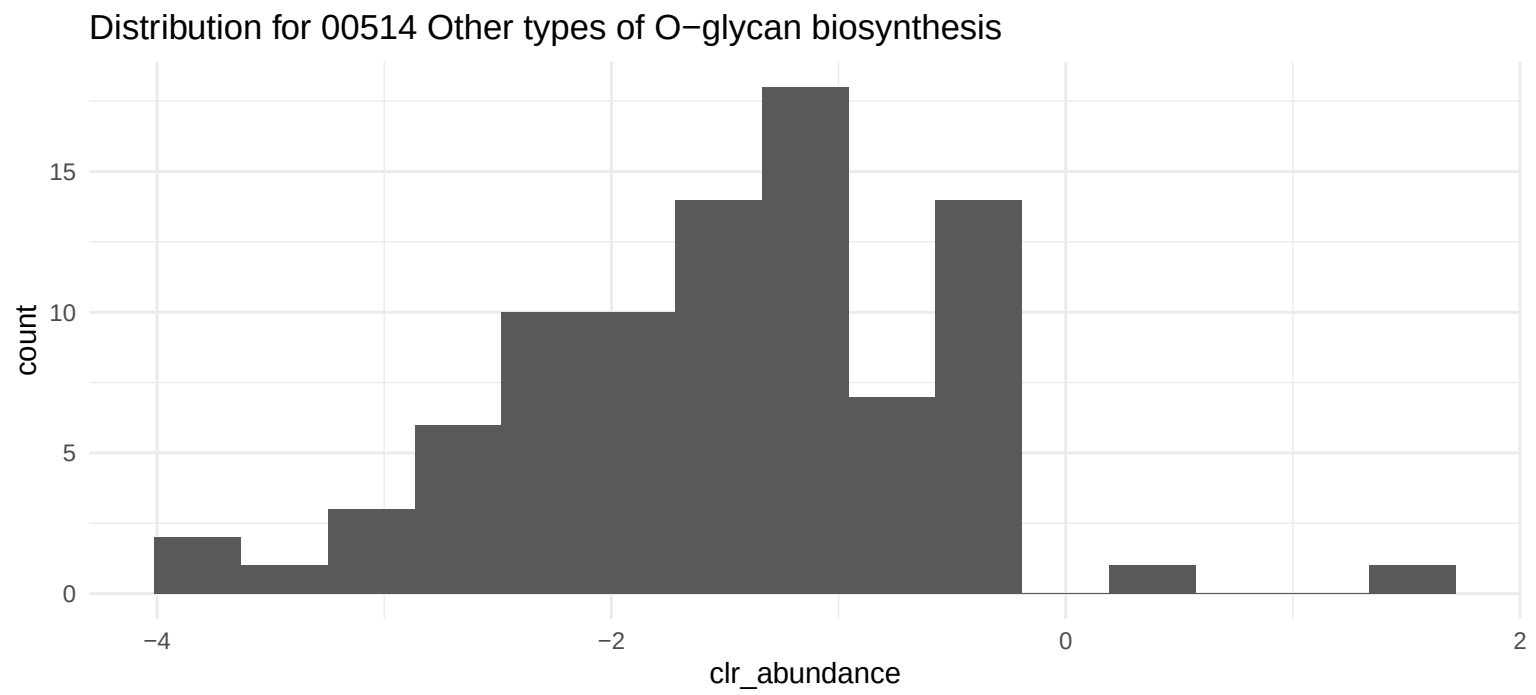

Distribution for 00480 Glutathione metabolism
count
clr_abundance
10.0
7.5
5.0
2.5
0.0
−20
0
20
40
60

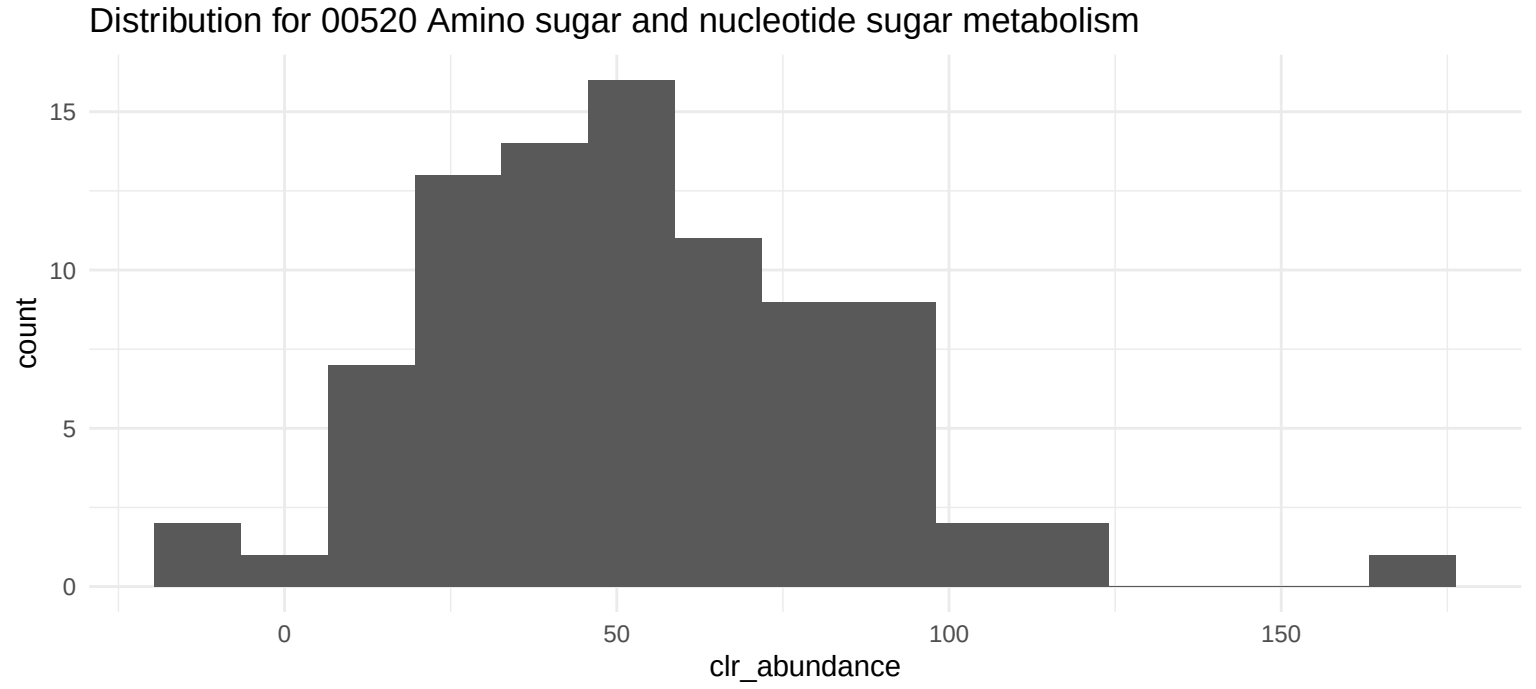

Distribution for 00500 Starch and sucrose metabolism
count
clr_abundance
15
10
5
0
−50
−25
0
25
50

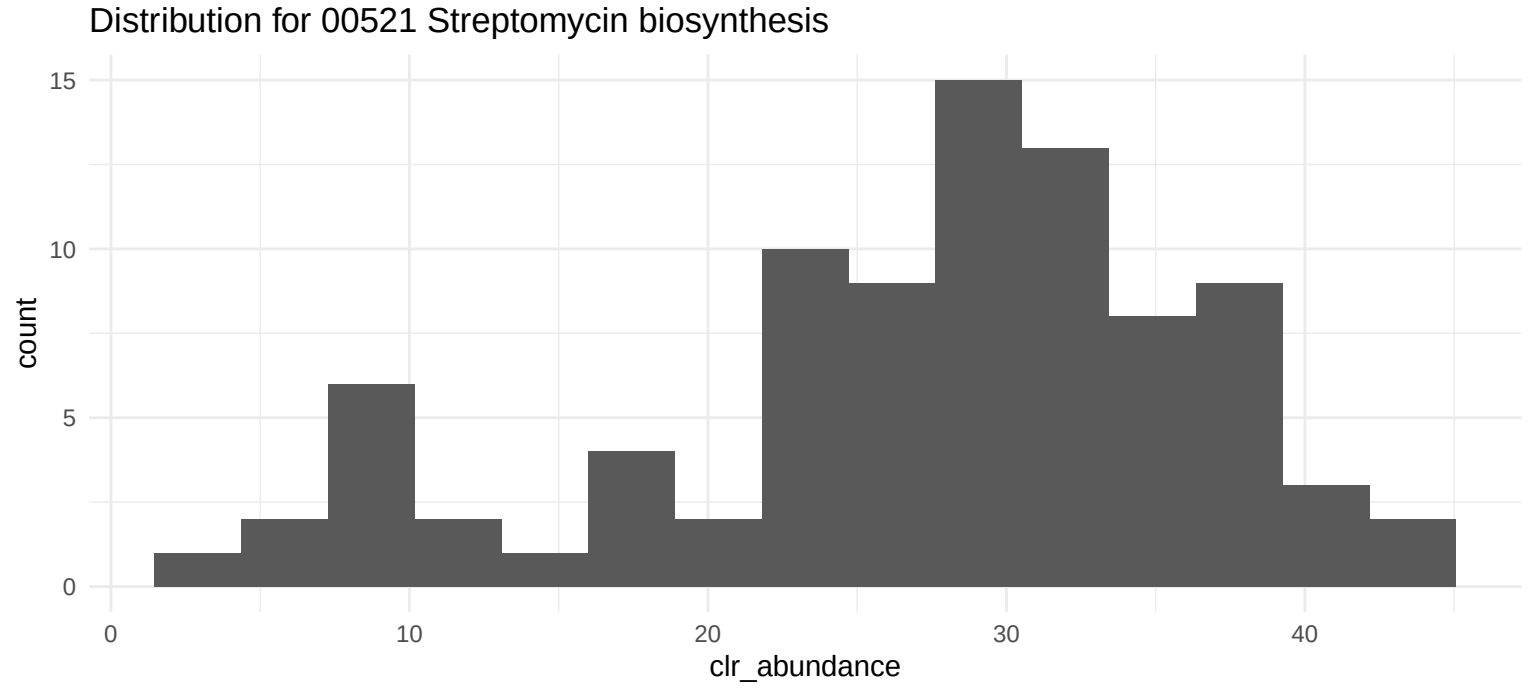

Distribution for 00510 N−Glycan biosynthesis
count
15
10
5
0
−30
−20
−10
0
clr_abundance


Distribution for 00511 Other glycan degradation
count
10
5
0
−20
0
20
40
clr_abundance

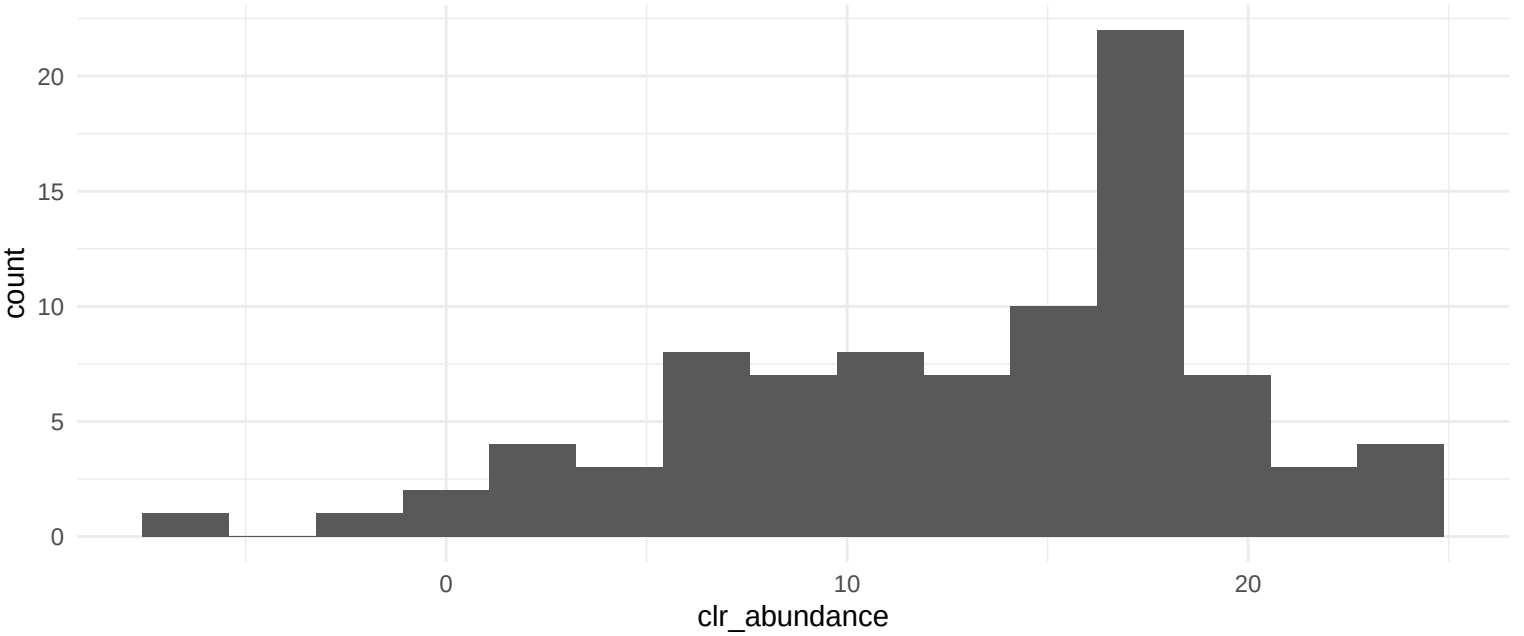

Distribution for 00513 Various types of N−glycan biosynthesis
count
15
10
5
0
−20
−10
0
10
clr_abundance


Distribution for 00514 Other types of O−glycan biosynthesis
count
15
10
5
0
−4
−2
0
2
clr_abundance

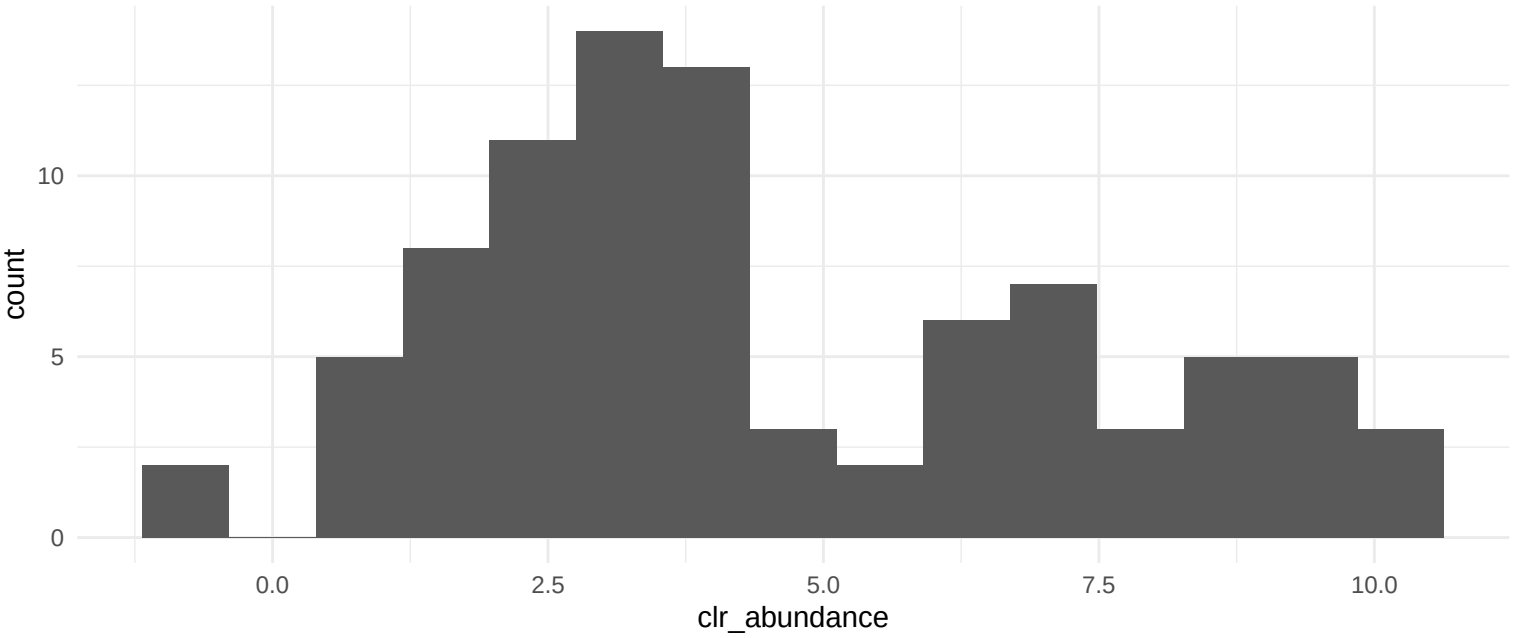

Distribution for 00520 Amino sugar and nucleotide sugar metabolism
count
15
10
5
0
0
50
100
150
clr_abundance


Distribution for 00521 Streptomycin biosynthesis
count
15
10
5
0
0
10
20
30
40
clr_abundance

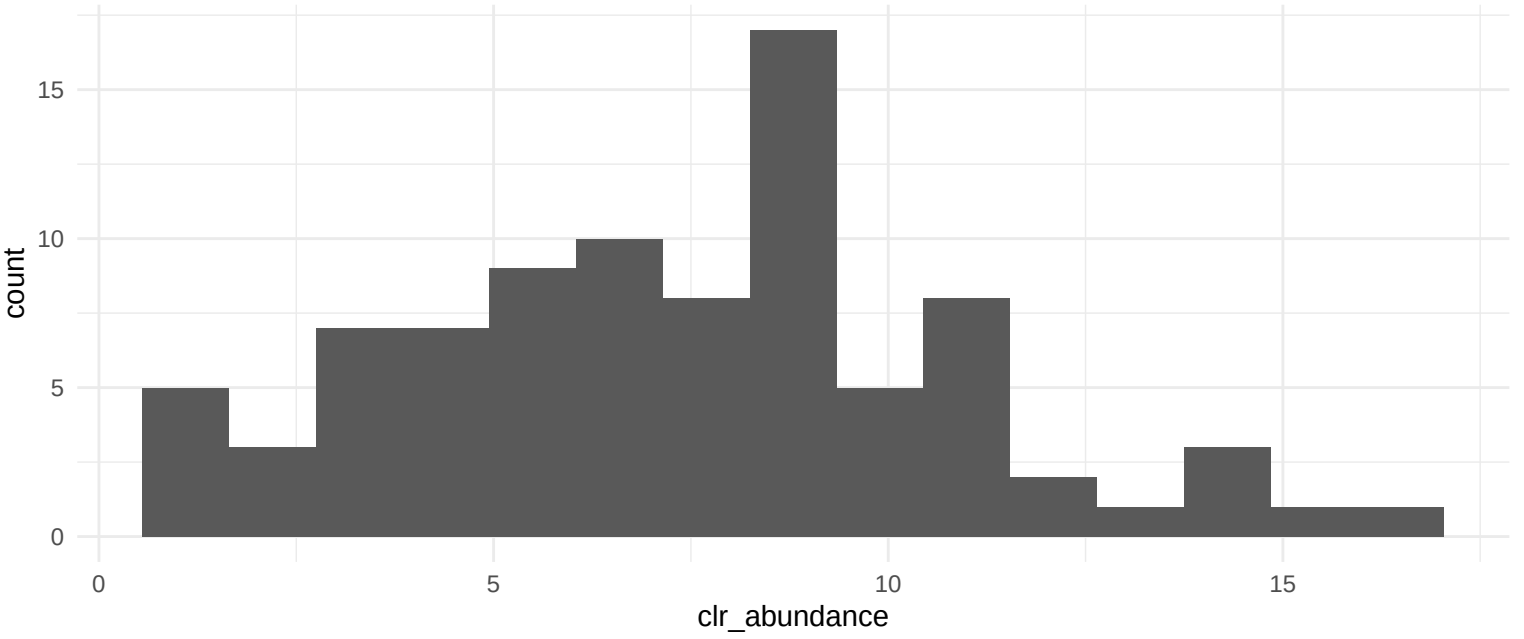
Distribution for 00523 Polyketide sugar unit biosynthesis

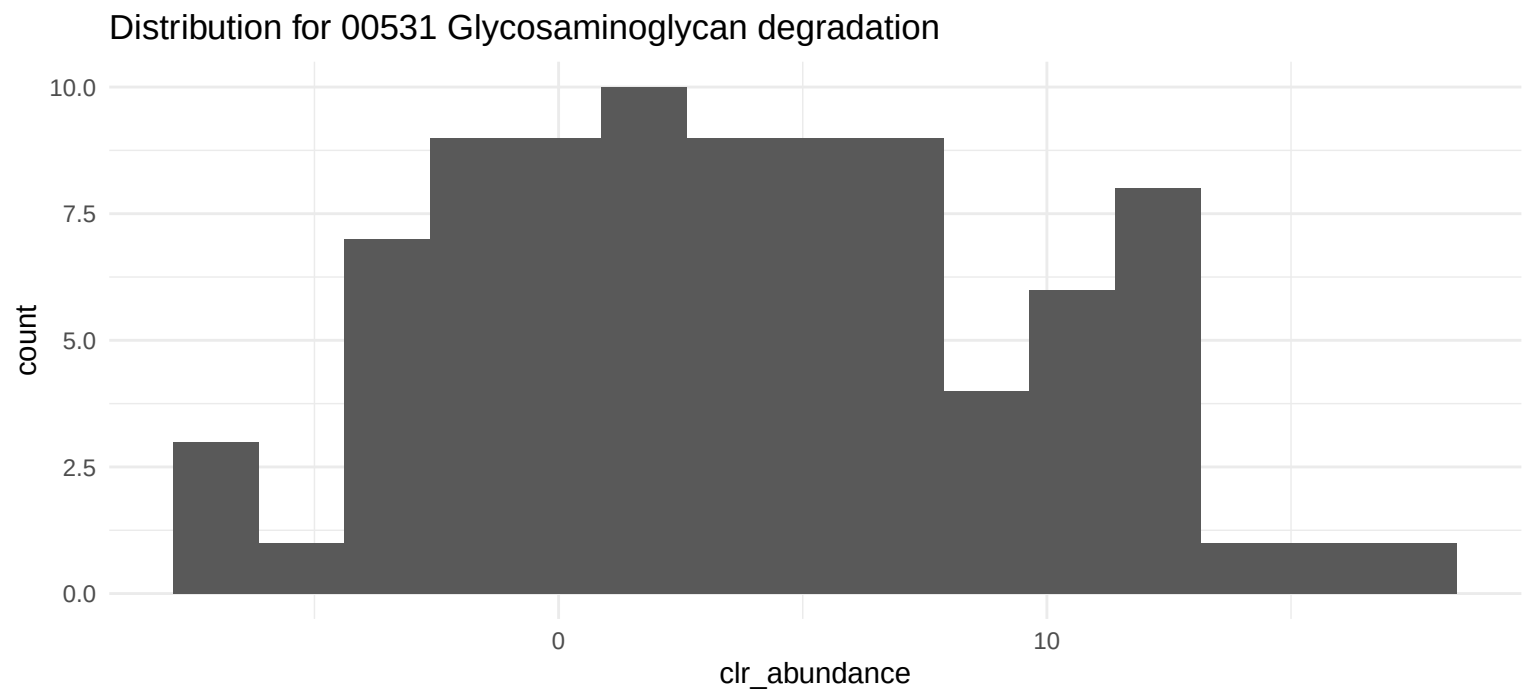


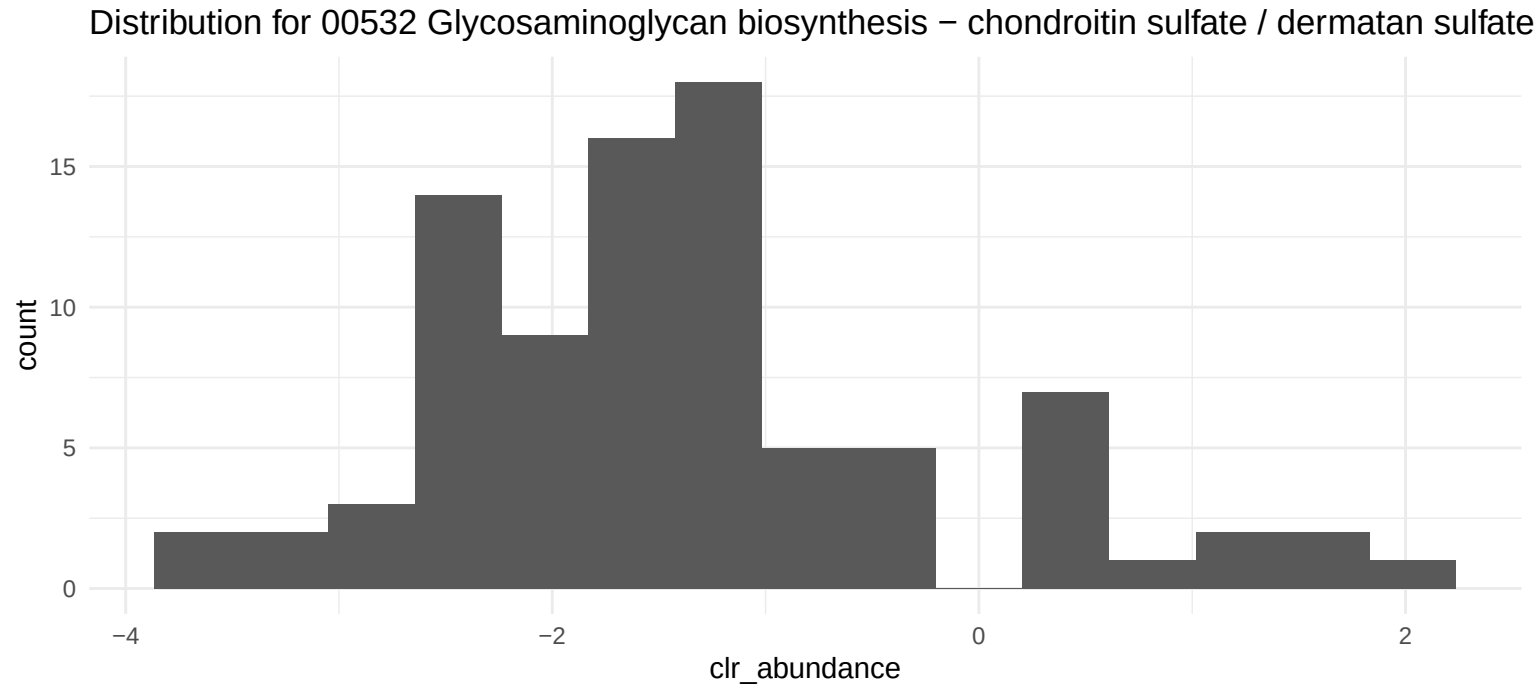
Distribution for 00524 Neomycin, kanamycin and gentamicin biosynthesis

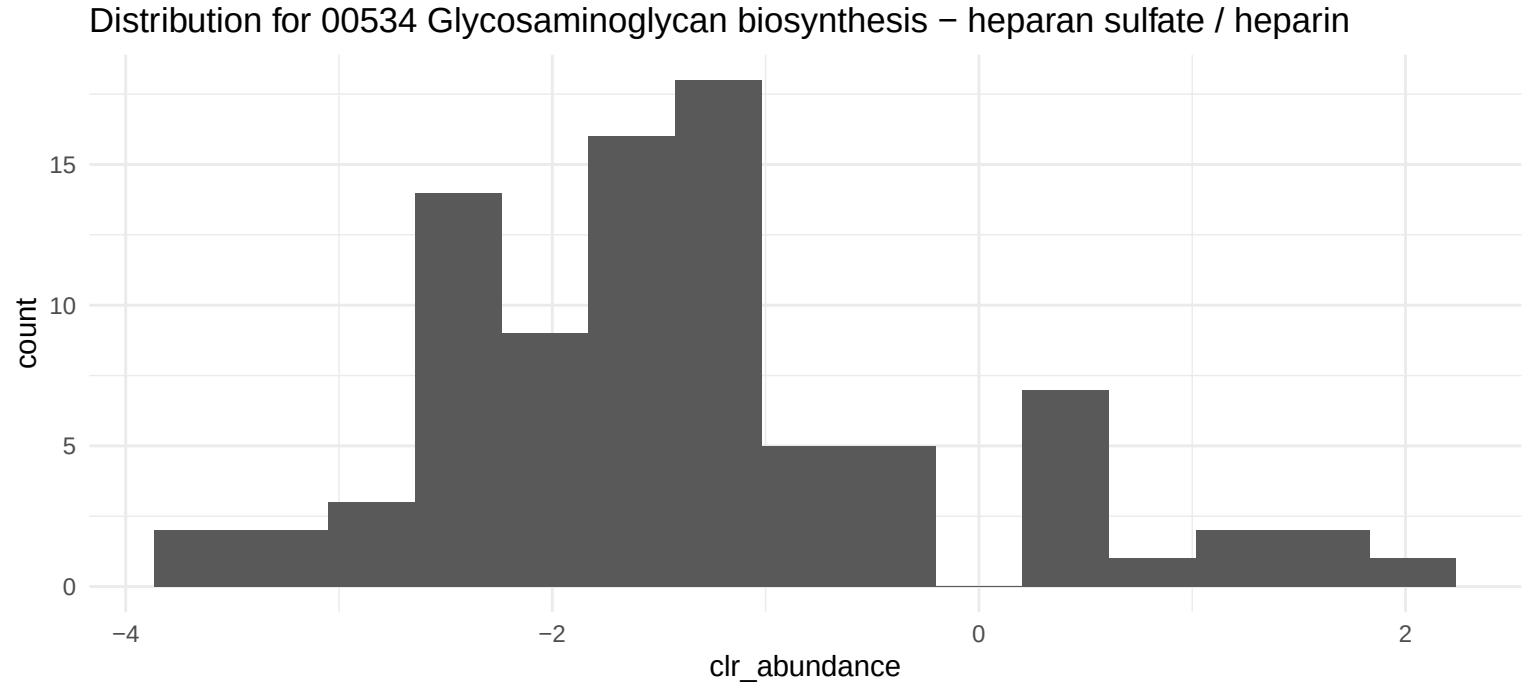


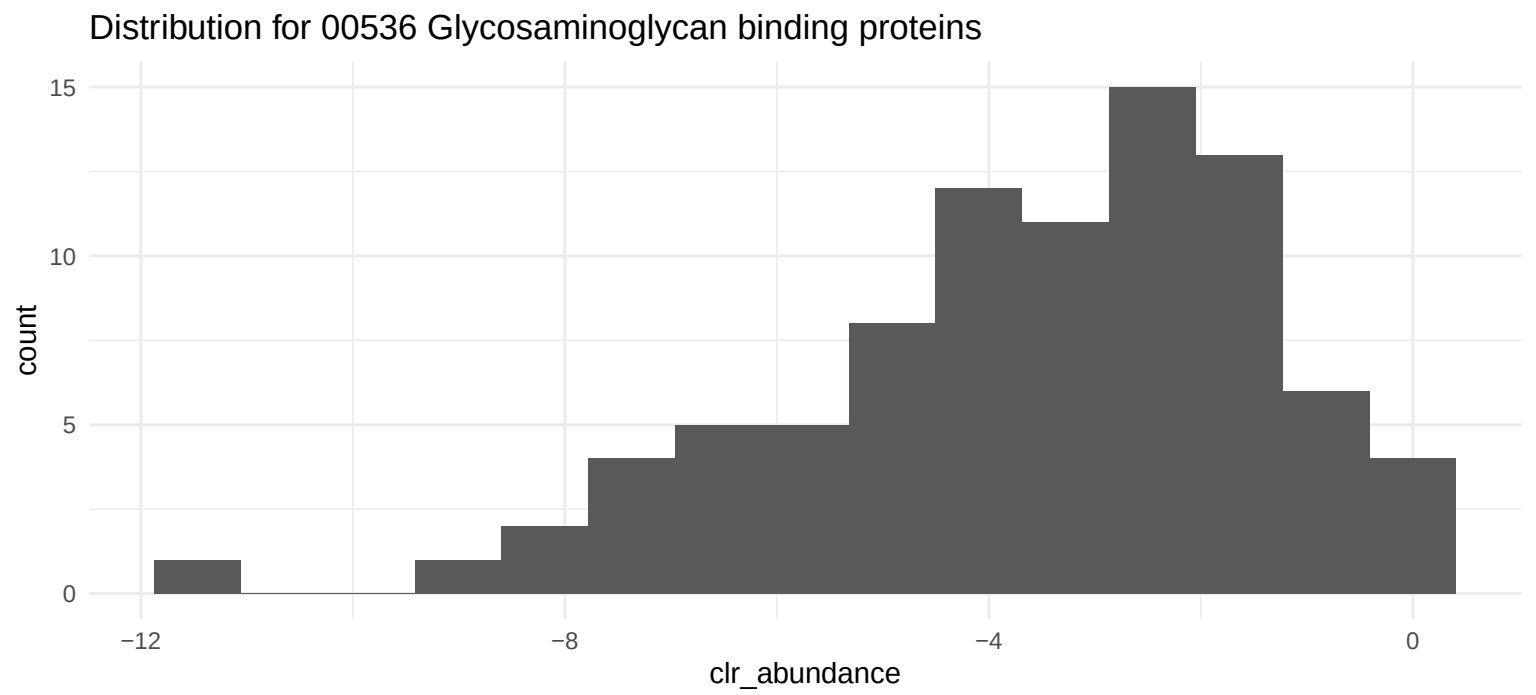
Distribution for 00525 Acarbose and validamycin biosynthesis

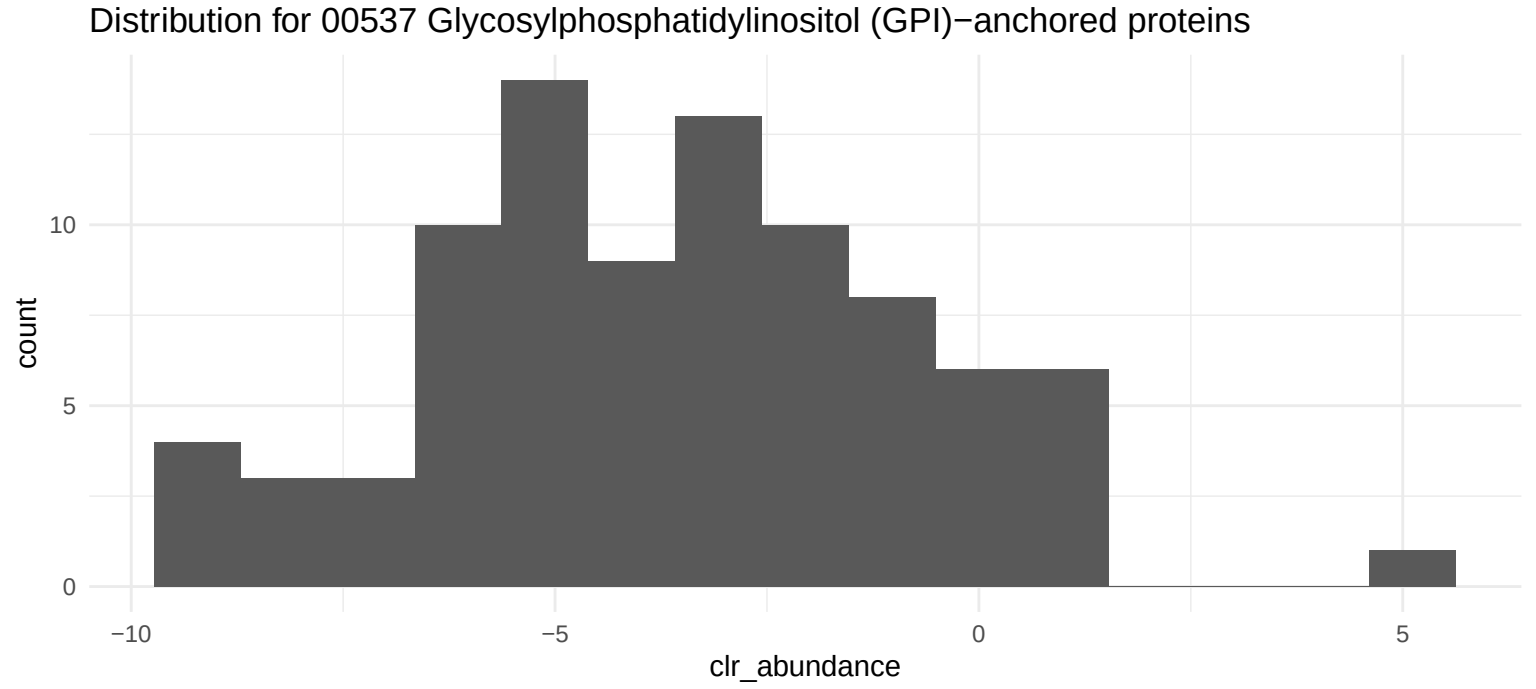


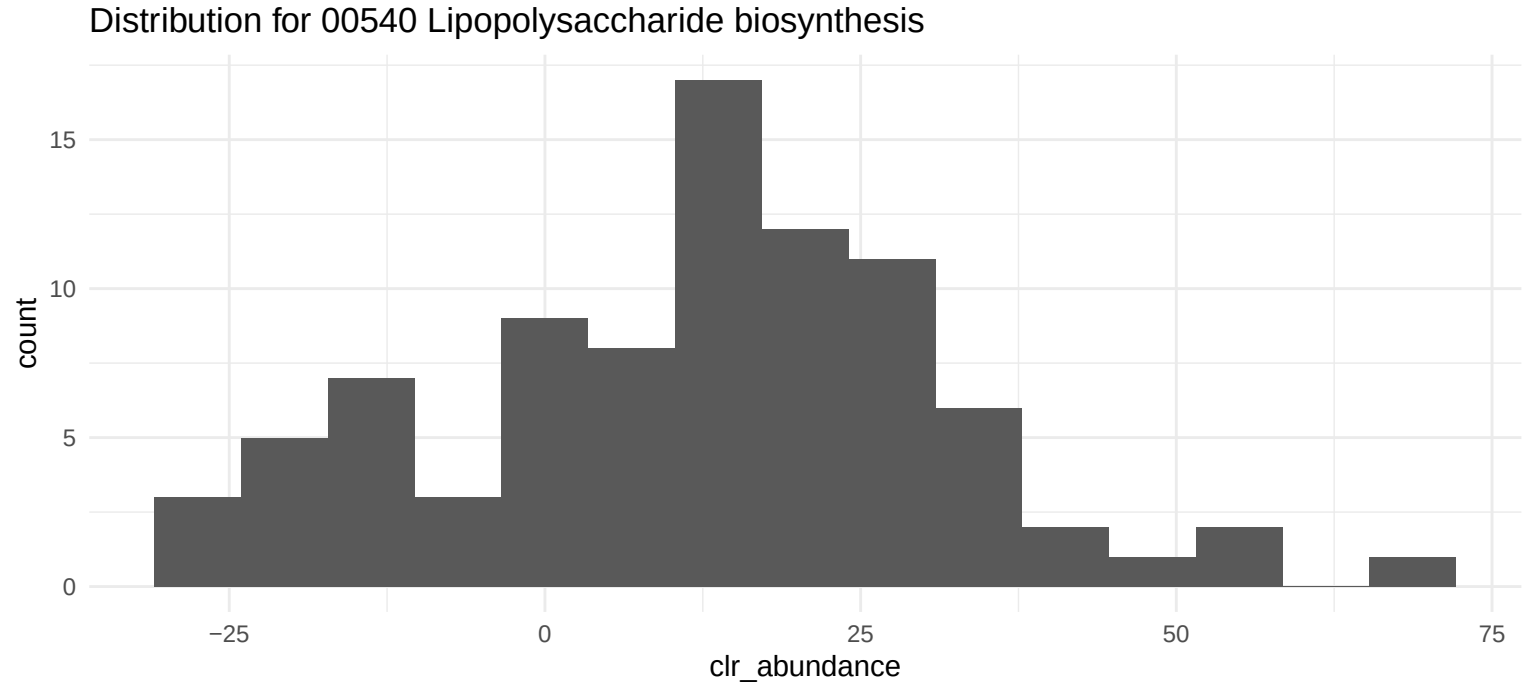

Distribution for 00531 Glycosaminoglycan degradation
count
10.0
7.5
5.0
2.5
0.0
0
10
clr_abundance

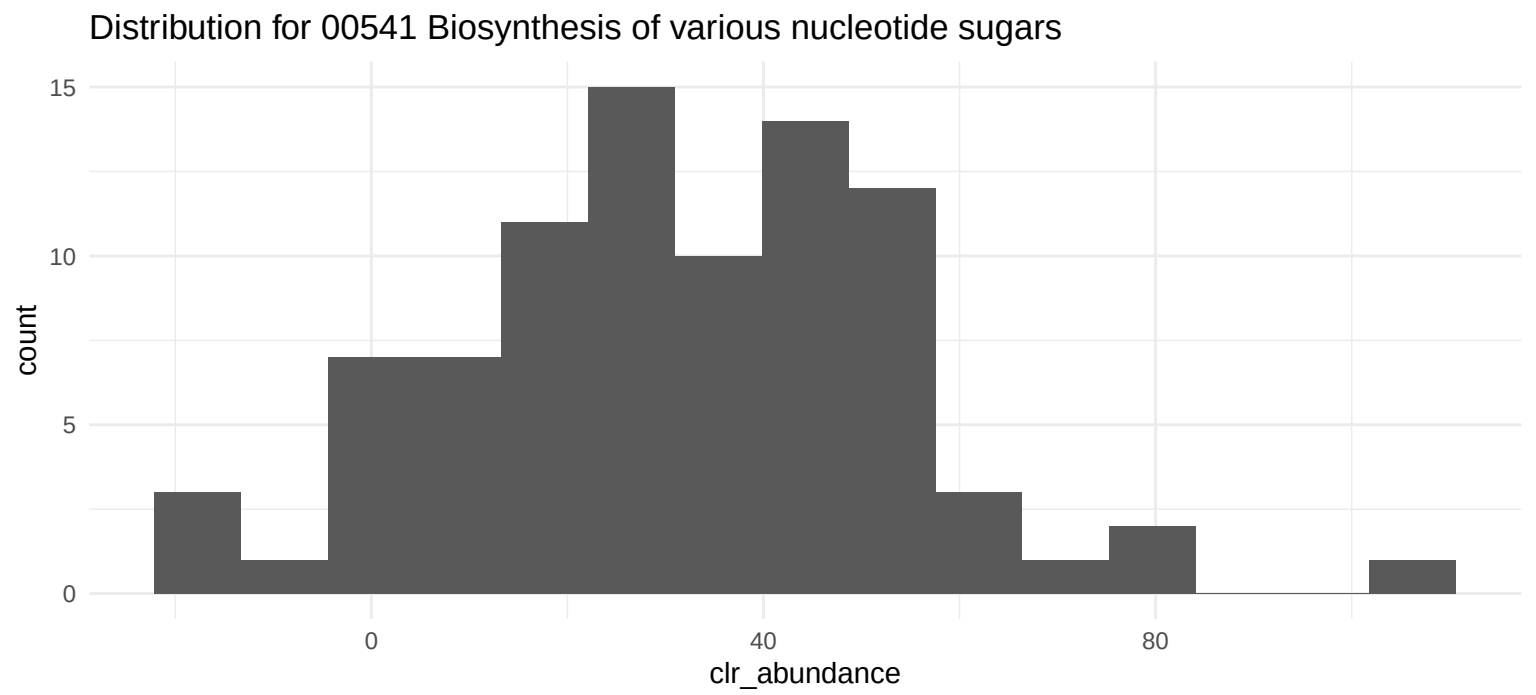

Distribution for 00532 Glycosaminoglycan biosynthesis − chondroitin sulfate / dermatan sulfate
count
15
10
5
0
−4
−2
0
2
clr_abundance

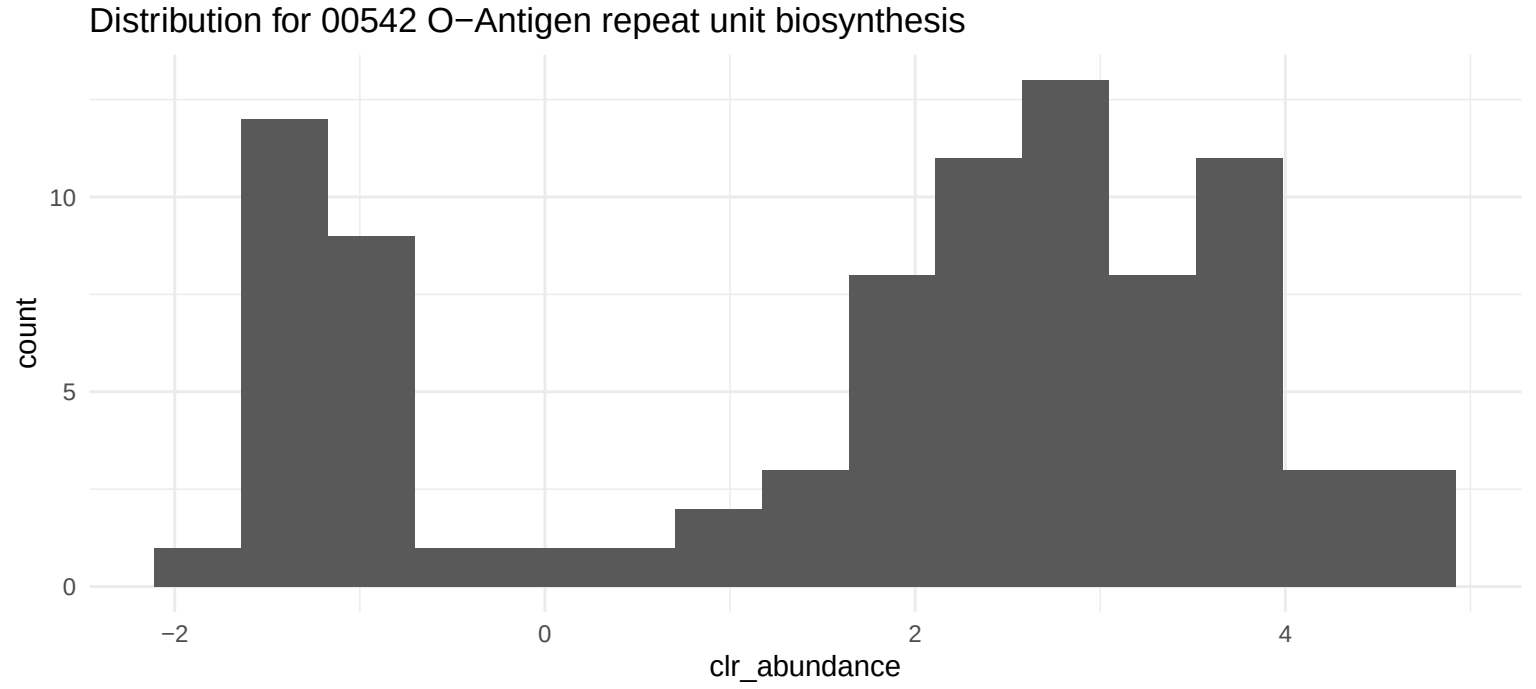

Distribution for 00534 Glycosaminoglycan biosynthesis − heparan sulfate / heparin
count
15
10
5
0
−4
−2
0
2
clr_abundance

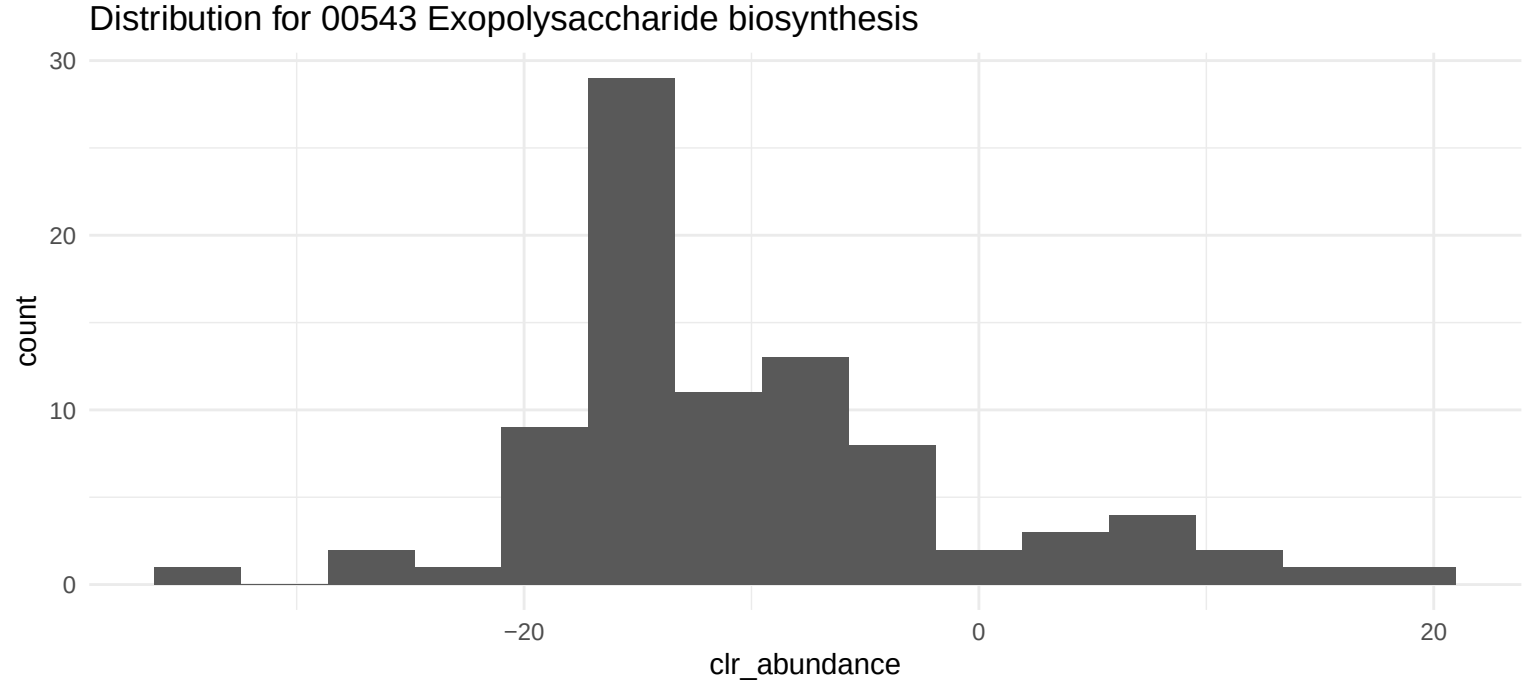

Distribution for 00536 Glycosaminoglycan binding proteins
count
15
10
5
0
−12
−8
−4
0
clr_abundance

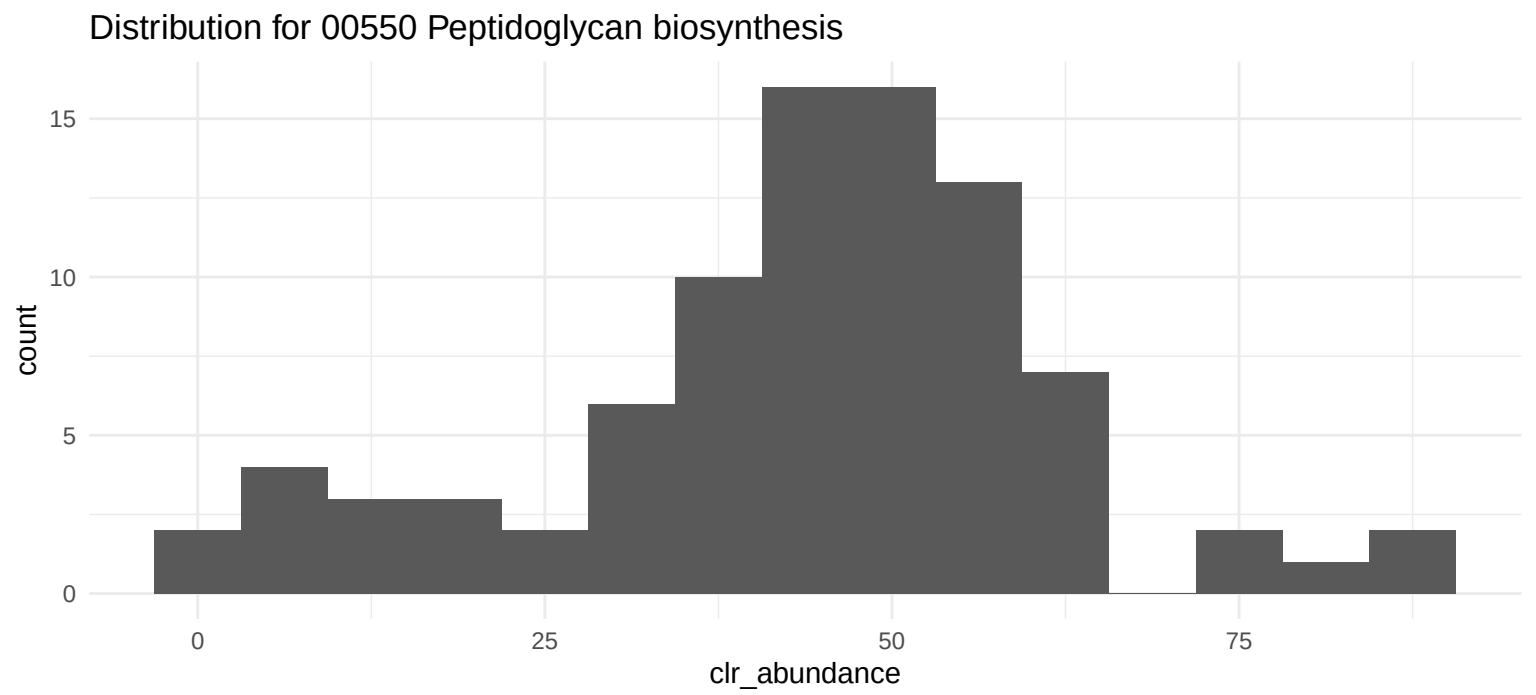

Distribution for 00537 Glycosylphosphatidylinositol (GPI)−anchored proteins
count
10
5
0
−10
−5
0
5
clr_abundance

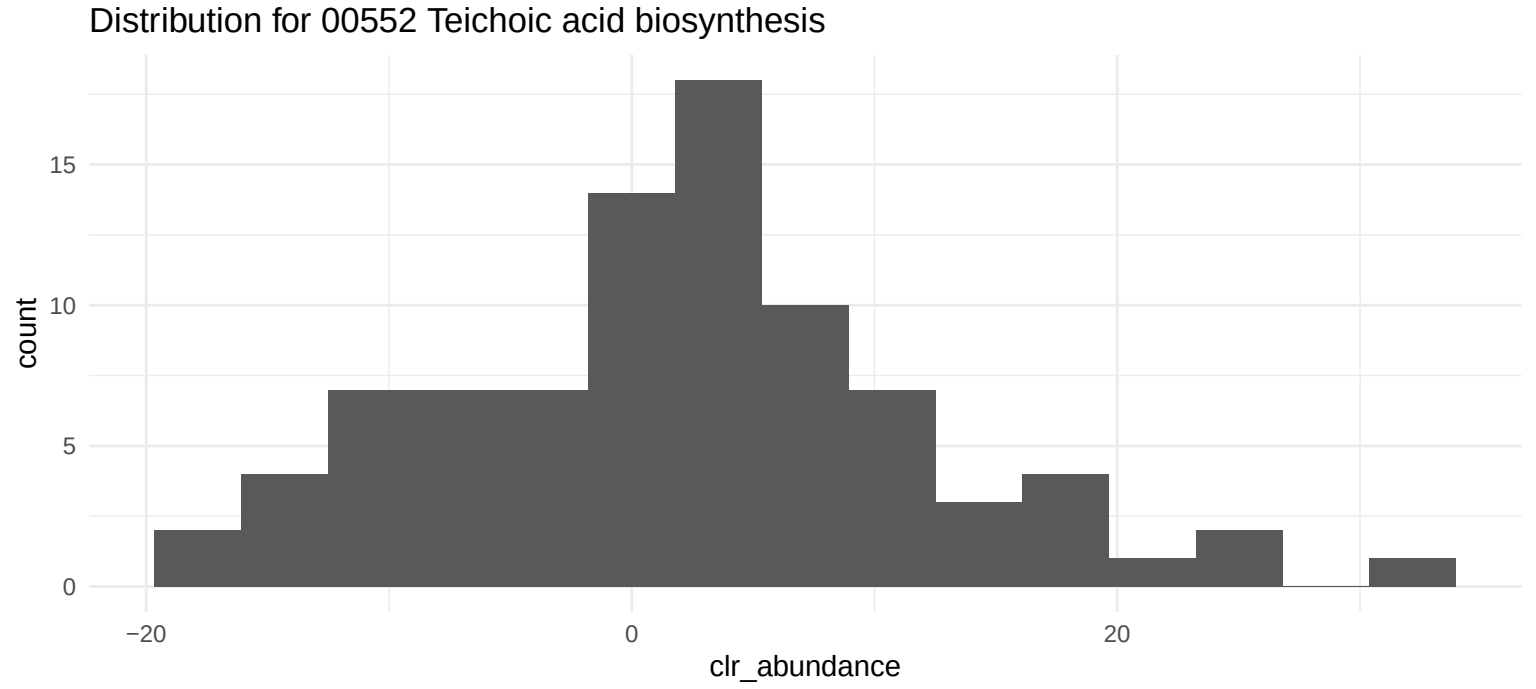

Distribution for 00540 Lipopolysaccharide biosynthesis
count
15
10
5
0
−25
0
25
50
75
clr_abundance

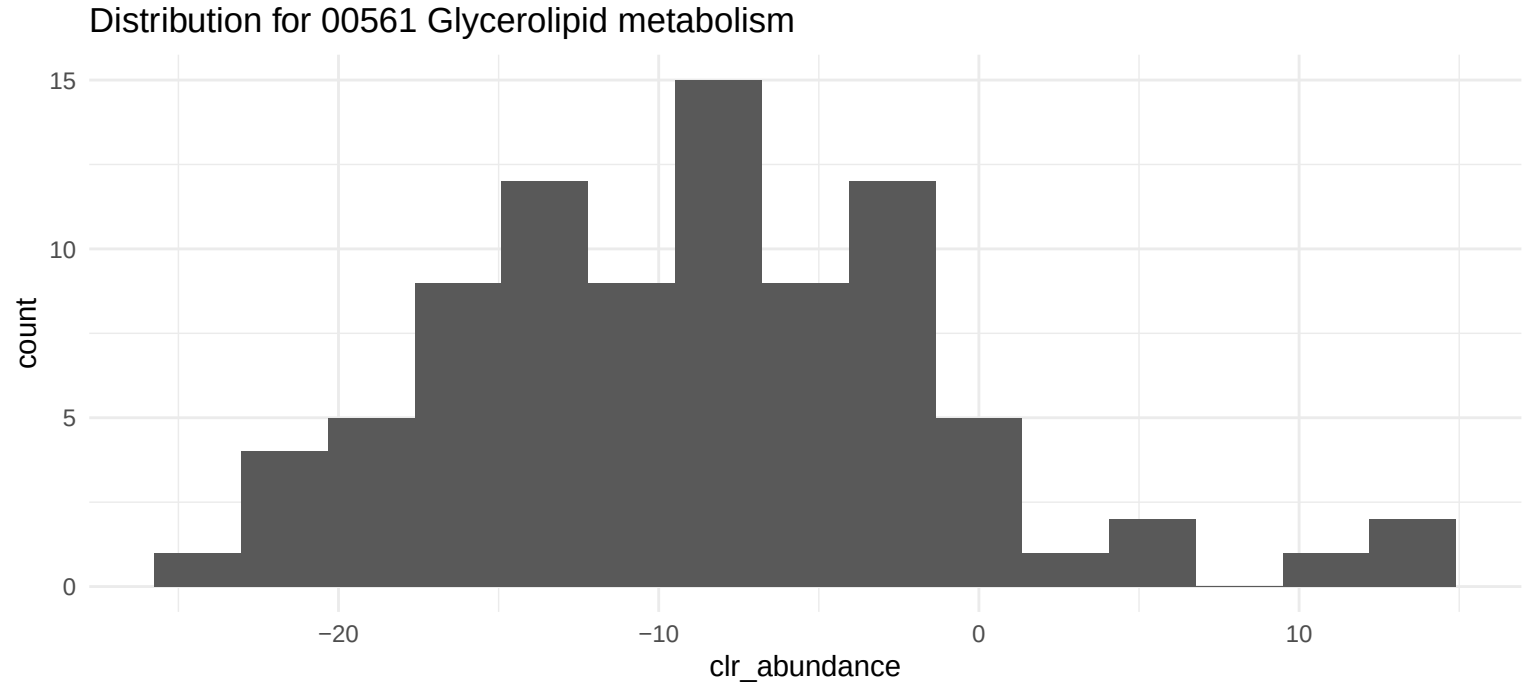

Distribution for 00541 Biosynthesis of various nucleotide sugars
count
clr_abundance
0
5
10
15
0
40
80

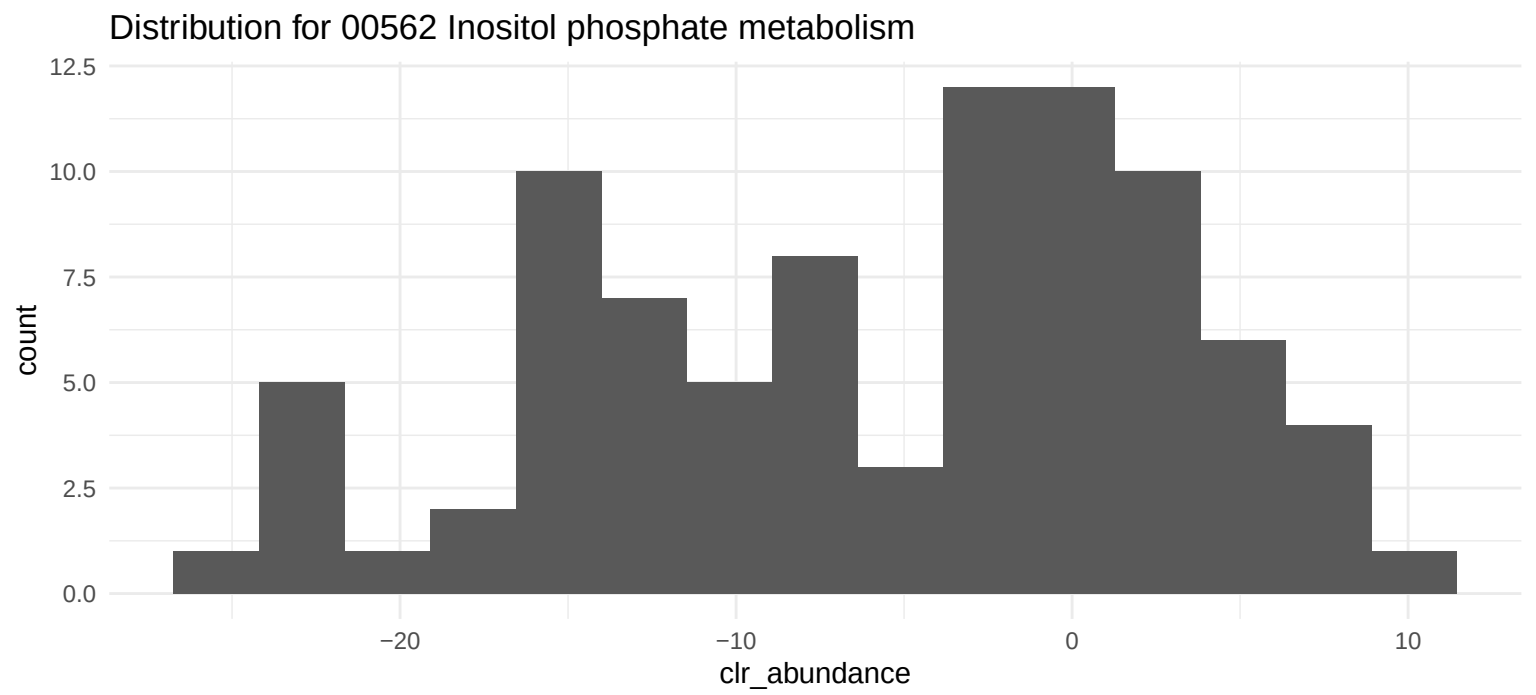

Distribution for 00542 O−Antigen repeat unit biosynthesis
count
clr_abundance
0
5
10
−2
0
2
4

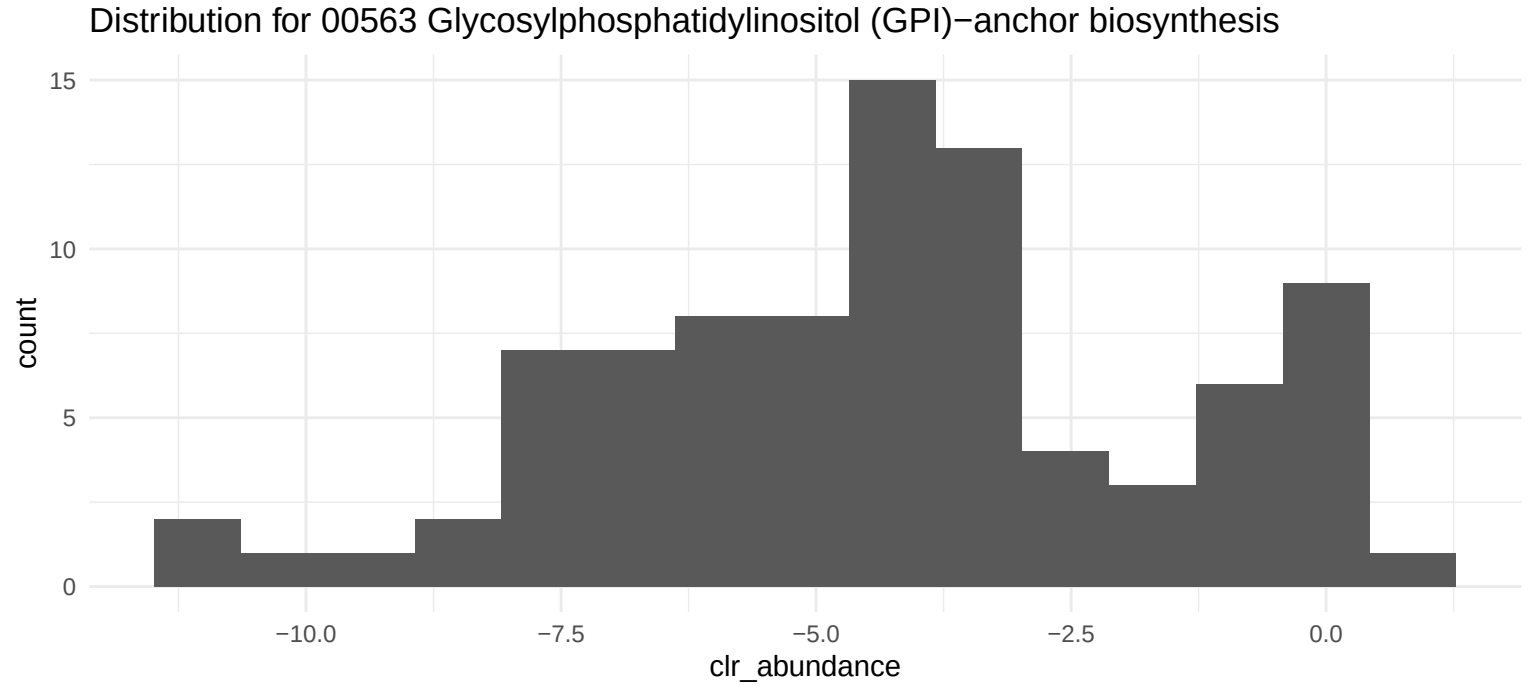

Distribution for 00543 Exopolysaccharide biosynthesis
count
clr_abundance
0
10
20
30
−20
0
20

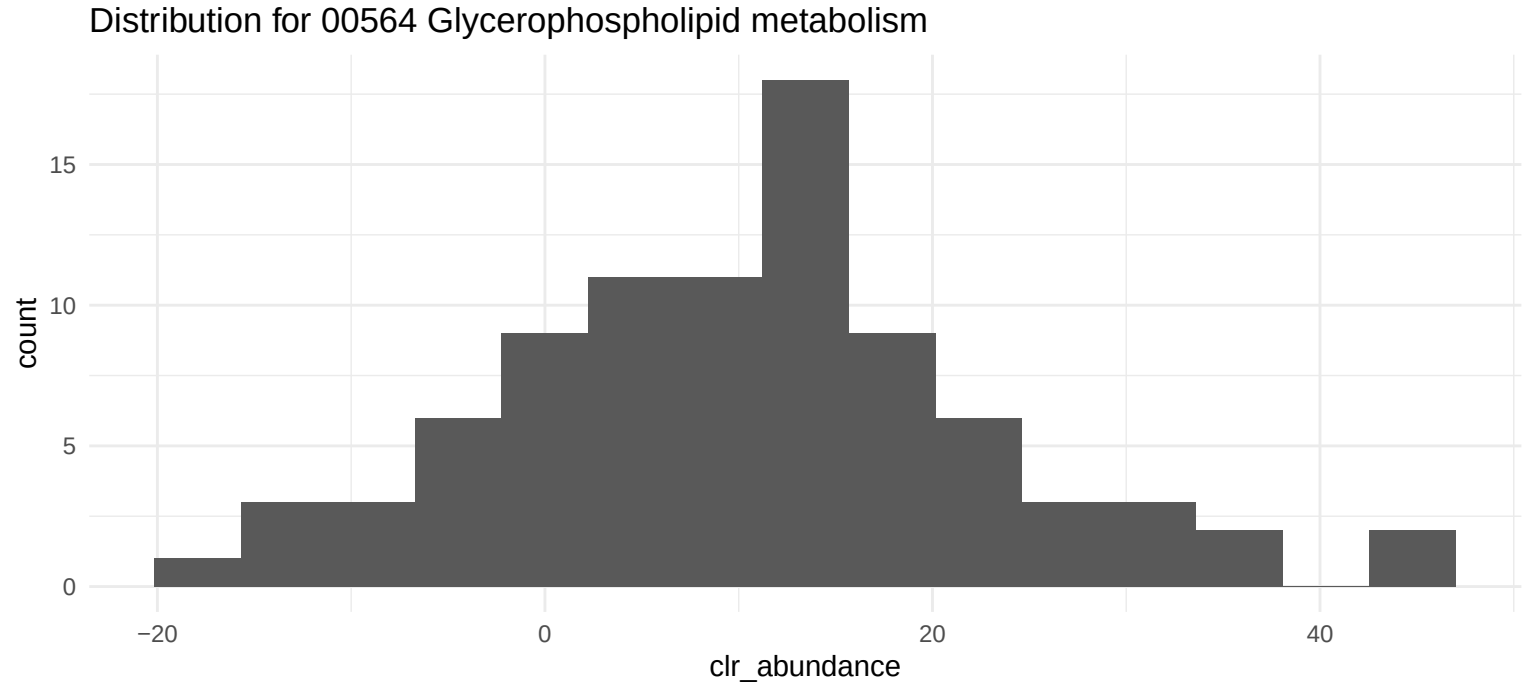

Distribution for 00550 Peptidoglycan biosynthesis
count
15
10
5
0
0
25
50
75
clr_abundance


Distribution for 00552 Teichoic acid biosynthesis
count
15
10
5
0
−20
0
20
clr_abundance

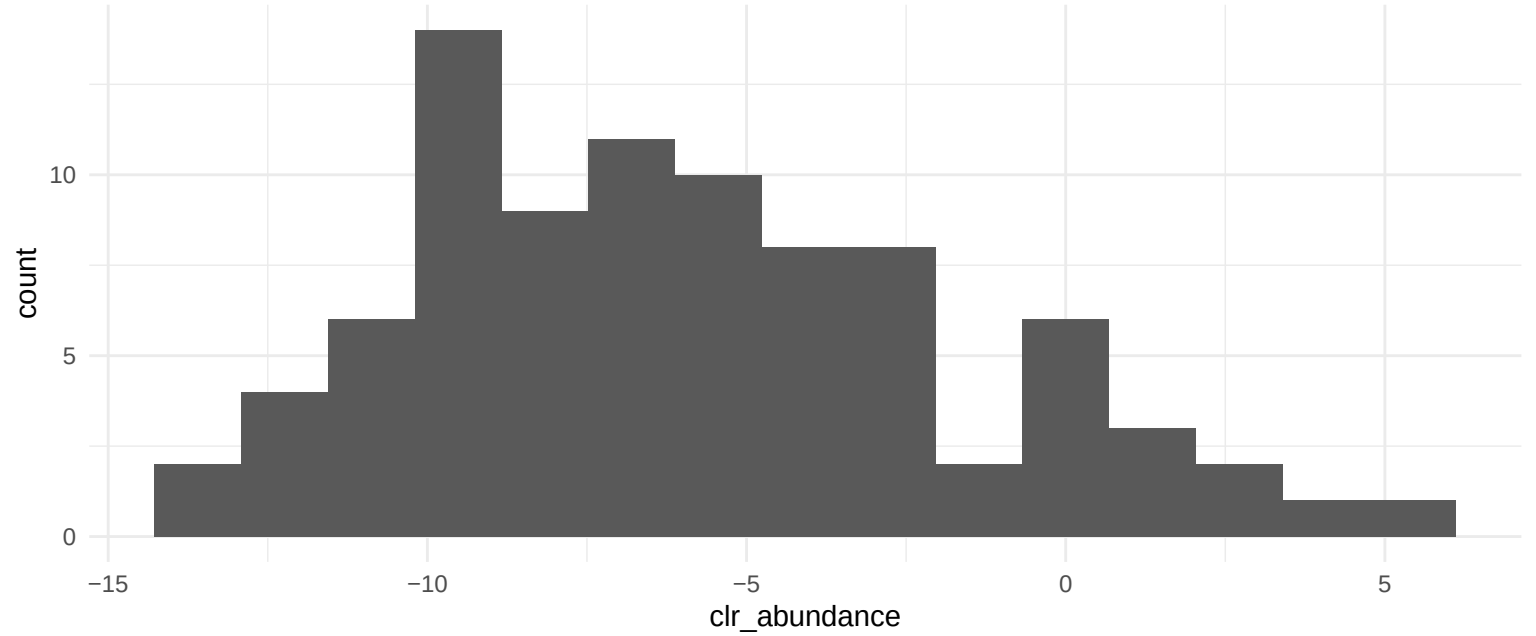

Distribution for 00561 Glycerolipid metabolism
count
15
10
5
0
−20
−10
0
10
clr_abundance


Distribution for 00562 Inositol phosphate metabolism
12.5
10.0
7.5
5.0
2.5
0.0
count
−20
−10
0
10
clr_abundance

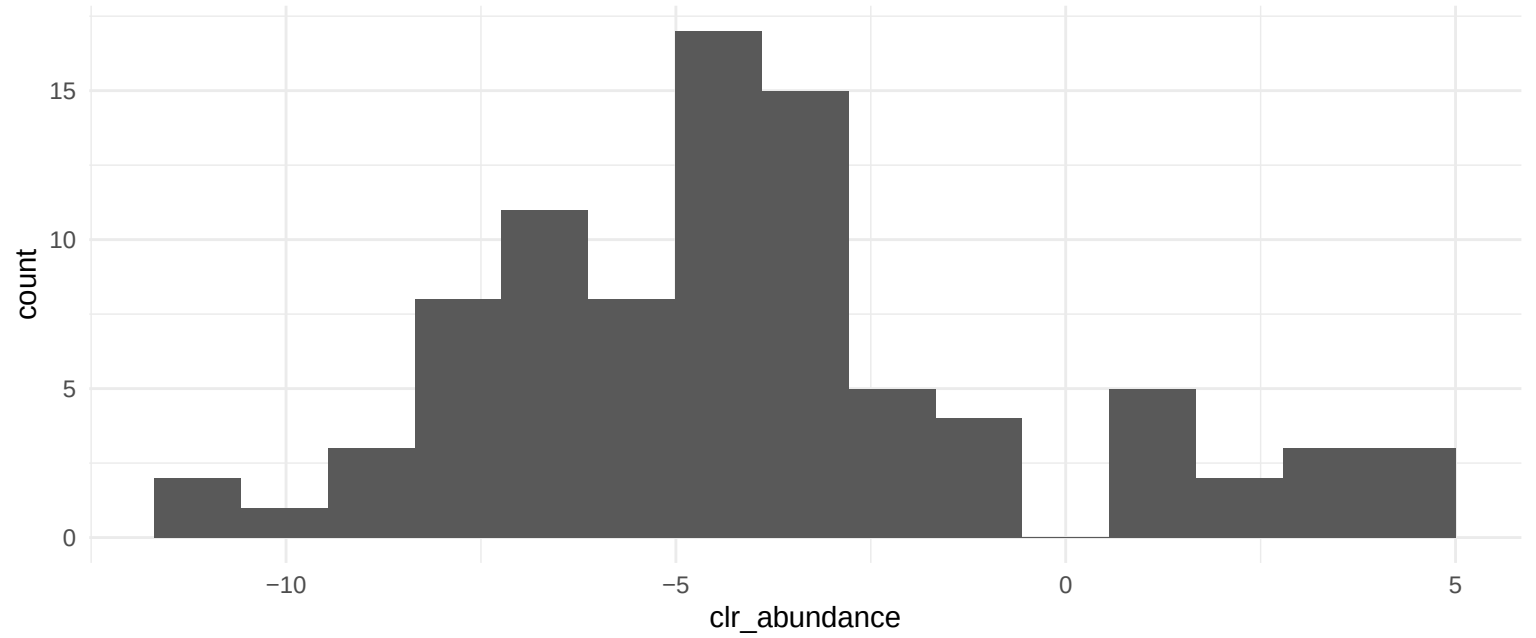

Distribution for 00563 Glycosylphosphatidylinositol (GPI)−anchor biosynthesis
15
10
5
0
count
−10.0
−7.5
−5.0
−2.5
0.0
clr_abundance


Distribution for 00564 Glycerophospholipid metabolism
15
10
5
0
count
−20
0
20
40
clr_abundance

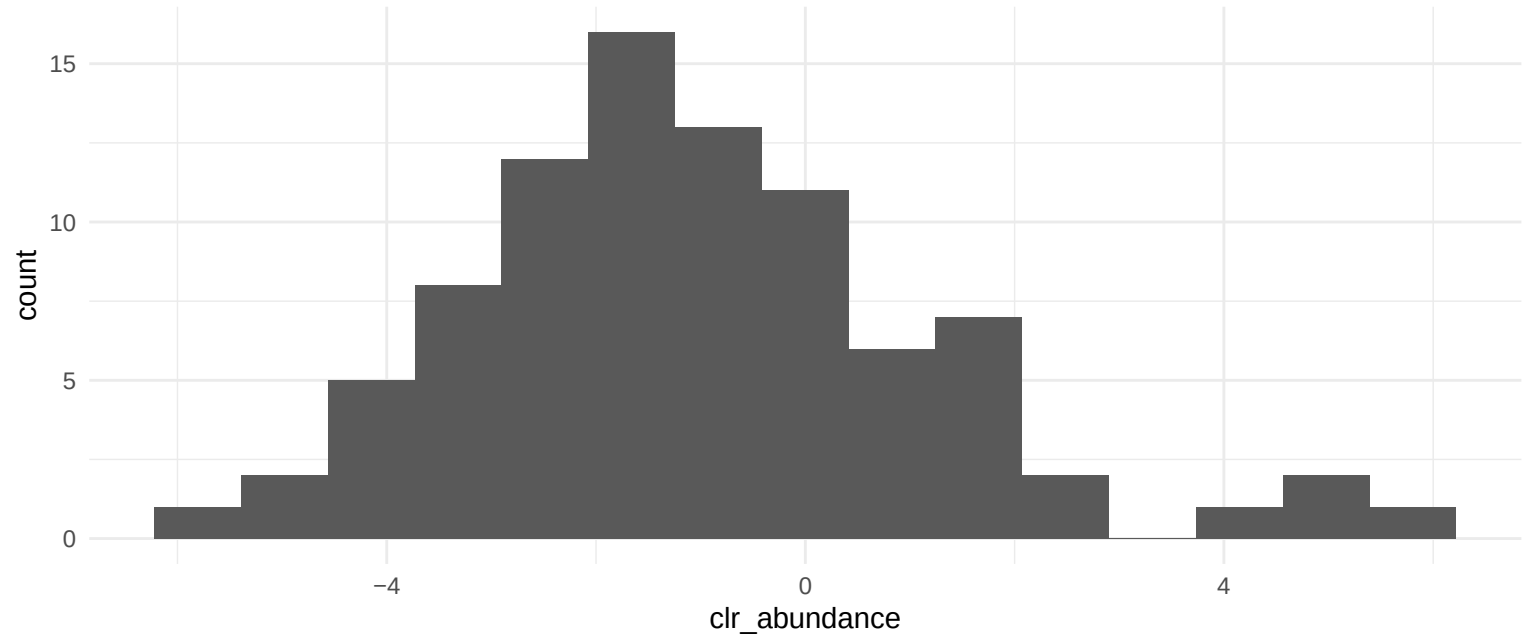
Distribution for 00565 Ether lipid metabolism

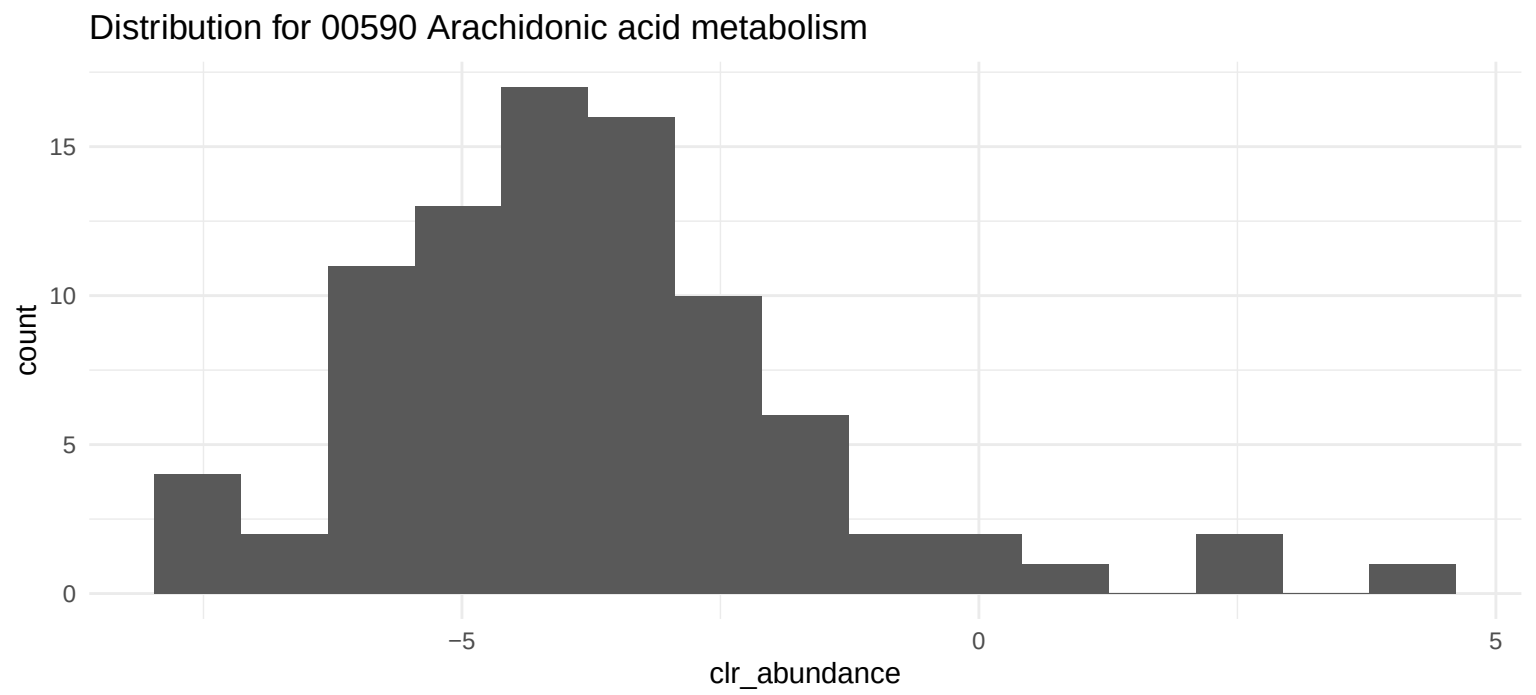


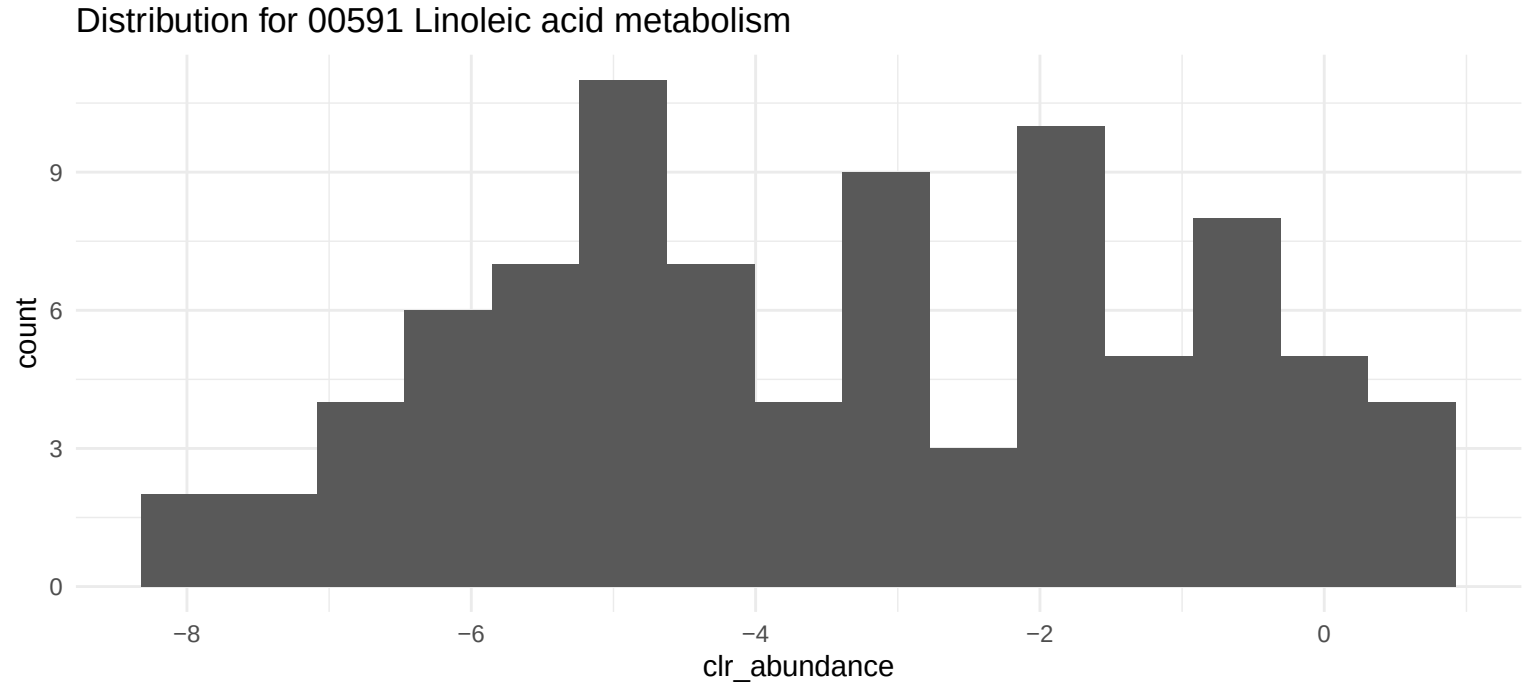
Distribution for 00571 Lipoarabinomannan (LAM) biosynthesis

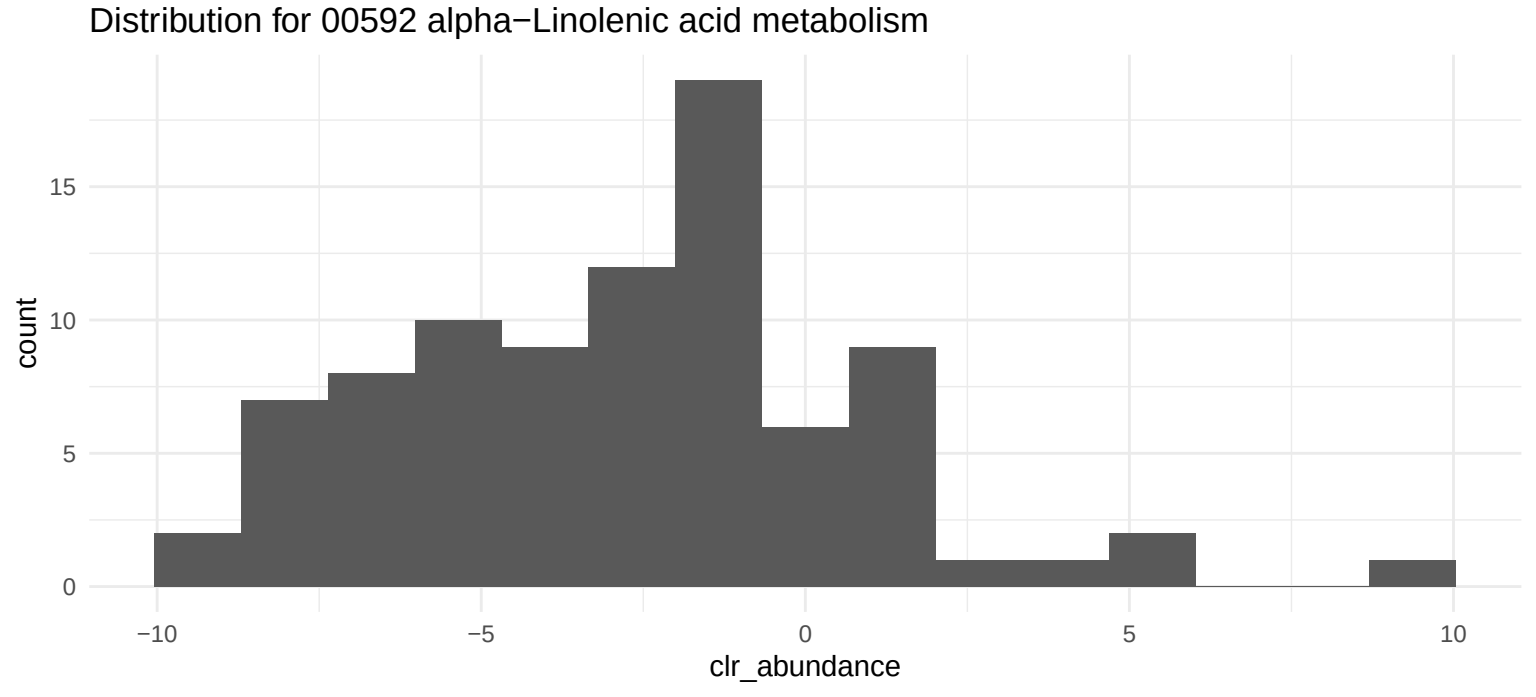


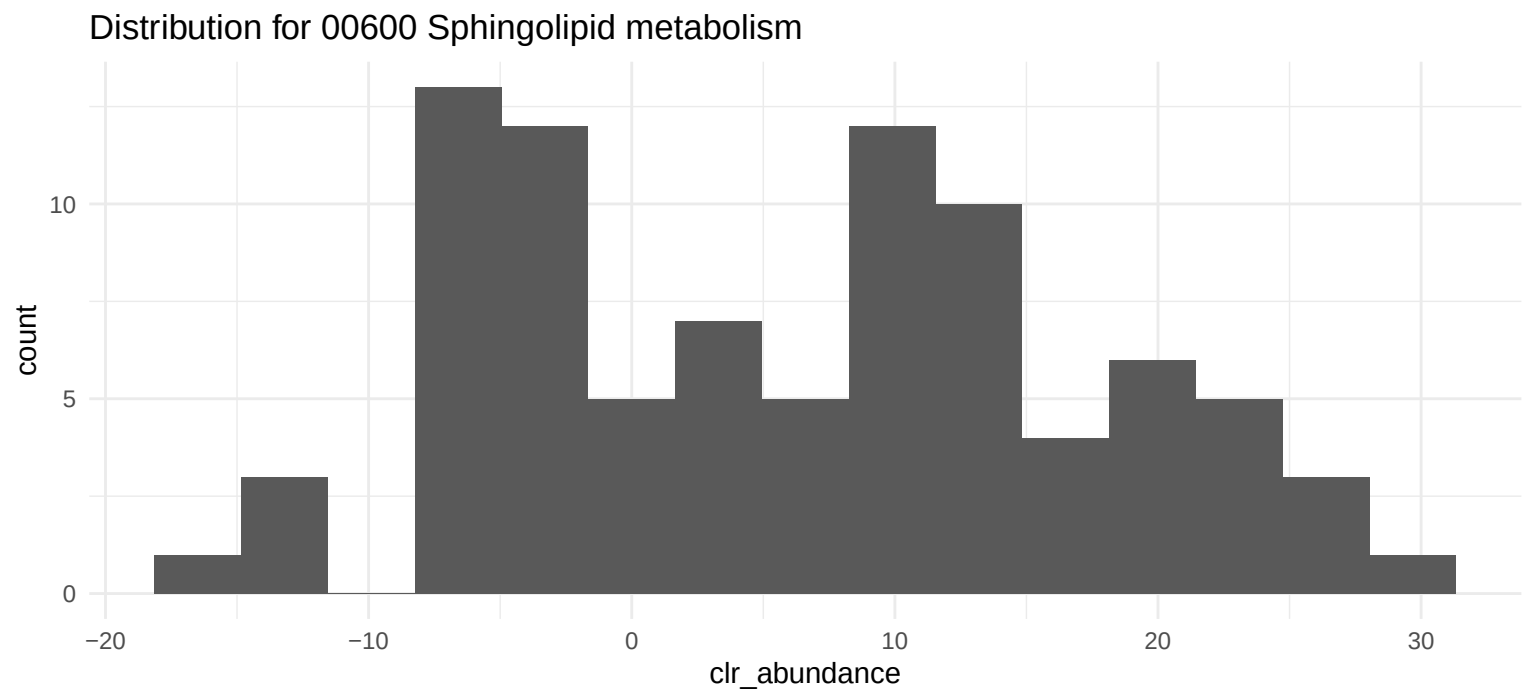
Distribution for 00572 Arabinogalactan biosynthesis − Mycobacterium

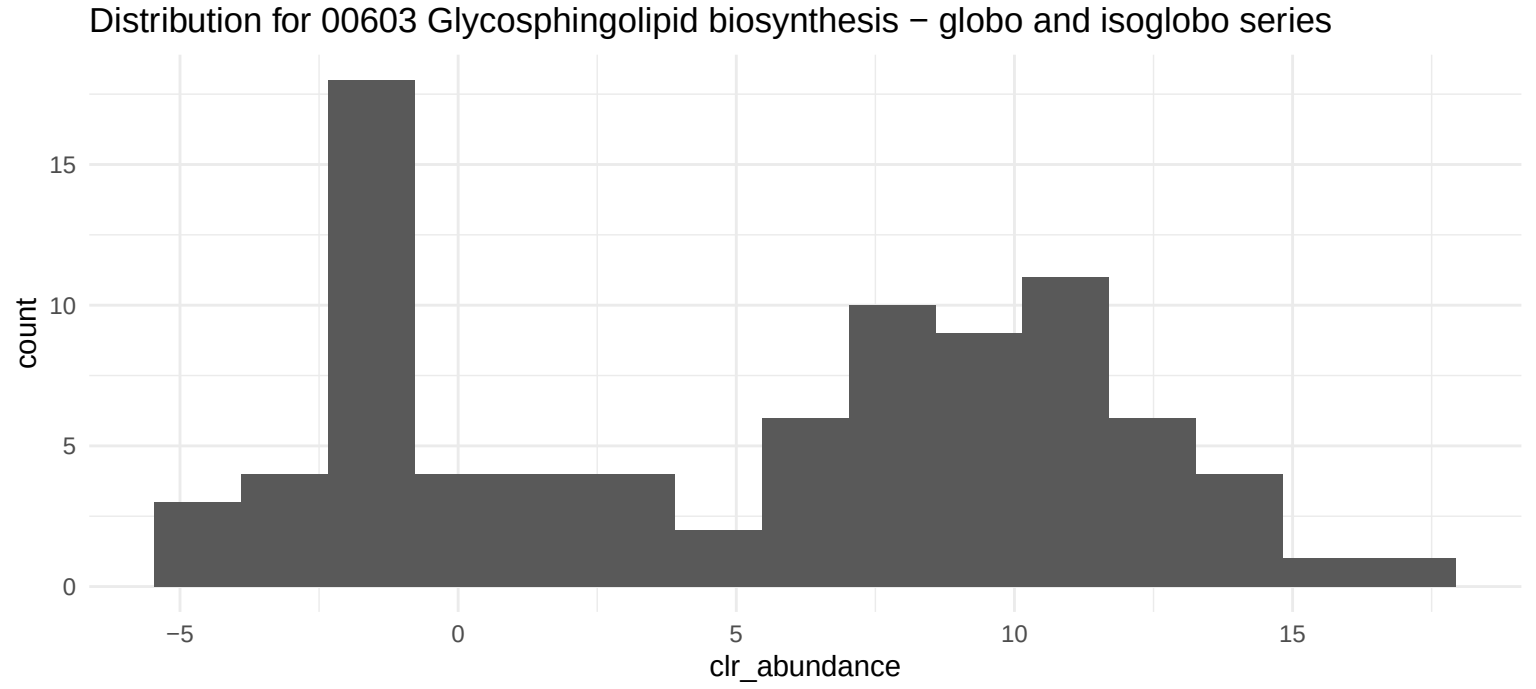


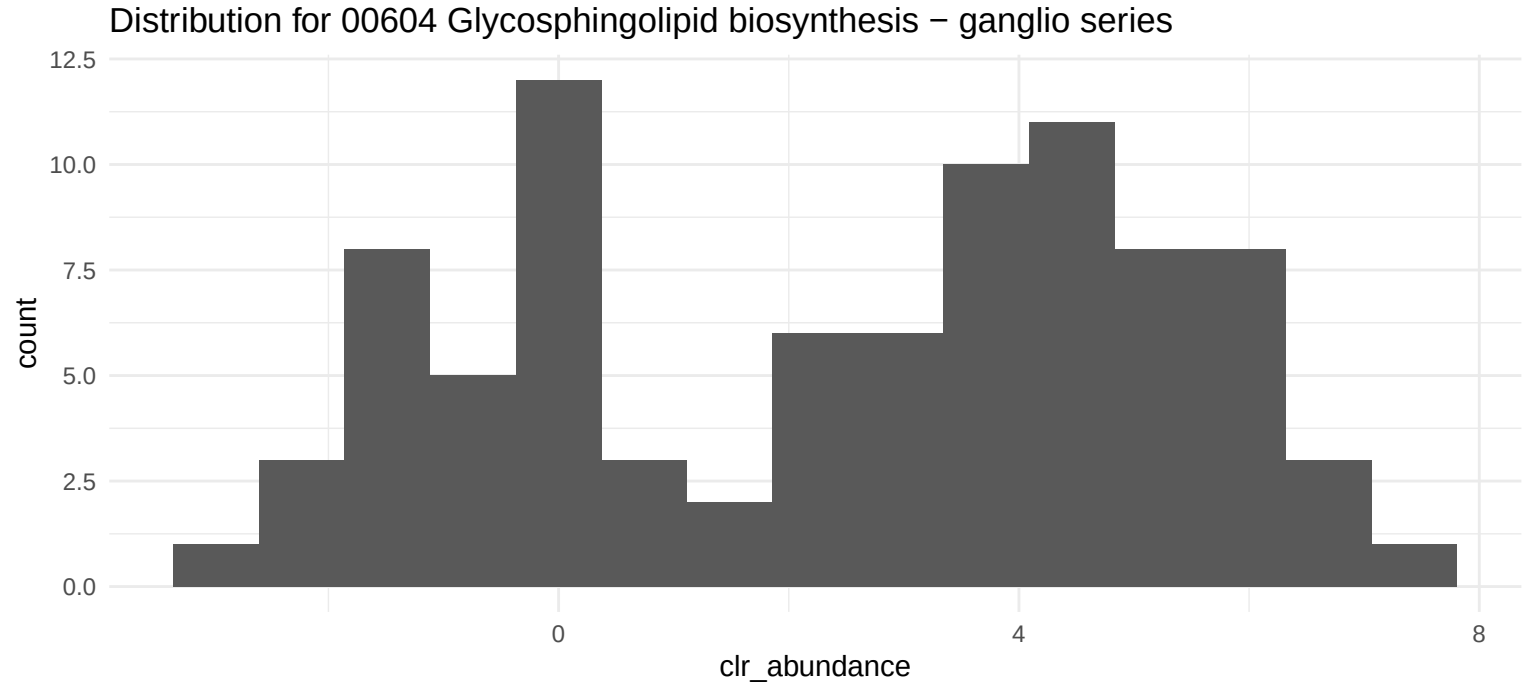

Distribution for 00590 Arachidonic acid metabolism
count
15
10
5
0
−5
0
5
clr_abundance


Distribution for 00591 Linoleic acid metabolism
count
9
6
3
0
−8
−6
−4
−2
0
clr_abundance

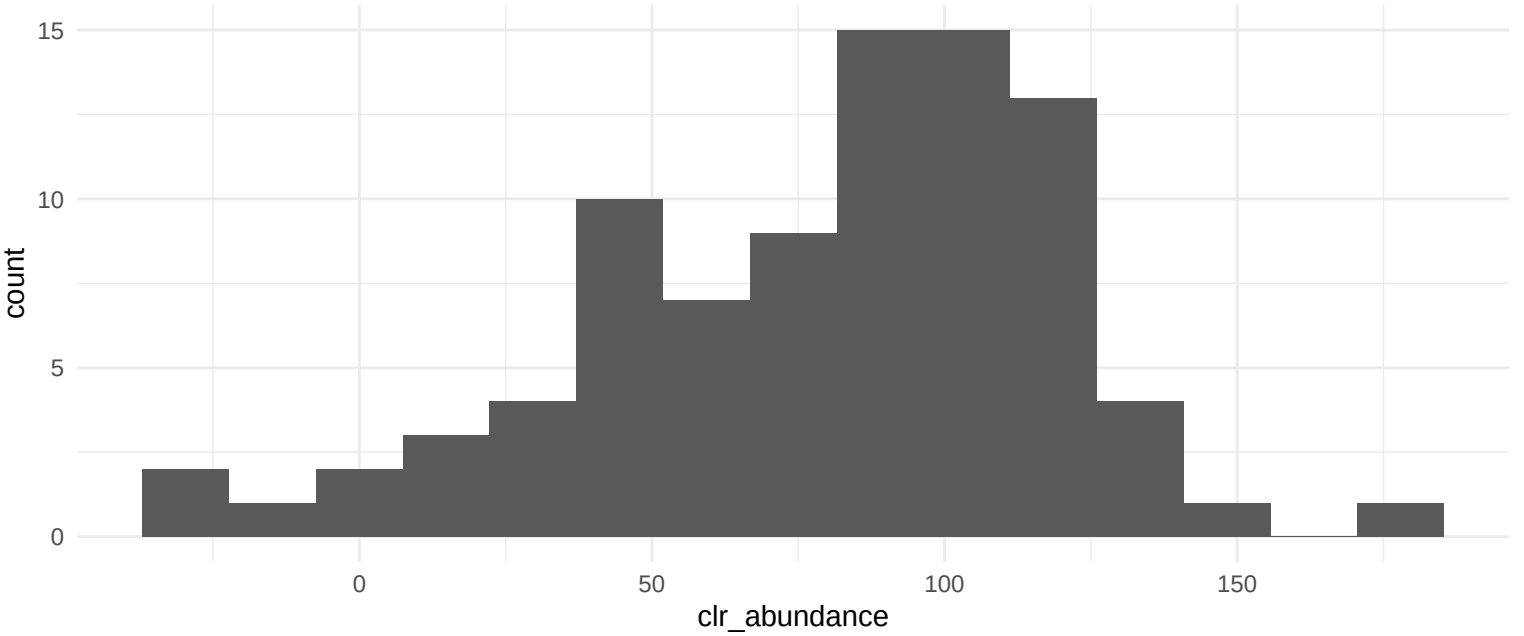

Distribution for 00592 alpha−Linolenic acid metabolism
count
15
10
5
0
−10
−5
0
5
10
clr_abundance


Distribution for 00600 Sphingolipid metabolism
count
clr_abundance

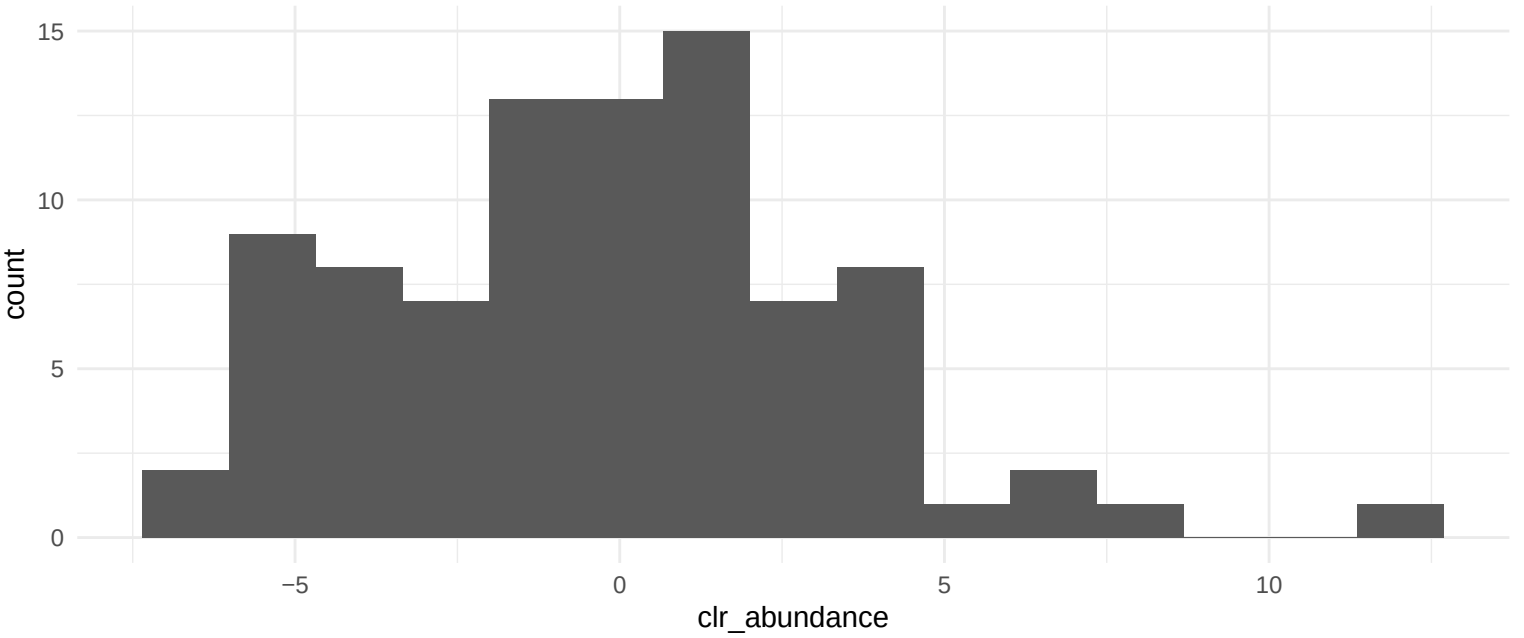

Distribution for 00603 Glycosphingolipid biosynthesis − globo and isoglobo series
count
clr_abundance


Distribution for 00604 Glycosphingolipid biosynthesis − ganglio series
count
clr_abundance

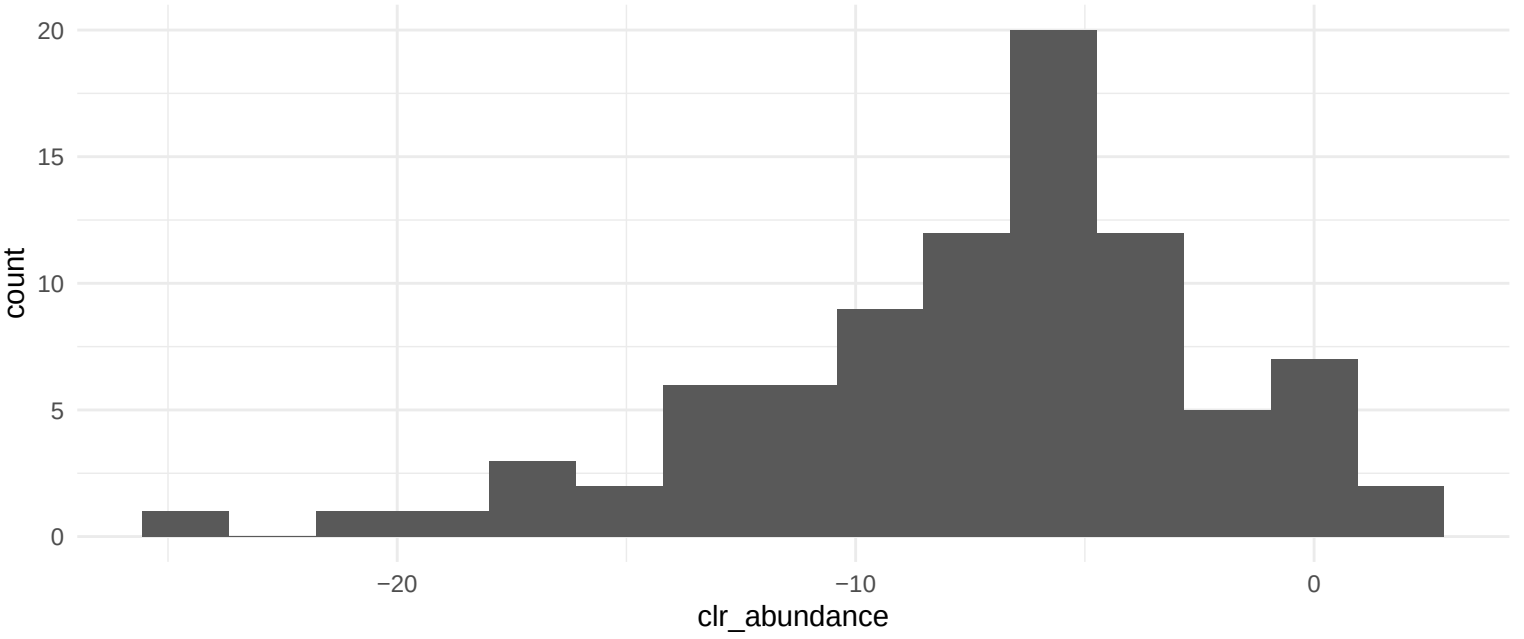
Distribution for 00620 Pyruvate metabolism

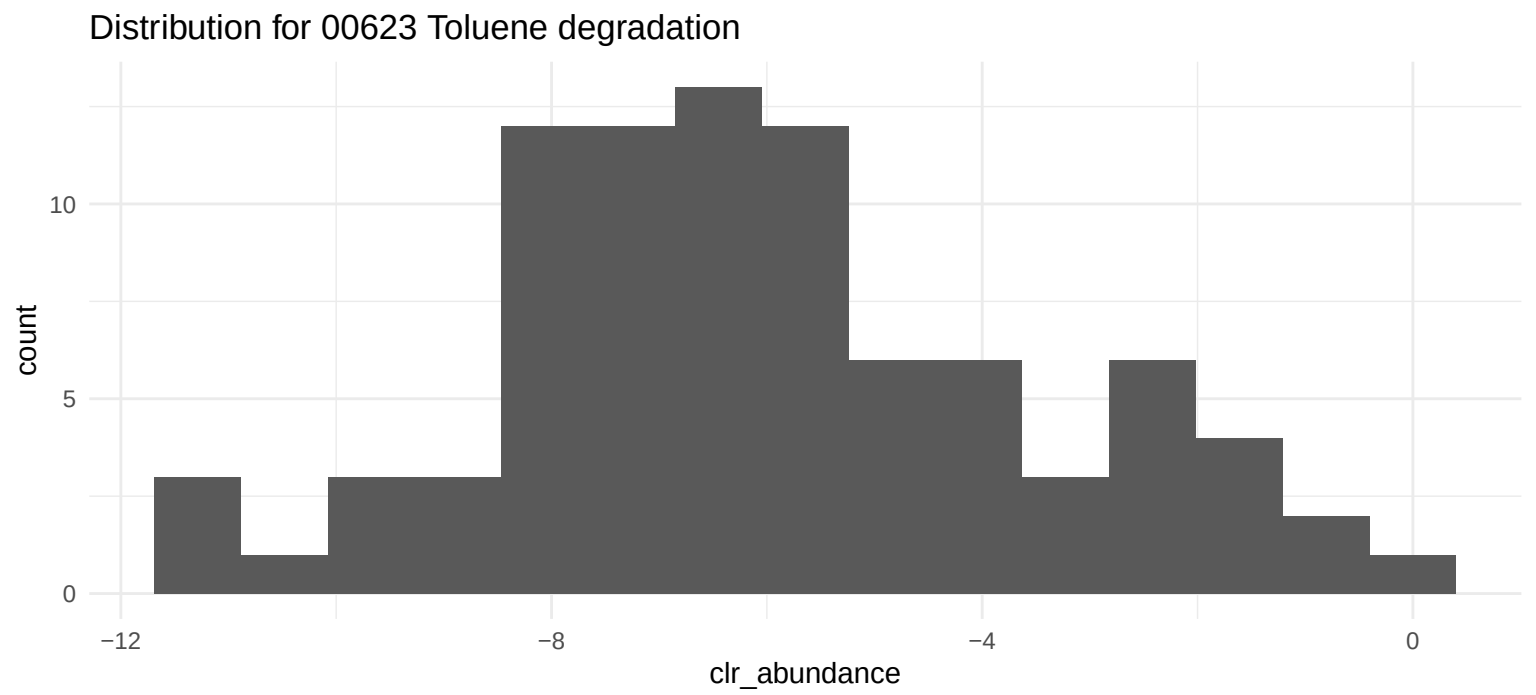


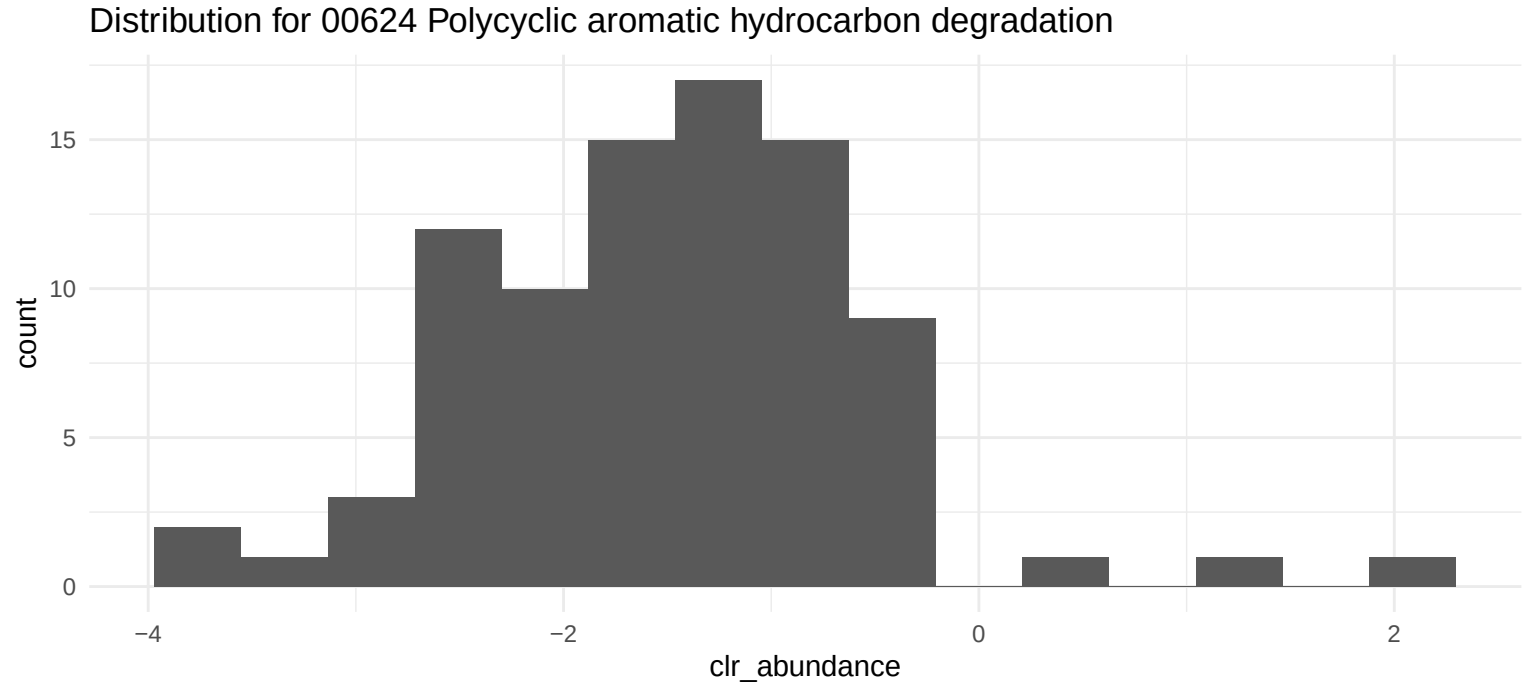
Distribution for 00621 Dioxin degradation

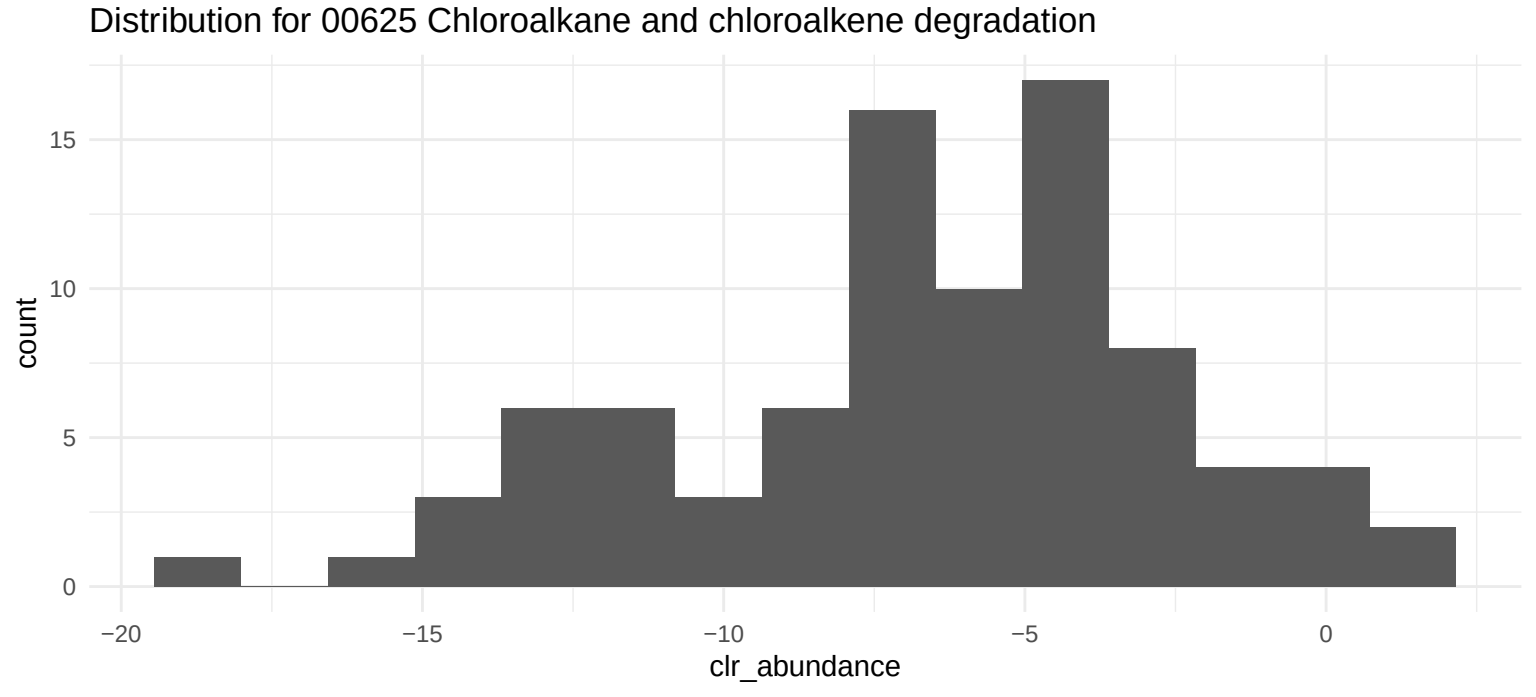


Distribution for 00622 Xylene degradation

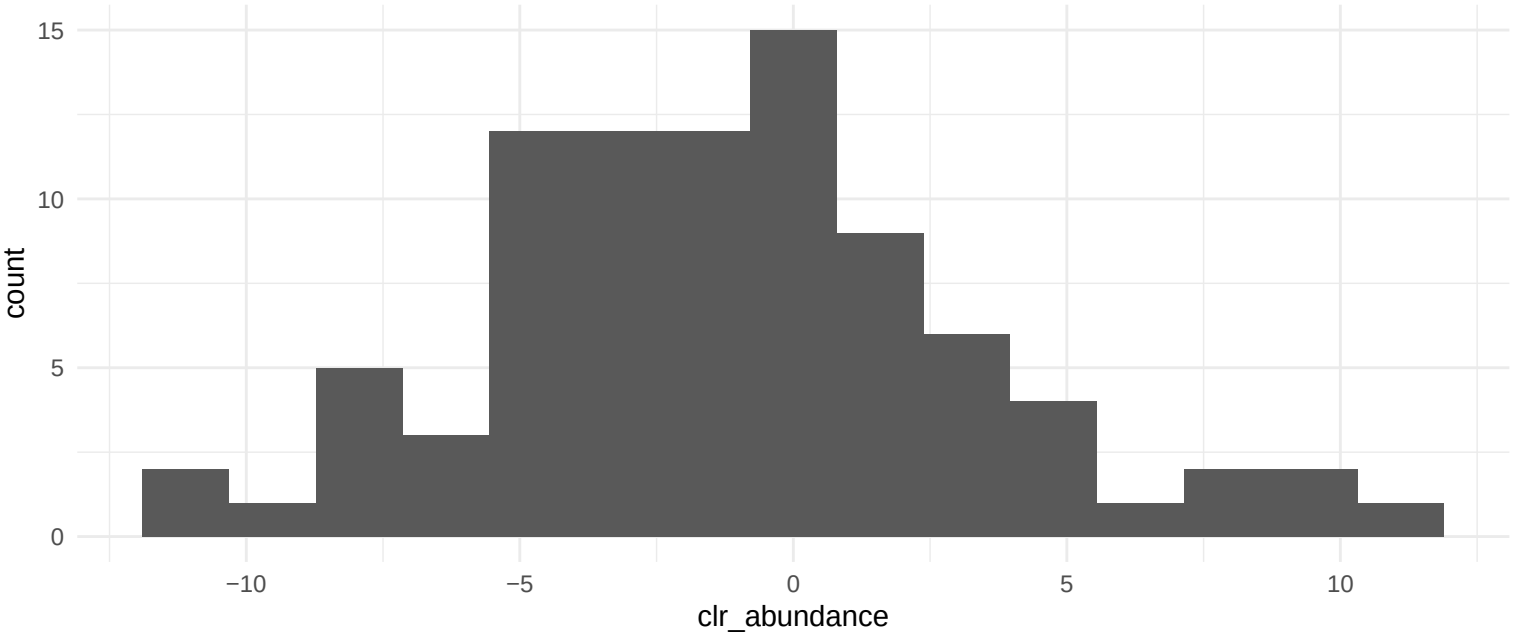



Distribution for 00623 Toluene degradation
count
10
5
0
−12
−8
−4
0
clr_abundance

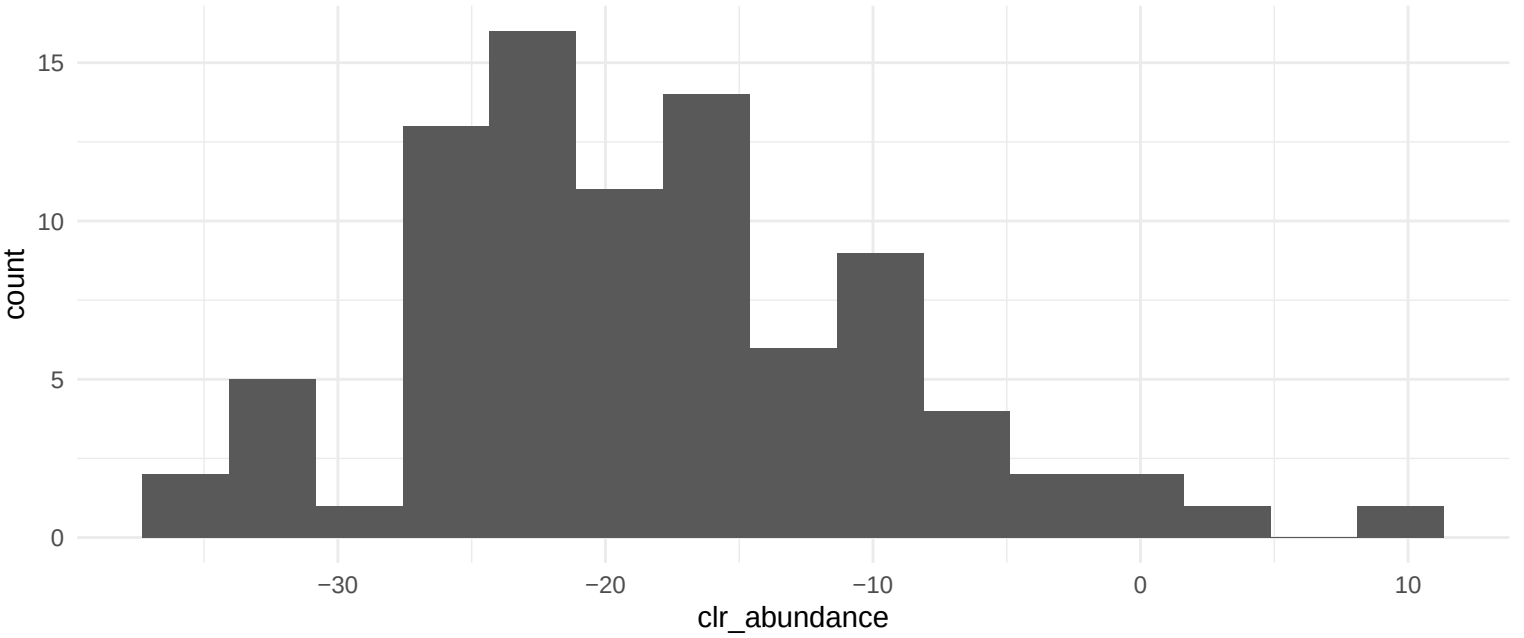

Distribution for 00624 Polycyclic aromatic hydrocarbon degradation
count
15
10
5
0
−4
−2
0
2
clr_abundance


Distribution for 00625 Chloroalkane and chloroalkene degradation
count
15
10
5
0
−20
−15
−10
−5
0
clr_abundance

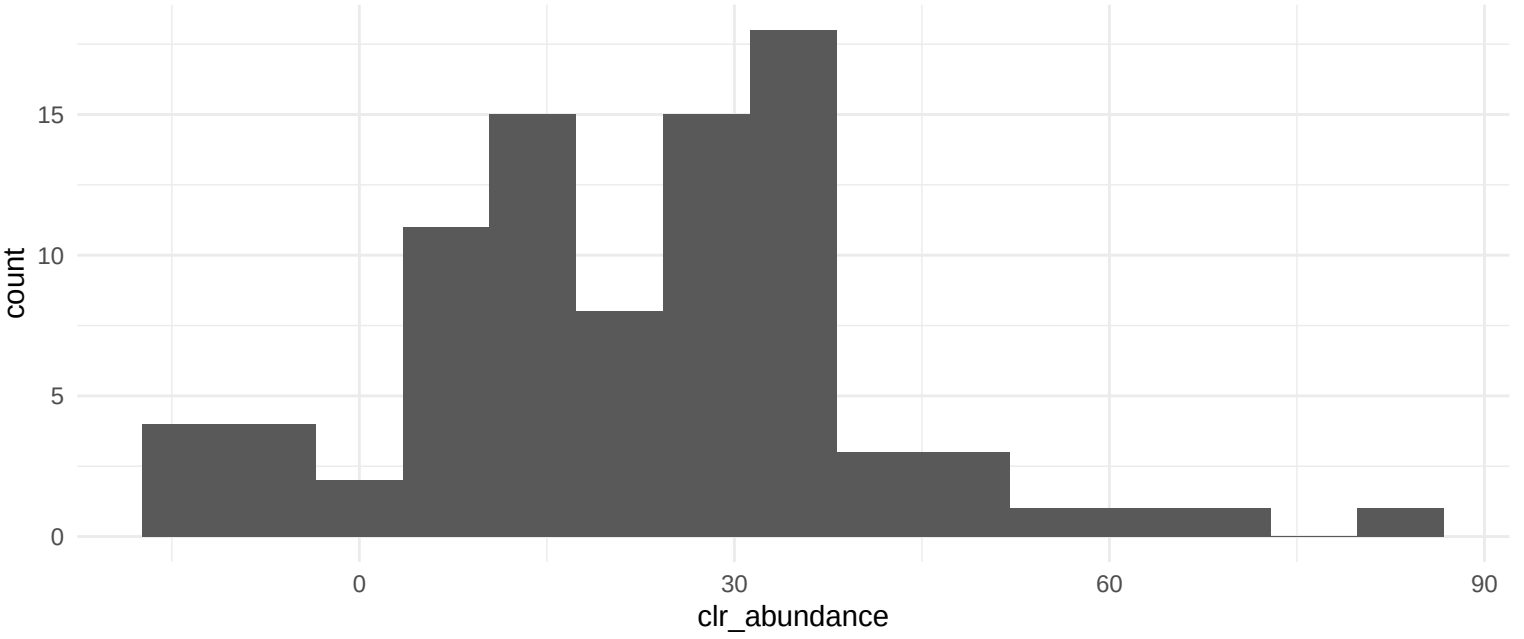
Distribution for 00626 Naphthalene degradation

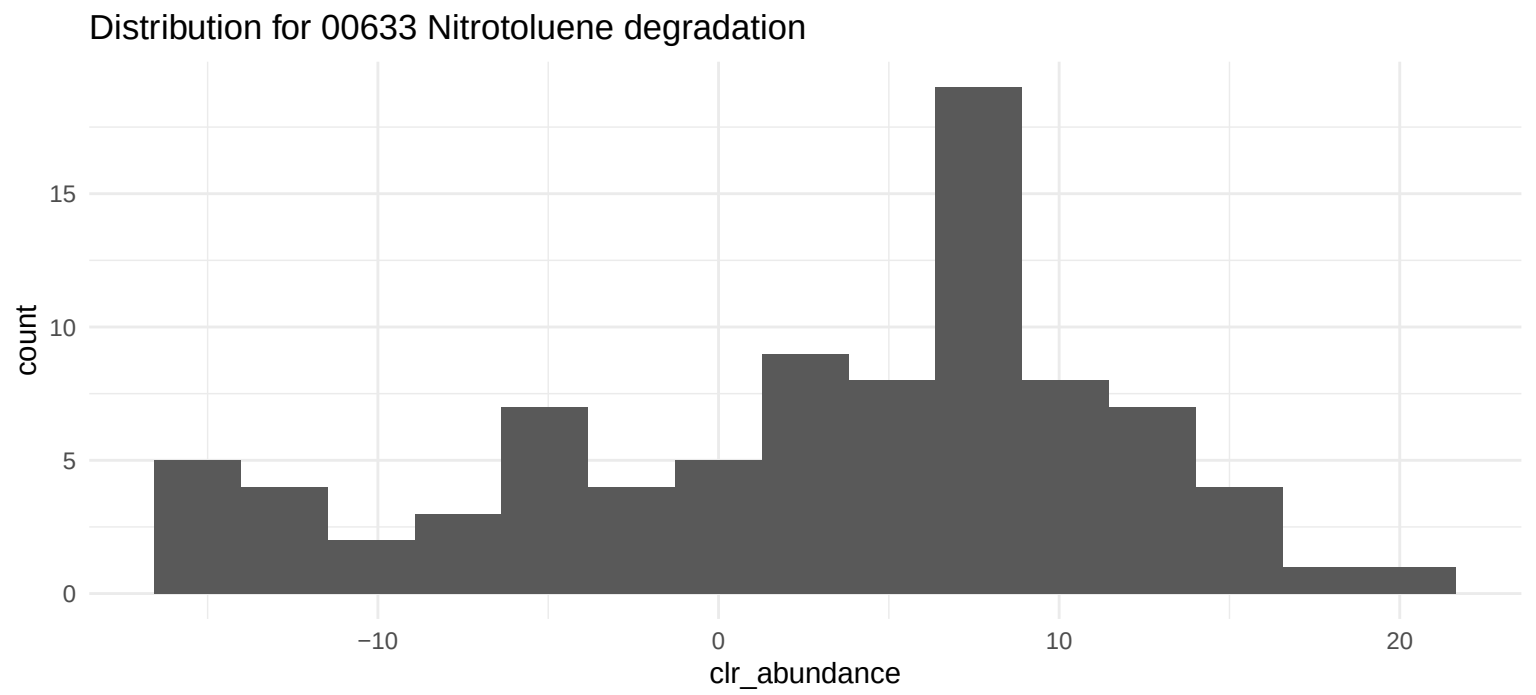


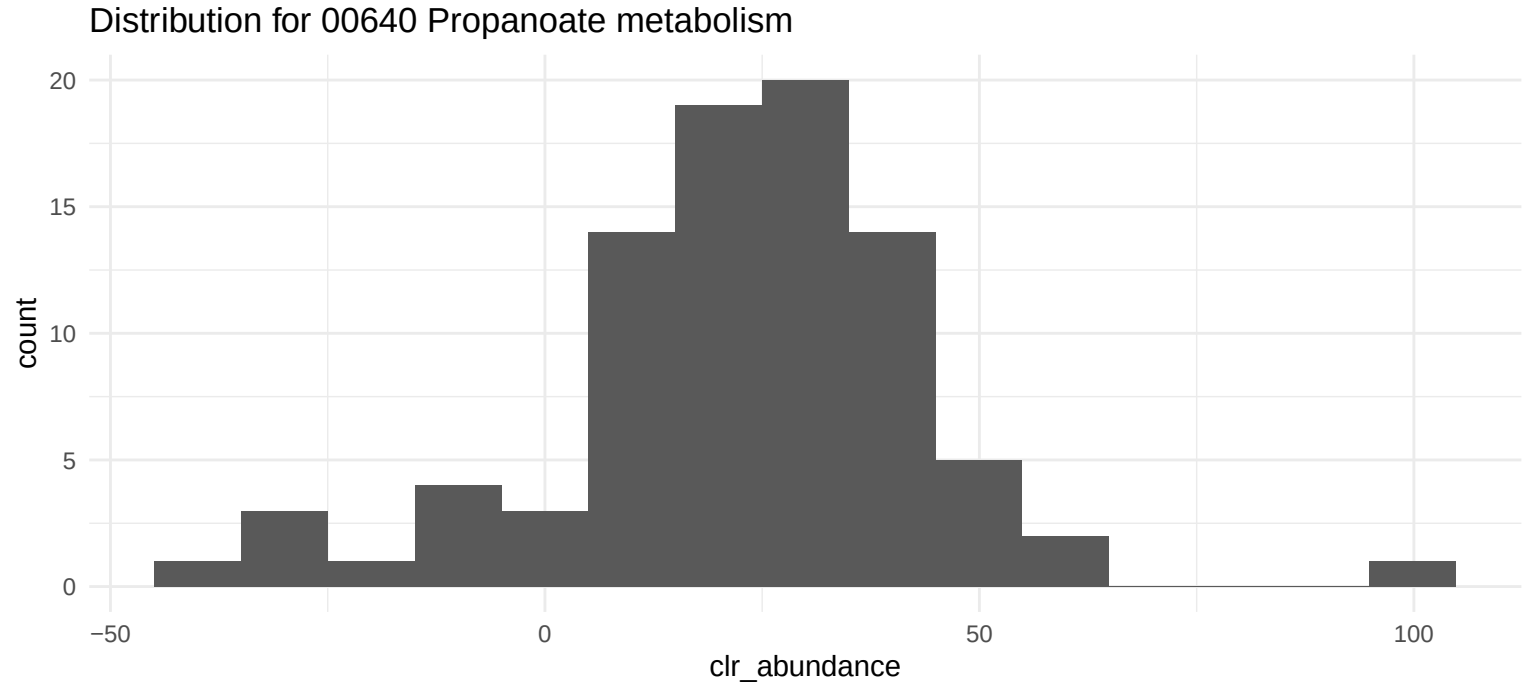
Distribution for 00627 Aminobenzoate degradation

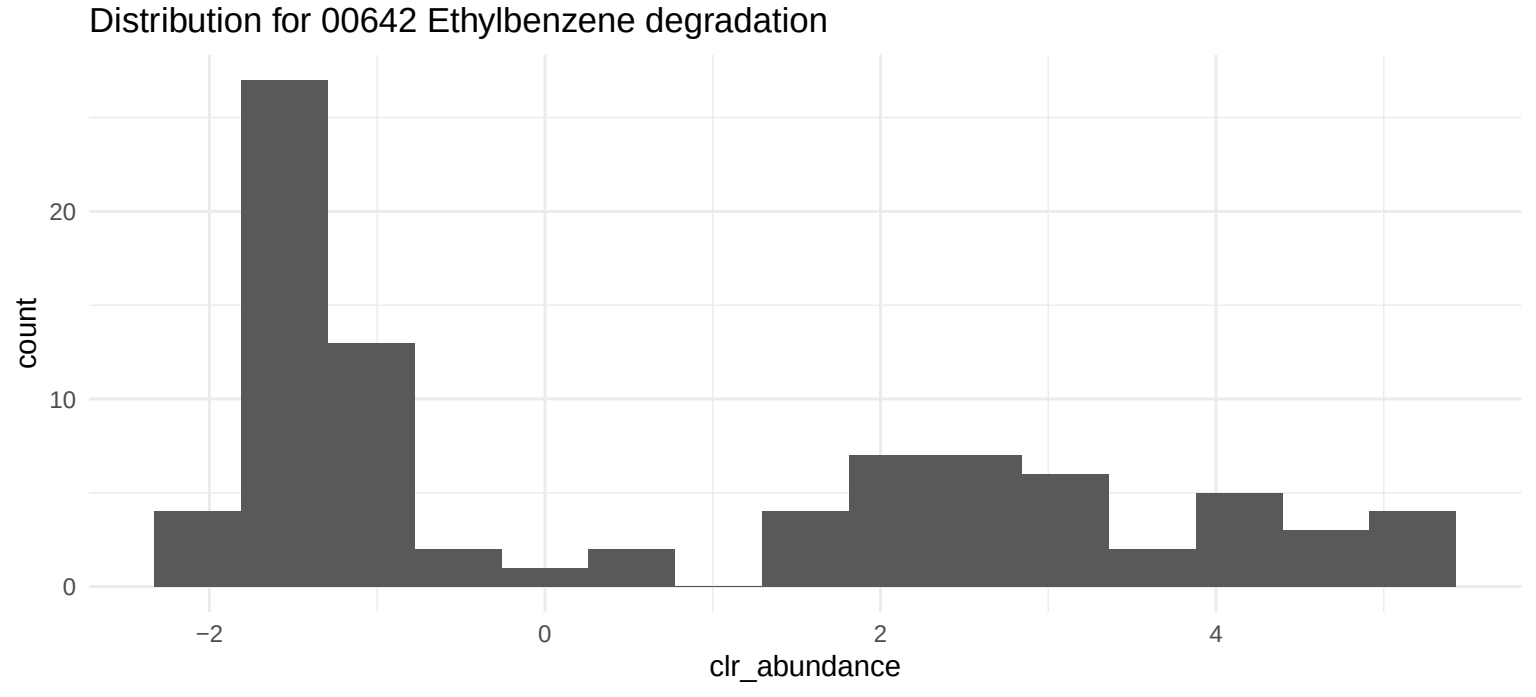


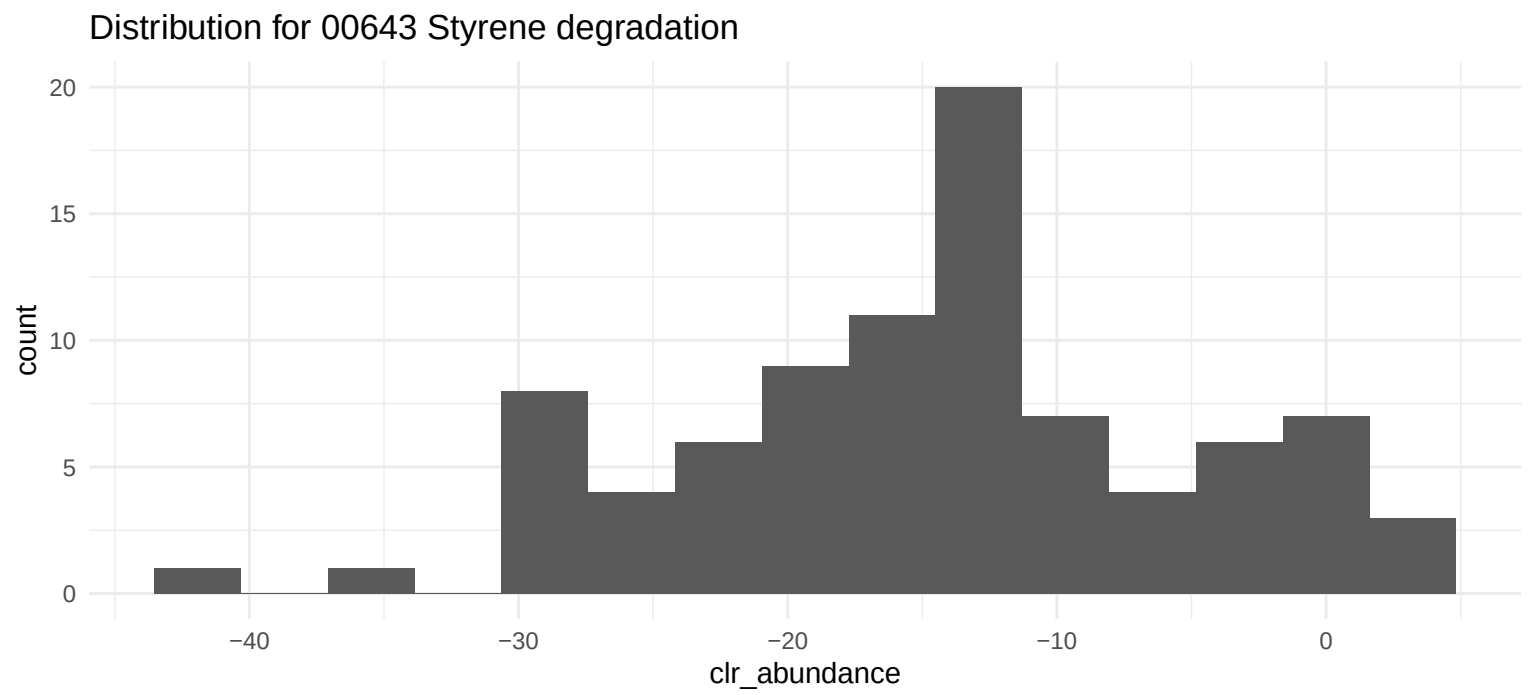
Distribution for 00630 Glyoxylate and dicarboxylate metabolism

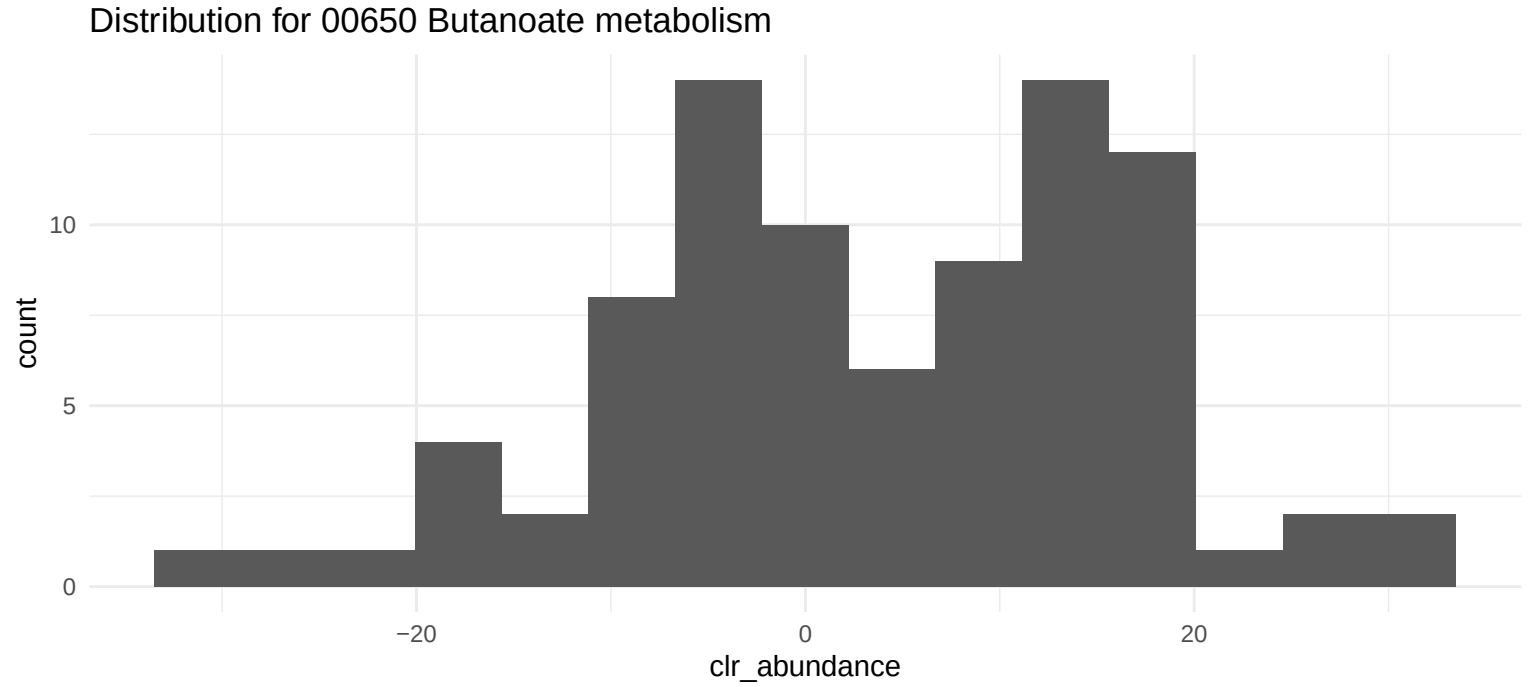


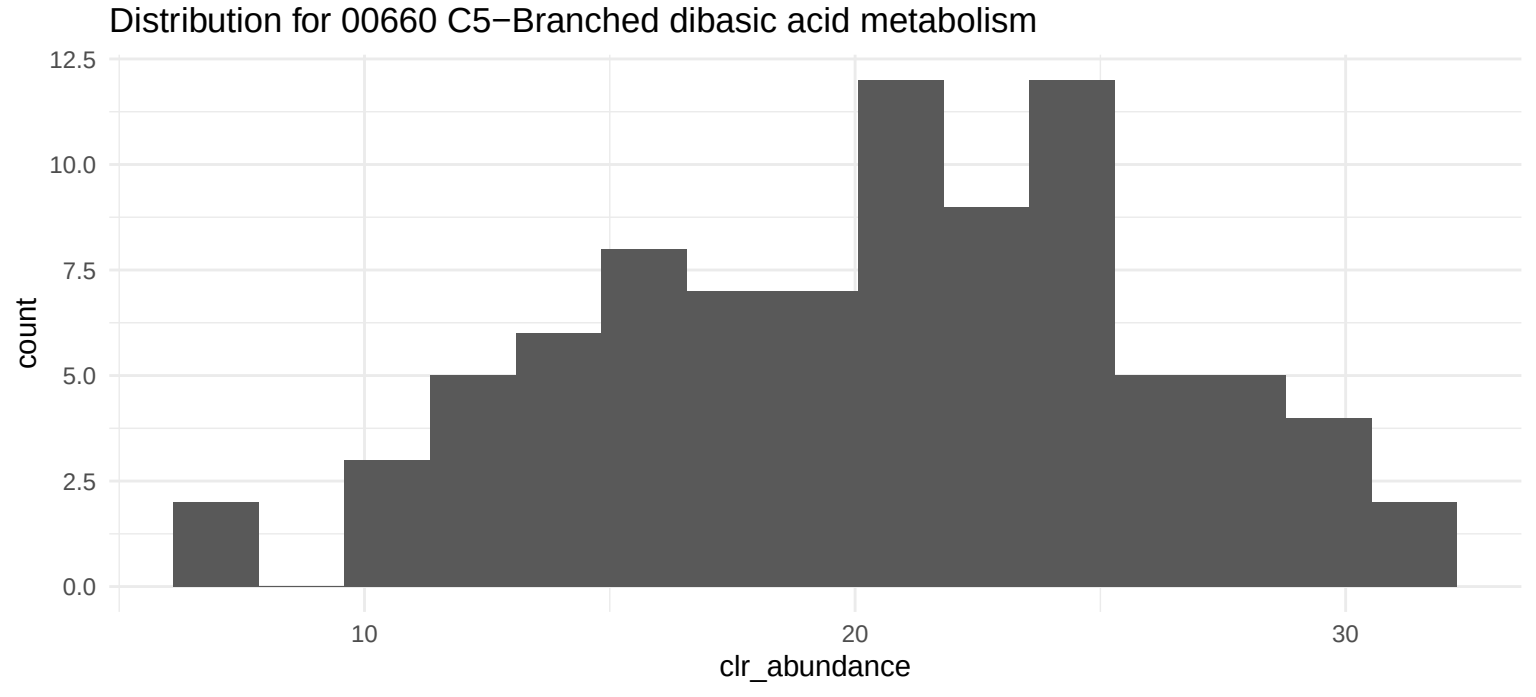

Distribution for 00633 Nitrotoluene degradation
count
15
10
5
0
−10
0
10
20
clr_abundance

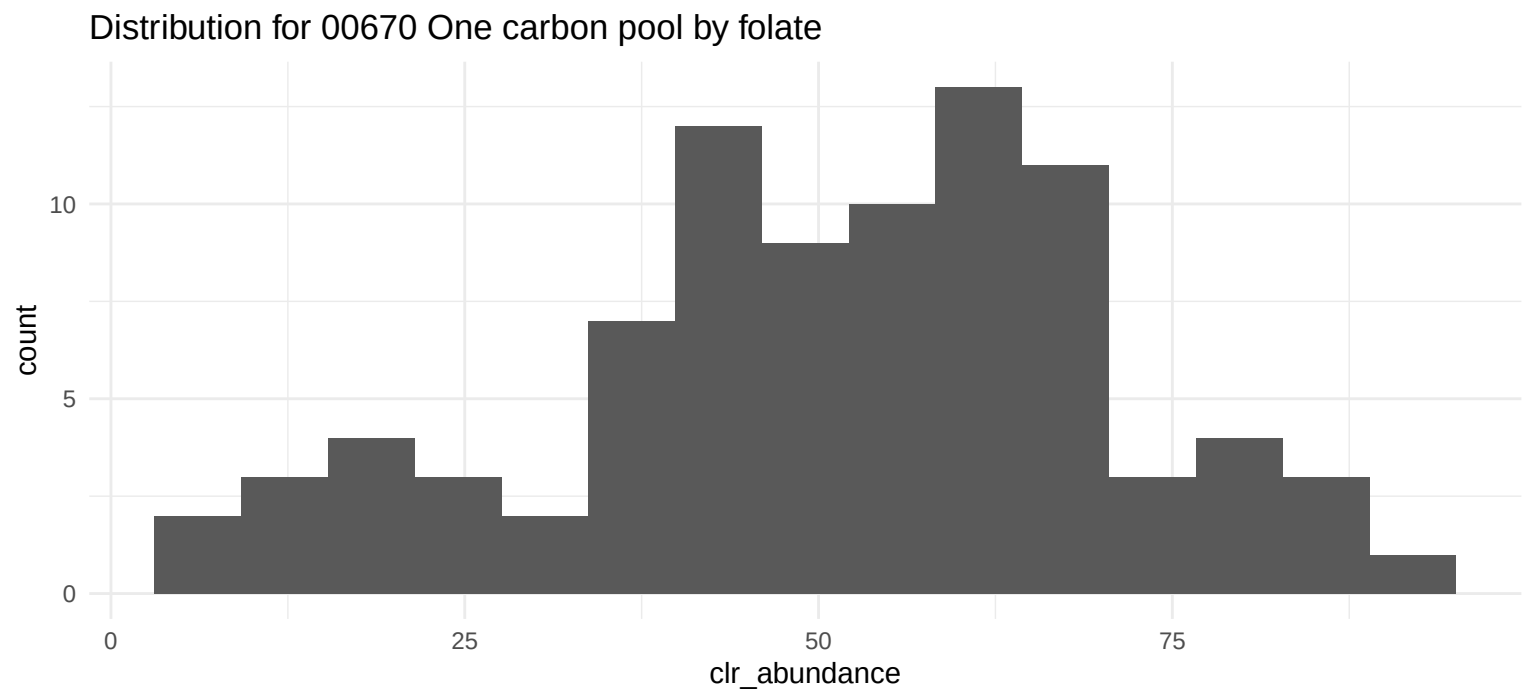

Distribution for 00640 Propanoate metabolism
count
20
15
10
5
0
−50
0
50
100
clr_abundance

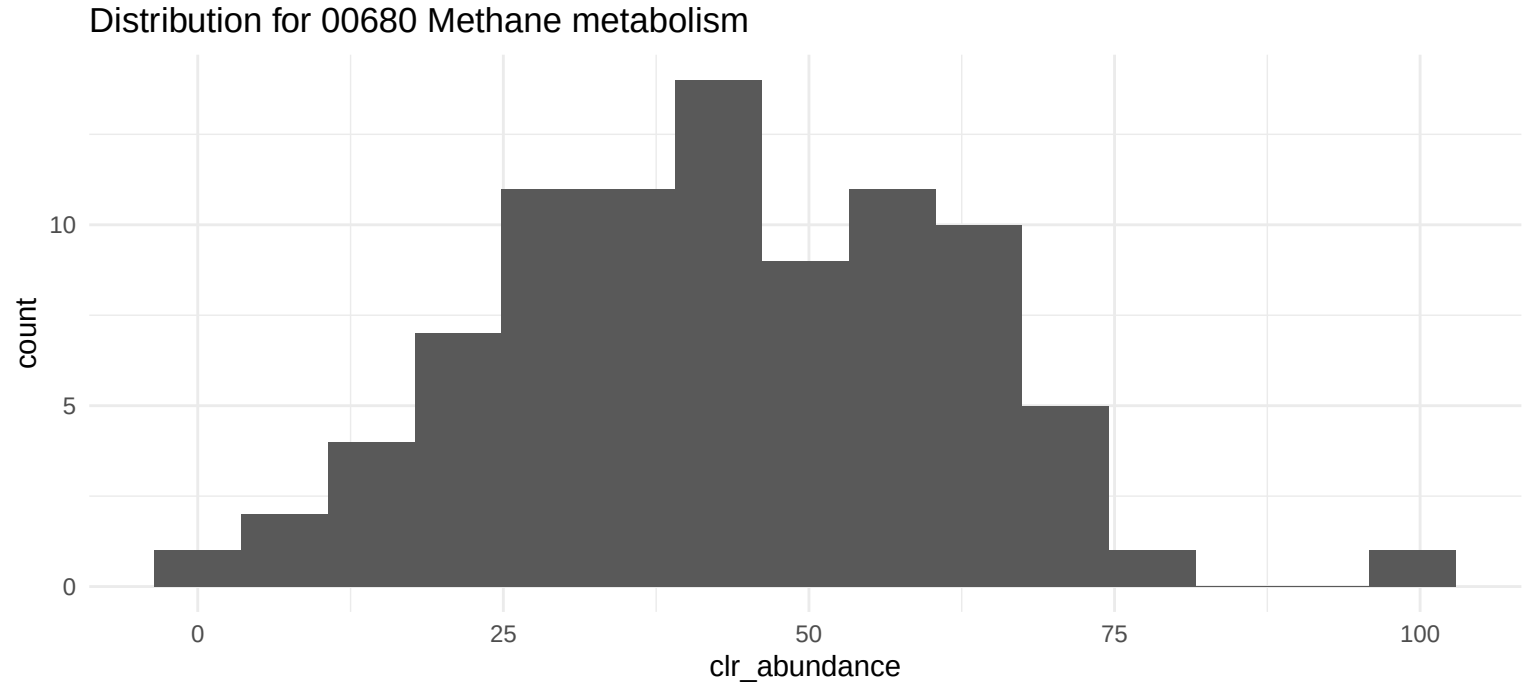

Distribution for 00642 Ethylbenzene degradation
count
20
10
0
−2
0
2
4
clr_abundance

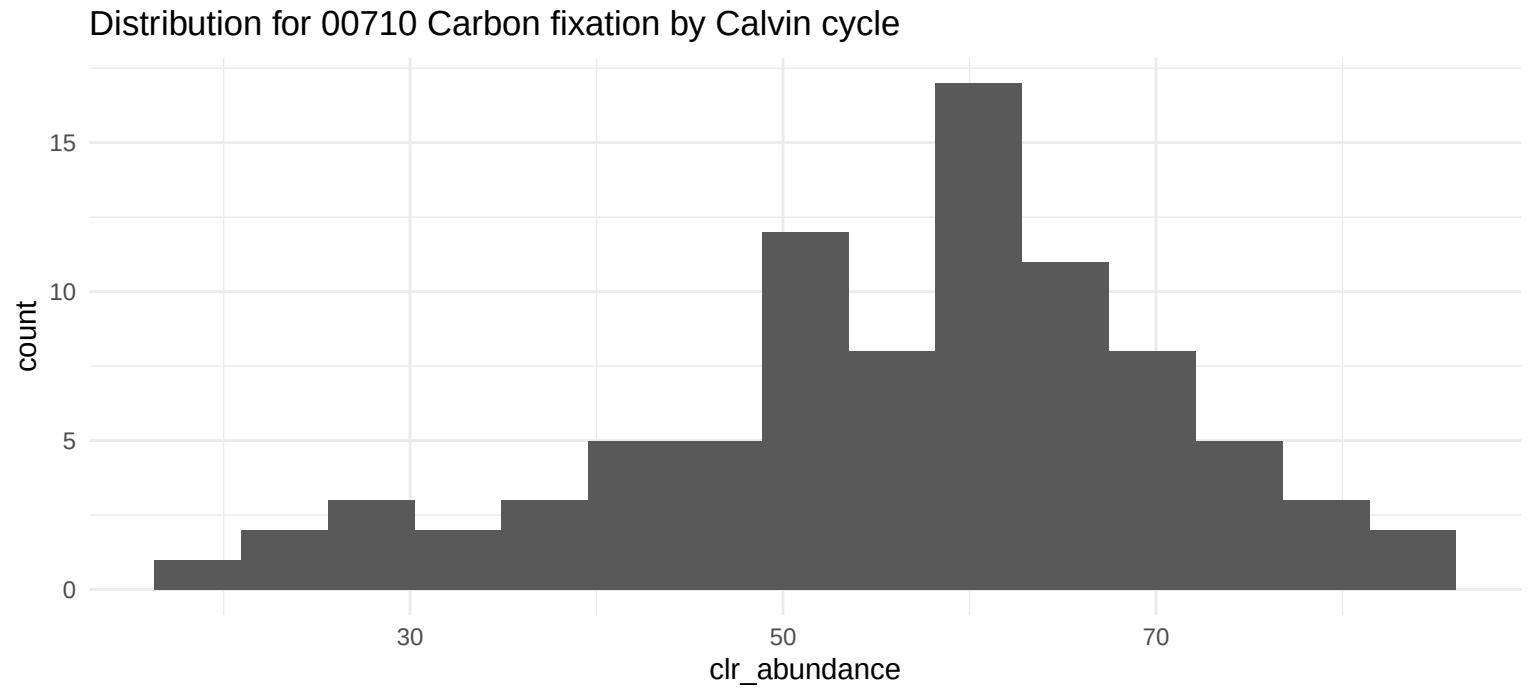

Distribution for 00643 Styrene degradation
count
clr_abundance
20
15
10
5
0
−40
−30
−20
−10
0

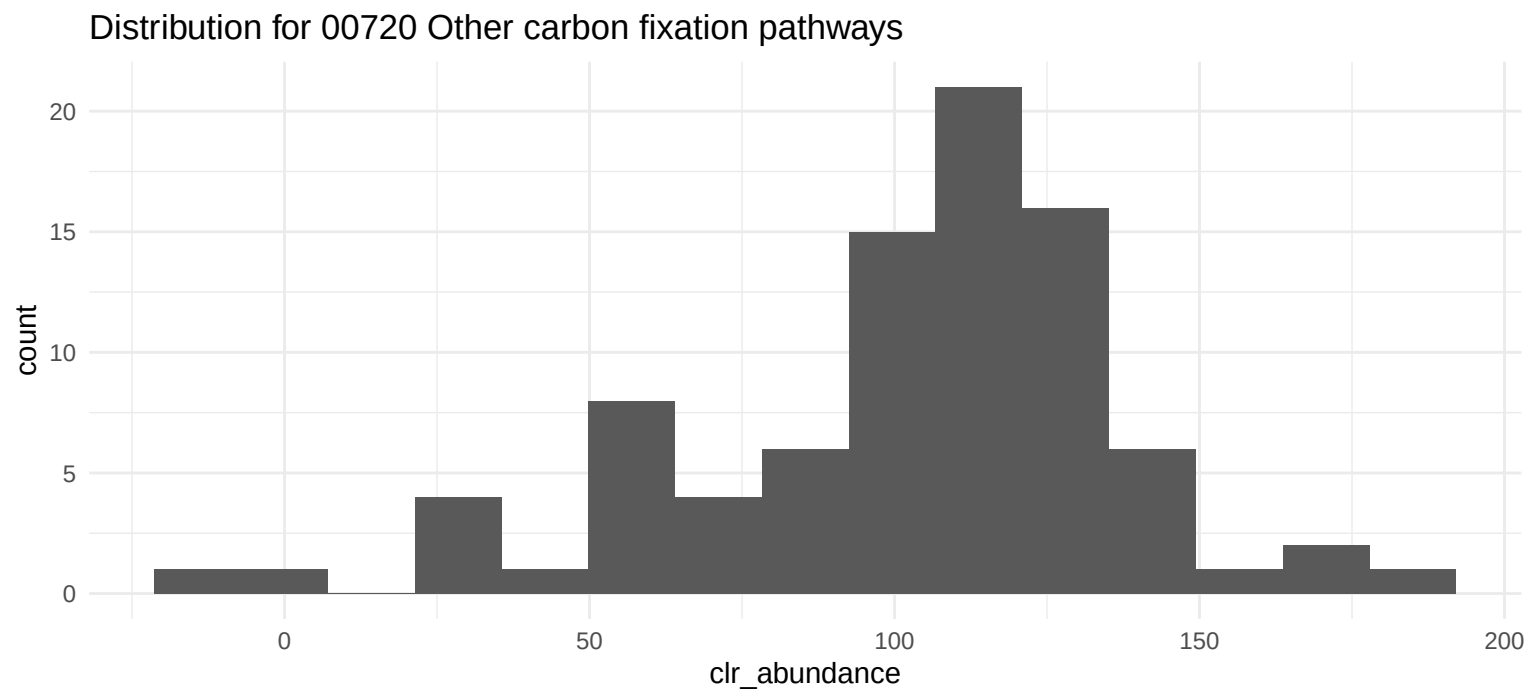

Distribution for 00650 Butanoate metabolism
count
clr_abundance
10
5
0
−20
0
20

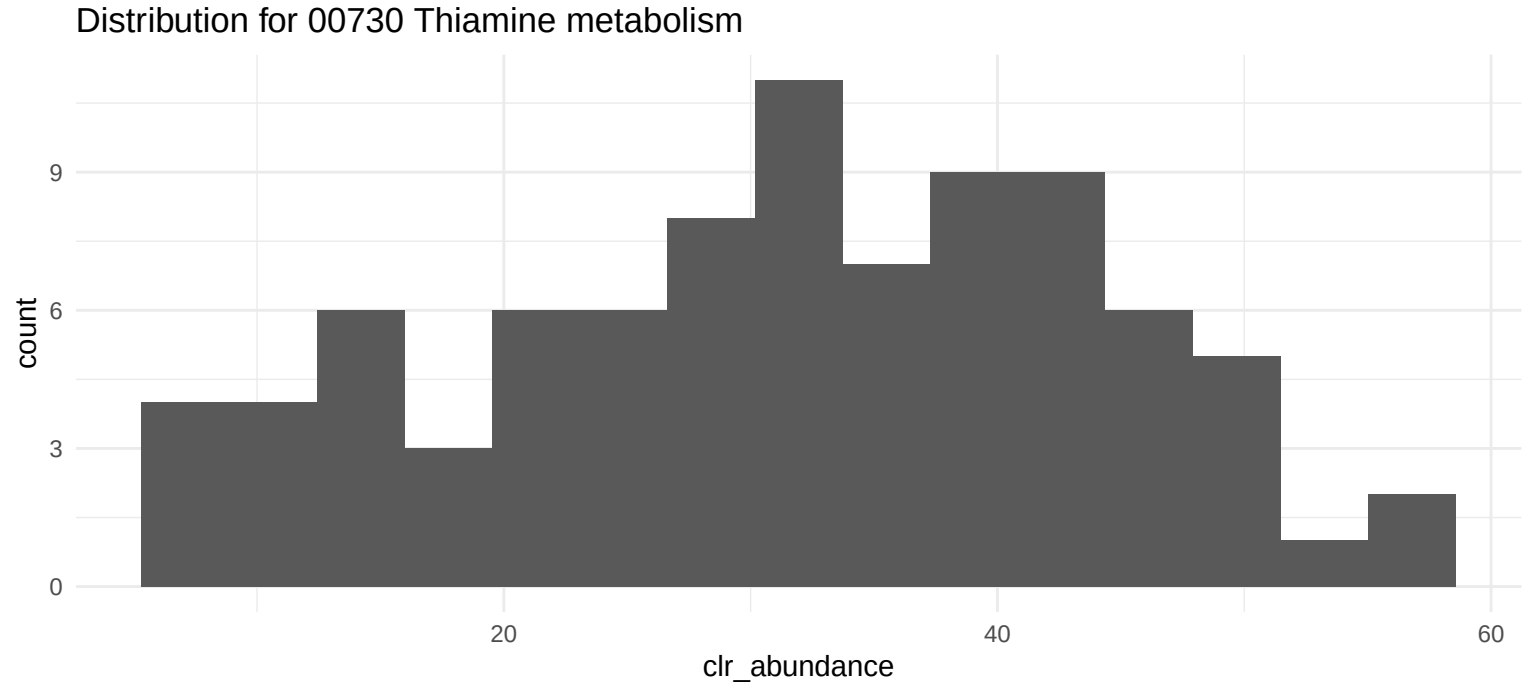

Distribution for 00660 C5−Branched dibasic acid metabolism
count
clr_abundance
12.5
10.0
7.5
5.0
2.5
0.0
10
20
30

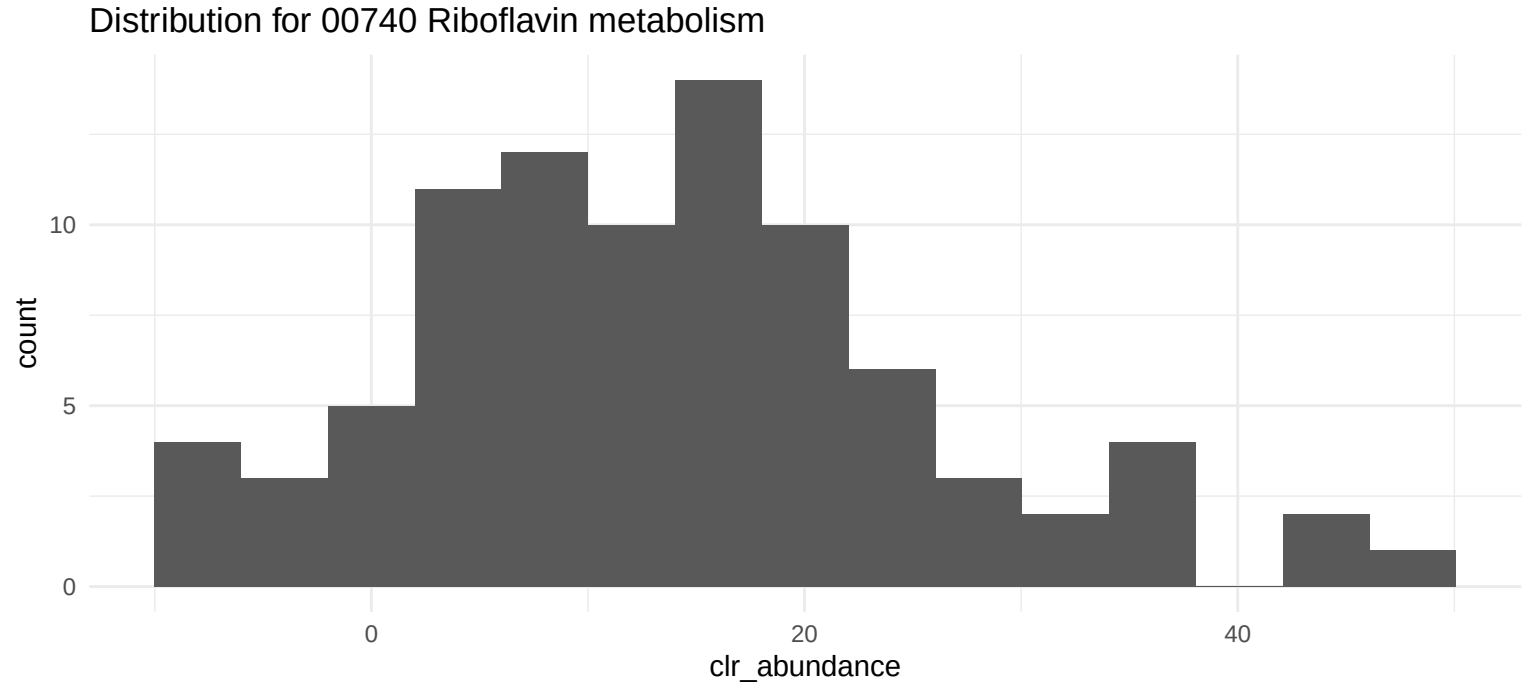

Distribution for 00670 One carbon pool by folate
count
clr_abundance
0
5
10
0
25
50
75

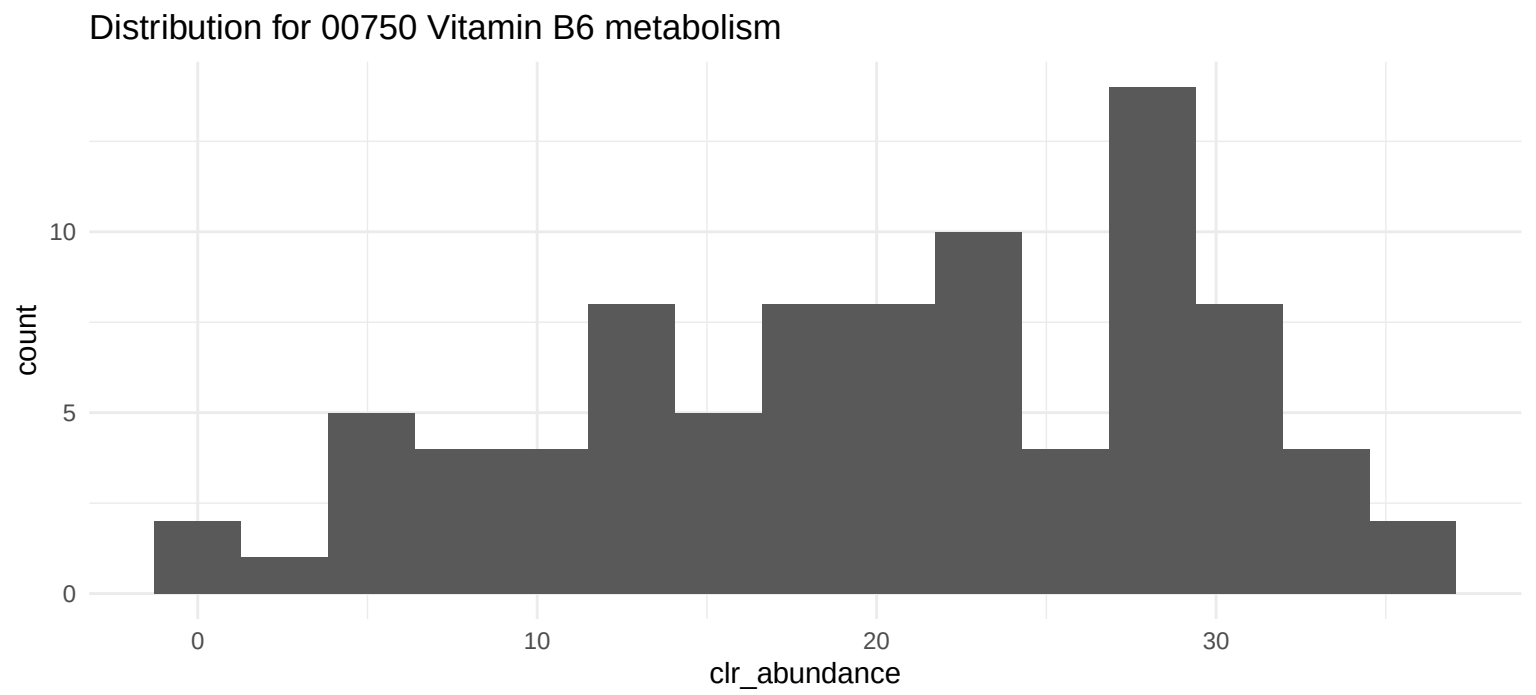

Distribution for 00680 Methane metabolism
count
clr_abundance
0
5
10
0
25
50
75
100

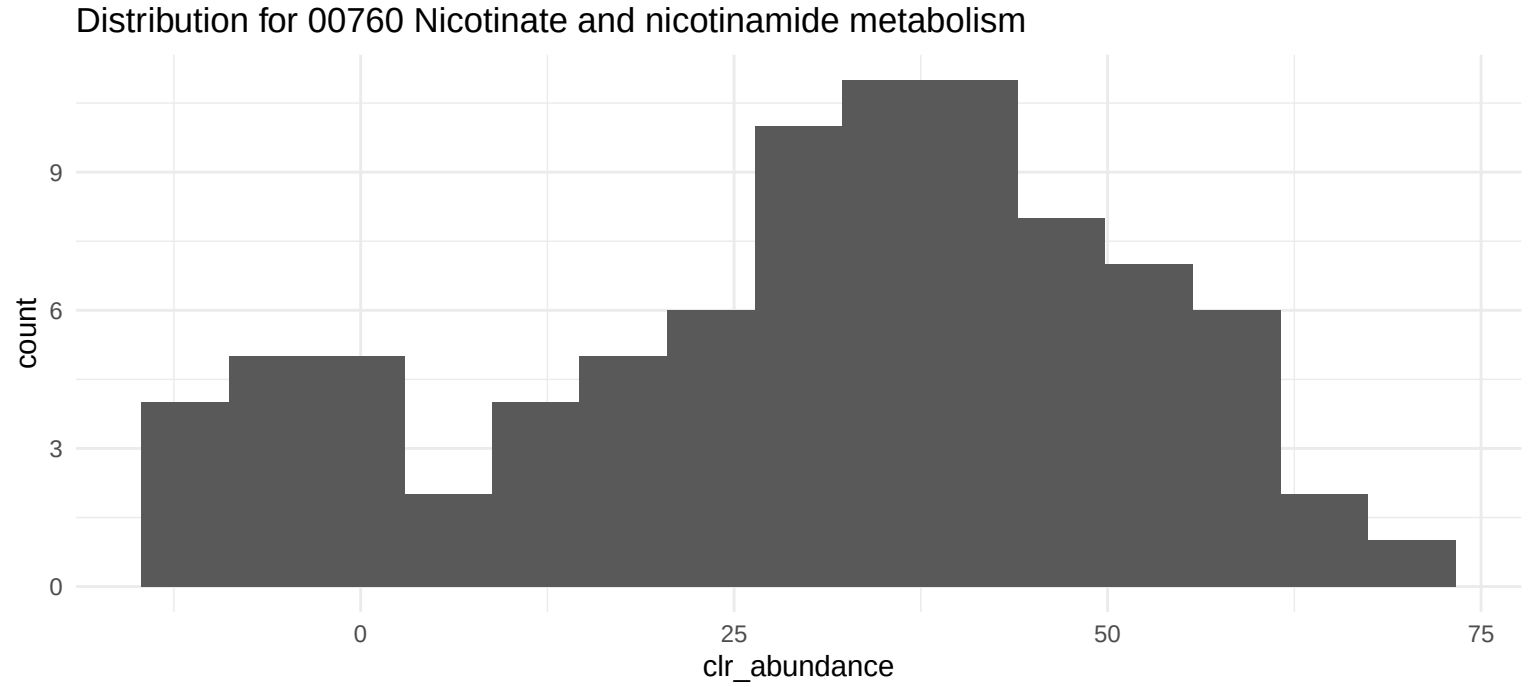

Distribution for 00710 Carbon fixation by Calvin cycle
count
clr_abundance
0
5
10
15
30
50
70

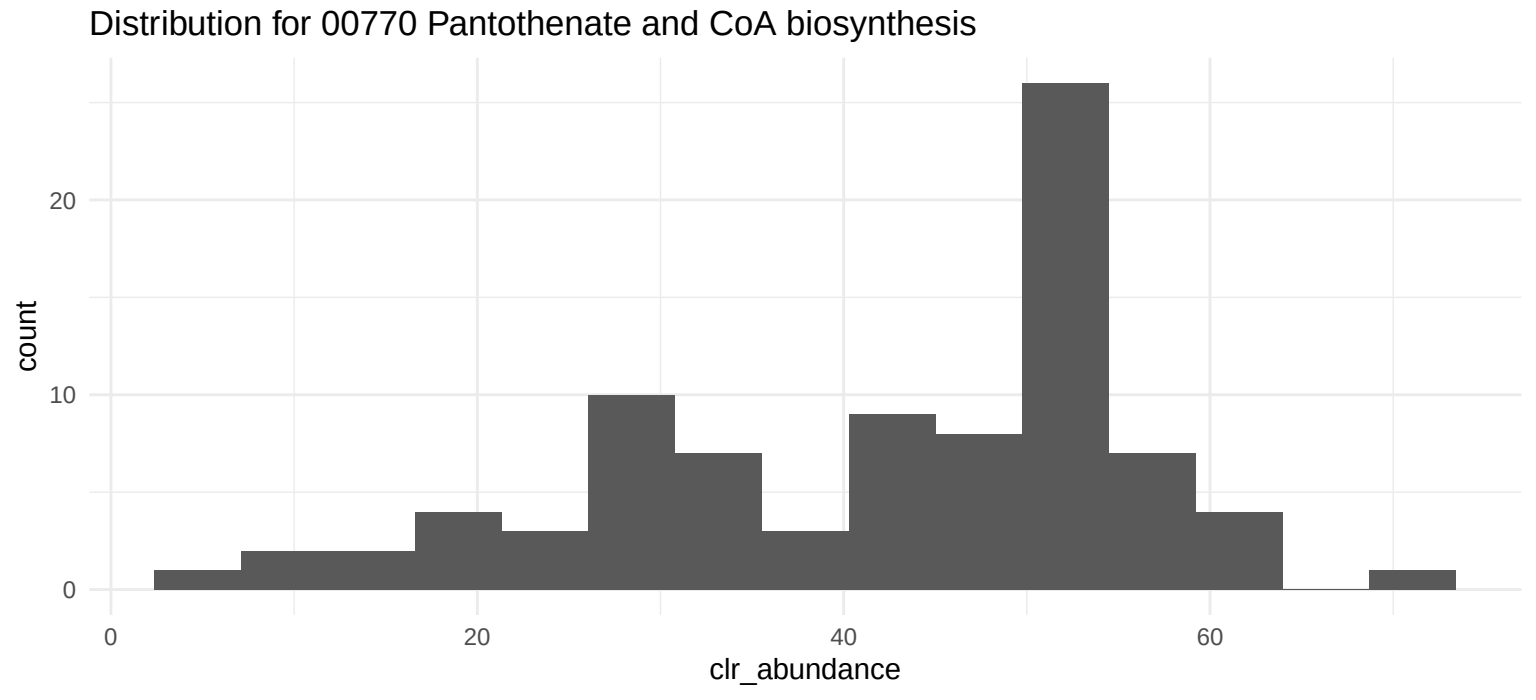

Distribution for 00720 Other carbon fixation pathways
count
clr_abundance
0
5
10
15
20
0
50
100
150
200

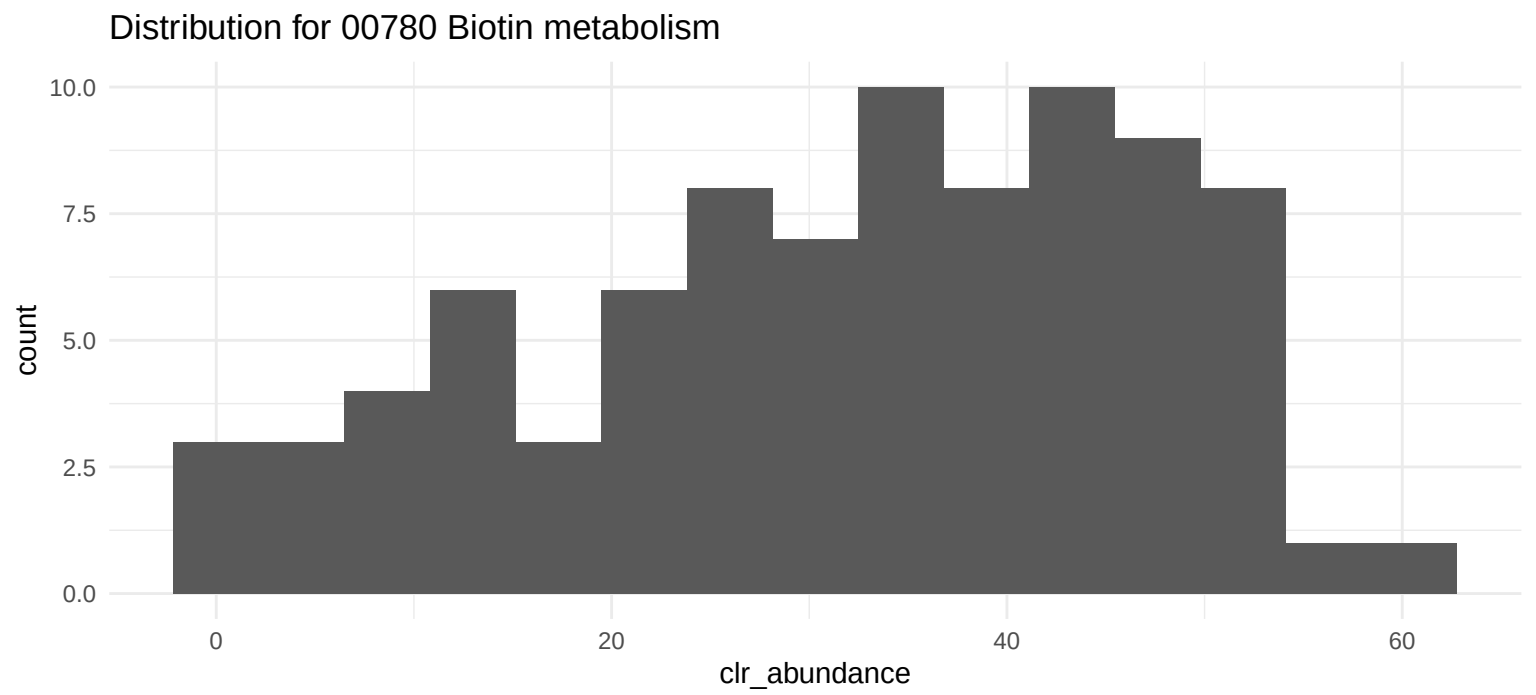

Distribution for 00730 Thiamine metabolism
count
clr_abundance
0
3
6
9
20
40
60

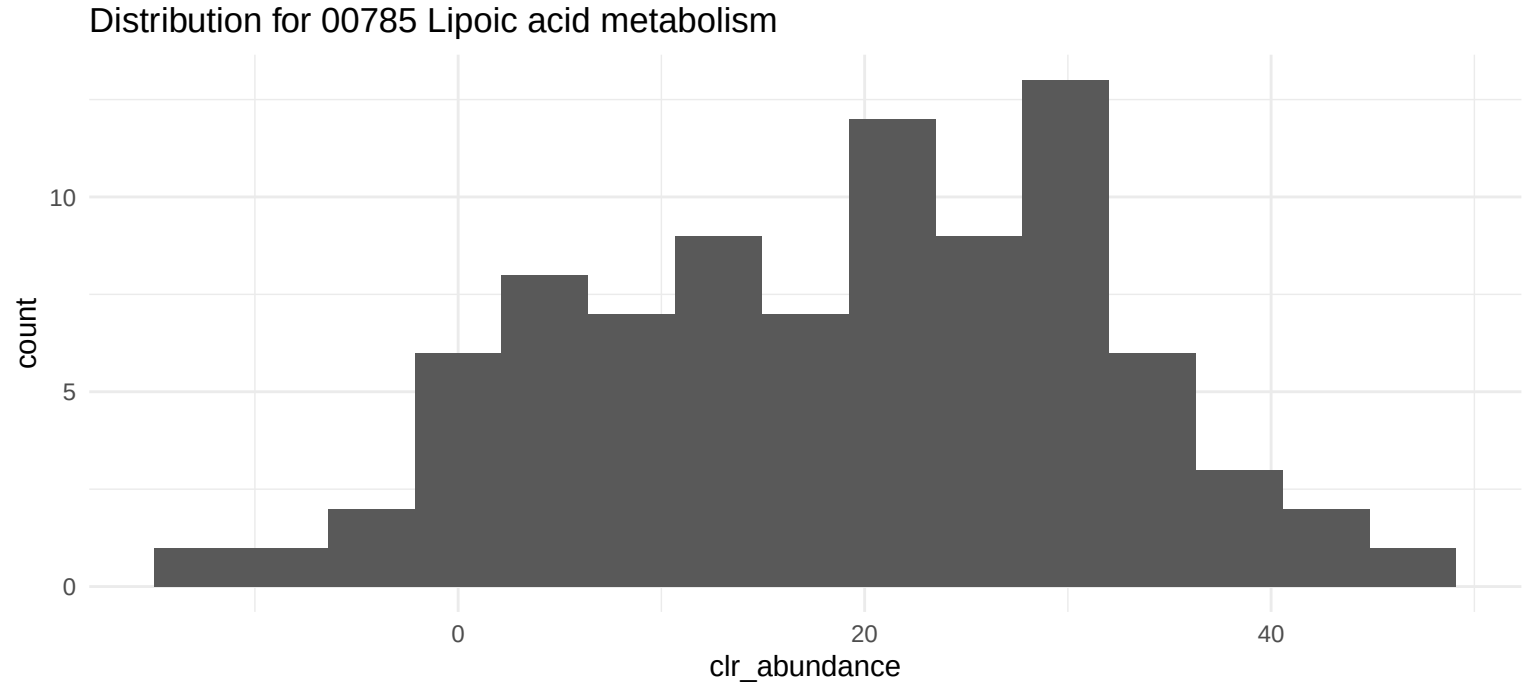

Distribution for 00740 Riboflavin metabolism
count
clr_abundance
0
5
10
0
20
40

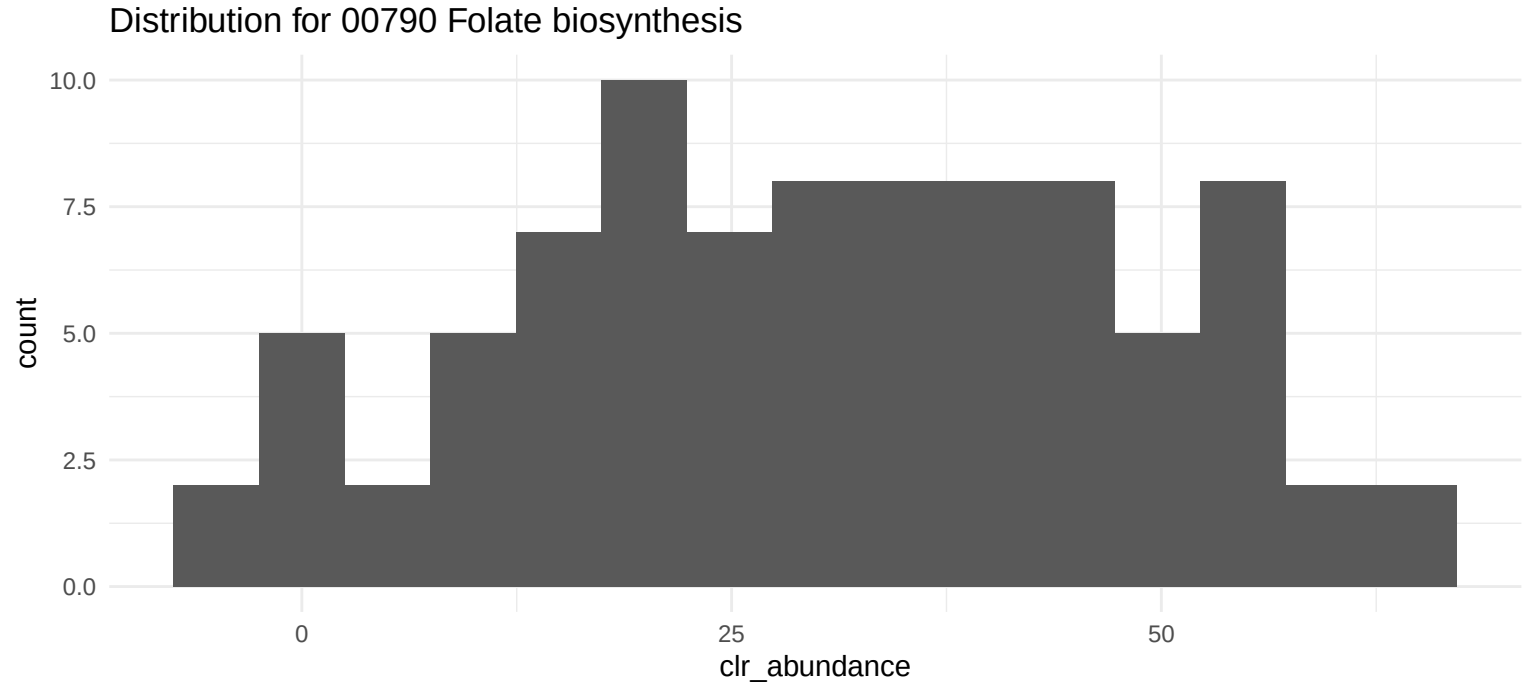

Distribution for 00750 Vitamin B6 metabolism
count
clr_abundance
0
5
10
0
10
20
30

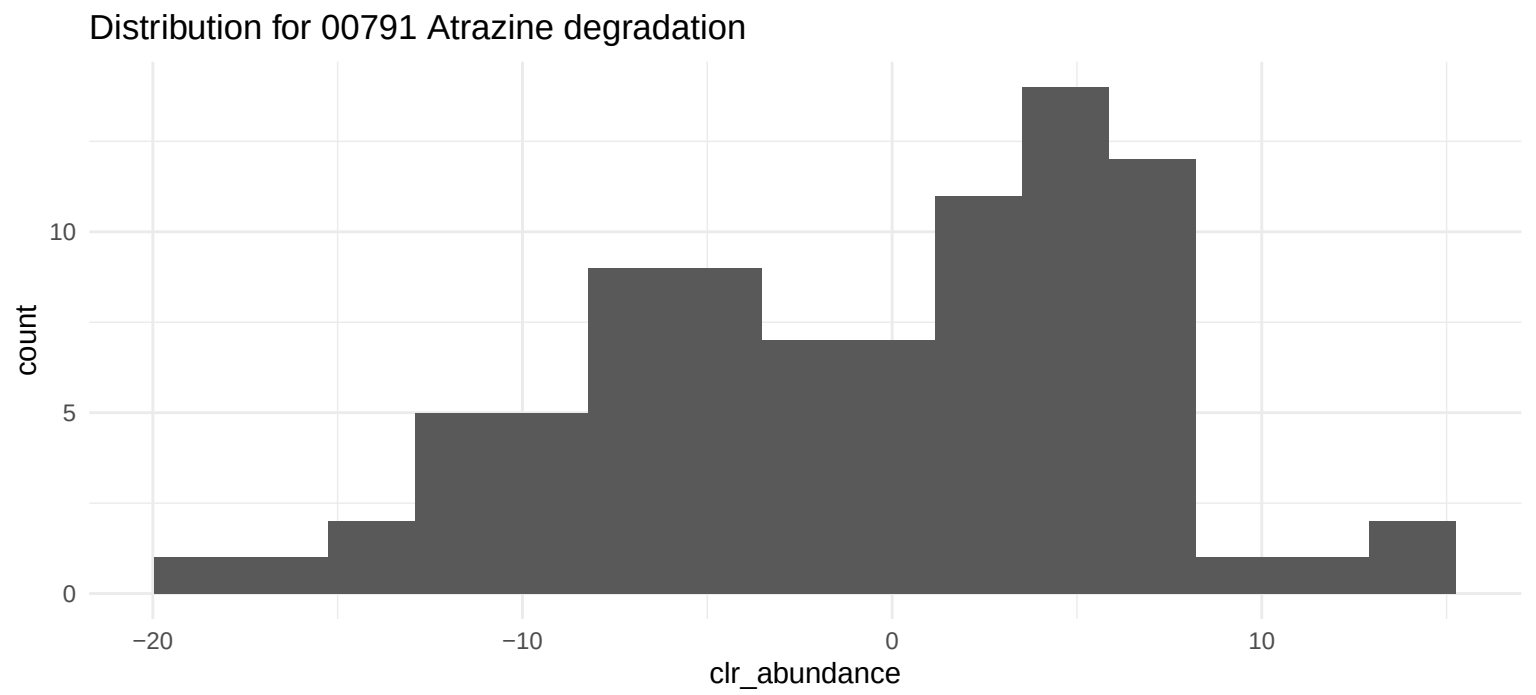

Distribution for 00760 Nicotinate and nicotinamide metabolism
count
clr_abundance
0
3
6
9
0
25
50
75

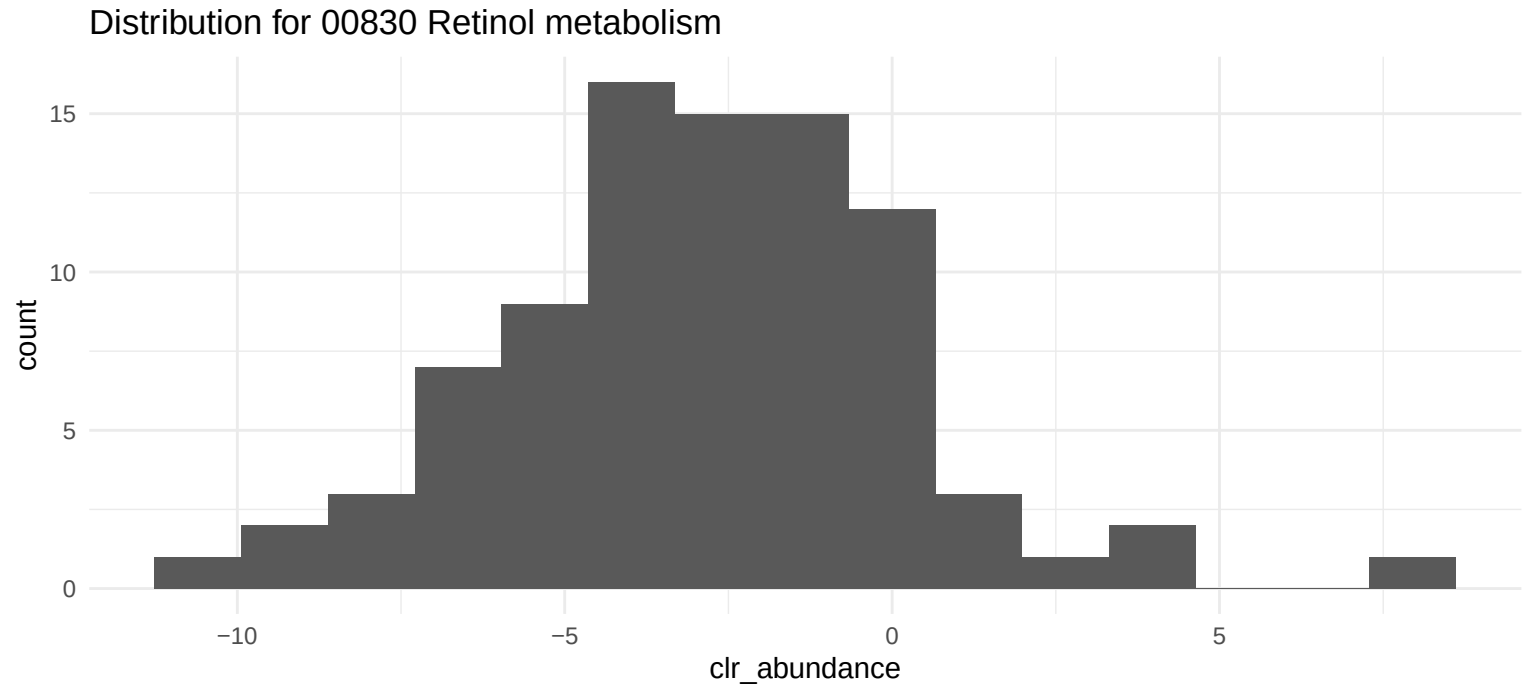

Distribution for 00770 Pantothenate and CoA biosynthesis
count
clr_abundance
0
10
20
0
20
40
60

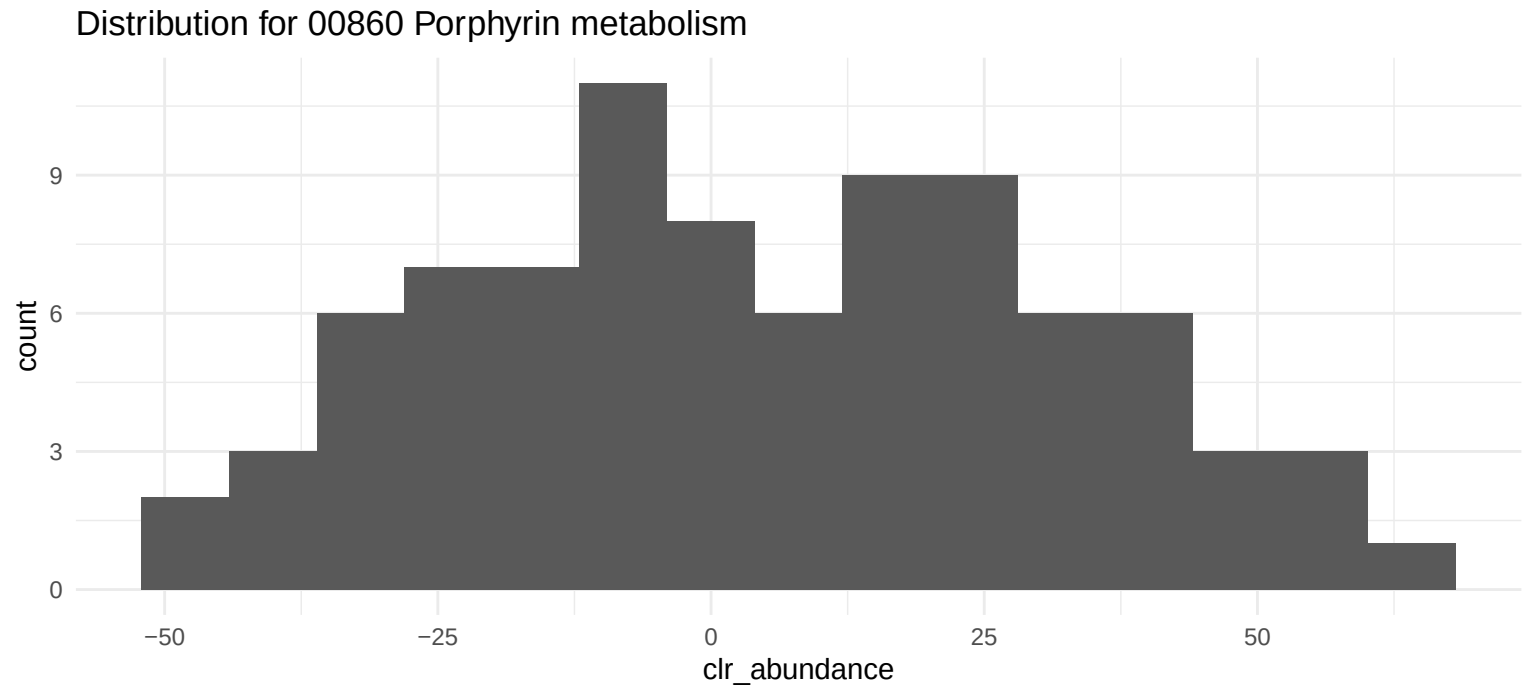

Distribution for 00780 Biotin metabolism
10.0
7.5
5.0
2.5
0.0
count
0
20
40
60
clr_abundance


Distribution for 00785 Lipoic acid metabolism
10
5
0
count
0
20
40
clr_abundance

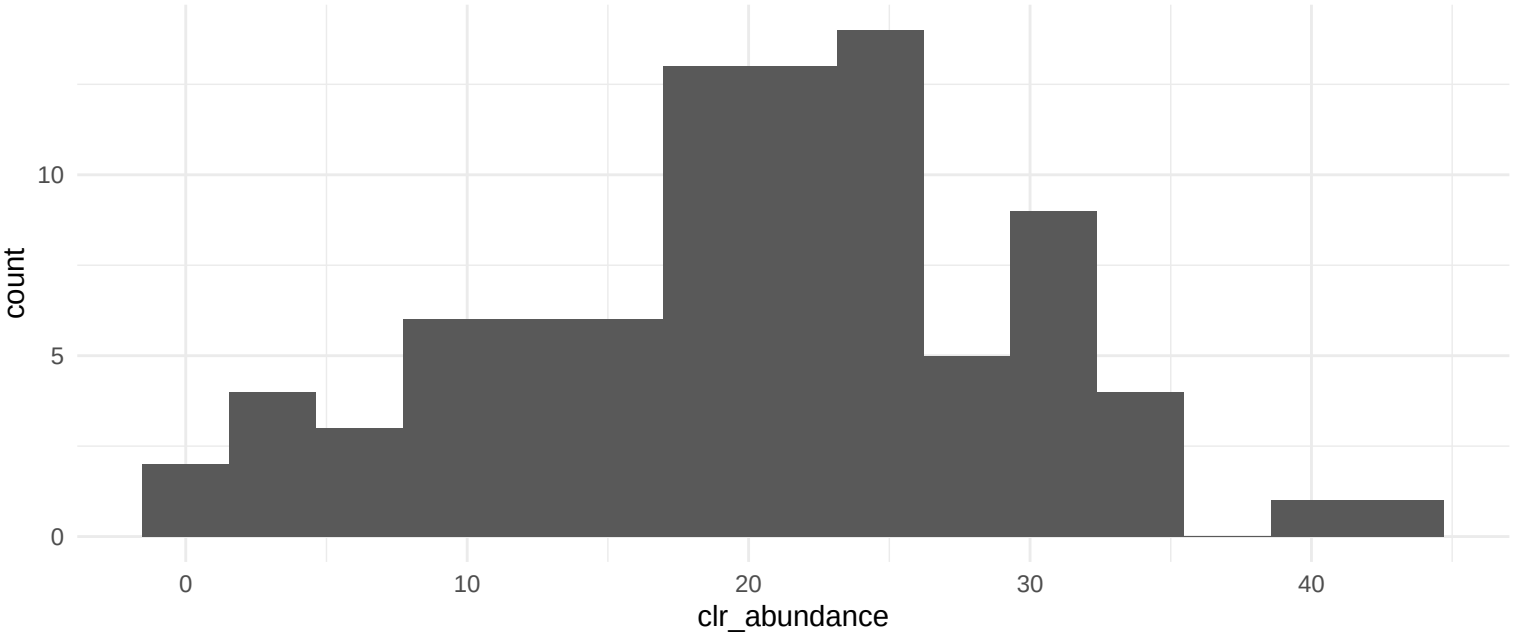

Distribution for 00790 Folate biosynthesis
10.0
7.5
5.0
2.5
0.0
count
0
25
50
clr_abundance


Distribution for 00791 Atrazine degradation
count
10
5
0
−20
−10
0
10
clr_abundance

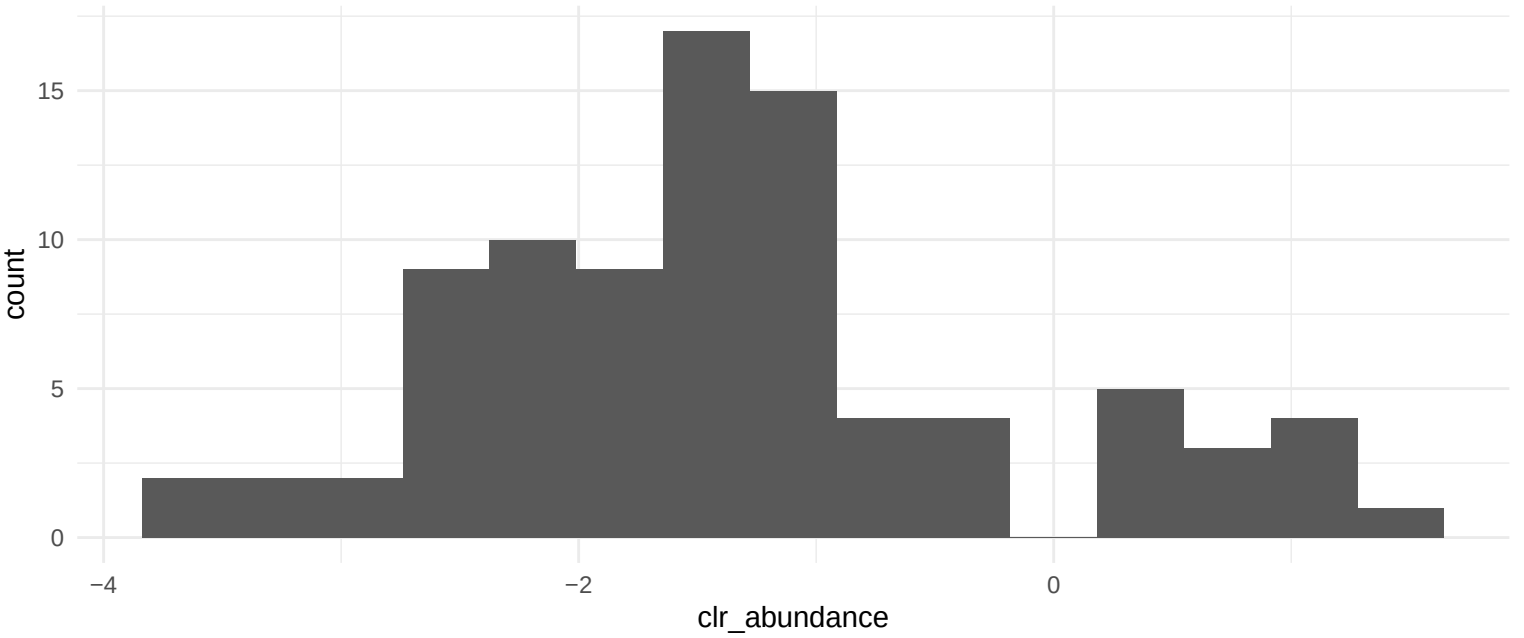

Distribution for 00830 Retinol metabolism
count
15
10
5
0
−10
−5
0
5
clr_abundance


Distribution for 00860 Porphyrin metabolism
count
9
6
3
0
−50
−25
0
25
50
clr_abundance

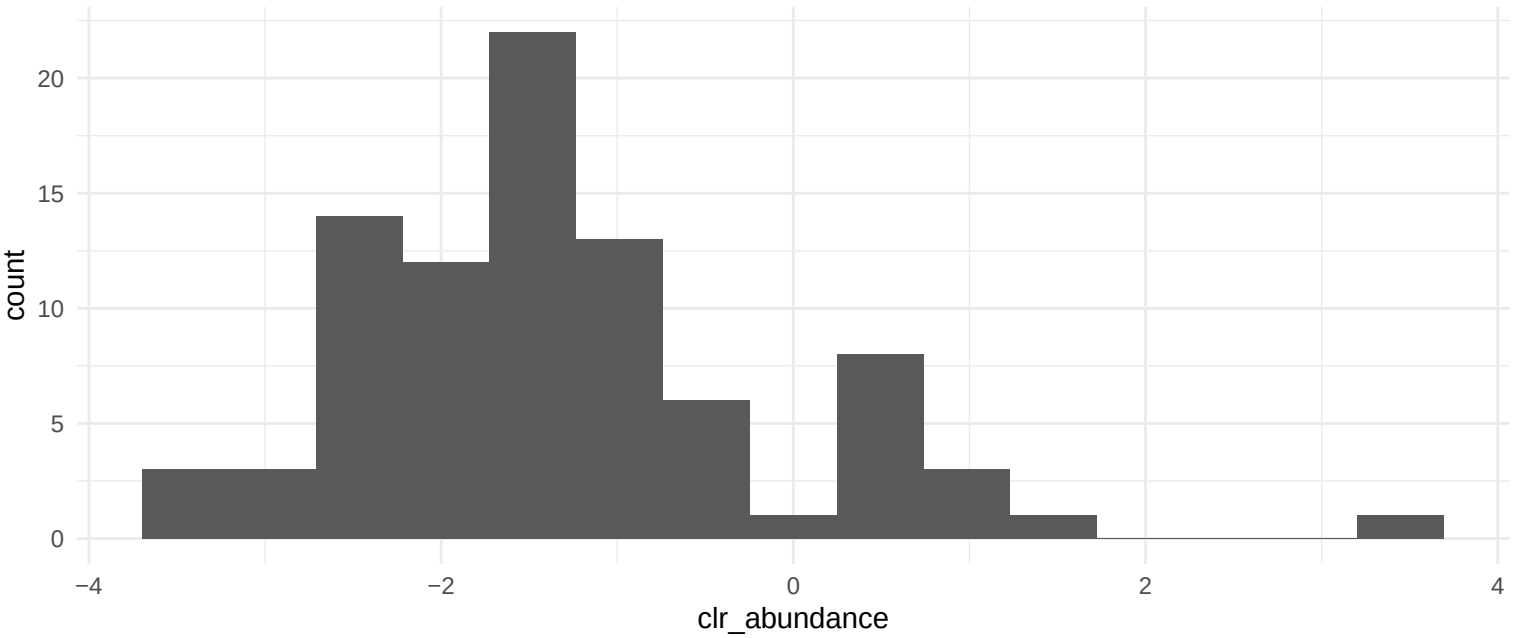
Distribution for 00900 Terpenoid backbone biosynthesis

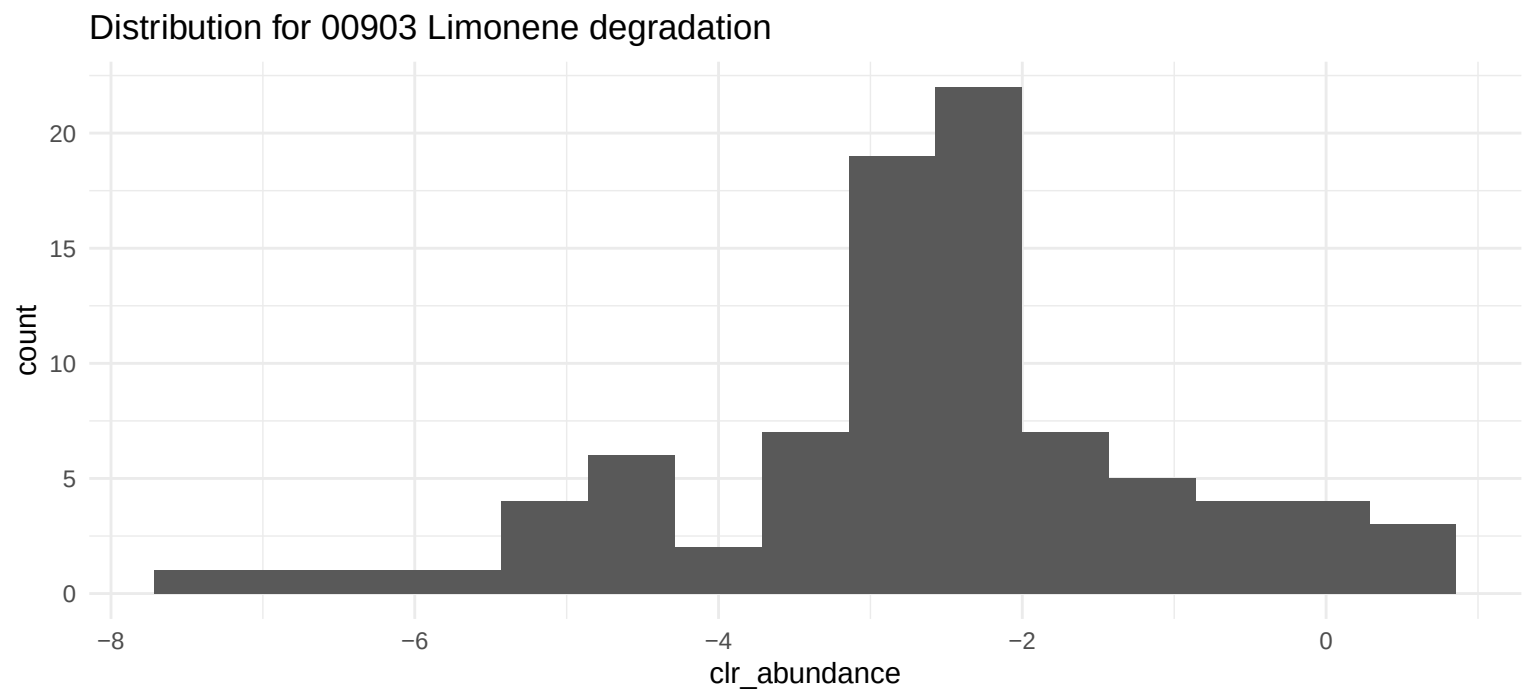


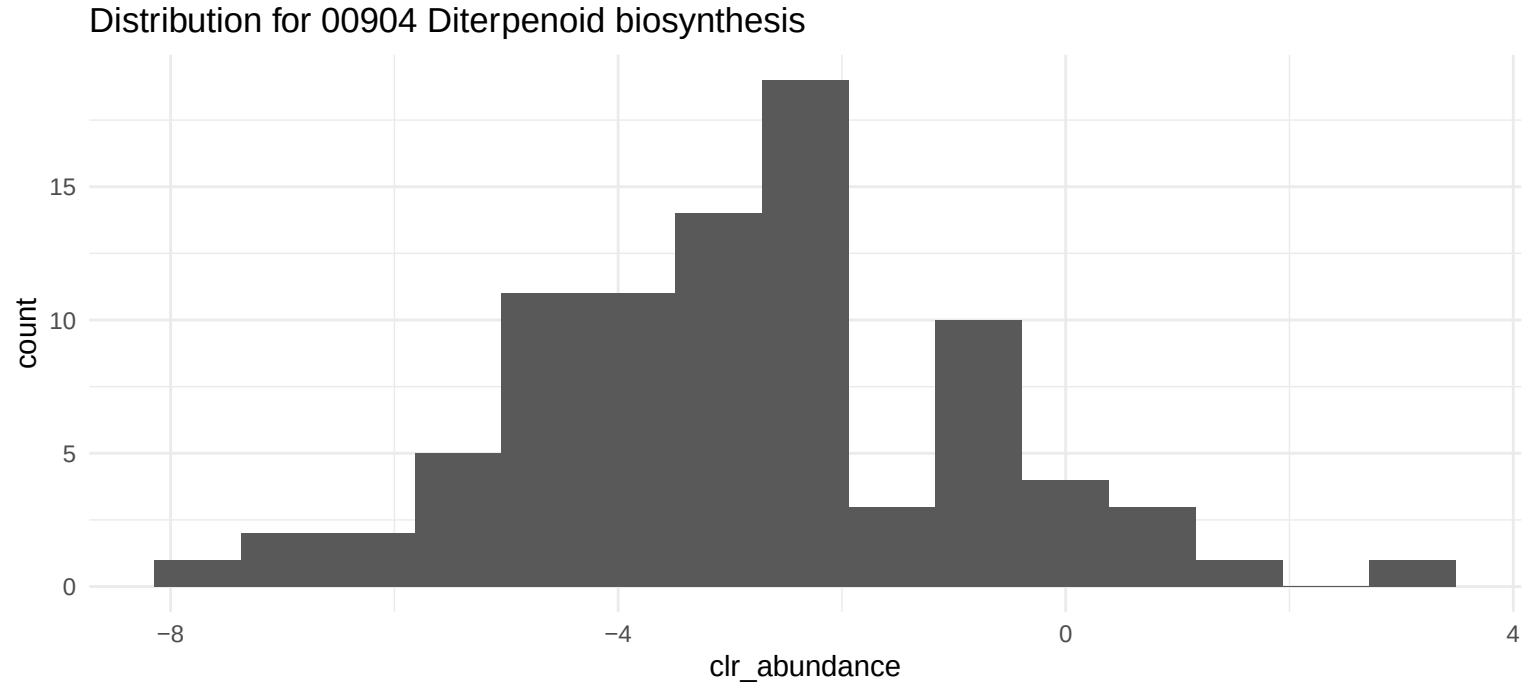
Distribution for 00901 Indole alkaloid biosynthesis

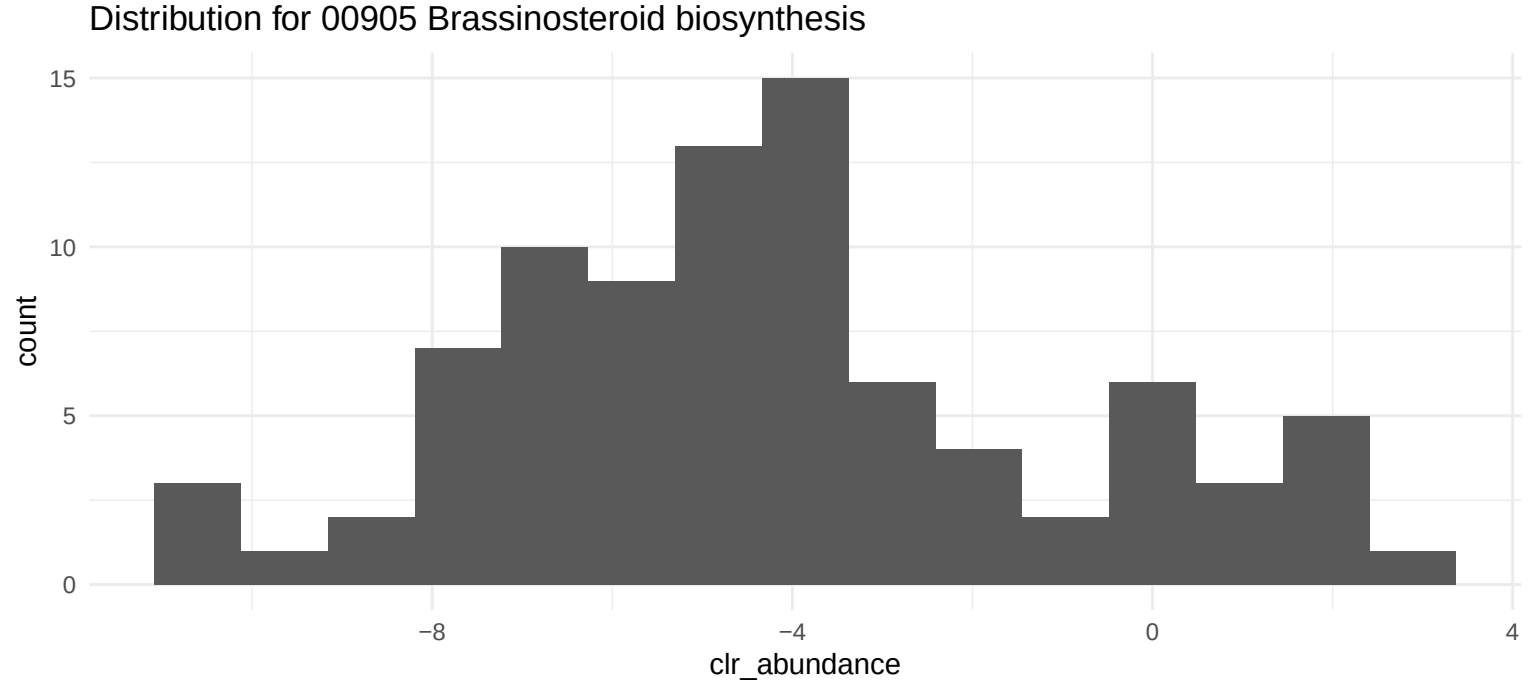


Distribution for 00902 Monoterpenoid biosynthesis

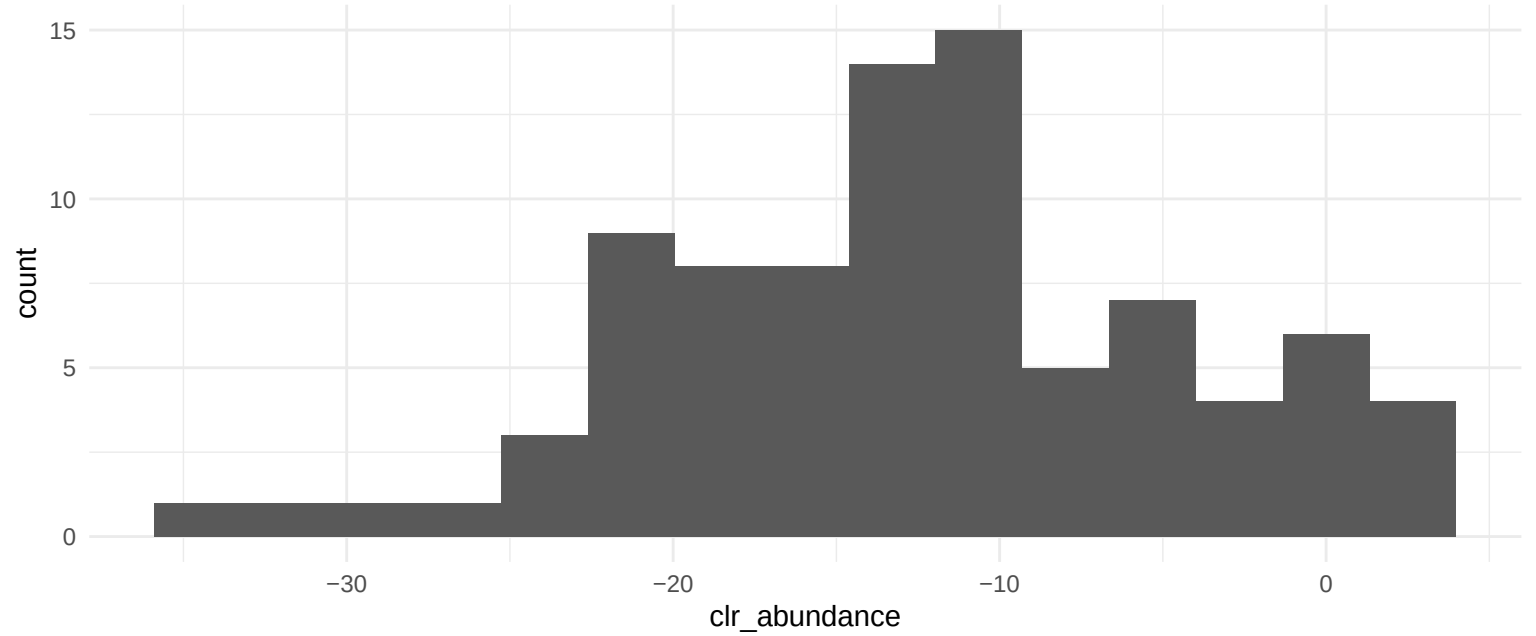



Distribution for 00903 Limonene degradation
count
clr_abundance

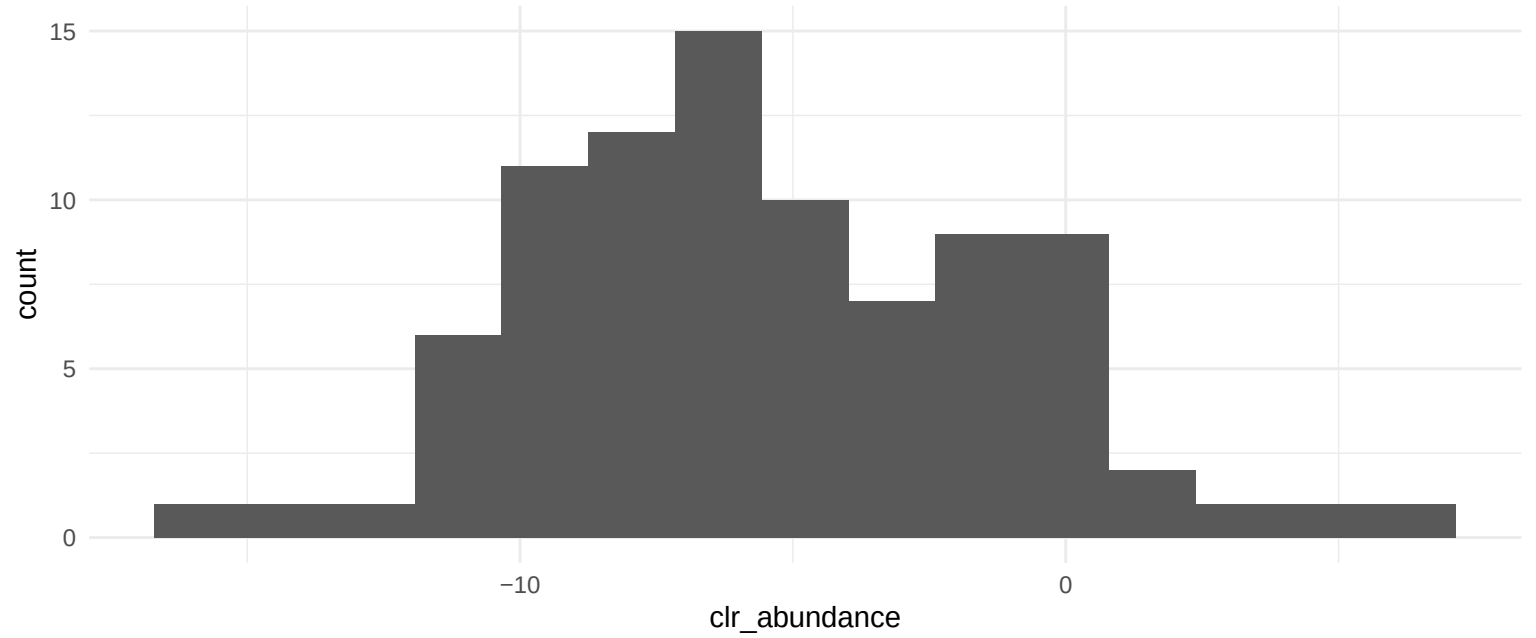

Distribution for 00904 Diterpenoid biosynthesis
count
clr_abundance


Distribution for 00905 Brassinosteroid biosynthesis
count
clr_abundance

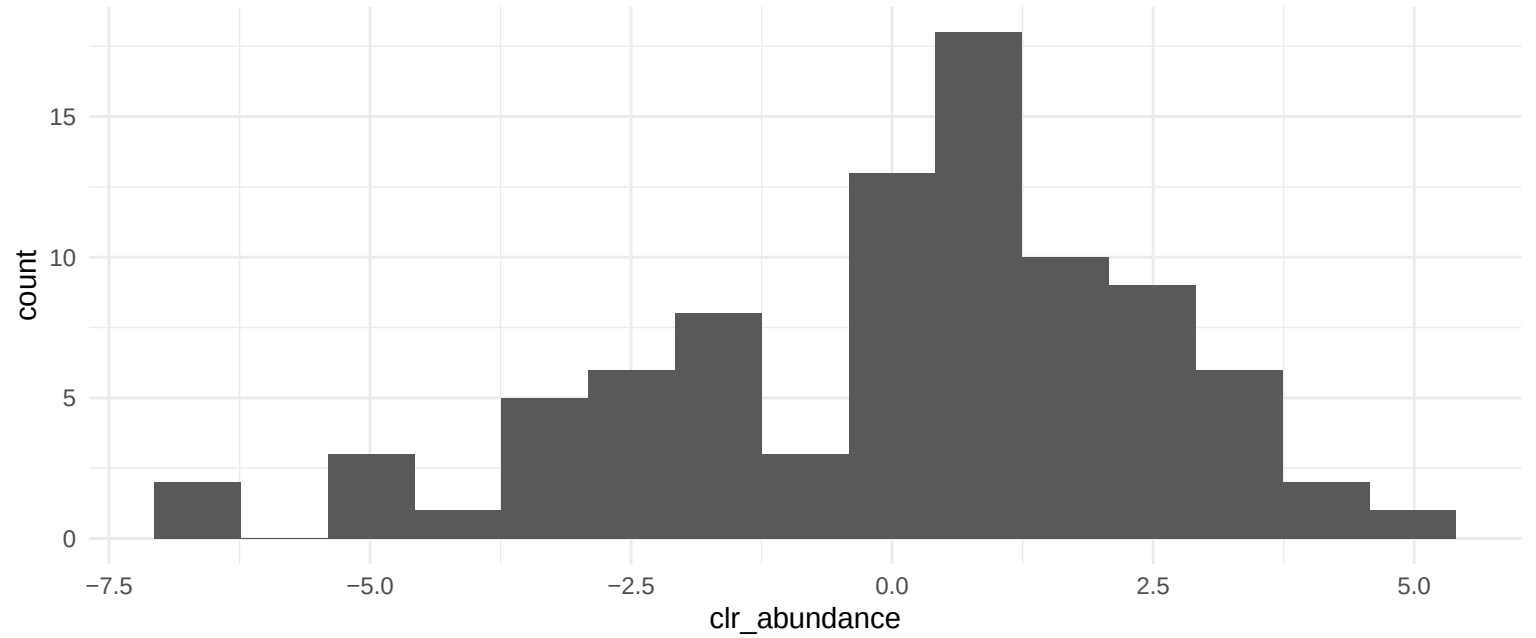
Distribution for 00906 Carotenoid biosynthesis

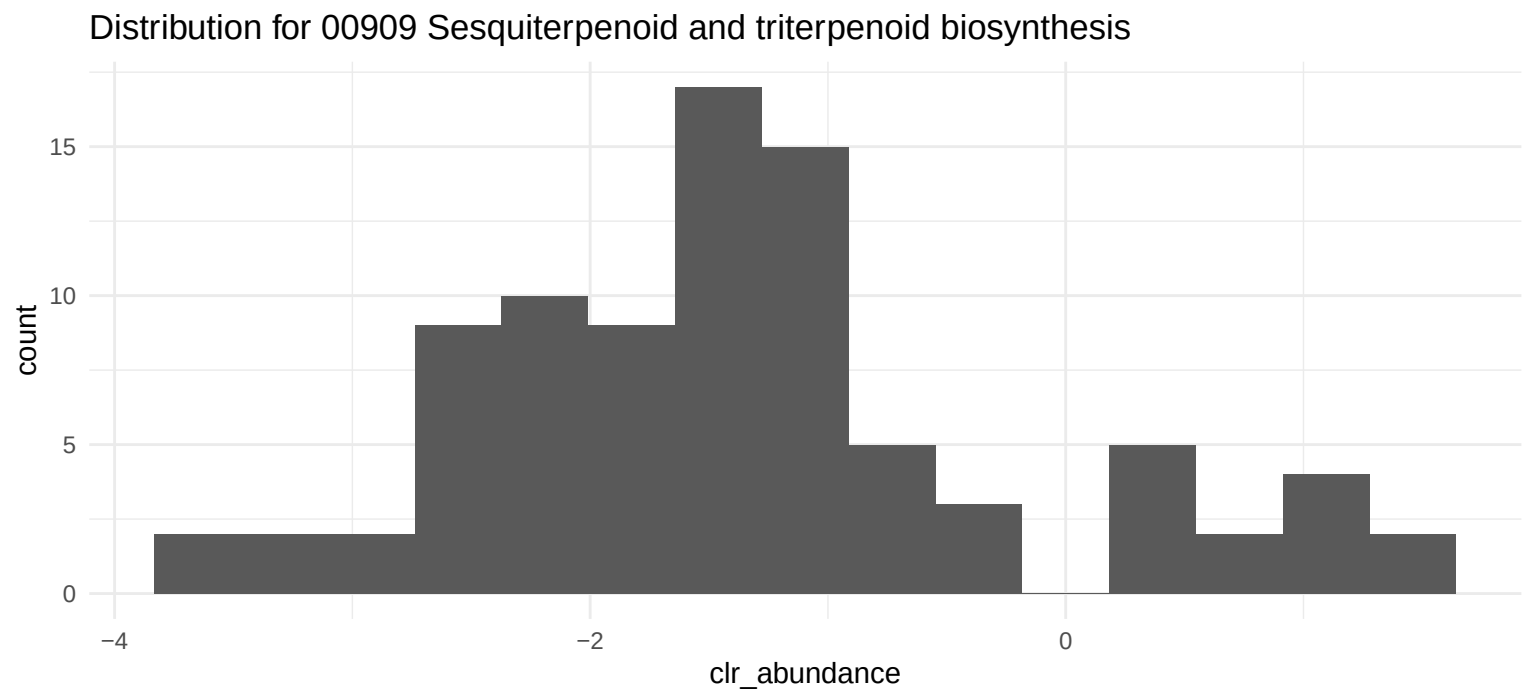


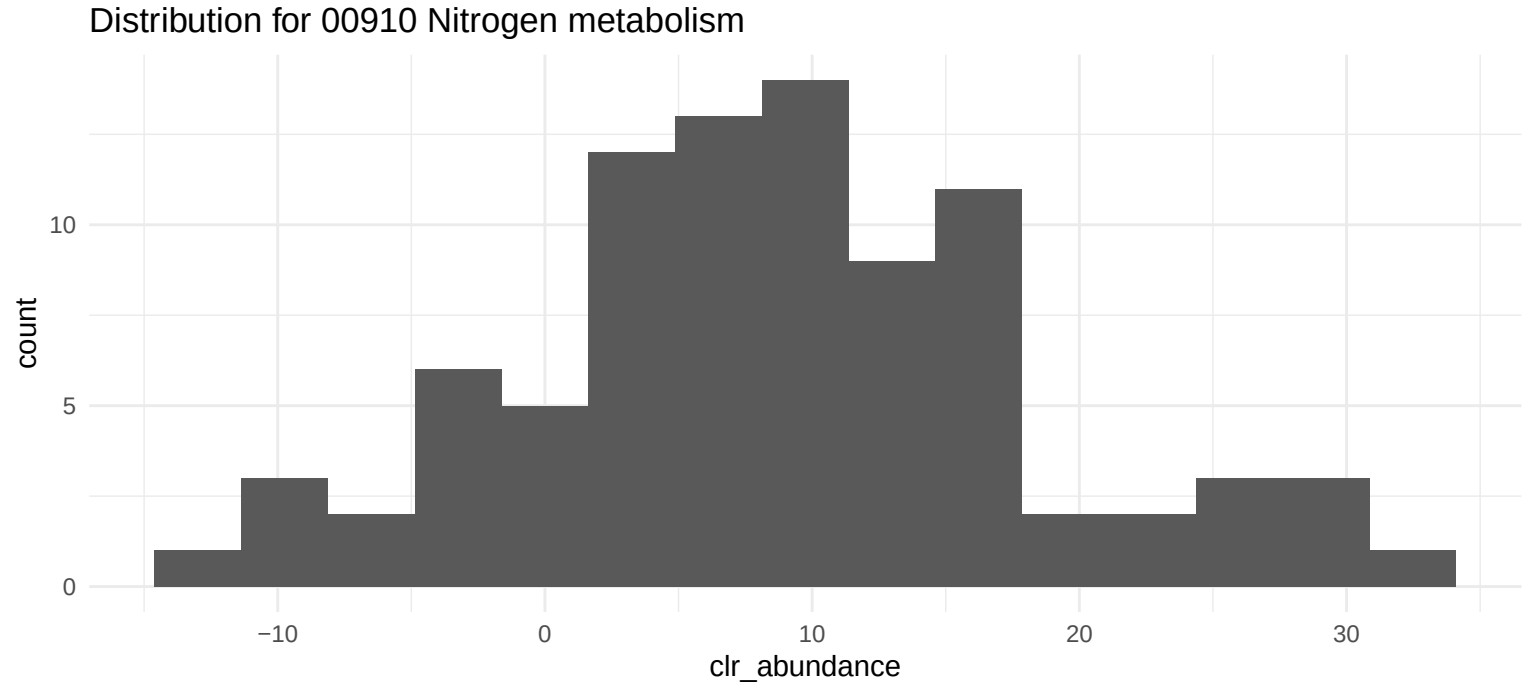
Distribution for 00907 Pinene, camphor and geraniol degradation

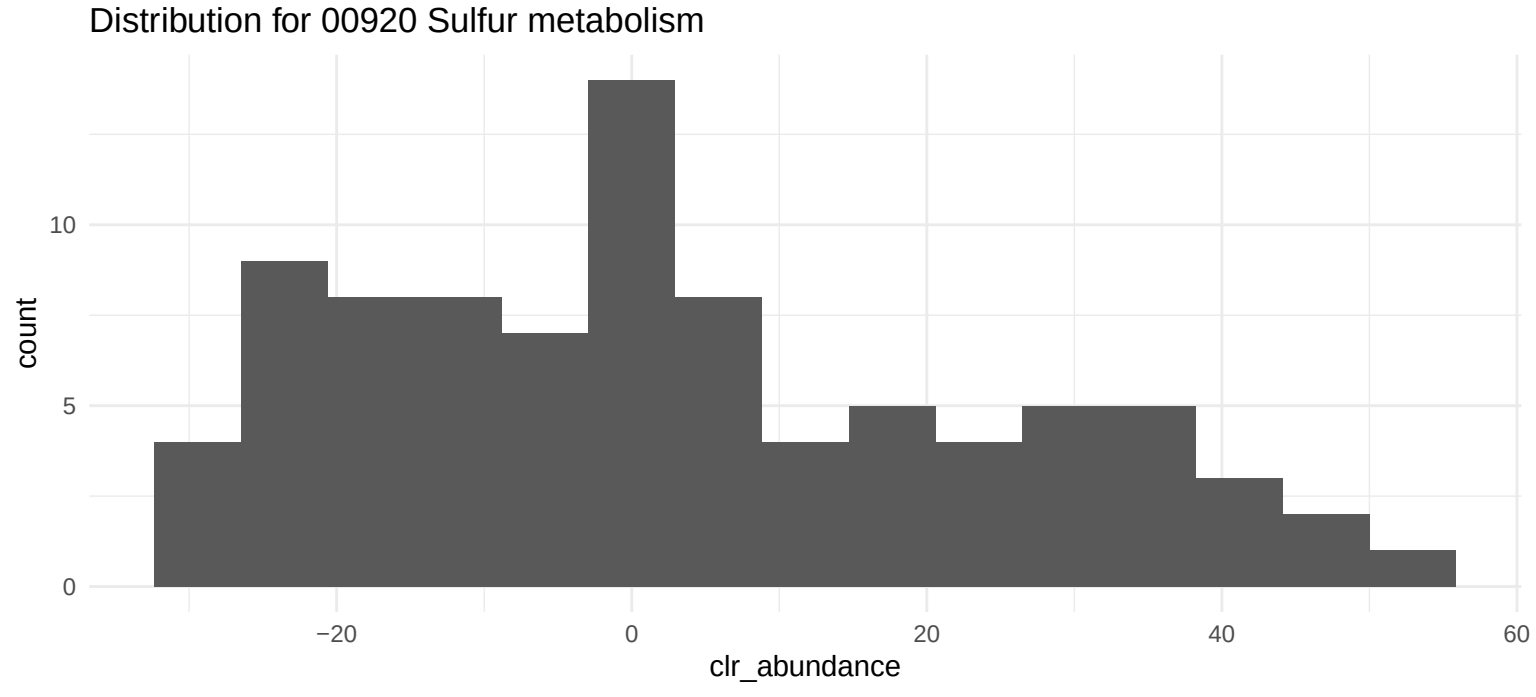


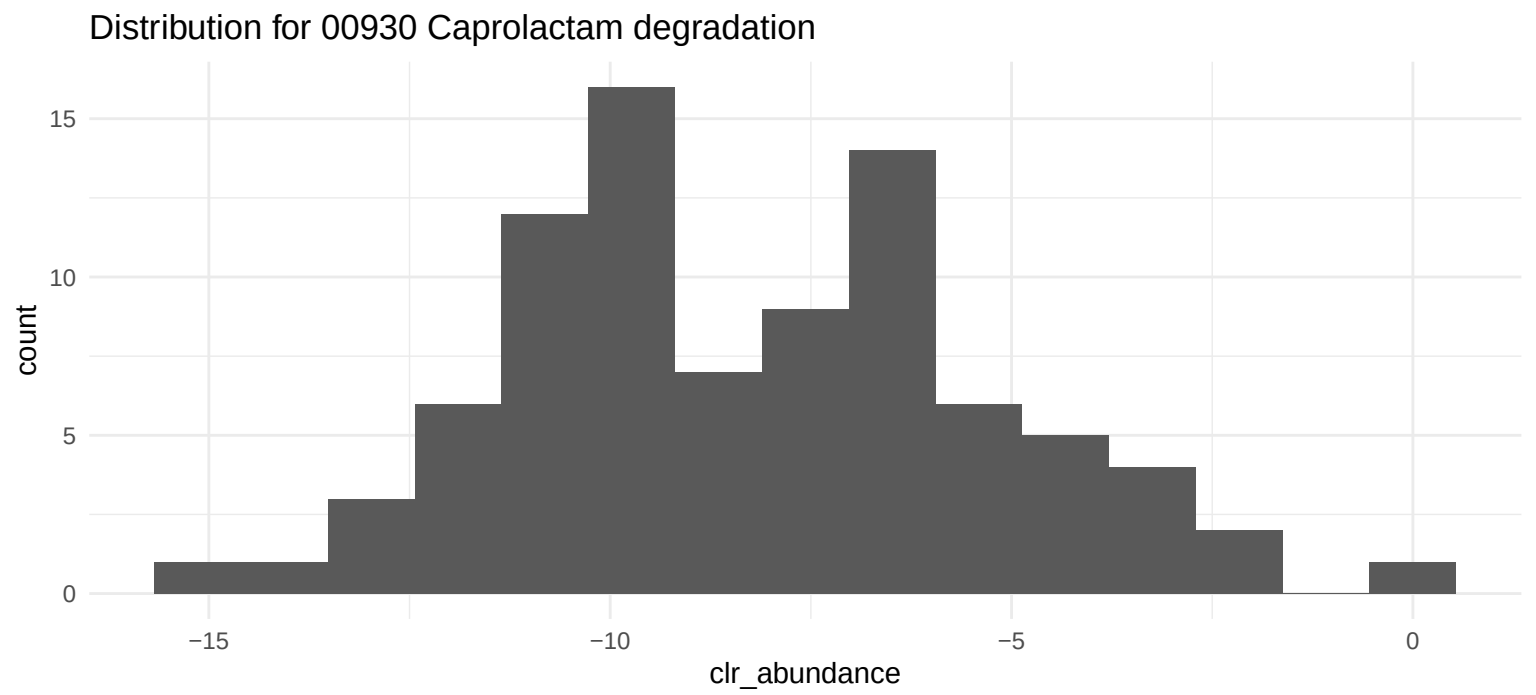
Distribution for 00908 Zeatin biosynthesis

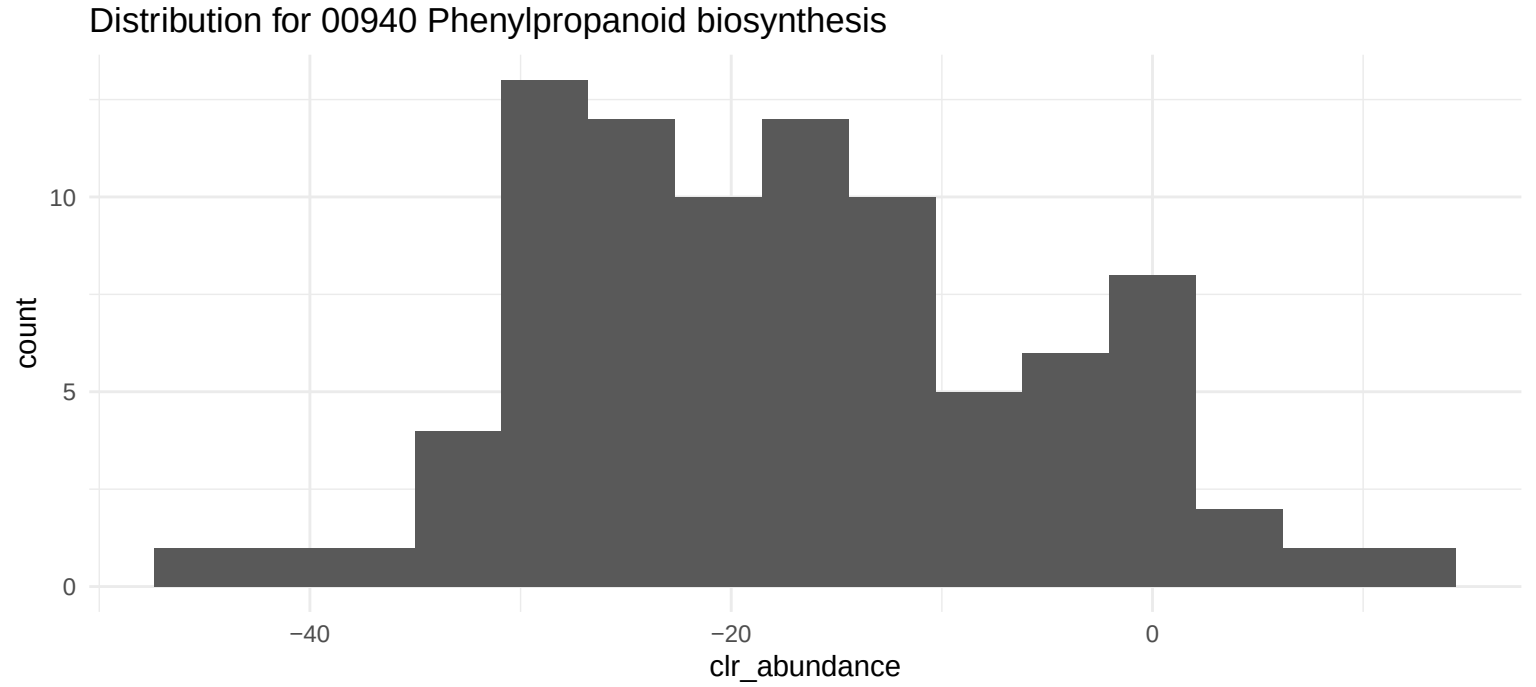


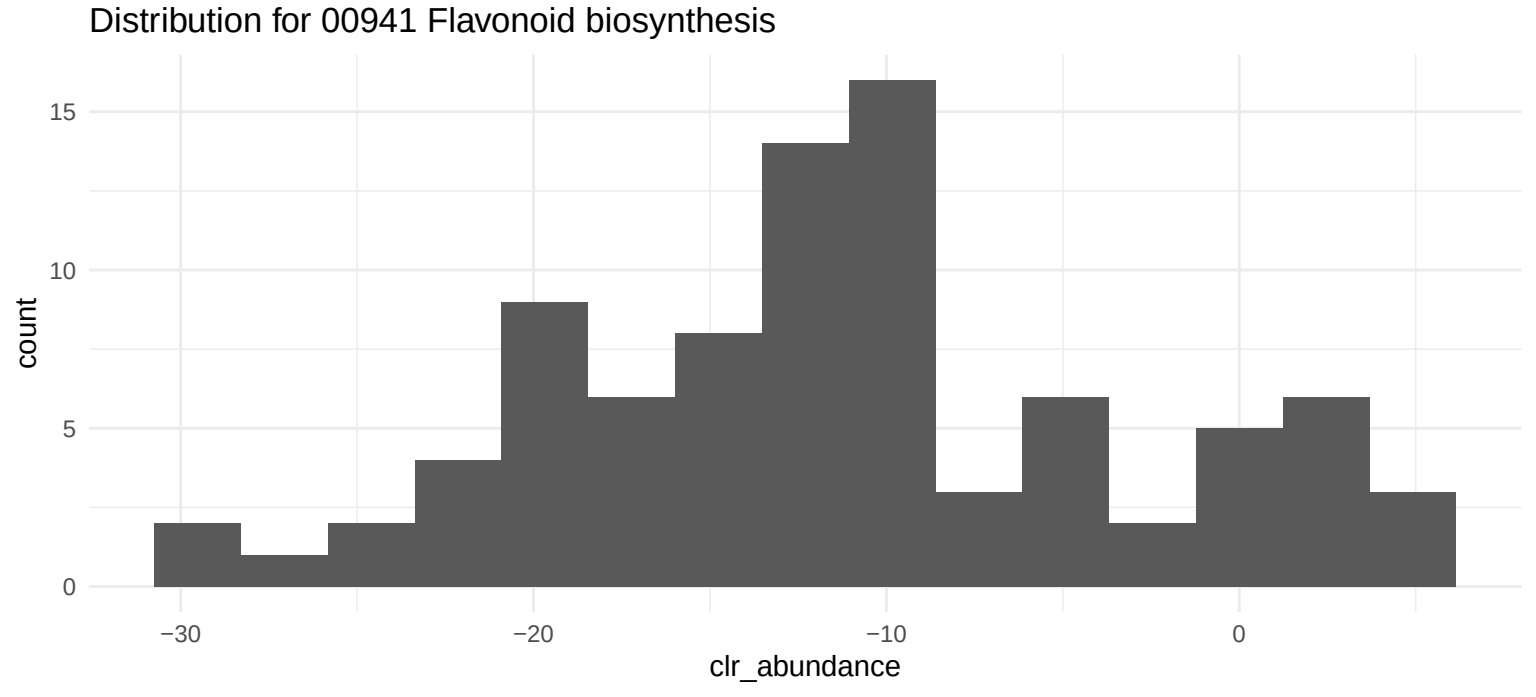

Distribution for 00909 Sesquiterpenoid and triterpenoid biosynthesis
count
15
10
5
0
−4
−2
0
clr_abundance


Distribution for 00910 Nitrogen metabolism
count
10
5
0
−10
0
10
20
30
clr_abundance

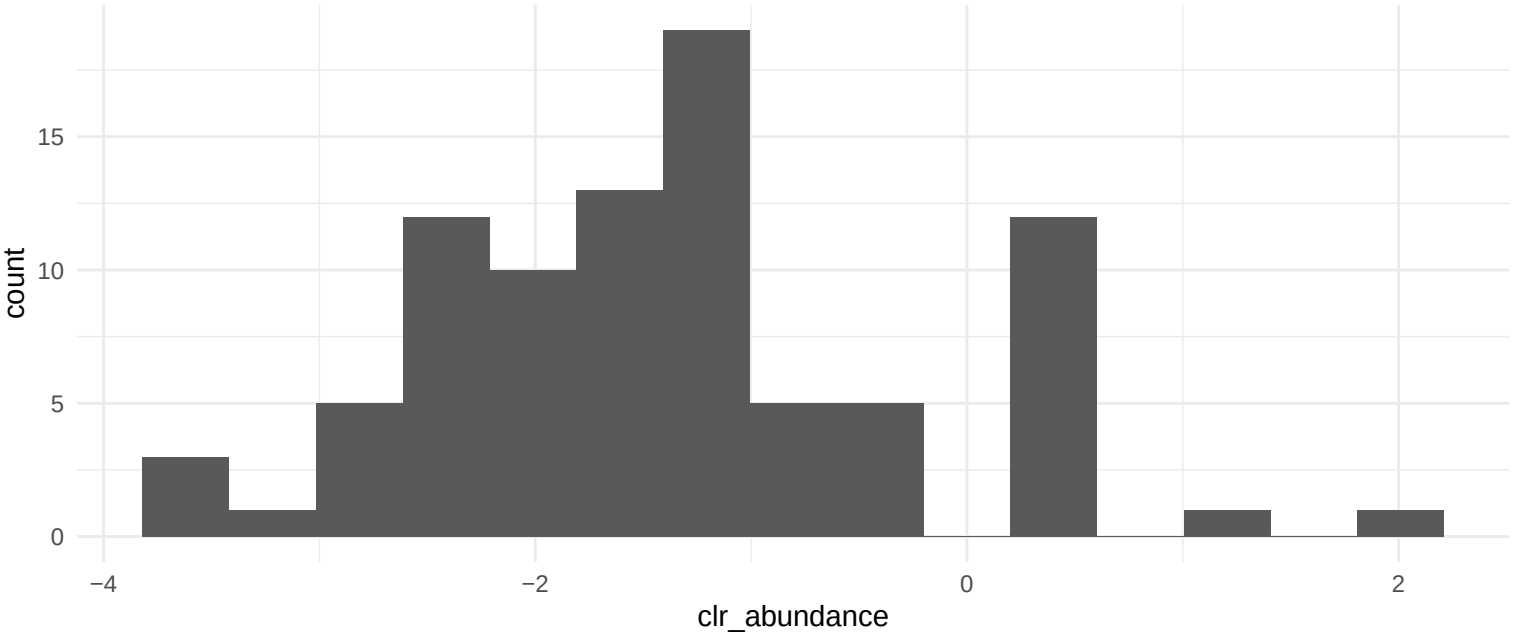

Distribution for 00920 Sulfur metabolism
count
10
5
0
−20
0
20
40
60
clr_abundance


Distribution for 00930 Caprolactam degradation
count
15
10
5
0
−15
−10
−5
0
clr_abundance

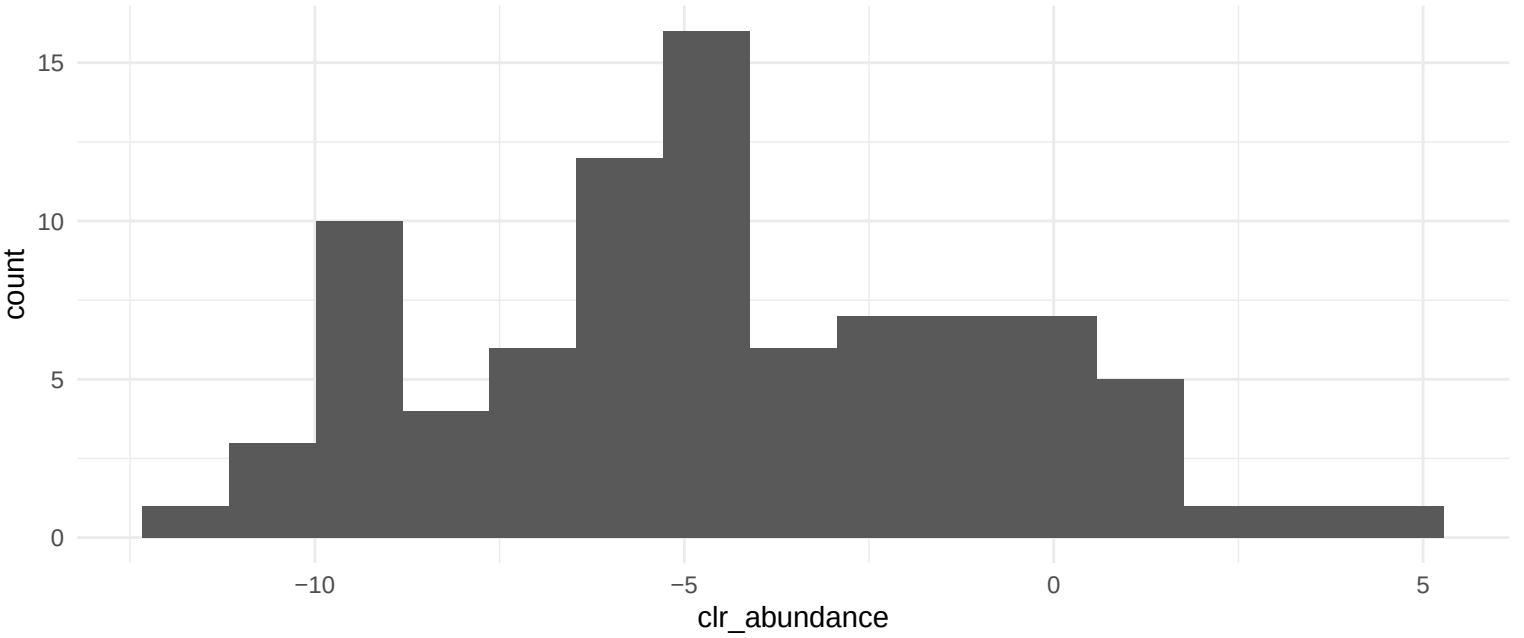

Distribution for 00940 Phenylpropanoid biosynthesis
count
10
5
0
−40
−20
0
clr_abundance


Distribution for 00941 Flavonoid biosynthesis
count
15
10
5
0
−30
−20
−10
0
clr_abundance

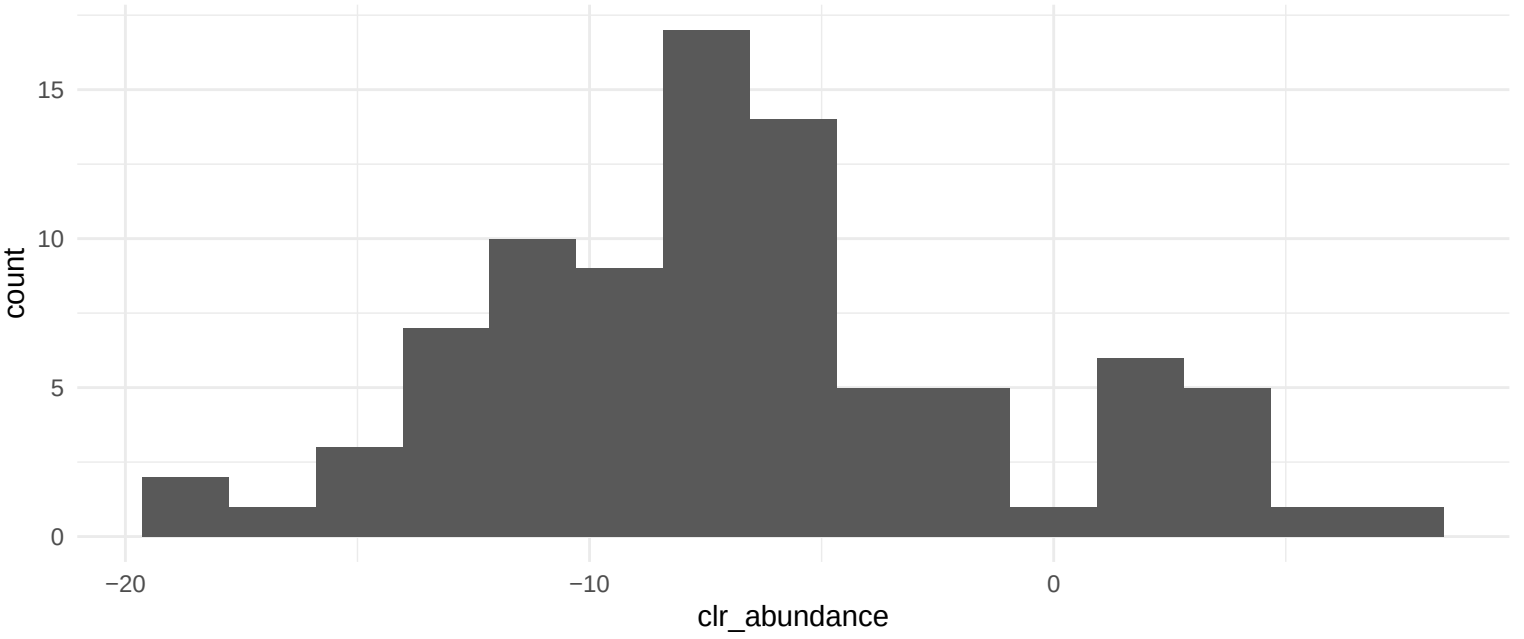
Distribution for 00943 Isoflavonoid biosynthesis



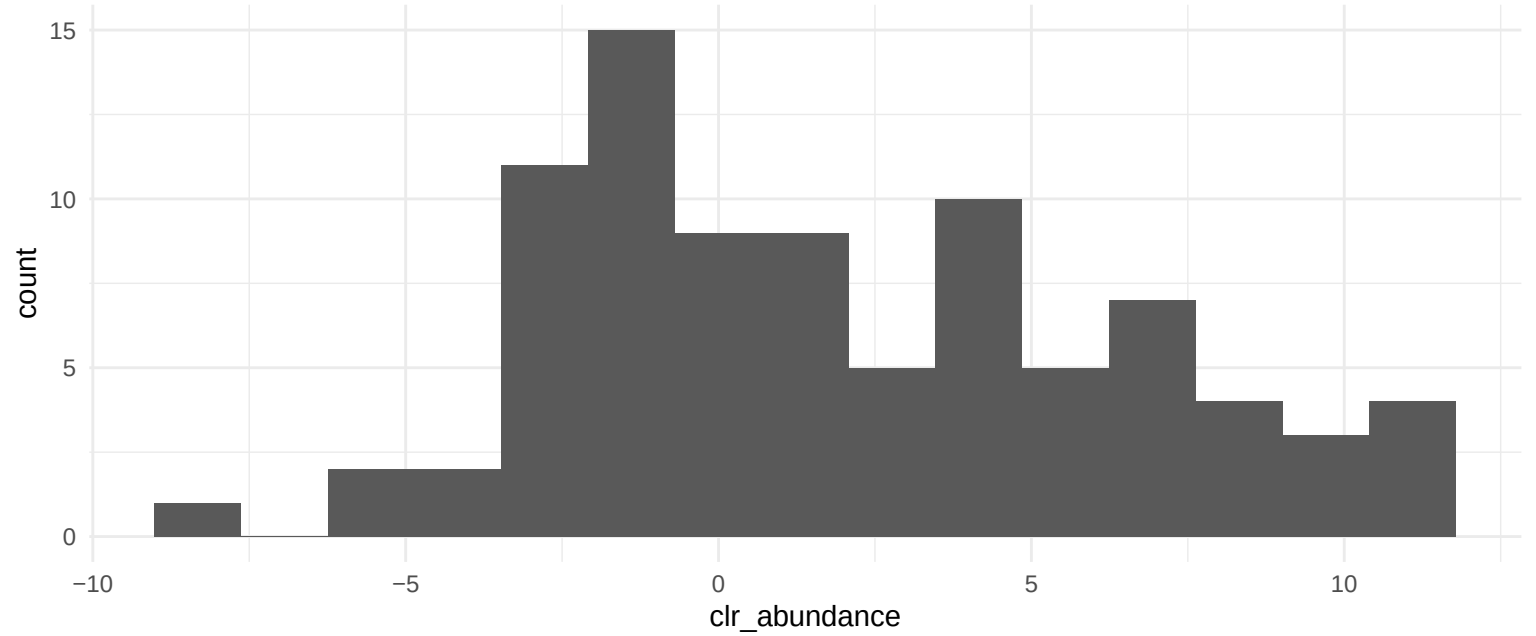
Distribution for 00944 Flavone and flavonol biosynthesis



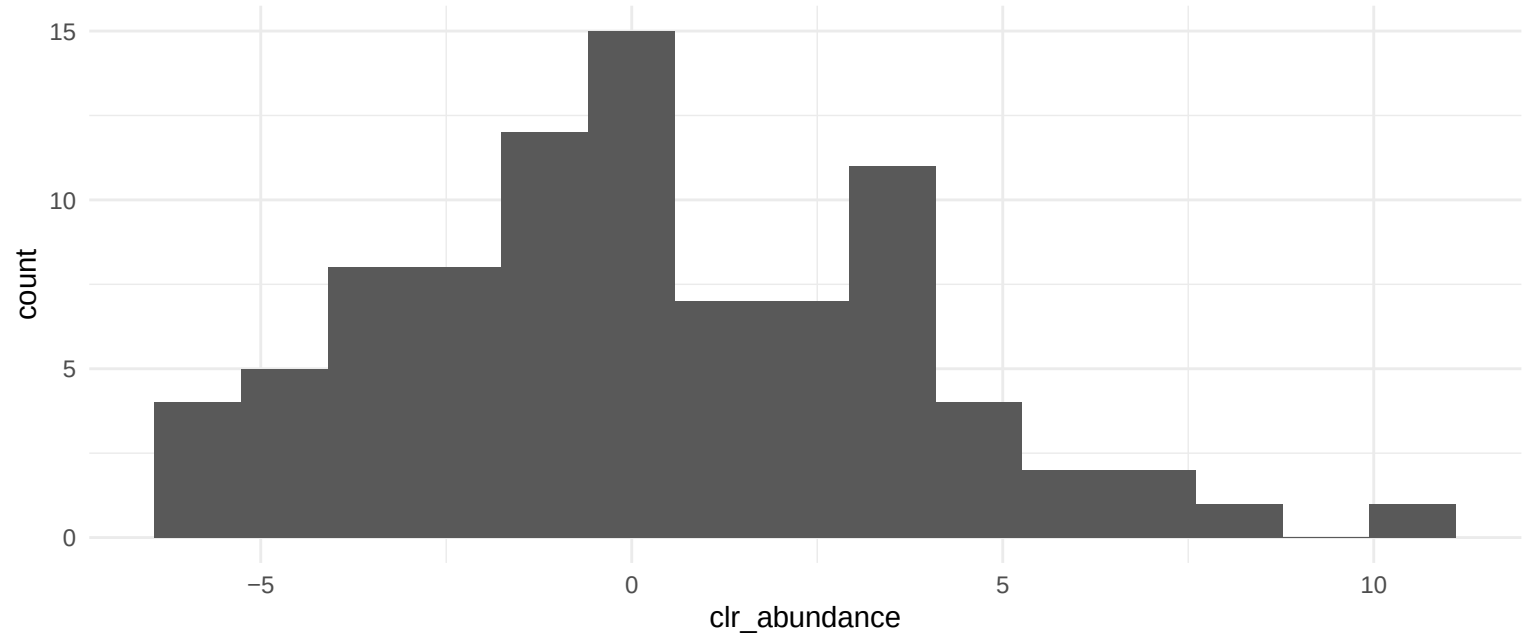
Distribution for 00945 Stilbenoid, diarylheptanoid and gingerol biosynthesis



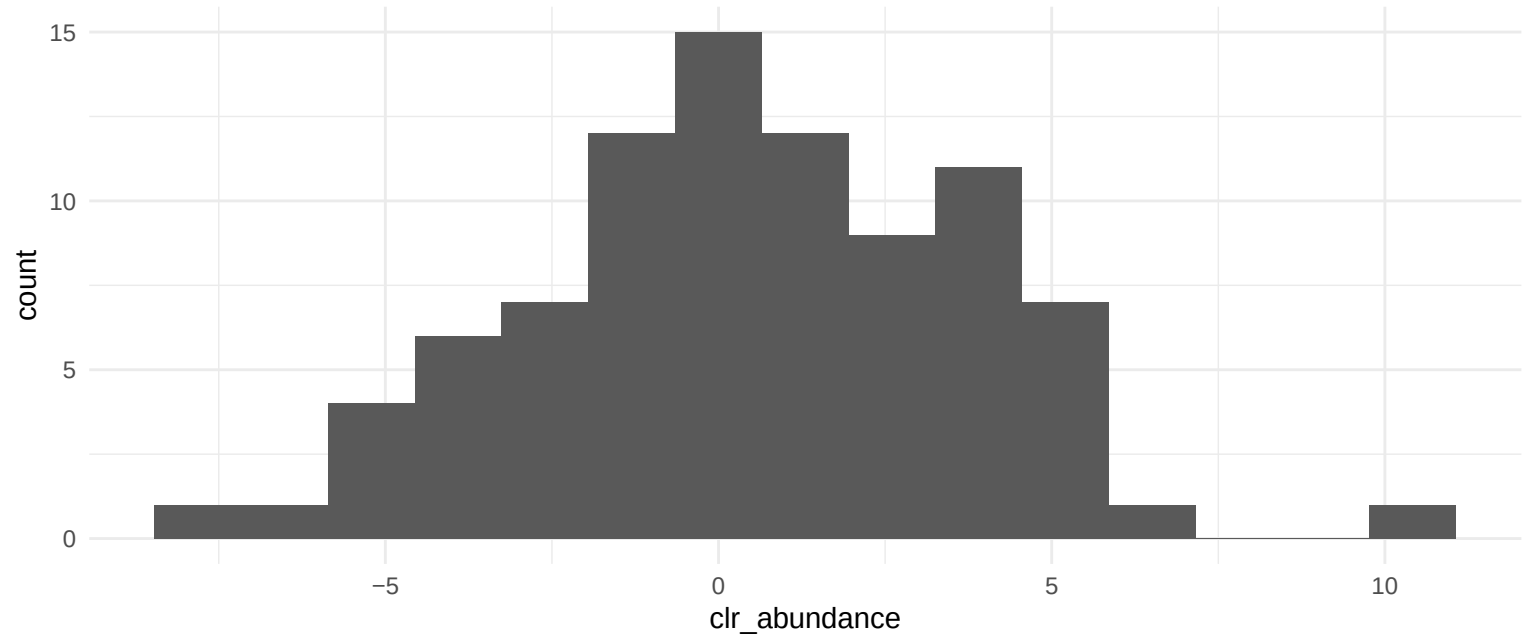
Distribution for 00946 Degradation of flavonoids

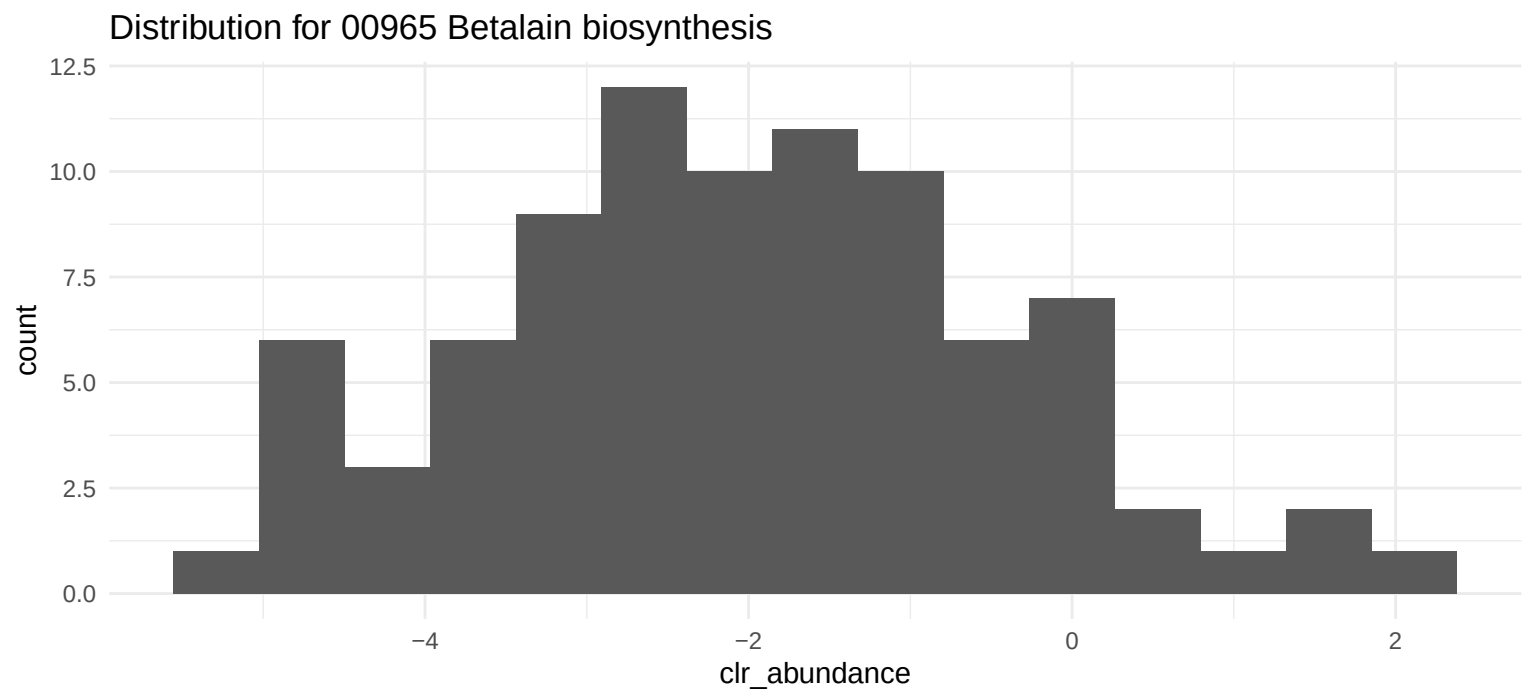


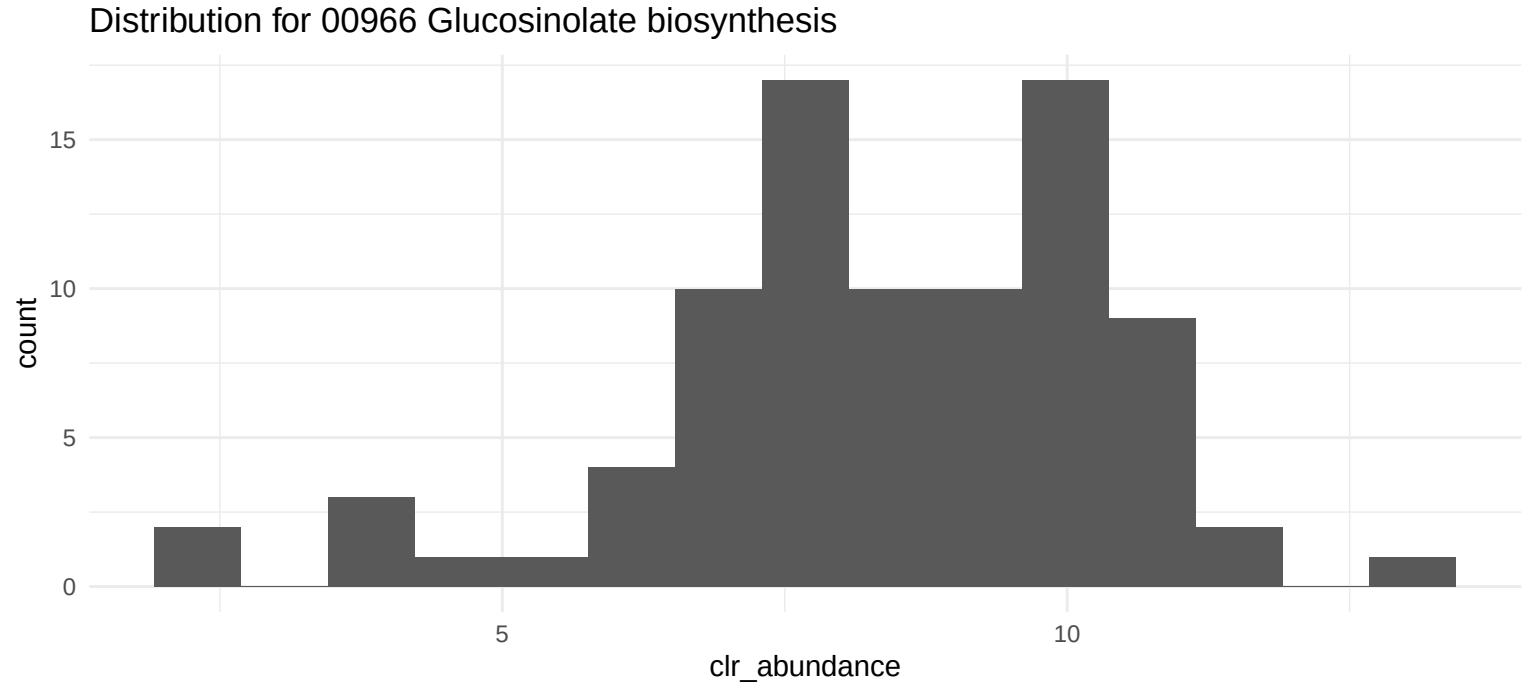
Distribution for 00950 Isoquinoline alkaloid biosynthesis

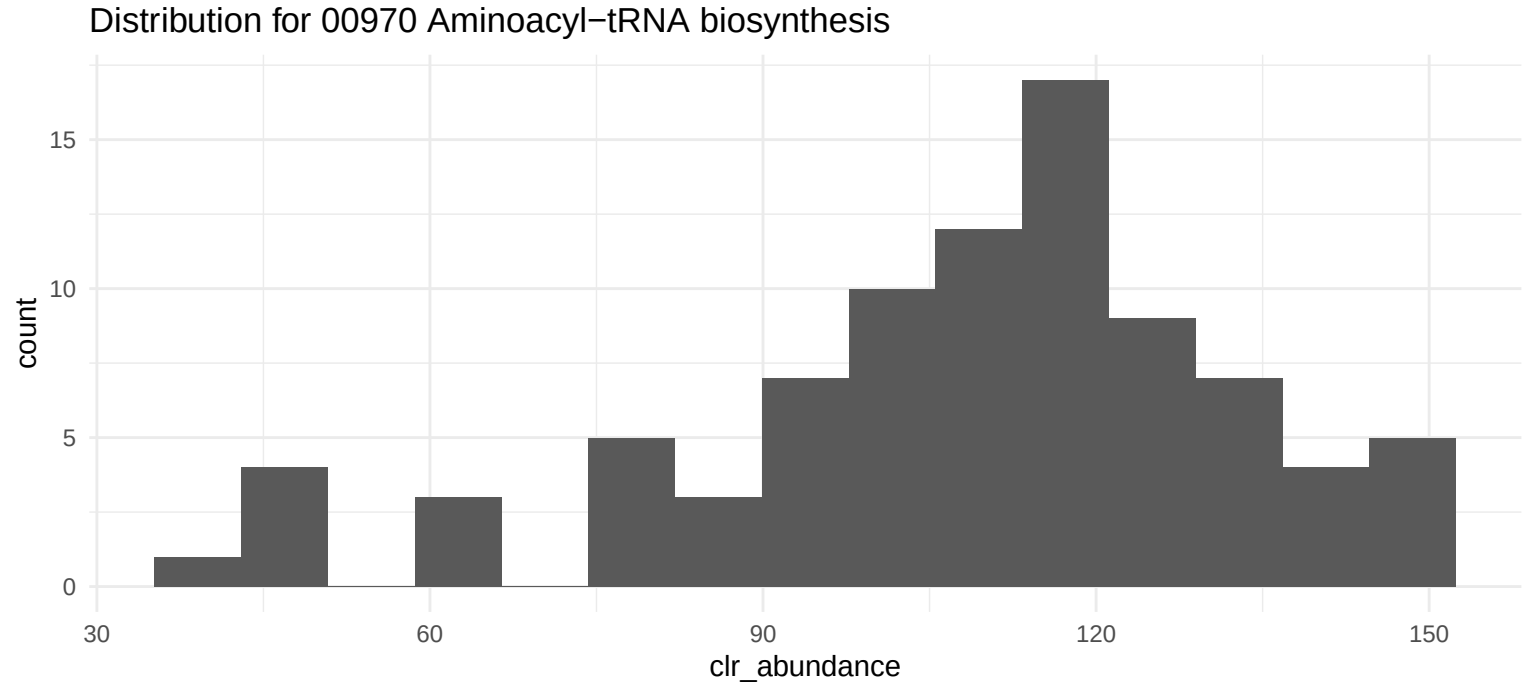


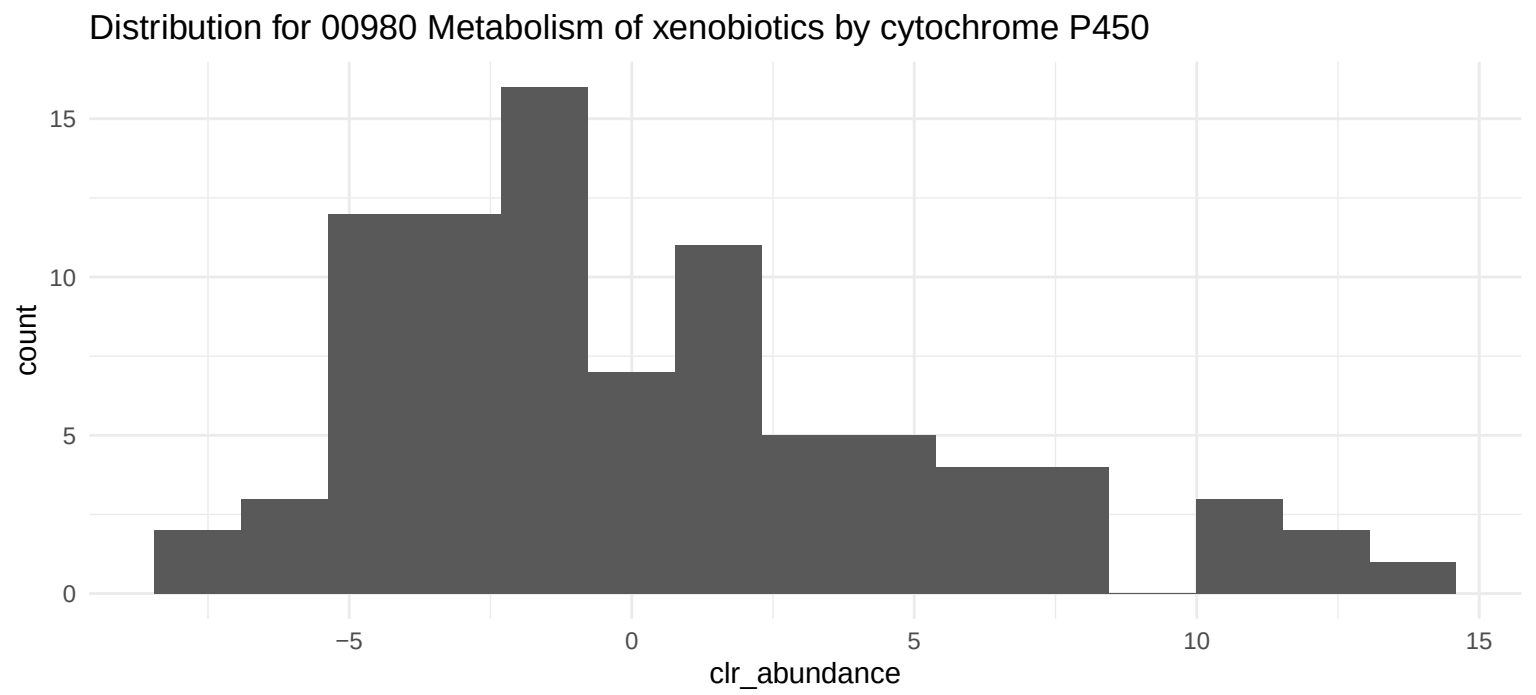
Distribution for 00960 Tropane, piperidine and pyridine alkaloid biosynthesis

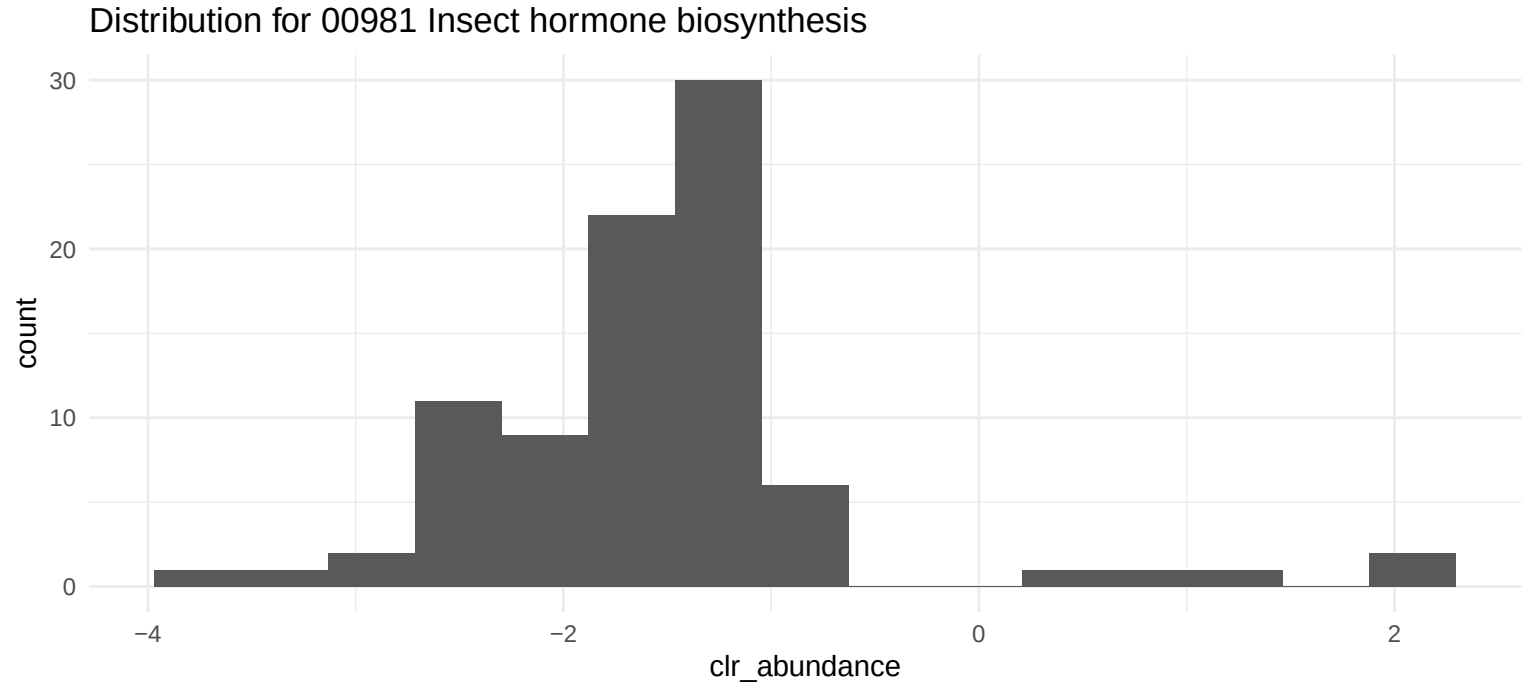


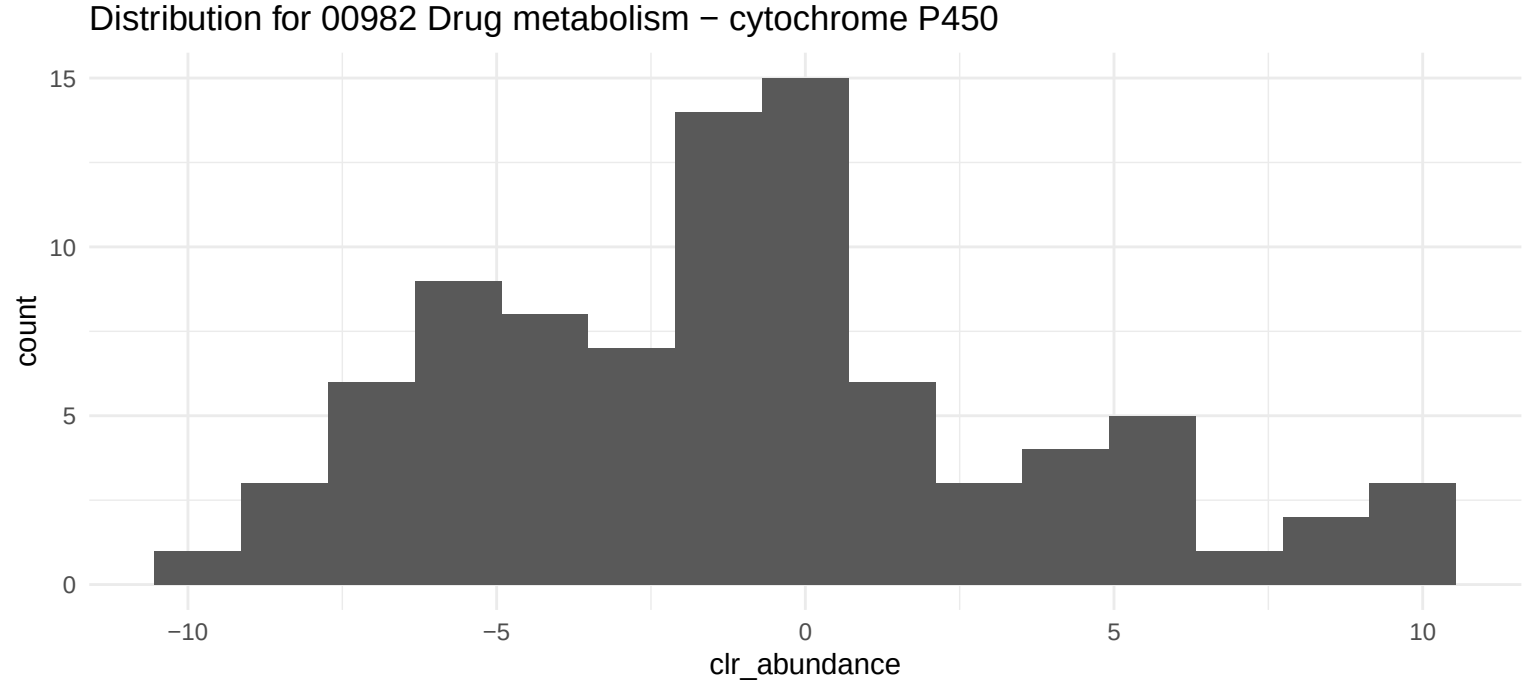

Distribution for 00965 Betalain biosynthesis
12.5
10.0
7.5
5.0
2.5
0.0
count
−4
−2
0
2
clr_abundance


Distribution for 00966 Glucosinolate biosynthesis
15
10
5
0
count
5
10
clr_abundance

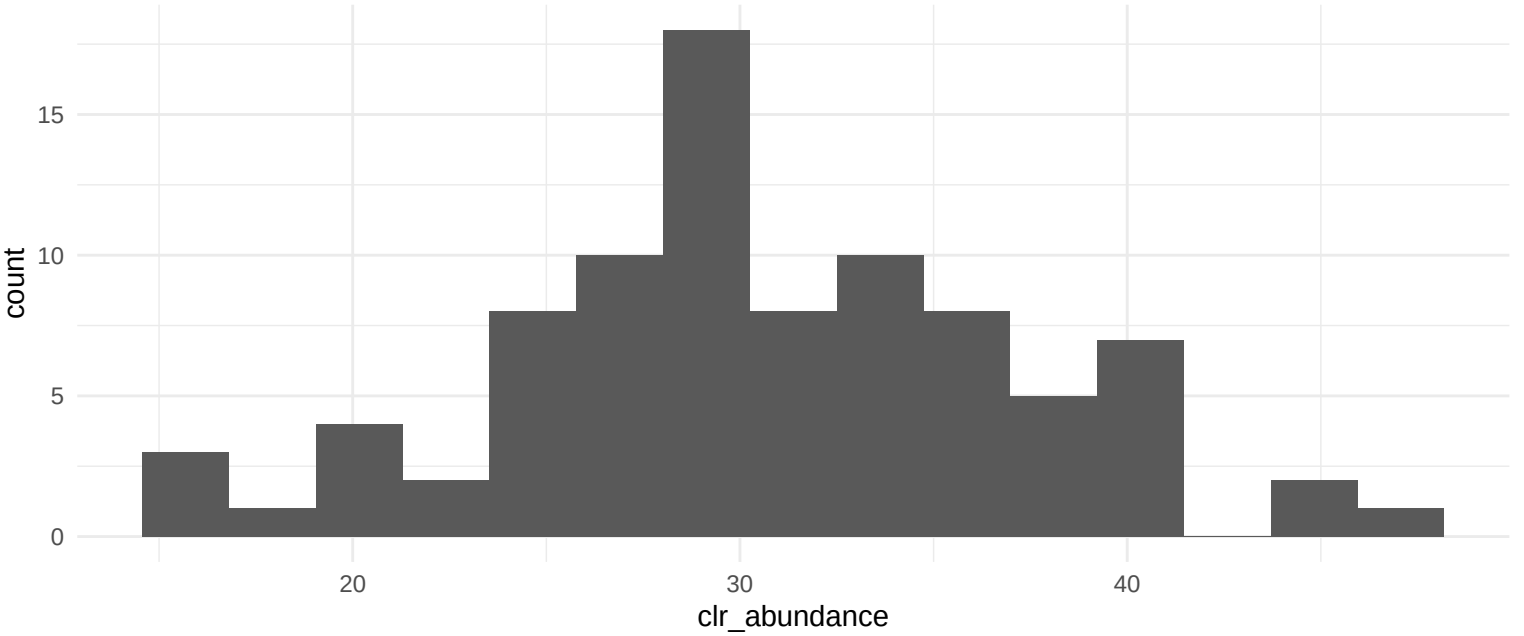

Distribution for 00970 Aminoacyl−tRNA biosynthesis
15
10
5
0
count
30
60
90
120
150
clr_abundance


Distribution for 00980 Metabolism of xenobiotics by cytochrome P450
count
clr_abundance
0
5
10
15
−5
0
5
10
15

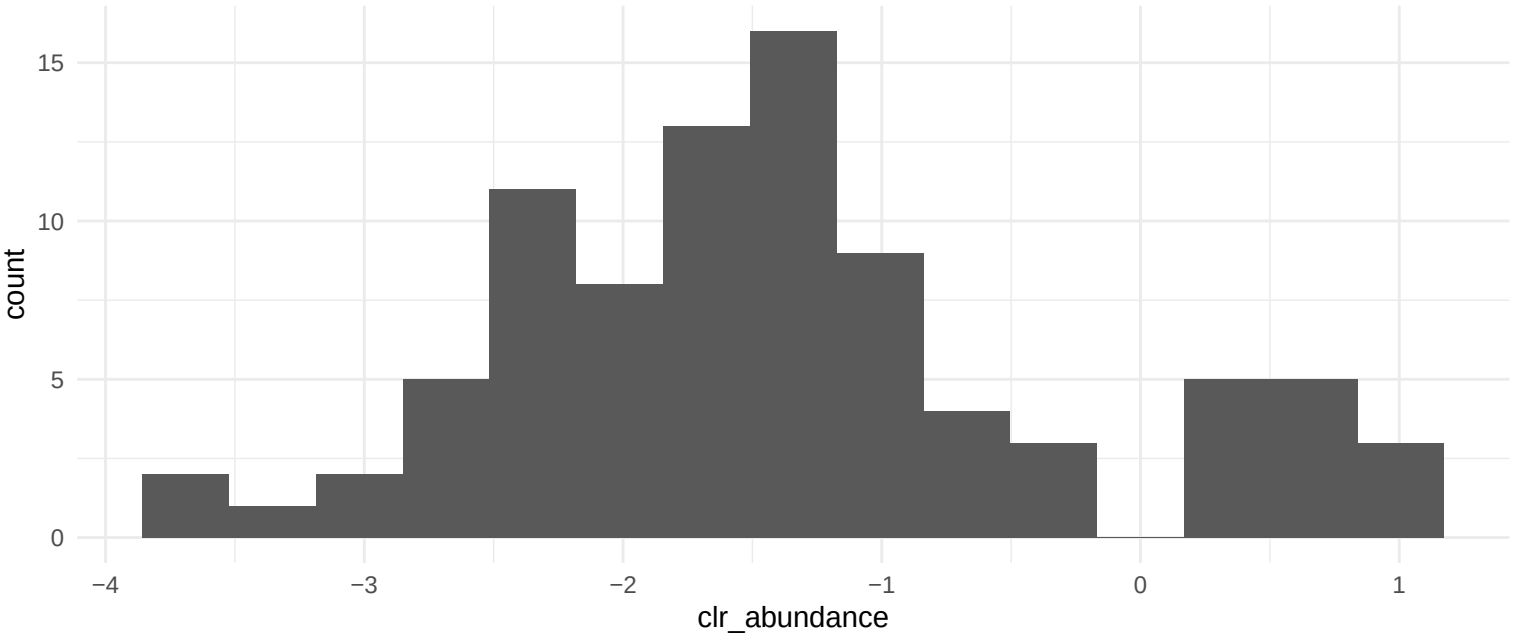

Distribution for 00981 Insect hormone biosynthesis
count
clr_abundance
0
10
20
30
−4
−2
0
2


Distribution for 00982 Drug metabolism − cytochrome P450
count
clr_abundance
0
5
10
15
−10
−5
0
5
10

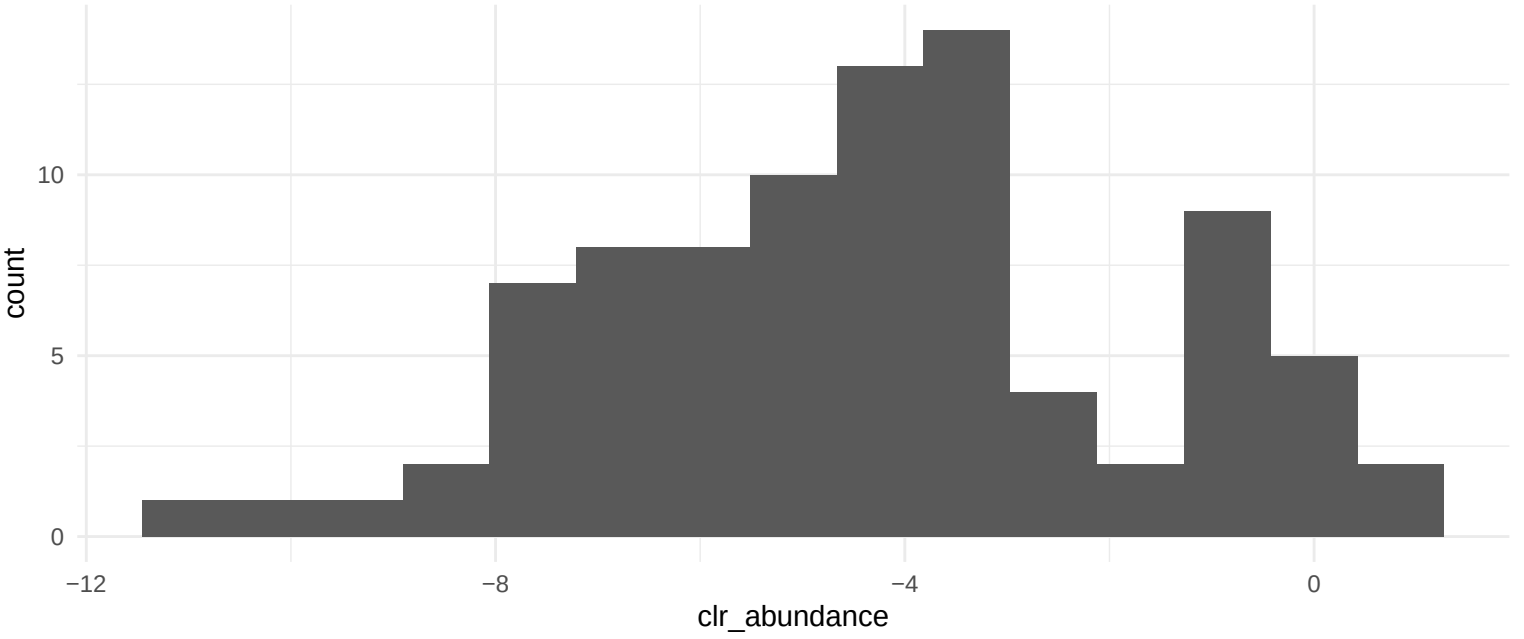
Distribution for 00983 Drug metabolism − other enzymes

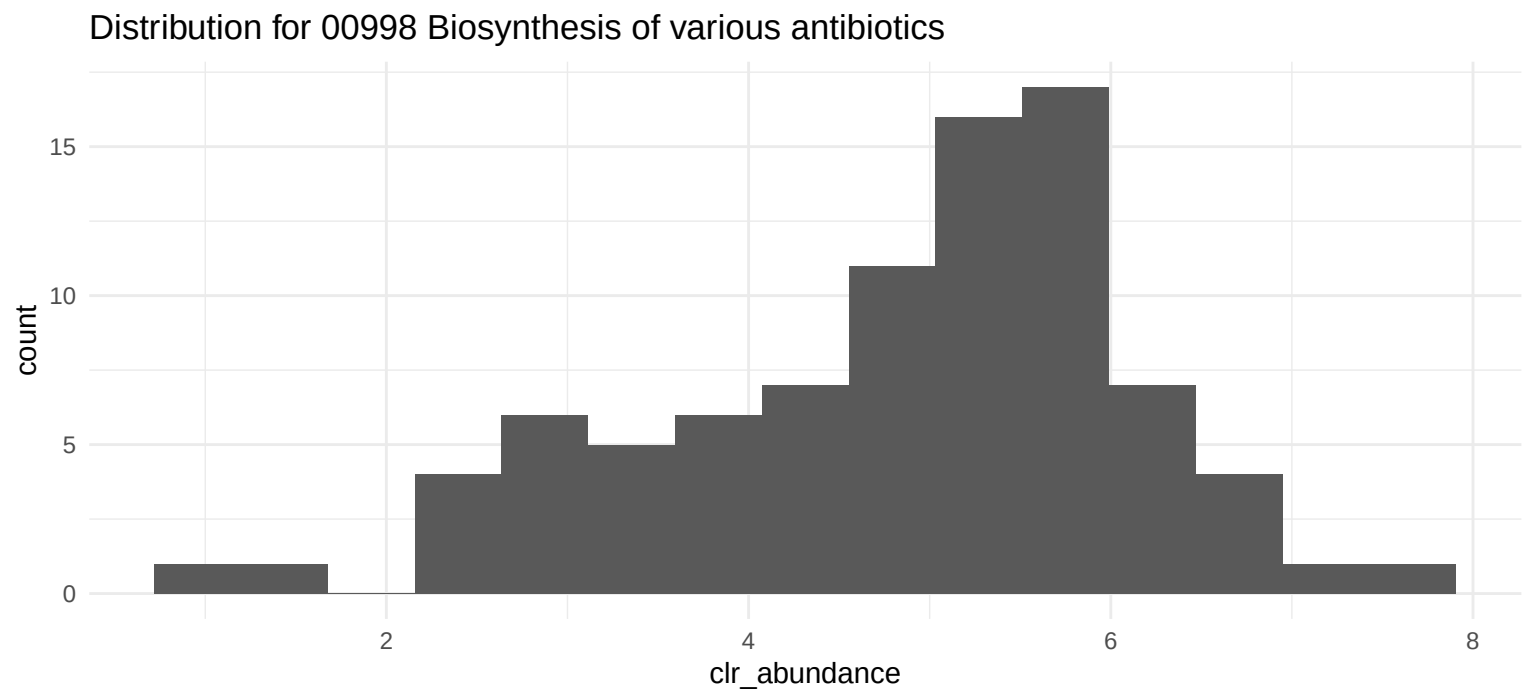


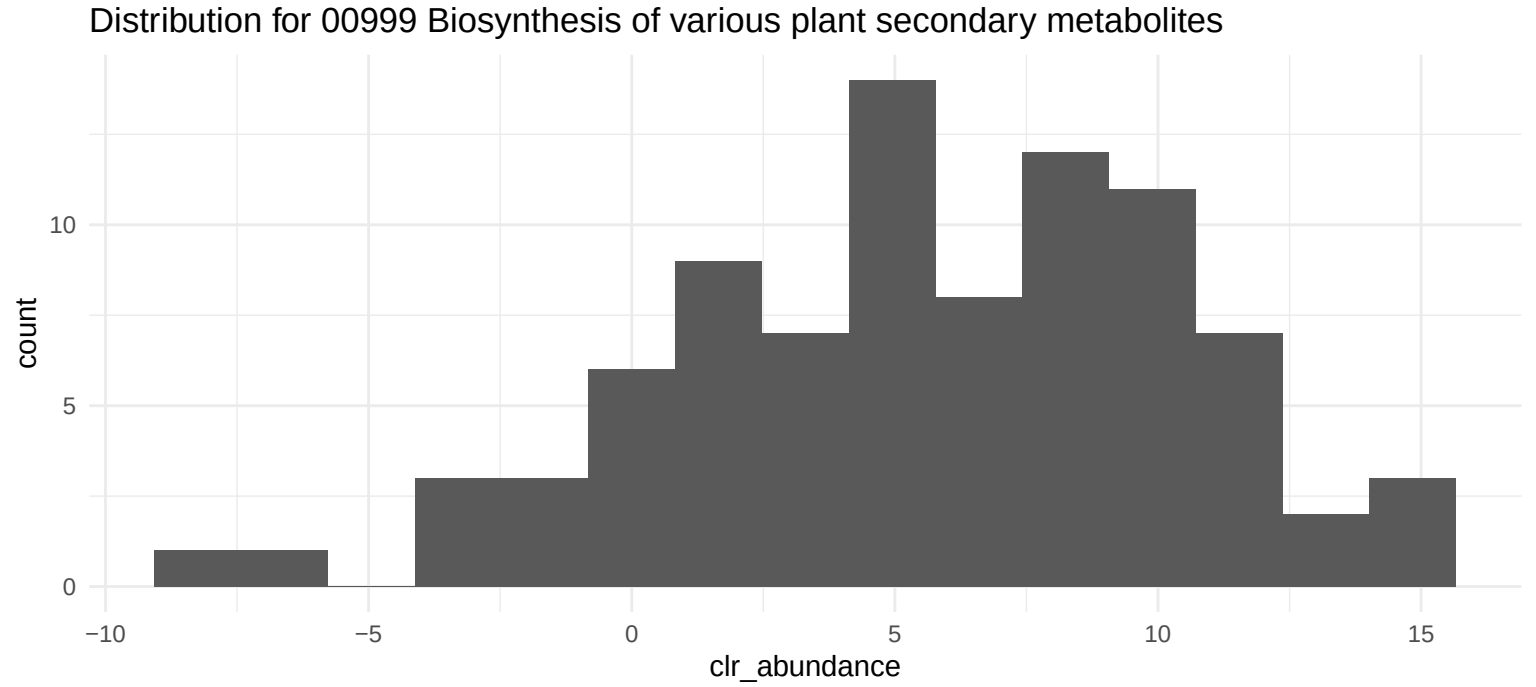
Distribution for 00996 Biosynthesis of various alkaloids

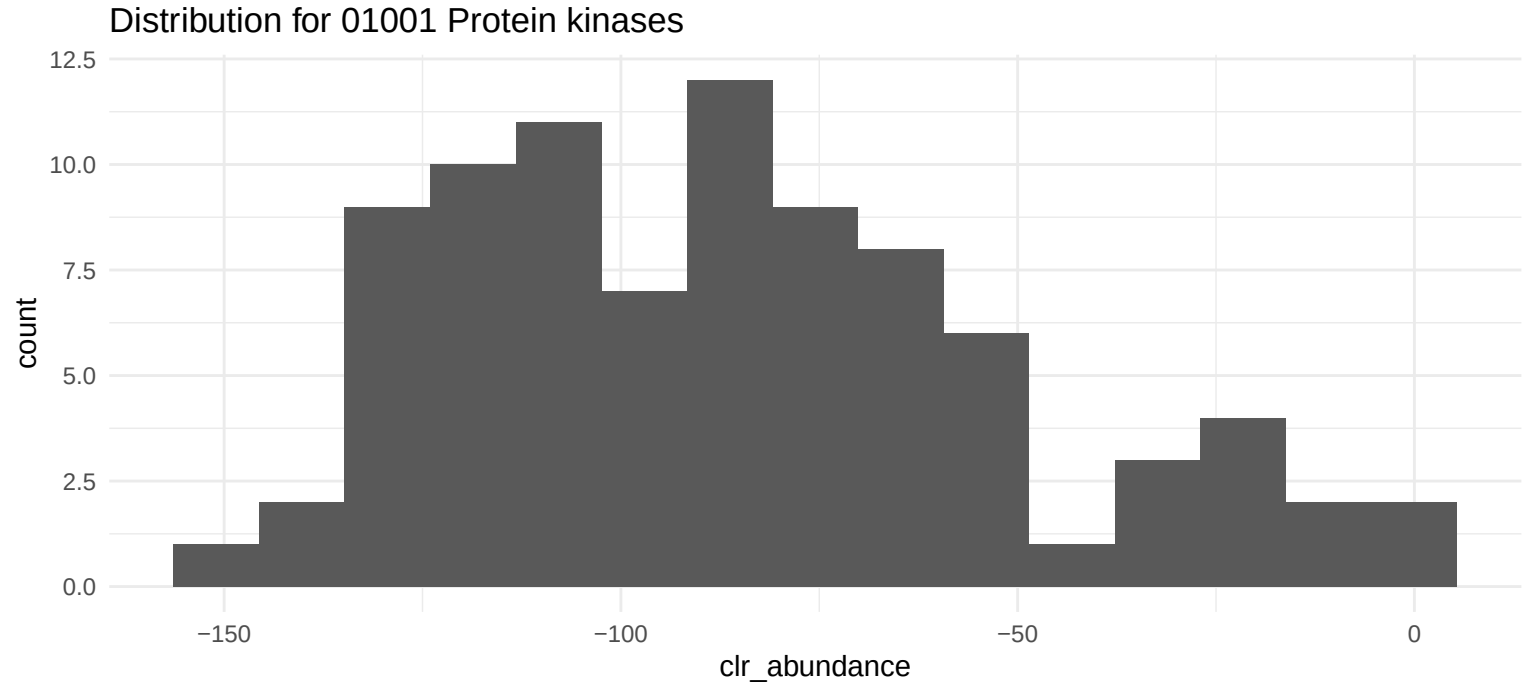


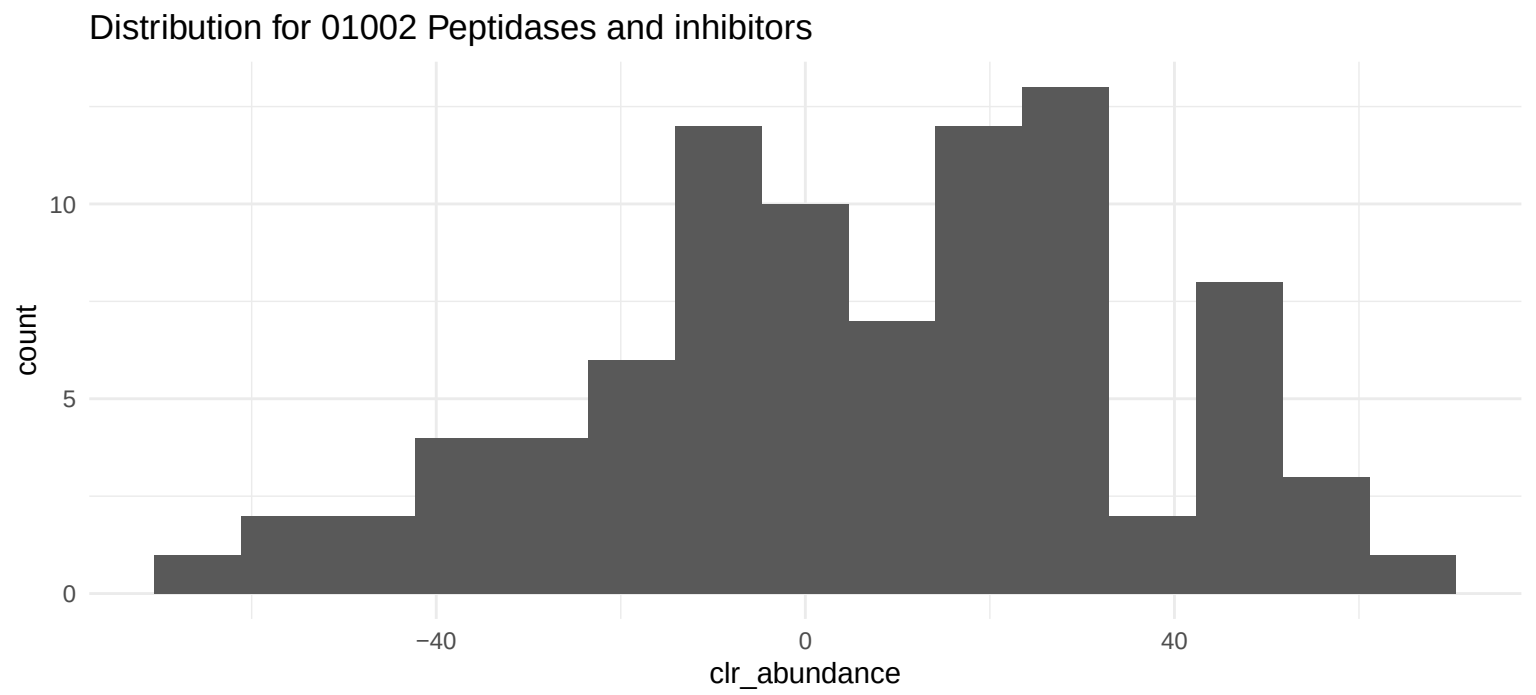
Distribution for 00997 Biosynthesis of various other secondary metabolites

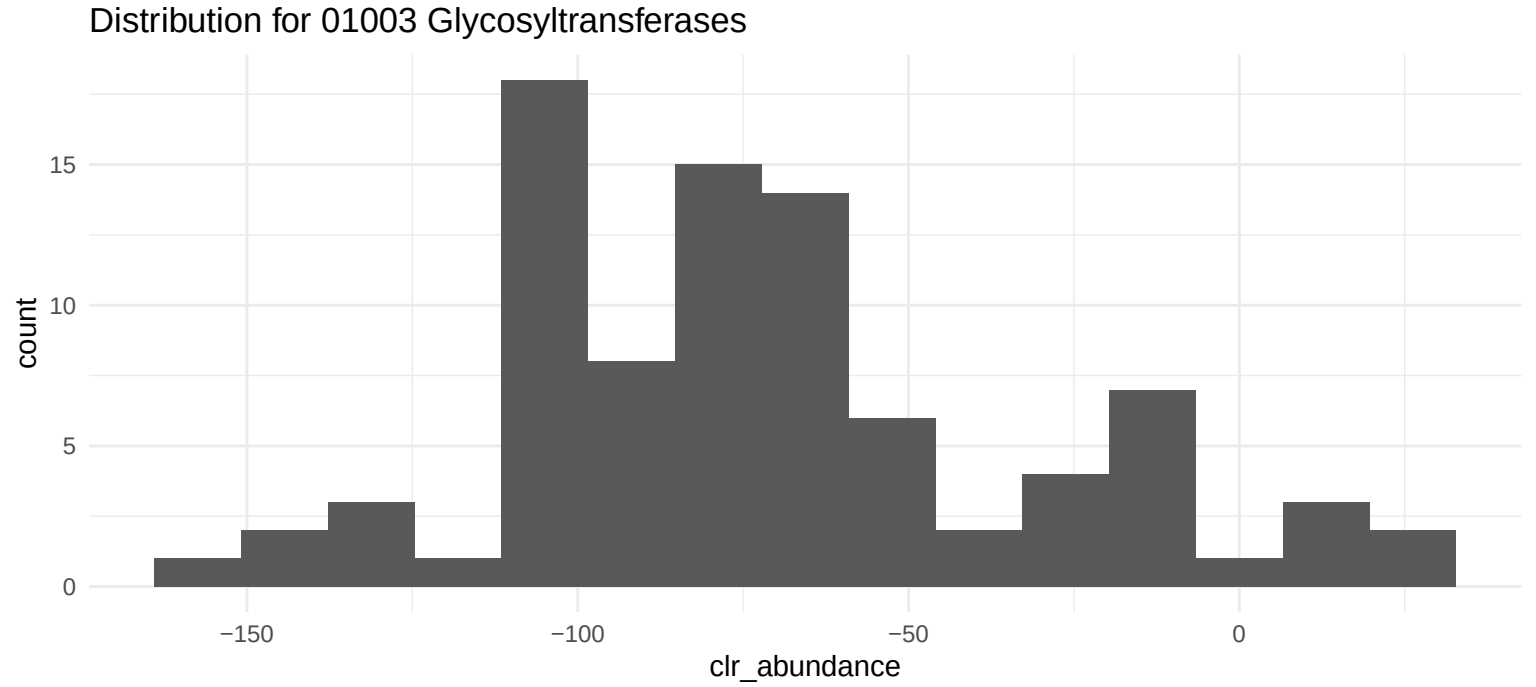


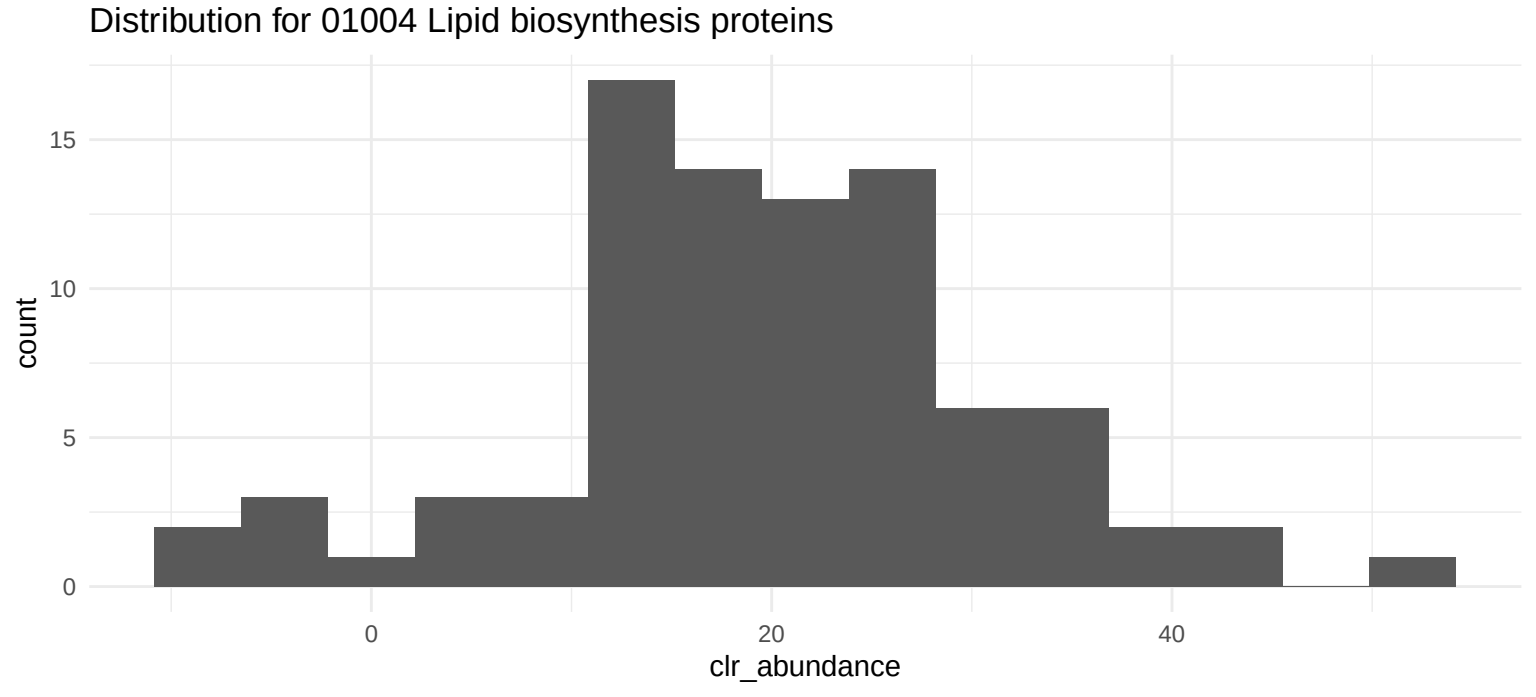

Distribution for 00998 Biosynthesis of various antibiotics
count
15
10
5
0
2
4
6
8
clr_abundance

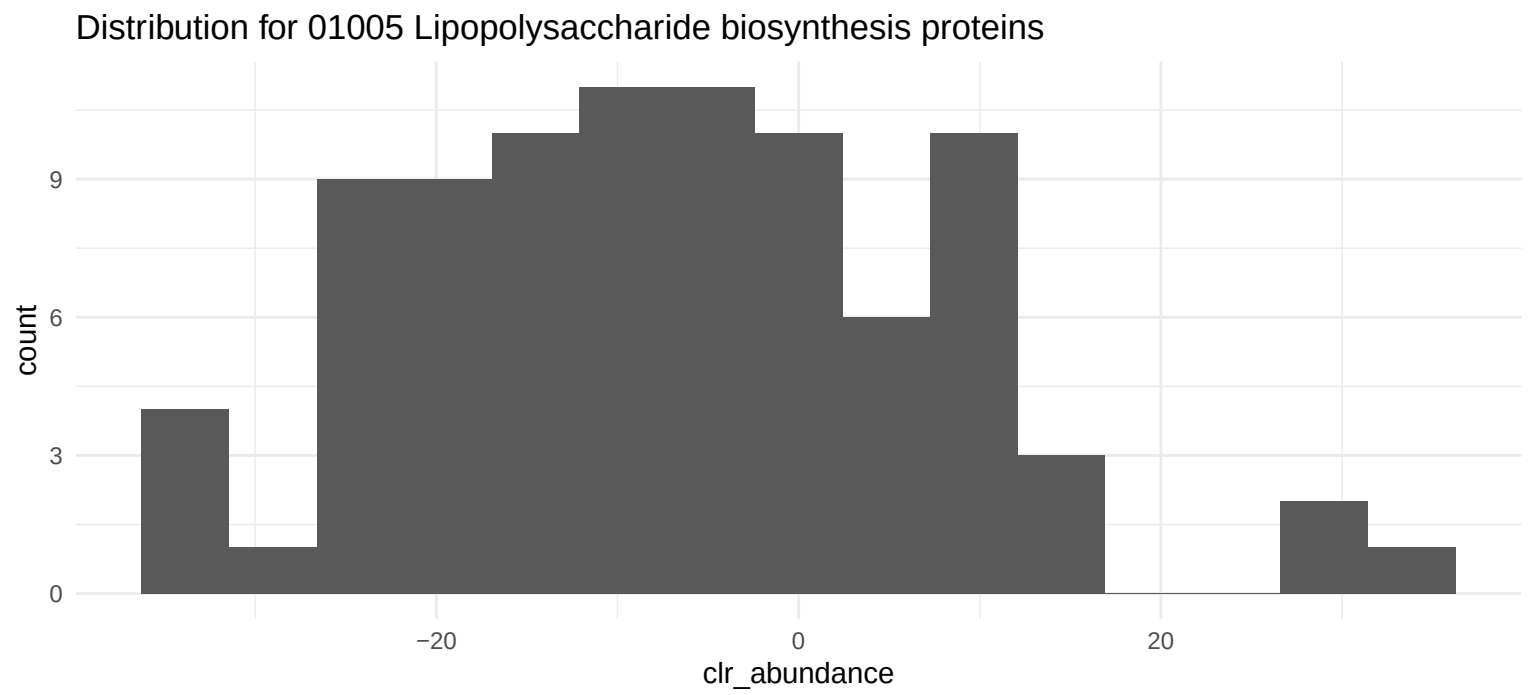

Distribution for 00999 Biosynthesis of various plant secondary metabolites
count
10
5
0
−10
−5
0
5
10
15
clr_abundance

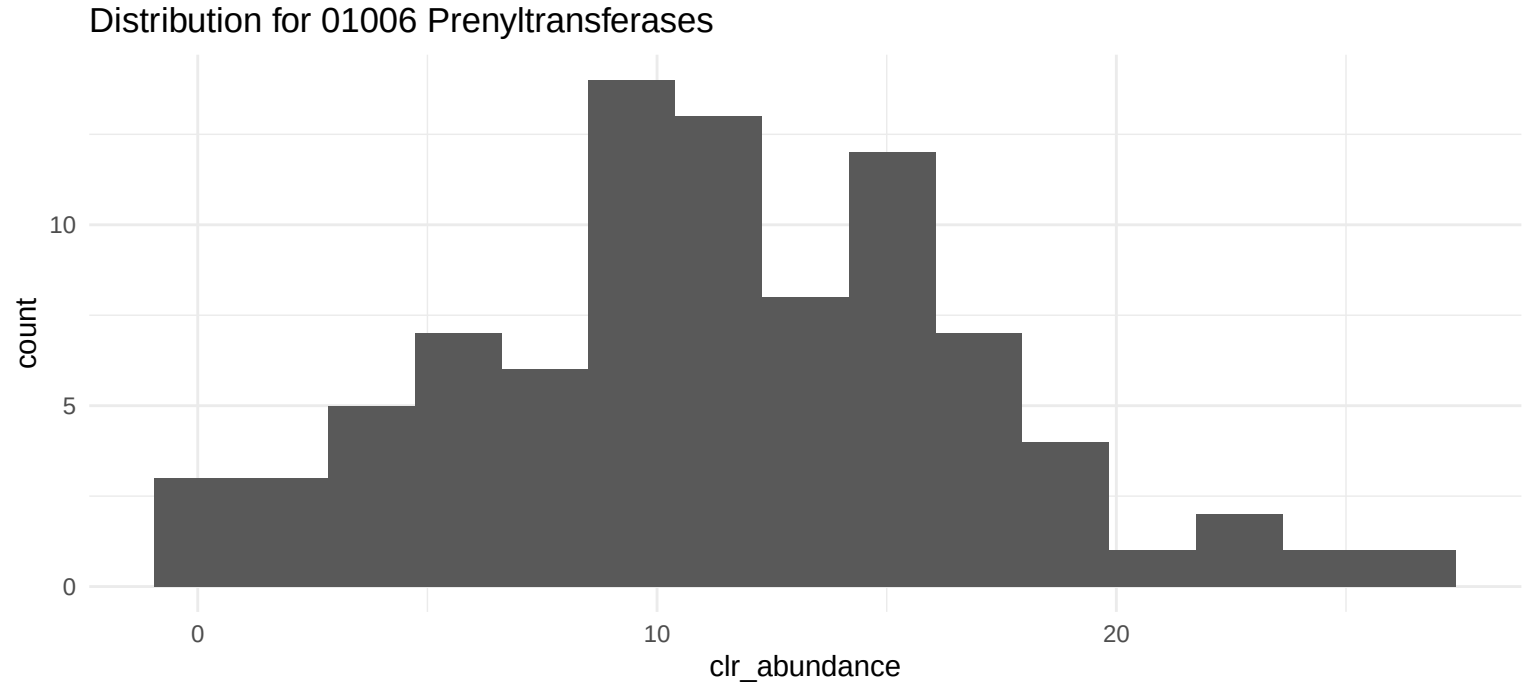

Distribution for 01001 Protein kinases
count
12.5
10.0
7.5
5.0
2.5
0.0
−150
−100
−50
0
clr_abundance

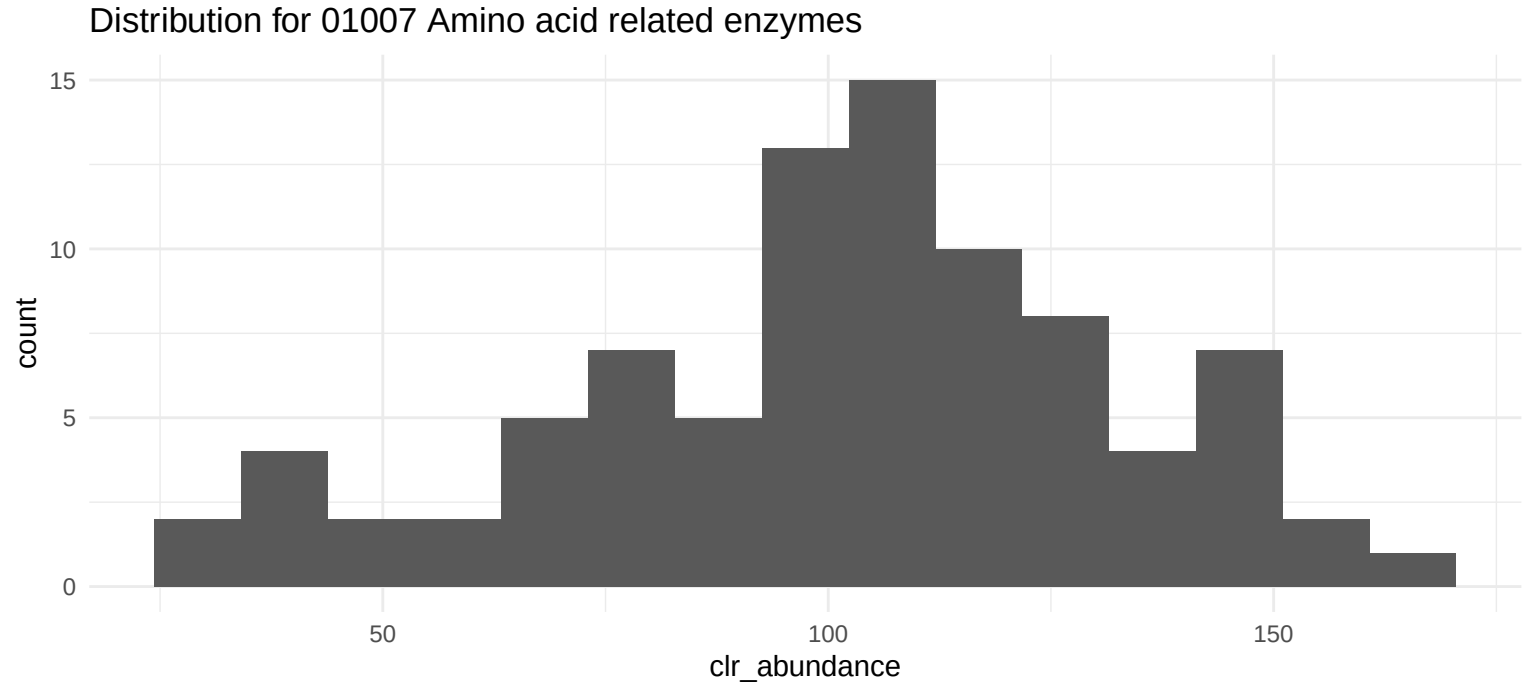

Distribution for 01002 Peptidases and inhibitors
count
clr_abundance
0
5
10
−40
0
40


Distribution for 01003 Glycosyltransferases
count
clr_abundance
0
5
10
15
−150
−100
−50
0

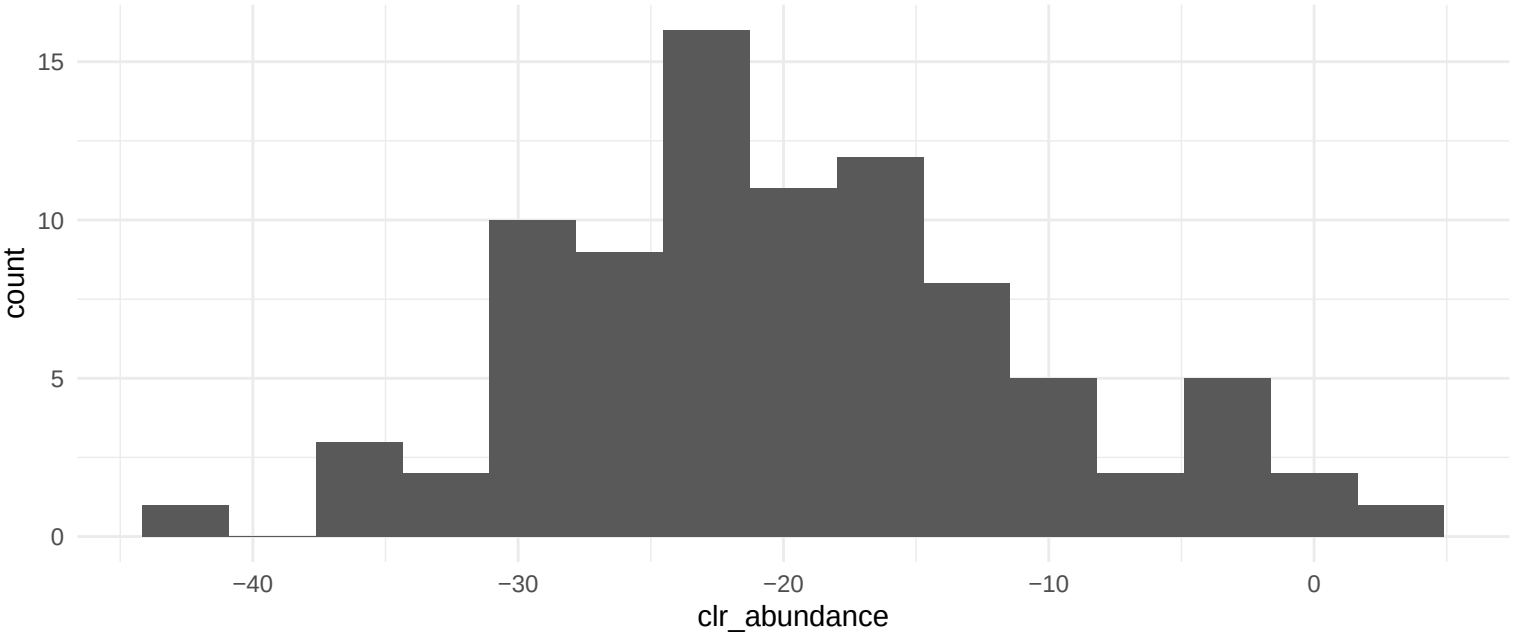

Distribution for 01004 Lipid biosynthesis proteins
count
clr_abundance
0
5
10
15
0
20
40


Distribution for 01005 Lipopolysaccharide biosynthesis proteins
count
clr_abundance
0
3
6
9
−20
0
20

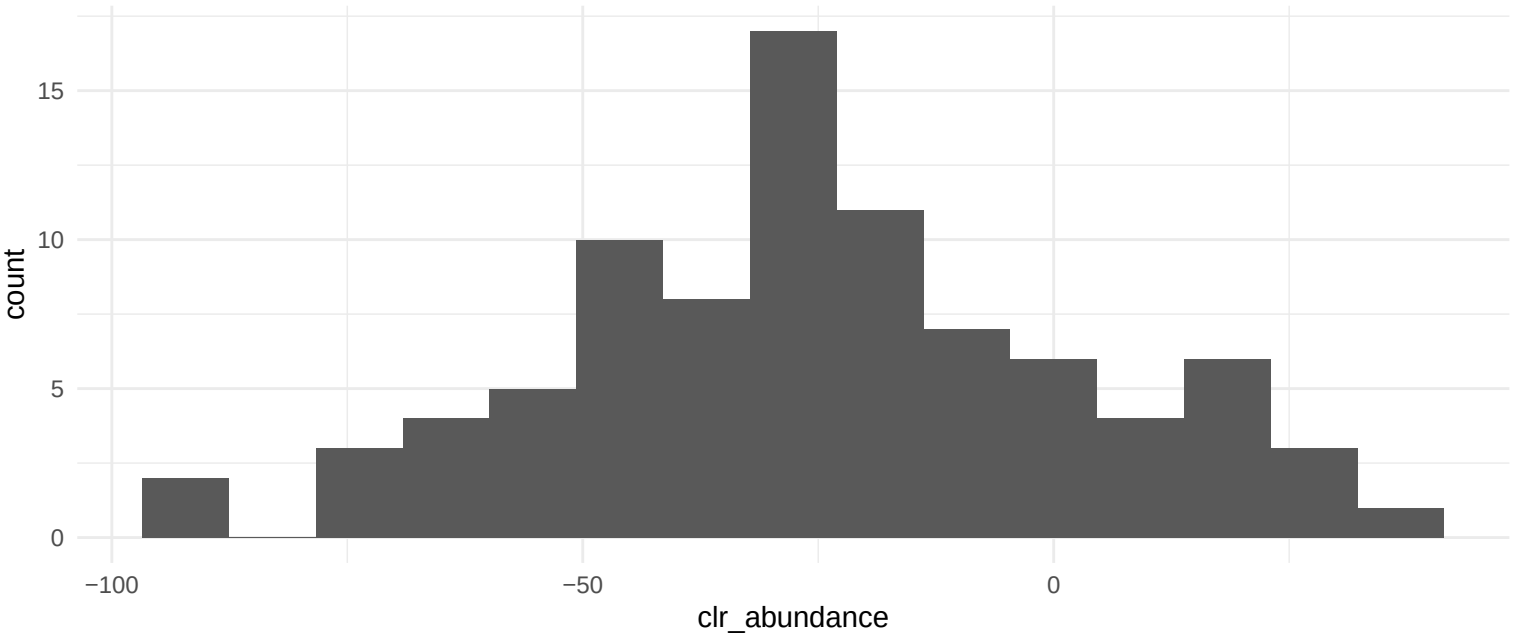

Distribution for 01006 Prenyltransferases
count
clr_abundance
0
5
10
0
10
20


Distribution for 01007 Amino acid related enzymes
count
clr_abundance
0
5
10
15
50
100
150

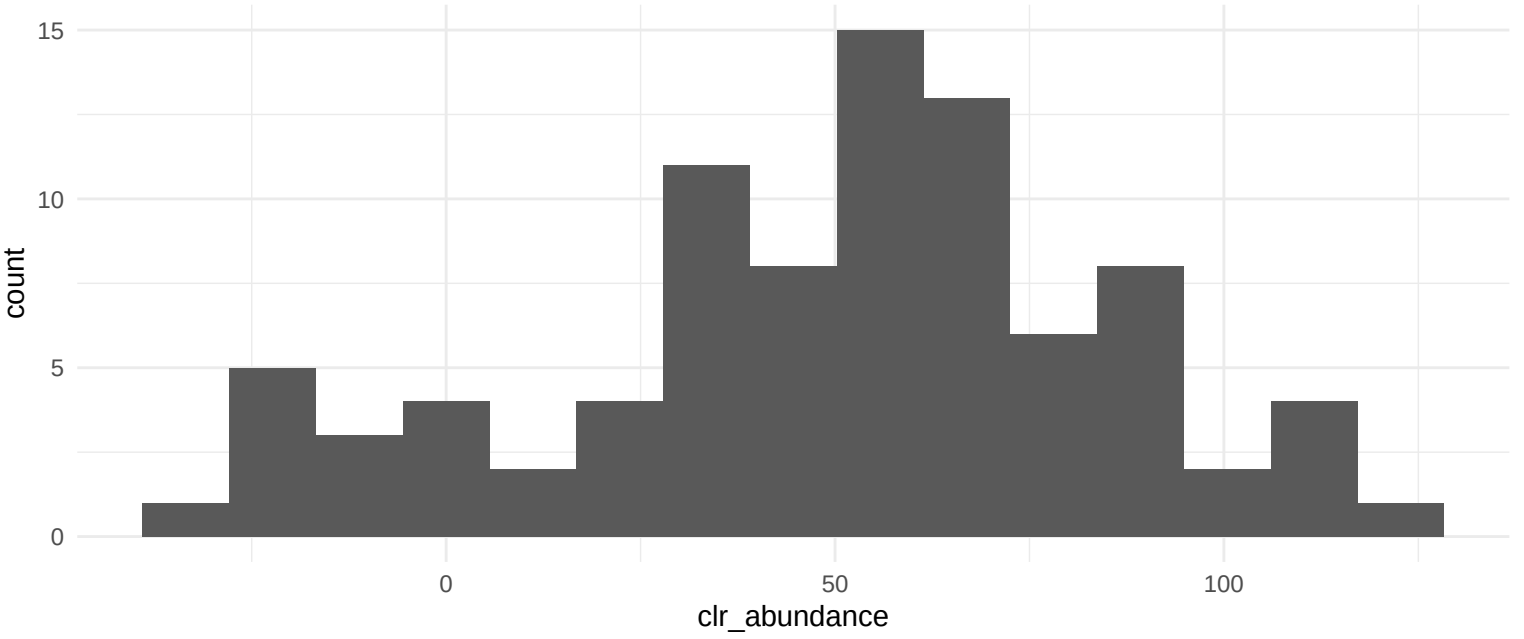
Distribution for 01008 Polyketide biosynthesis proteins



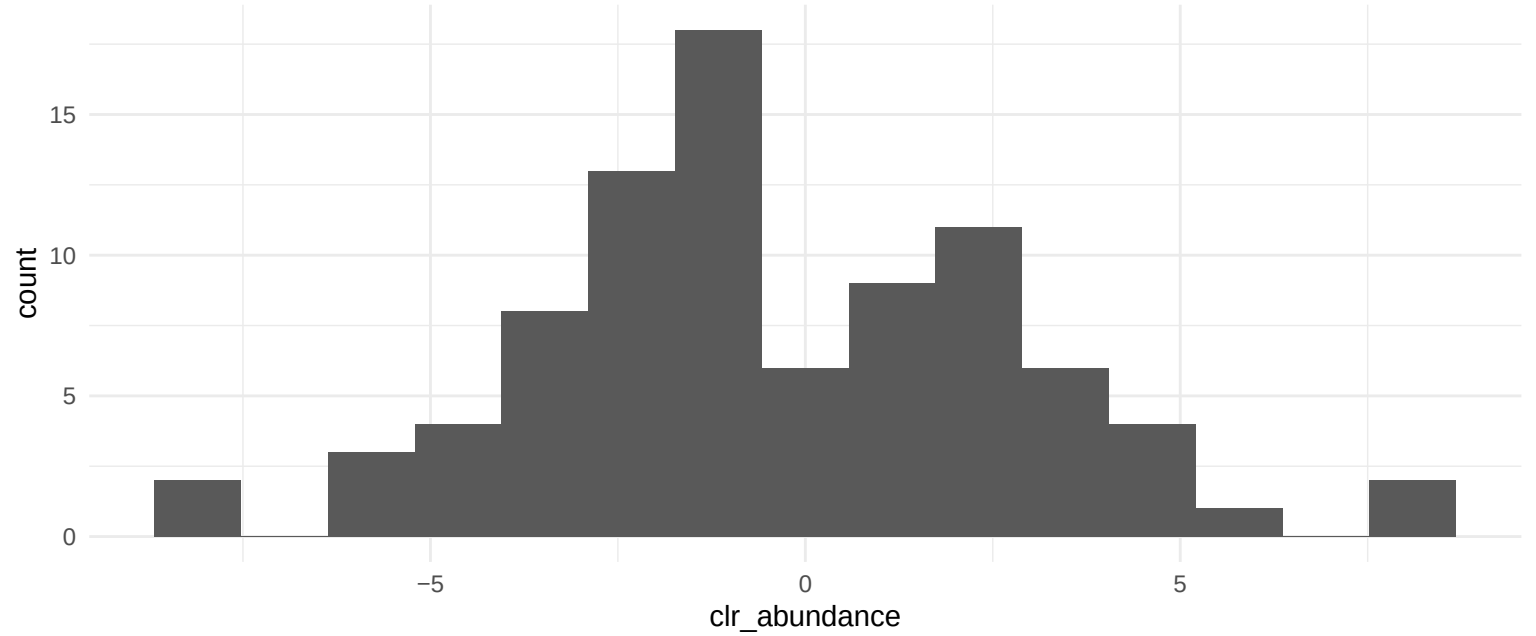
Distribution for 01009 Protein phosphatases and associated proteins



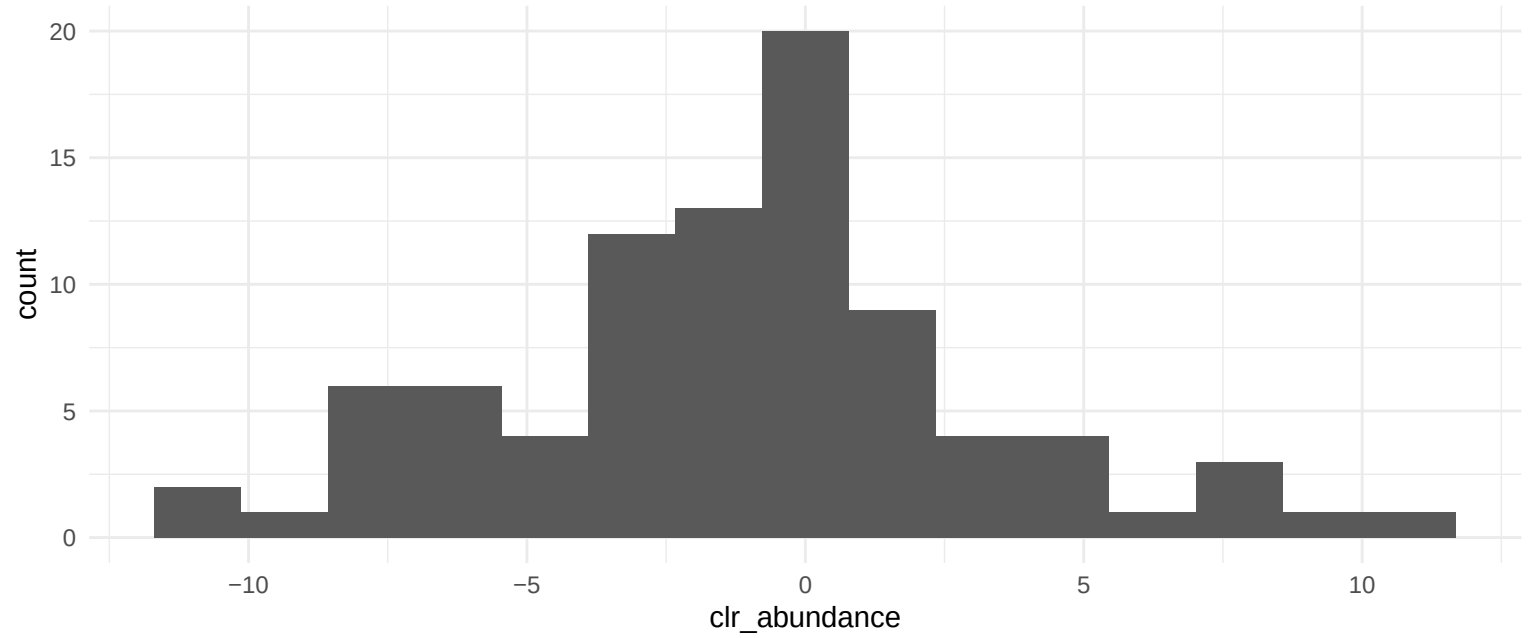
Distribution for 01011 Peptidoglycan biosynthesis and degradation proteins



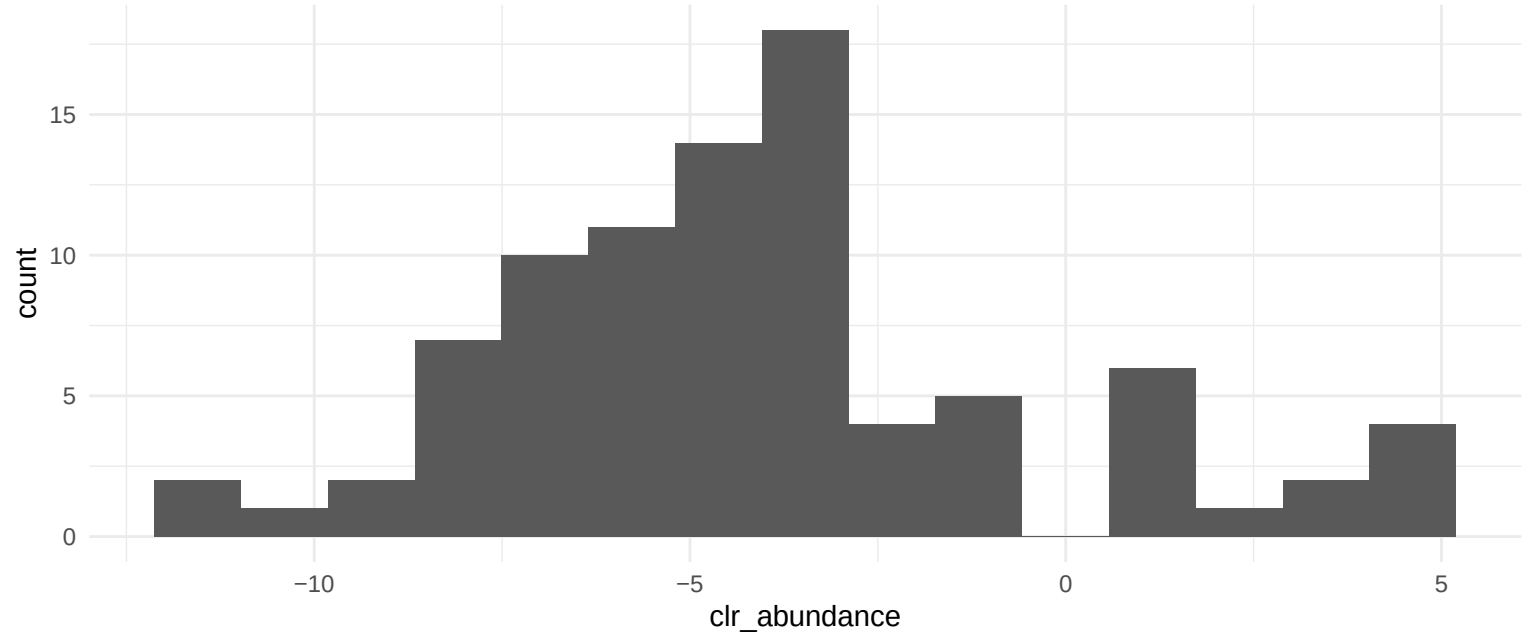
Distribution for 01040 Biosynthesis of unsaturated fatty acids

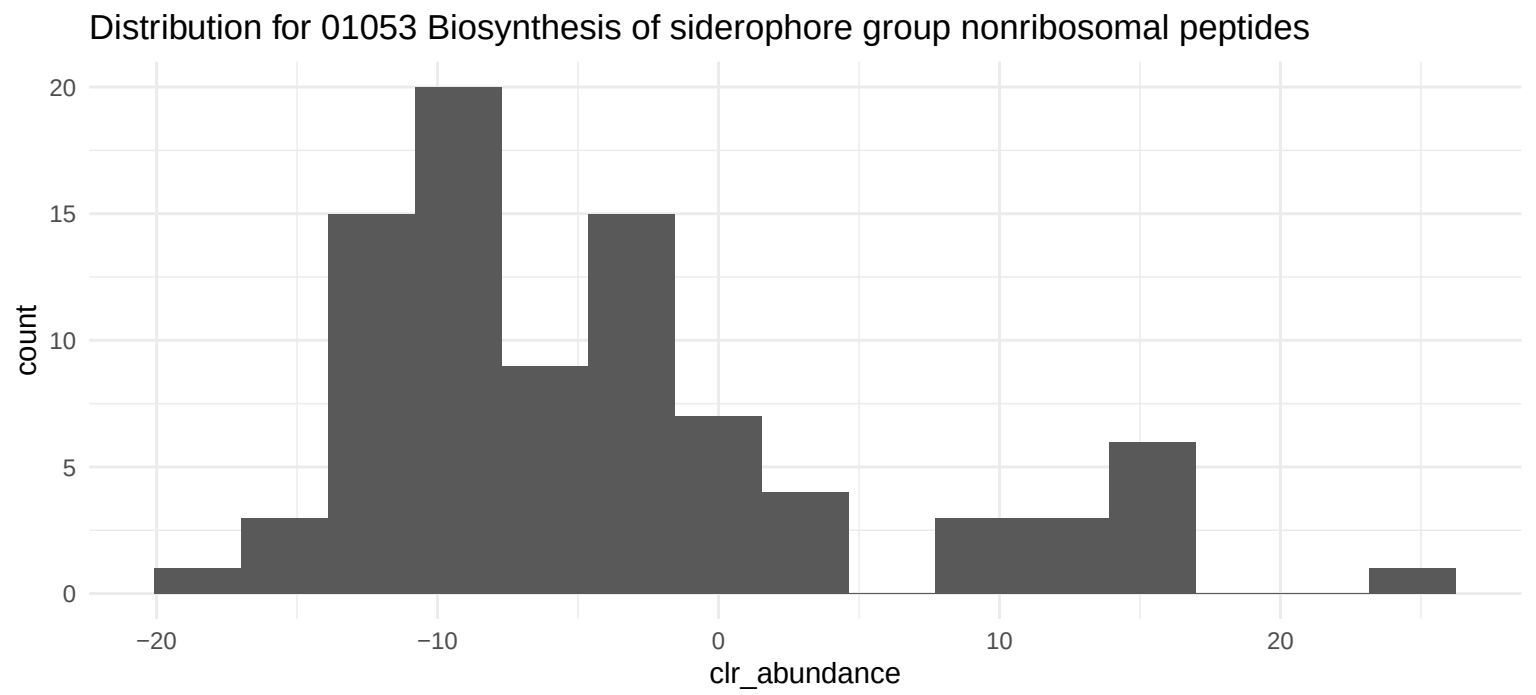


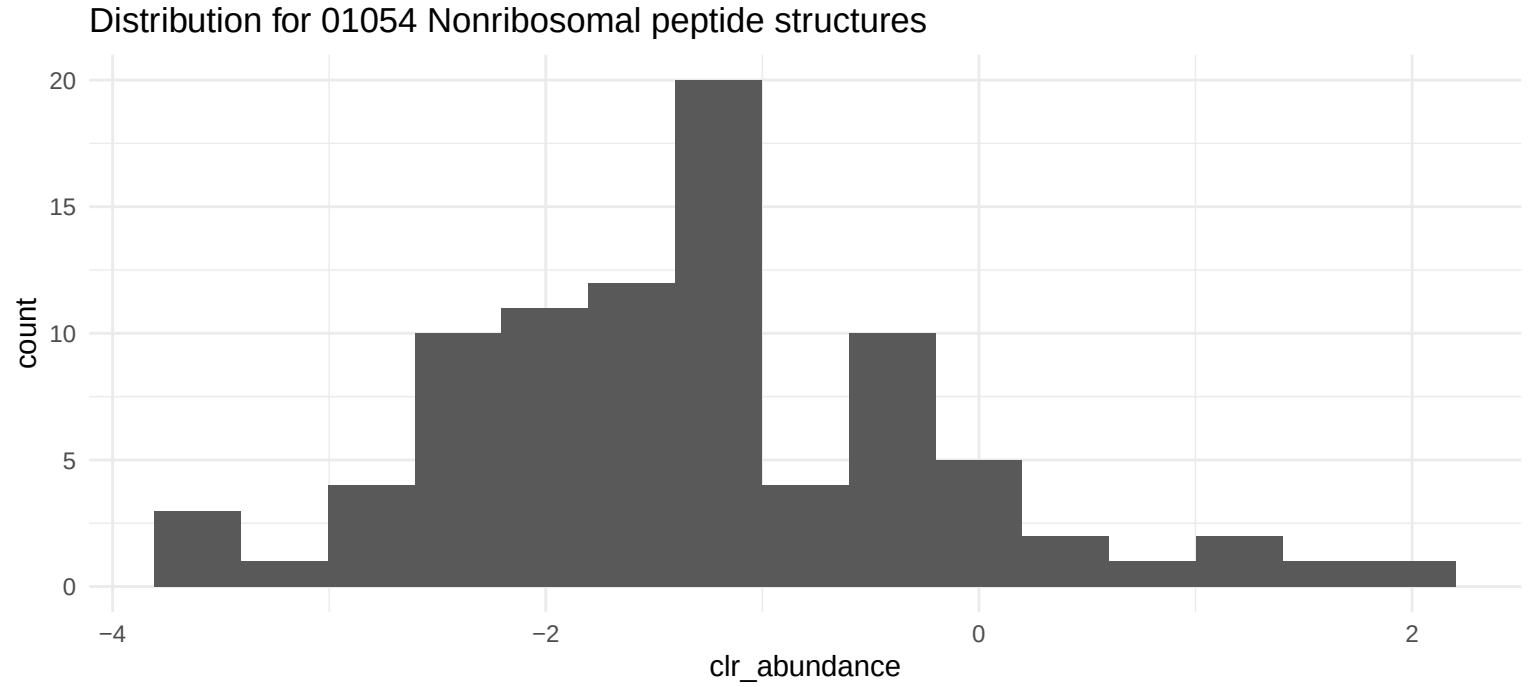
Distribution for 01051 Biosynthesis of ansamycins

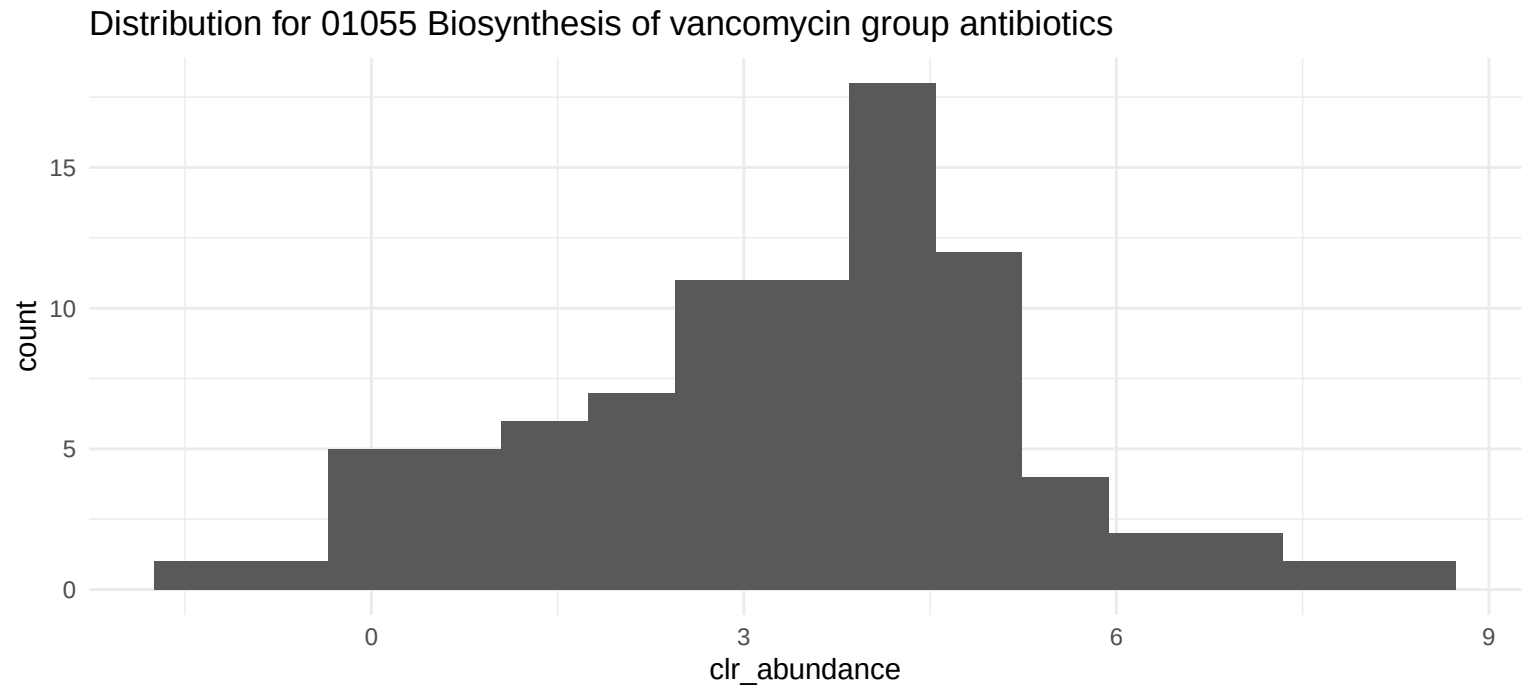


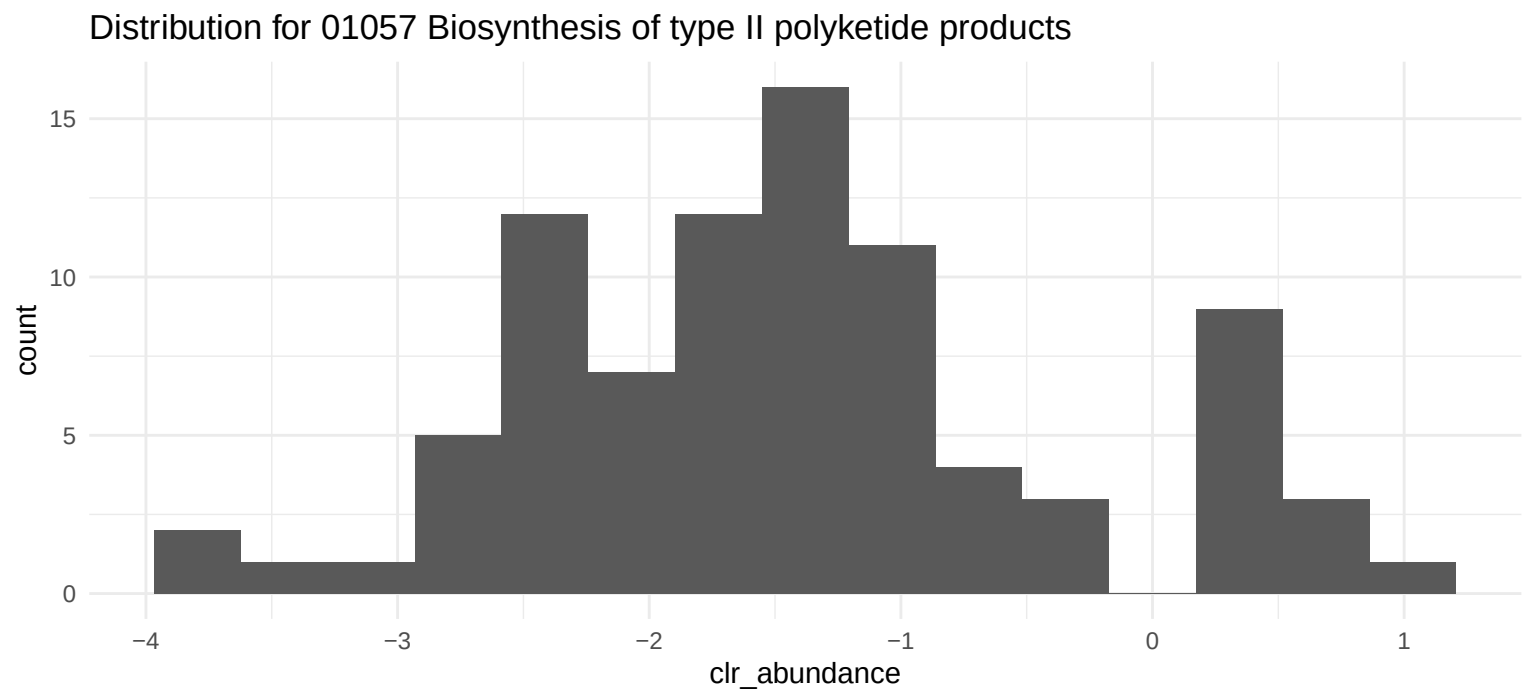
Distribution for 01052 Type I polyketide structures

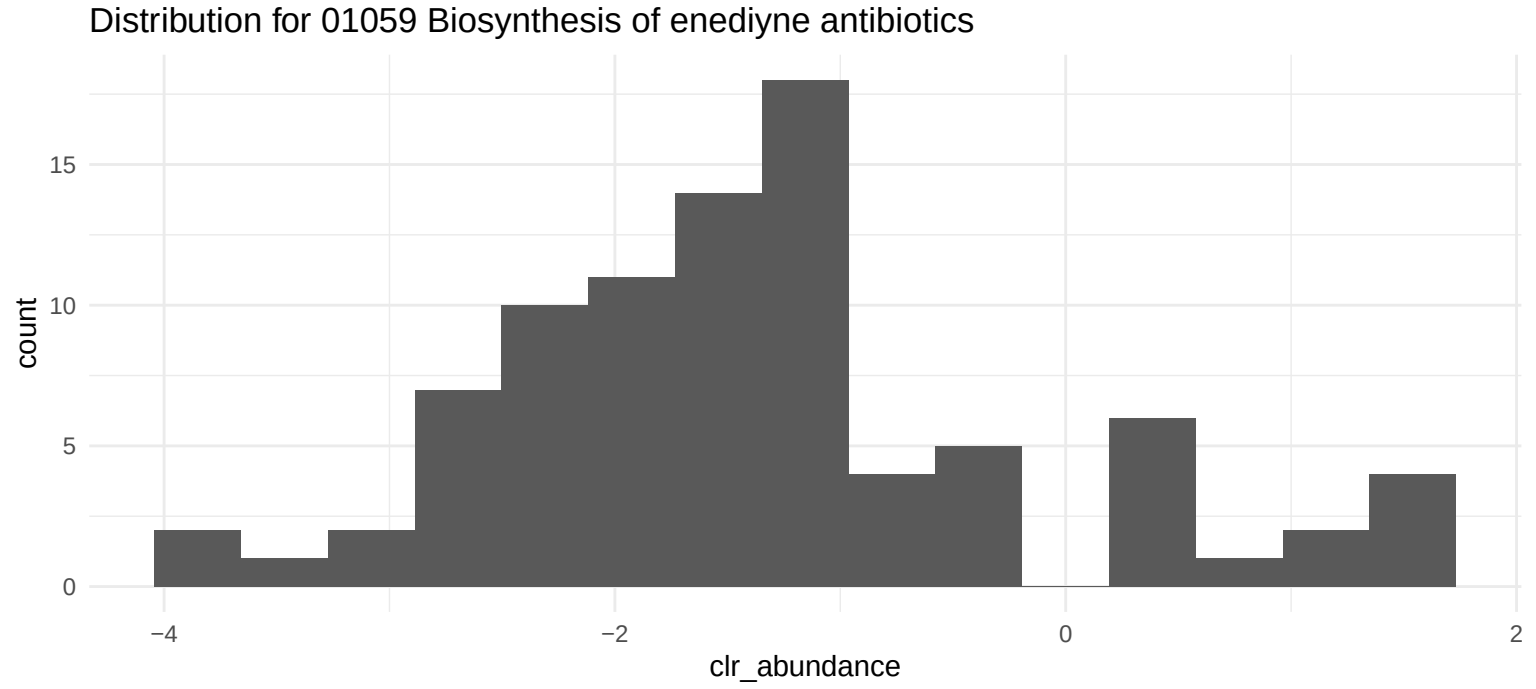


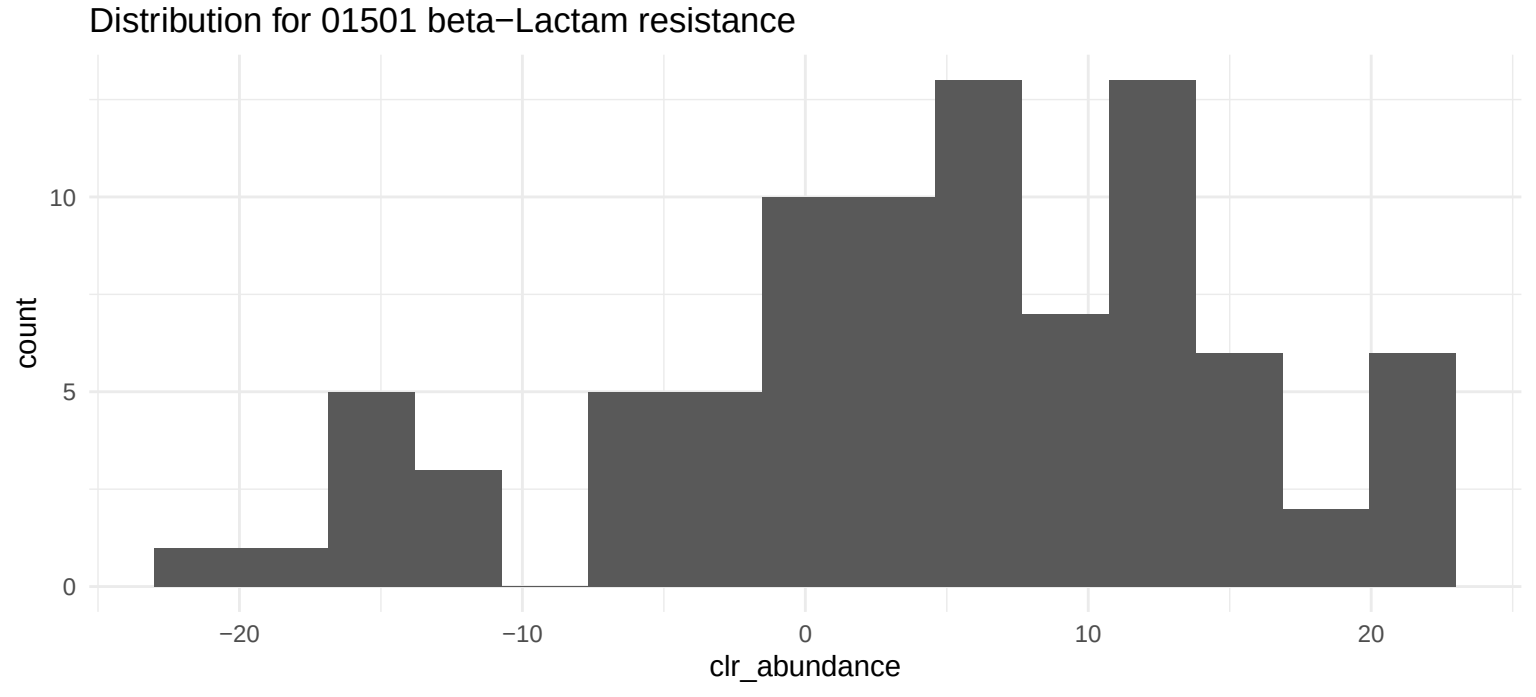

Distribution for 01053 Biosynthesis of siderophore group nonribosomal peptides
count
clr_abundance


Distribution for 01054 Nonribosomal peptide structures
count
clr_abundance

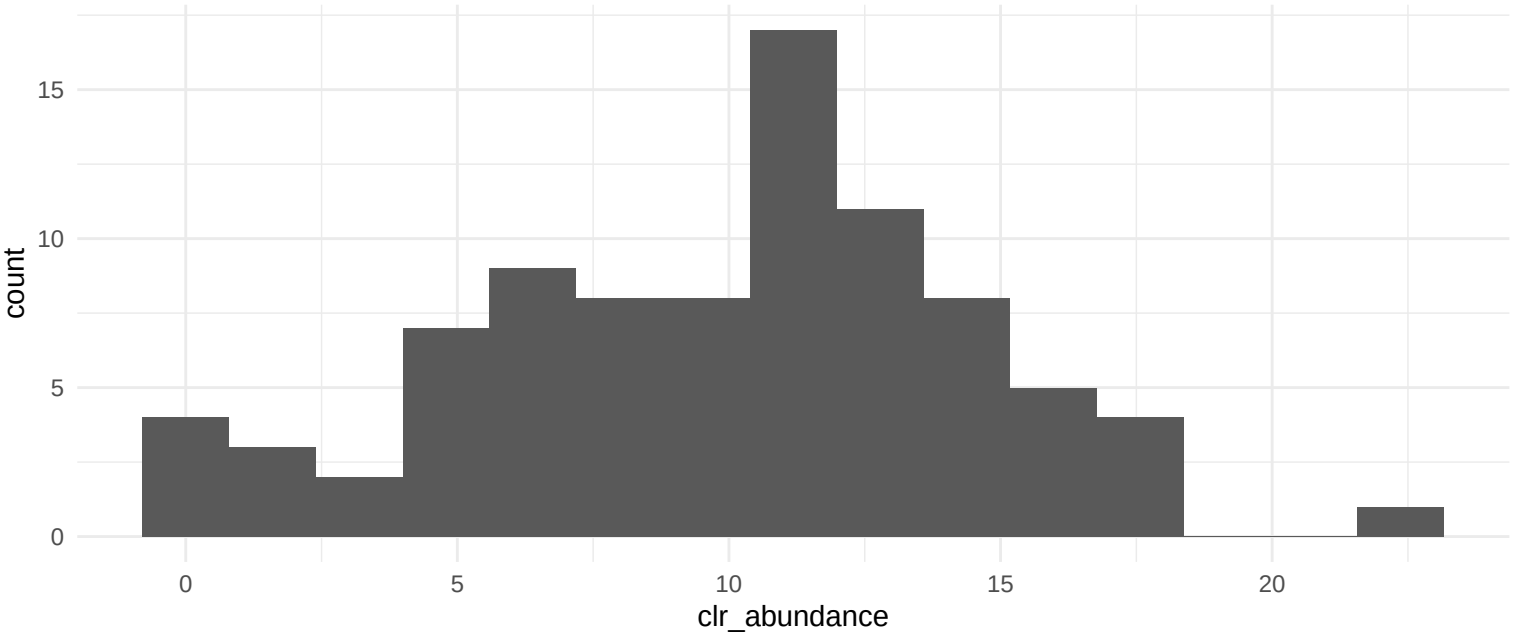

Distribution for 01055 Biosynthesis of vancomycin group antibiotics
count
clr_abundance


Distribution for 01057 Biosynthesis of type II polyketide products
count
clr_abundance

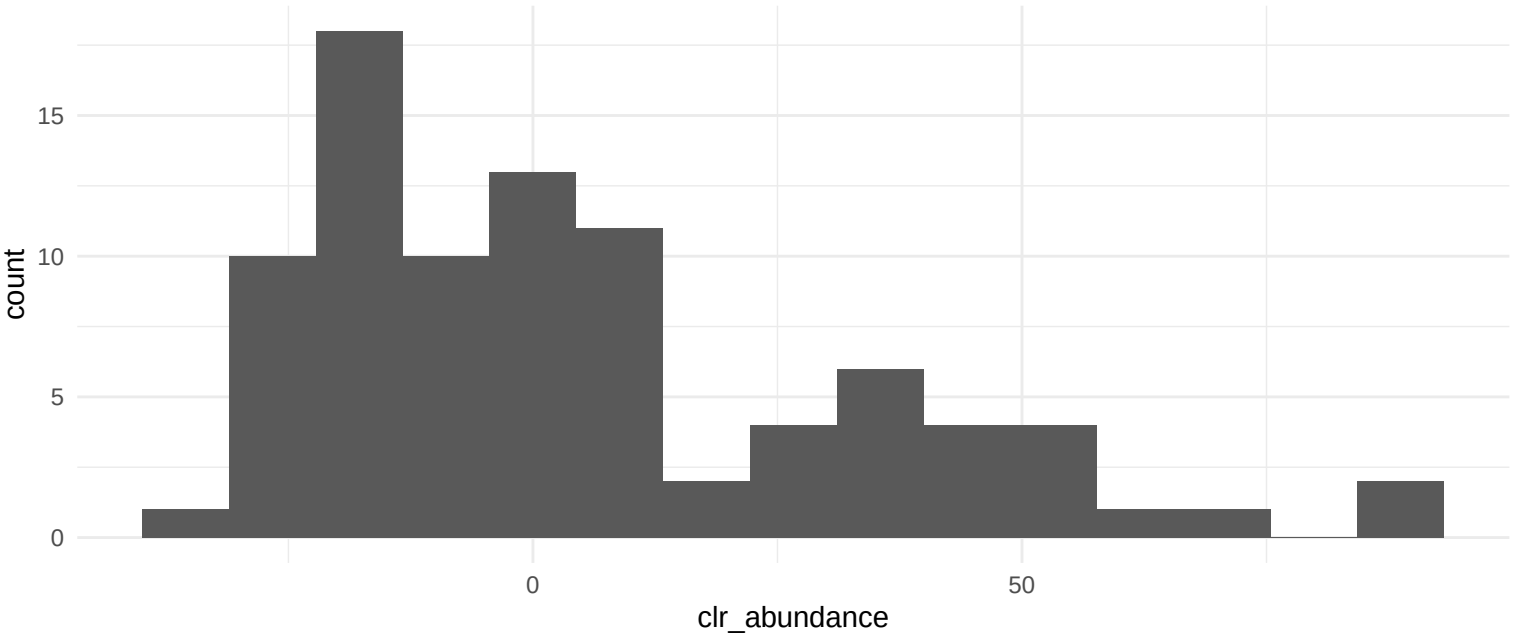

Distribution for 01059 Biosynthesis of enediyne antibiotics
count
clr_abundance


Distribution for 01501 beta−Lactam resistance
count
clr_abundance

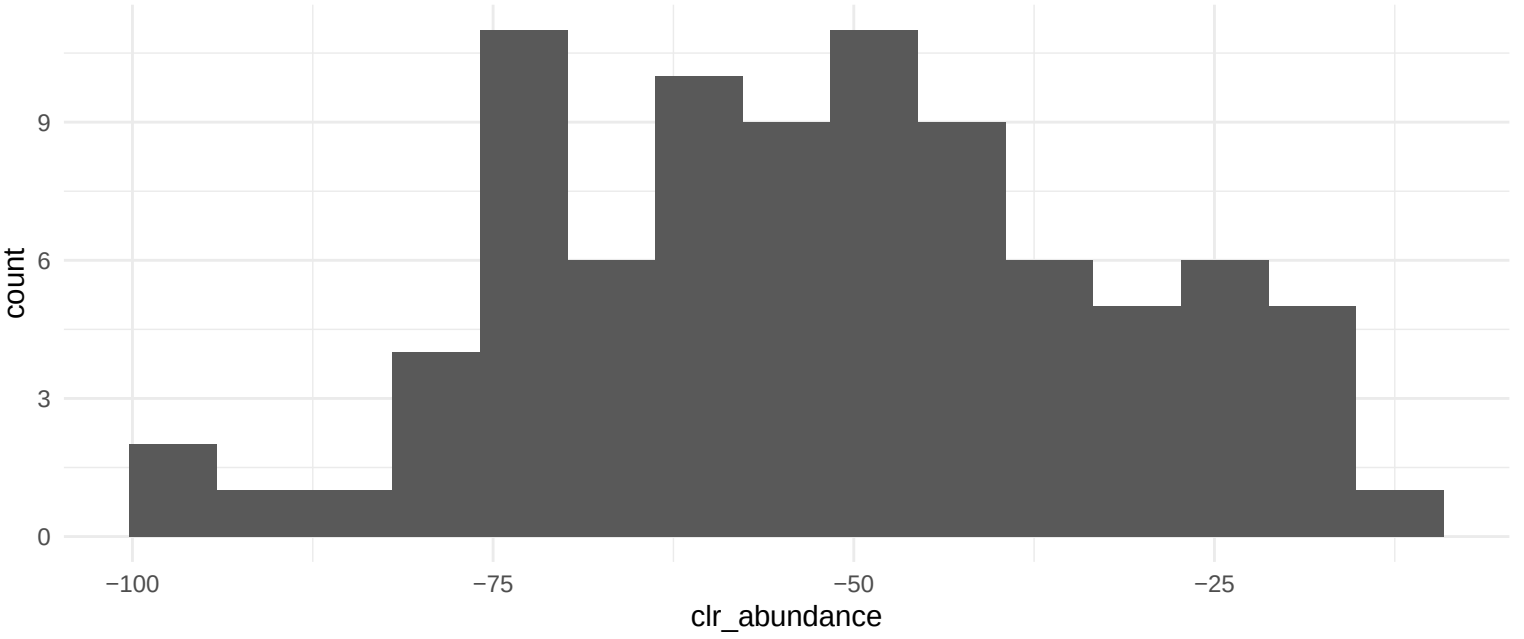
Distribution for 01502 Vancomycin resistance

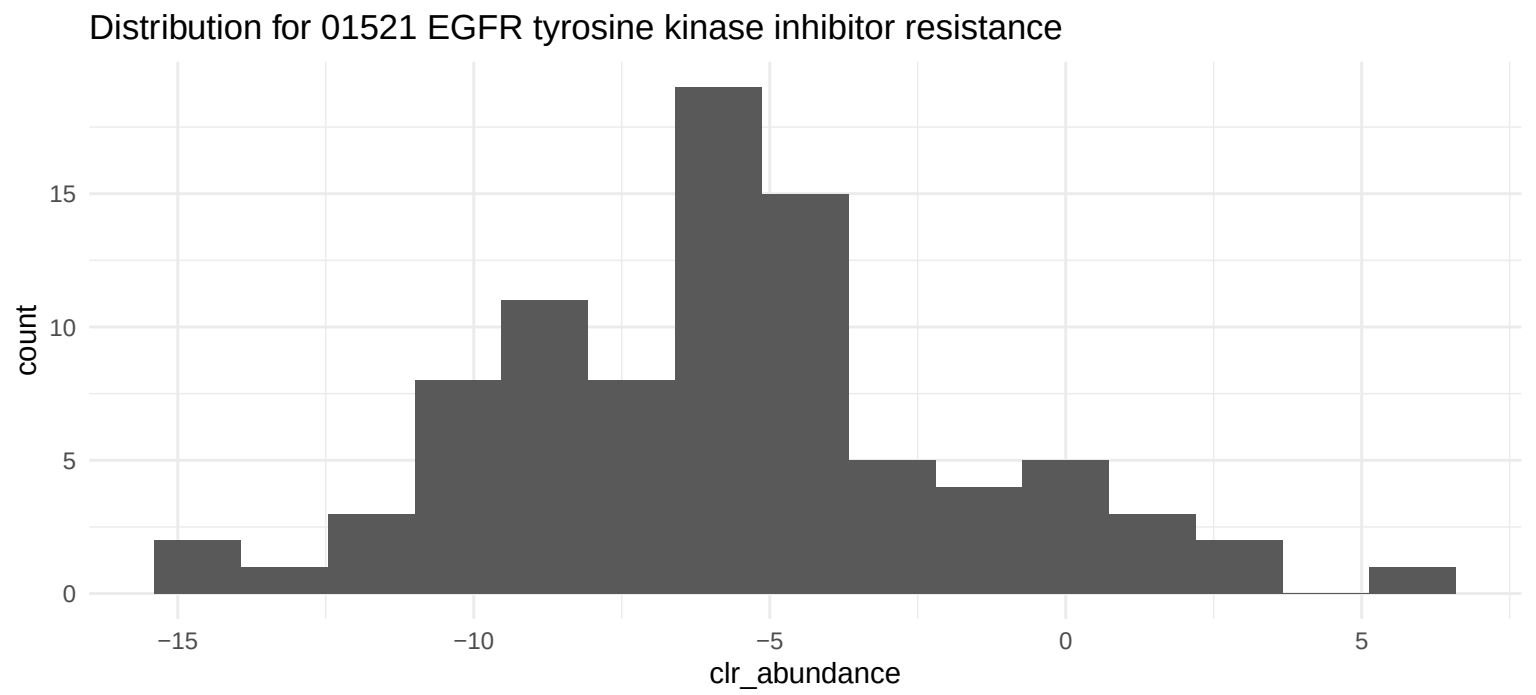


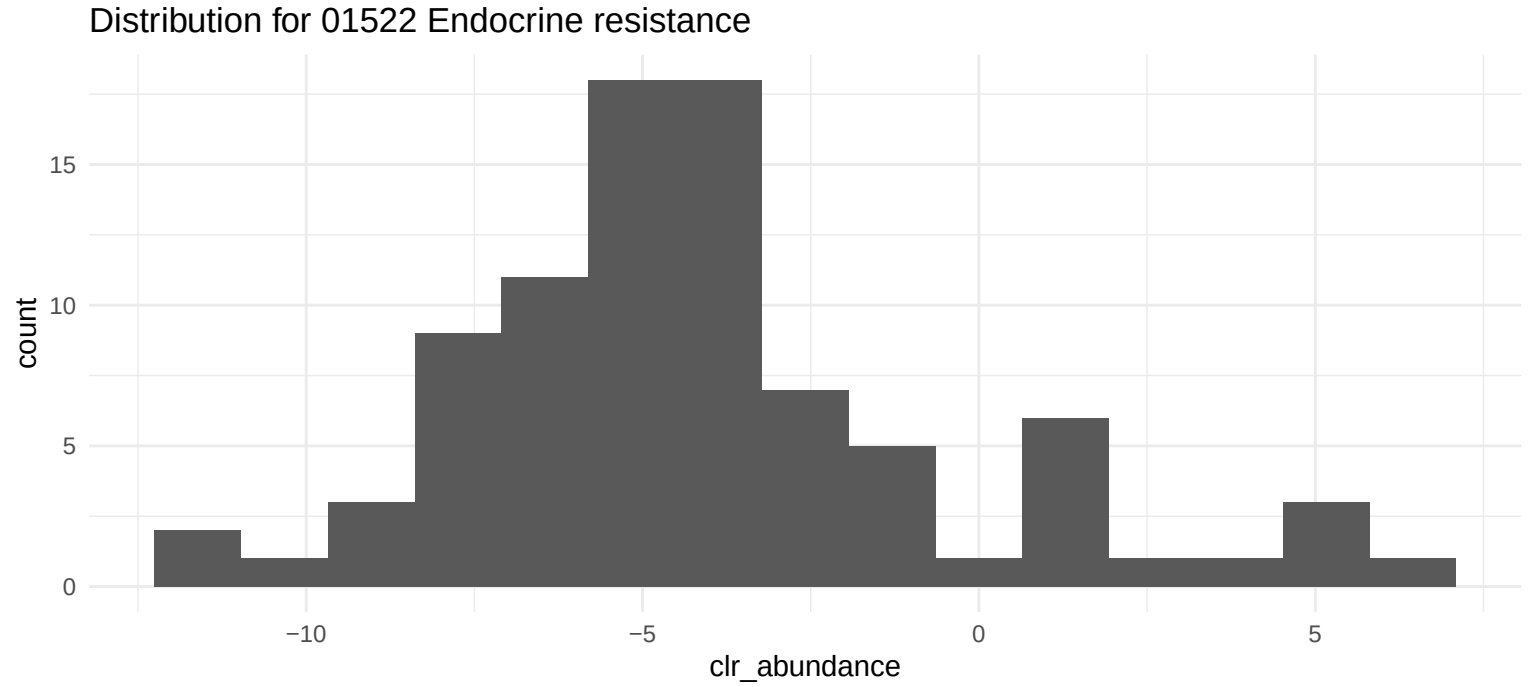
Distribution for 01503 Cationic antimicrobial peptide (CAMP) resistance

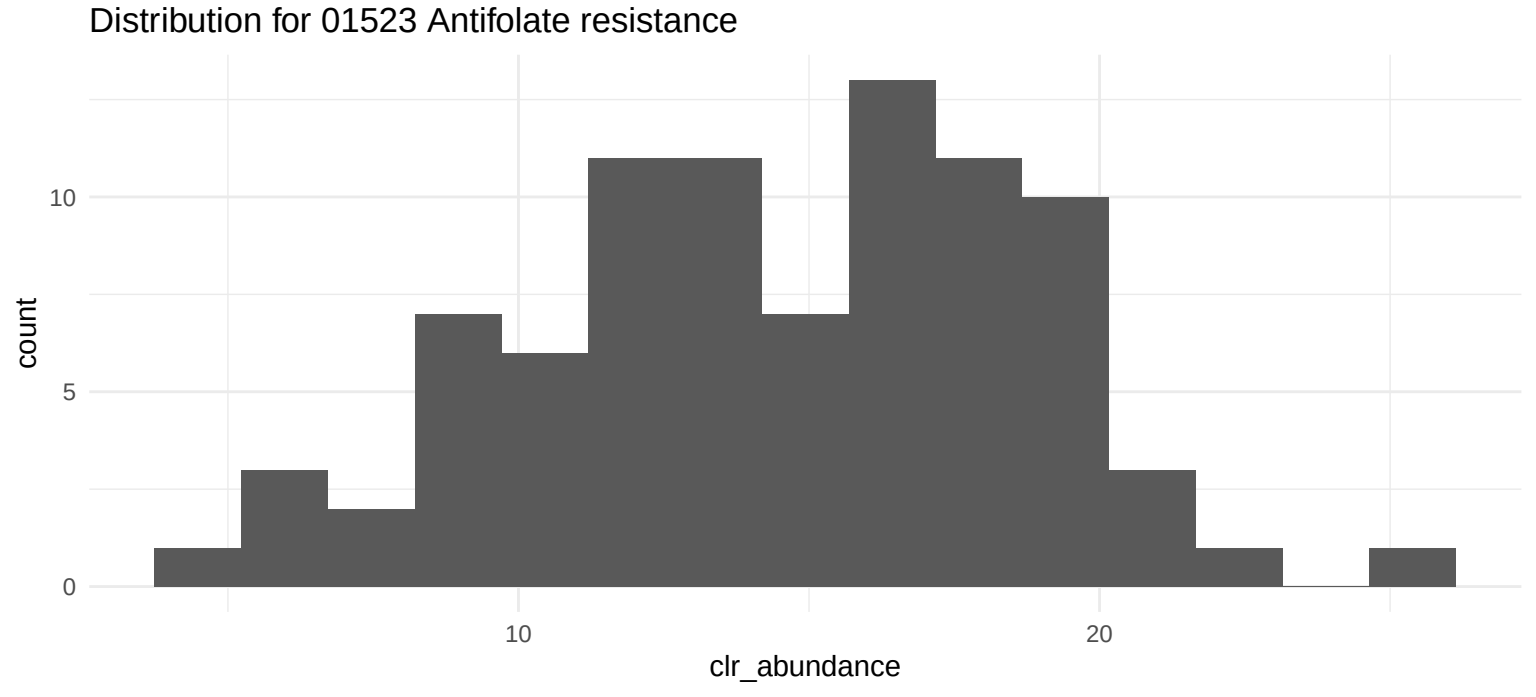


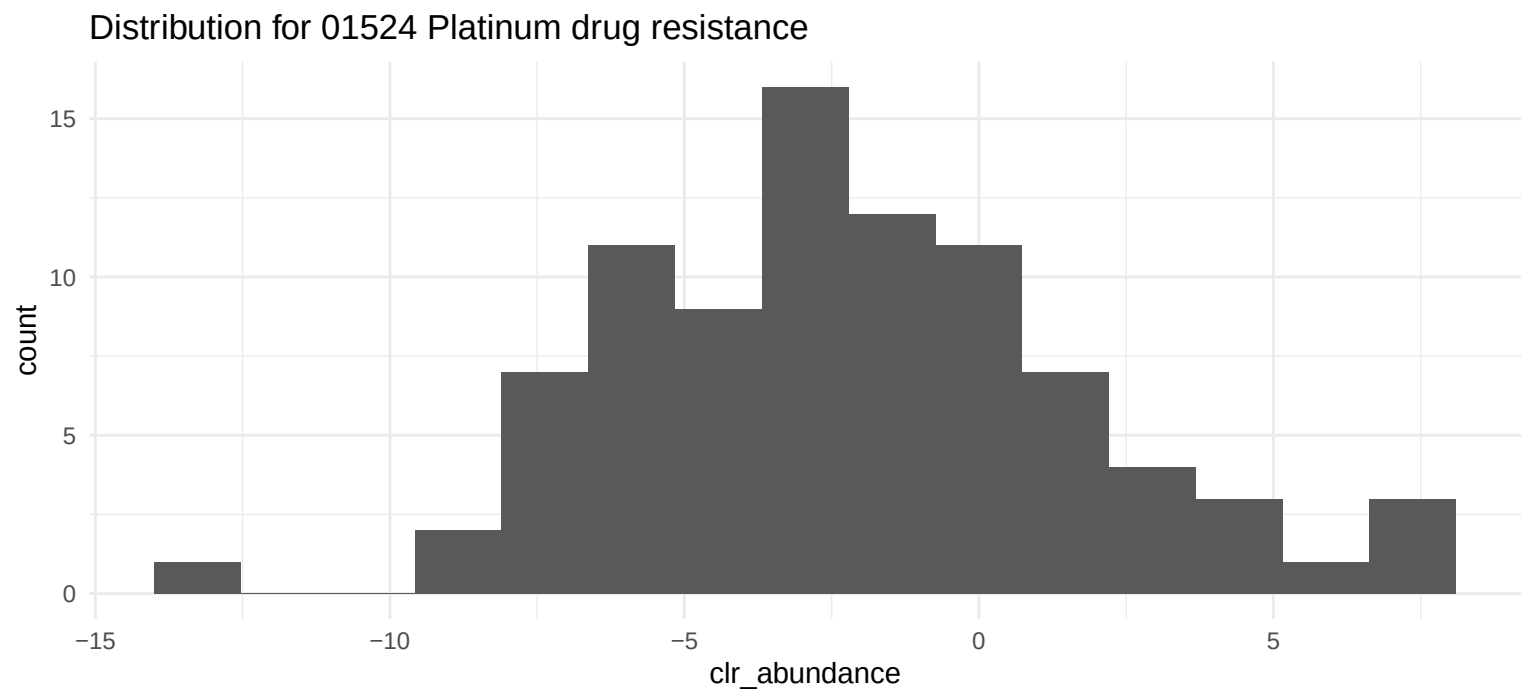
Distribution for 01504 Antimicrobial resistance genes

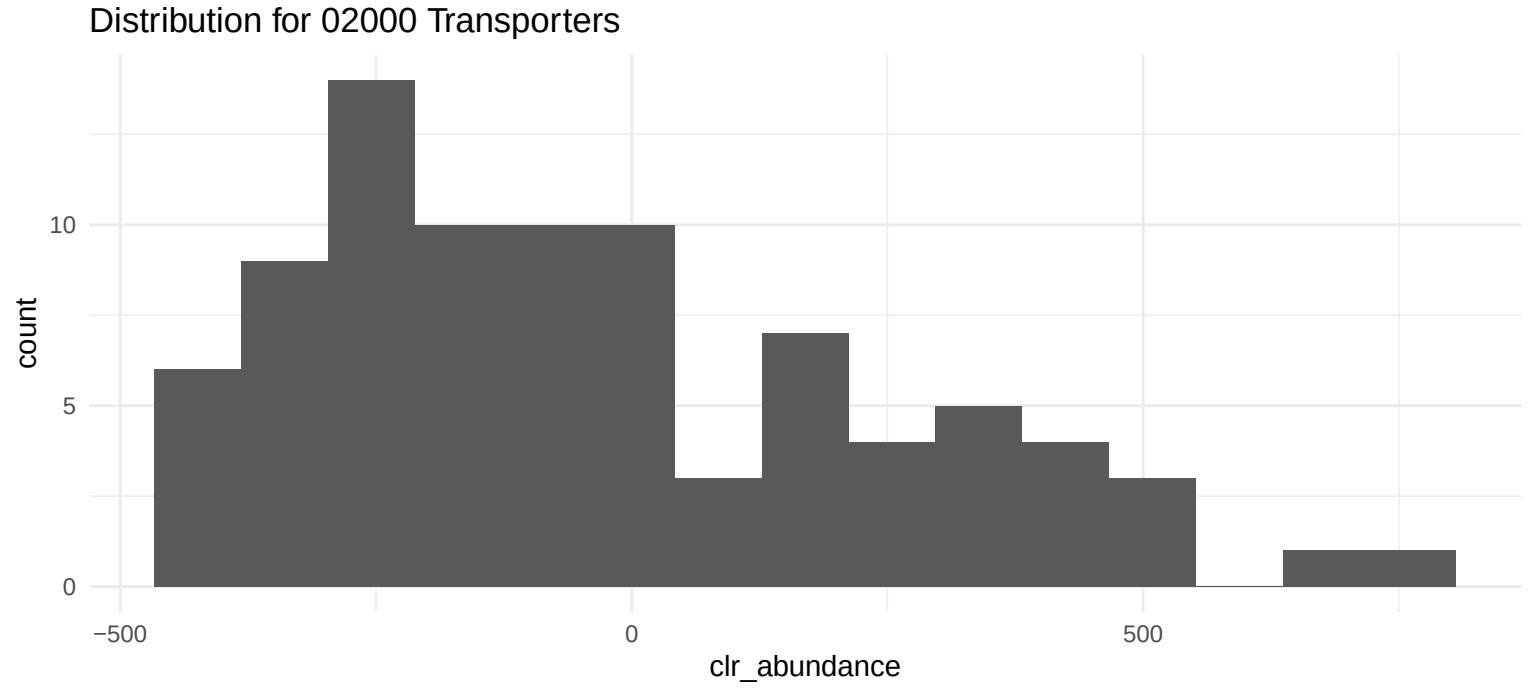


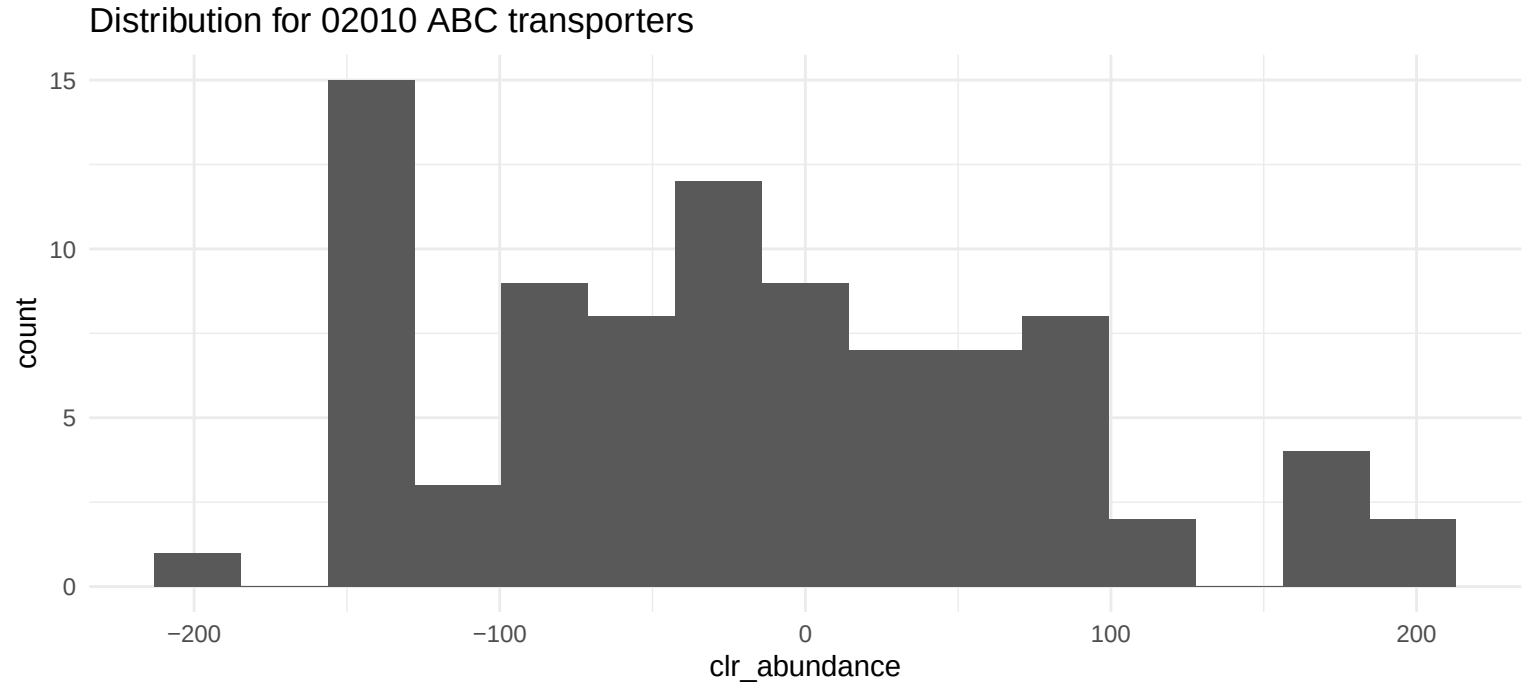

Distribution for 01521 EGFR tyrosine kinase inhibitor resistance
count
15
10
5
0
−15
−10
−5
0
5
clr_abundance

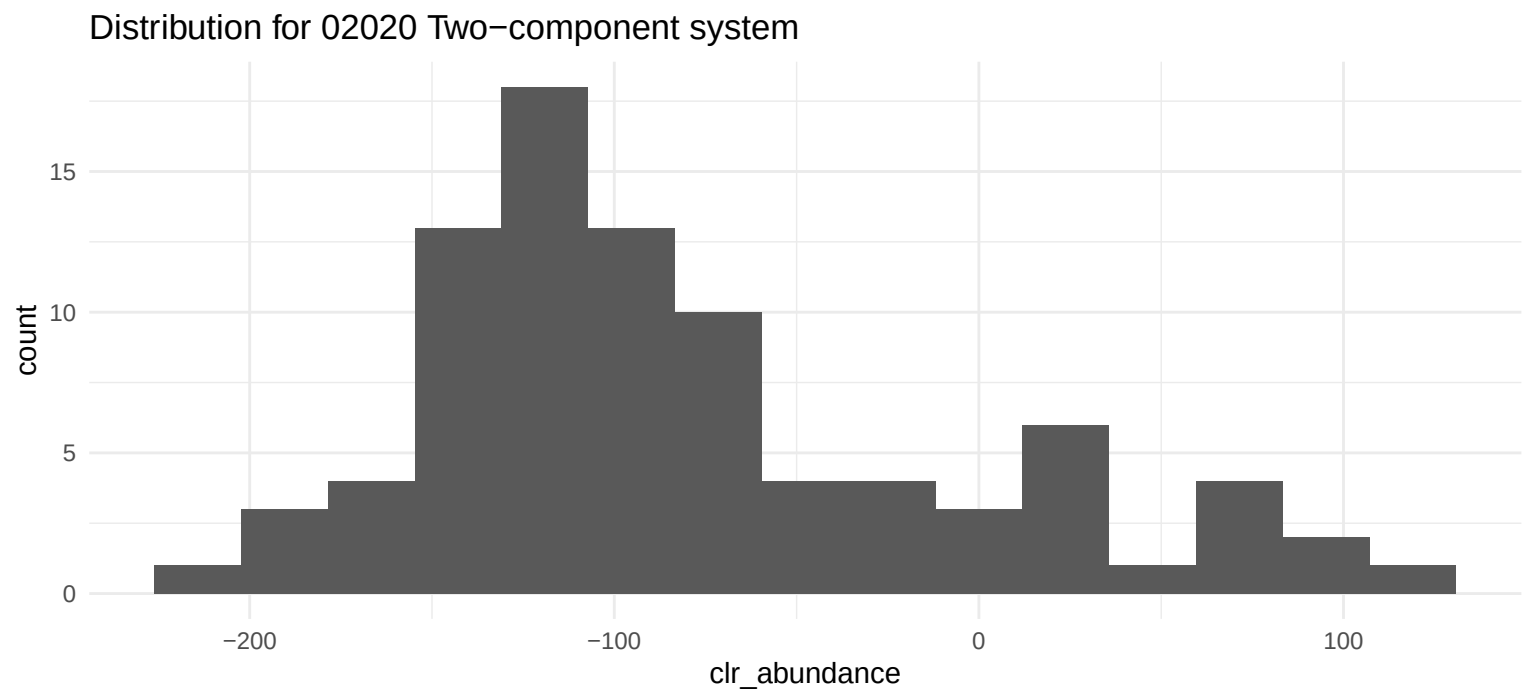

Distribution for 01522 Endocrine resistance
count
15
10
5
0
−10
−5
0
5
clr_abundance

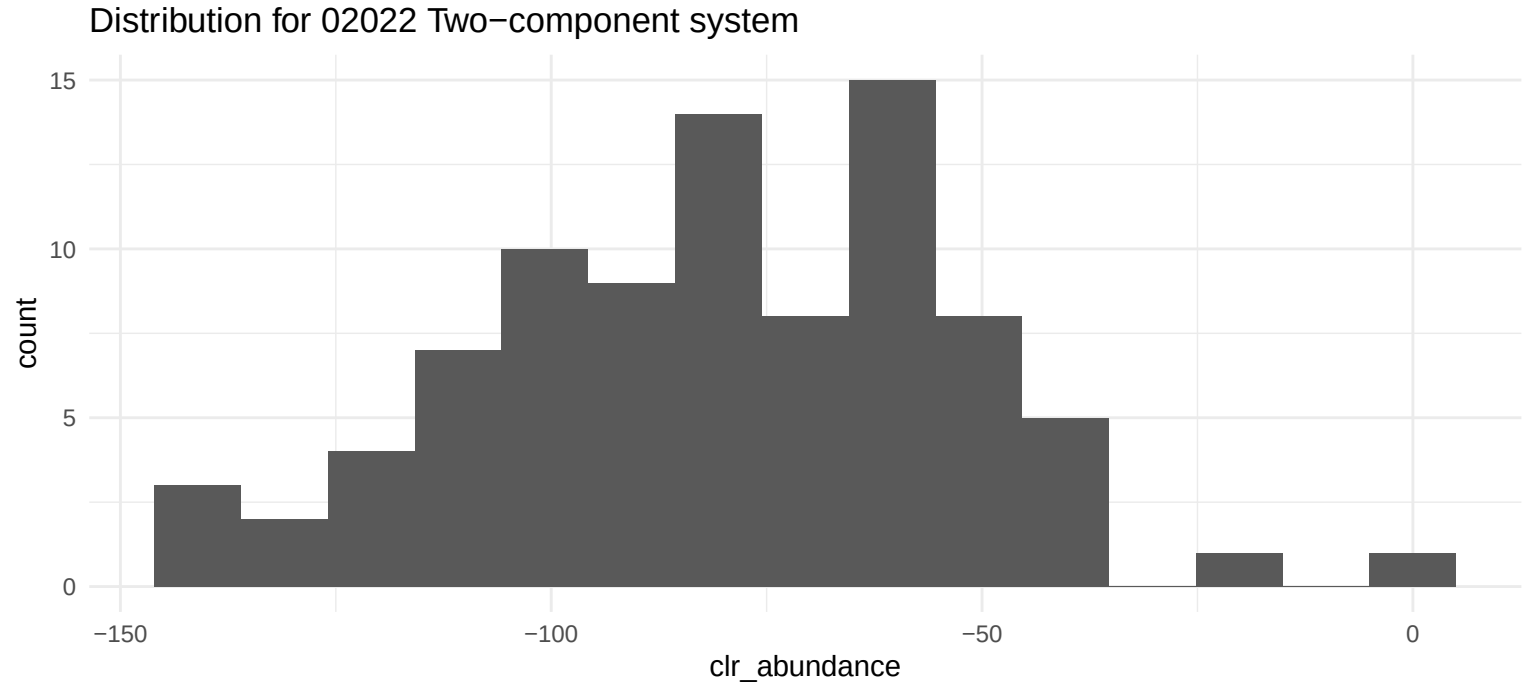

Distribution for 01523 Antifolate resistance
count
10
5
0
10
20
clr_abundance

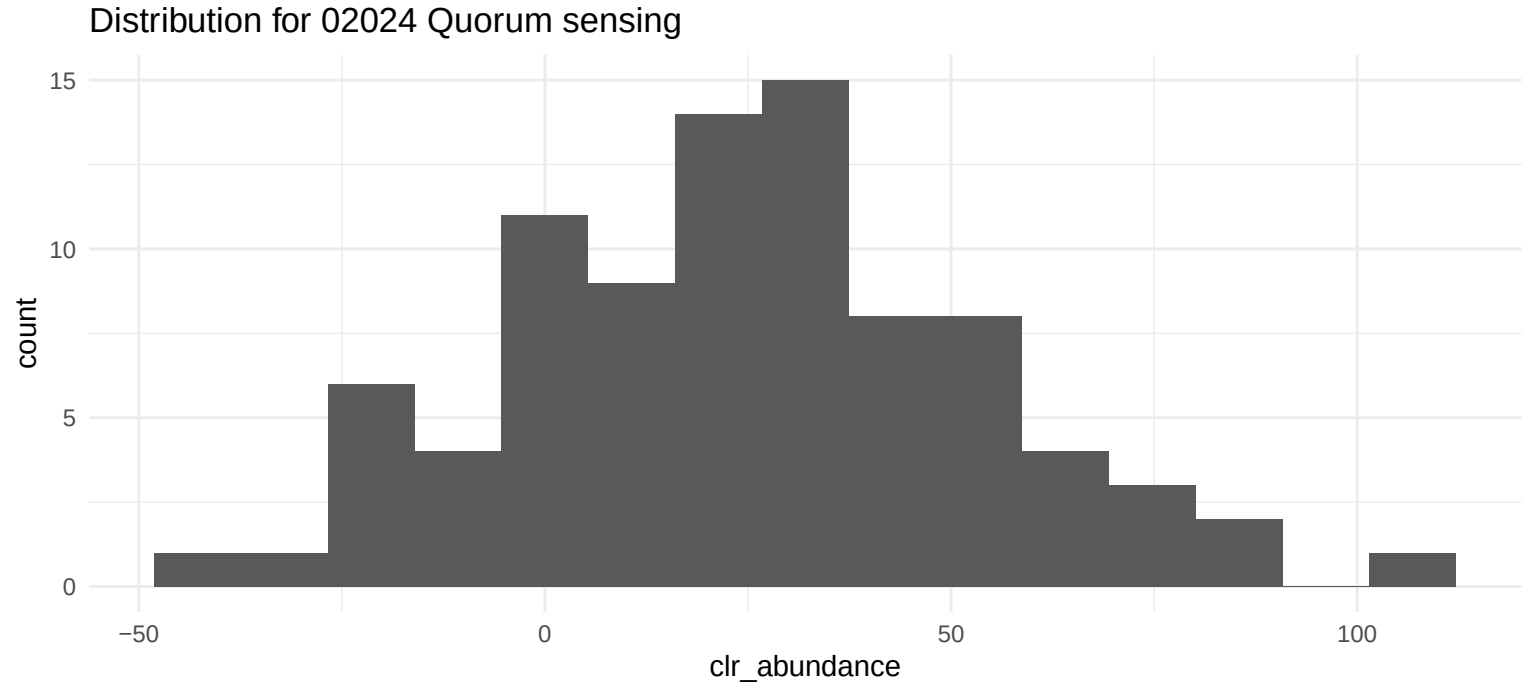

Distribution for 01524 Platinum drug resistance
count
15
10
5
0
−15
−10
−5
0
5
clr_abundance

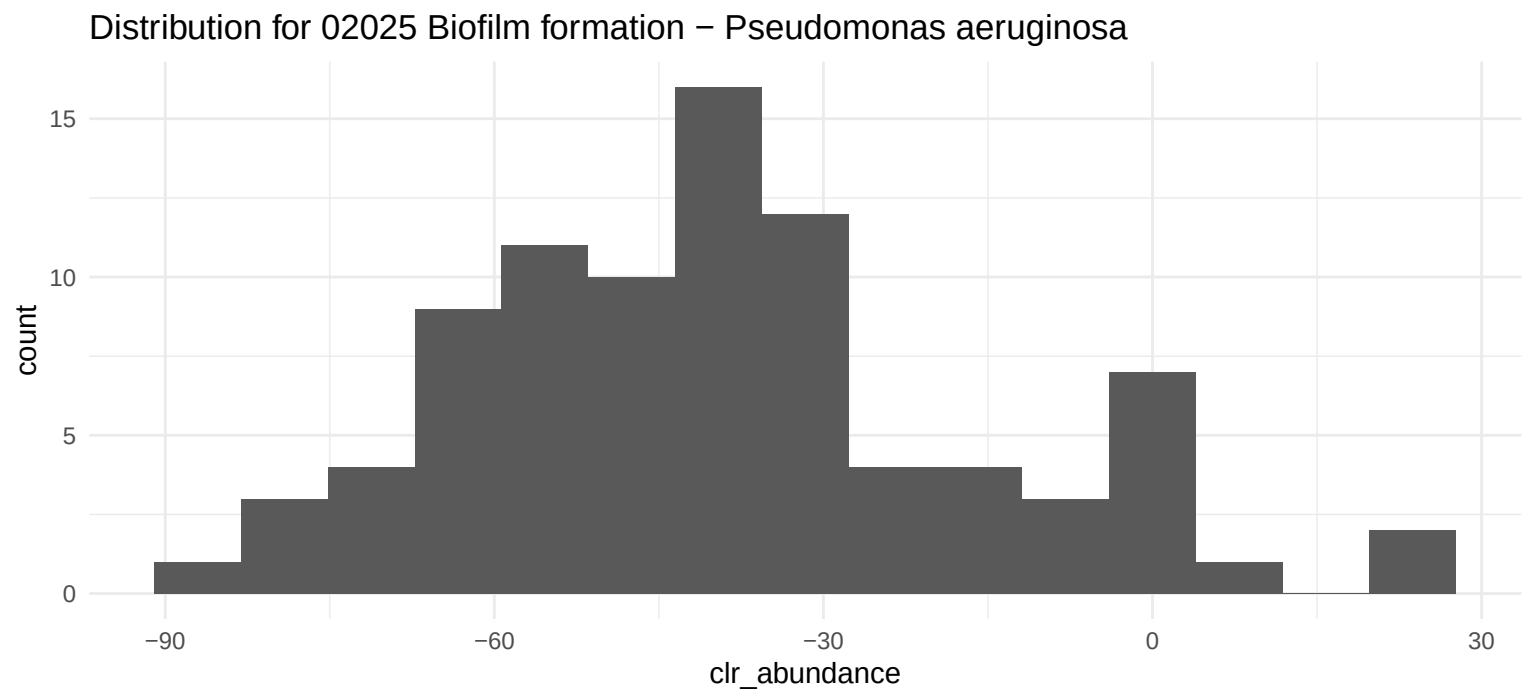

Distribution for 02000 Transporters
count
10
5
0
−500
0
500
clr_abundance

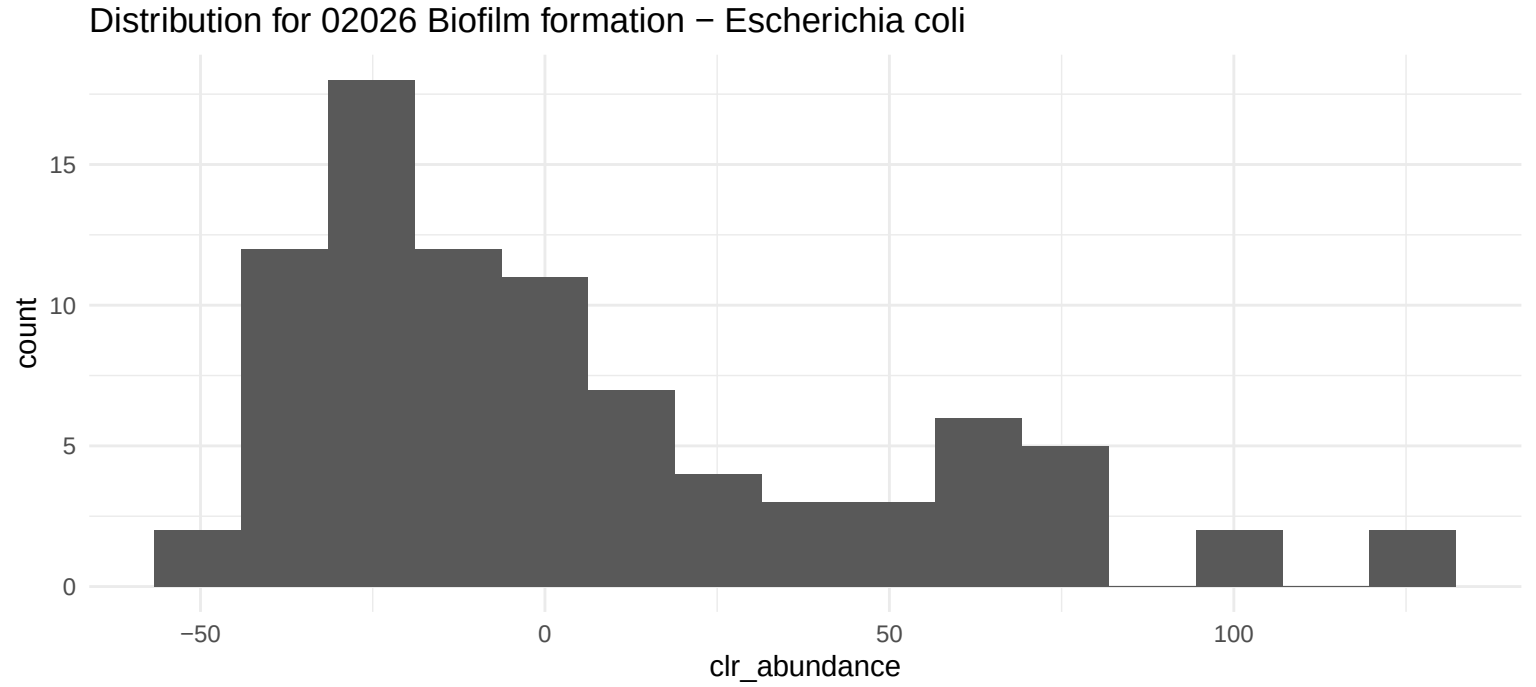

Distribution for 02010 ABC transporters
count
15
10
5
0
−200
−100
0
100
200
clr_abundance

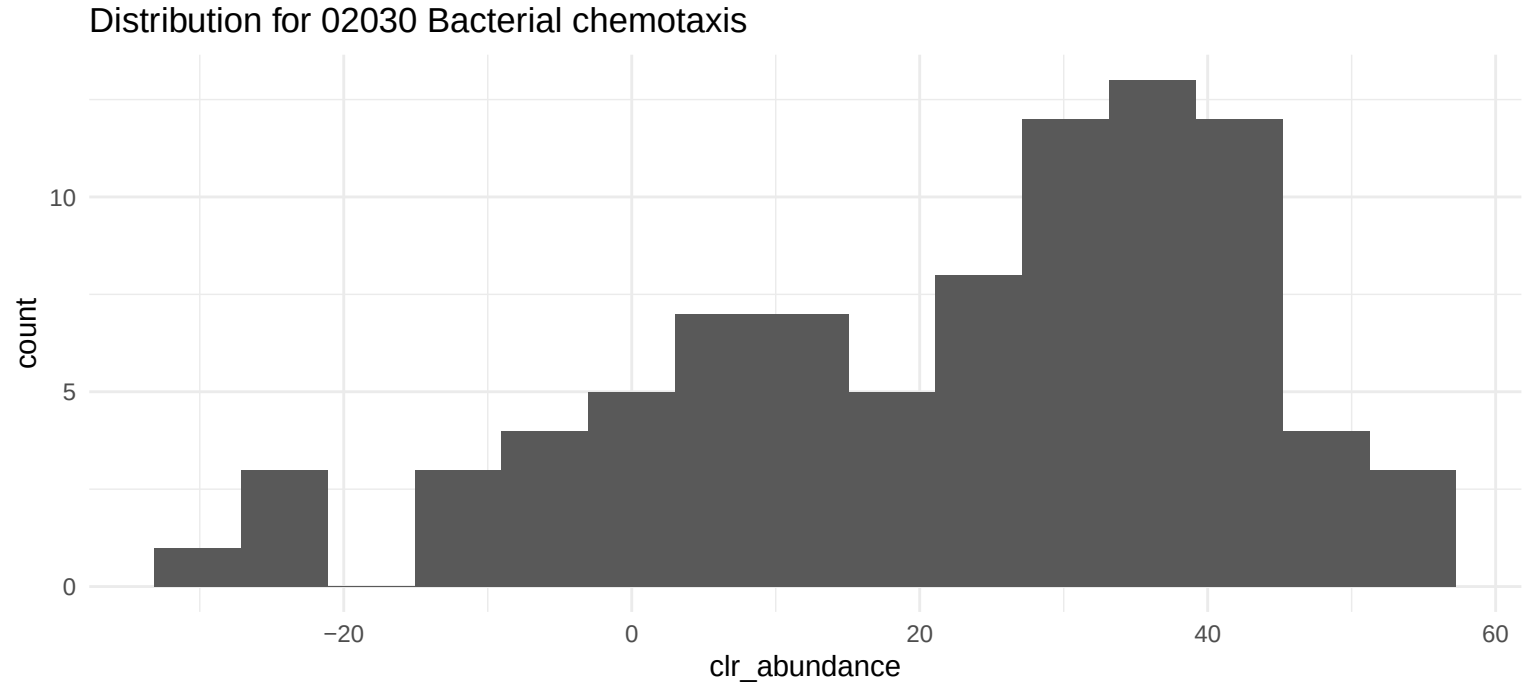

Distribution for 02020 Two−component system
count
clr_abundance
0
5
10
15
−200
−100
0
100

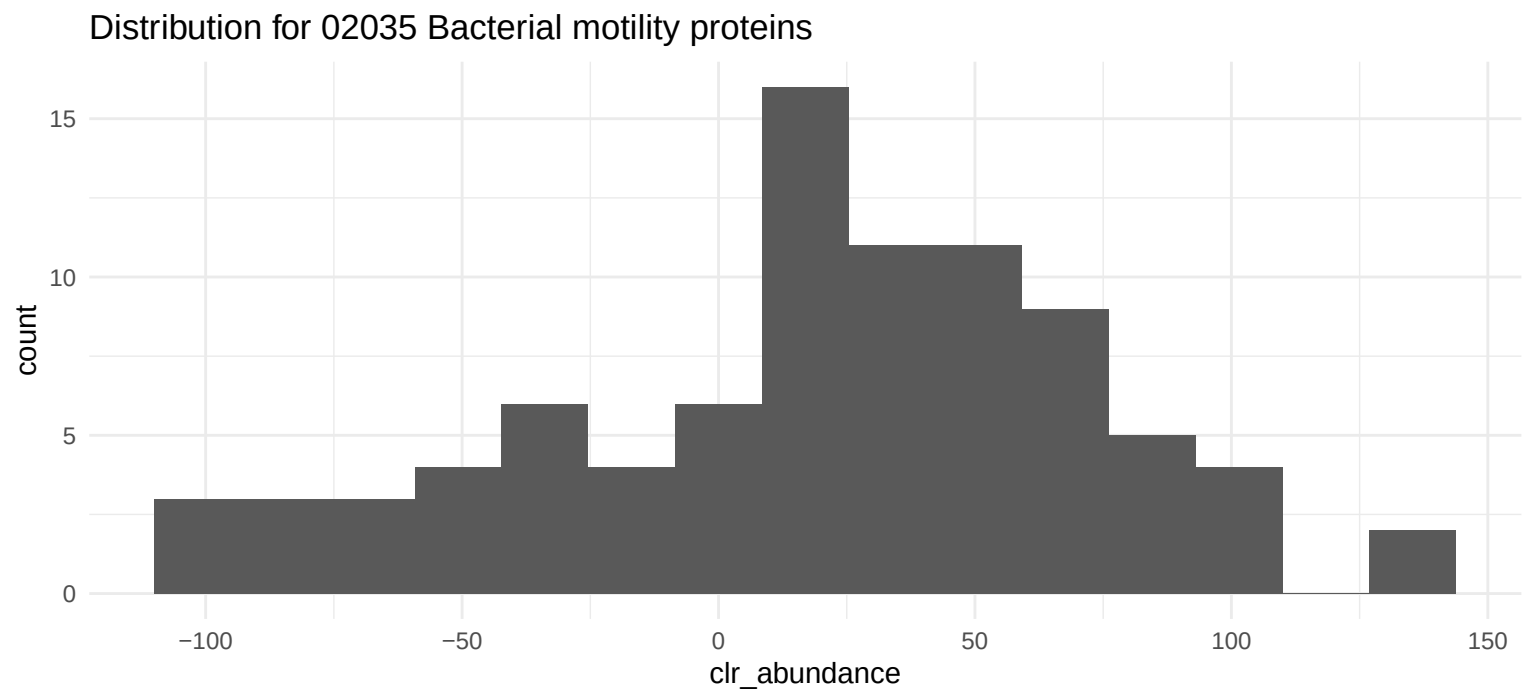

Distribution for 02022 Two−component system
count
clr_abundance
0
5
10
15
−150
−100
−50
0

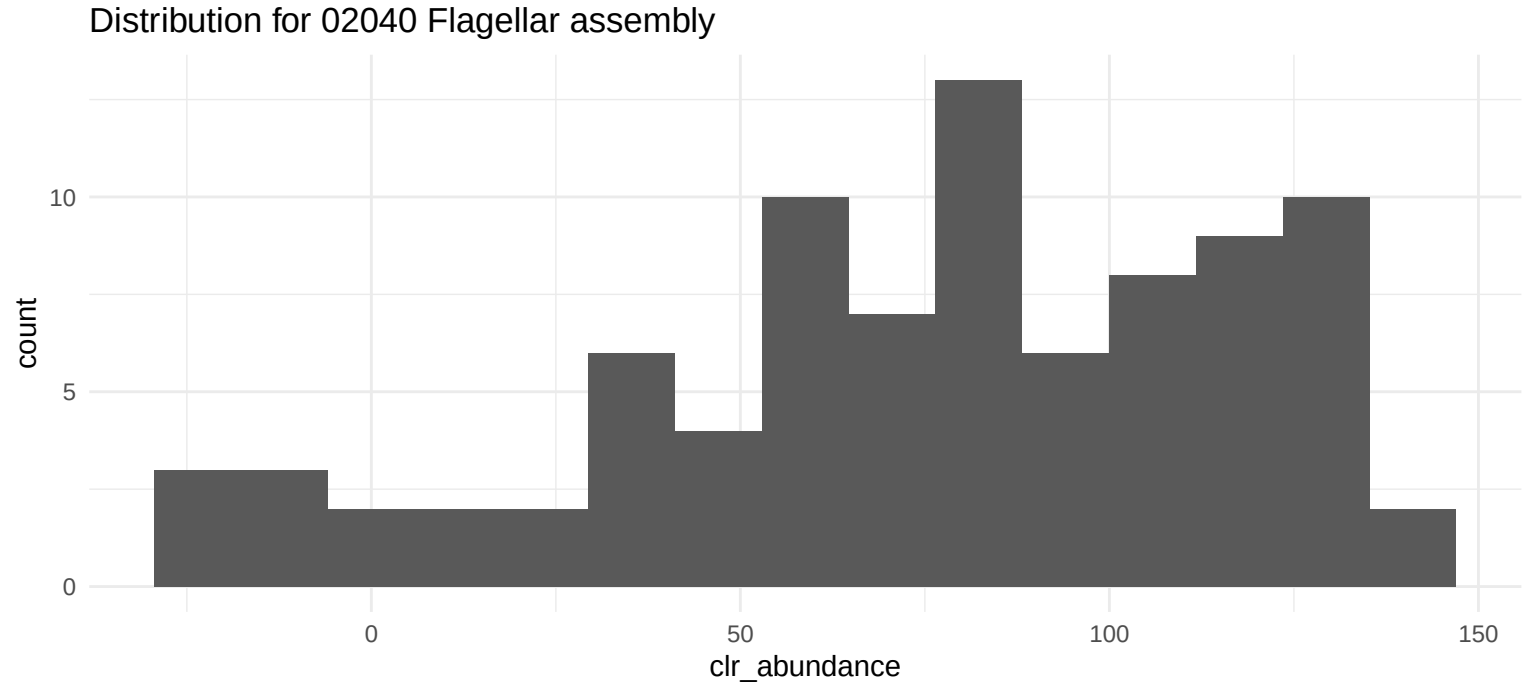

Distribution for 02024 Quorum sensing
count
clr_abundance
0
5
10
15
−50
0
50
100

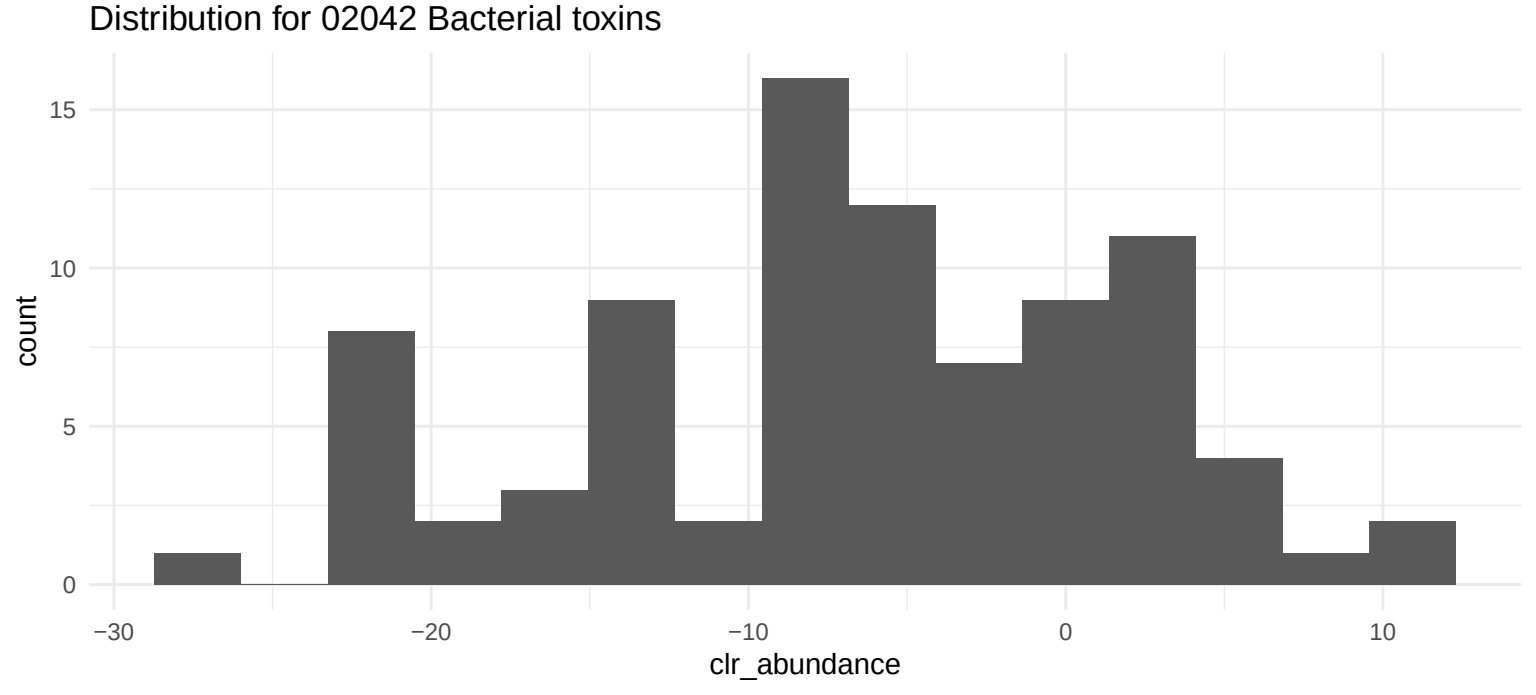

Distribution for 02025 Biofilm formation − Pseudomonas aeruginosa
count
clr_abundance
0
5
10
15
−90
−60
−30
0
30

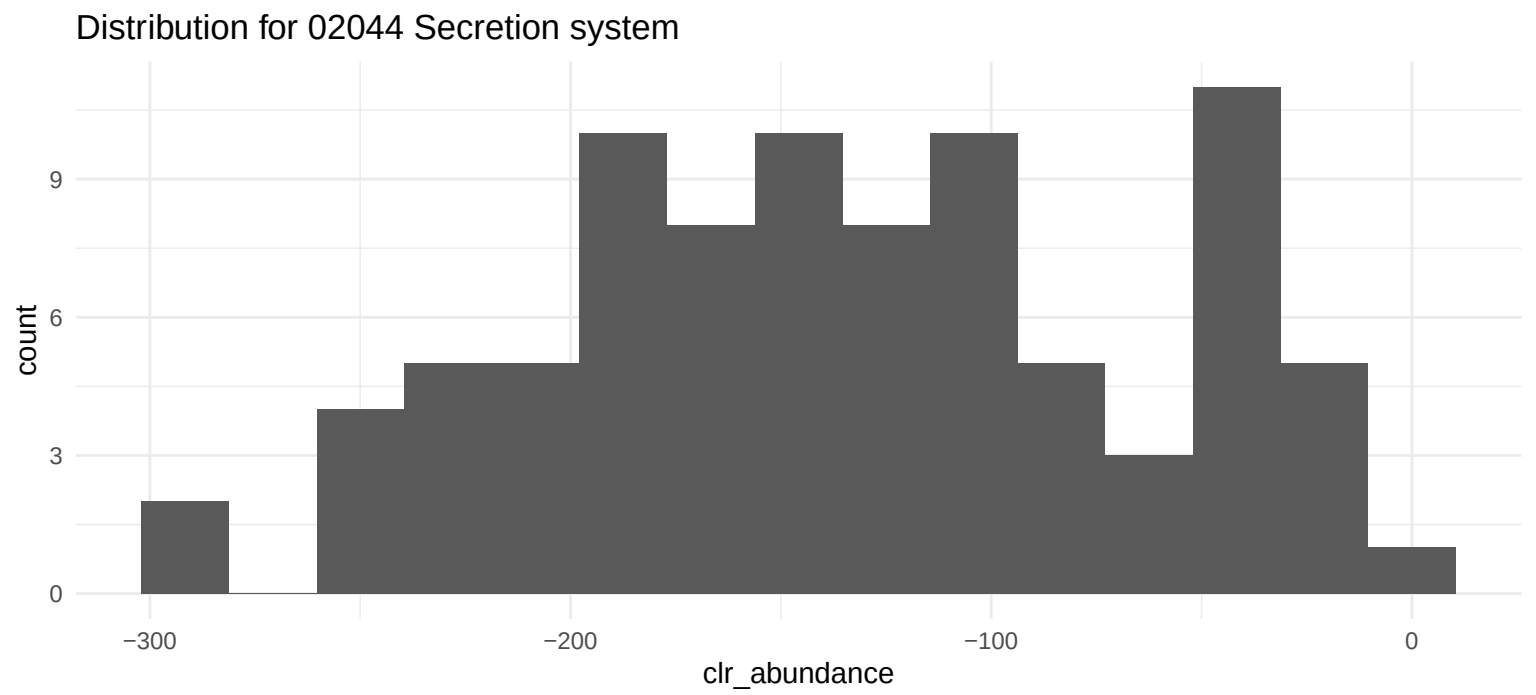

Distribution for 02026 Biofilm formation − Escherichia coli
count
clr_abundance
0
5
10
15
−50
0
50
100

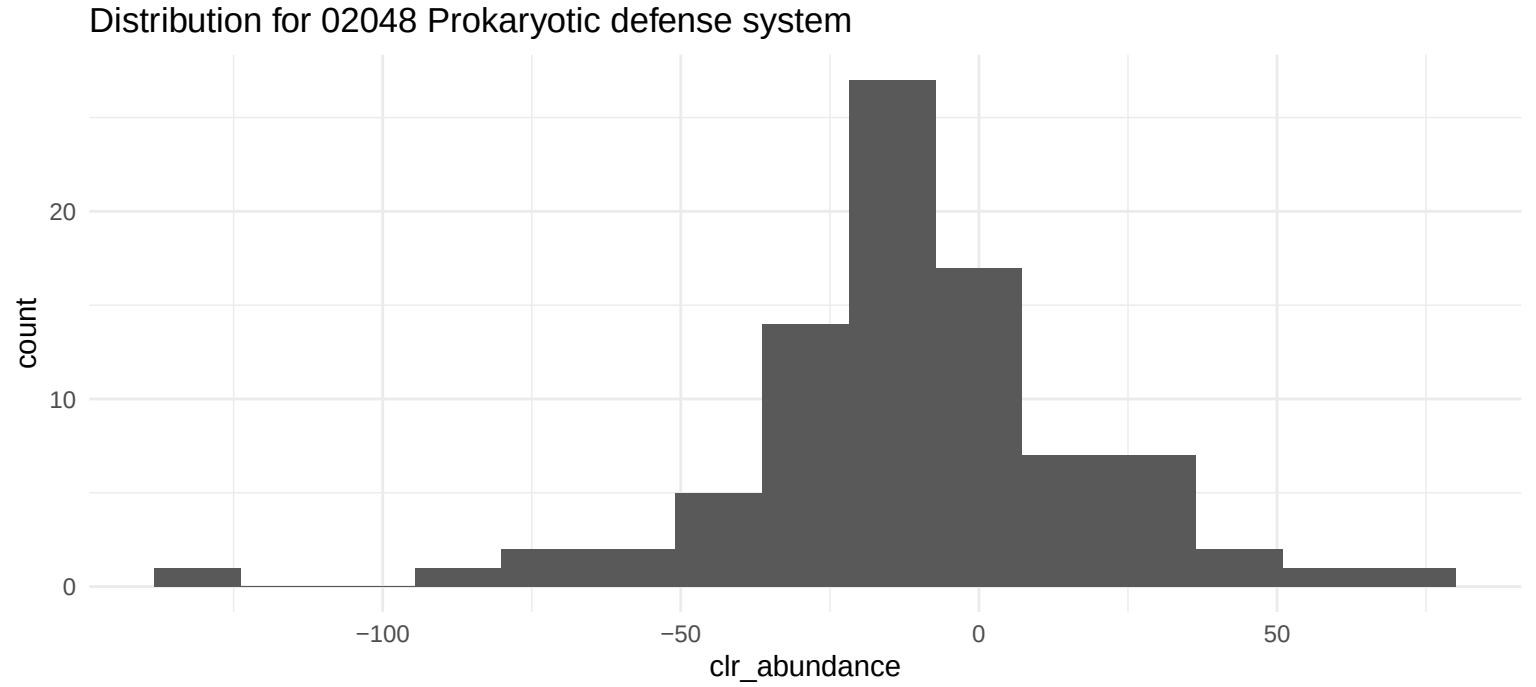

Distribution for 02030 Bacterial chemotaxis
count
clr_abundance
0
5
10
−20
0
20
40
60

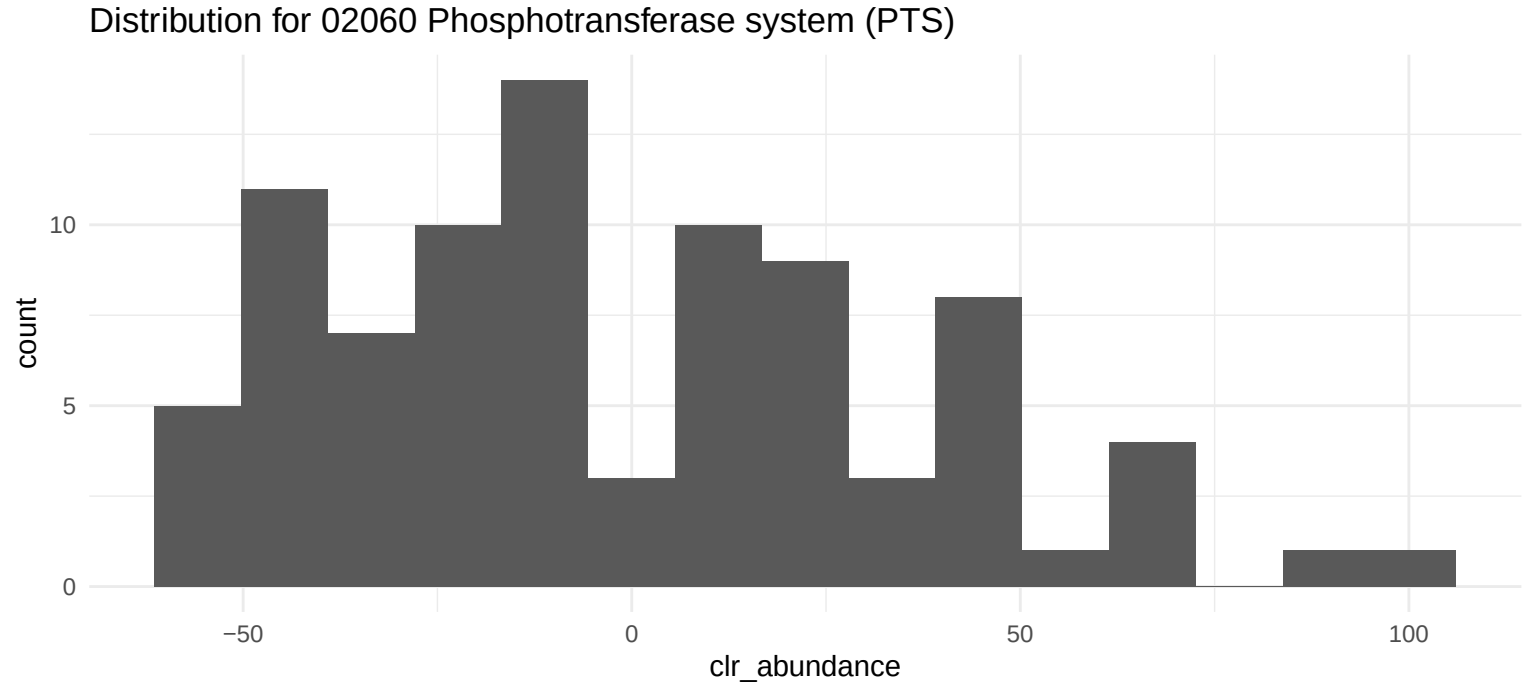

Distribution for 02035 Bacterial motility proteins
count
clr_abundance
15
10
5
0
−100
−50
0
50
100
150

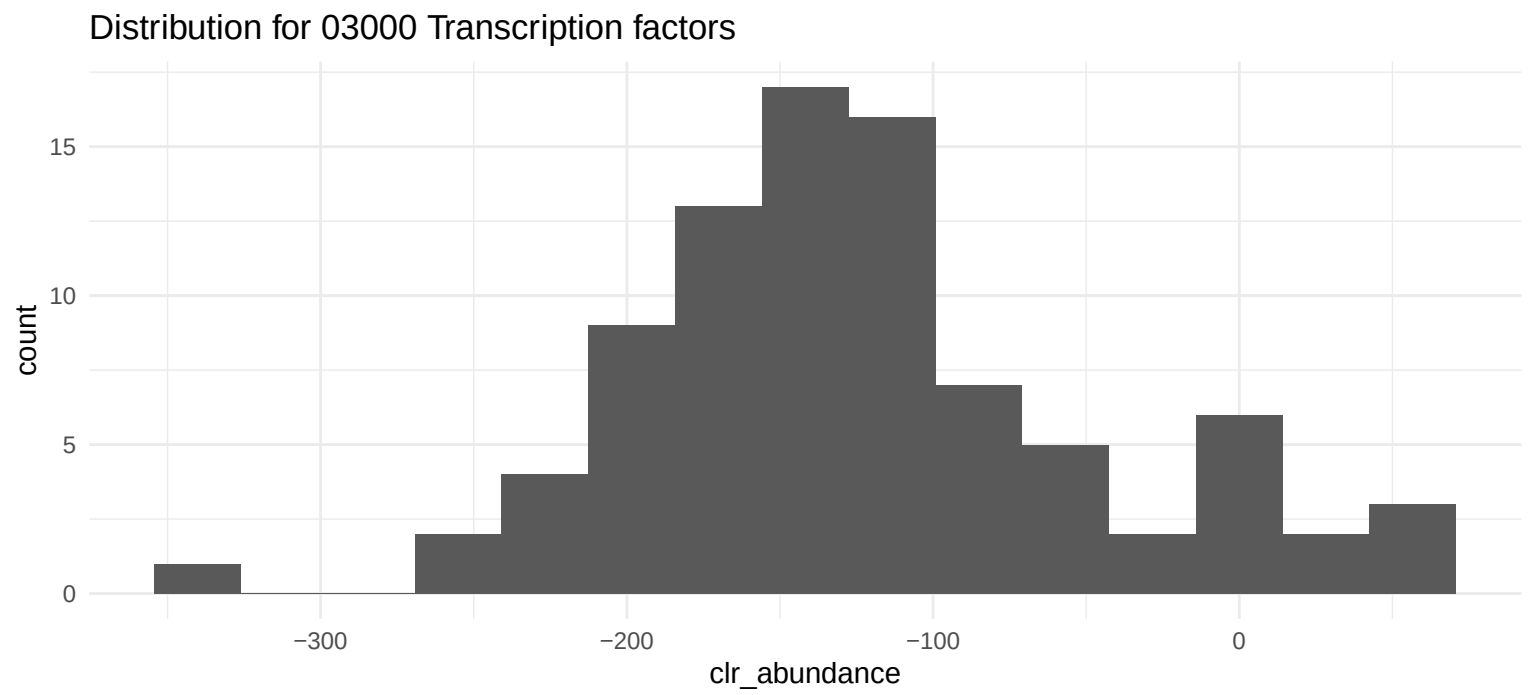

Distribution for 02040 Flagellar assembly
count
clr_abundance
10
5
0
0
50
100
150

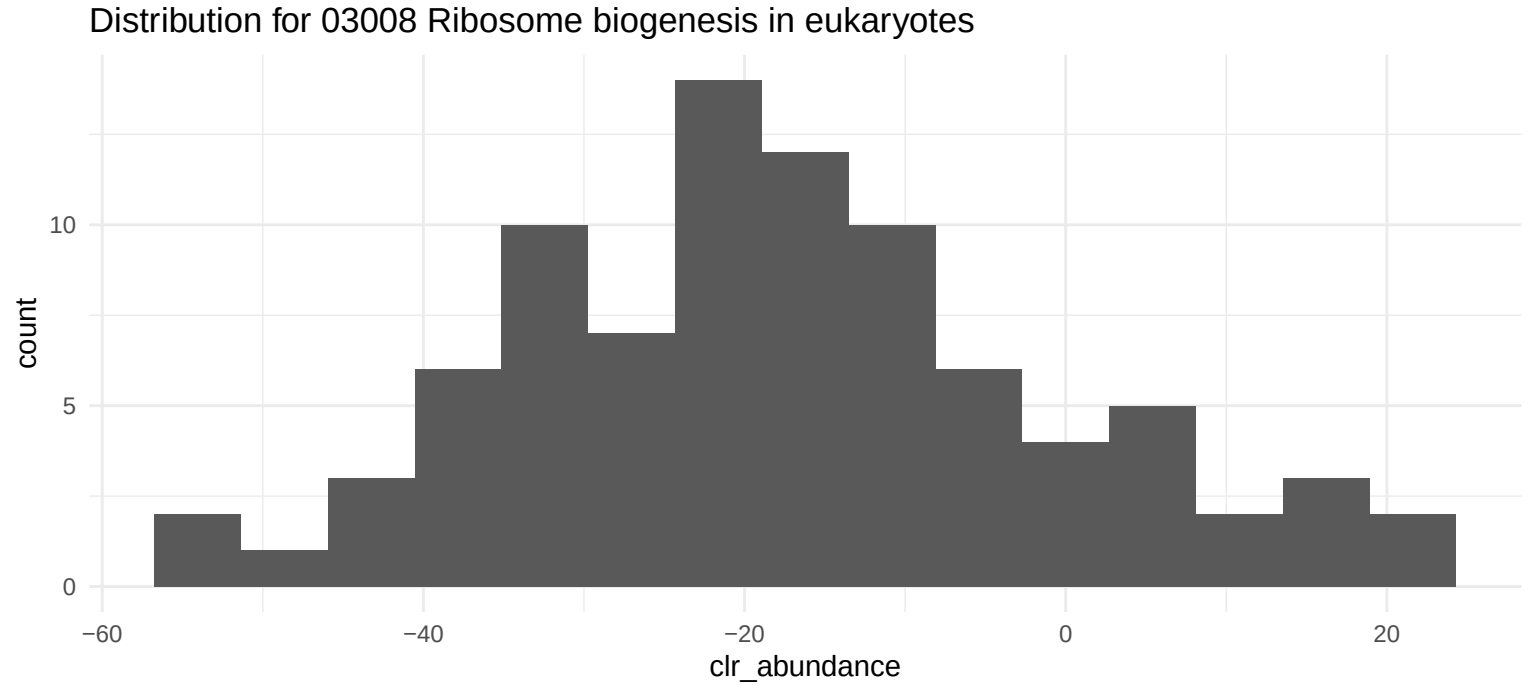

Distribution for 02042 Bacterial toxins
count
clr_abundance
15
10
5
0
−30
−20
−10
0
10

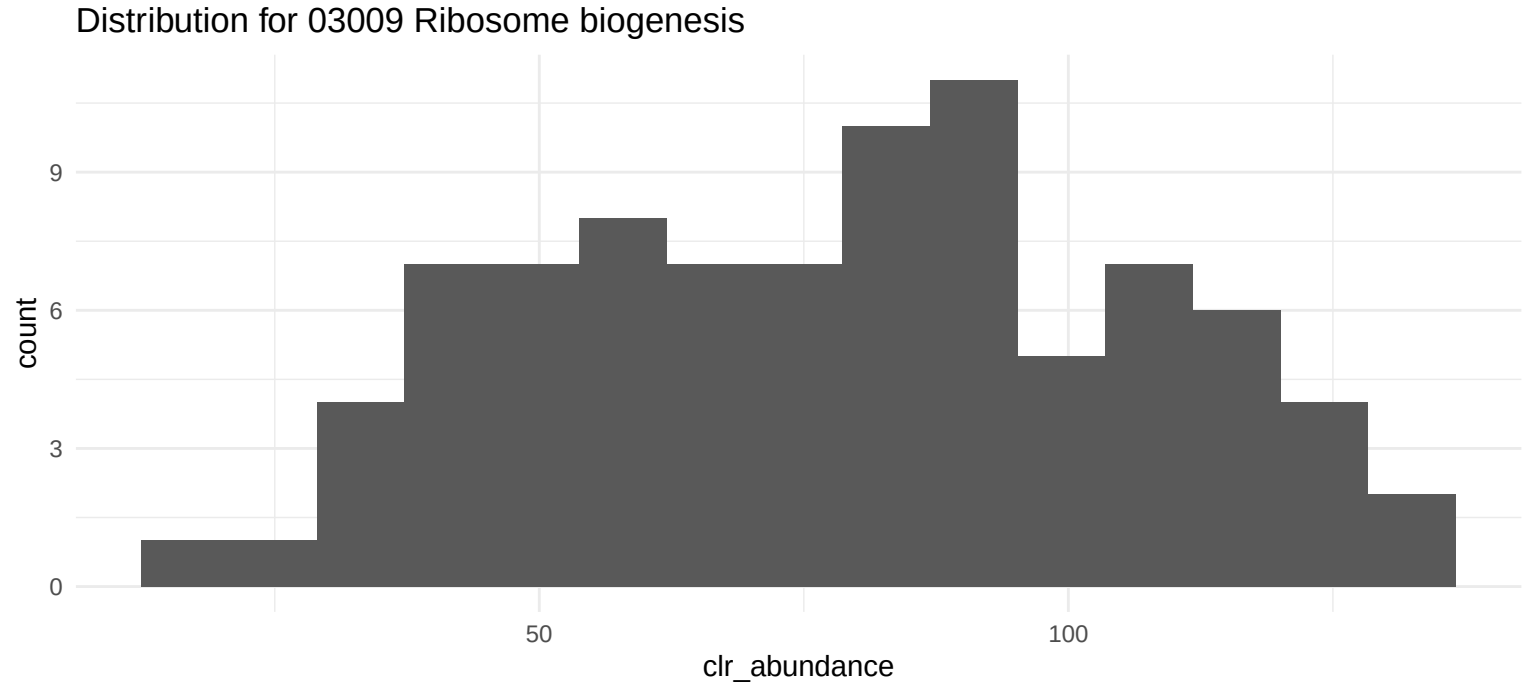

Distribution for 02044 Secretion system
count
clr_abundance
0
3
6
9
−300
−200
−100
0

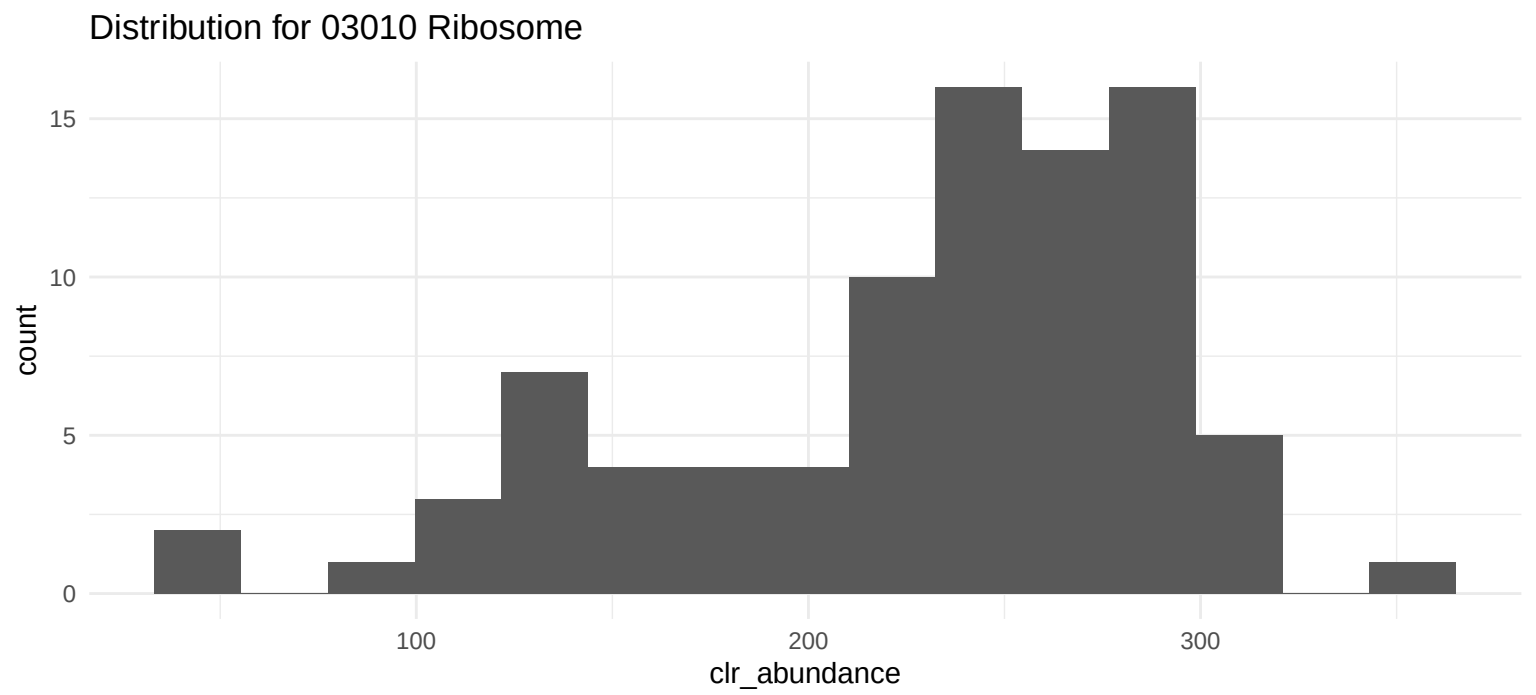

Distribution for 02048 Prokaryotic defense system
count
clr_abundance
0
10
20
−100
−50
0
50

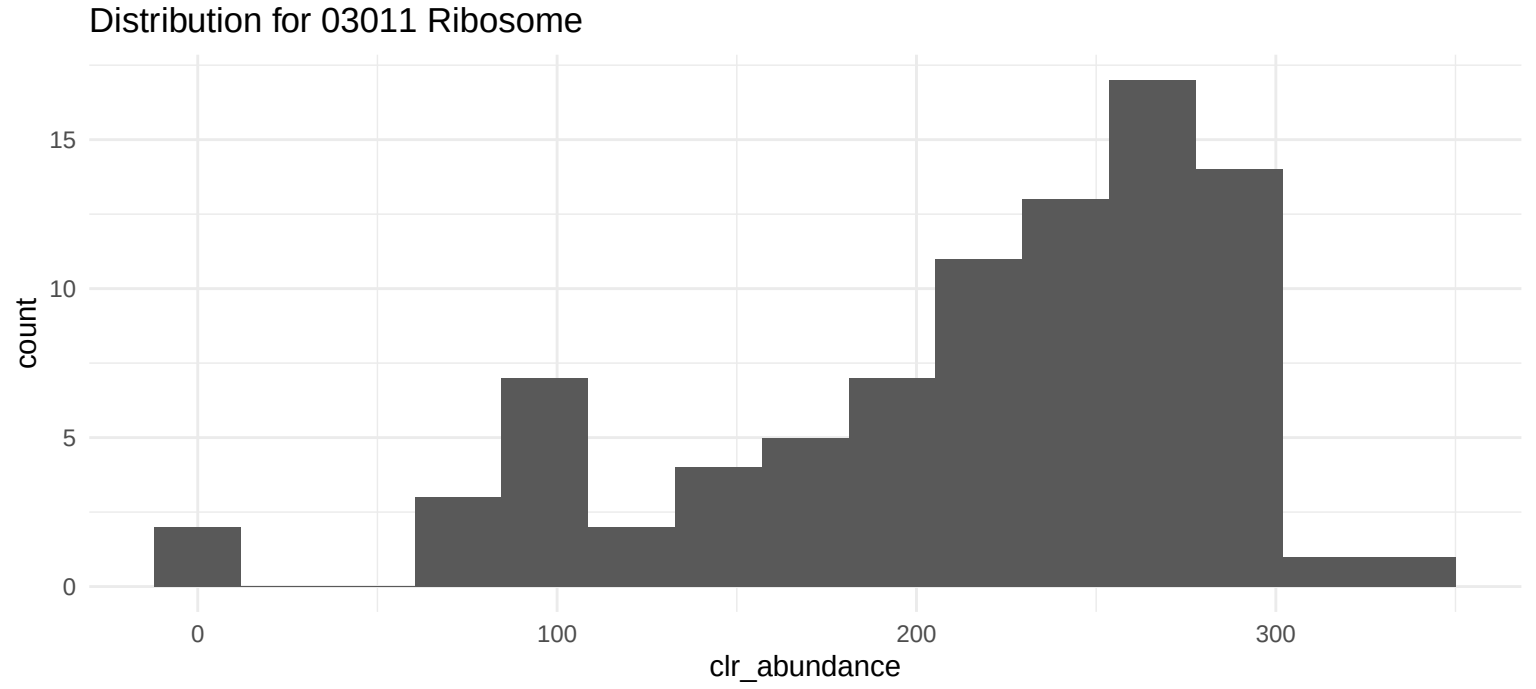

Distribution for 02060 Phosphotransferase system (PTS)
count
clr_abundance
0
5
10
−50
0
50
100

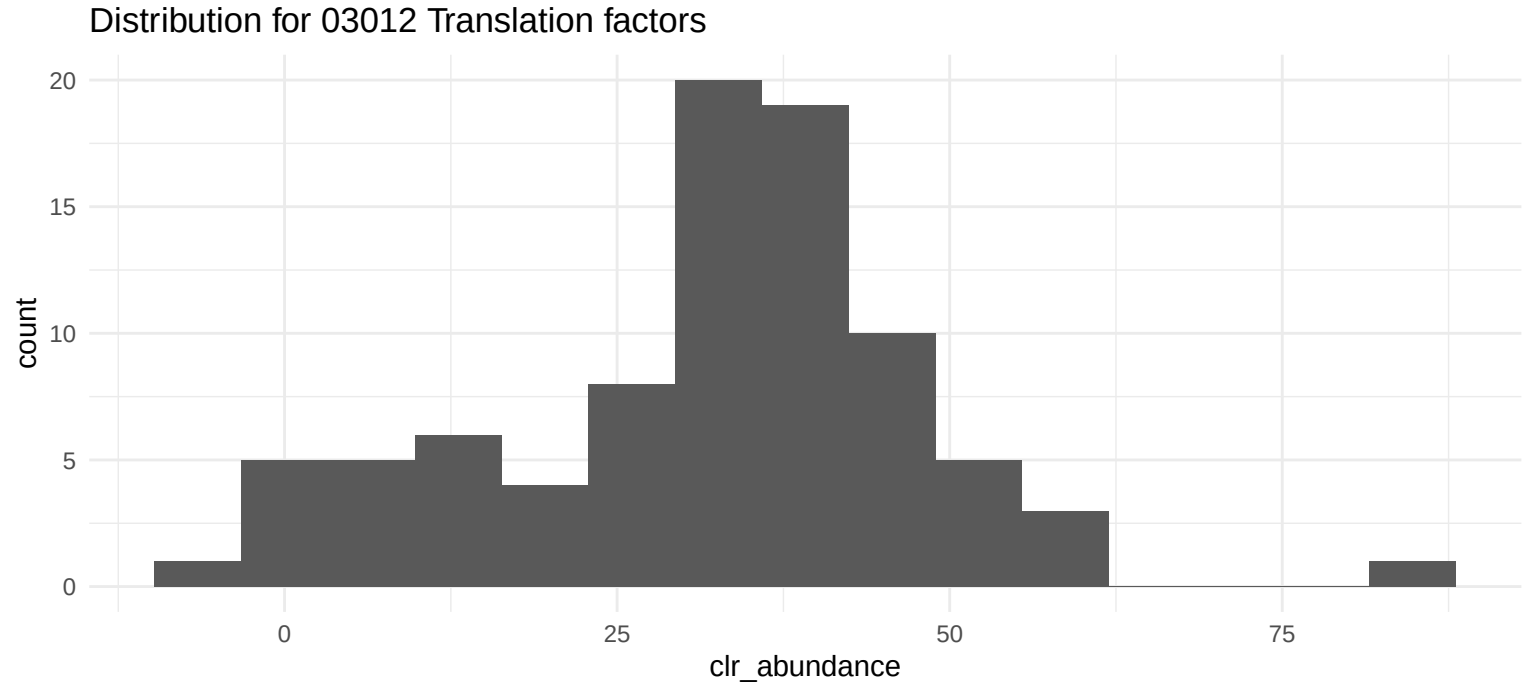

Distribution for 03000 Transcription factors
count
15
10
5
0
−300
−200
−100
0
clr_abundance

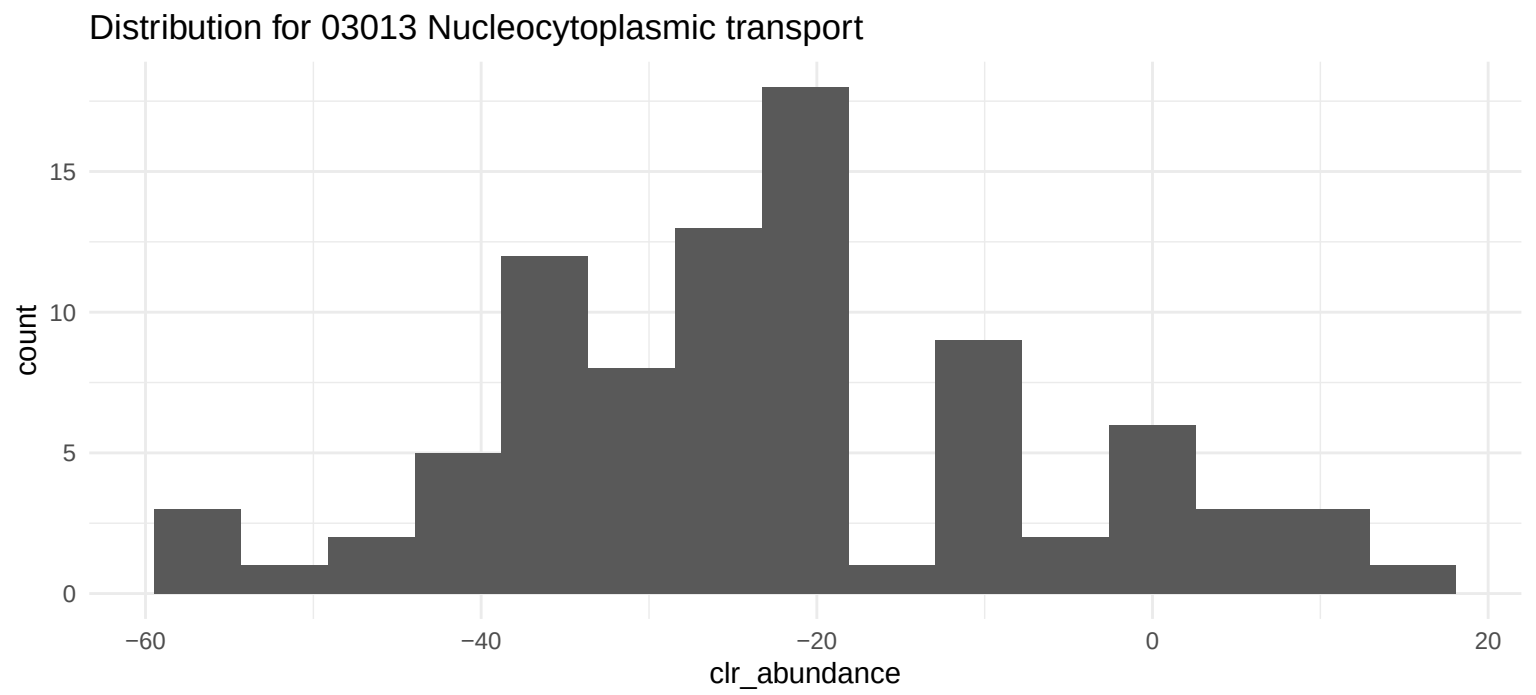

Distribution for 03008 Ribosome biogenesis in eukaryotes
count
10
5
0
−60
−40
−20
0
20
clr_abundance

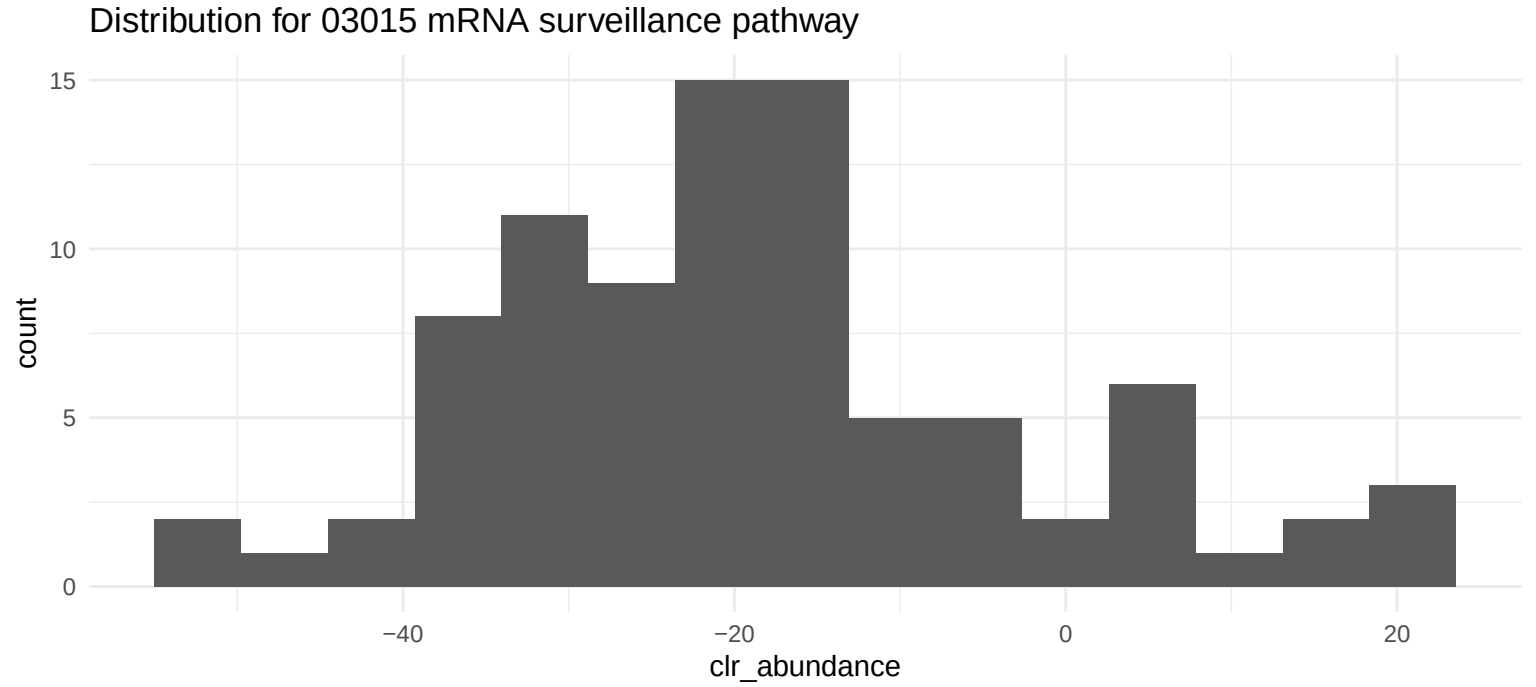

Distribution for 03009 Ribosome biogenesis
count
9
6
3
0
50
100
clr_abundance

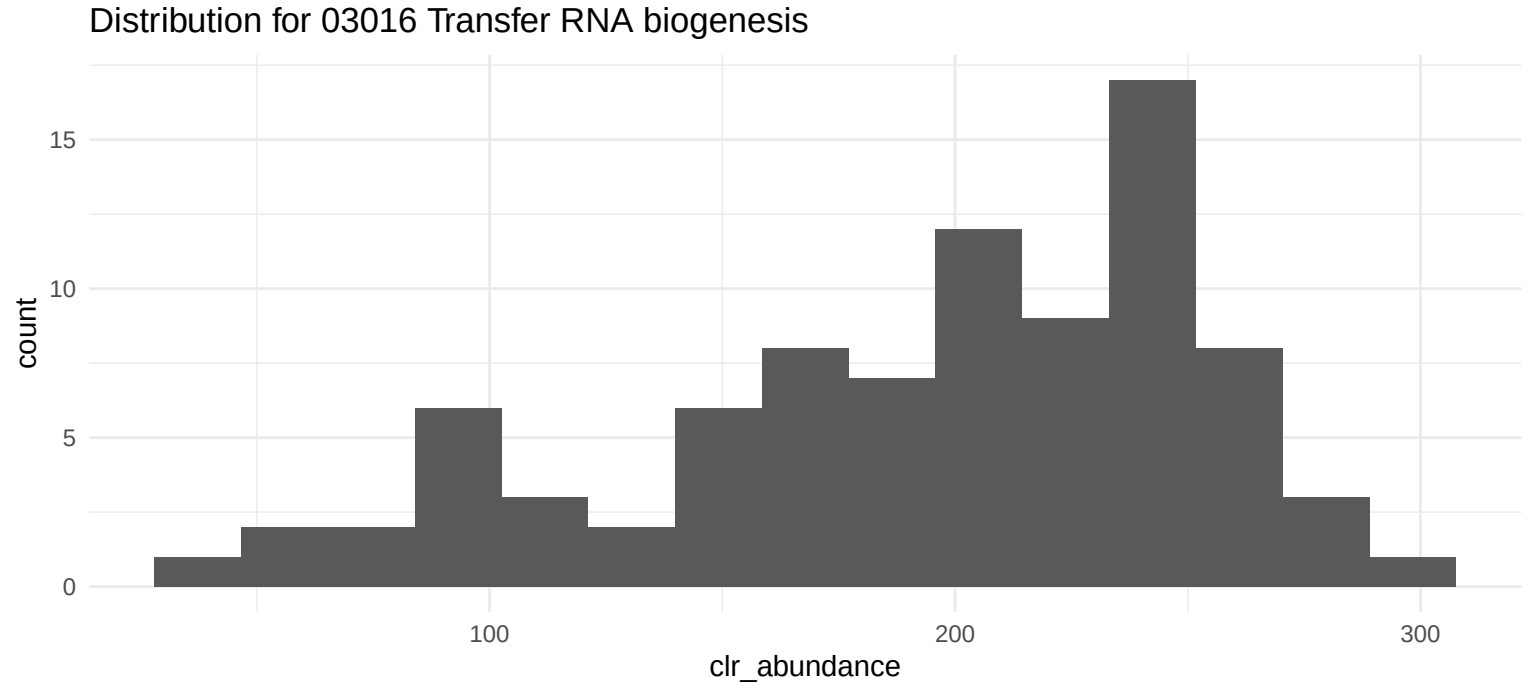

Distribution for 03010 Ribosome
count
clr_abundance
0
5
10
15
100
200
300

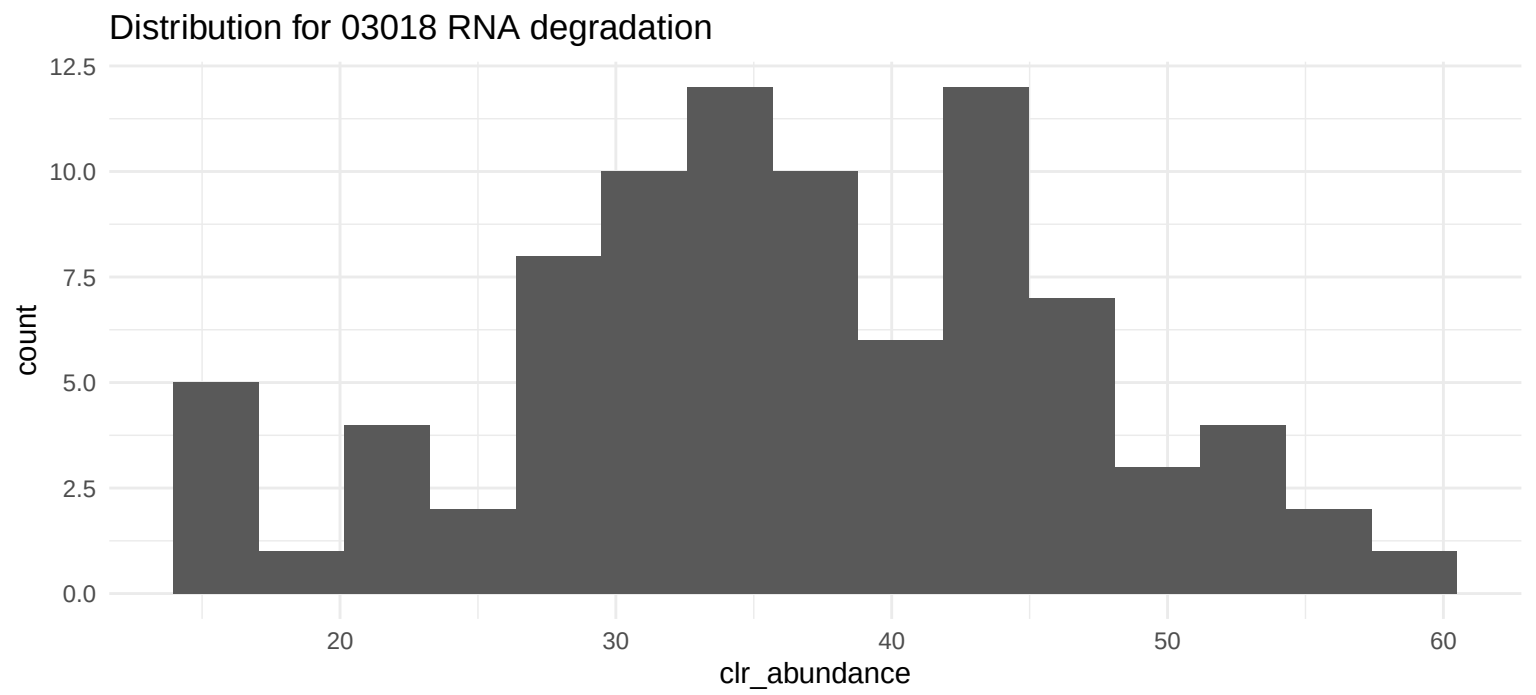

Distribution for 03011 Ribosome
count
clr_abundance
0
5
10
15
0
100
200
300

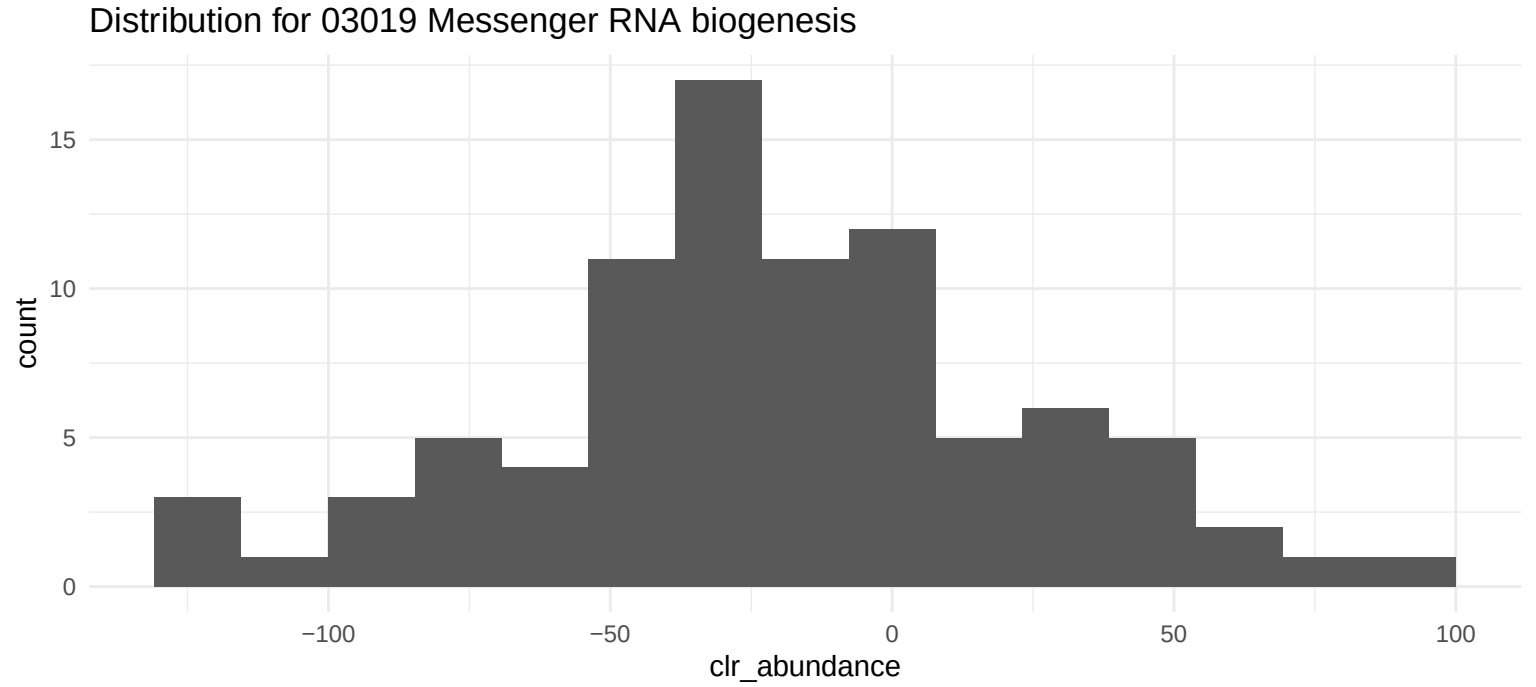

Distribution for 03012 Translation factors
count
clr_abundance
0
5
10
15
20
0
25
50
75

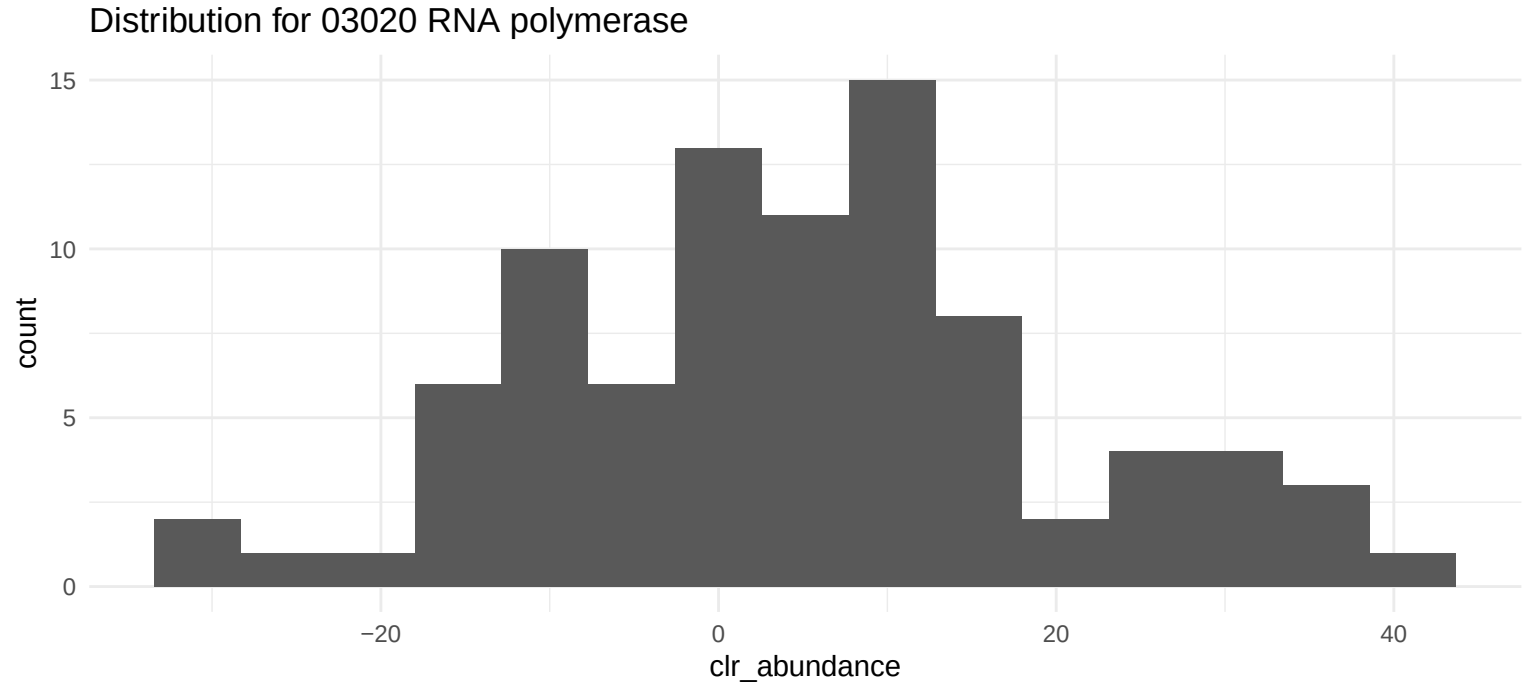

Distribution for 03013 Nucleocytoplasmic transport
count
clr_abundance
0
5
10
15
−60
−40
−20
0
20


Distribution for 03015 mRNA surveillance pathway
count
clr_abundance
0
5
10
15
−40
−20
0
20

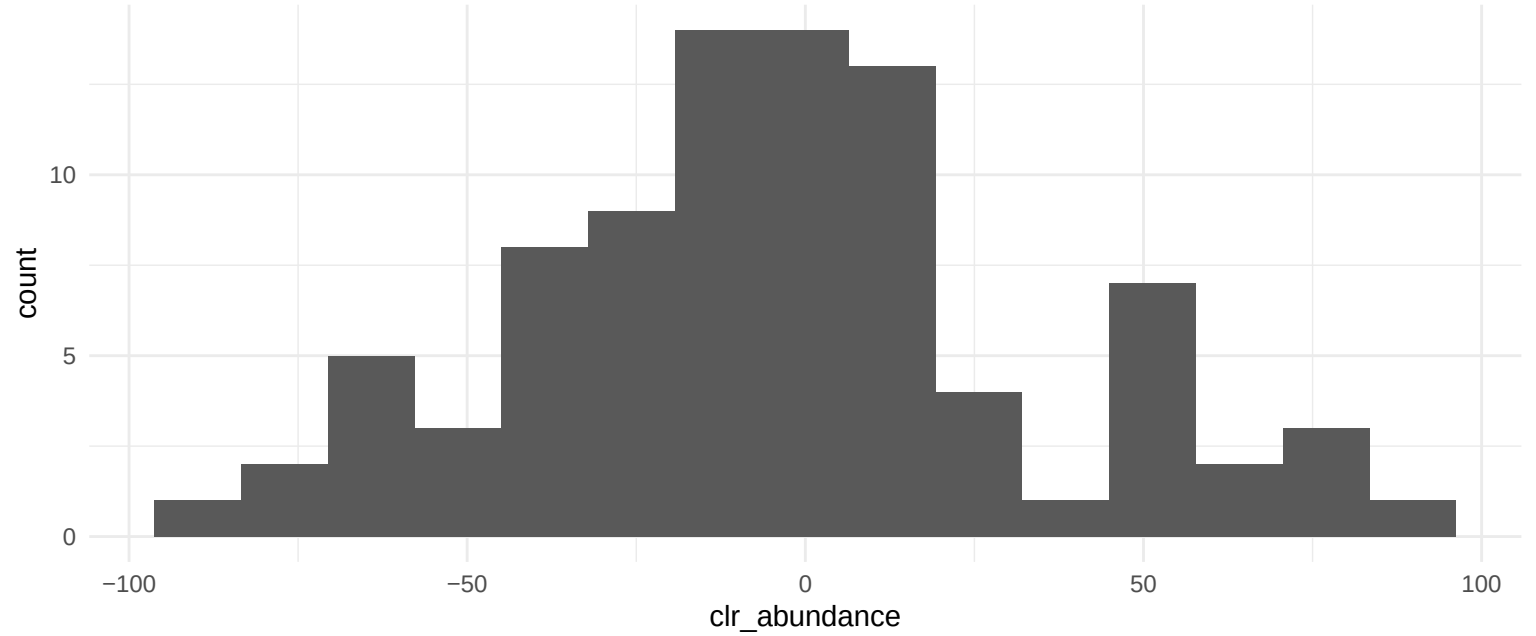

Distribution for 03016 Transfer RNA biogenesis
count
clr_abundance
0
5
10
15
100
200
300


Distribution for 03018 RNA degradation
12.5
10.0
7.5
5.0
2.5
0.0
count
20
30
40
50
60
clr_abundance

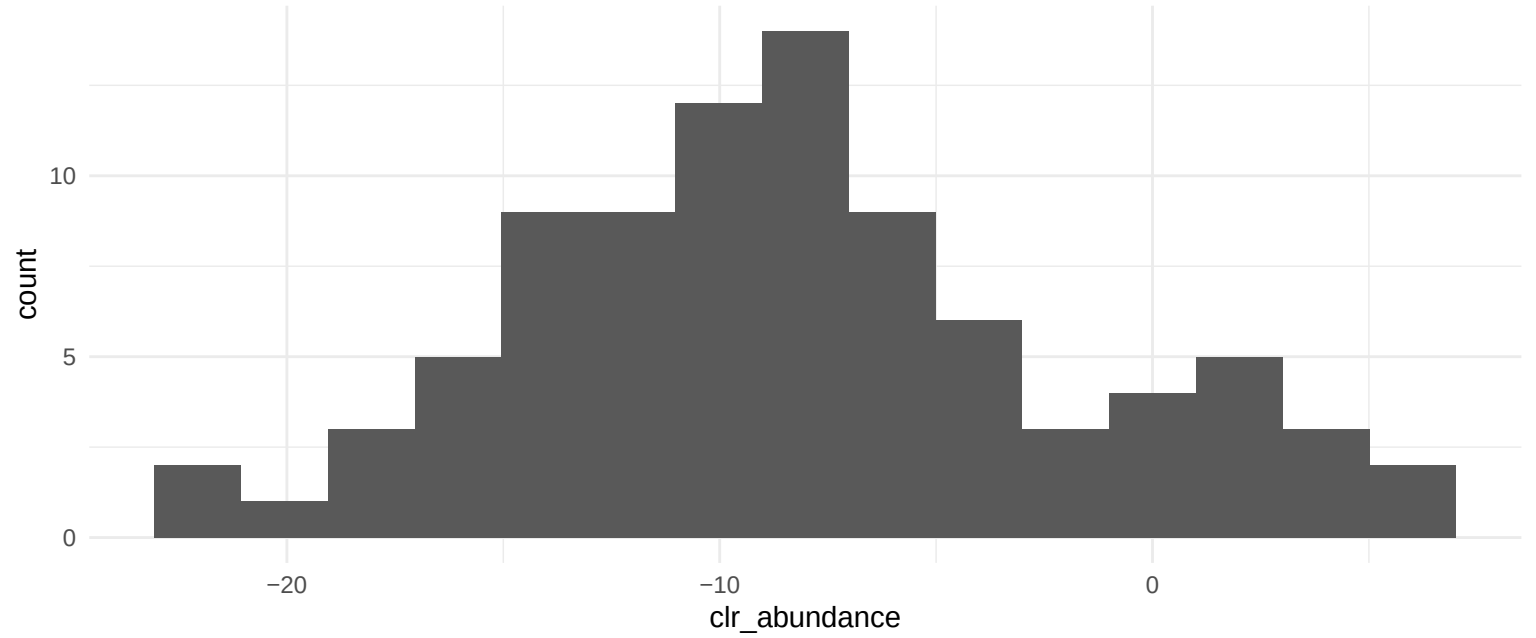

Distribution for 03019 Messenger RNA biogenesis
15
10
5
0
count
−100
−50
0
50
100
clr_abundance


Distribution for 03020 RNA polymerase
15
10
5
0
count
−20
0
20
40
clr_abundance

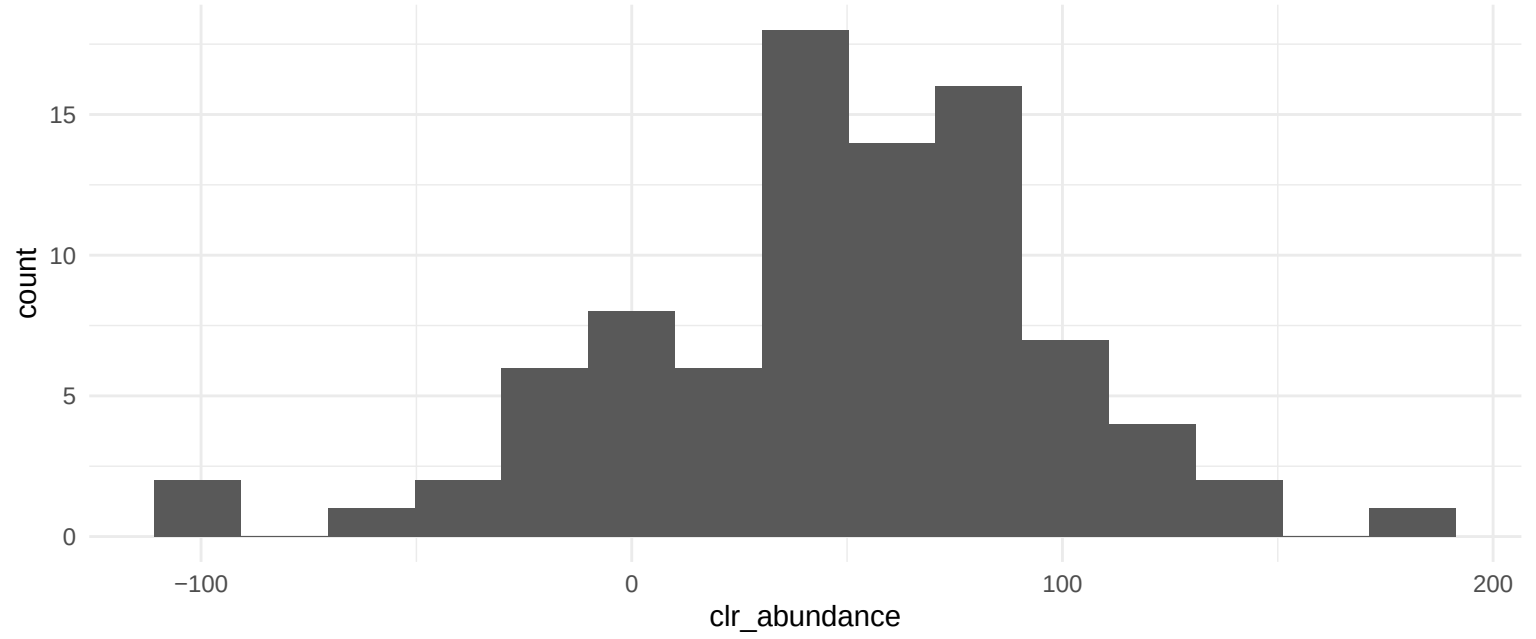
Distribution for 03021 Transcription machinery



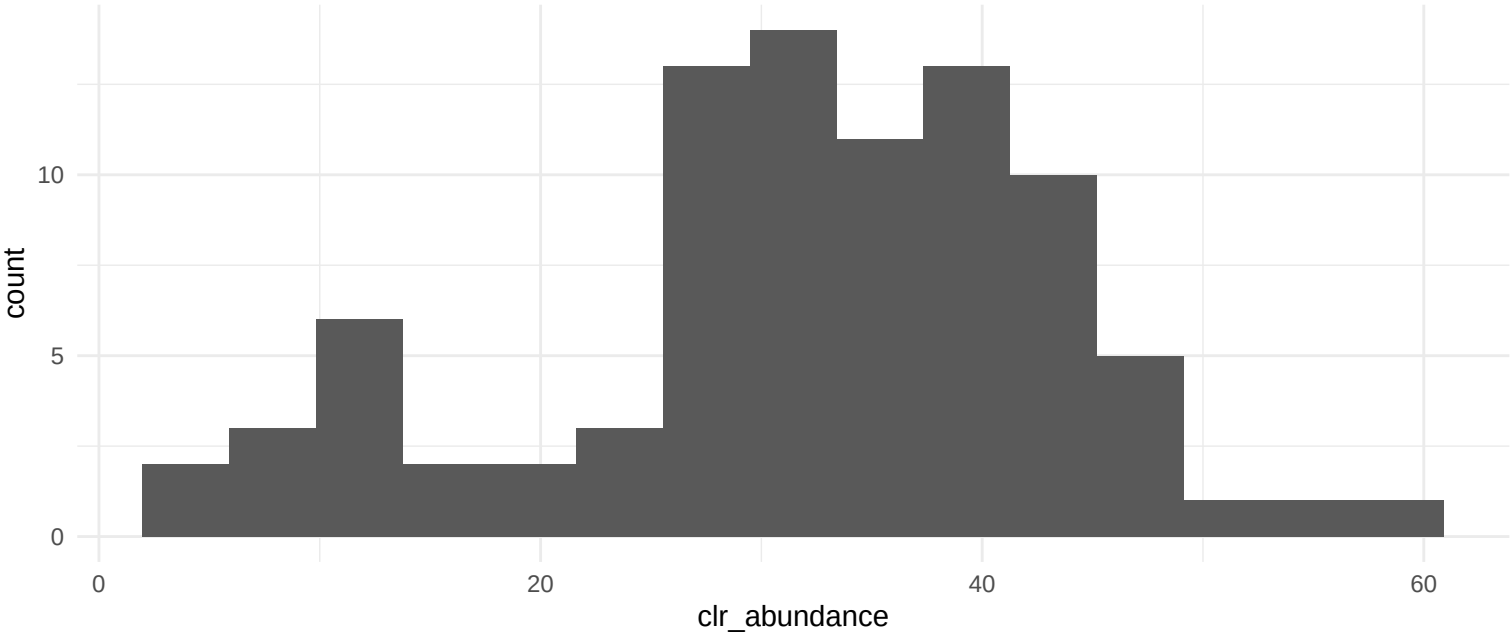
Distribution for 03022 Basal transcription factors



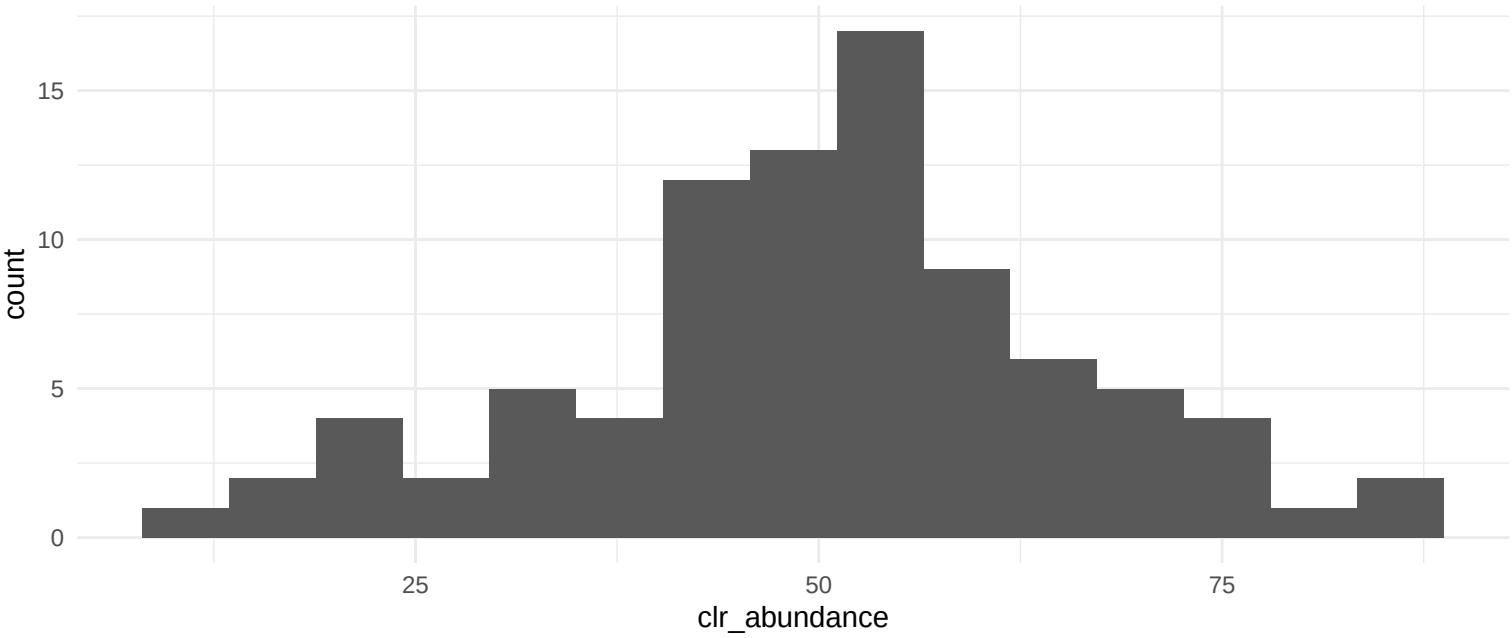
Distribution for 03029 Mitochondrial biogenesis



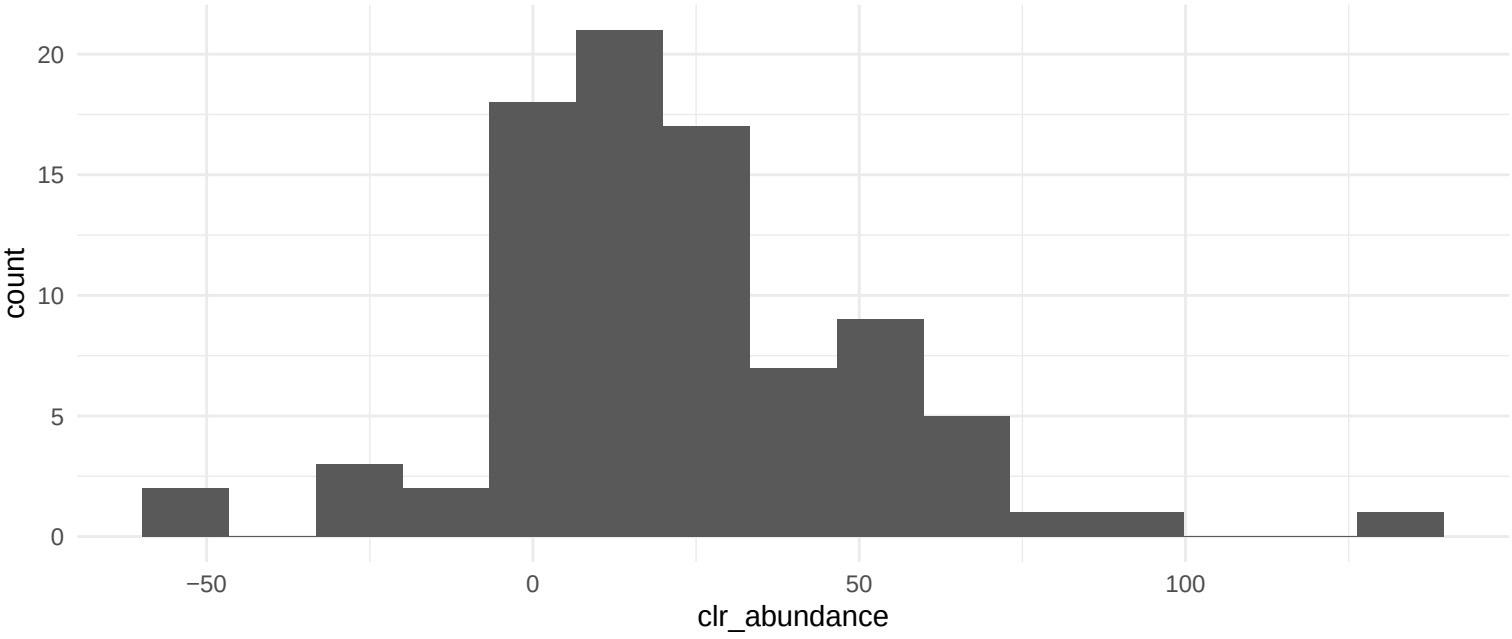
Distribution for 03030 DNA replication

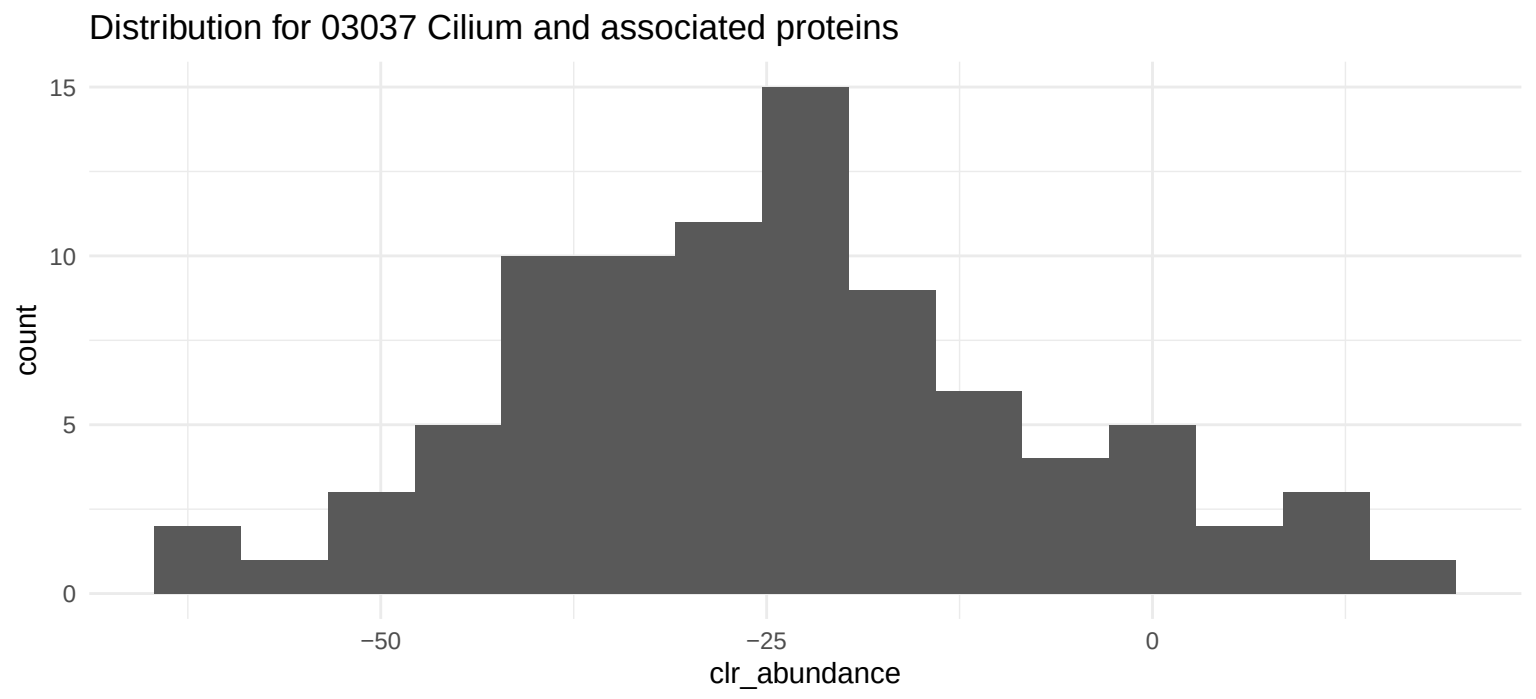


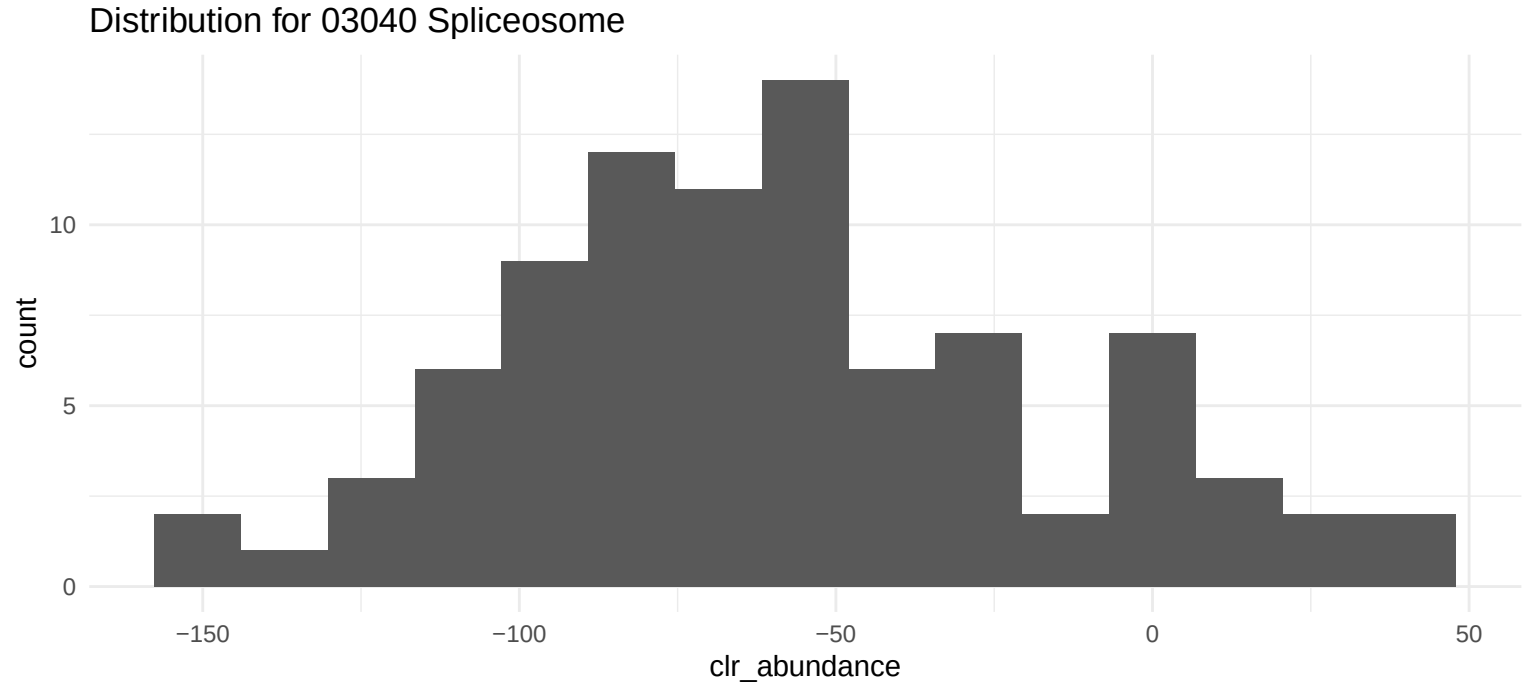
Distribution for 03032 DNA replication proteins

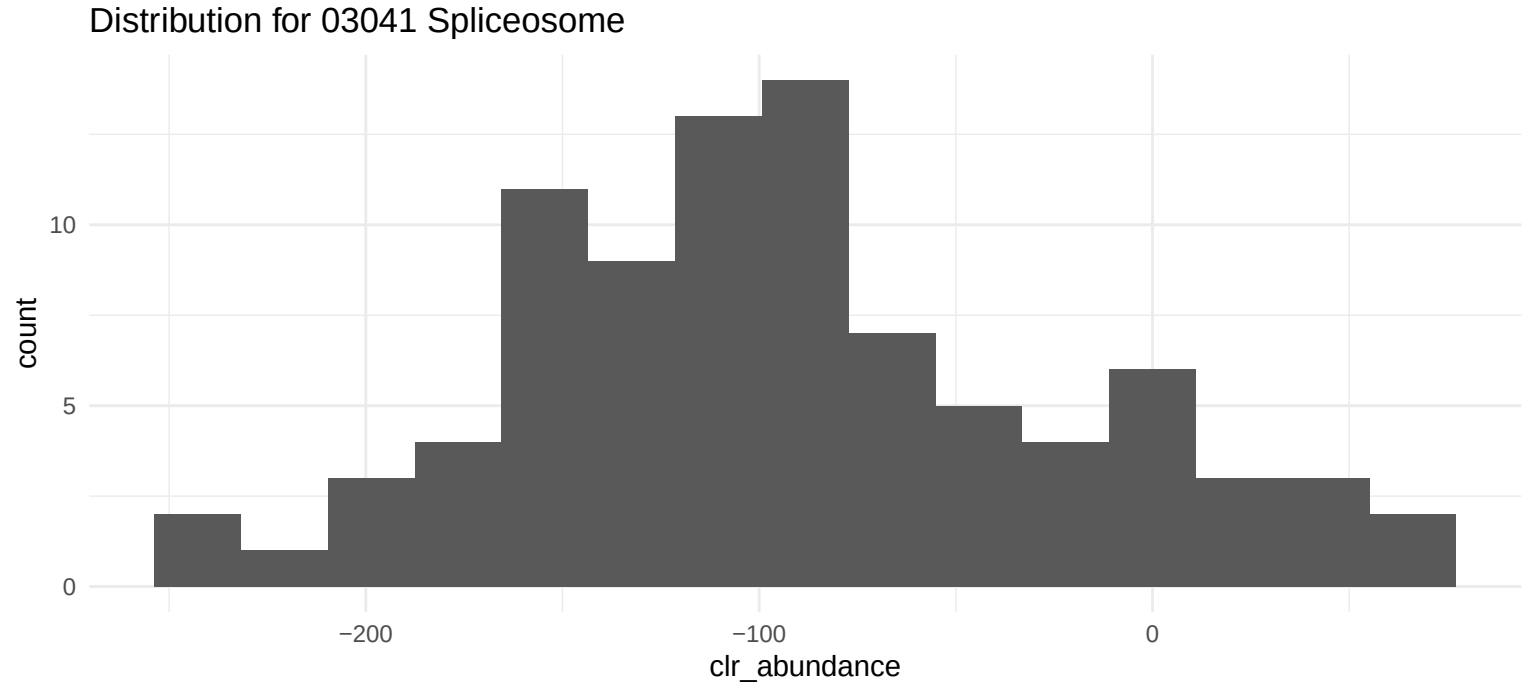


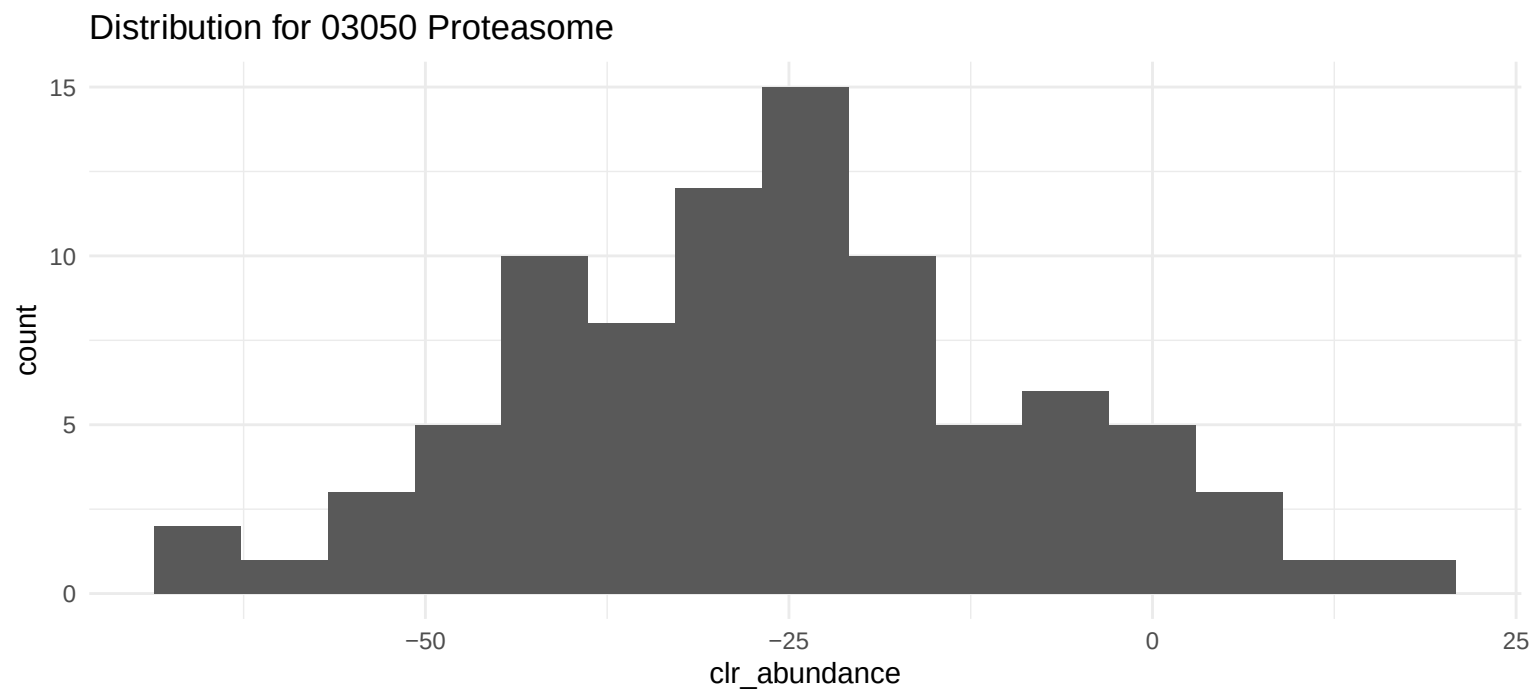
Distribution for 03036 Chromosome and associated proteins

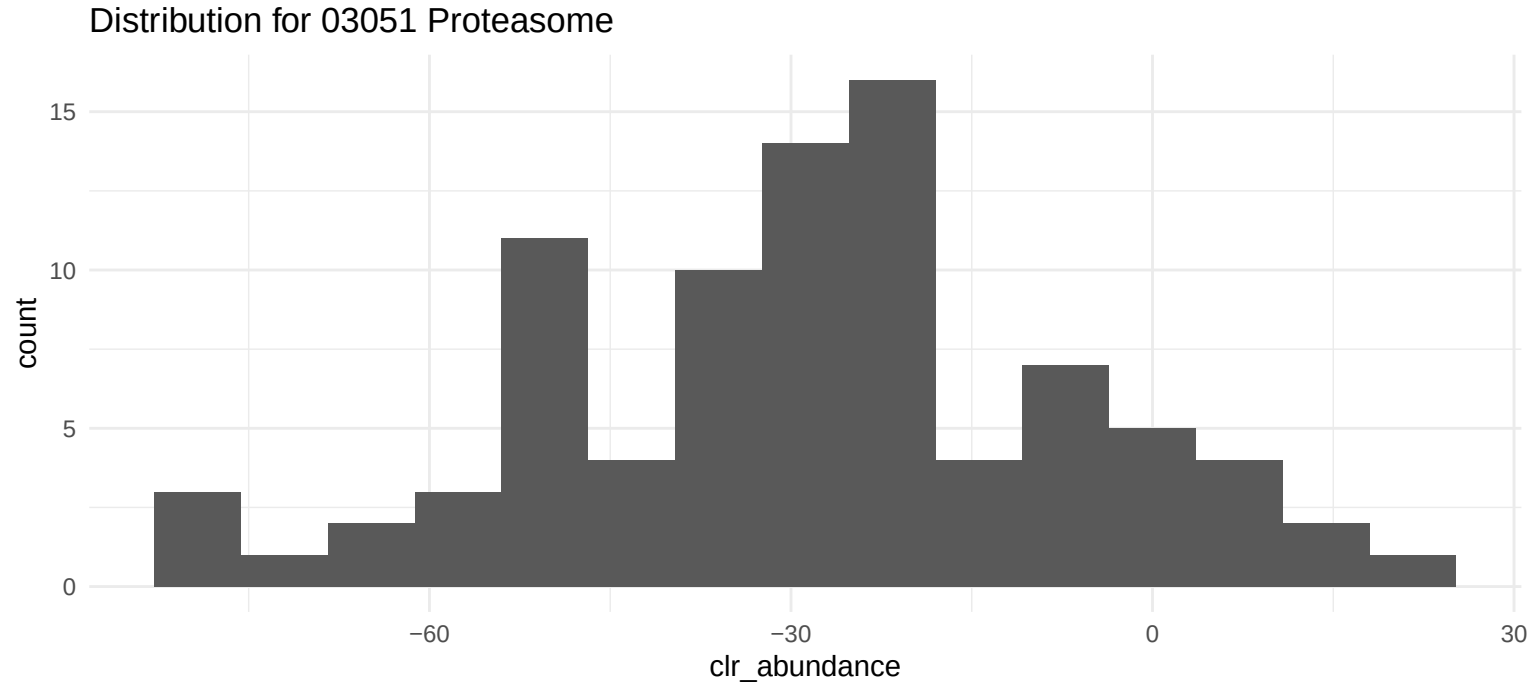


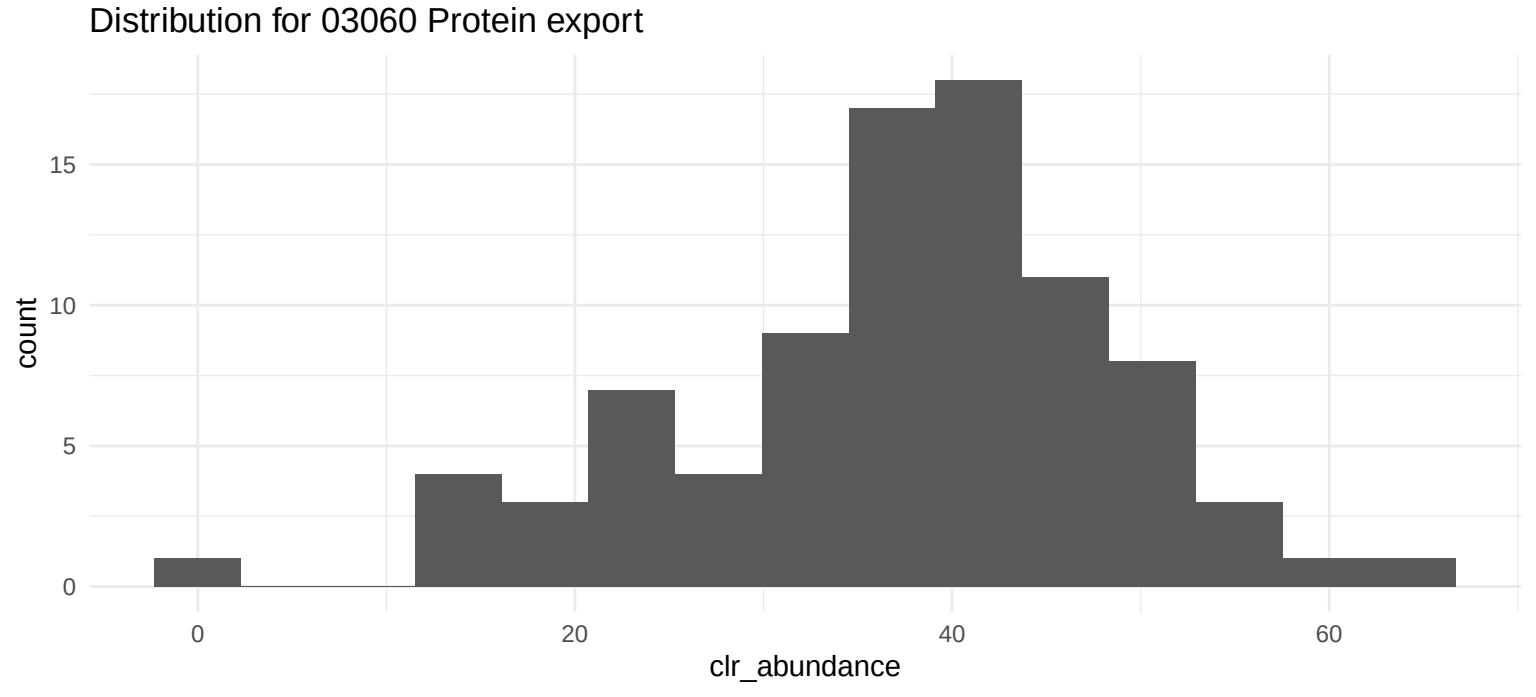

Distribution for 03037 Cilium and associated proteins
count
15
10
5
0
−50
−25
0
clr_abundance


Distribution for 03040 Spliceosome
count
10
5
0
−150
−100
−50
0
50
clr_abundance

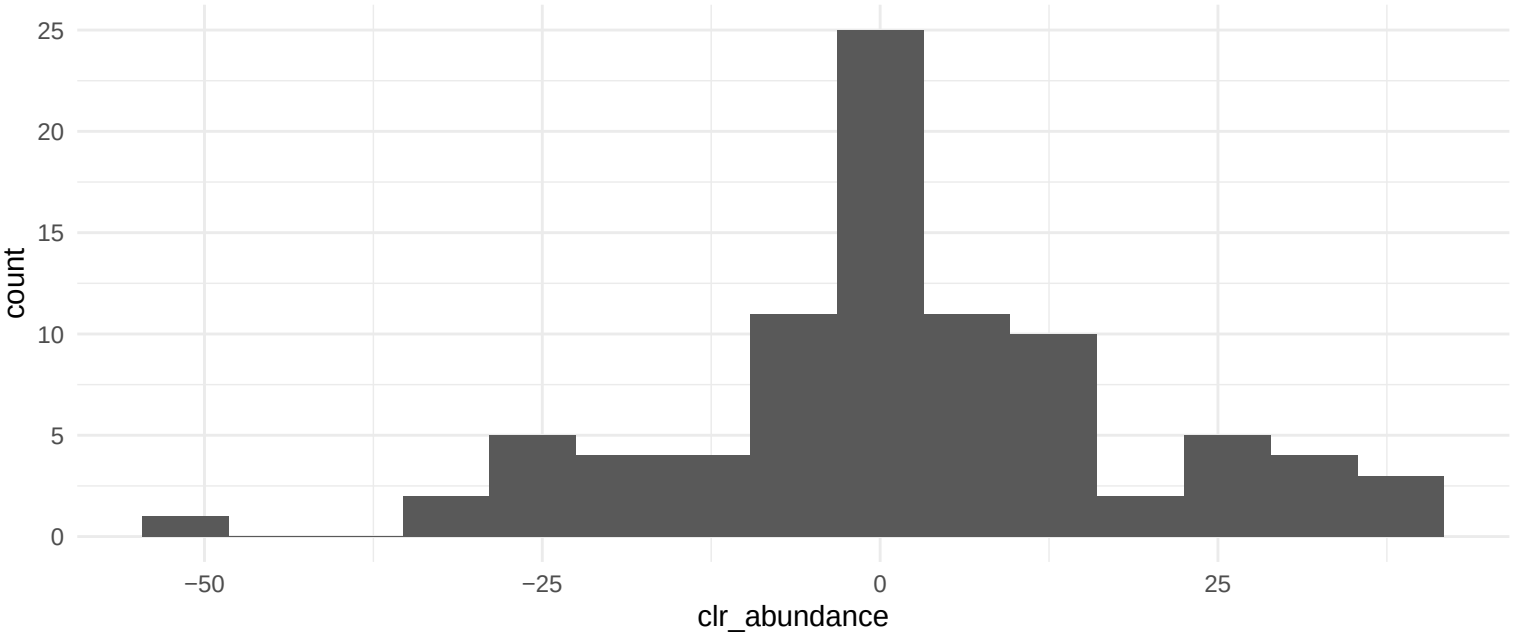

Distribution for 03041 Spliceosome
count
10
5
0
−200
−100
0
clr_abundance


Distribution for 03050 Proteasome
count
clr_abundance
15
10
5
0
−50
−25
0
25

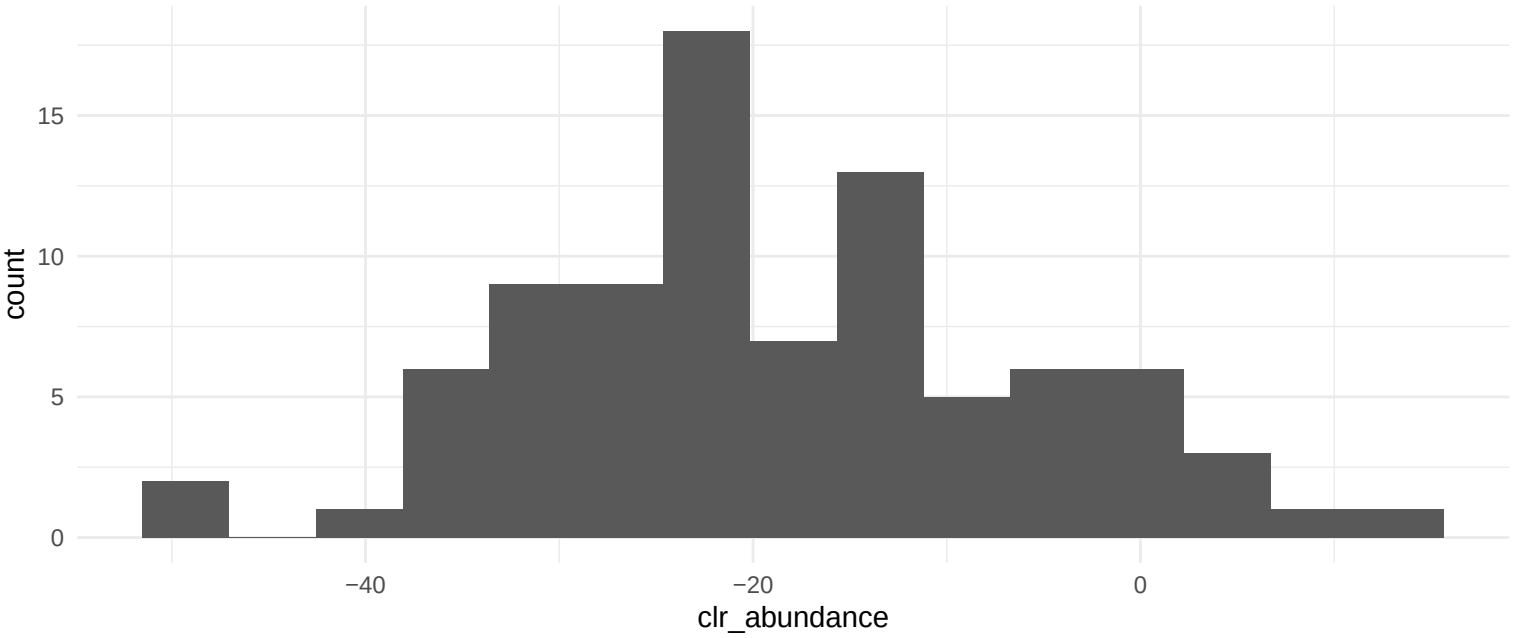

Distribution for 03051 Proteasome
count
clr_abundance
15
10
5
0
−60
−30
0
30


Distribution for 03060 Protein export
count
clr_abundance
15
10
5
0
0
20
40
60

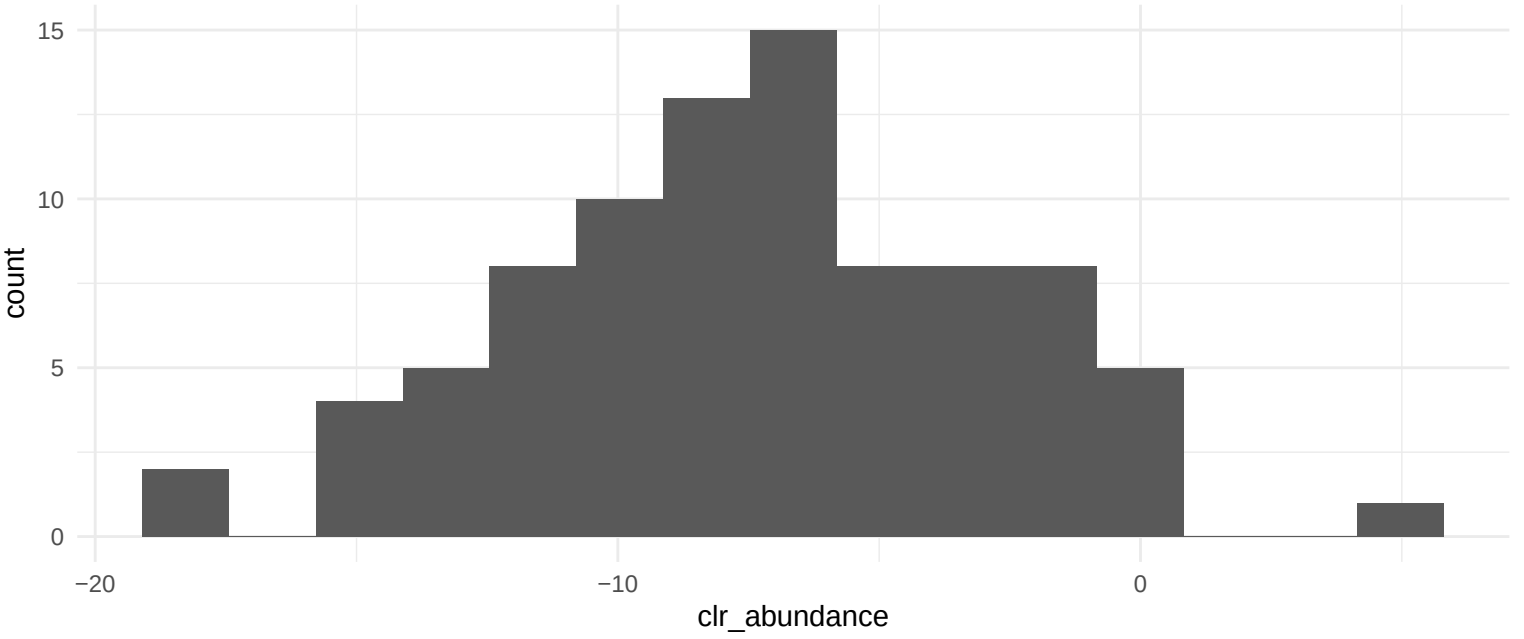
Distribution for 03070 Bacterial secretion system

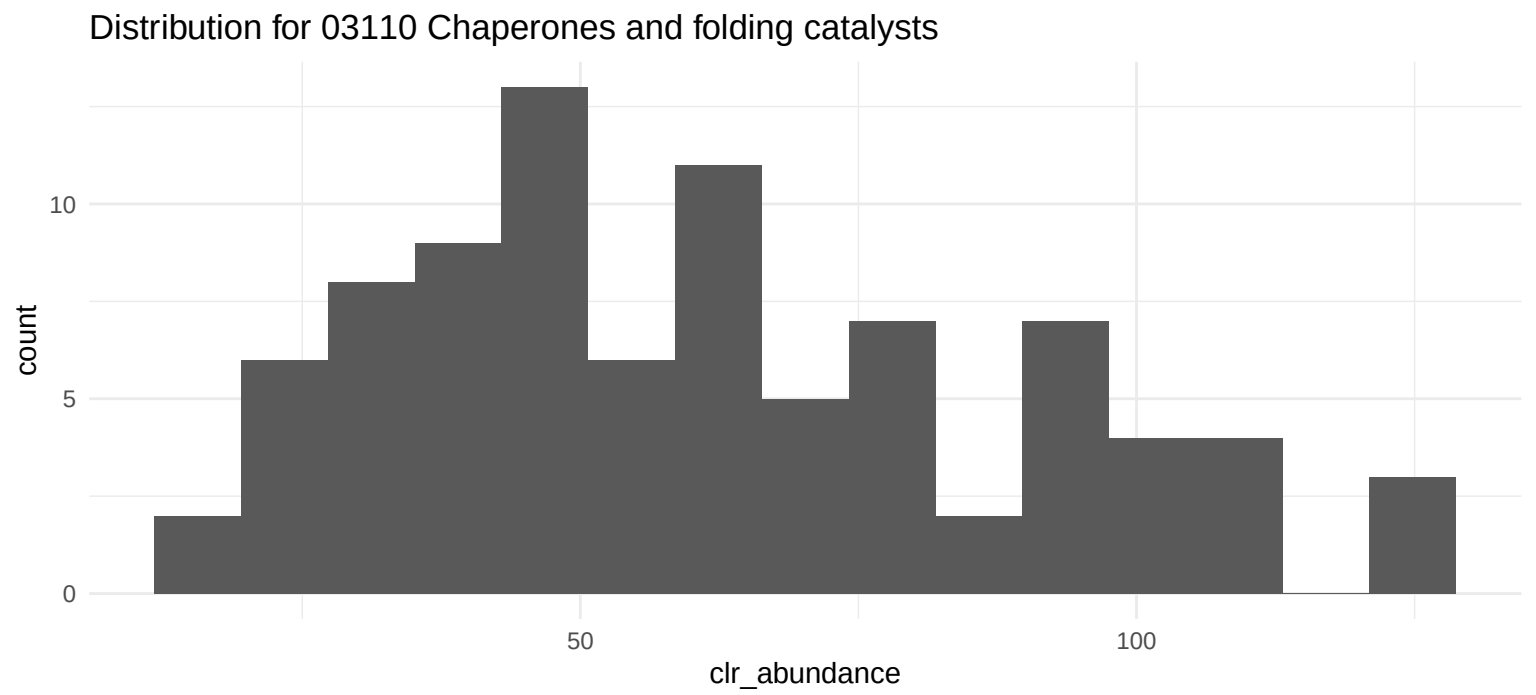


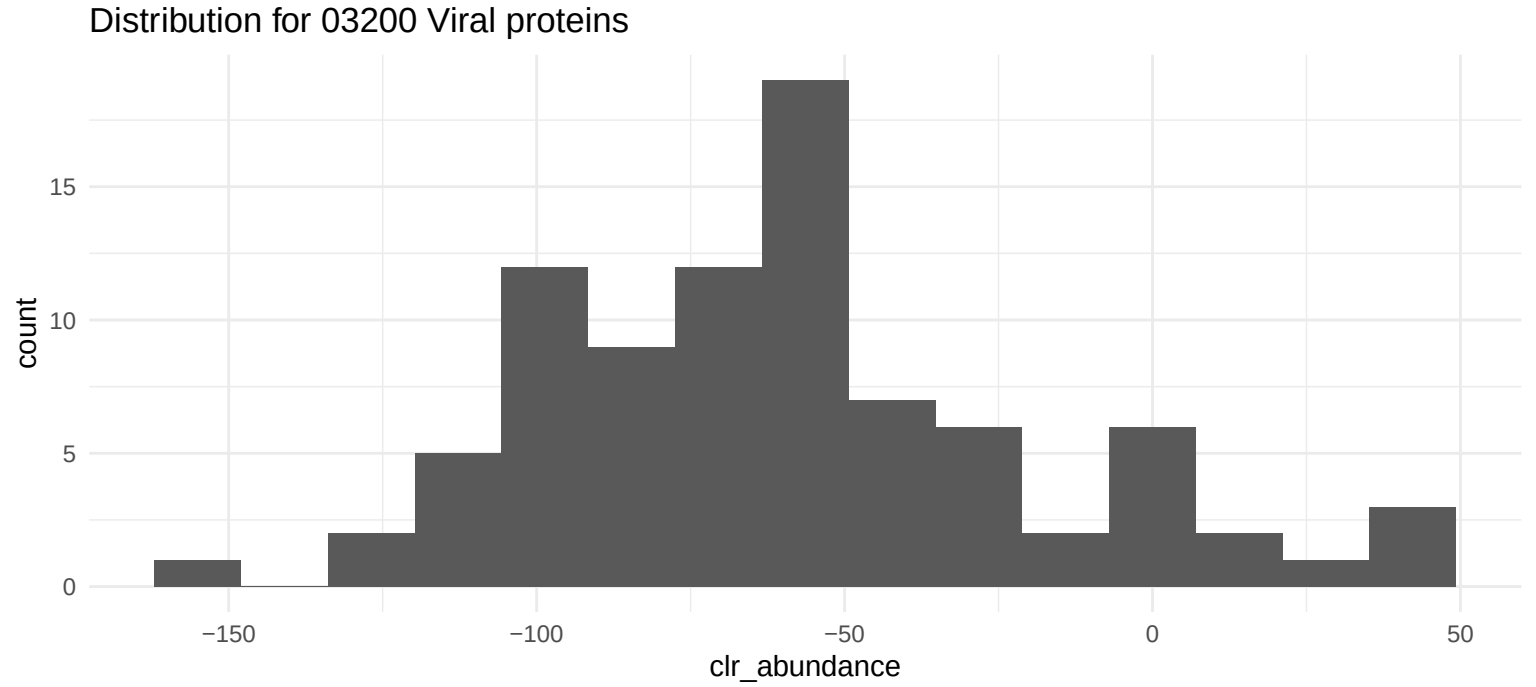
Distribution for 03082 ATP−dependent chromatin remodeling

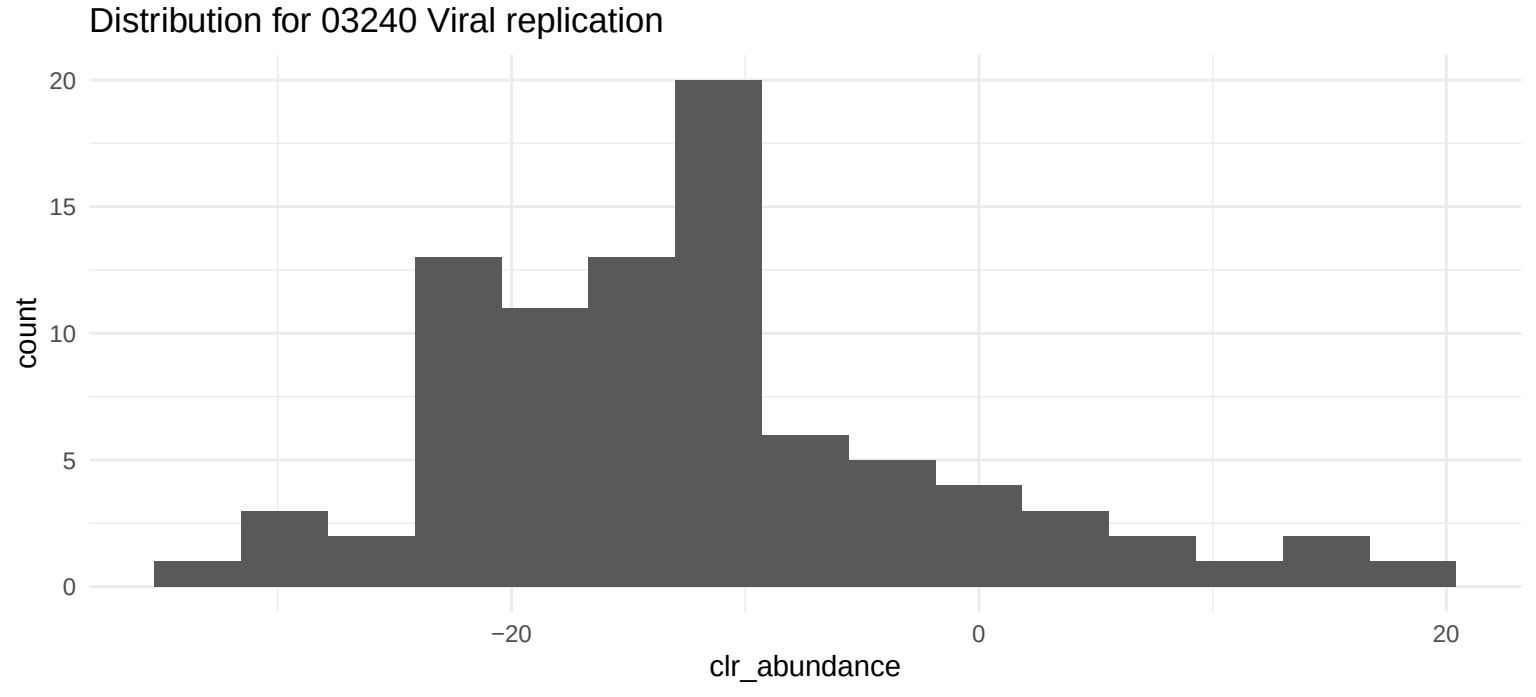


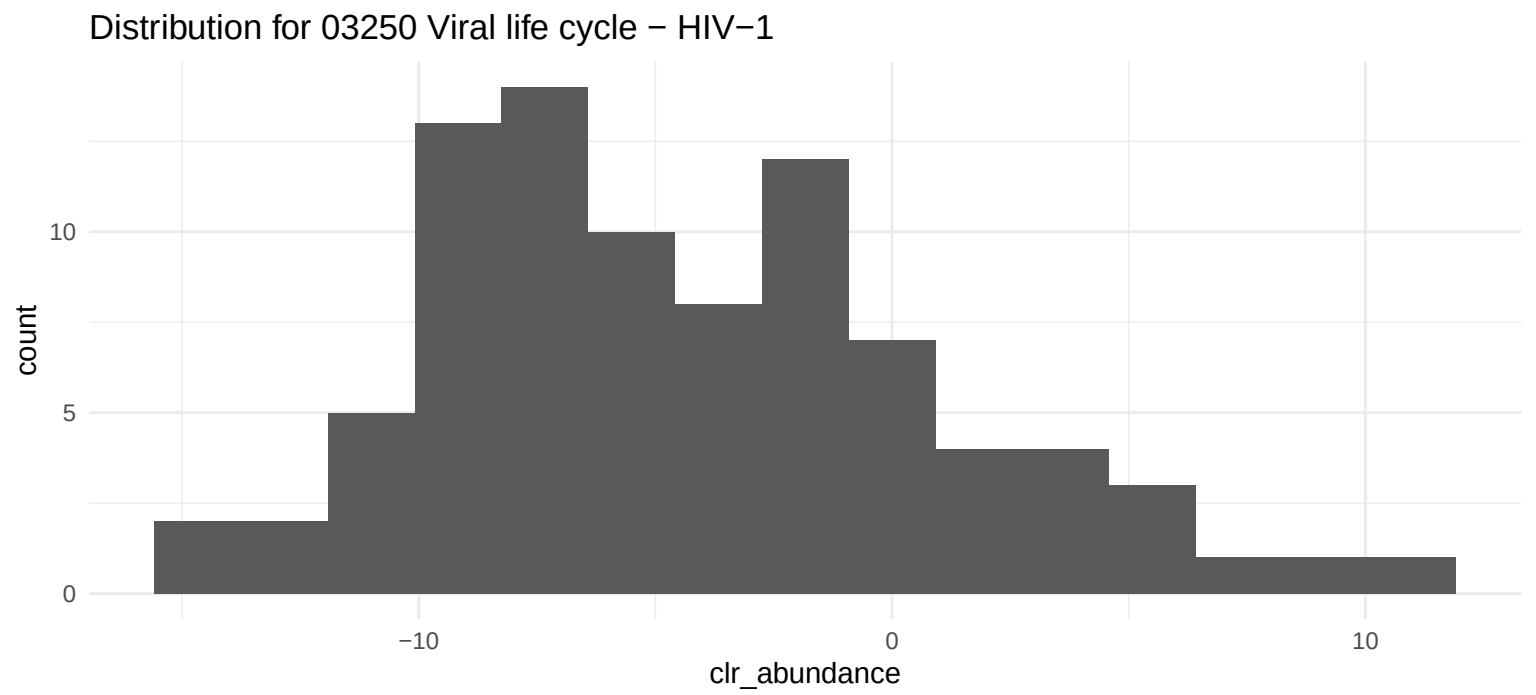
Distribution for 03083 Polycomb repressive complex

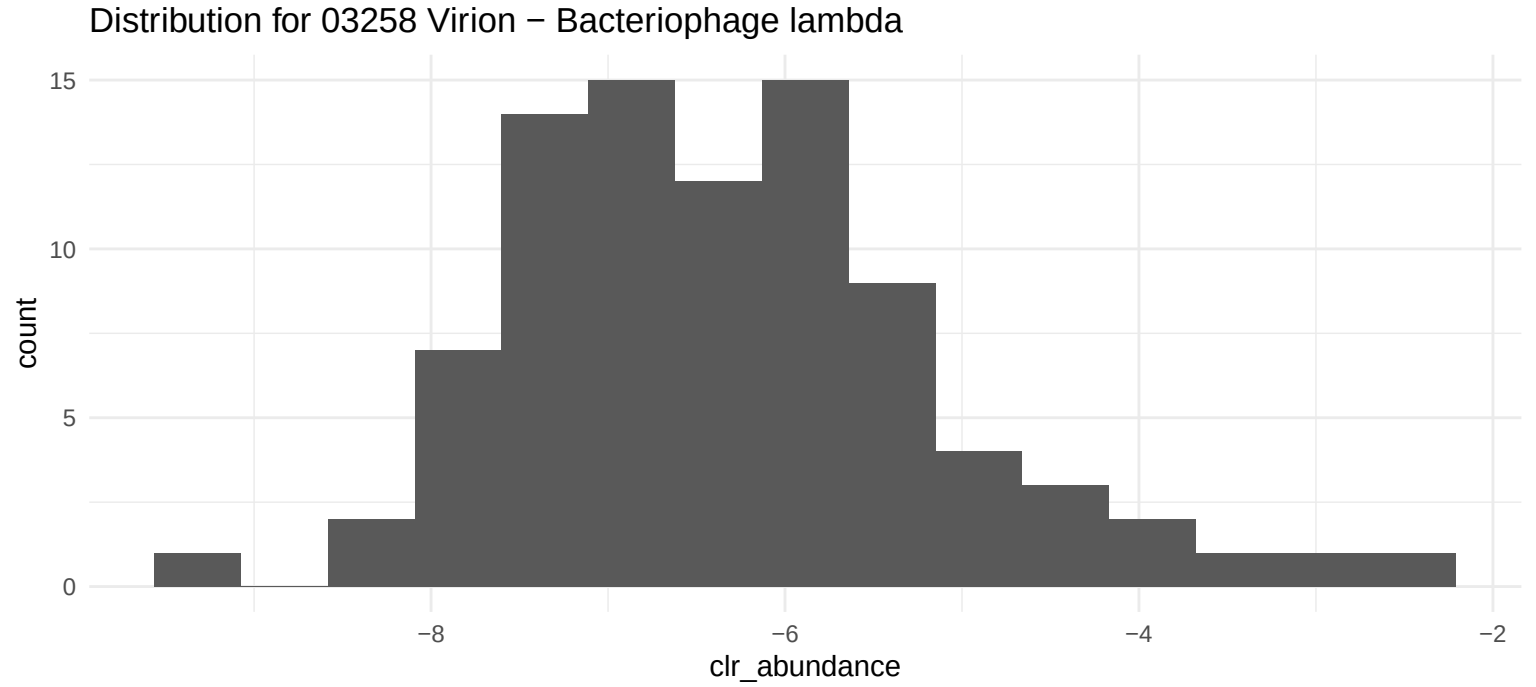


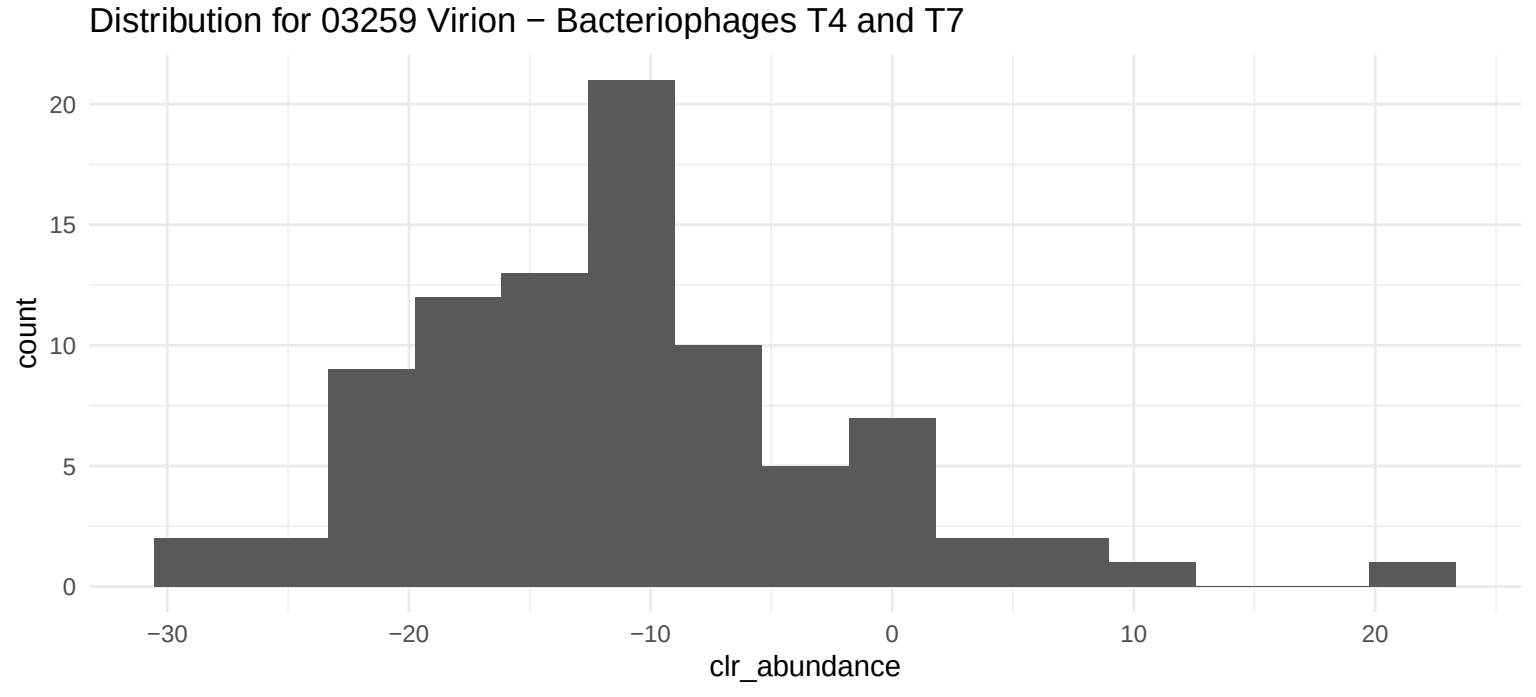

Distribution for 03110 Chaperones and folding catalysts
count
0
5
10
50
100
clr_abundance

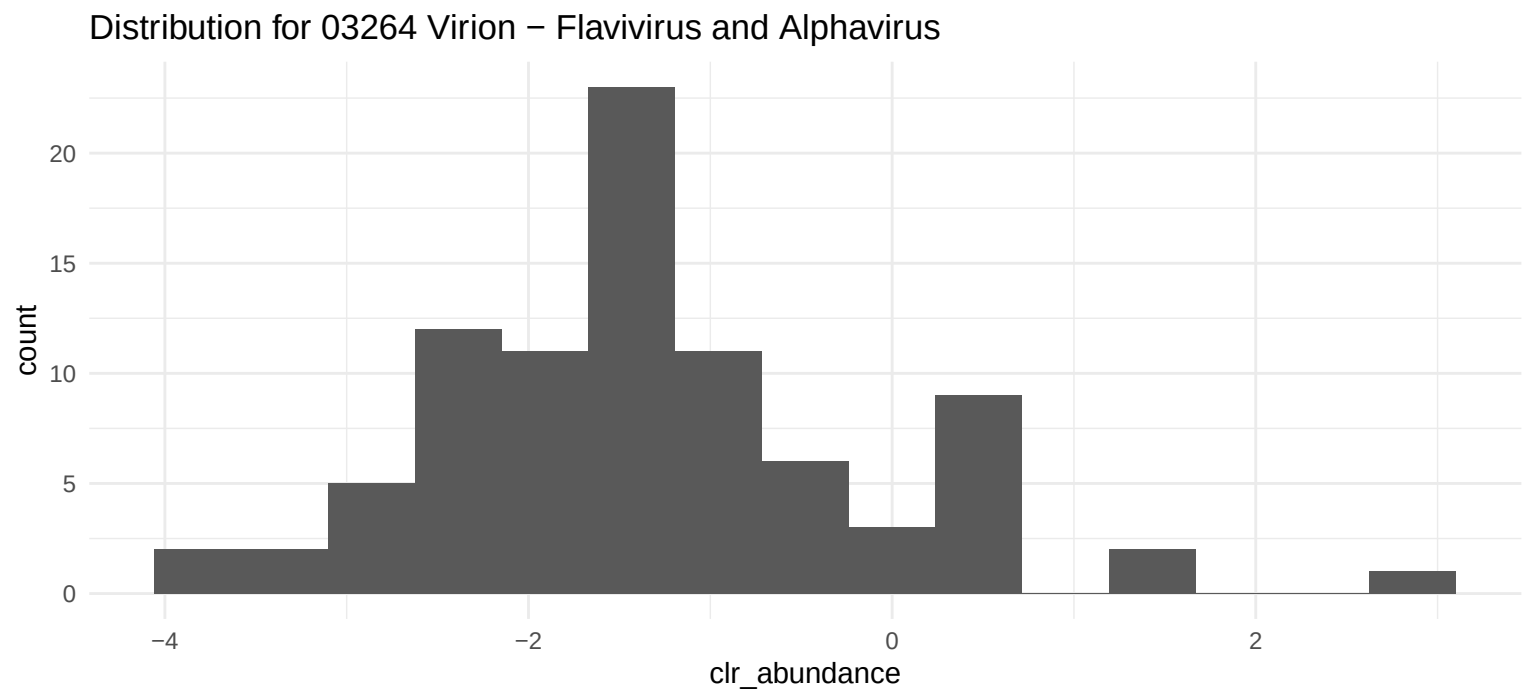

Distribution for 03200 Viral proteins
count
0
5
10
15
−150
−100
−50
0
50
clr_abundance

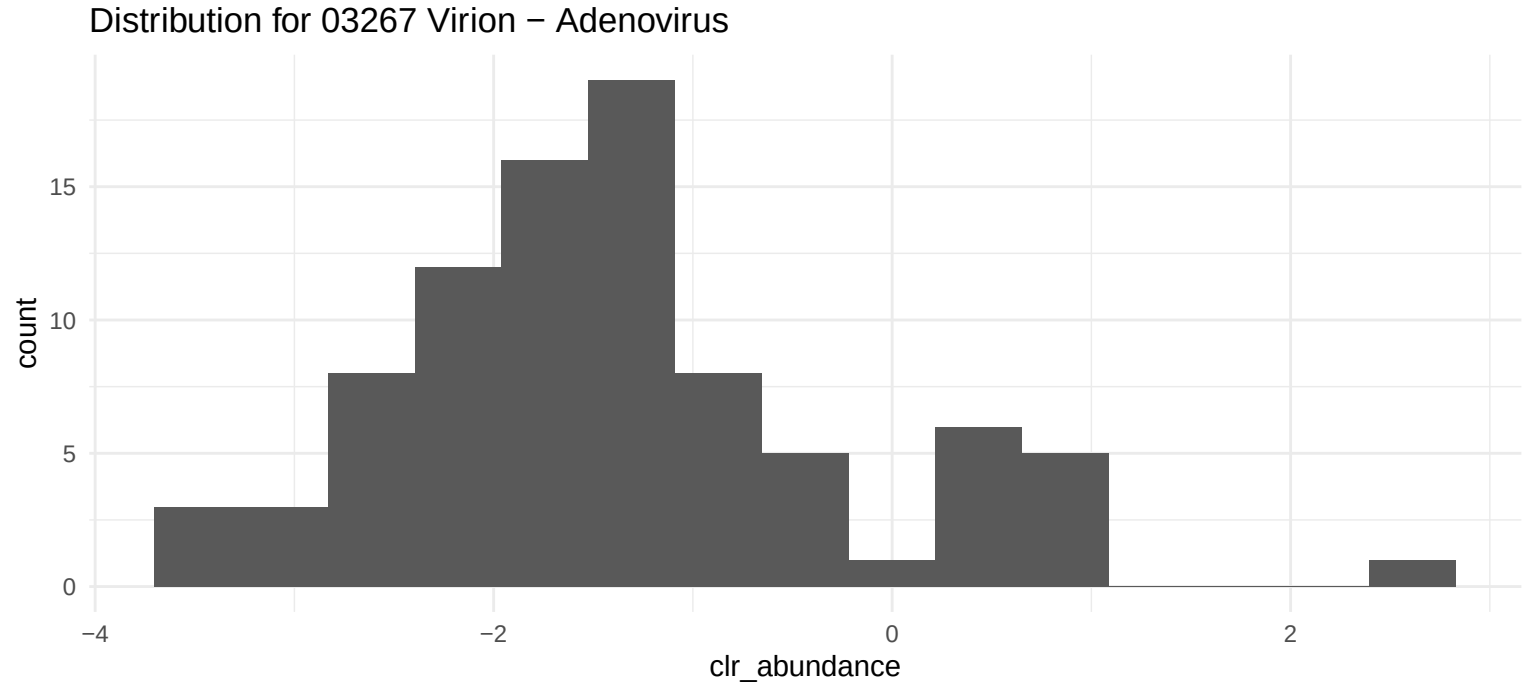

Distribution for 03240 Viral replication
count
0
5
10
15
20
−20
0
20
clr_abundance

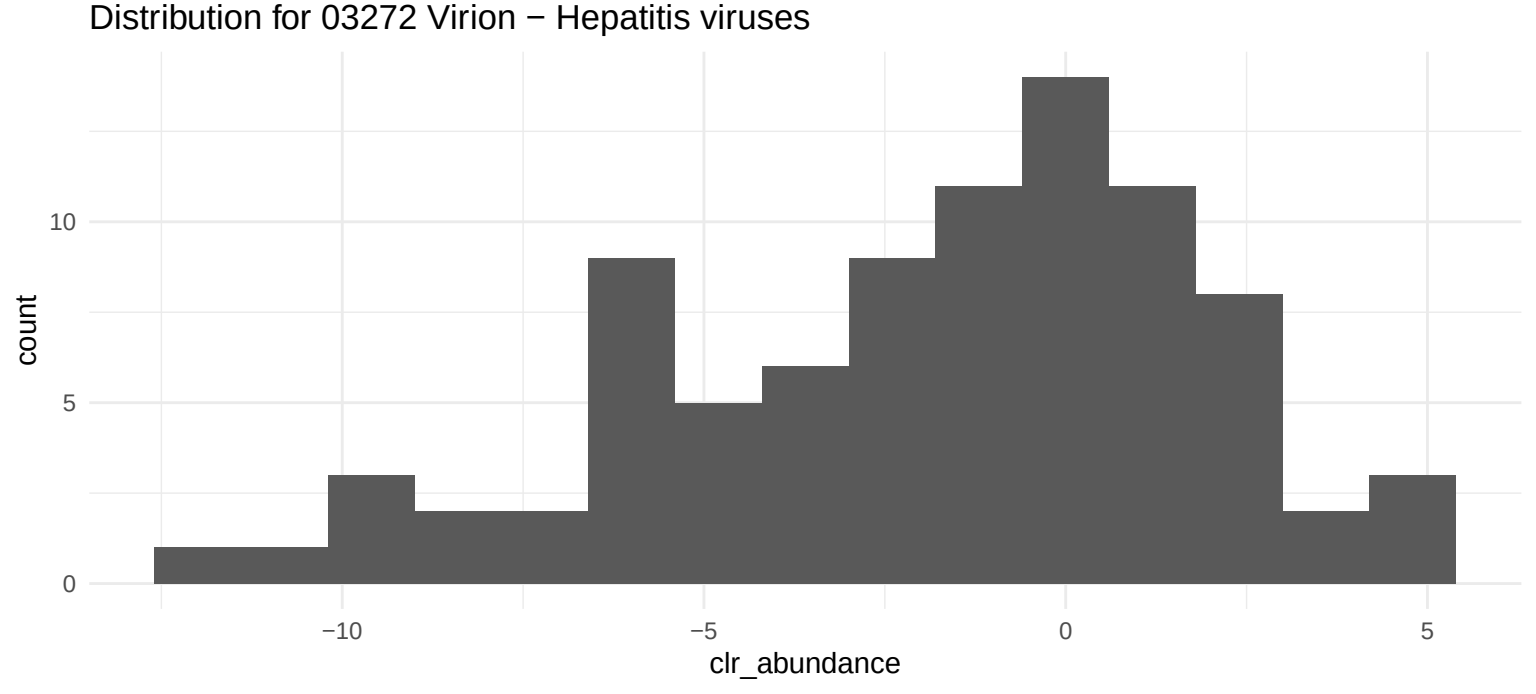

Distribution for 03250 Viral life cycle − HIV−1
count
clr_abundance
0
5
10
−10
0
10

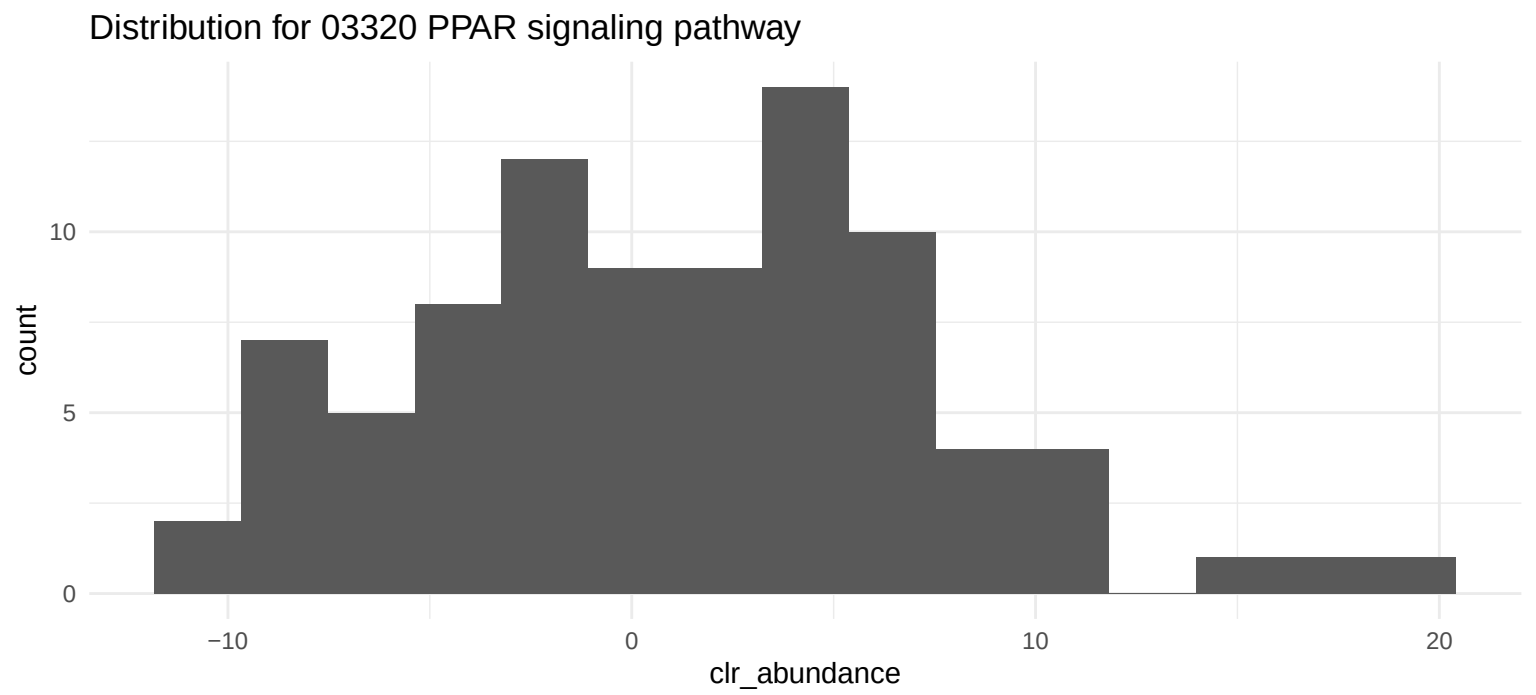

Distribution for 03258 Virion − Bacteriophage lambda
count
clr_abundance
0
5
10
15
−8
−6
−4
−2

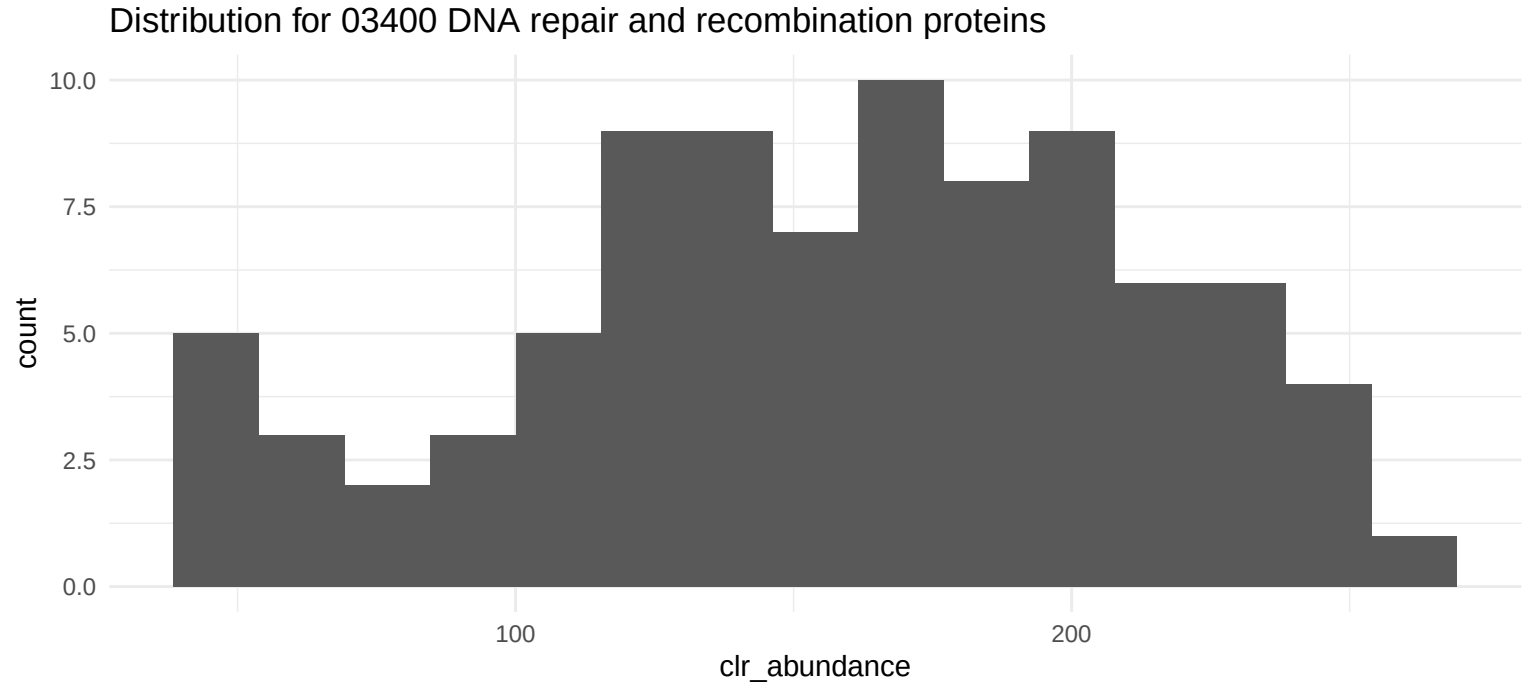

Distribution for 03259 Virion − Bacteriophages T4 and T7
count
clr_abundance
0
5
10
15
20
−30
−20
−10
0
10
20

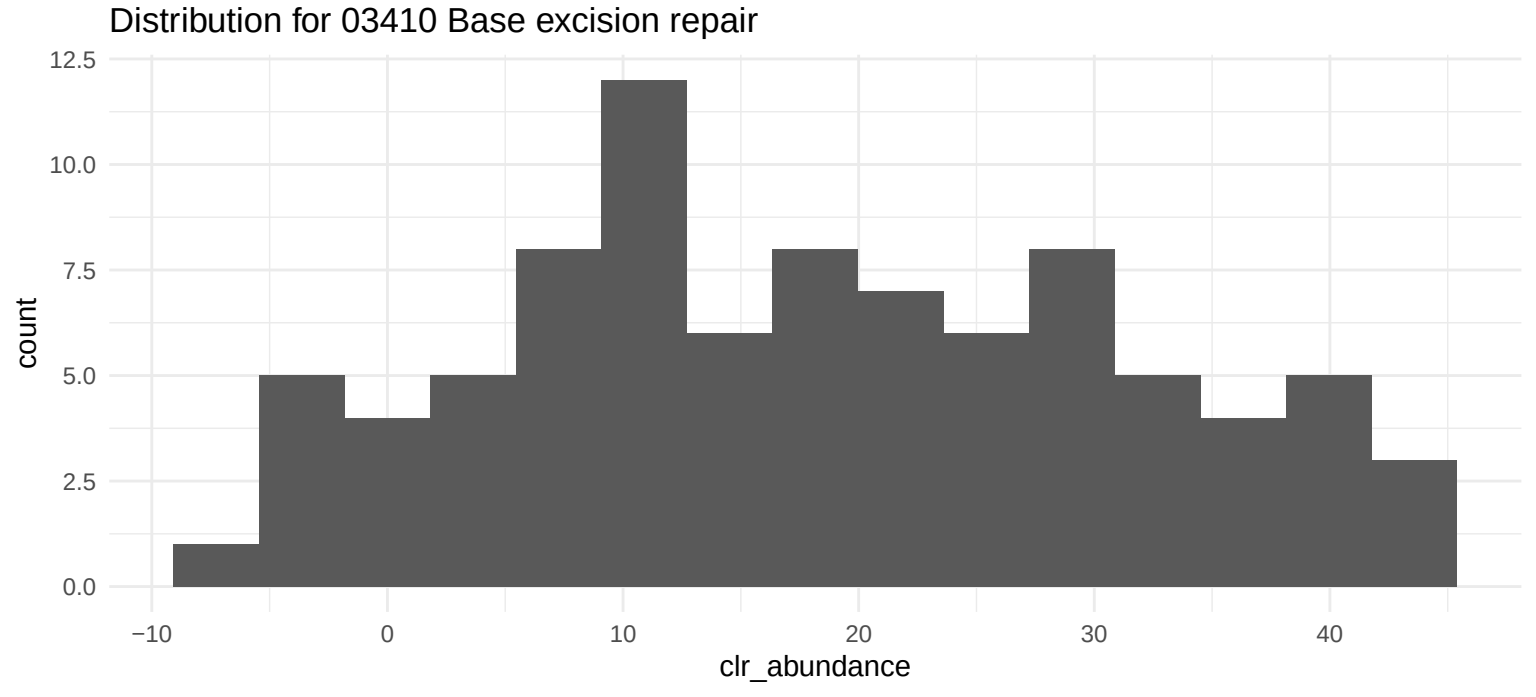

Distribution for 03264 Virion − Flavivirus and Alphavirus
count
clr_abundance
0
5
10
15
20
−4
−2
0
2

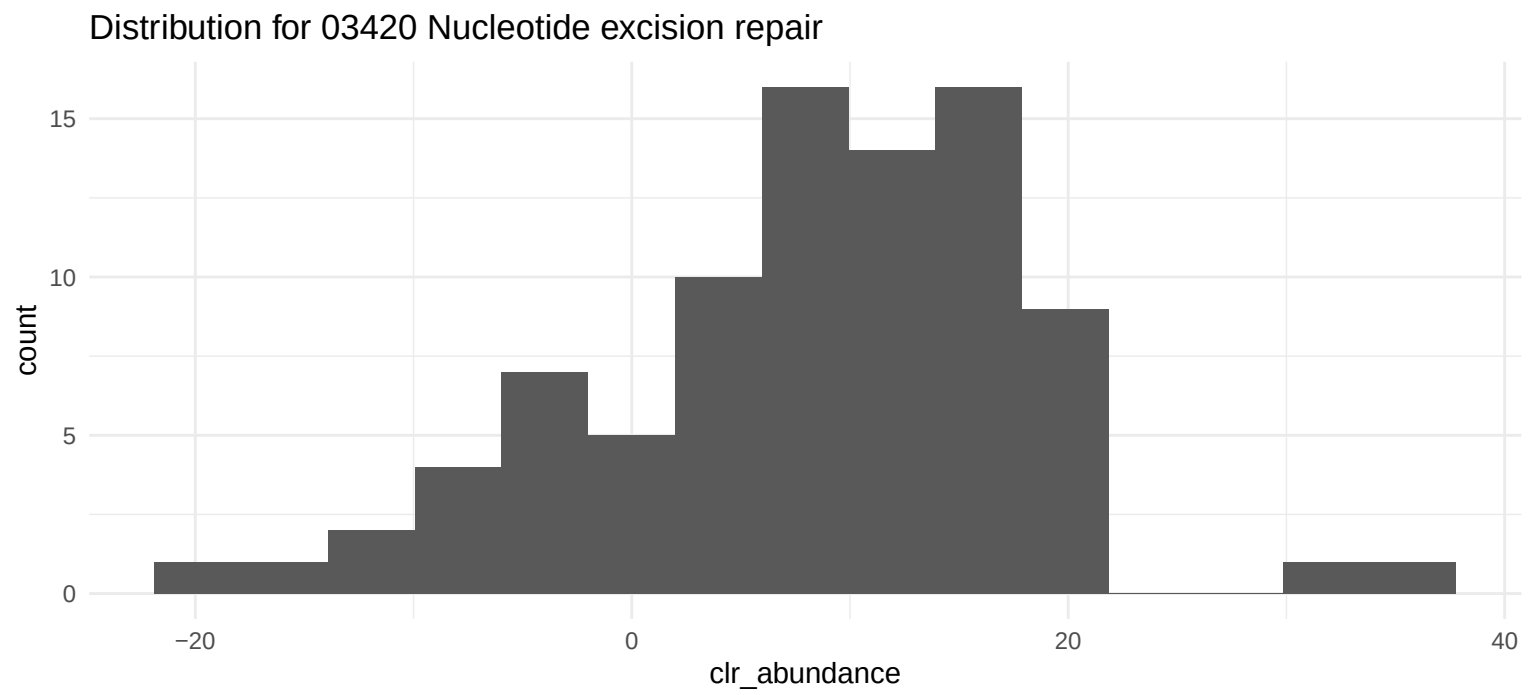

Distribution for 03267 Virion − Adenovirus
count
clr_abundance
0
5
10
15
−4
−2
0
2

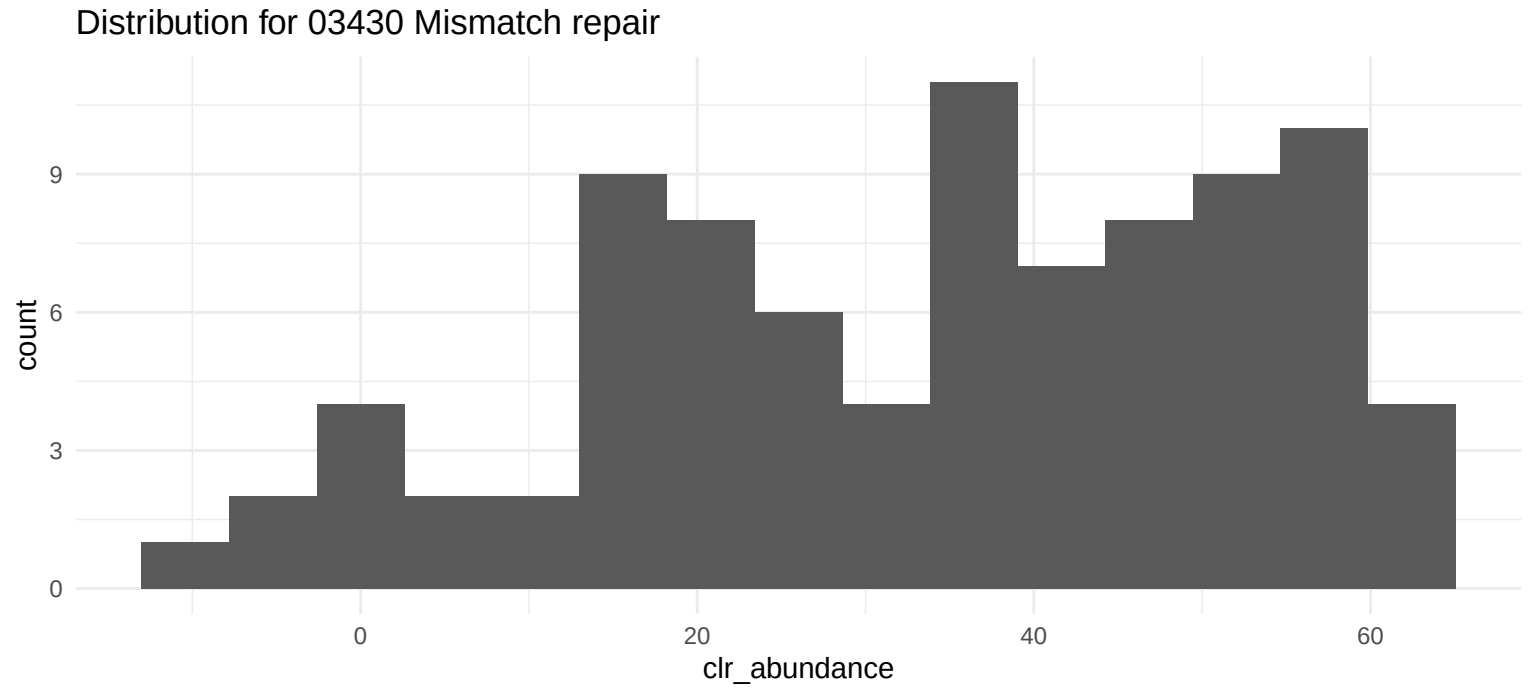

Distribution for 03272 Virion − Hepatitis viruses
count
clr_abundance
0
5
10
−10
−5
0
5

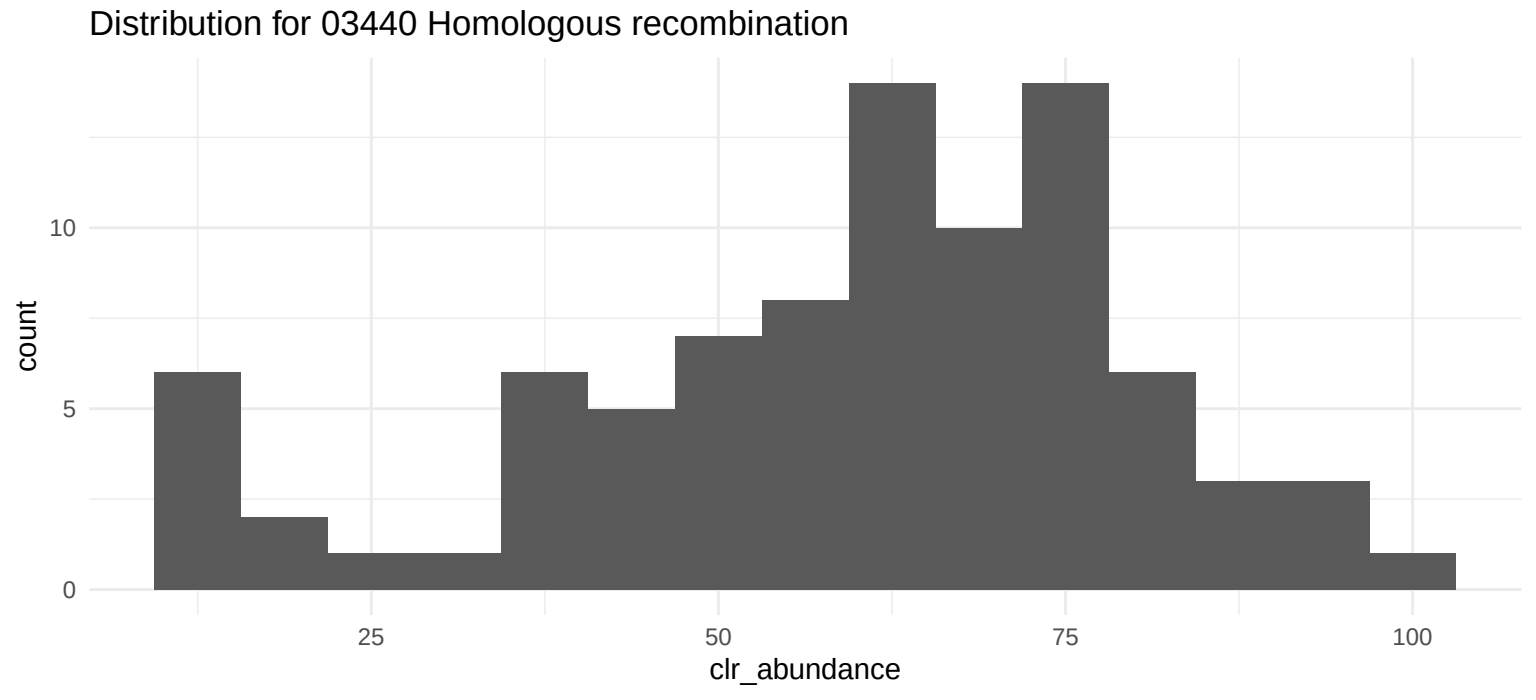

Distribution for 03320 PPAR signaling pathway
count
10
5
0
−10
0
10
20
clr_abundance

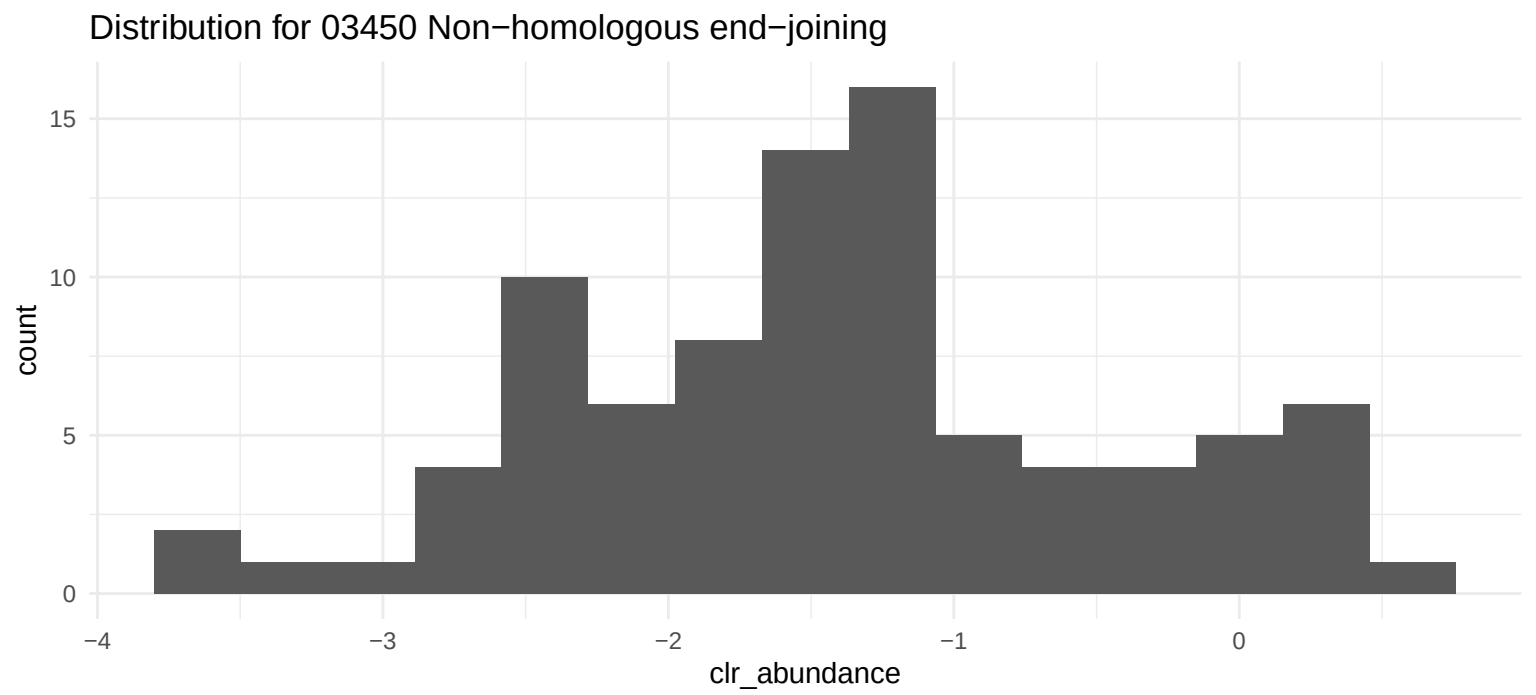

Distribution for 03400 DNA repair and recombination proteins
count
10.0
7.5
5.0
2.5
0.0
100
200
clr_abundance

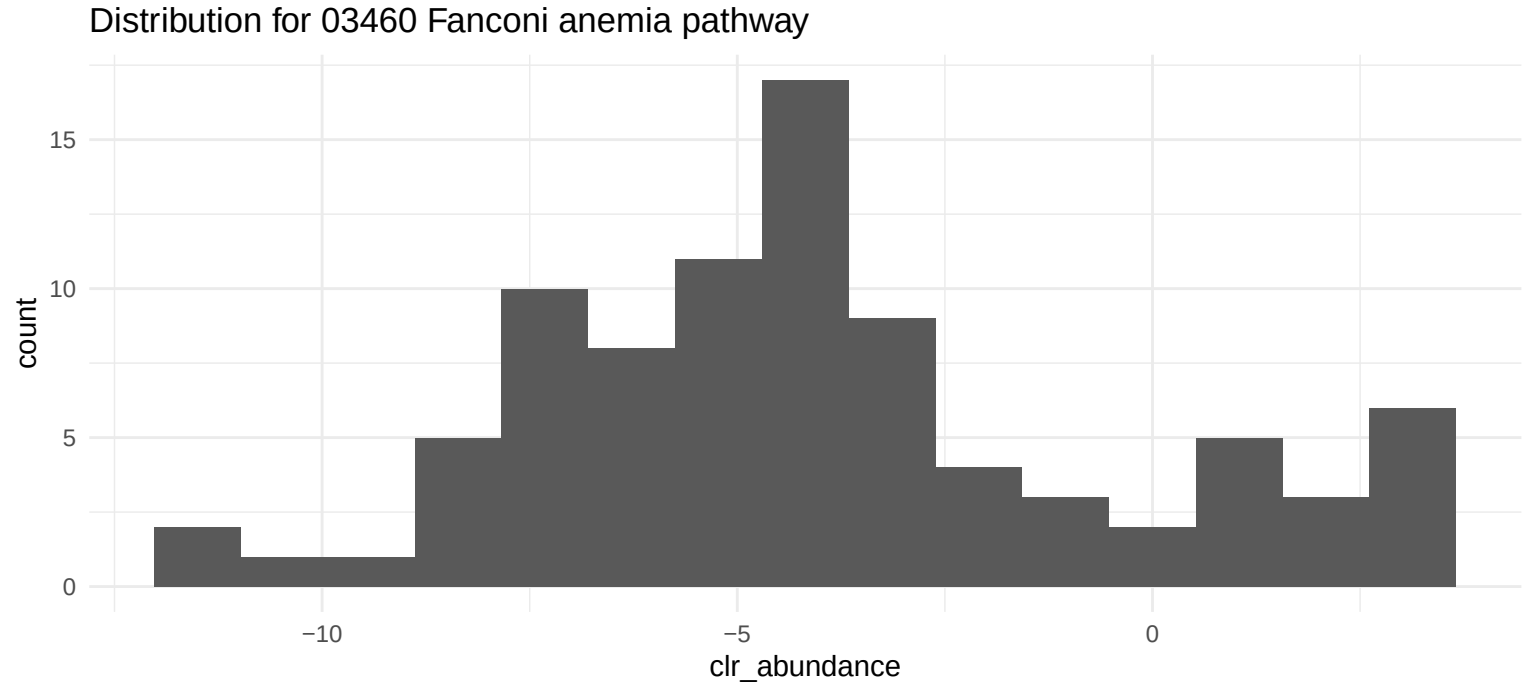

Distribution for 03410 Base excision repair
count
12.5
10.0
7.5
5.0
2.5
0.0
−10
0
10
20
30
40
clr_abundance

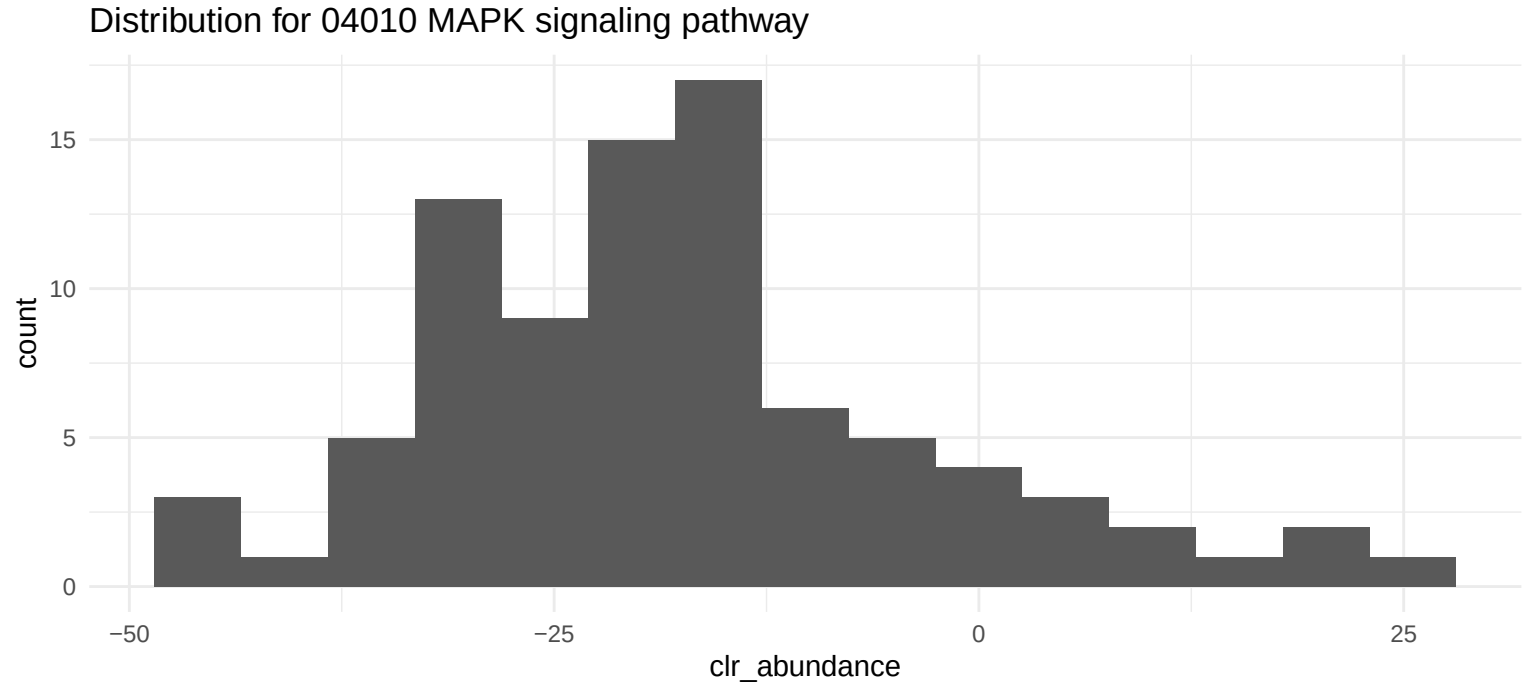

Distribution for 03420 Nucleotide excision repair
count
15
10
5
0
−20
0
20
40
clr_abundance

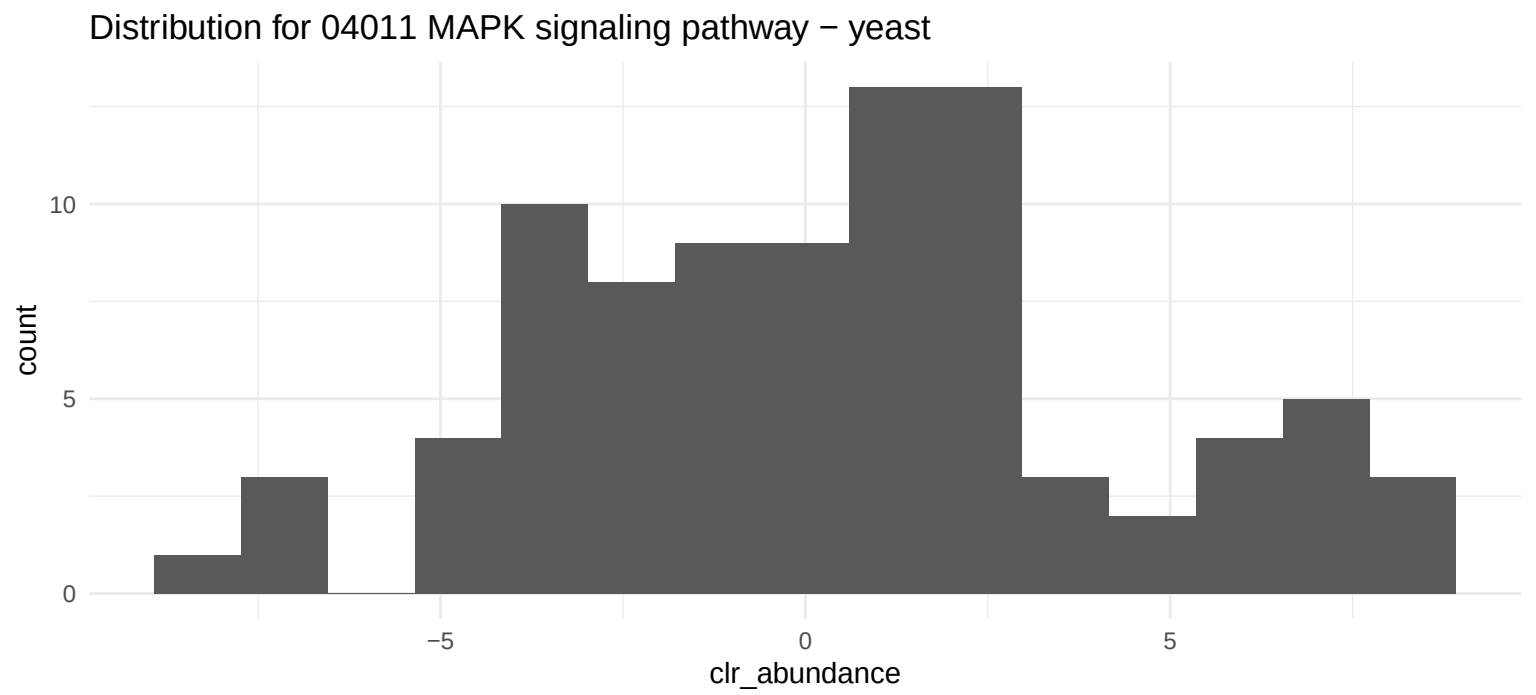

Distribution for 03430 Mismatch repair
count
9
6
3
0
0
20
40
60
clr_abundance

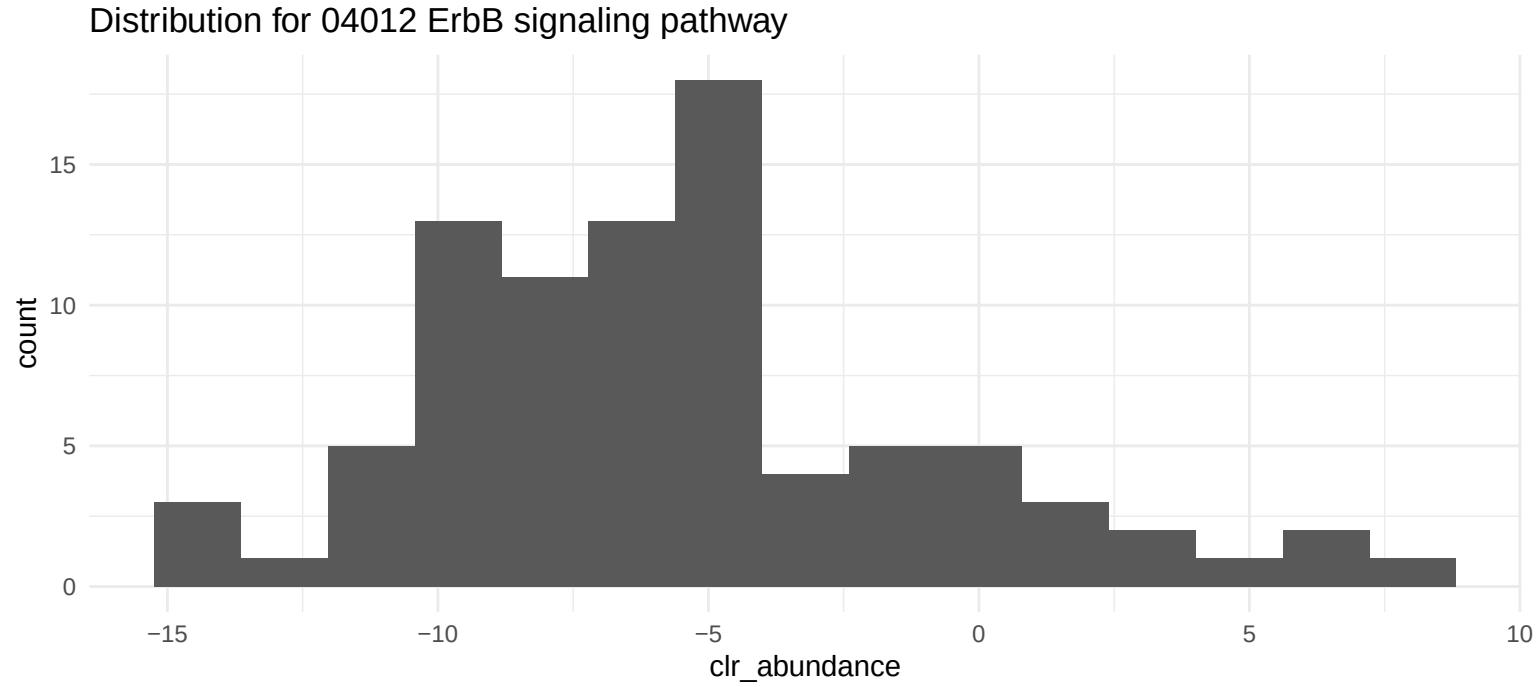

Distribution for 03440 Homologous recombination
count
10
5
0
25
50
75
100
clr_abundance

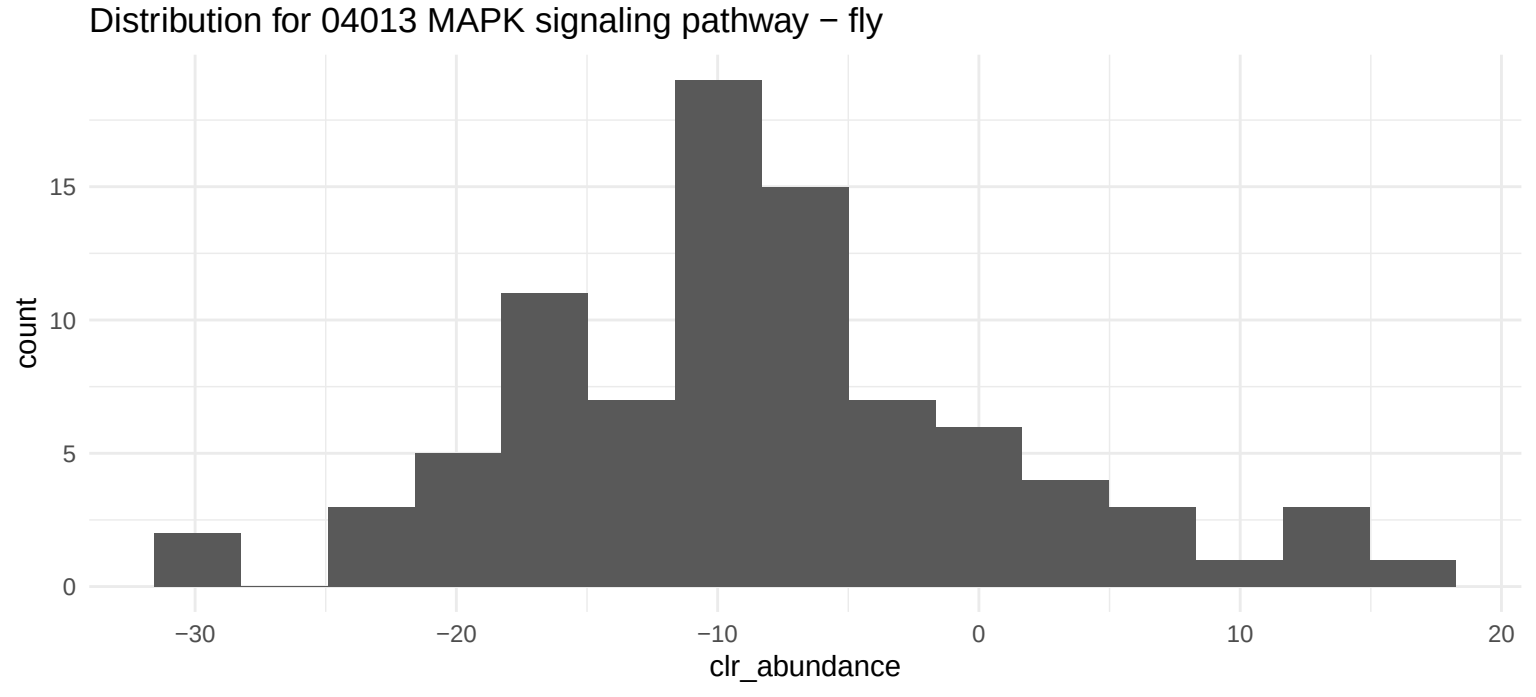

Distribution for 03450 Non−homologous end−joining
count
15
10
5
0
−4
−3
−2
−1
0
clr_abundance


Distribution for 03460 Fanconi anemia pathway
count
15
10
5
0
−10
−5
0
clr_abundance

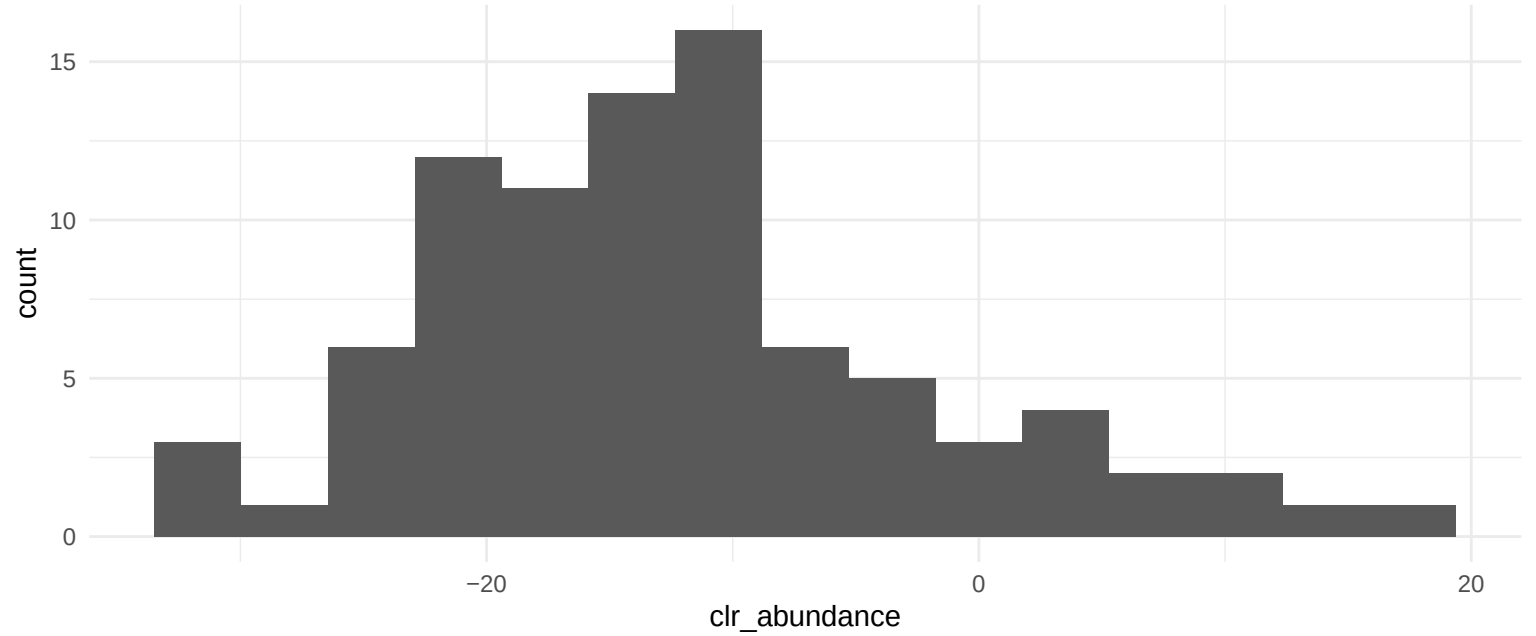

Distribution for 04010 MAPK signaling pathway
count
15
10
5
0
−50
−25
0
25
clr_abundance


Distribution for 04011 MAPK signaling pathway − yeast
count
clr_abundance
0
5
10
−5
0
5

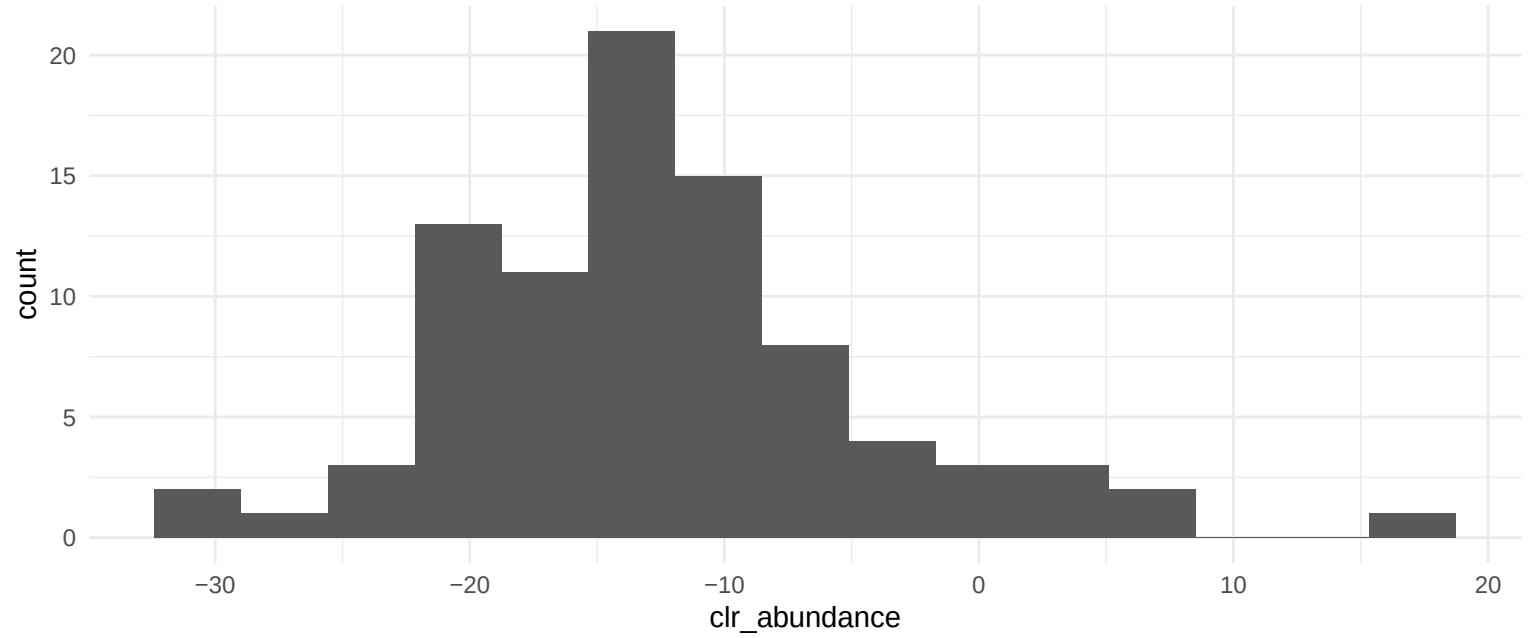

Distribution for 04012 ErbB signaling pathway
count
clr_abundance
0
5
10
15
−15
−10
−5
0
5
10


Distribution for 04013 MAPK signaling pathway − fly
count
clr_abundance
0
5
10
15
−30
−20
−10
0
10
20

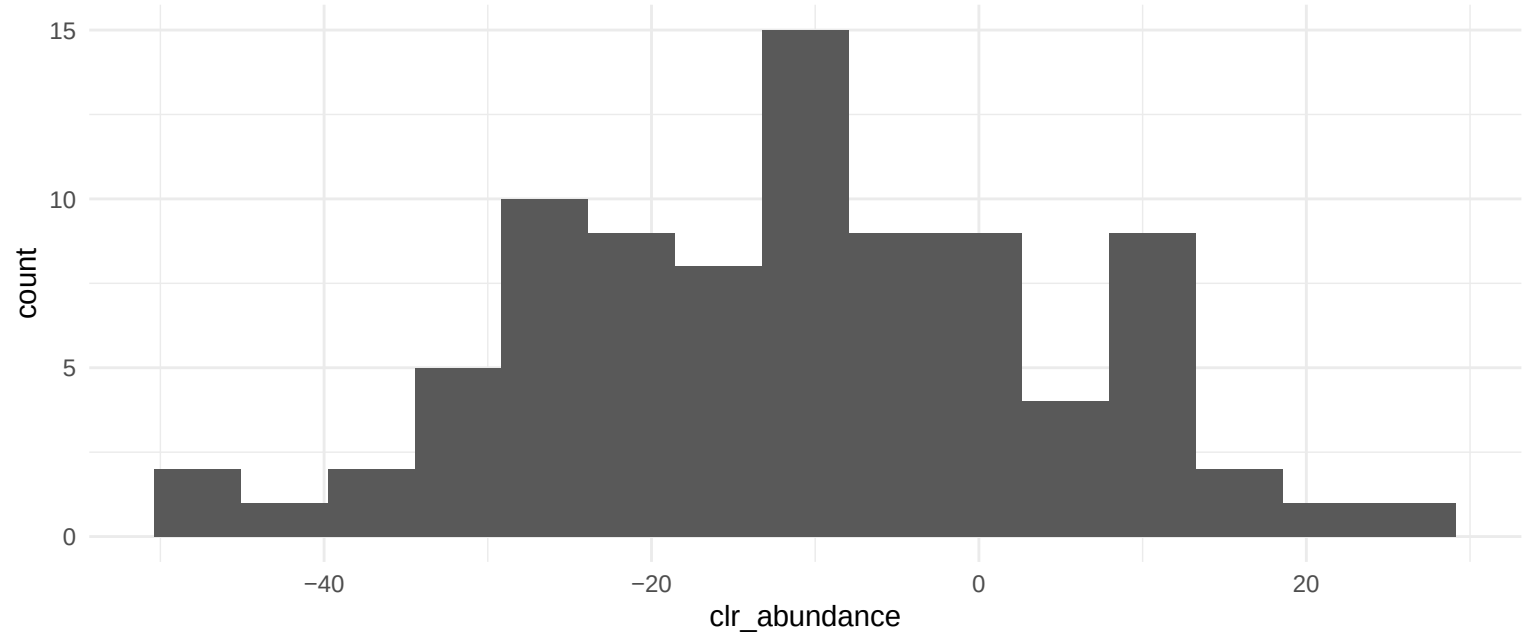
Distribution for 04014 Ras signaling pathway

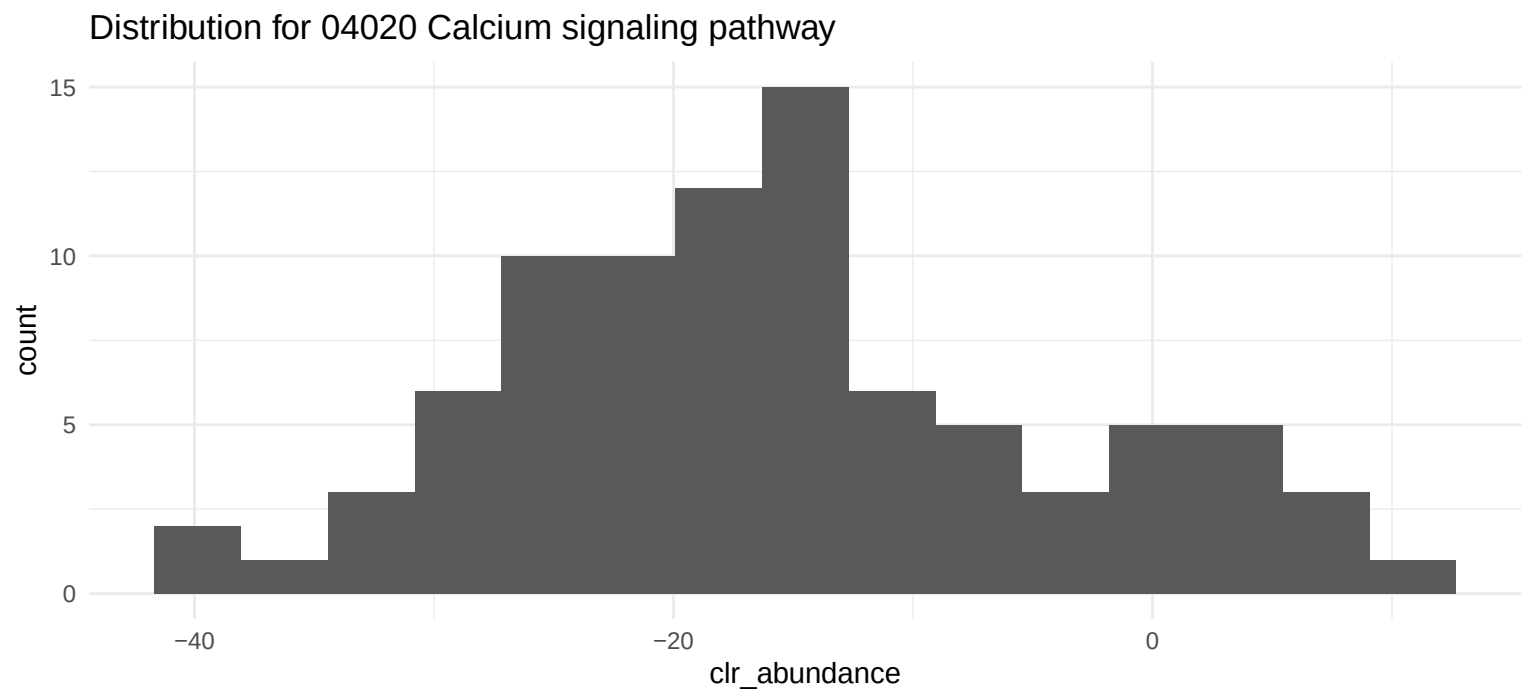


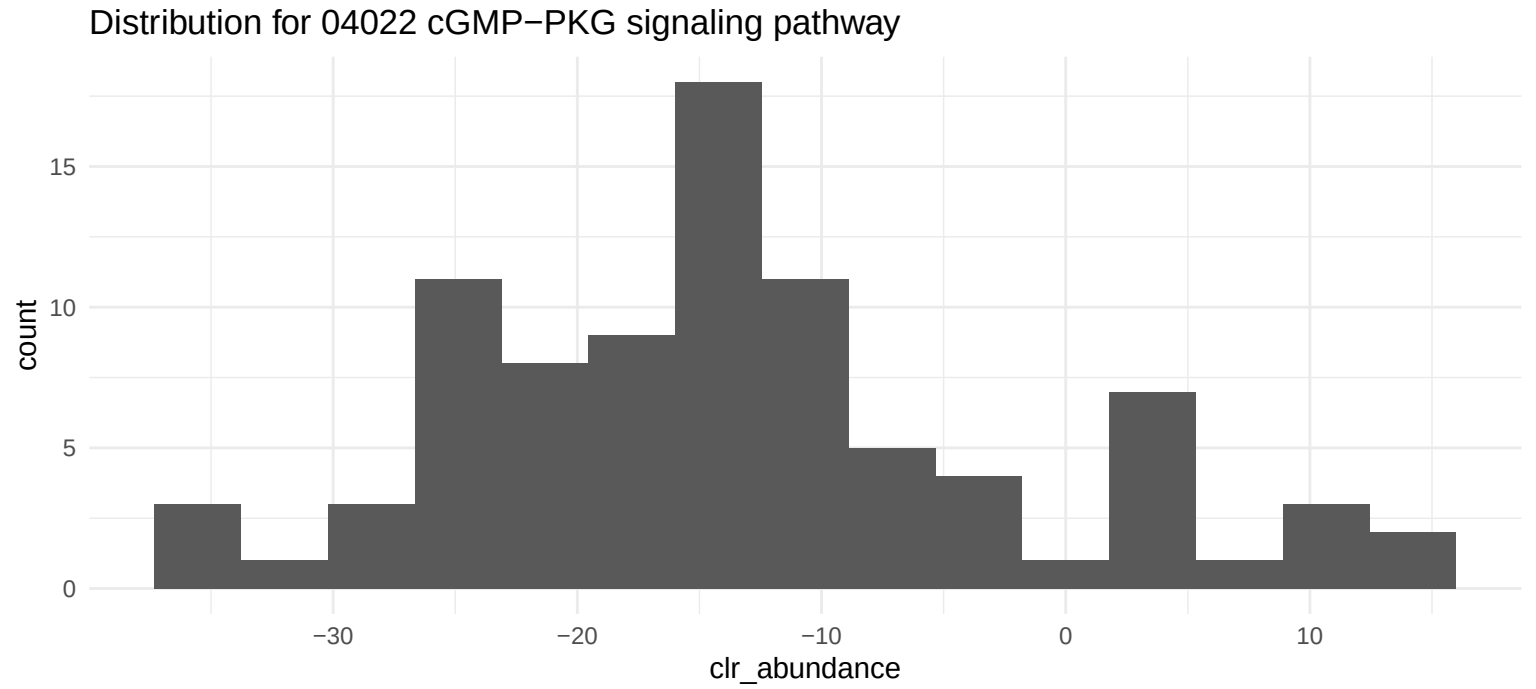
Distribution for 04015 Rap1 signaling pathway

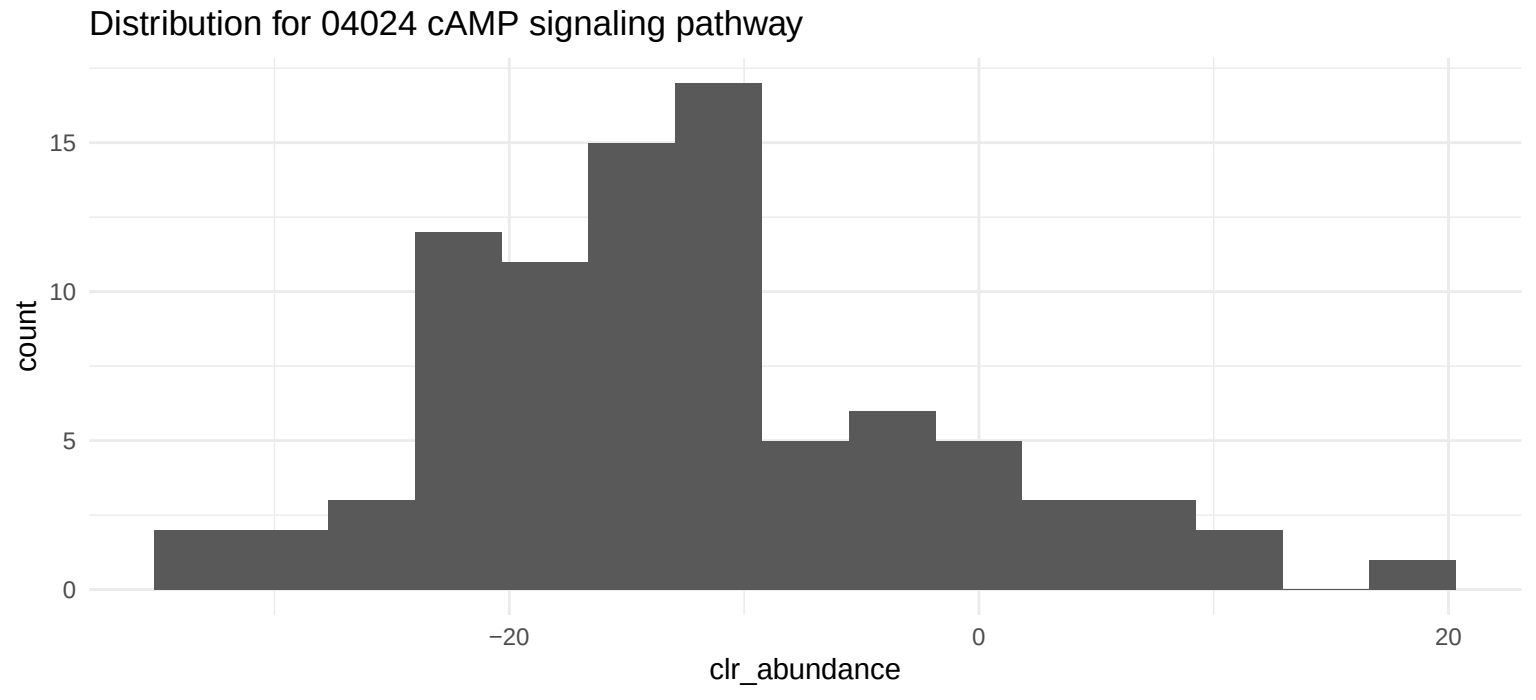


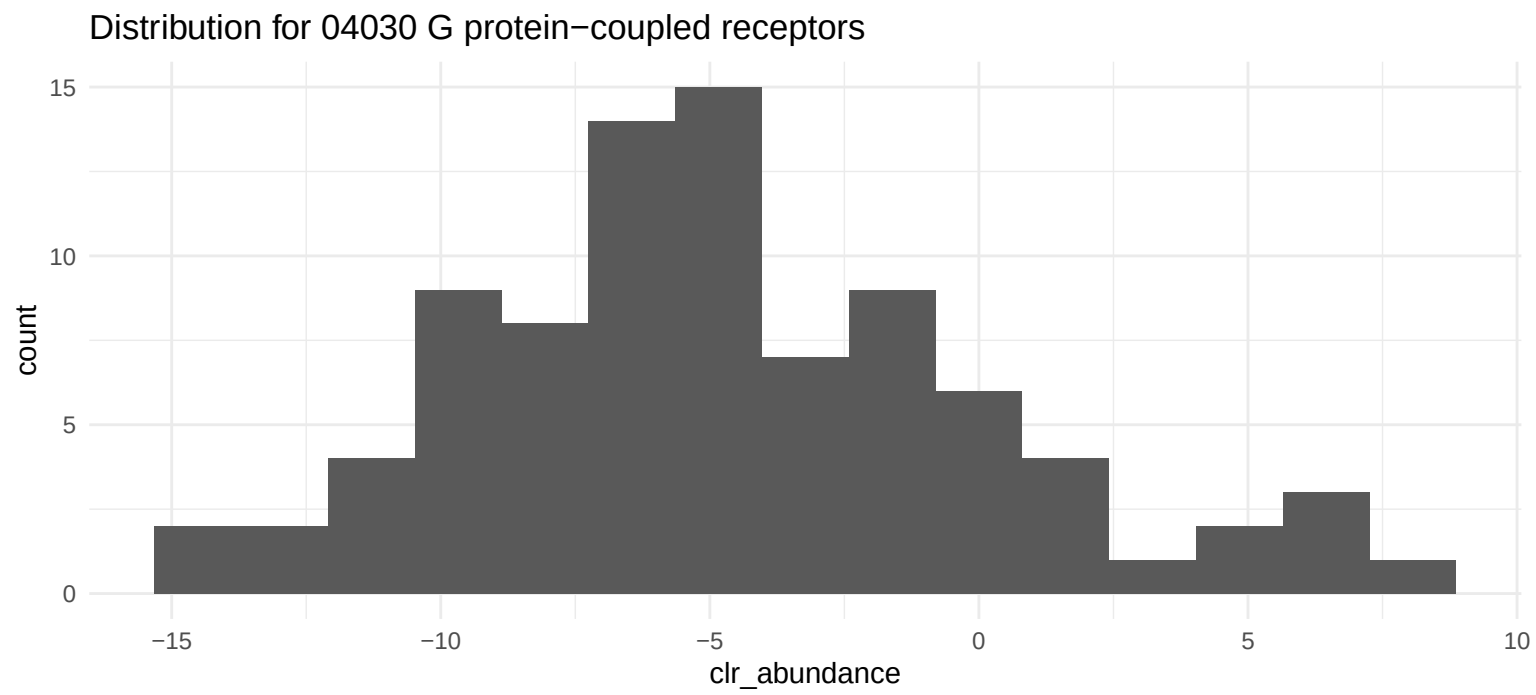
Distribution for 04016 MAPK signaling pathway − plant

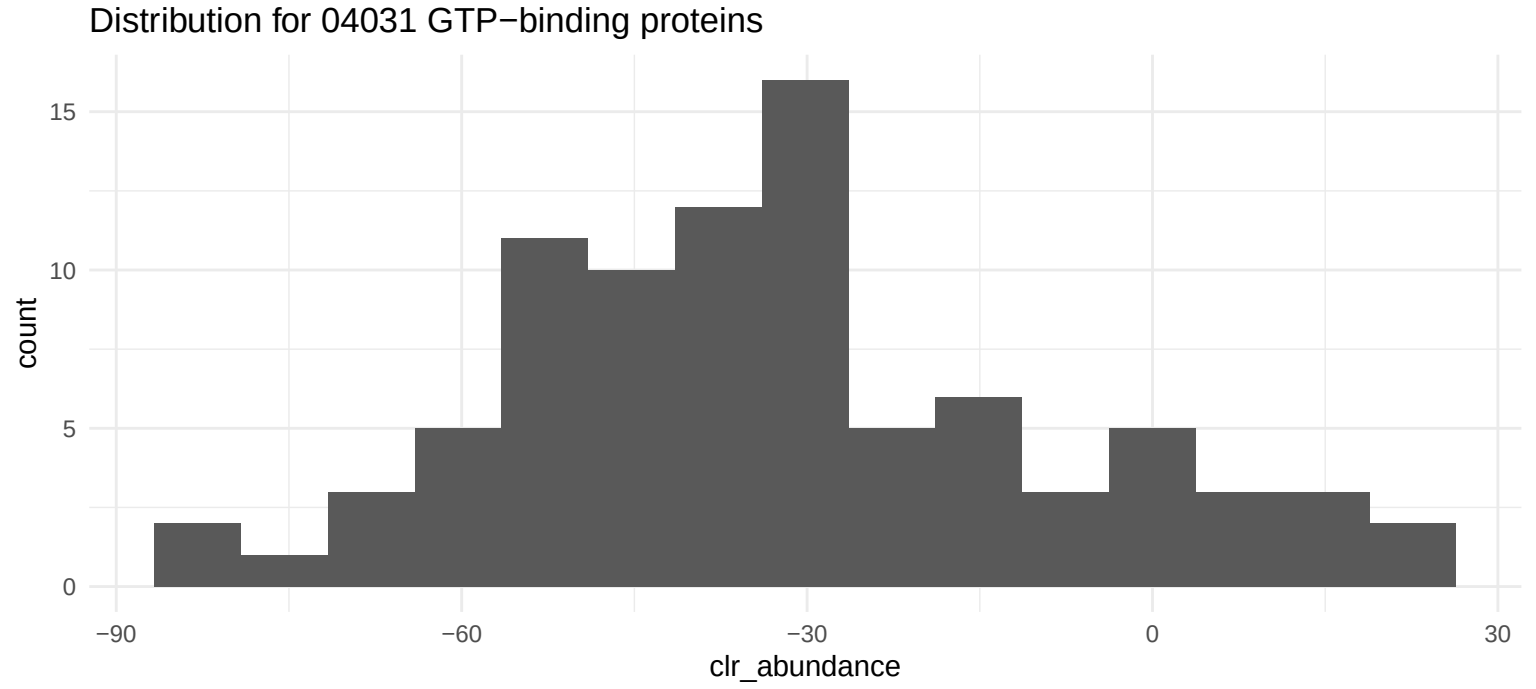


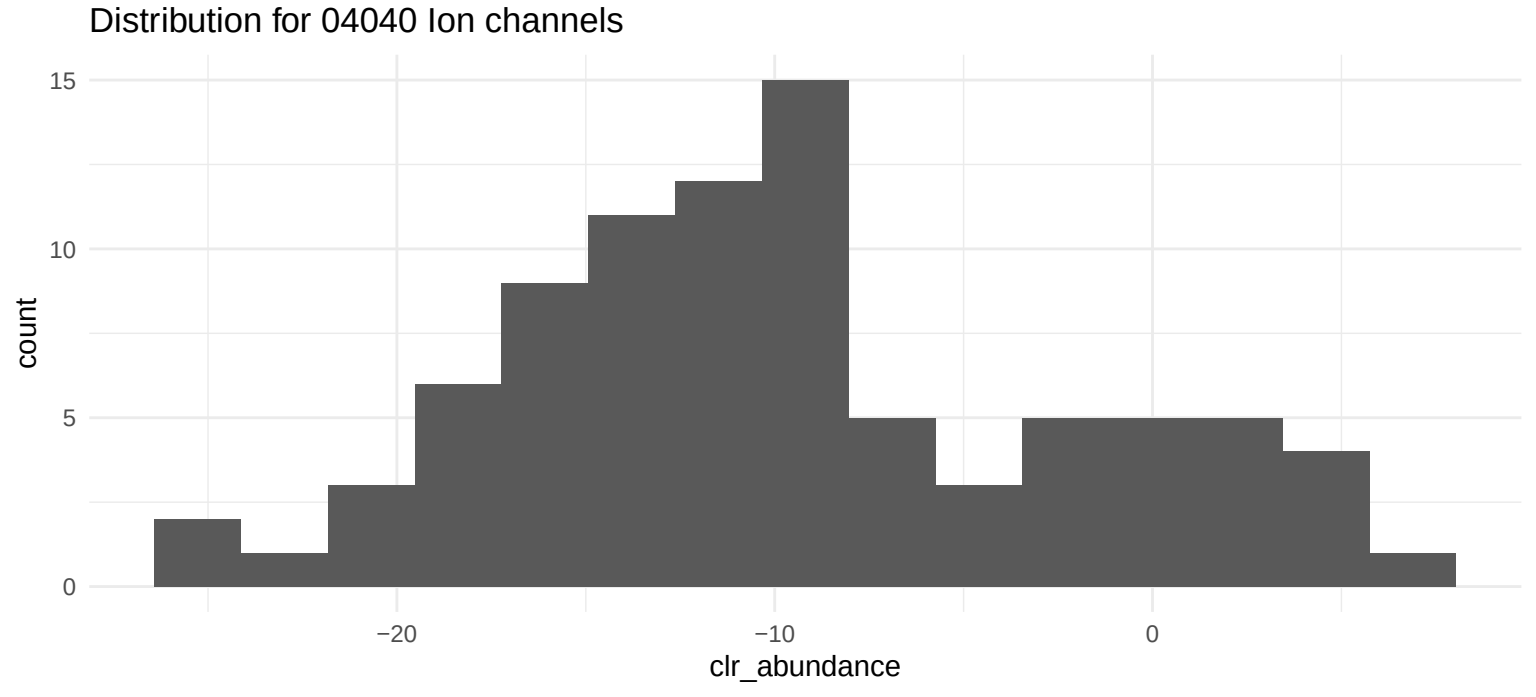

Distribution for 04020 Calcium signaling pathway
count
15
10
5
0
−40
−20
0
clr_abundance


Distribution for 04022 cGMP−PKG signaling pathway
count
15
10
5
0
−30
−20
−10
0
10
clr_abundance

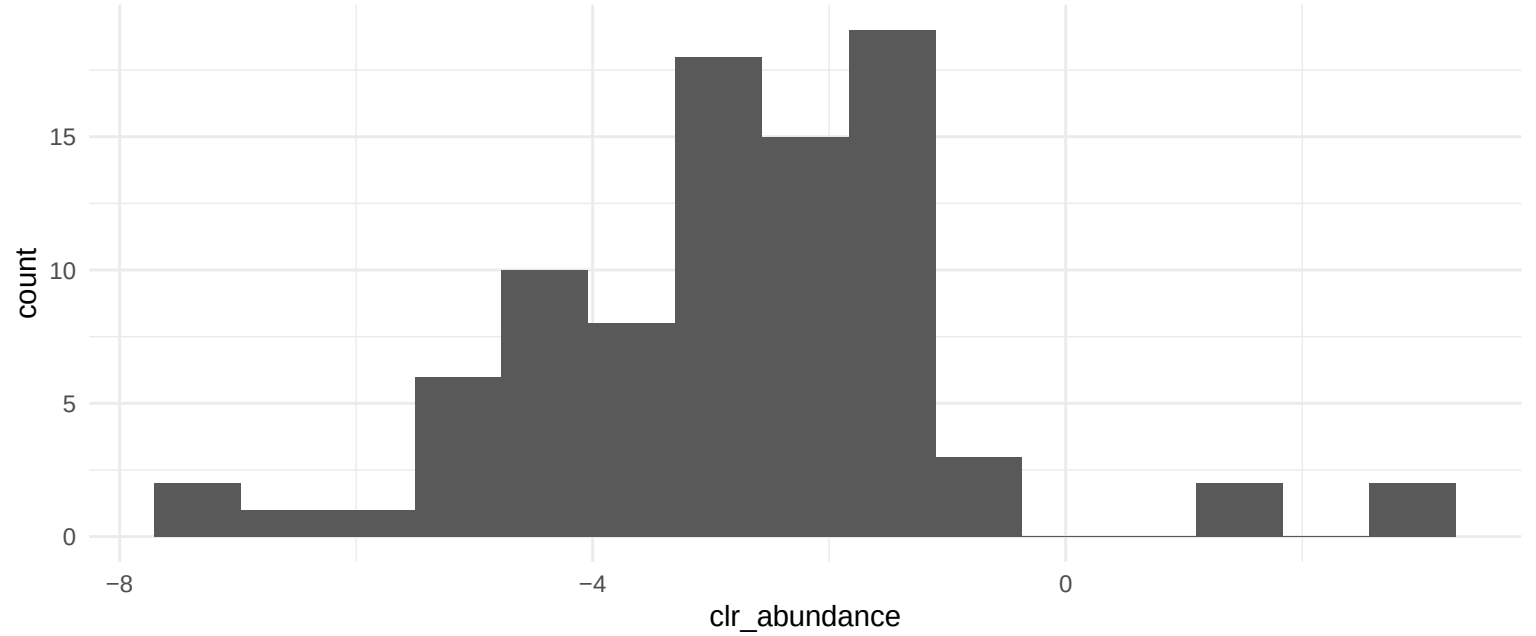

Distribution for 04024 cAMP signaling pathway
count
15
10
5
0
−20
0
20
clr_abundance


Distribution for 04030 G protein−coupled receptors
count
clr_abundance
15
10
5
0
−15
−10
−5
0
5
10

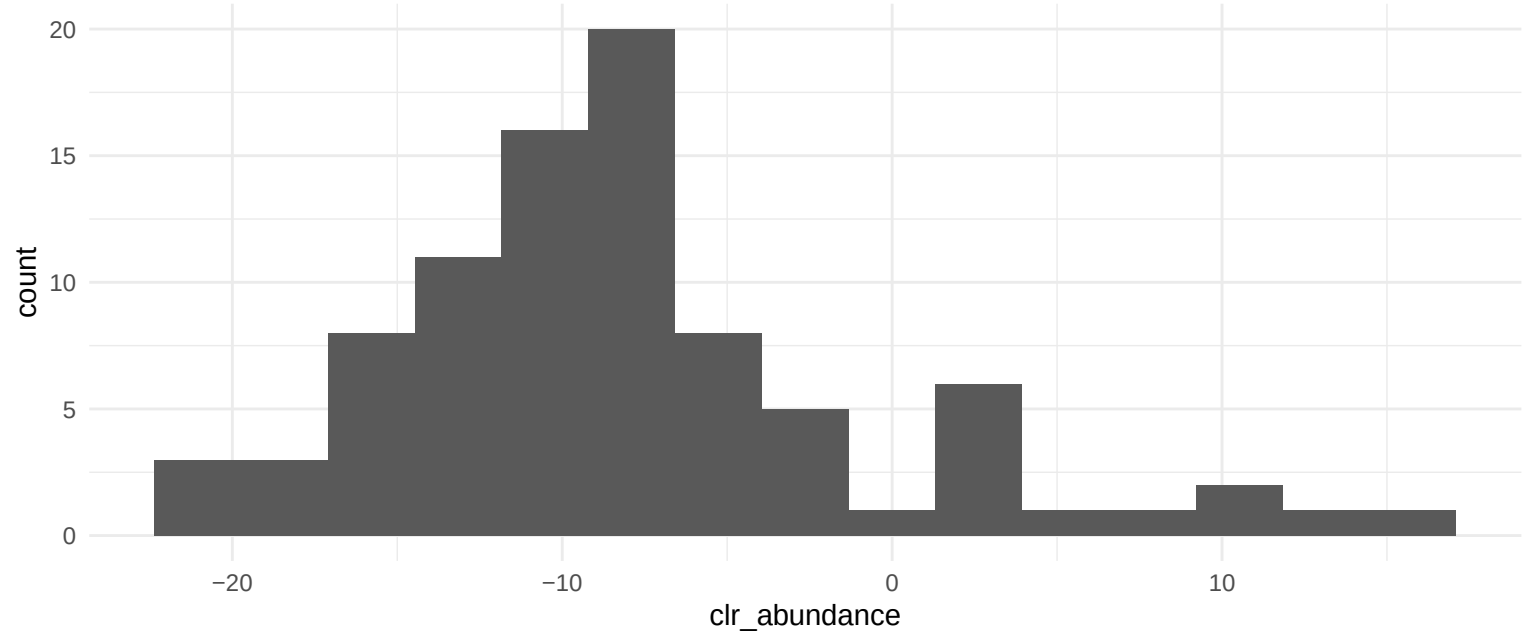

Distribution for 04031 GTP−binding proteins
count
clr_abundance
15
10
5
0
−90
−60
−30
0
30


Distribution for 04040 Ion channels
count
clr_abundance
15
10
5
0
−20
−10
0

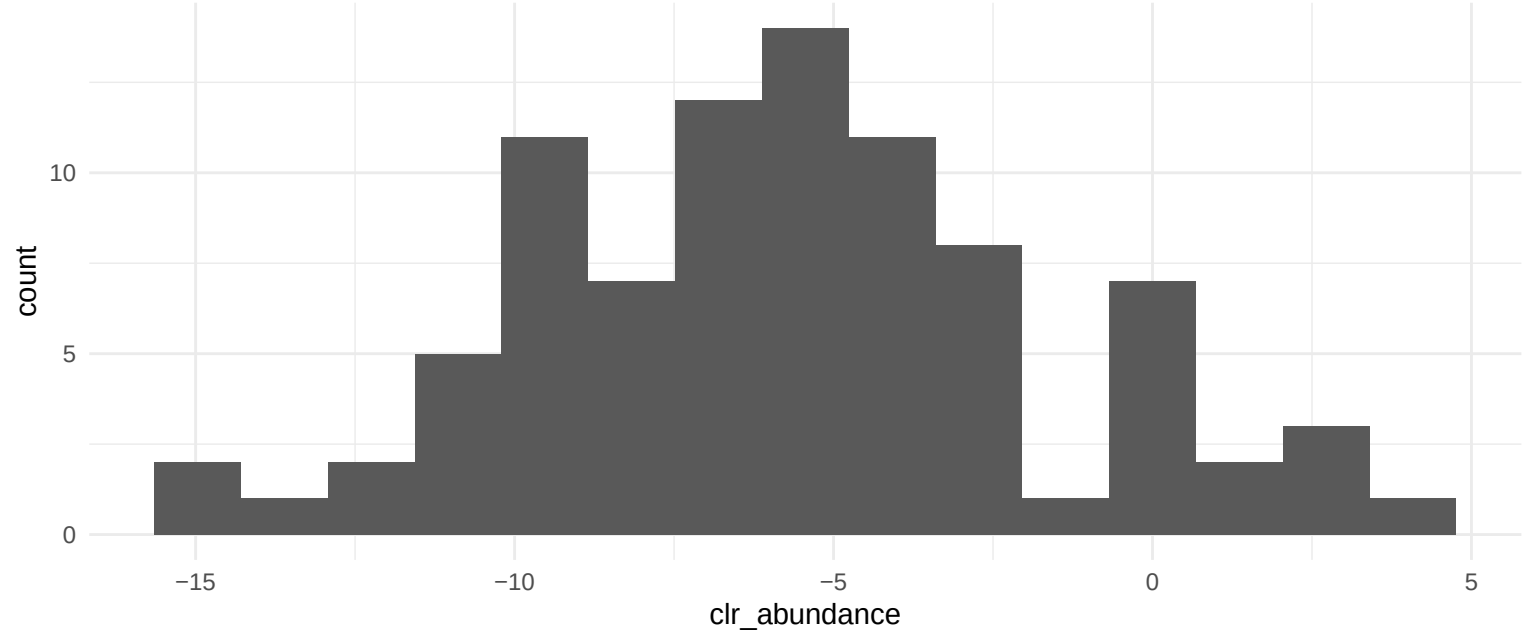
Distribution for 04052 Cytokines and neuropeptides

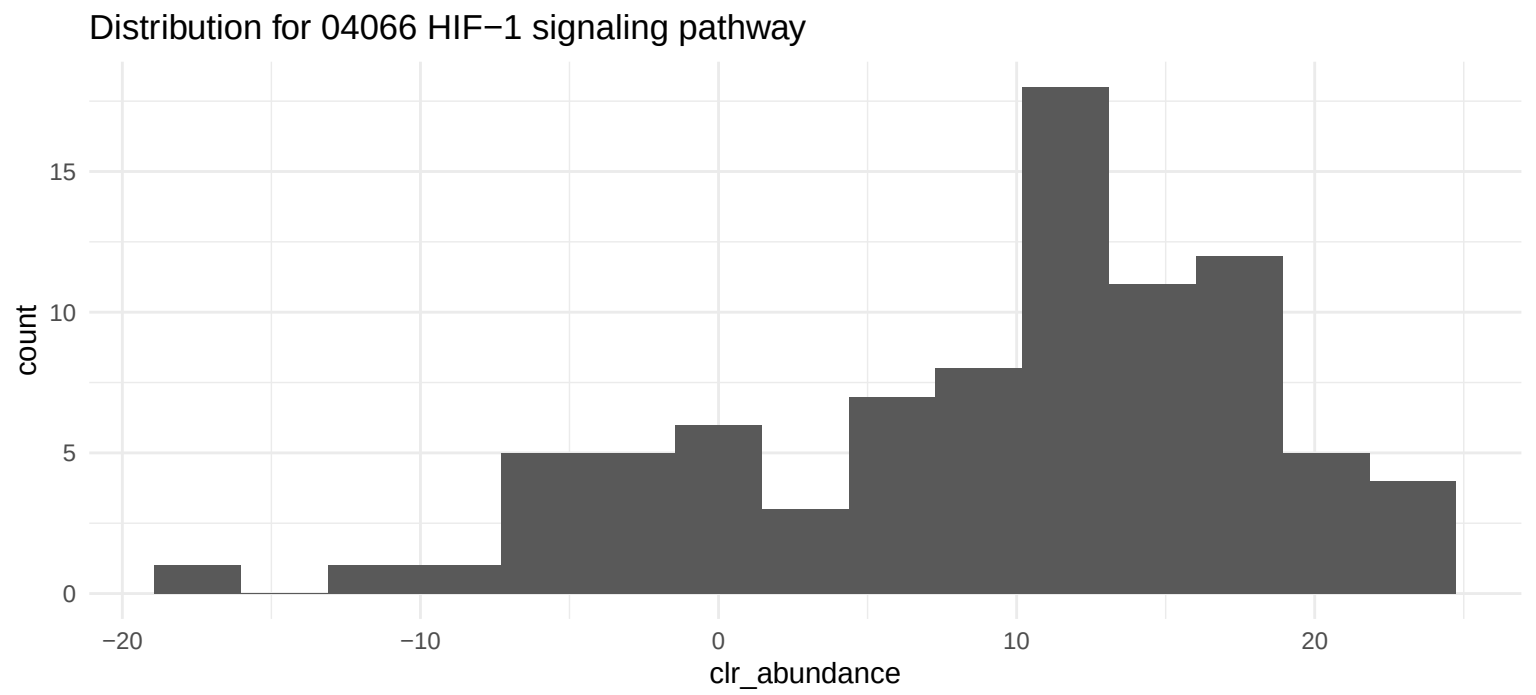


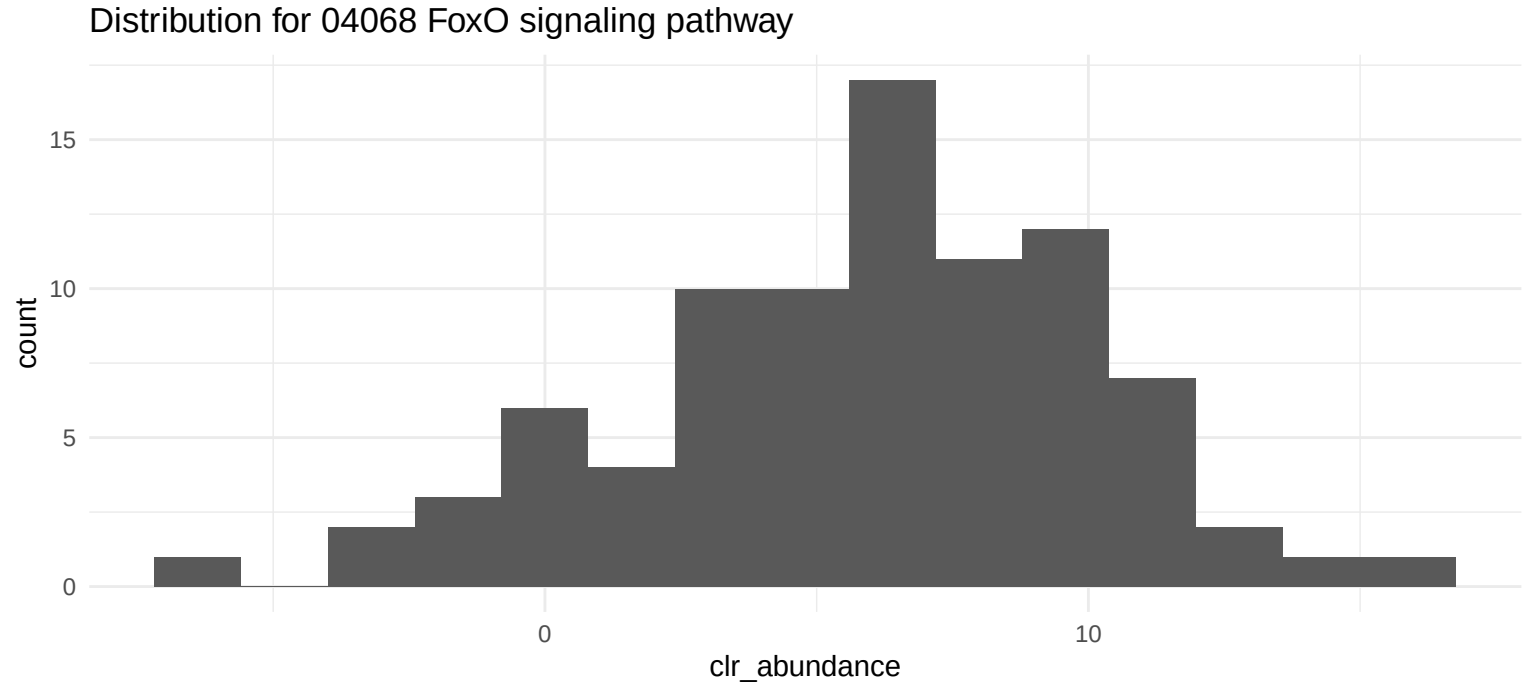
Distribution for 04062 Chemokine signaling pathway

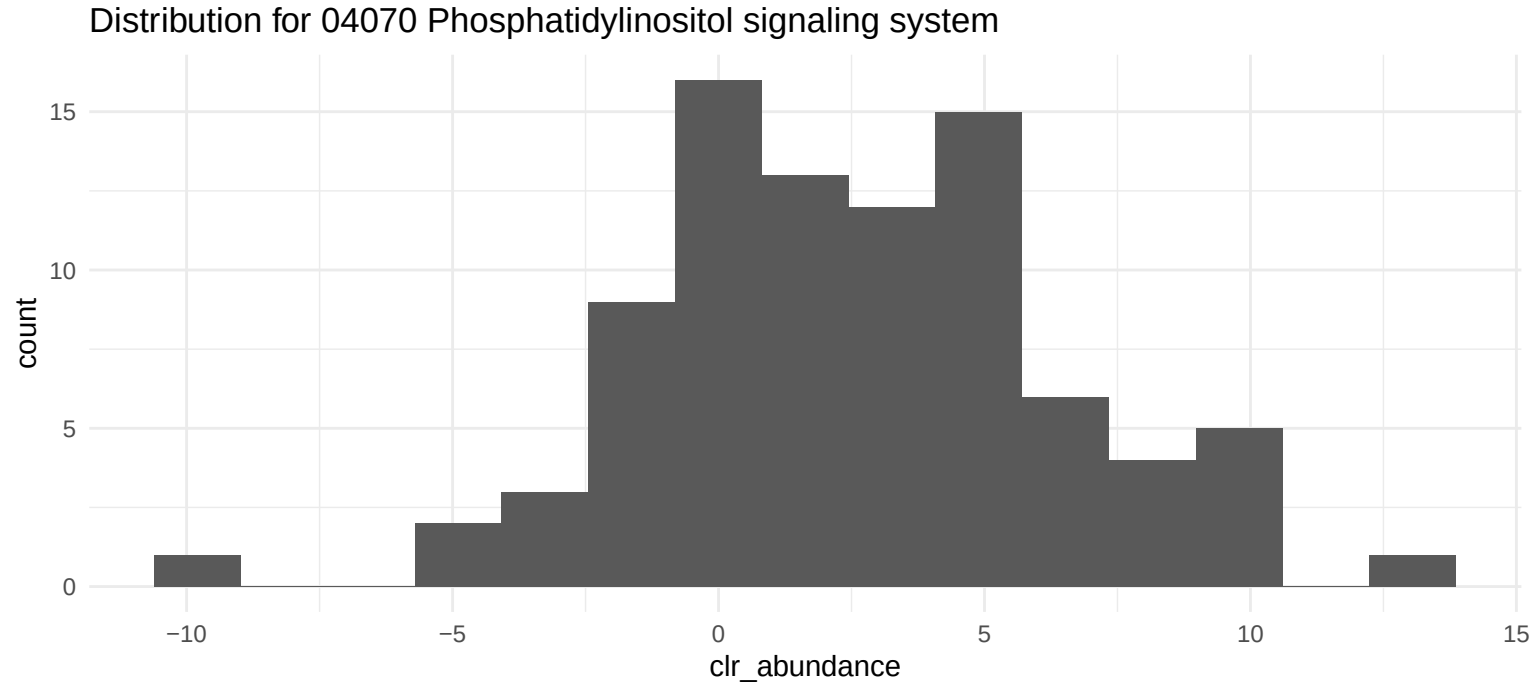


Distribution for 04064 NF−kappa B signaling pathway

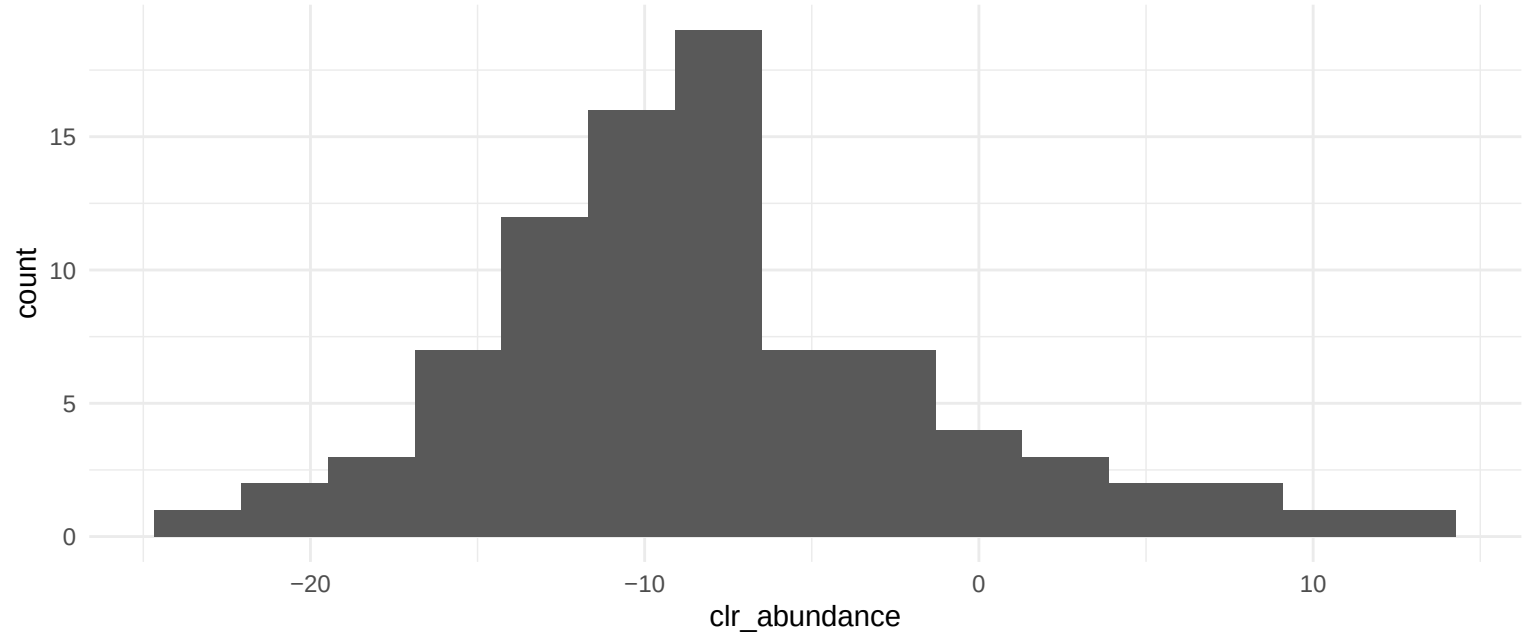



Distribution for 04066 HIF−1 signaling pathway
count
clr_abundance
0
5
10
15
−20
−10
0
10
20

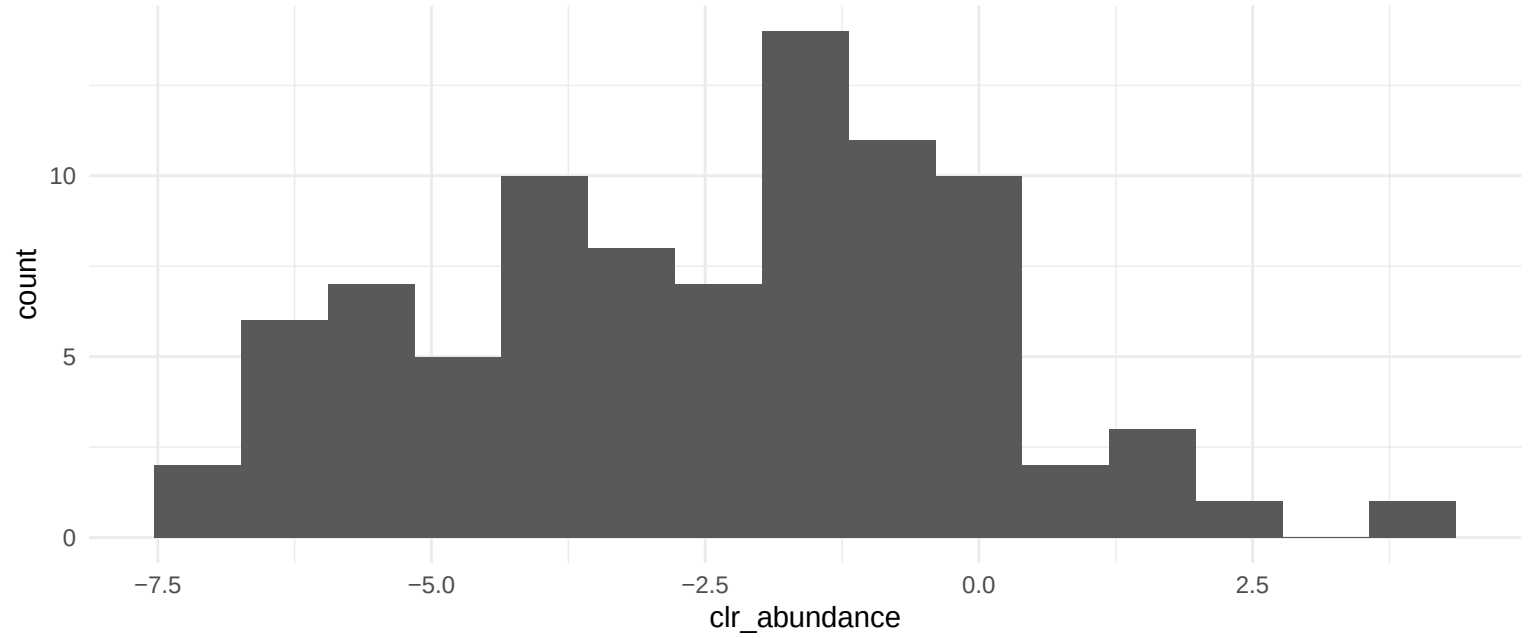

Distribution for 04068 FoxO signaling pathway
count
clr_abundance
0
5
10
15
0
10


Distribution for 04070 Phosphatidylinositol signaling system
count
clr_abundance
0
5
10
15
−10
−5
0
5
10
15

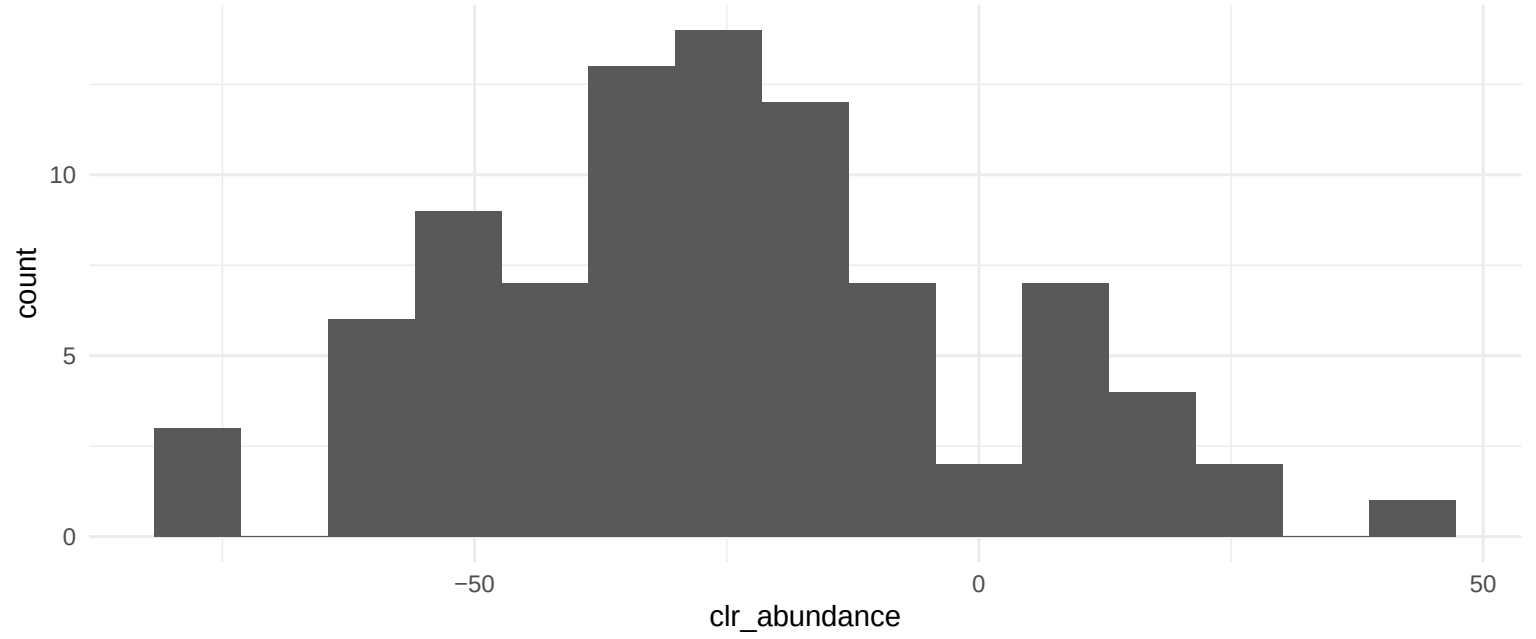
Distribution for 04071 Sphingolipid signaling pathway

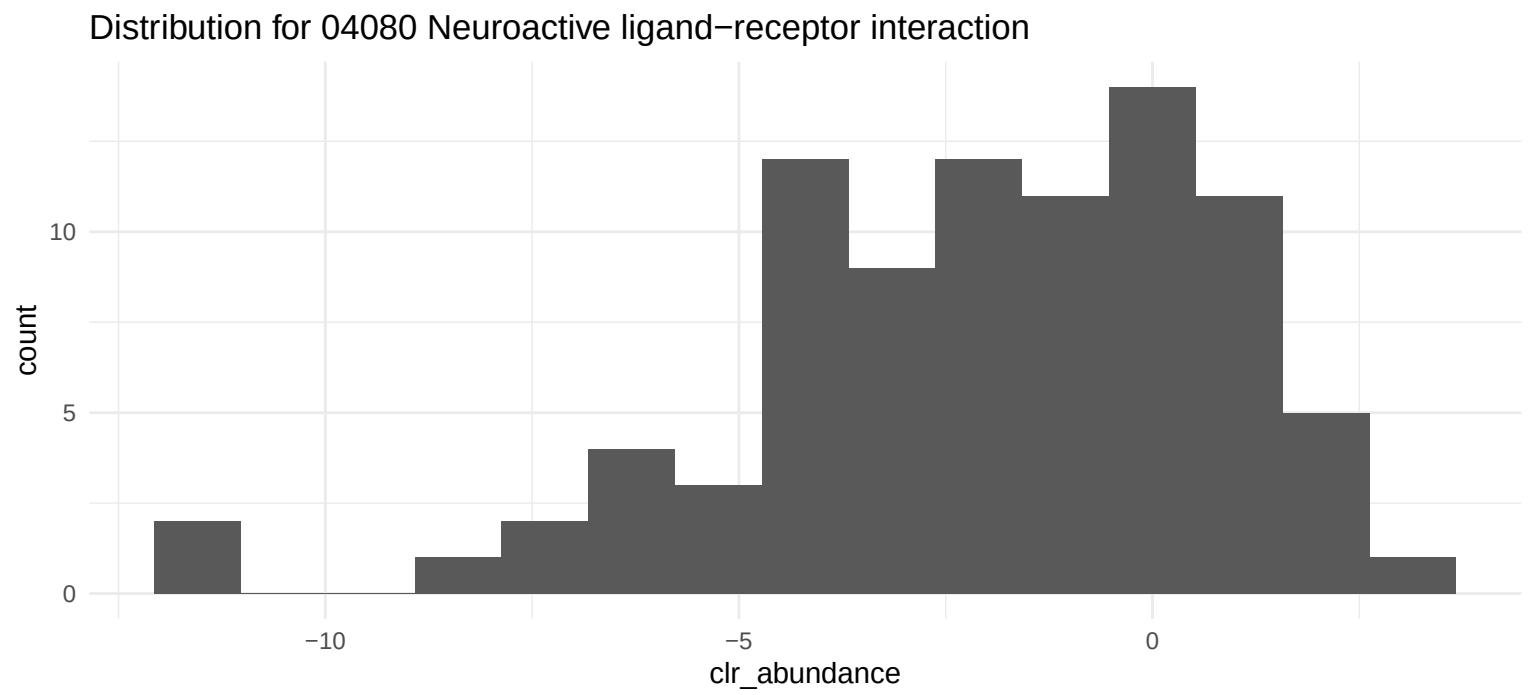


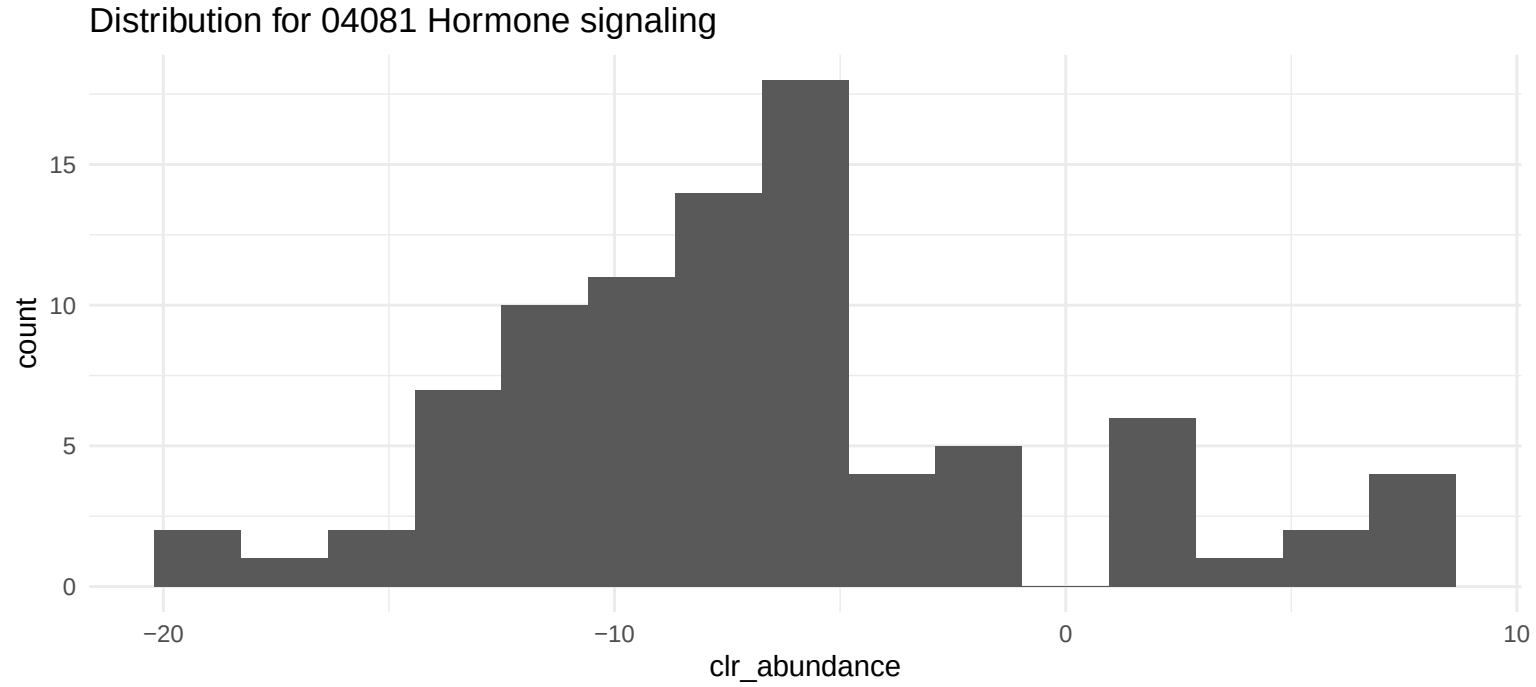
Distribution for 04072 Phospholipase D signaling pathway

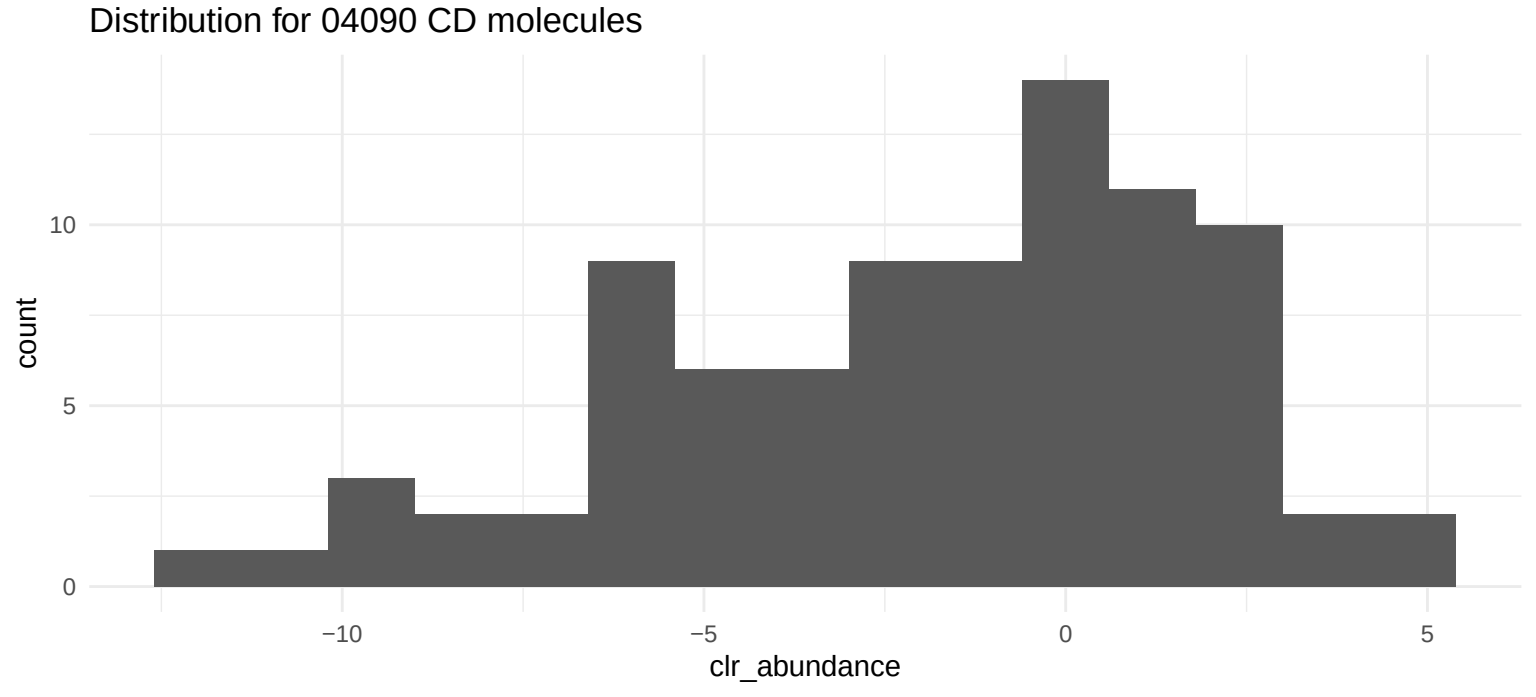


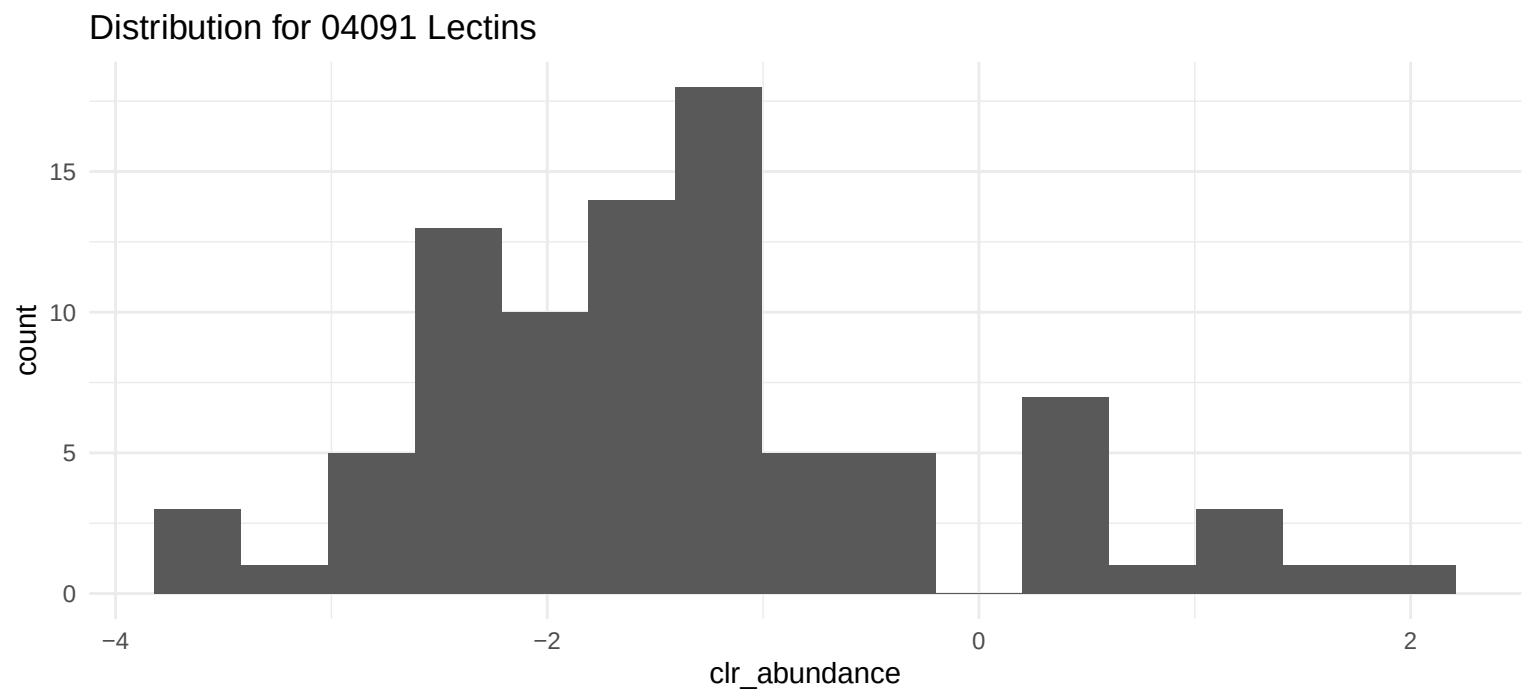
Distribution for 04075 Plant hormone signal transduction

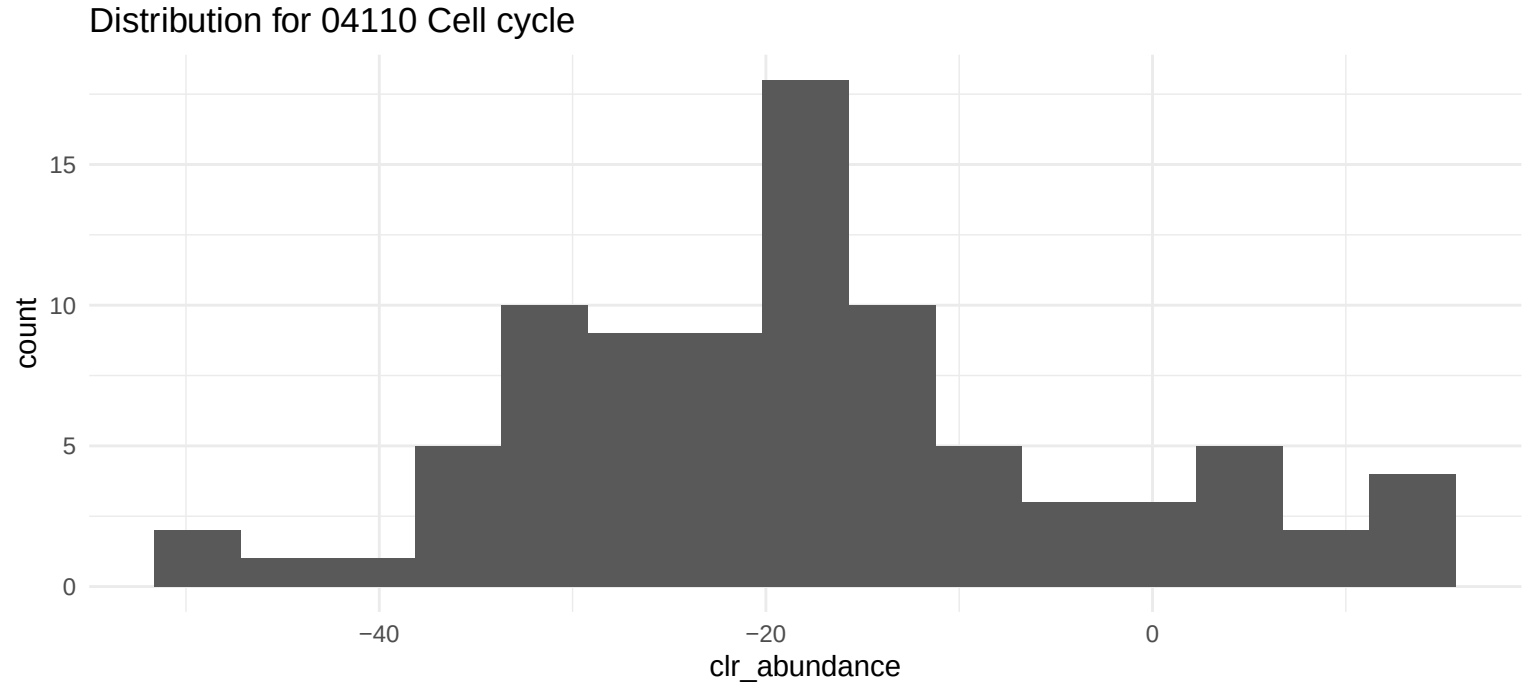


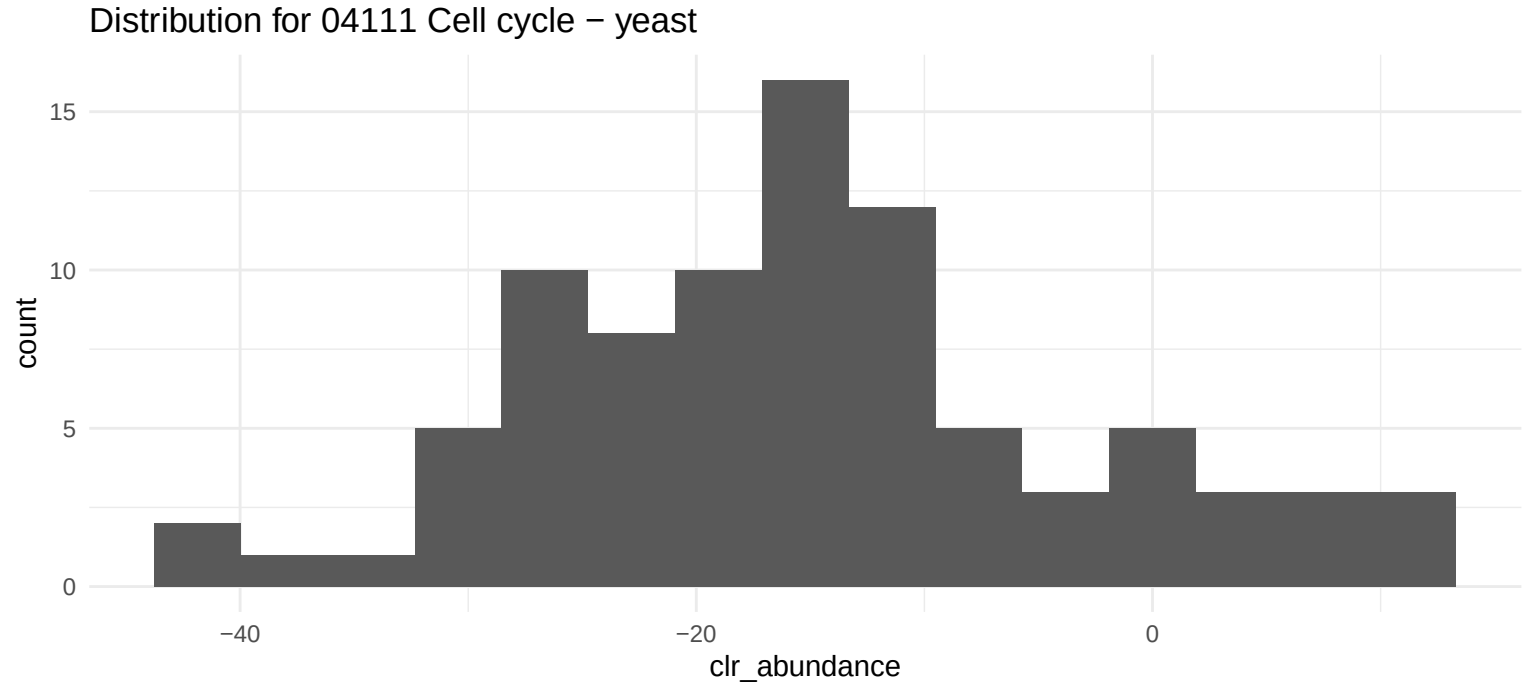

Distribution for 04080 Neuroactive ligand−receptor interaction
count
clr_abundance
0
5
10
−10
−5
0

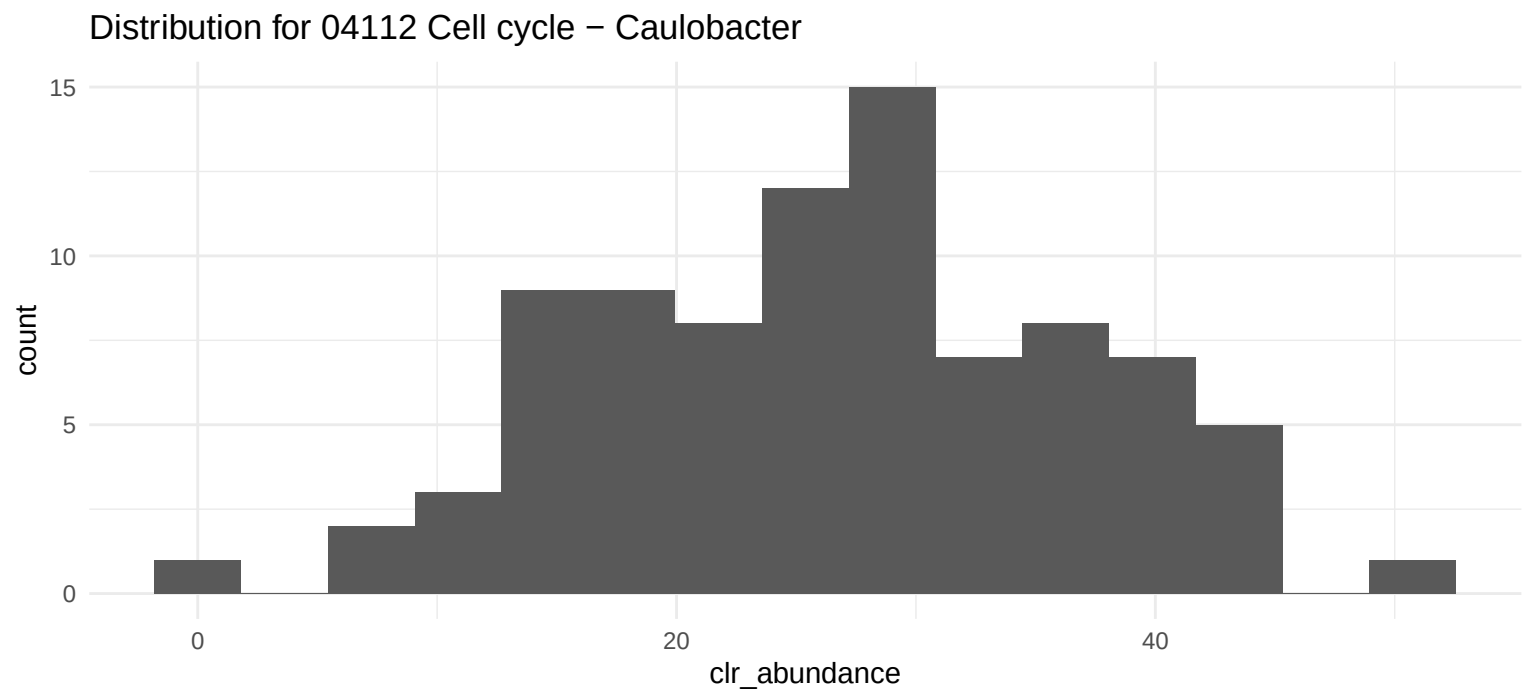

Distribution for 04081 Hormone signaling
count
clr_abundance
0
5
10
15
−20
−10
0
10

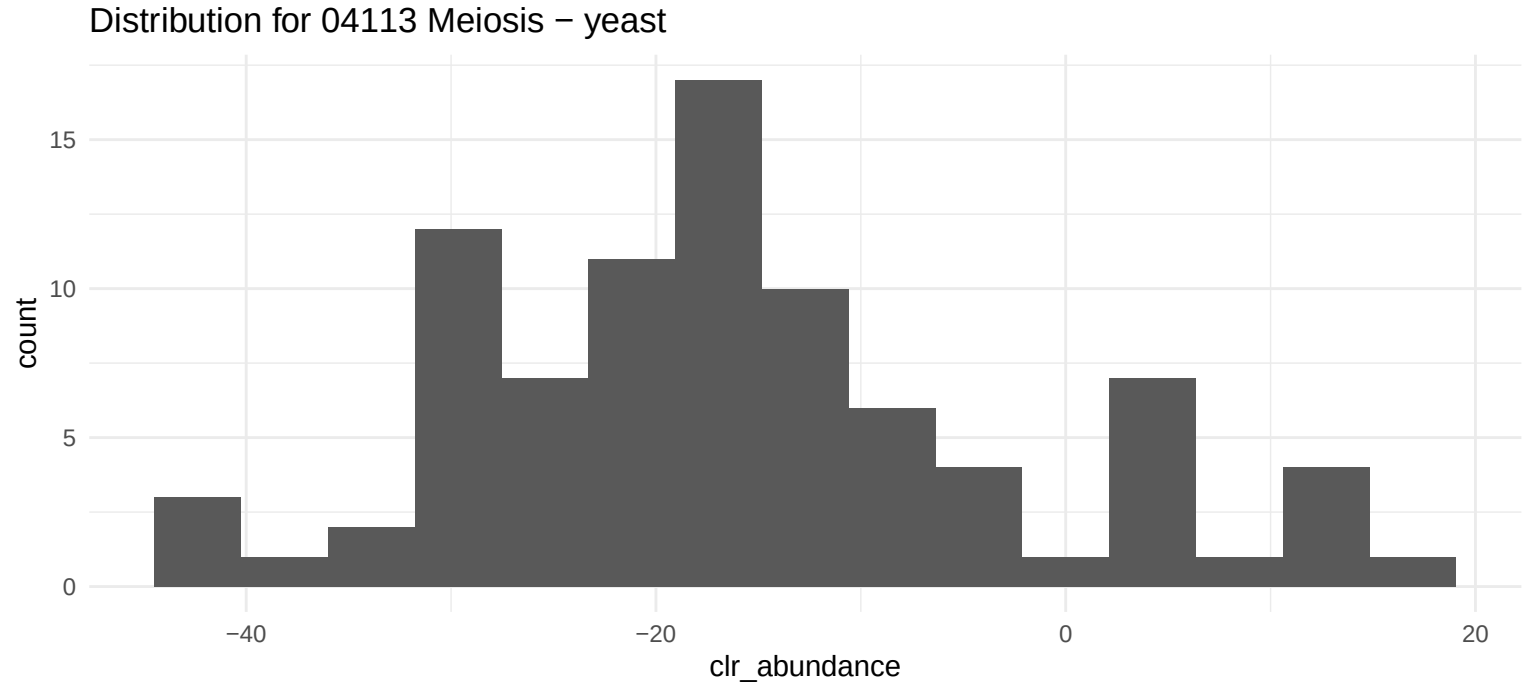

Distribution for 04090 CD molecules
count
clr_abundance
0
5
10
−10
−5
0
5

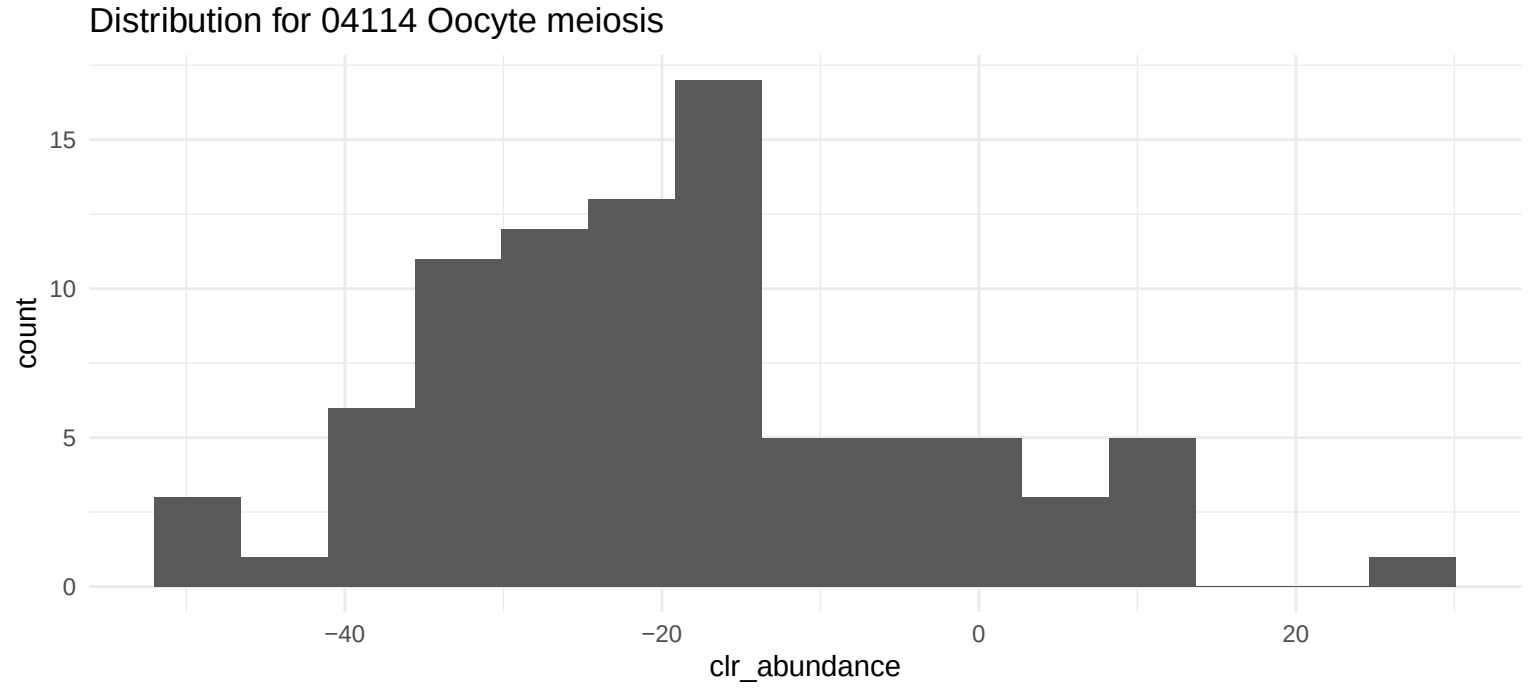

Distribution for 04091 Lectins
count
15
10
5
0
−4
−2
0
2
clr_abundance

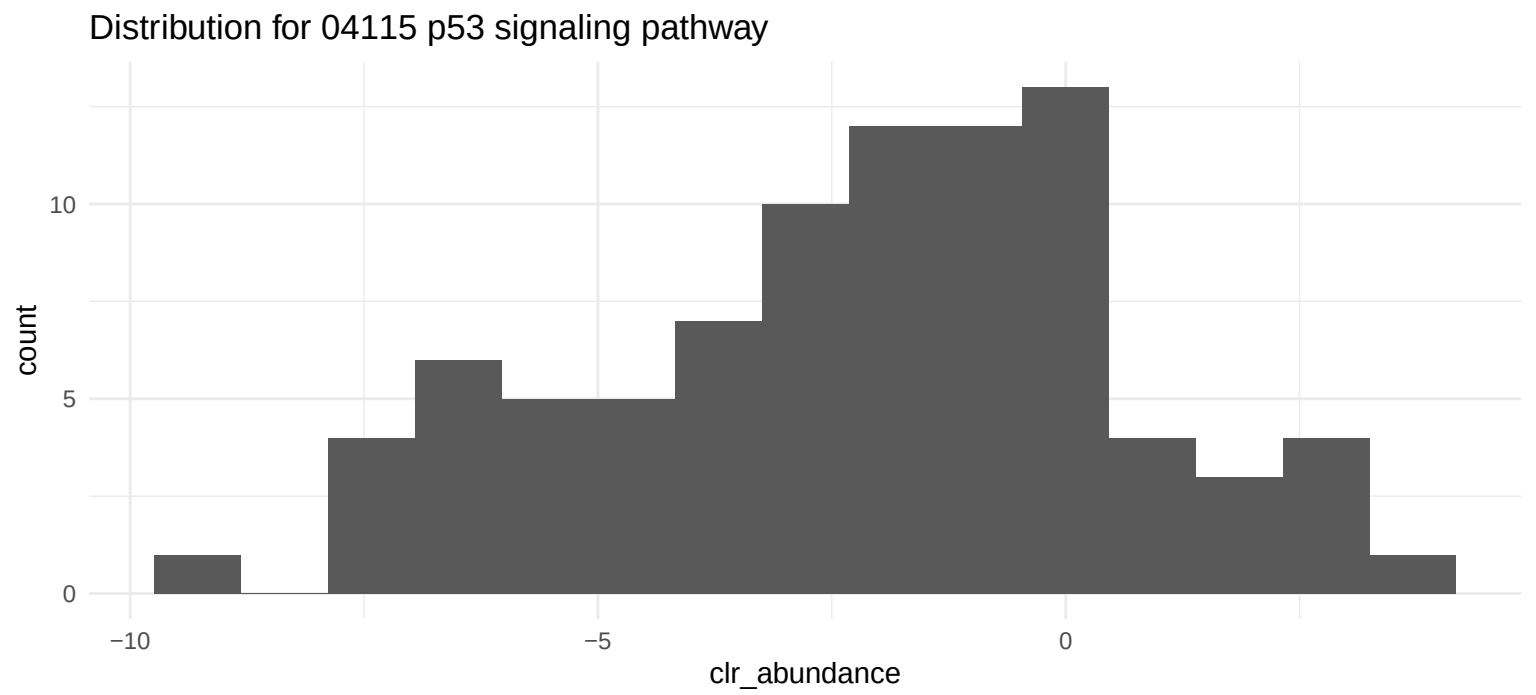

Distribution for 04110 Cell cycle
count
15
10
5
0
−40
−20
0
clr_abundance

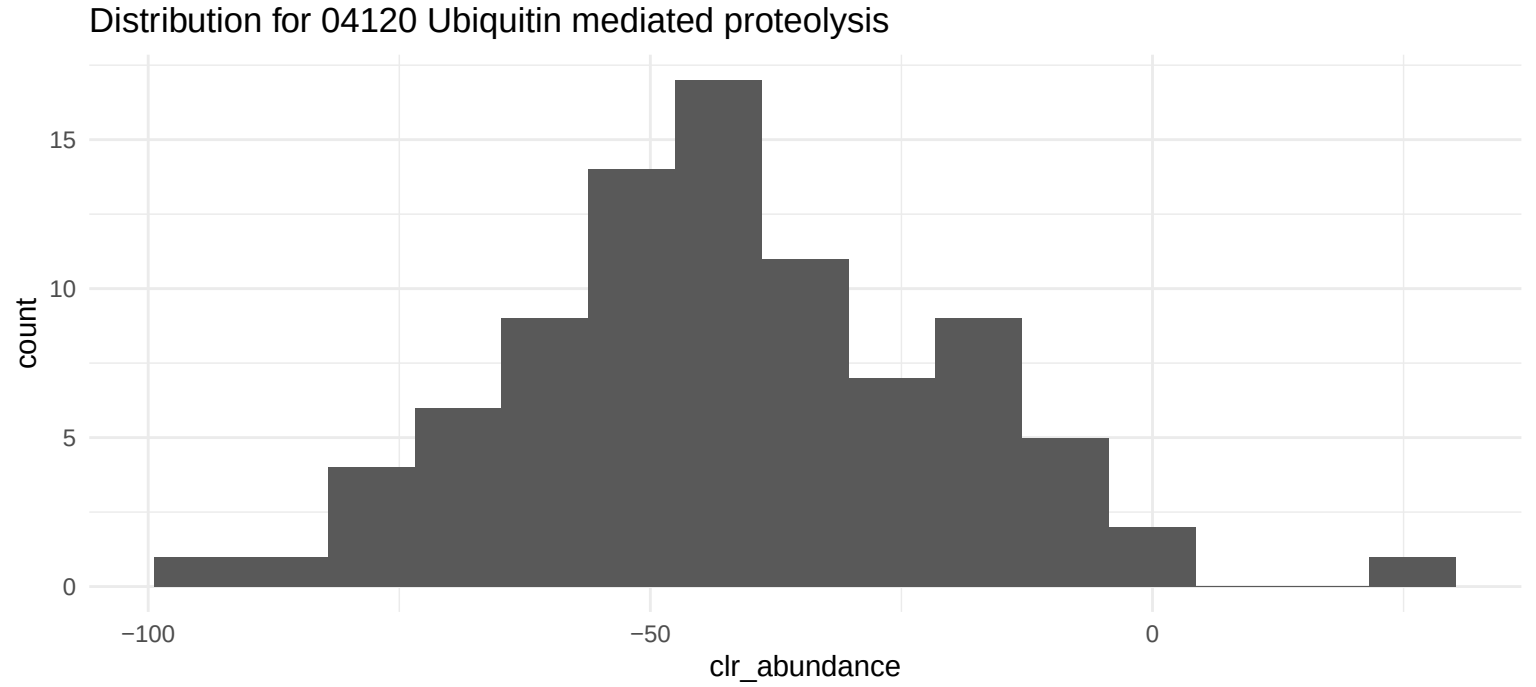

Distribution for 04111 Cell cycle − yeast
count
15
10
5
0
−40
−20
0
clr_abundance

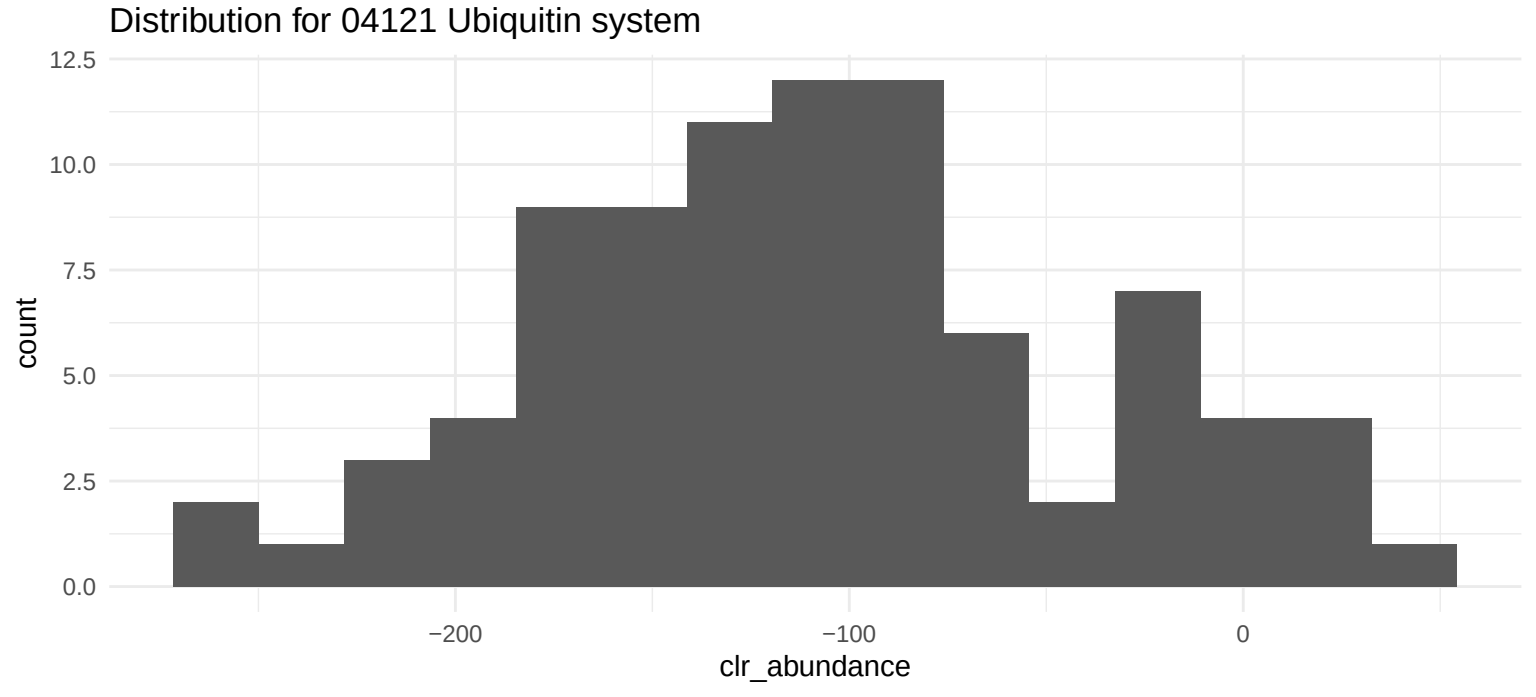

Distribution for 04112 Cell cycle − Caulobacter
count
clr_abundance
0
5
10
15
0
20
40


Distribution for 04113 Meiosis − yeast
count
clr_abundance
0
5
10
15
−40
−20
0
20

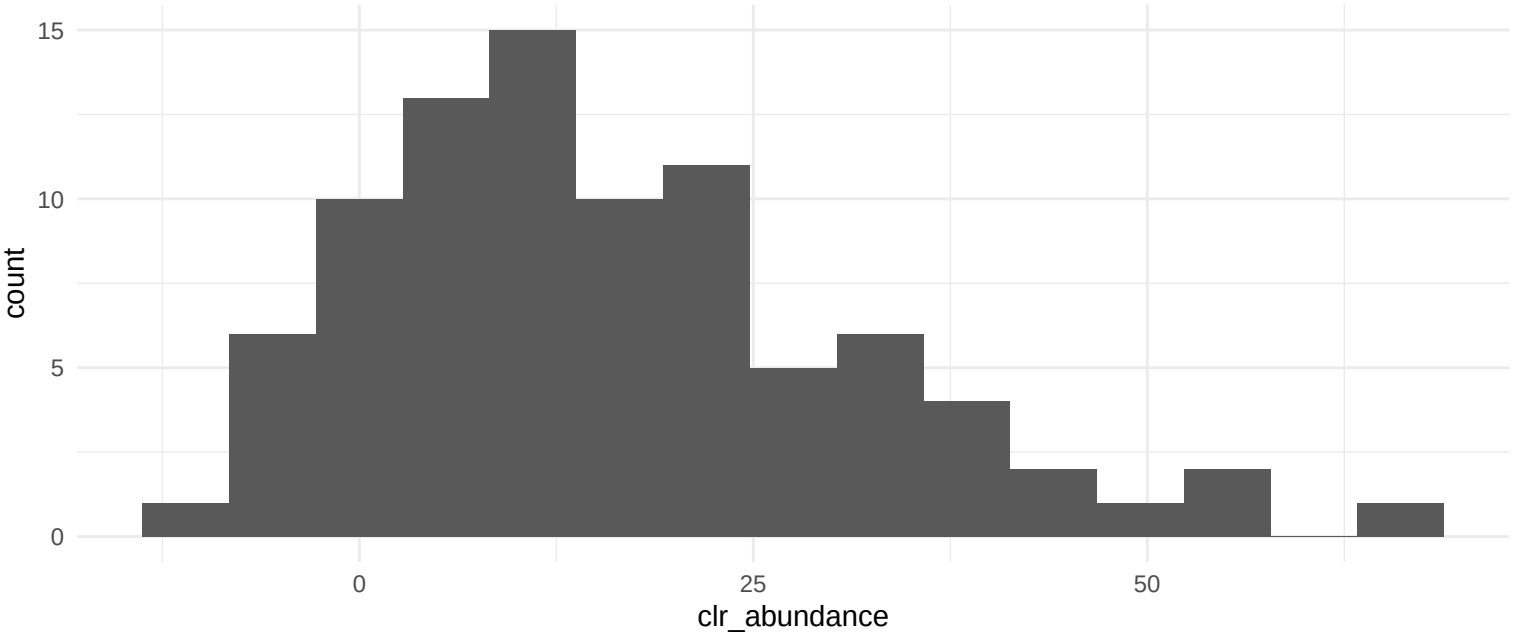

Distribution for 04114 Oocyte meiosis
count
clr_abundance
0
5
10
15
−40
−20
0
20


Distribution for 04115 p53 signaling pathway
count
10
5
0
−10
−5
0
clr_abundance

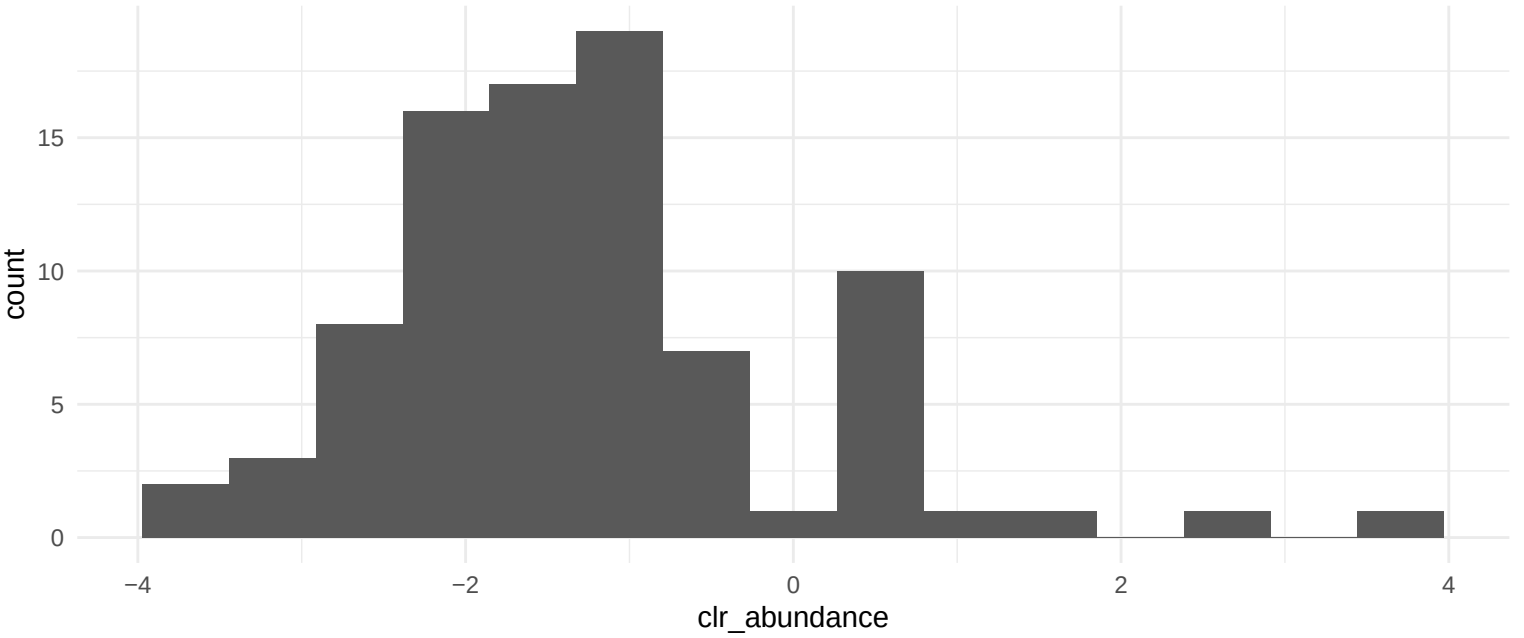

Distribution for 04120 Ubiquitin mediated proteolysis
count
15
10
5
0
−100
−50
0
clr_abundance


Distribution for 04121 Ubiquitin system
count
12.5
10.0
7.5
5.0
2.5
0.0
−200
−100
0
clr_abundance

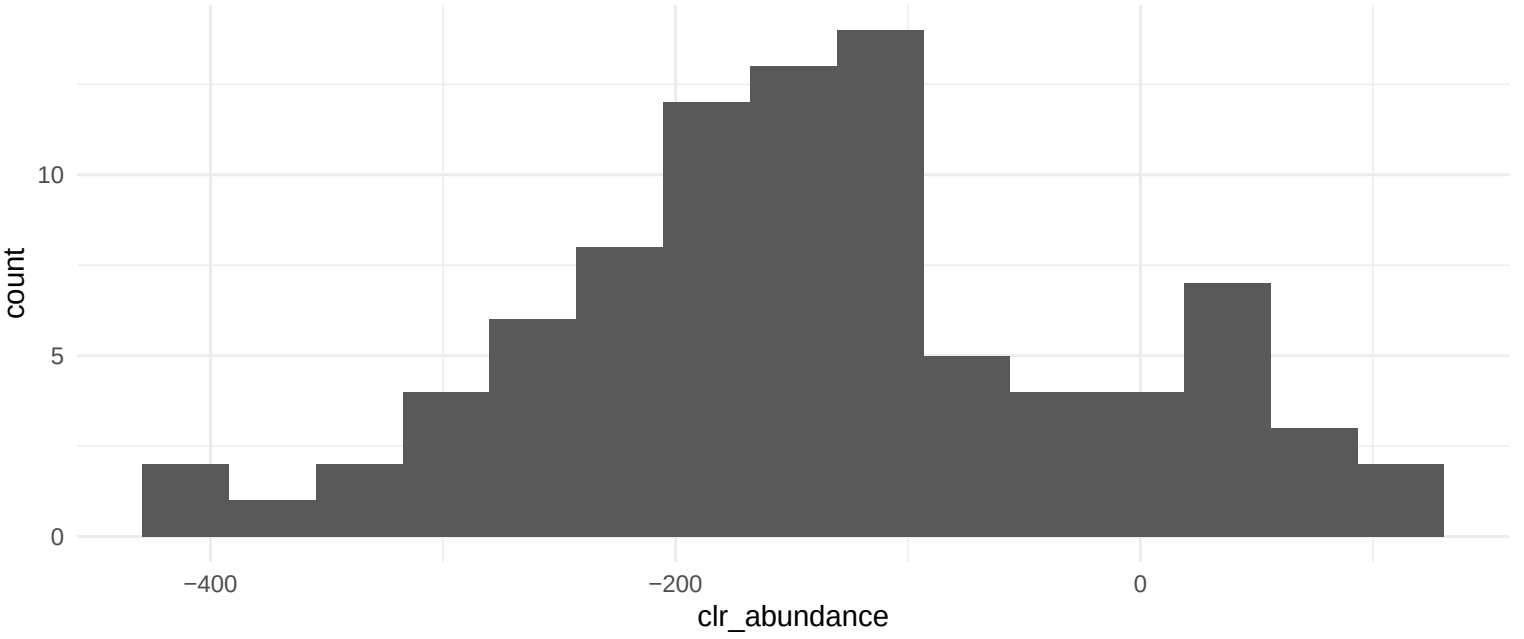
Distribution for 04122 Sulfur relay system

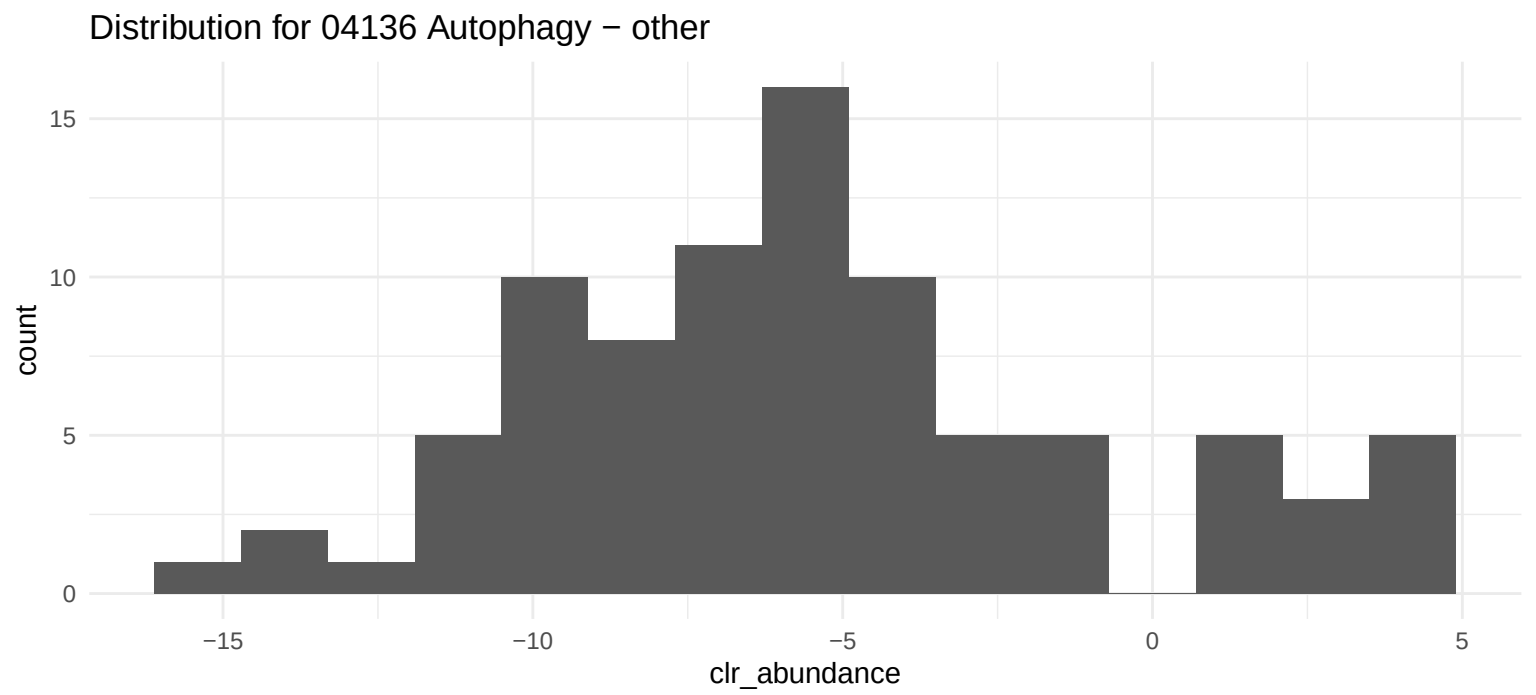


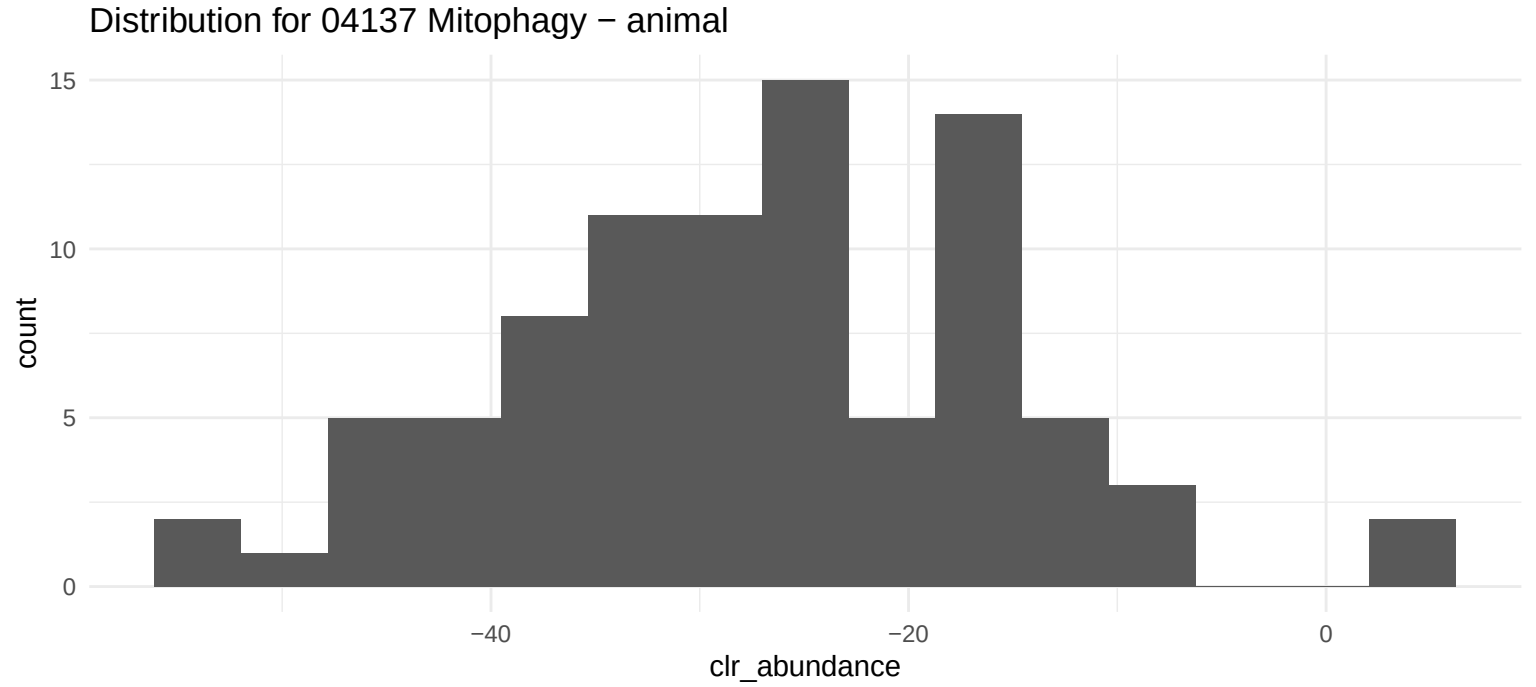
Distribution for 04130 SNARE interactions in vesicular transport

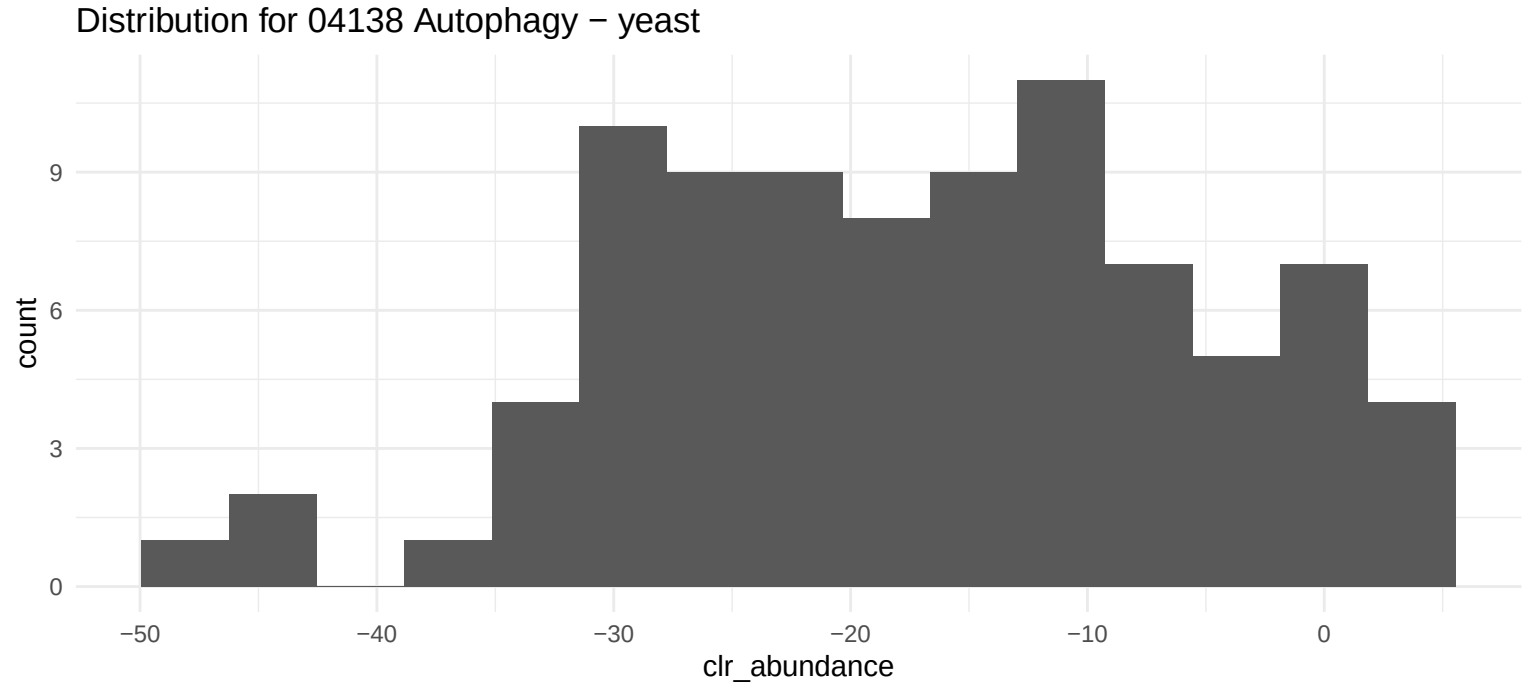


Distribution for 04131 Membrane trafficking

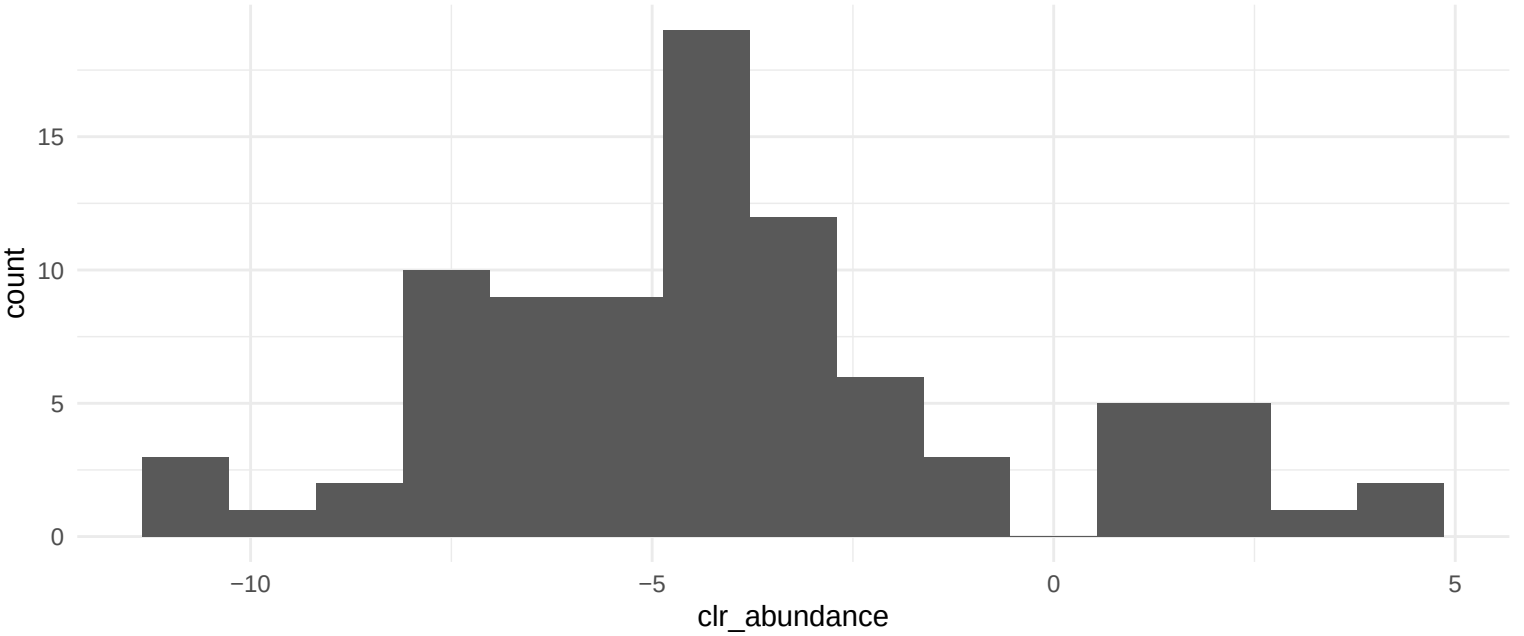



Distribution for 04136 Autophagy − other
count
clr_abundance
15
10
5
0
−15
−10
−5
0
5

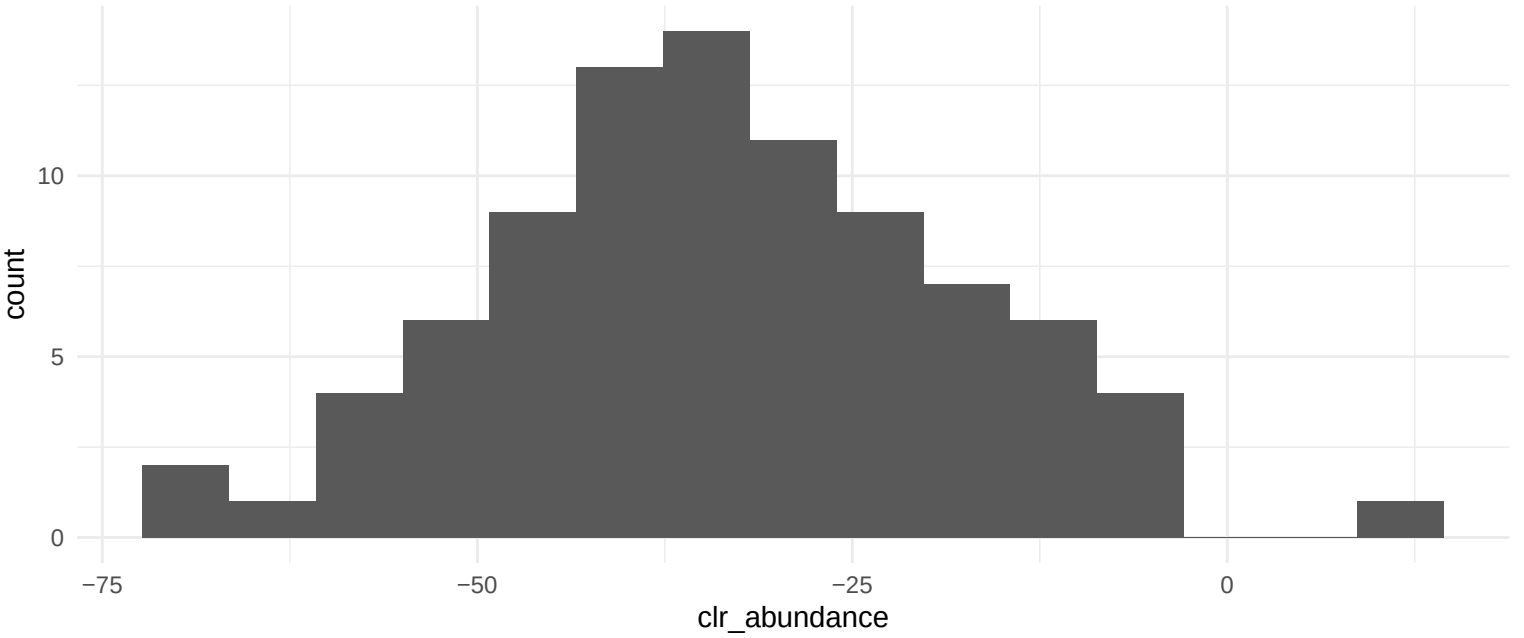

Distribution for 04137 Mitophagy − animal
count
clr_abundance
15
10
5
0
−40
−20
0


Distribution for 04138 Autophagy − yeast
count
clr_abundance
9
6
3
0
−50
−40
−30
−20
−10
0

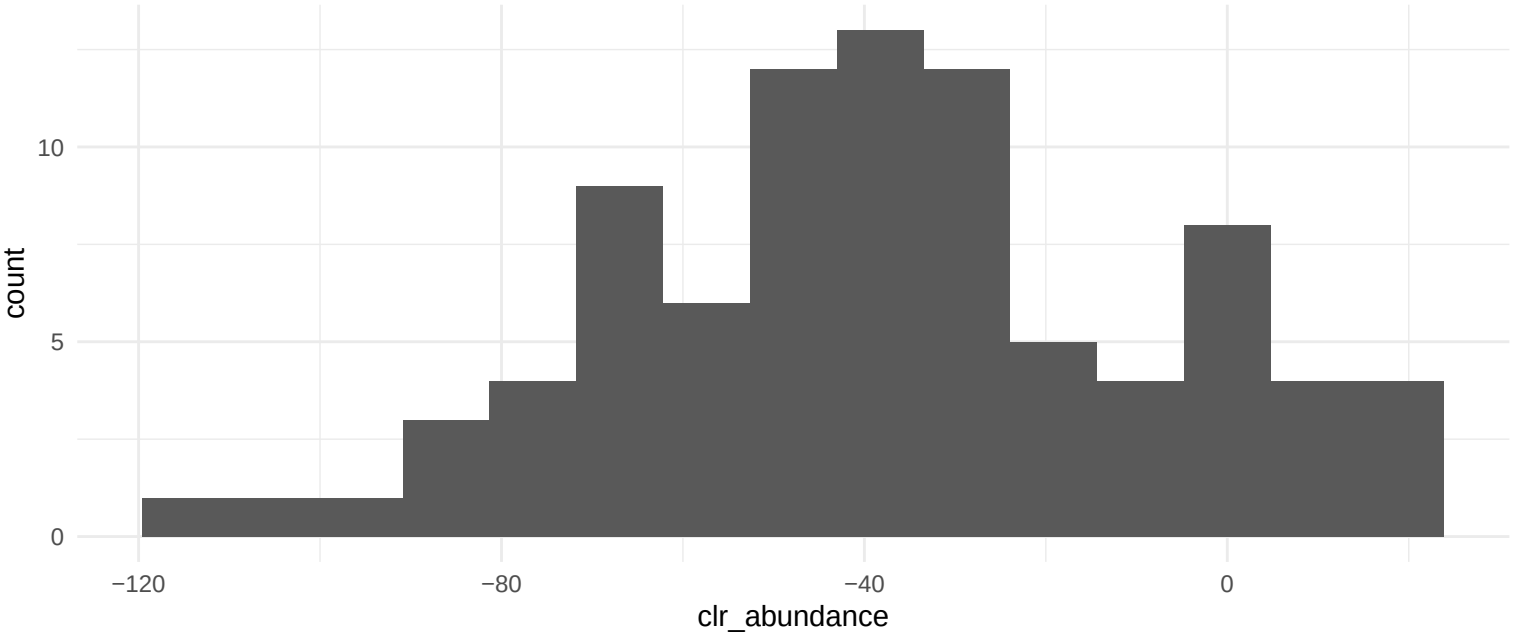
Distribution for 04139 Mitophagy − yeast

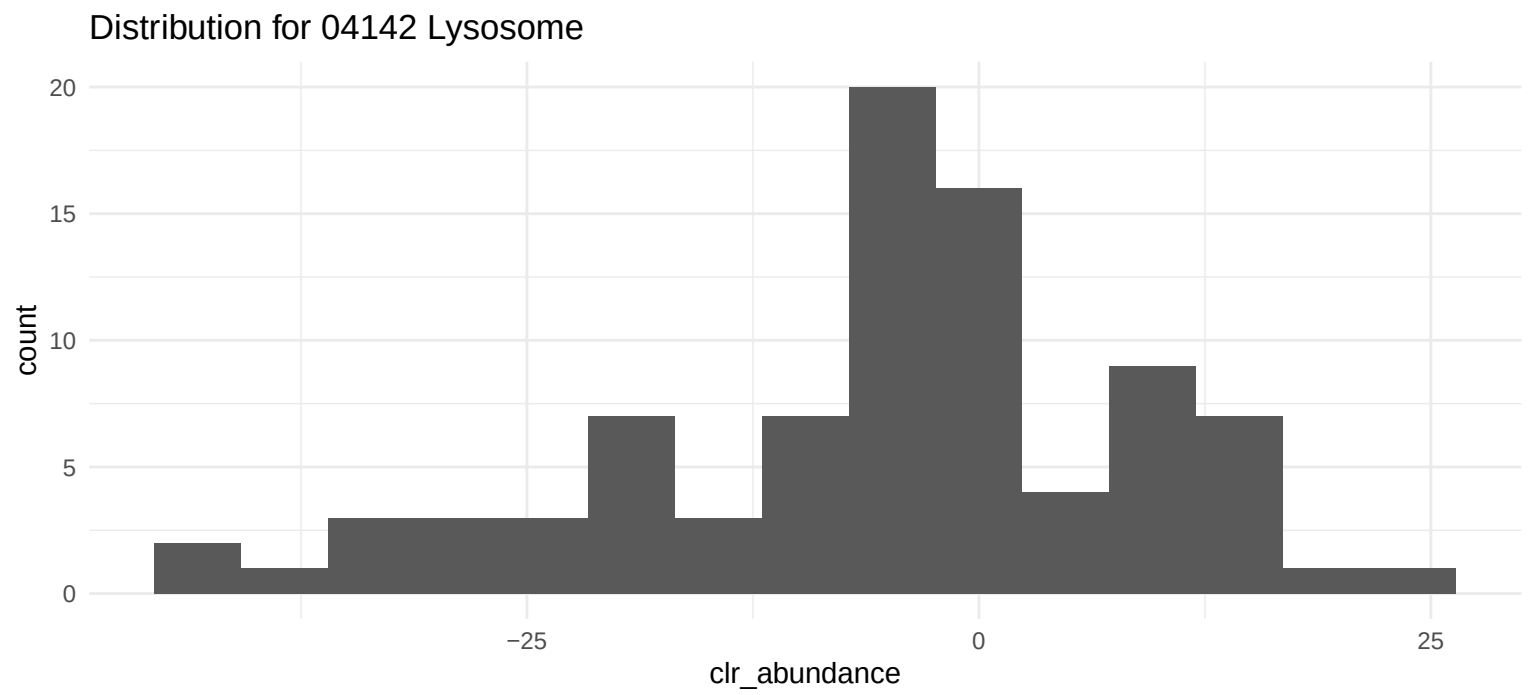


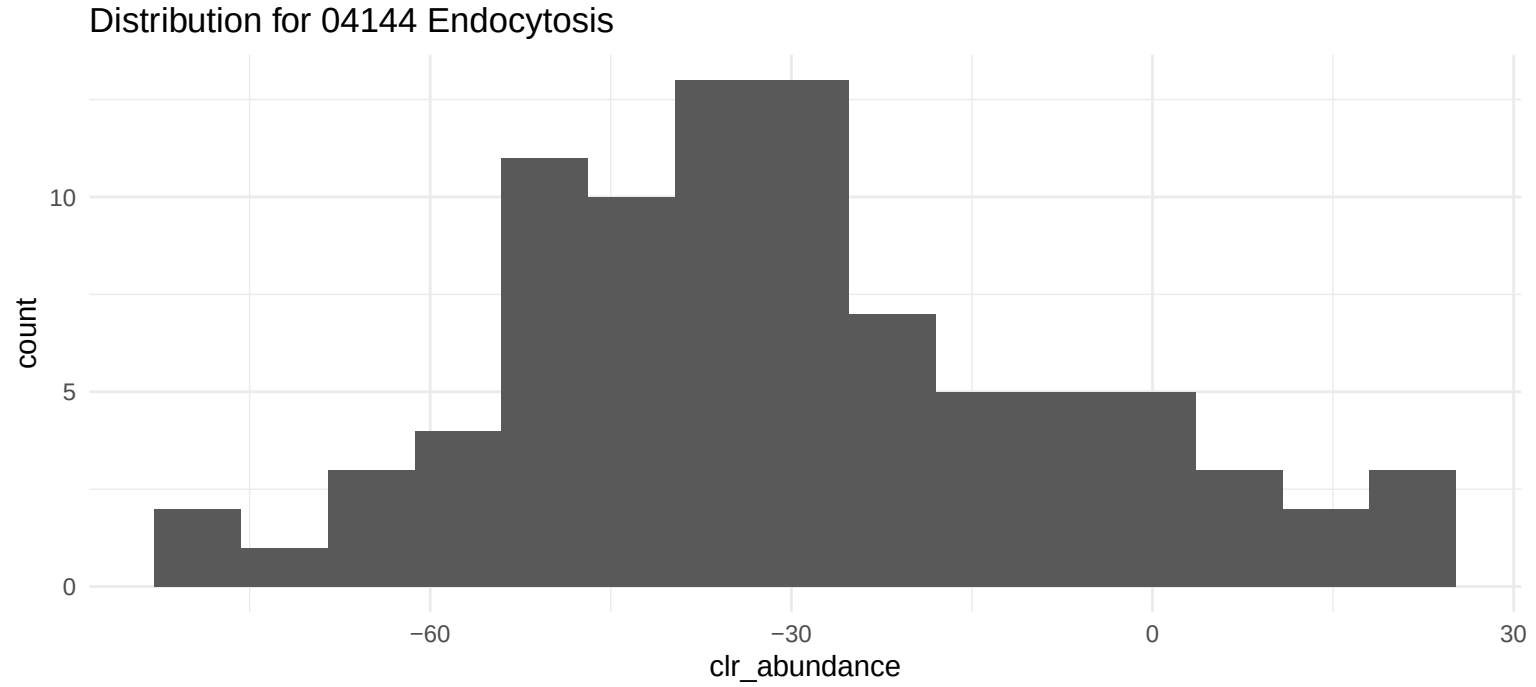
Distribution for 04140 Autophagy − animal

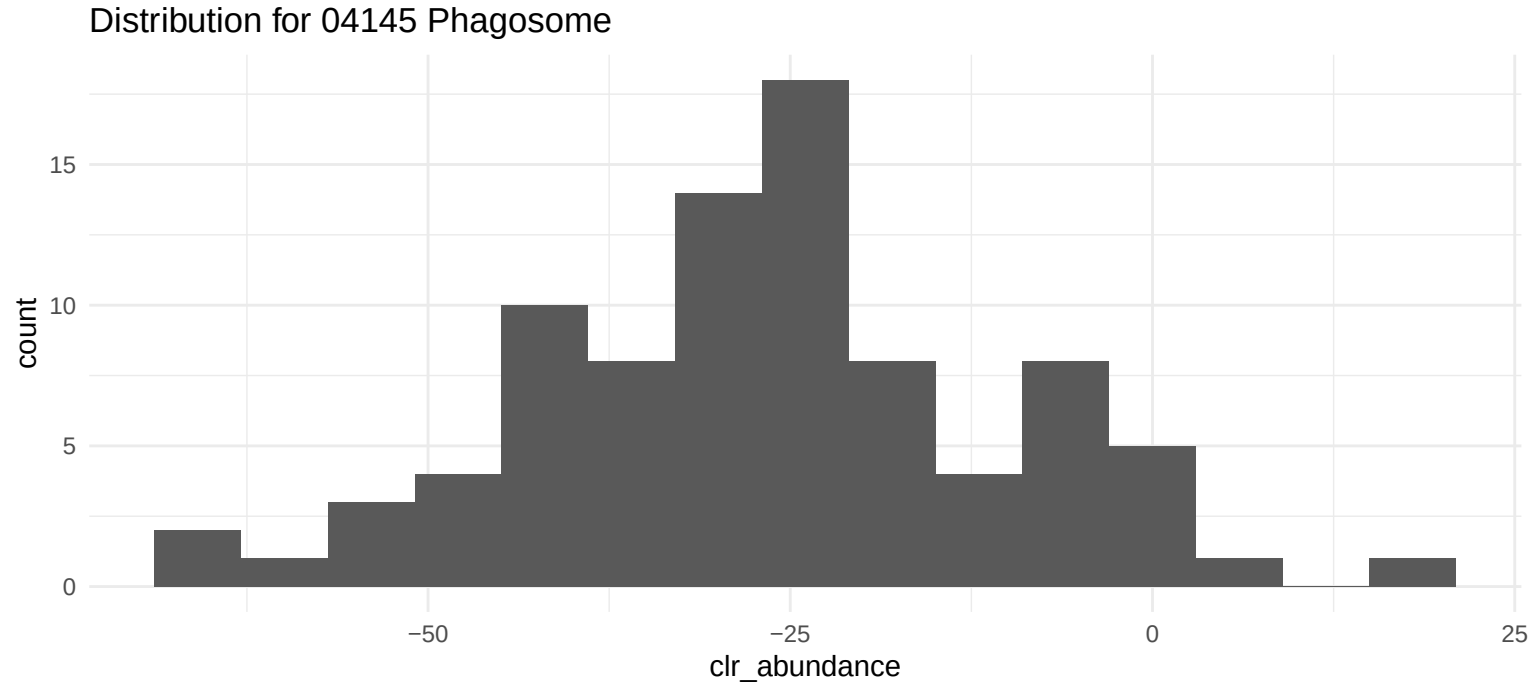


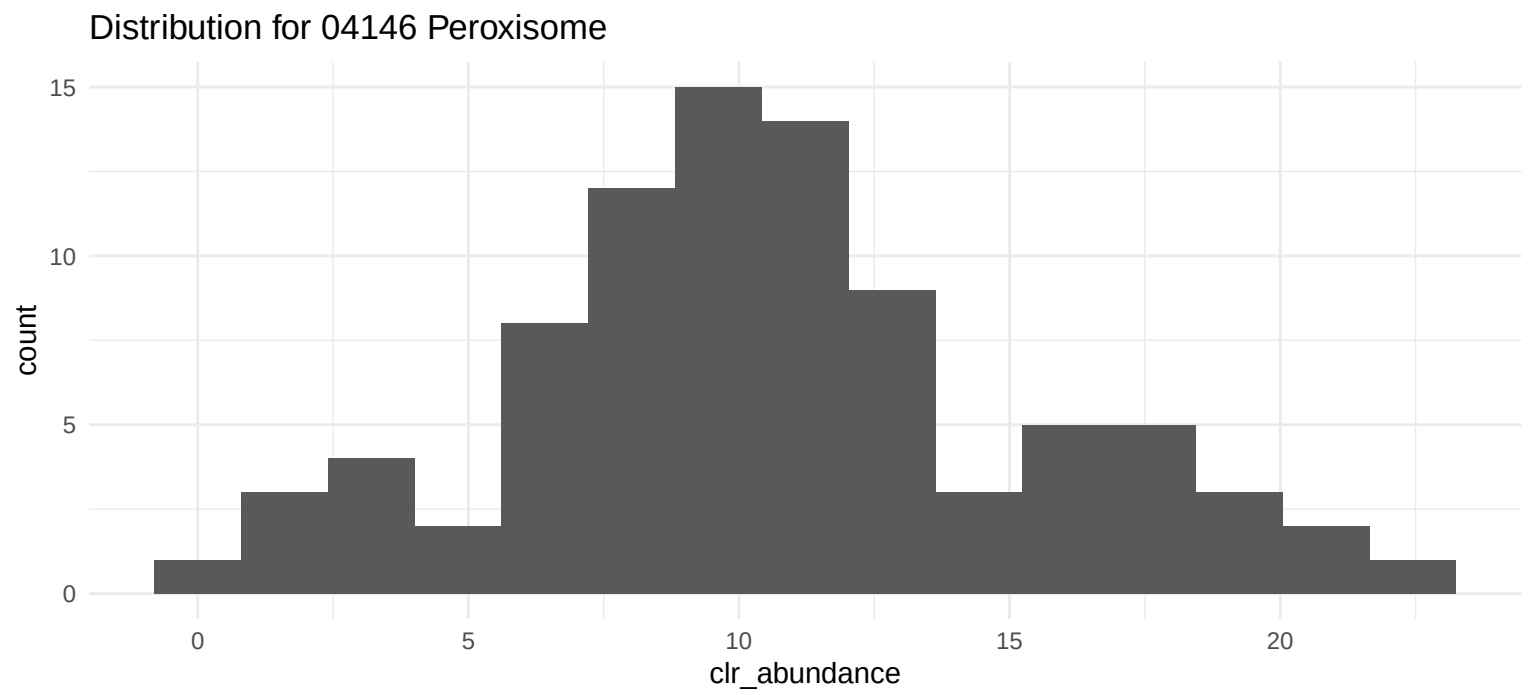
Distribution for 04141 Protein processing in endoplasmic reticulum

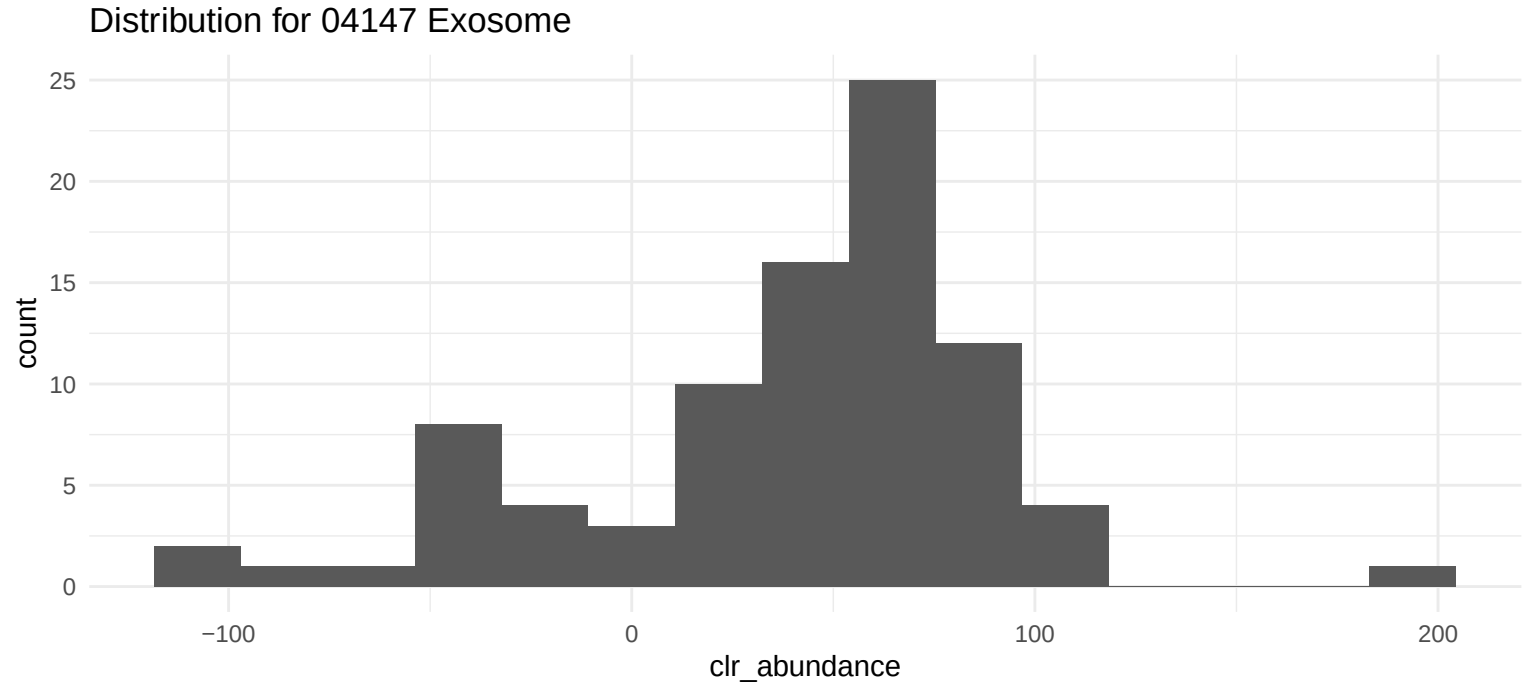


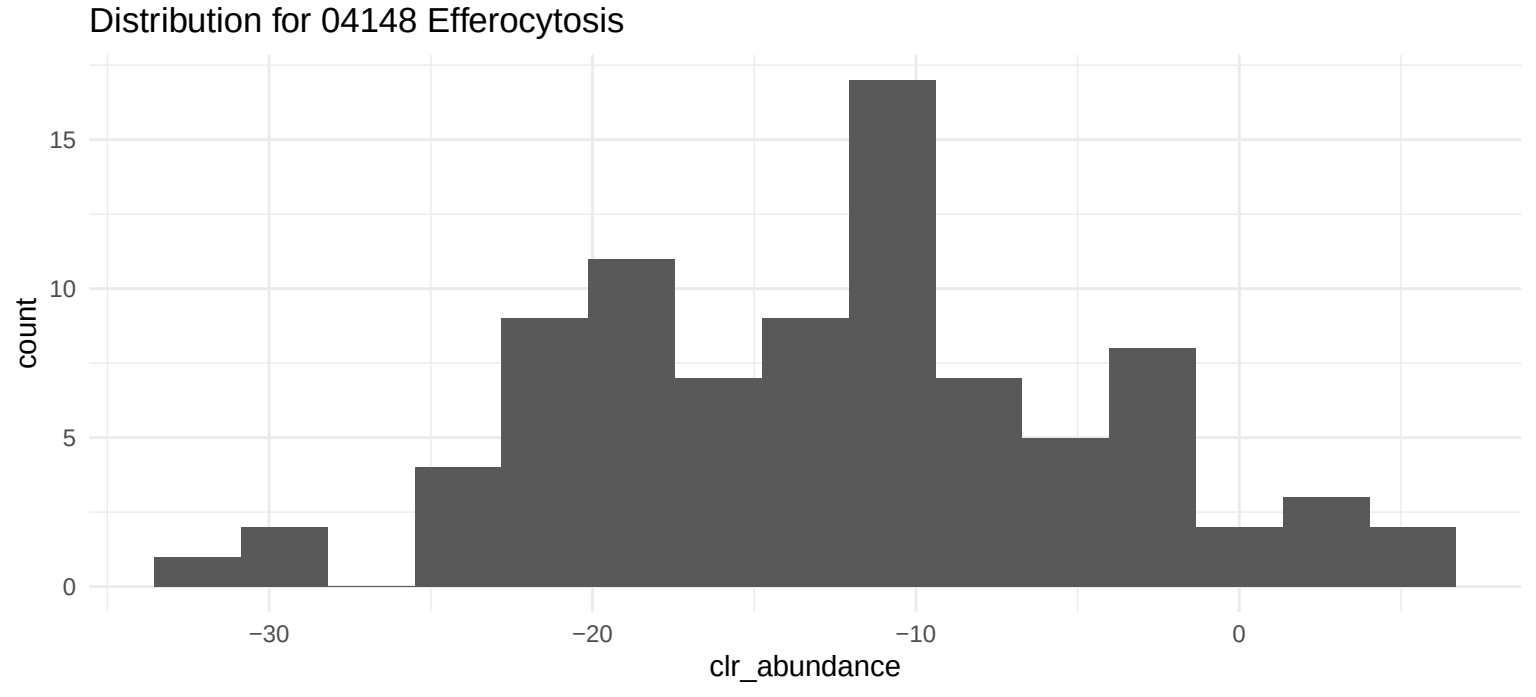

Distribution for 04142 Lysosome
count
clr_abundance
0
5
10
15
20
−25
0
25

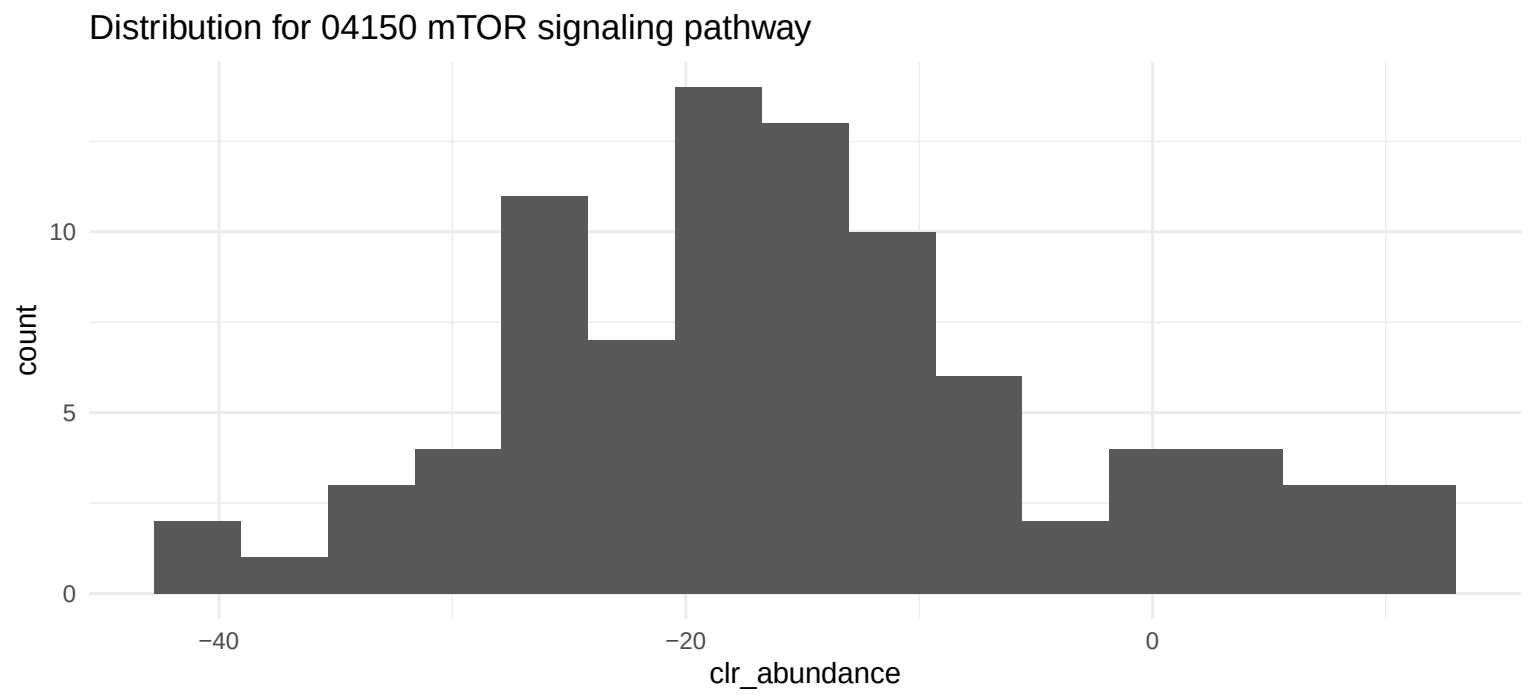

Distribution for 04144 Endocytosis
count
clr_abundance
0
5
10
−60
−30
0
30

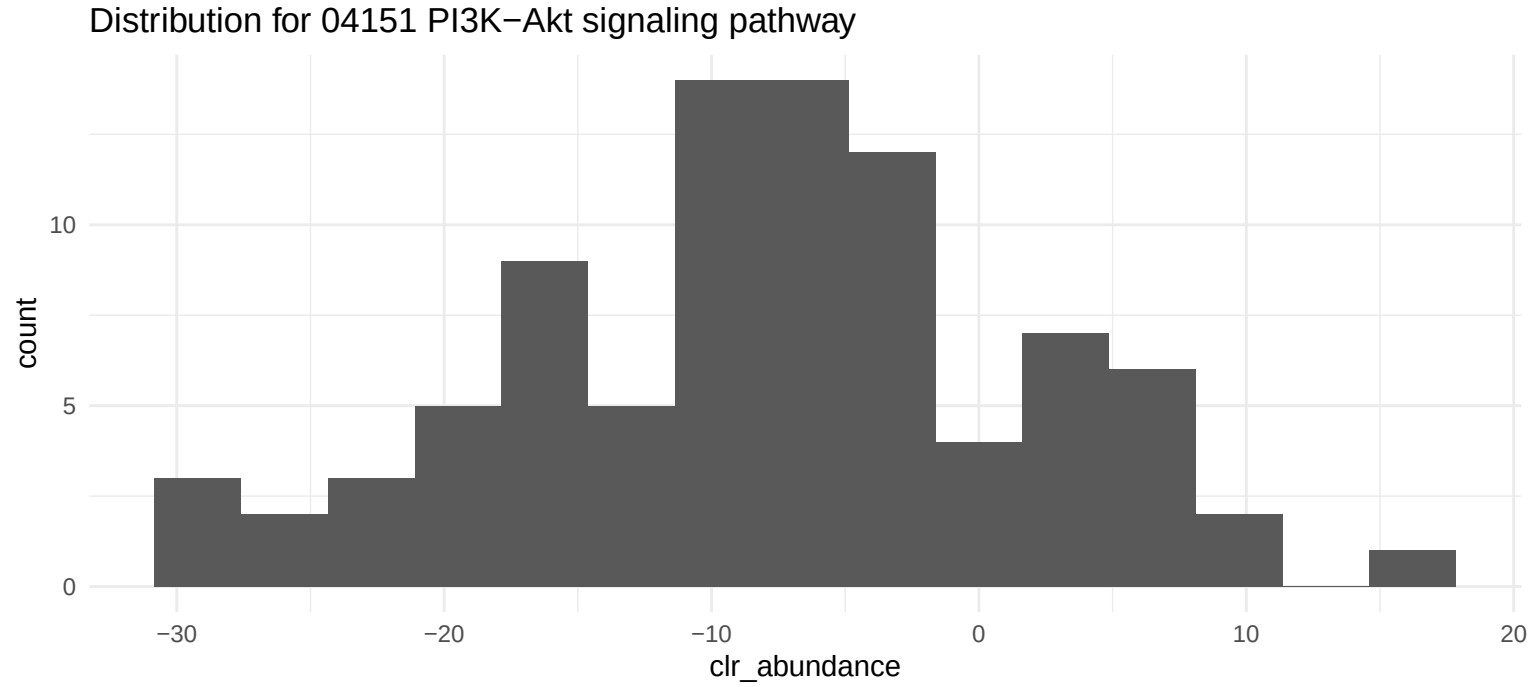

Distribution for 04145 Phagosome
count
clr_abundance
0
5
10
15
−50
−25
0
25

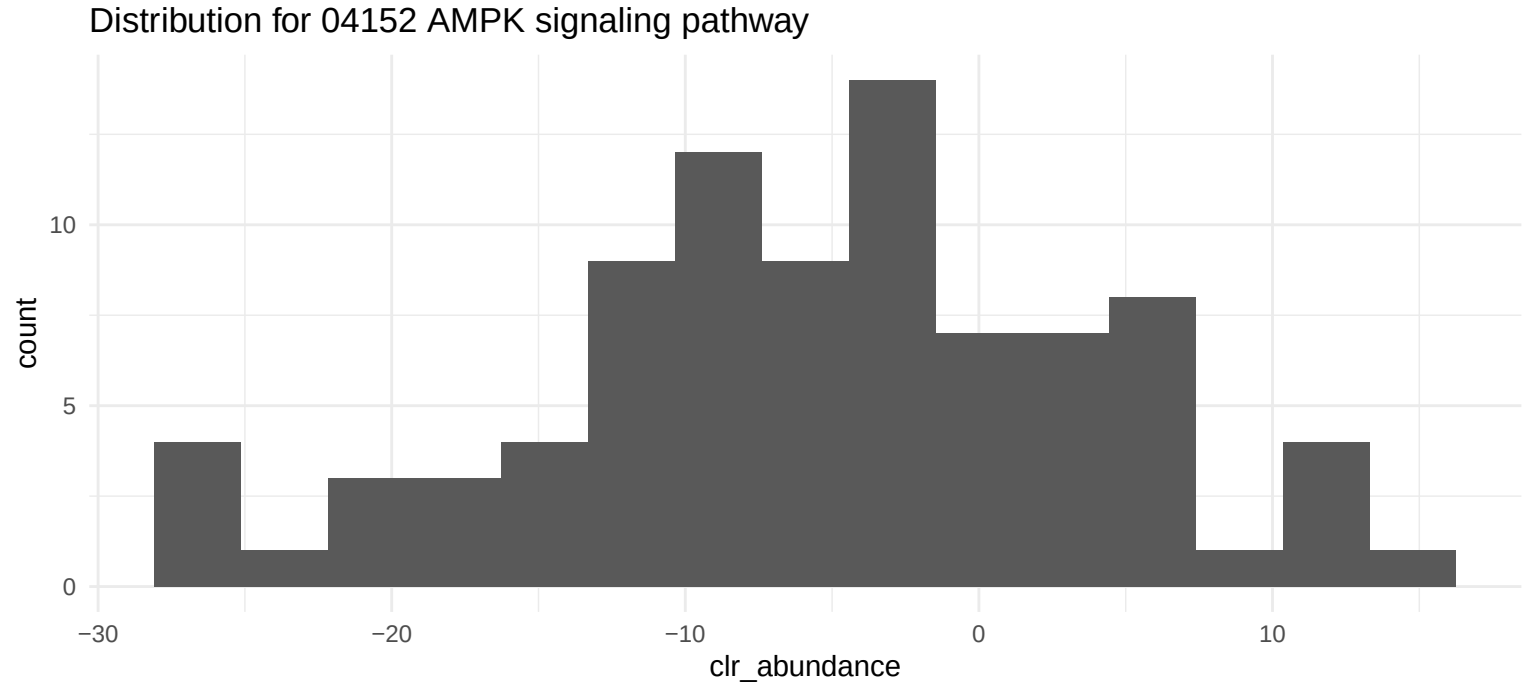

Distribution for 04146 Peroxisome
count
clr_abundance
0
5
10
15
20


Distribution for 04147 Exosome
count
clr_abundance
−100
0
100
200

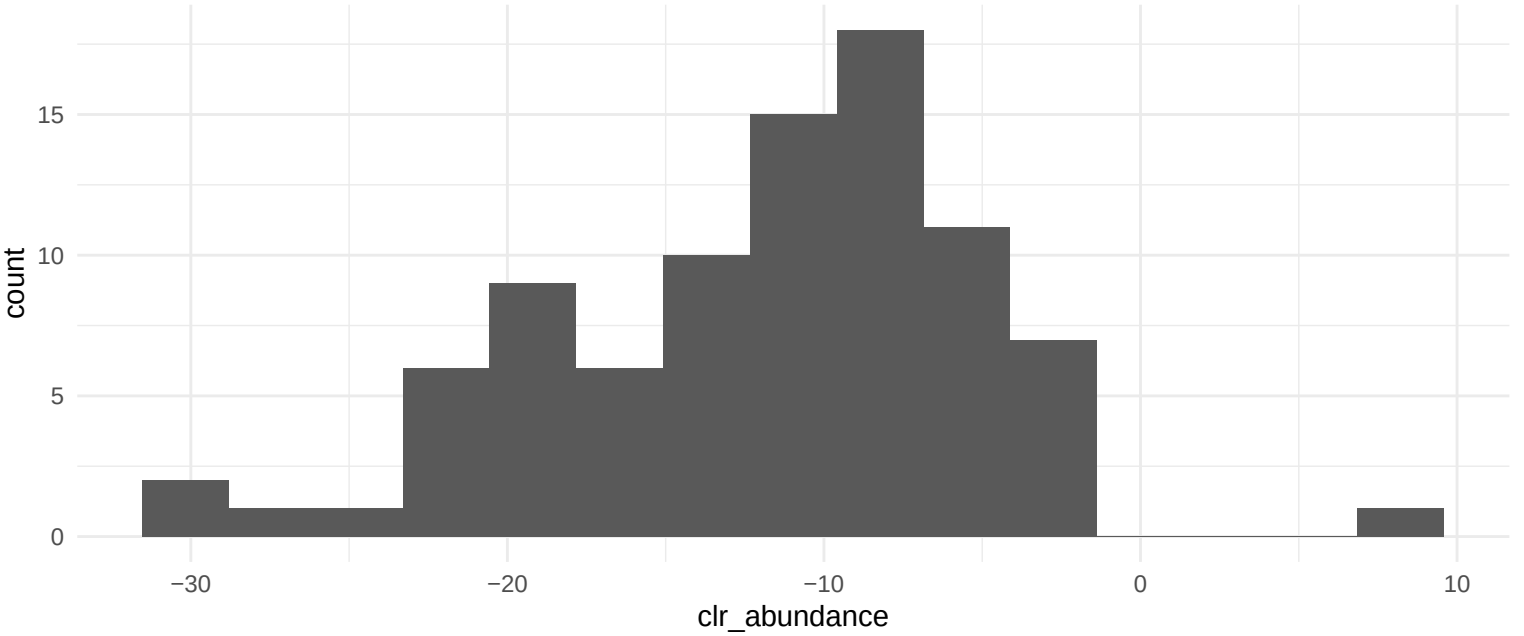

Distribution for 04148 Efferocytosis
count
clr_abundance
−30
−20
−10
0


Distribution for 04150 mTOR signaling pathway
count
10
5
0
−40
−20
0
clr_abundance

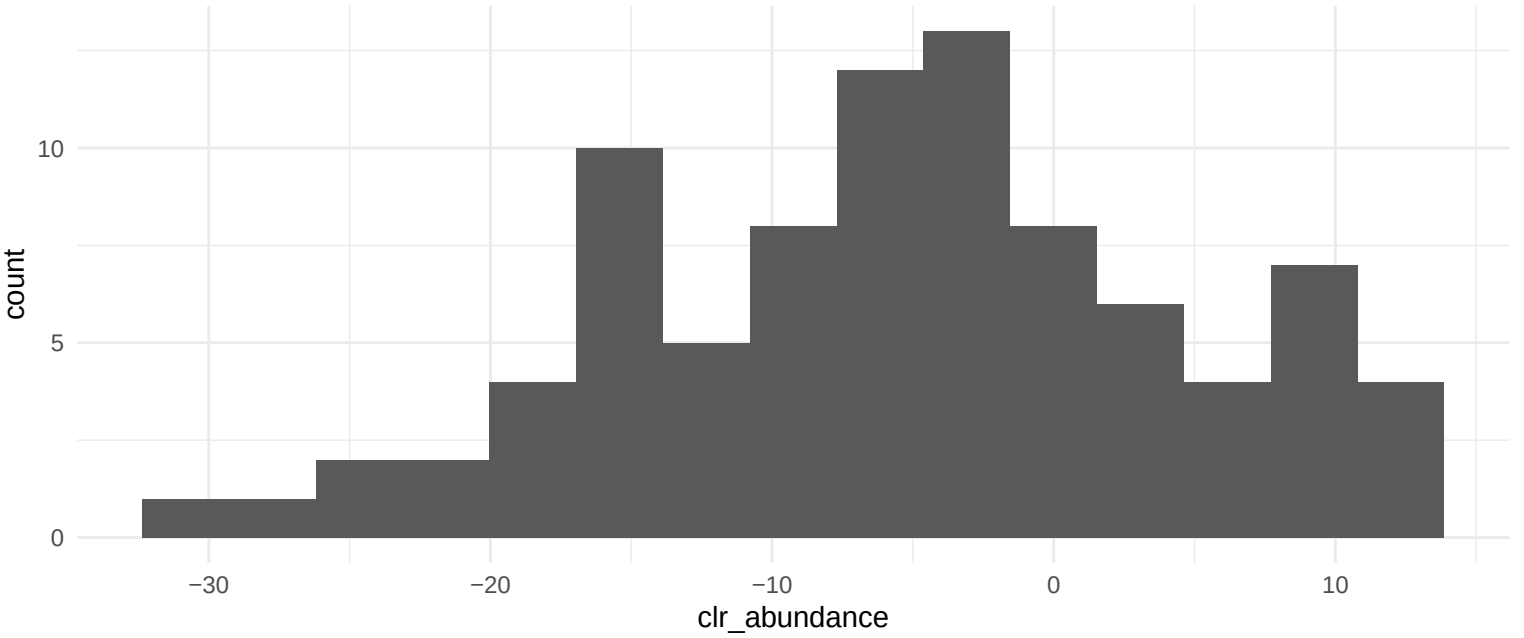

Distribution for 04151 PI3K−Akt signaling pathway
count
10
5
0
−30
−20
−10
0
10
20
clr_abundance


Distribution for 04152 AMPK signaling pathway
count
10
5
0
−30
−20
−10
0
10
clr_abundance

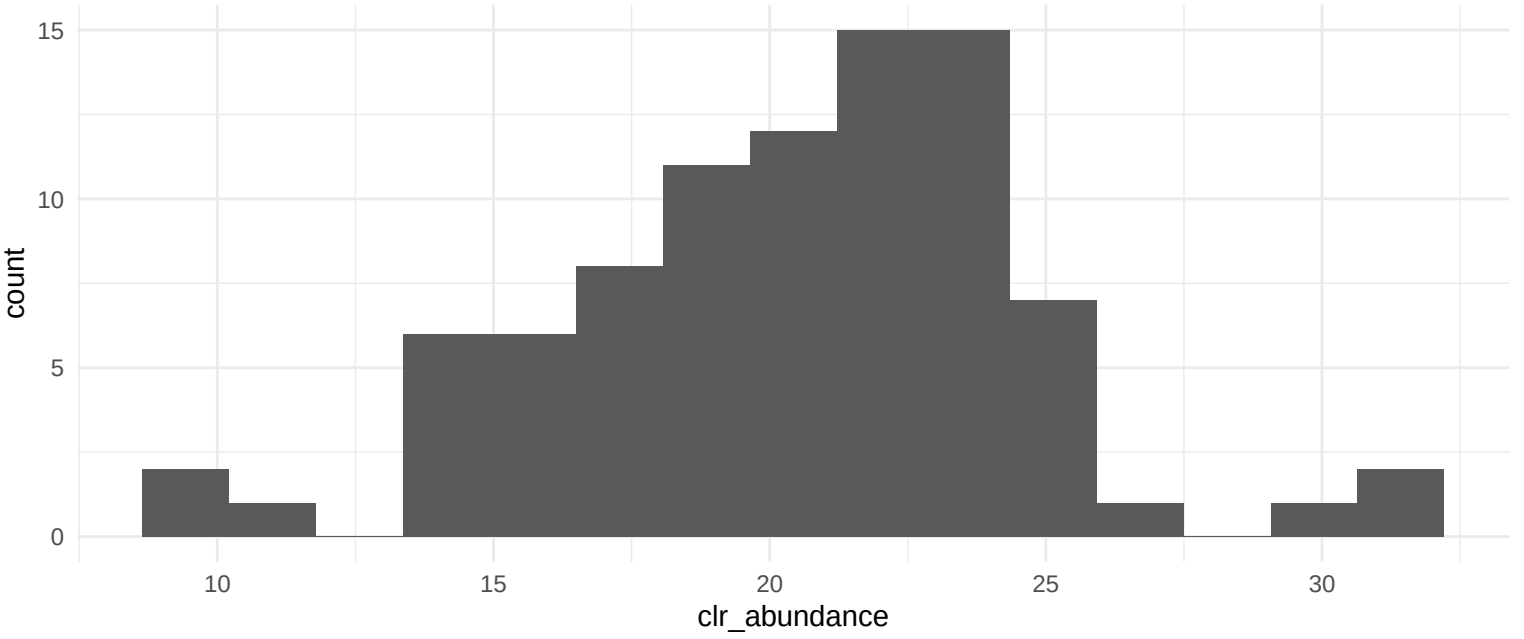
Distribution for 04210 Apoptosis

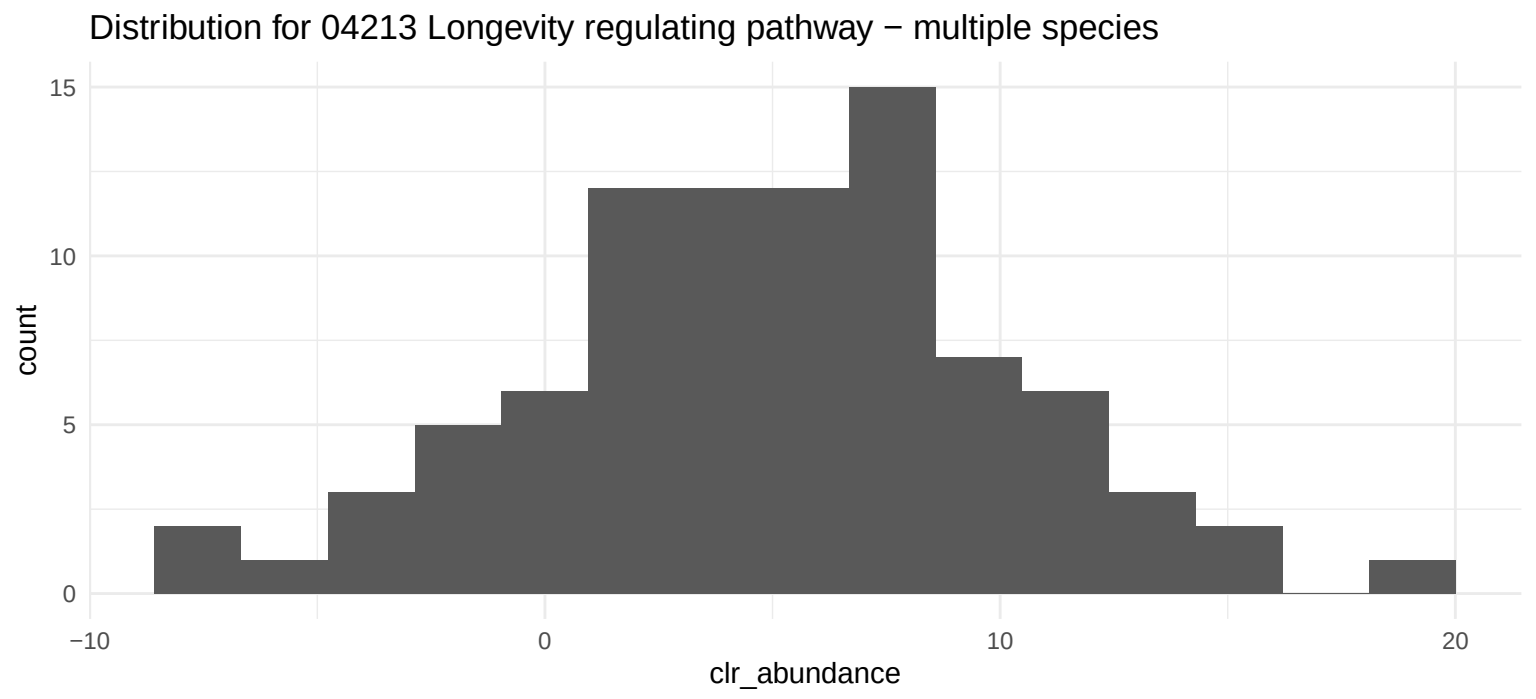


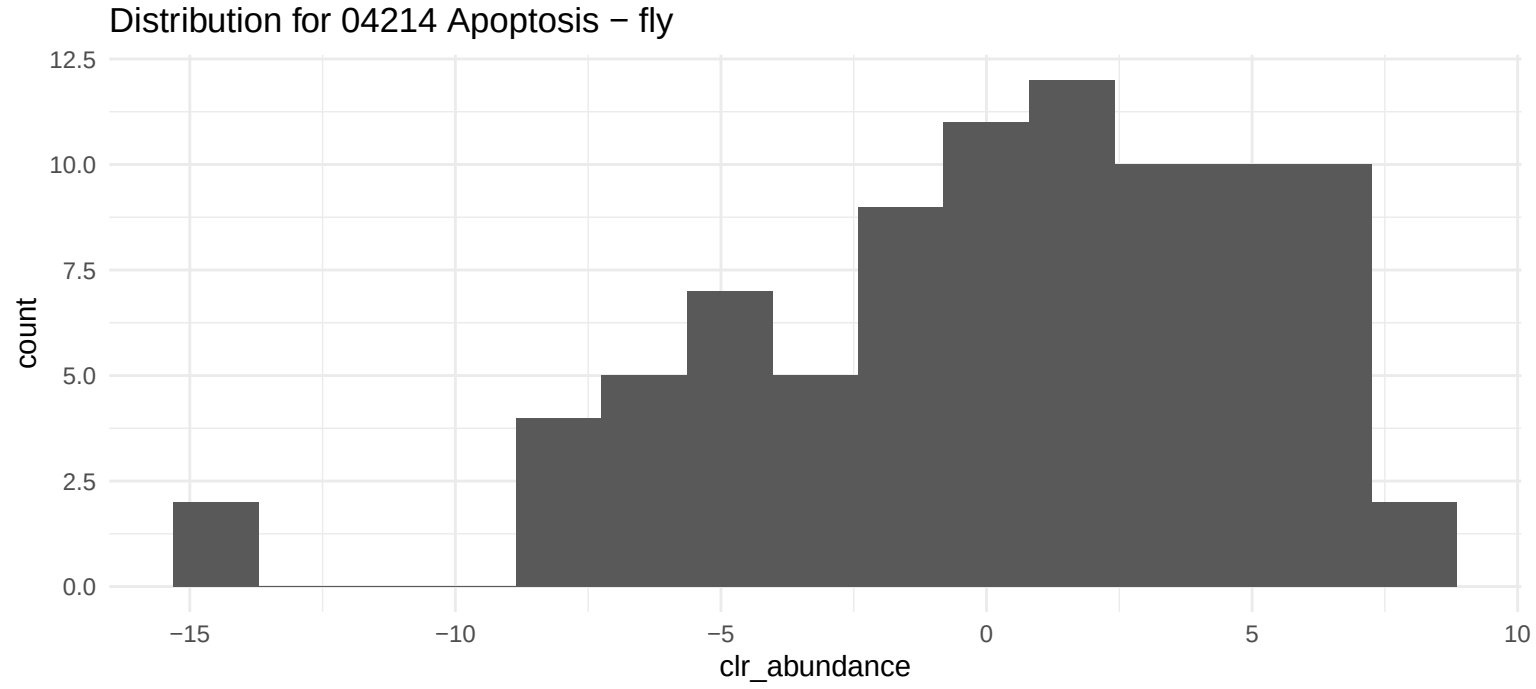
Distribution for 04211 Longevity regulating pathway

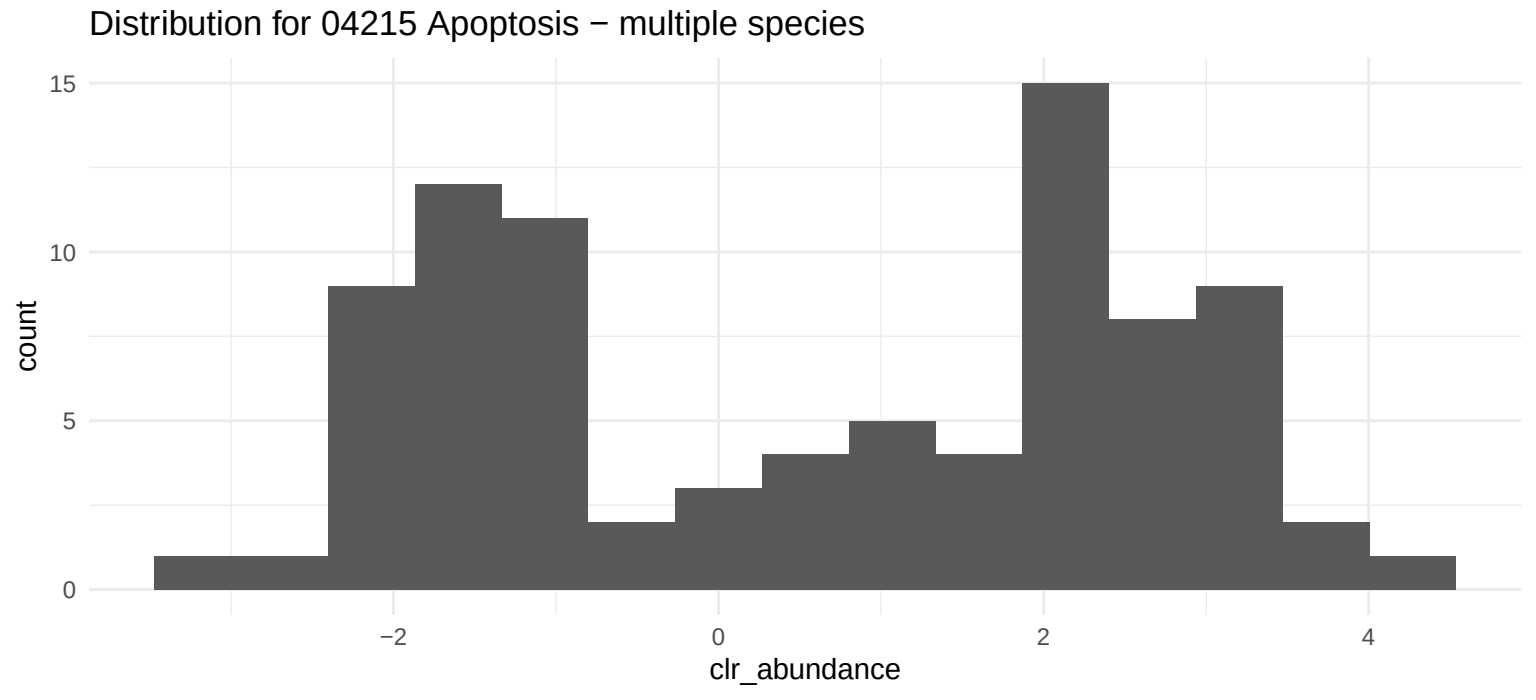


Distribution for 04212 Longevity regulating pathway − worm

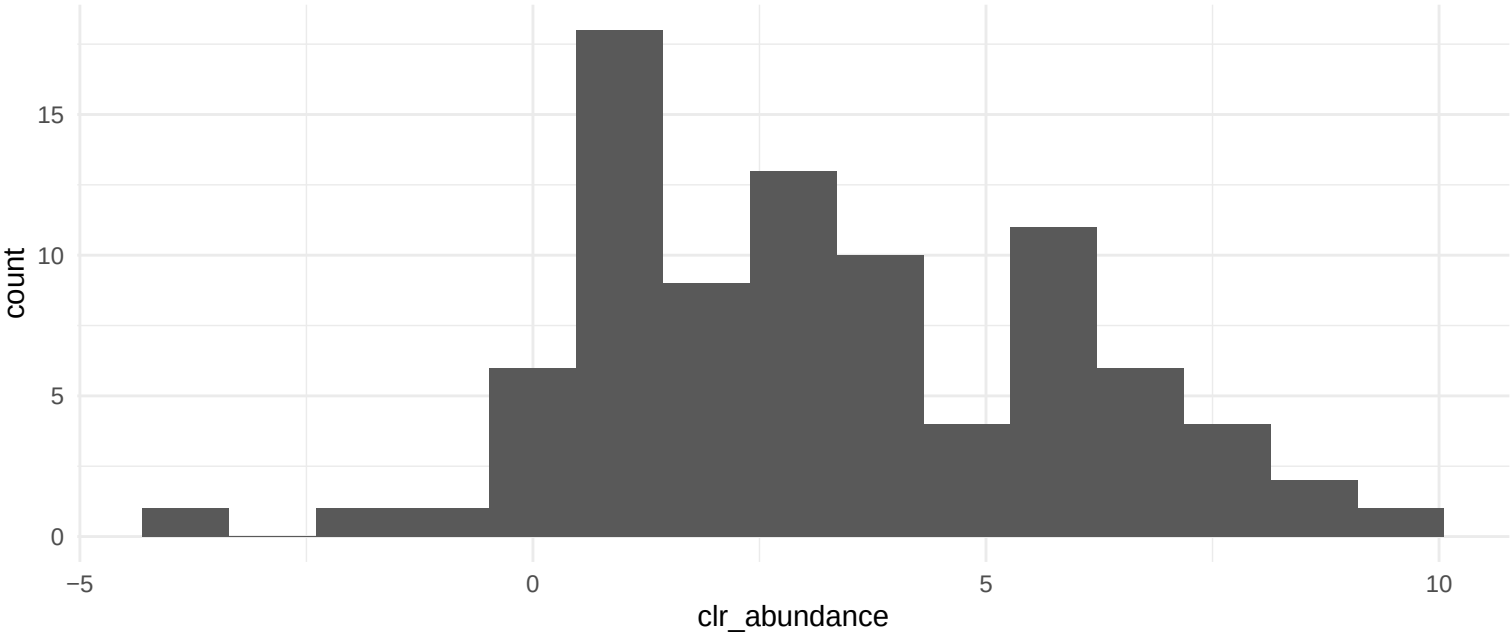



Distribution for 04213 Longevity regulating pathway − multiple species
count
clr_abundance
15
10
5
0
−10
0
10
20

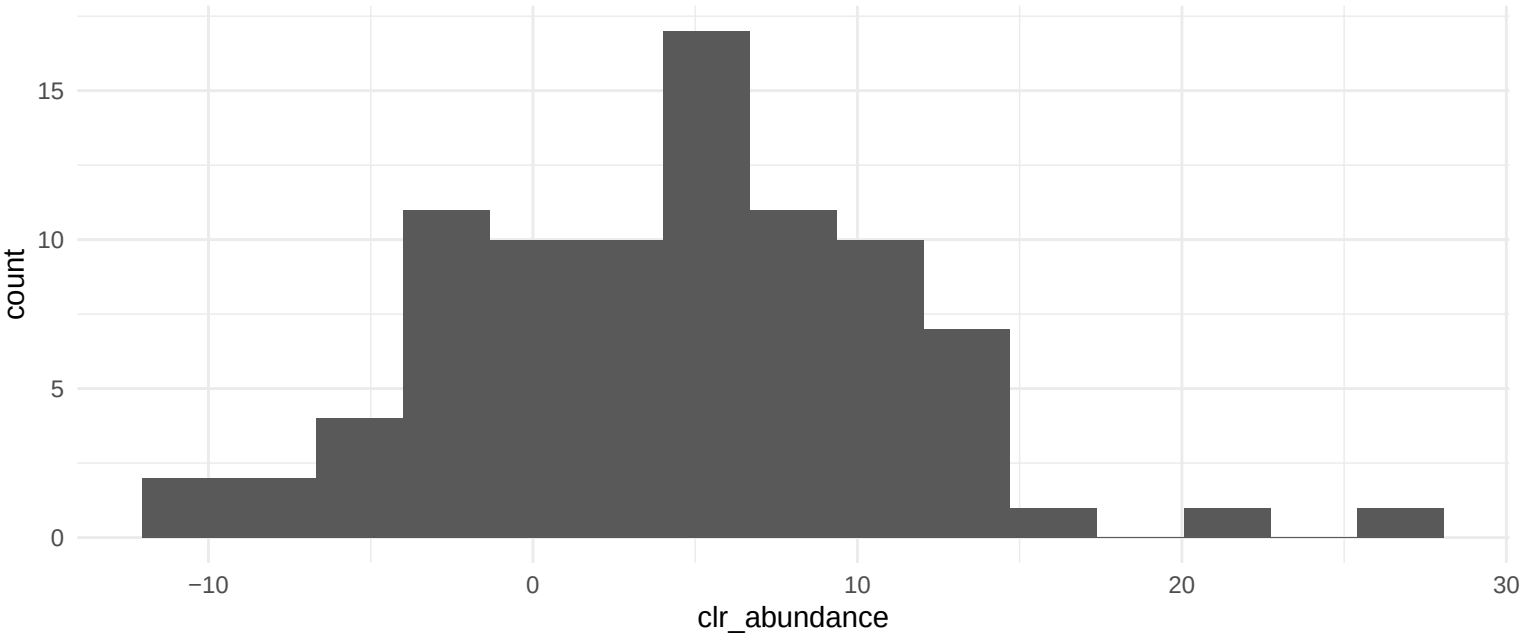

Distribution for 04214 Apoptosis − fly
count
clr_abundance
12.5
10.0
7.5
5.0
2.5
0.0
−15
−10
−5
0
5
10


Distribution for 04215 Apoptosis − multiple species
count
clr_abundance
15
10
5
0
−2
0
2
4

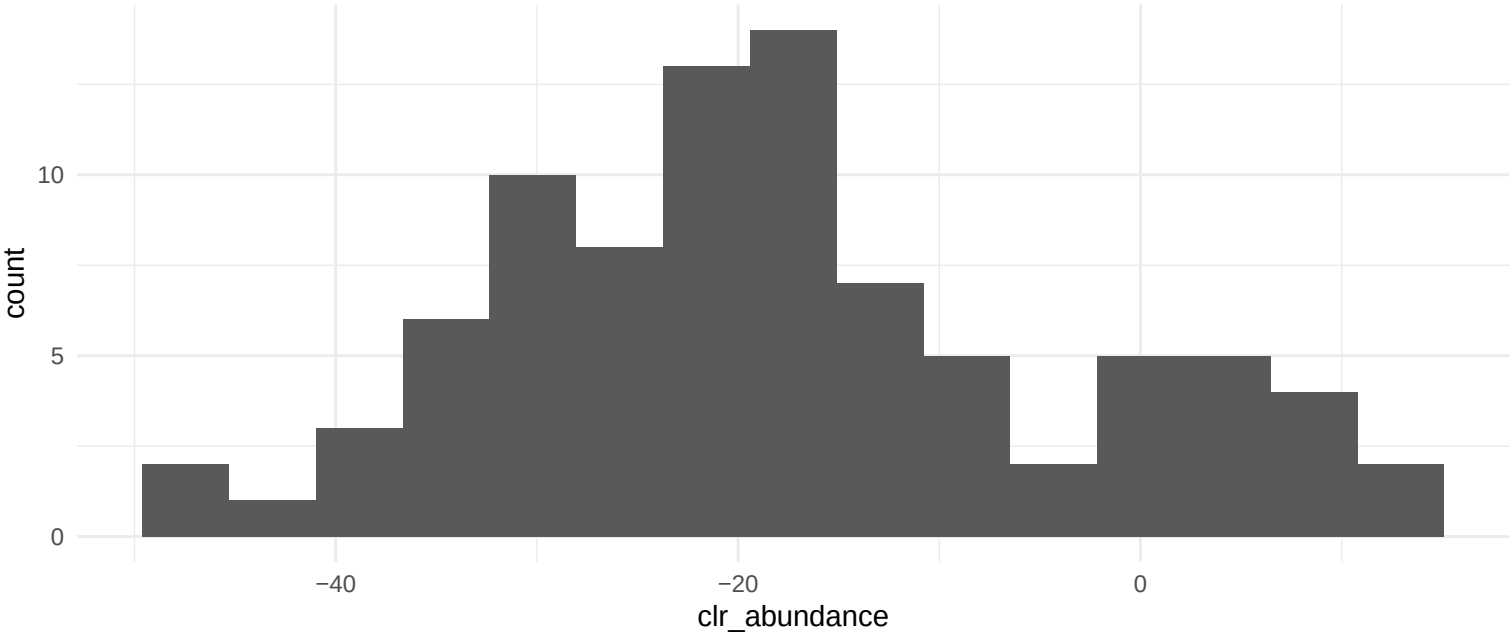
Distribution for 04216 Ferroptosis

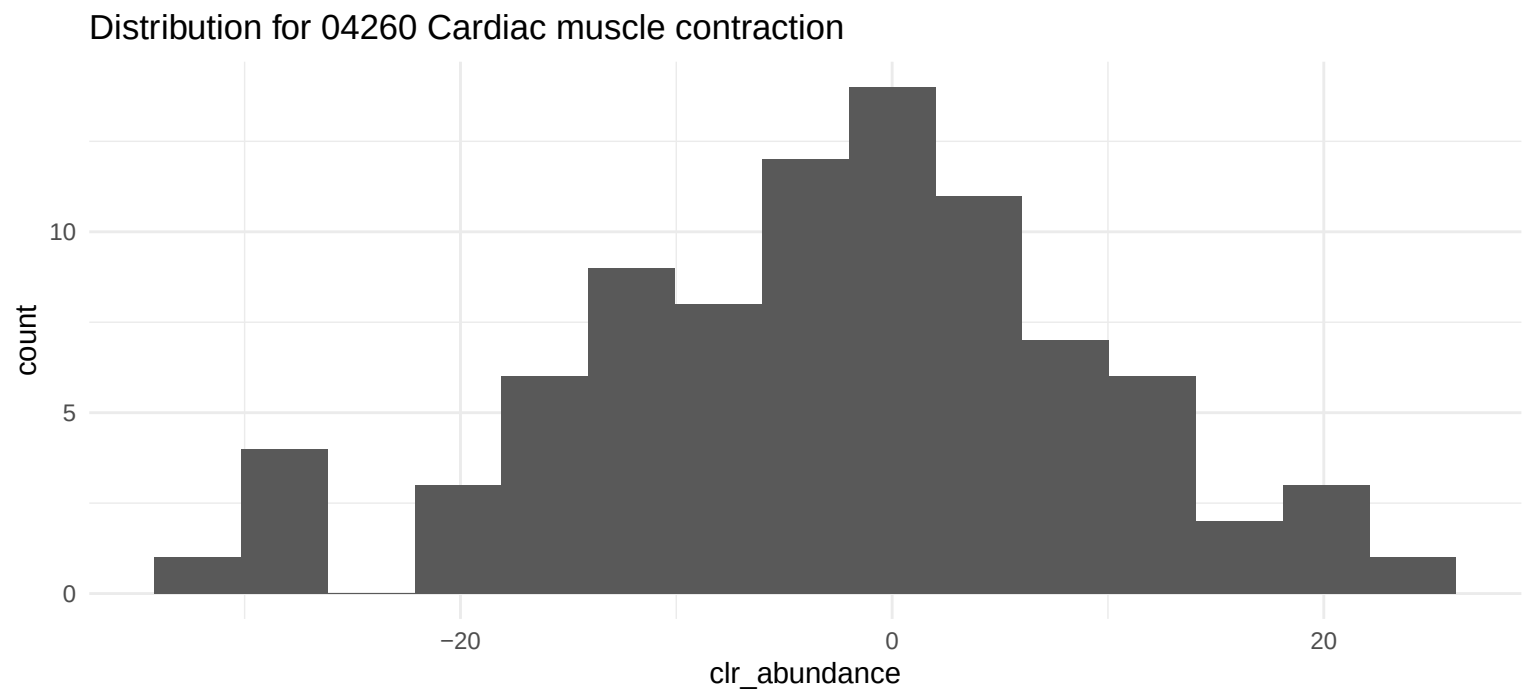


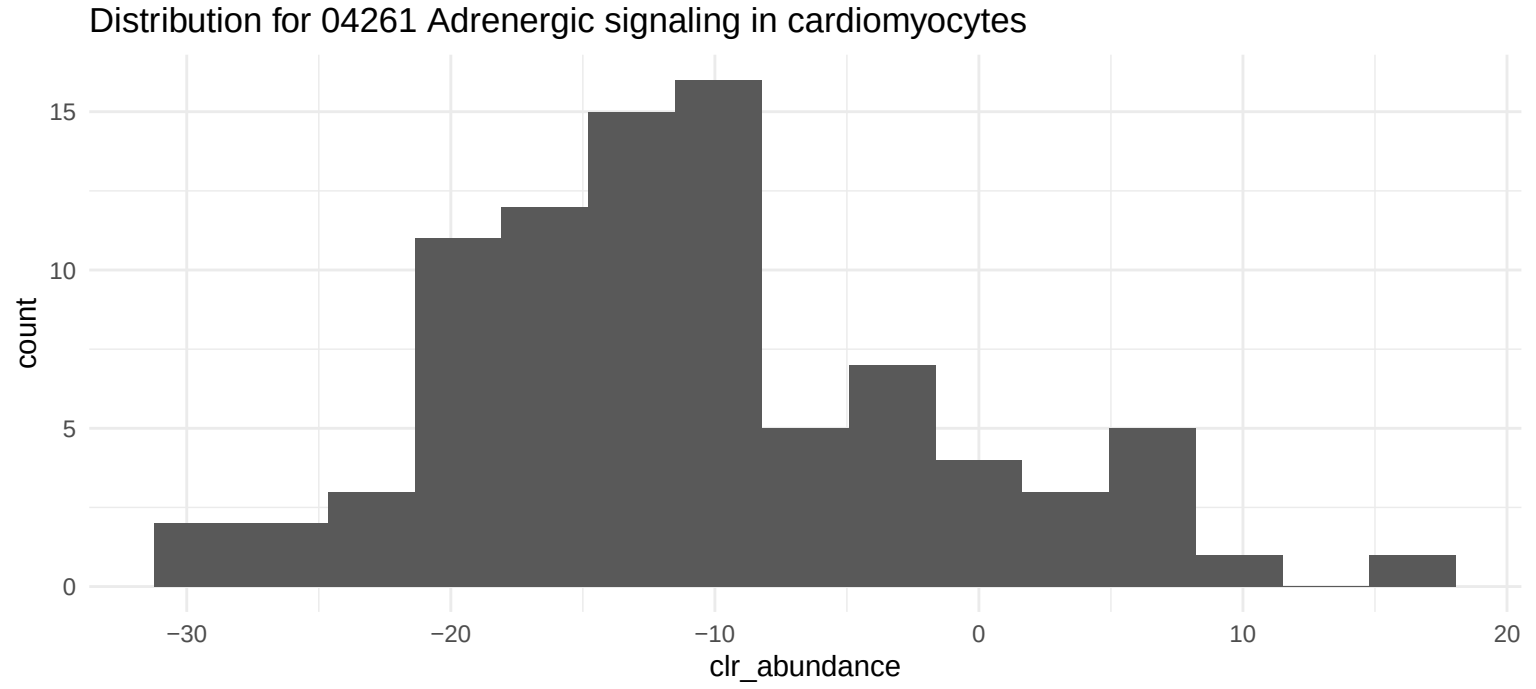
Distribution for 04217 Necroptosis

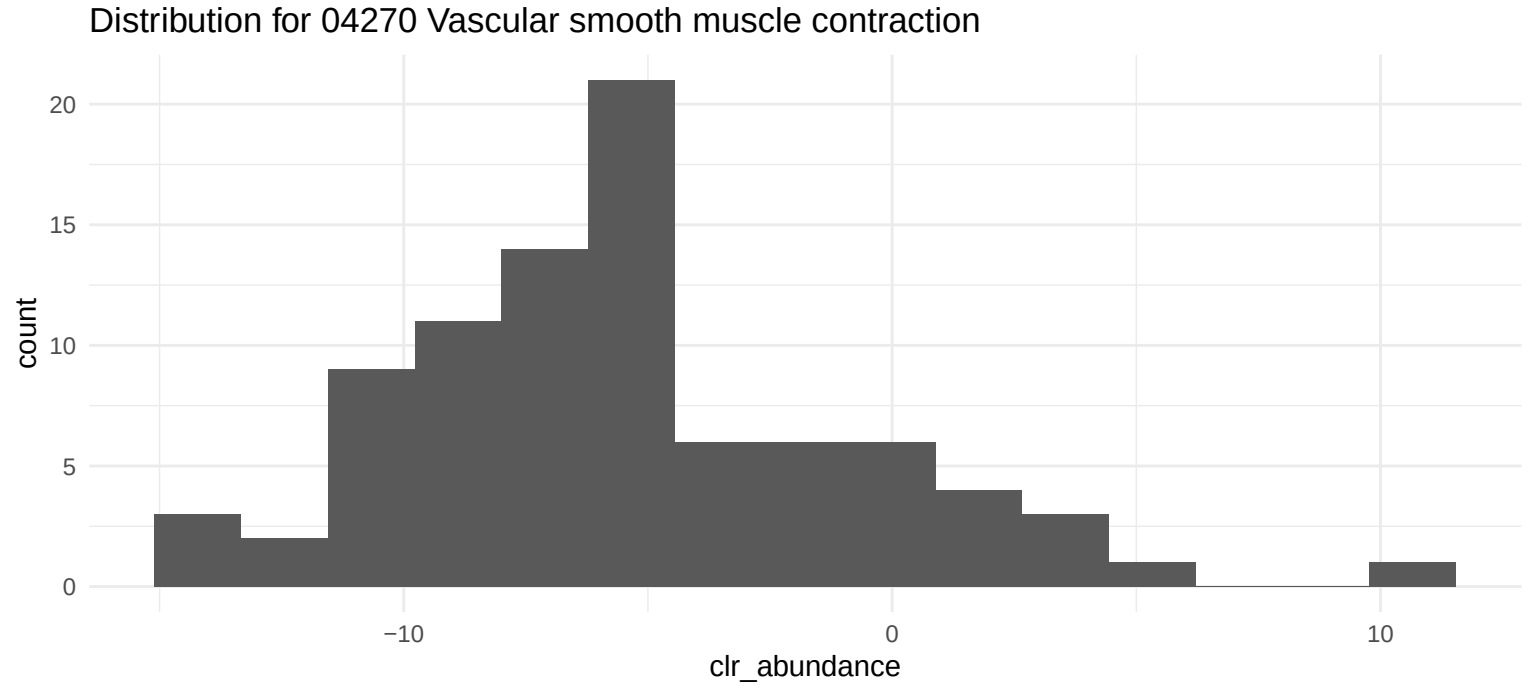


Distribution for 04218 Cellular senescence

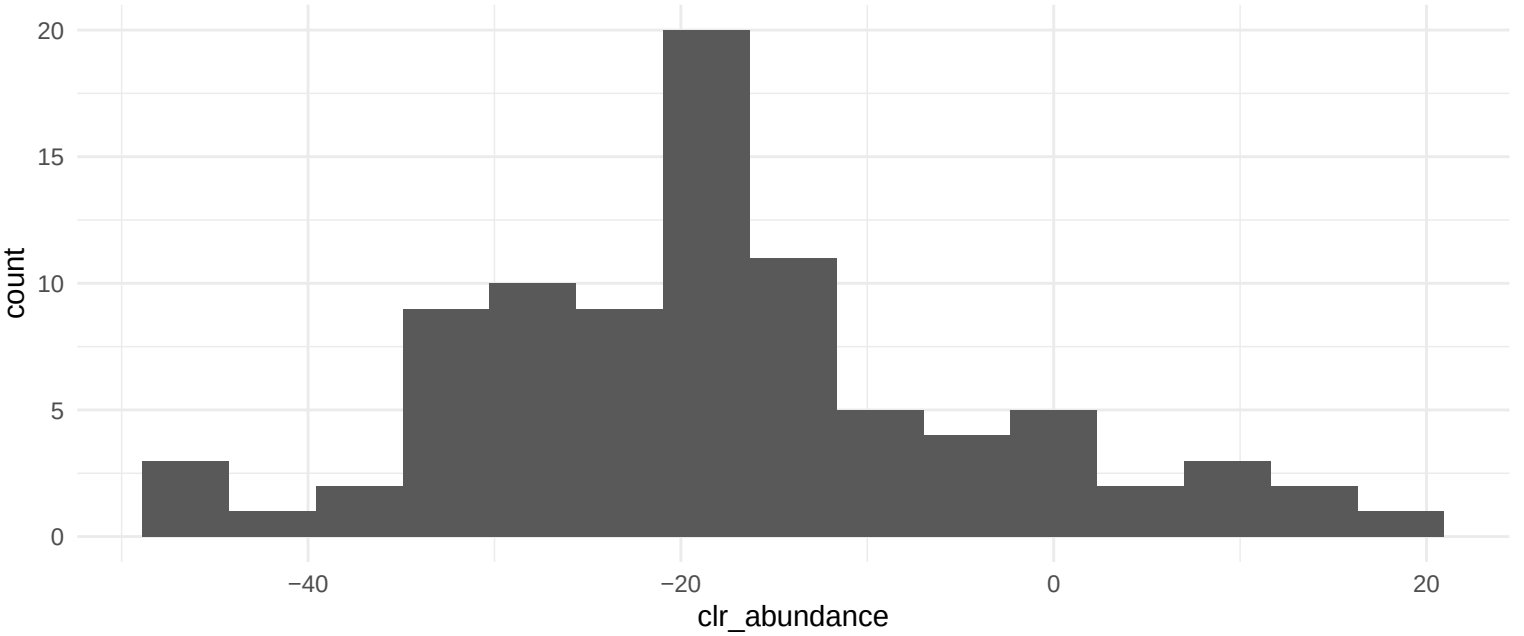



Distribution for 04260 Cardiac muscle contraction
count
clr_abundance
0
5
10
−20
0
20

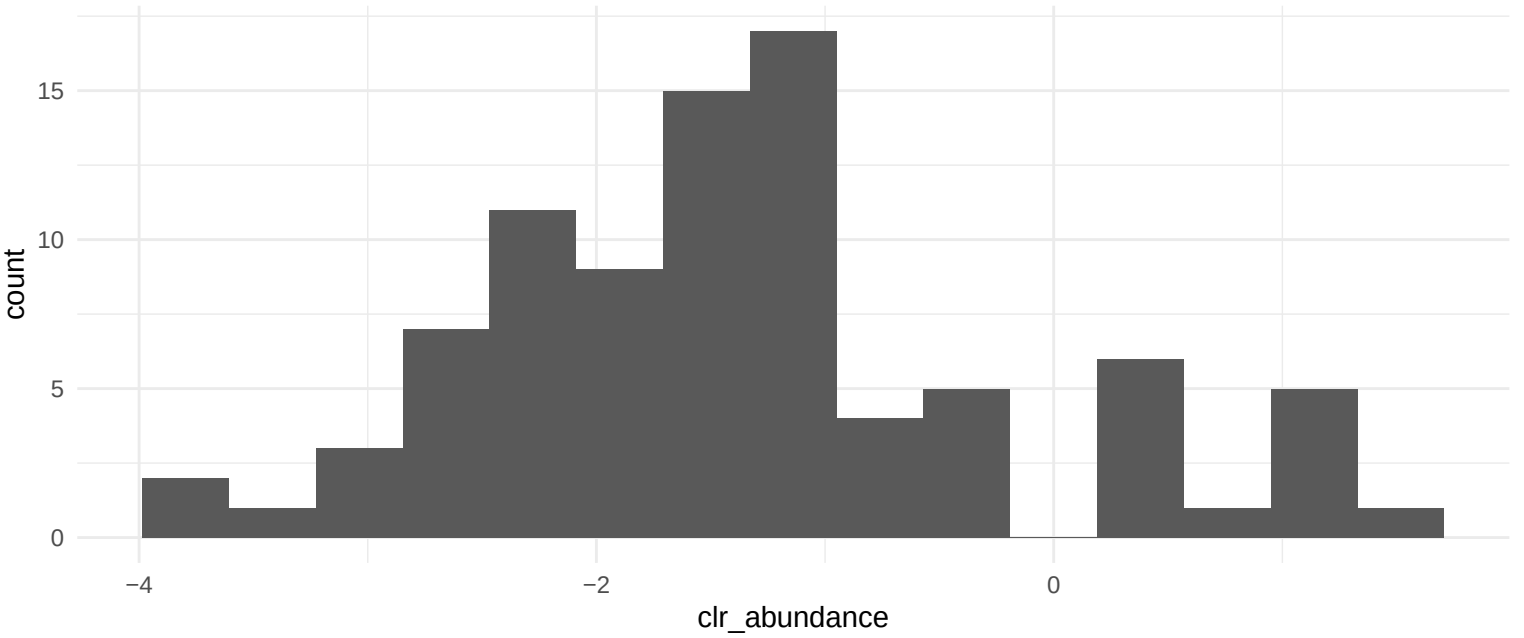

Distribution for 04261 Adrenergic signaling in cardiomyocytes
count
clr_abundance
0
5
10
15
−30
−20
−10
0
10
20


Distribution for 04270 Vascular smooth muscle contraction
count
clr_abundance
0
5
10
15
20
−10
0
10

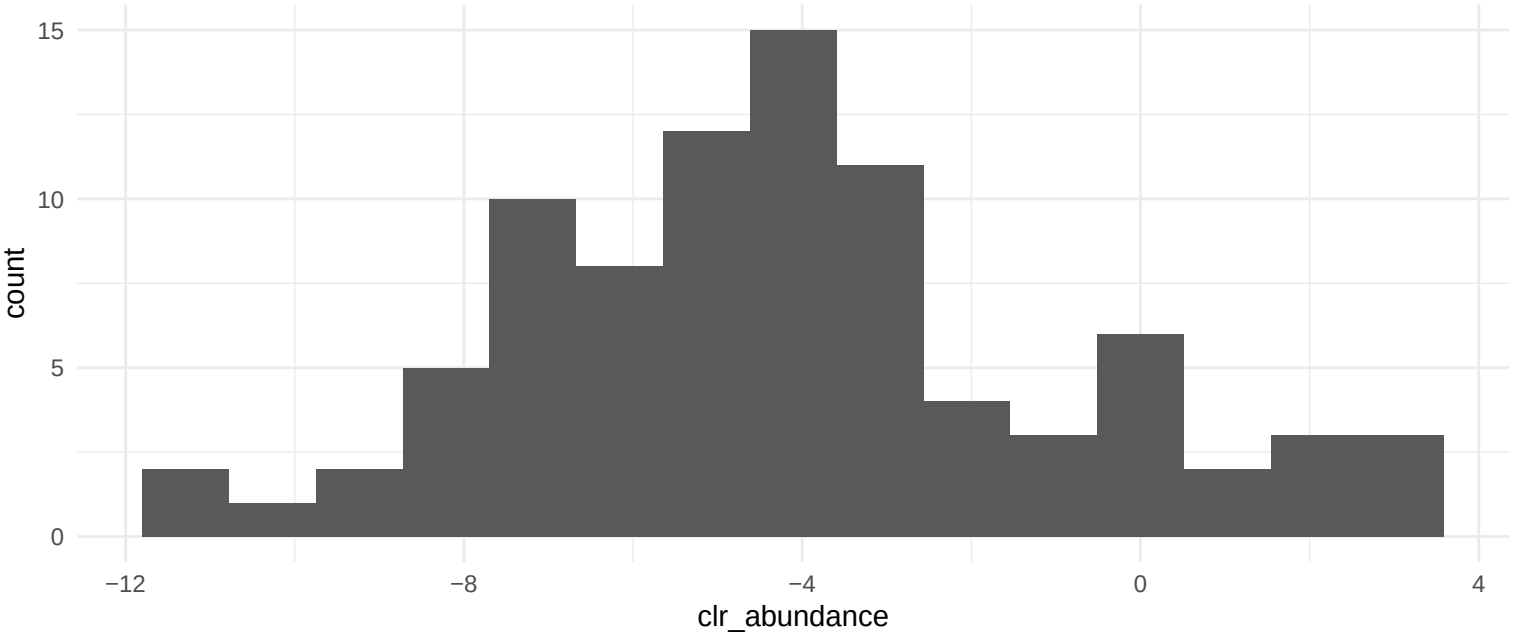
Distribution for 04310 Wnt signaling pathway



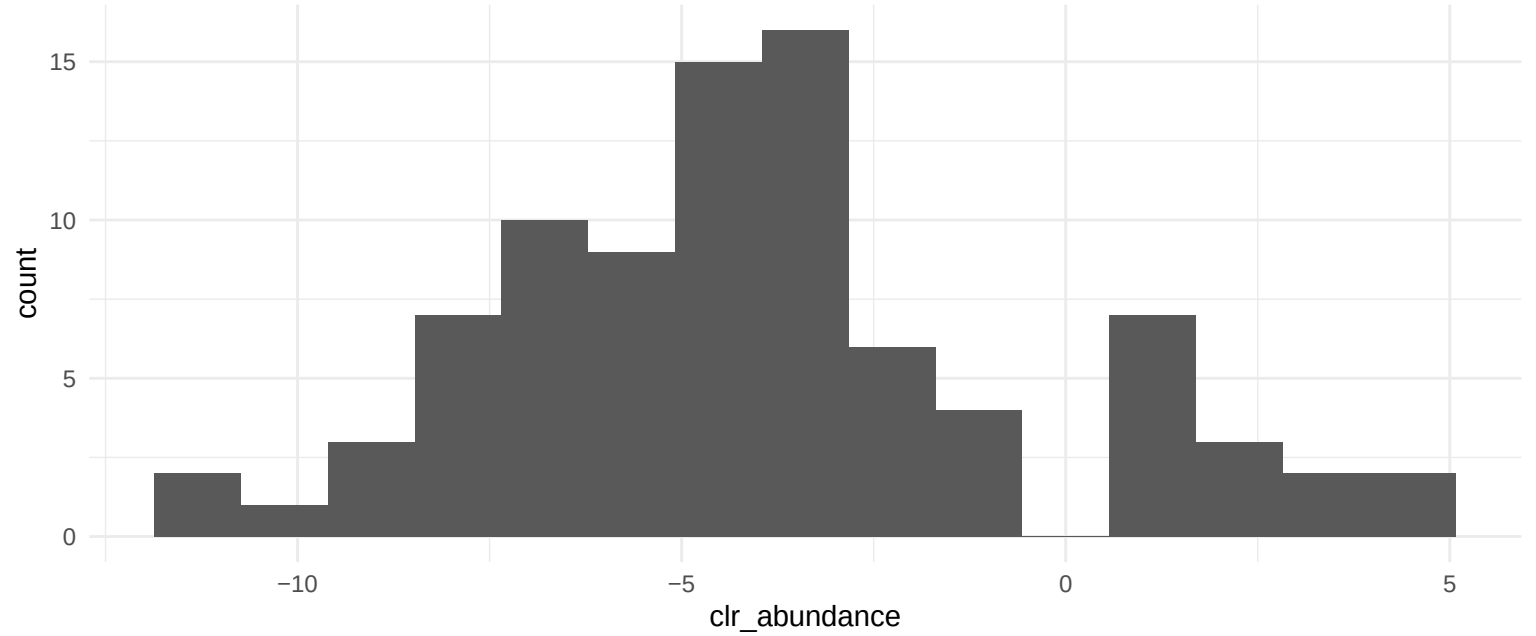
Distribution for 04320 Dorso−ventral axis formation



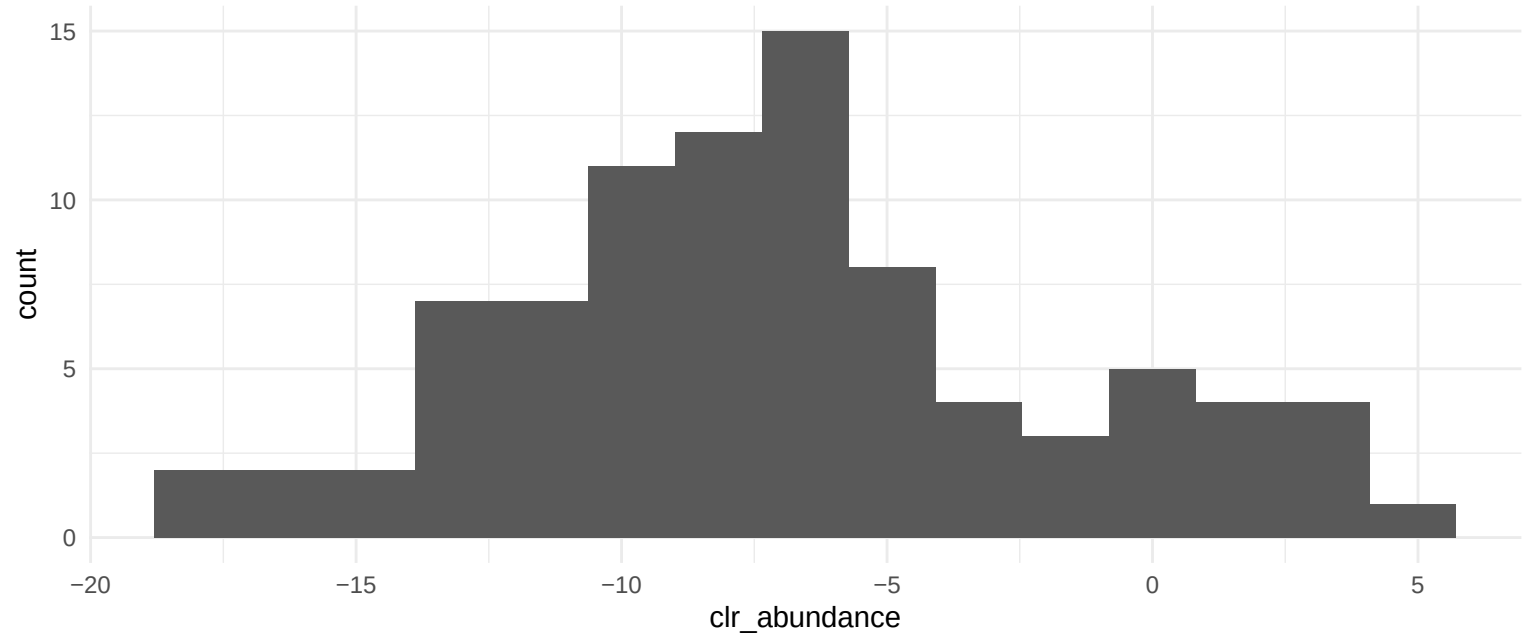
Distribution for 04330 Notch signaling pathway



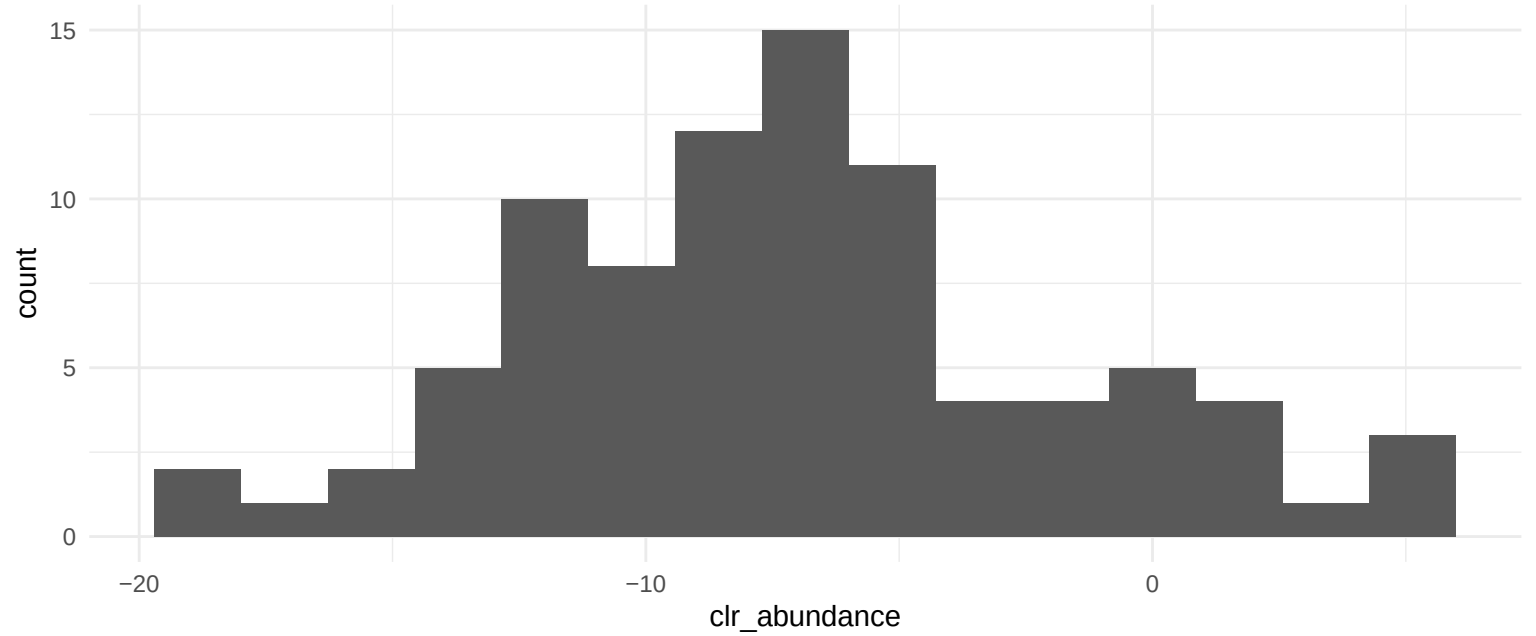
Distribution for 04340 Hedgehog signaling pathway



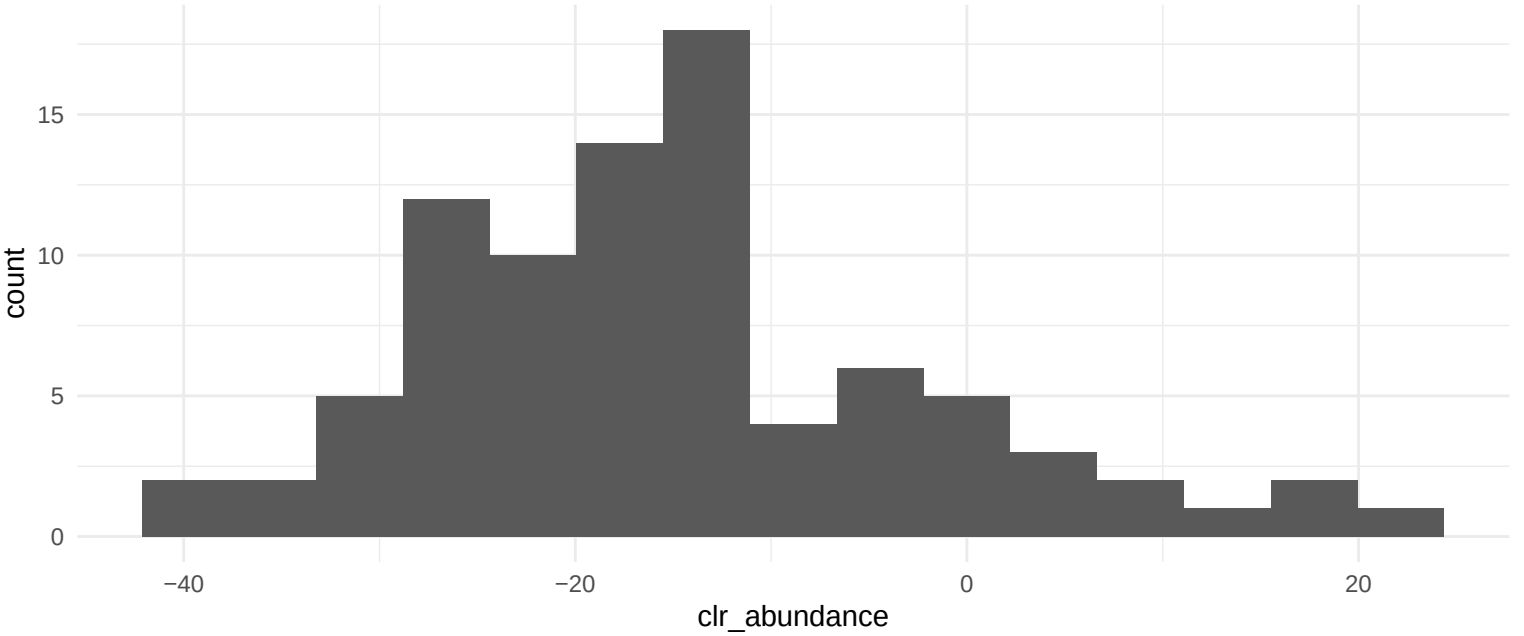
Distribution for 04341 Hedgehog signaling pathway − fly



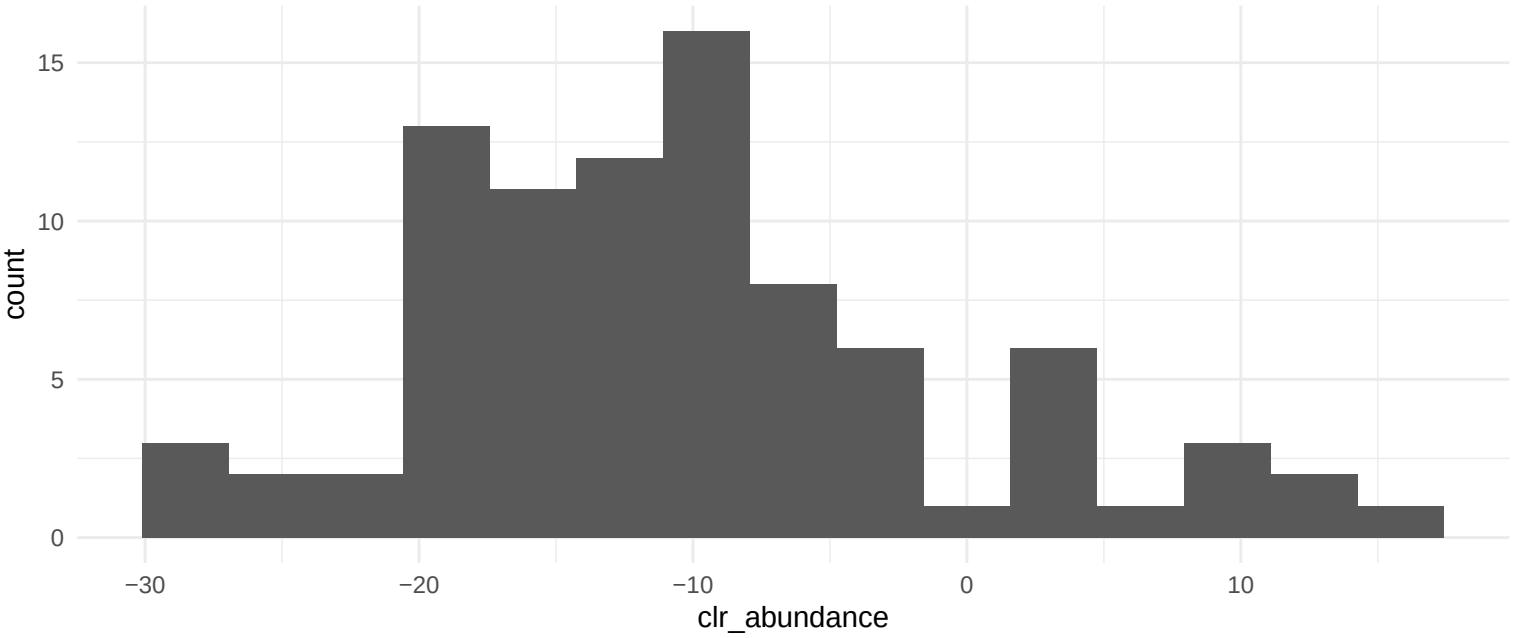
Distribution for 04350 TGF−beta signaling pathway



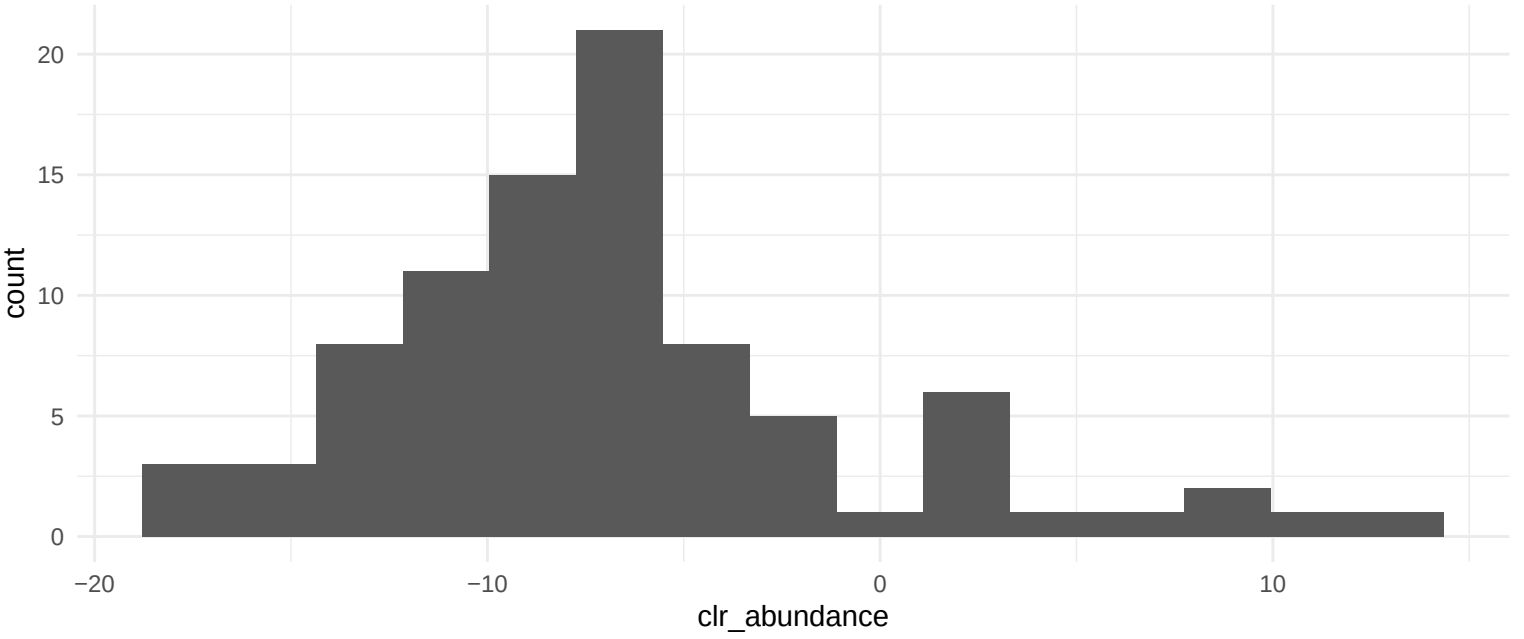
Distribution for 04360 Axon guidance



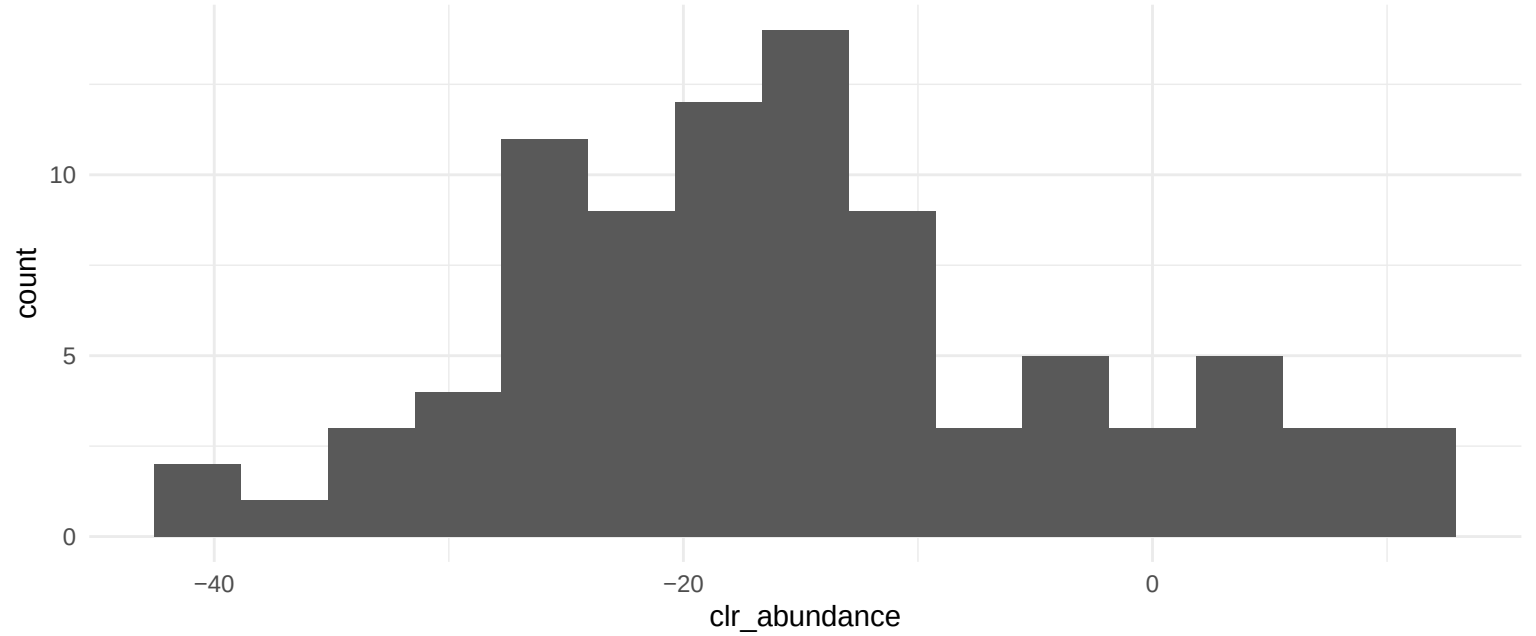
Distribution for 04361 Axon regeneration



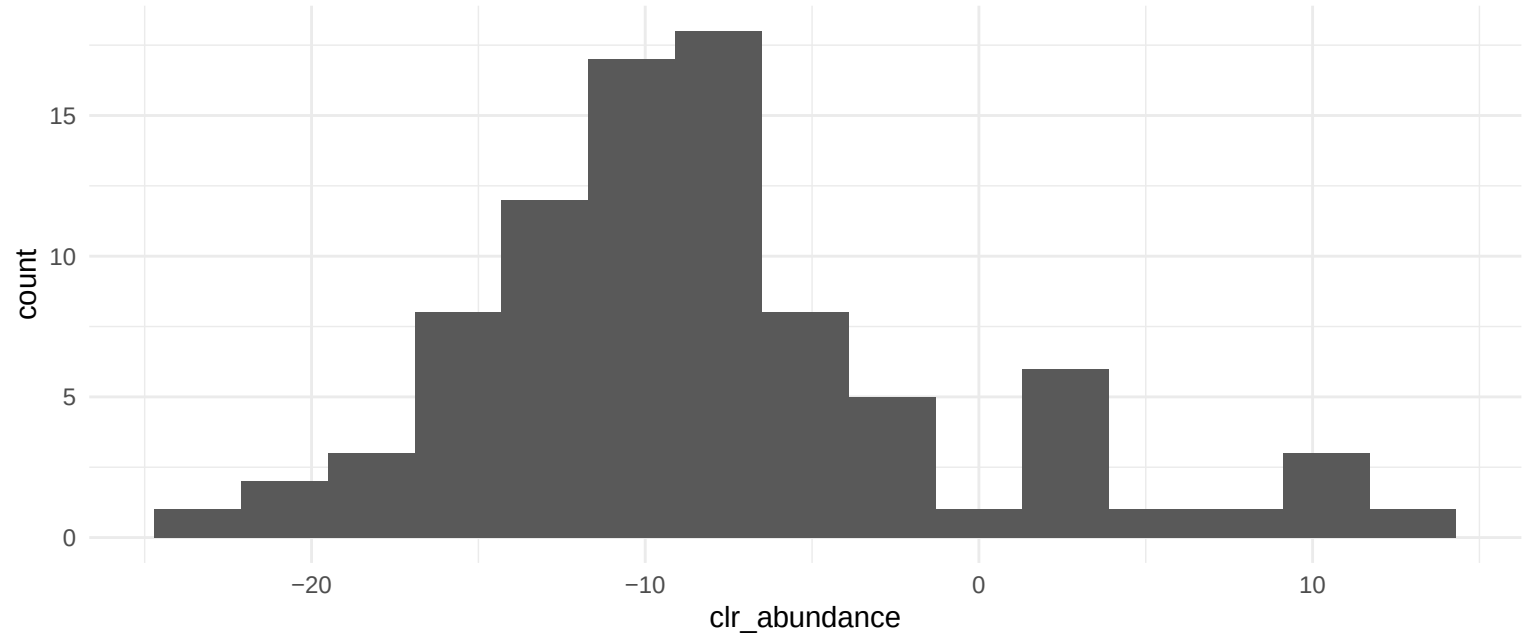
Distribution for 04370 VEGF signaling pathway



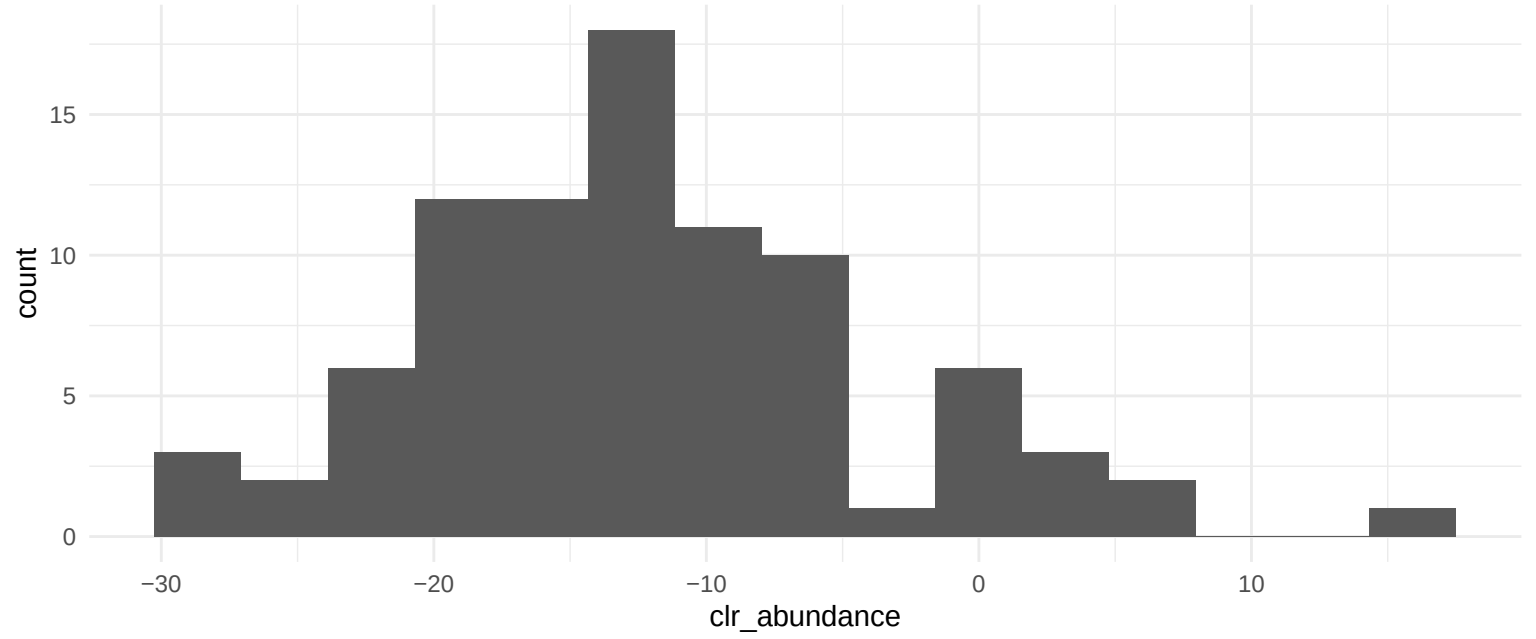
Distribution for 04371 Apelin signaling pathway



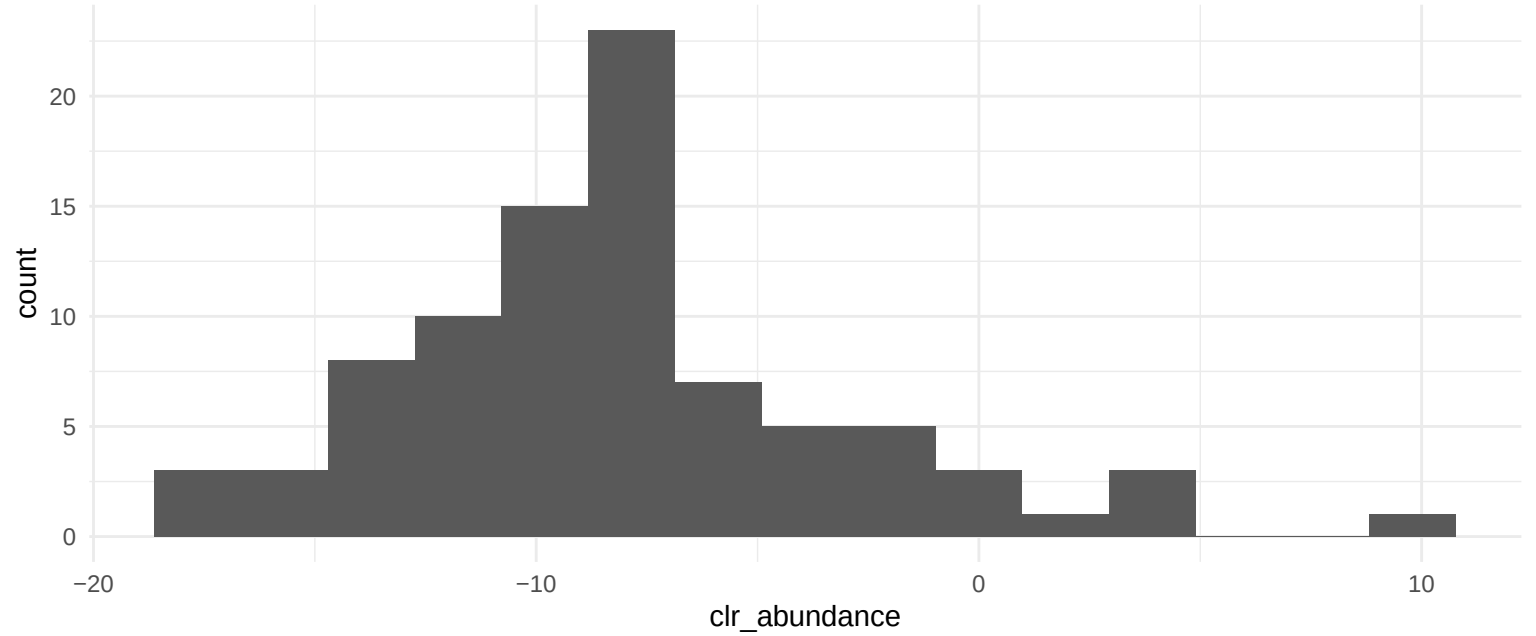
Distribution for 04380 Osteoclast differentiation



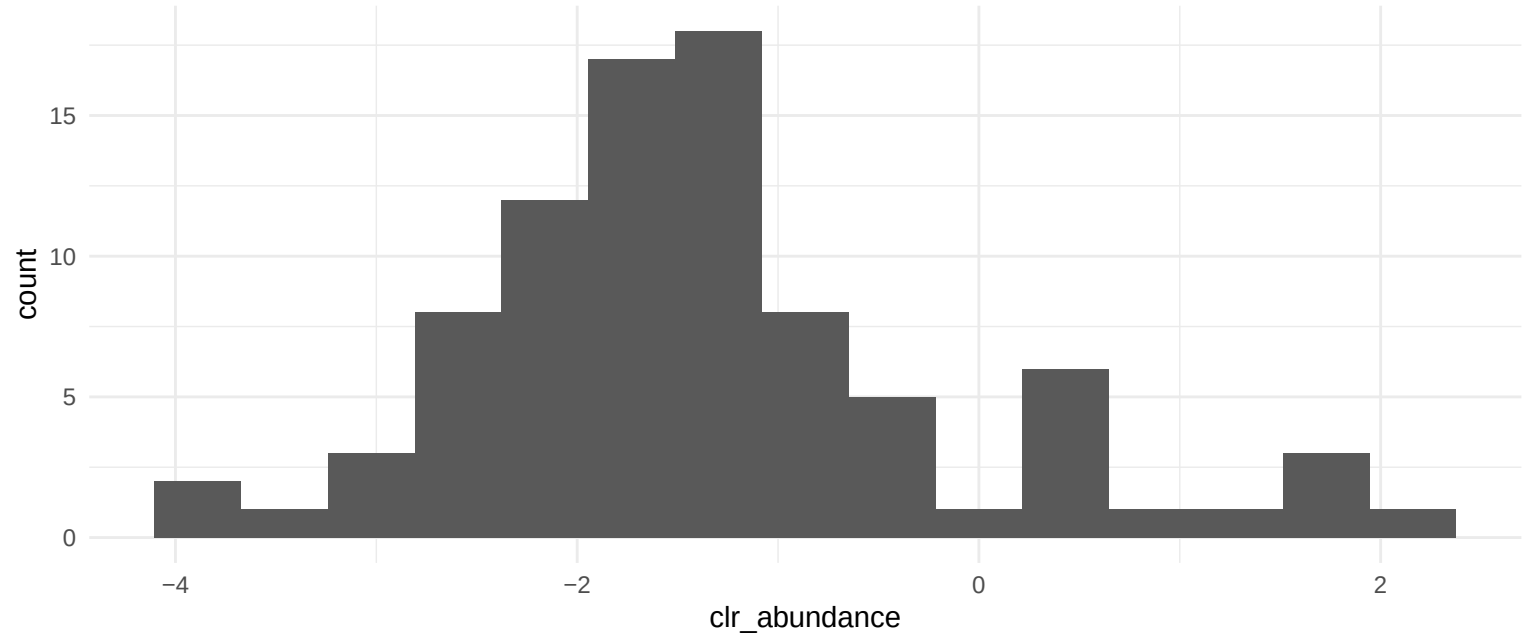
Distribution for 04390 Hippo signaling pathway



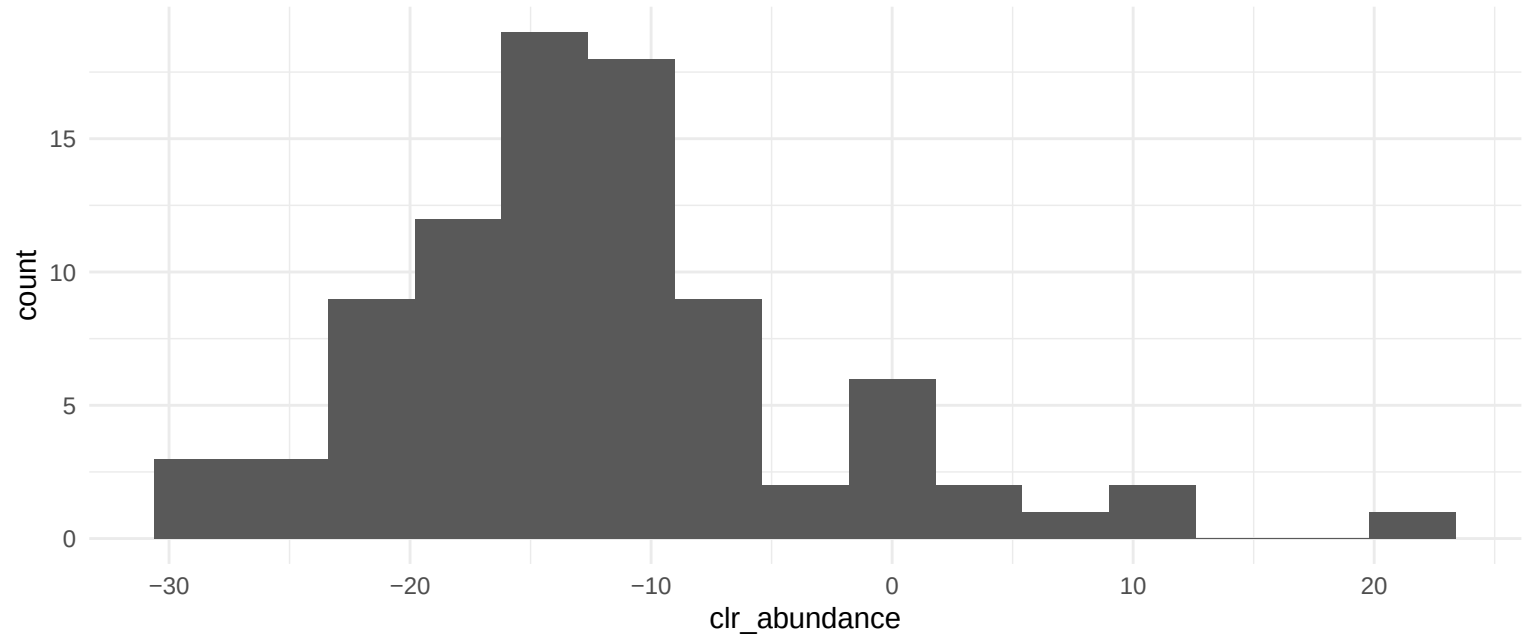
Distribution for 04391 Hippo signaling pathway − fly



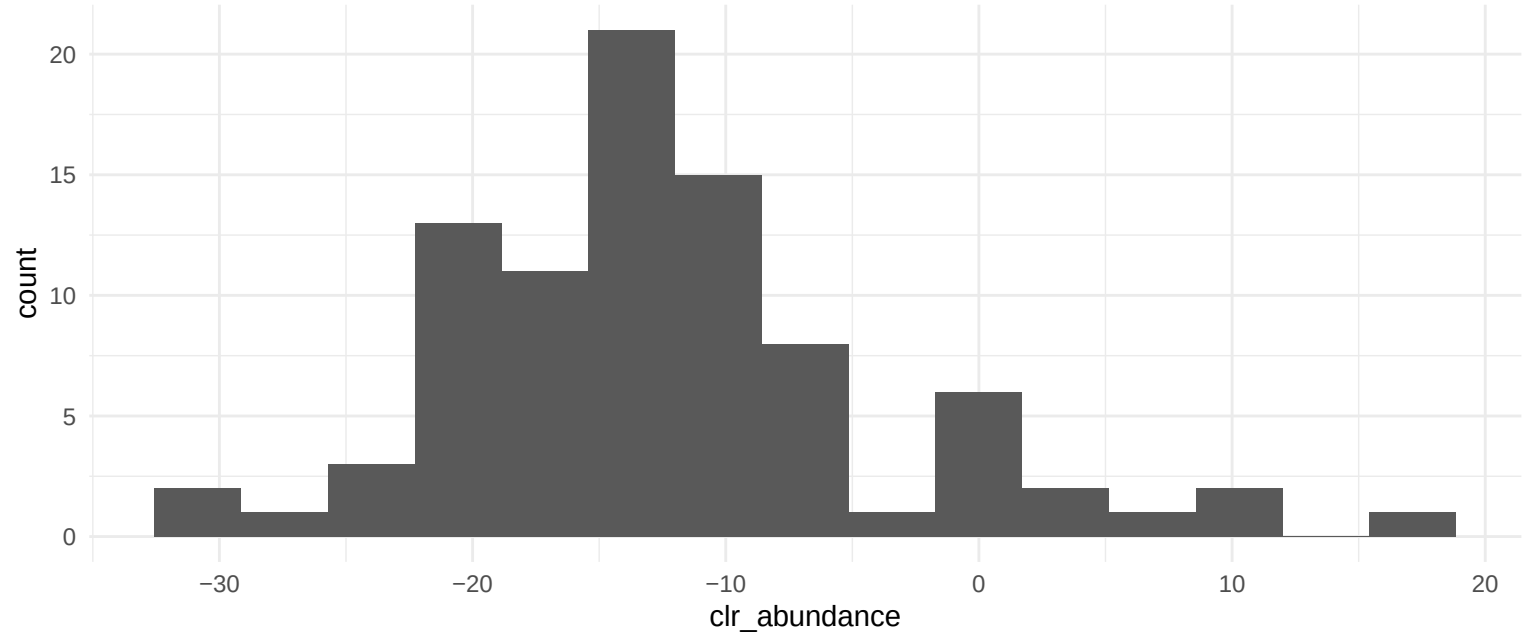
Distribution for 04392 Hippo signaling pathway − multiple species



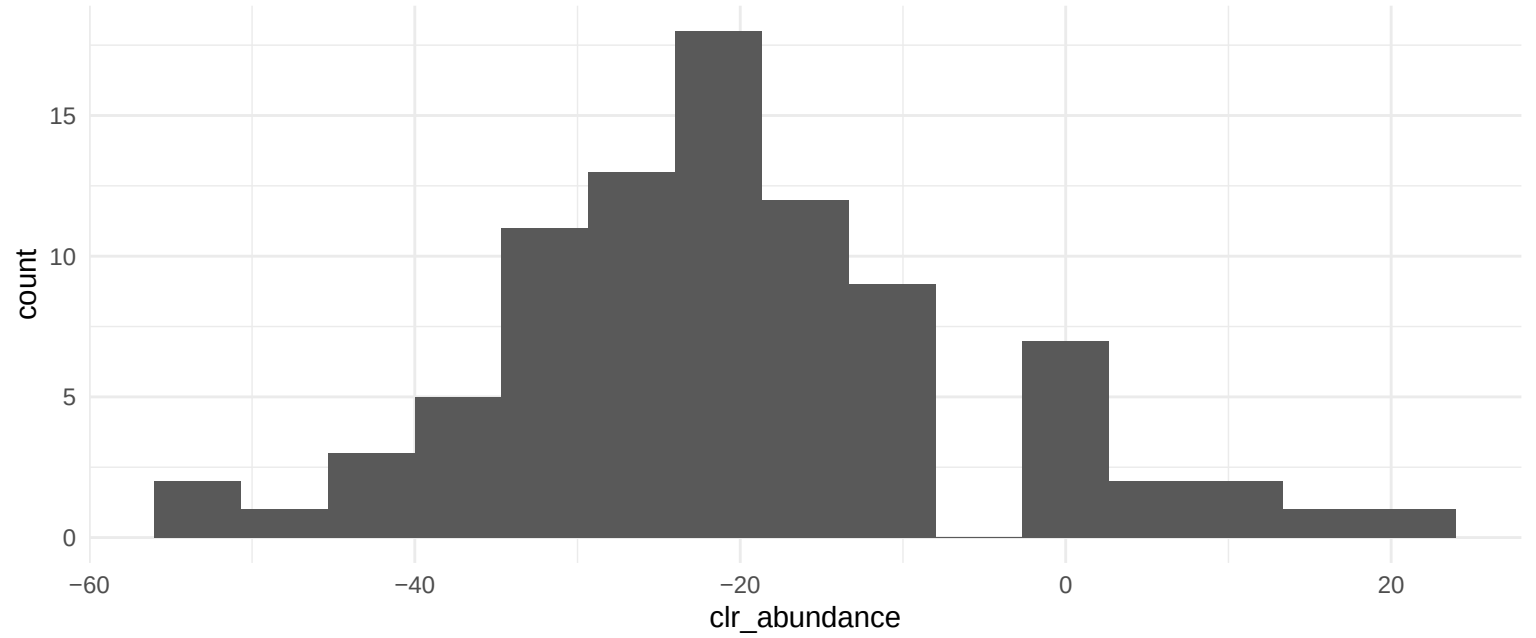
Distribution for 04510 Focal adhesion



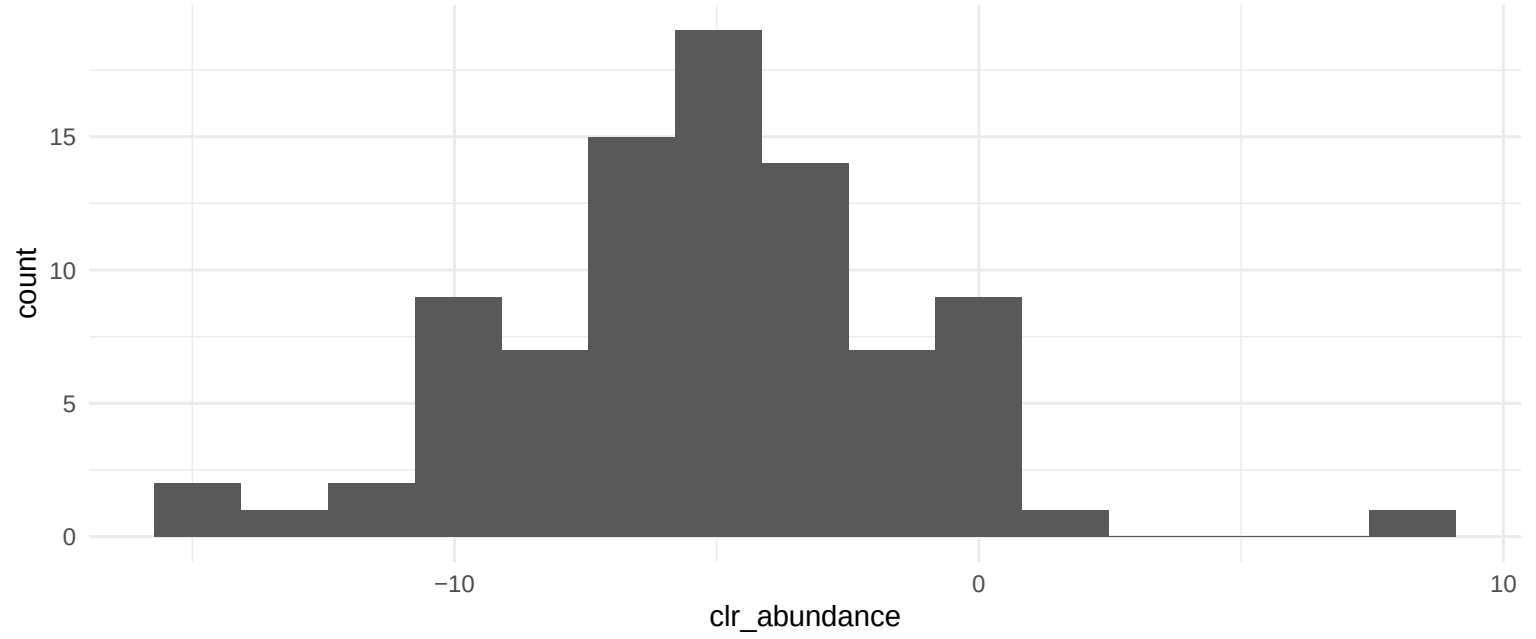
Distribution for 04520 Adherens junction



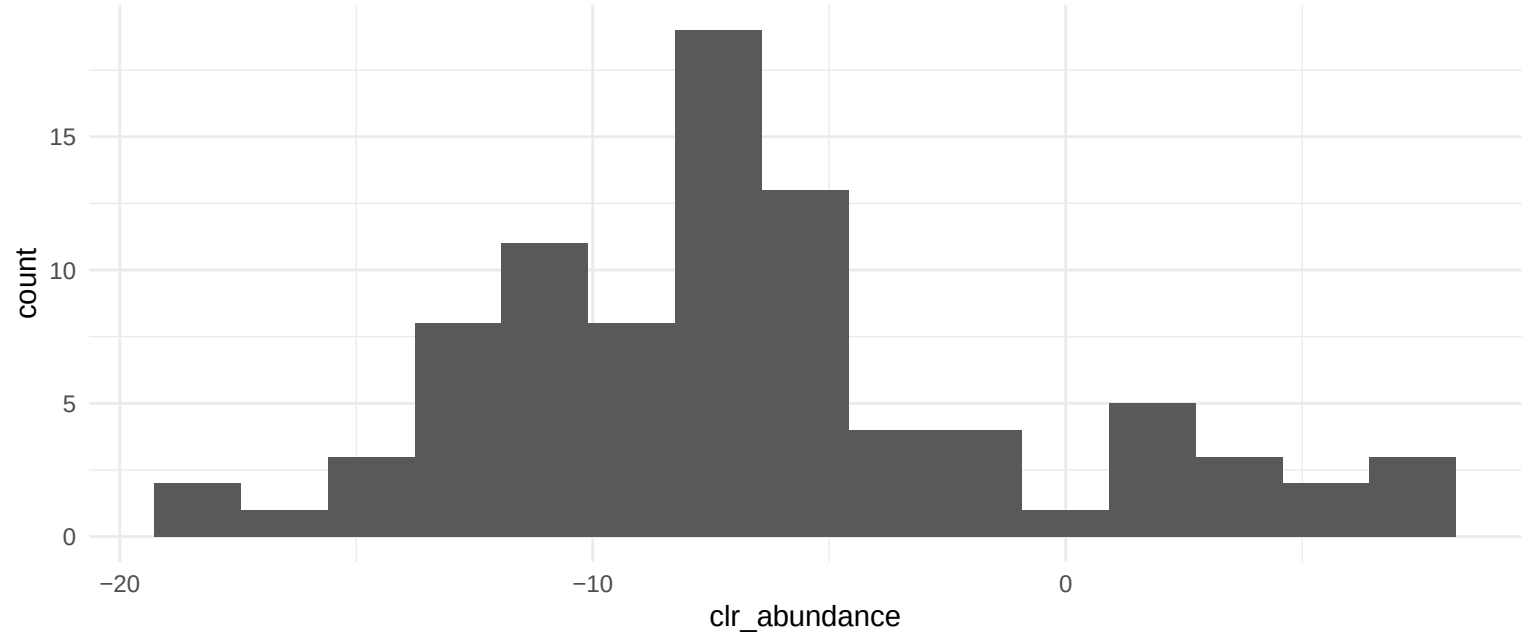
Distribution for 04530 Tight junction



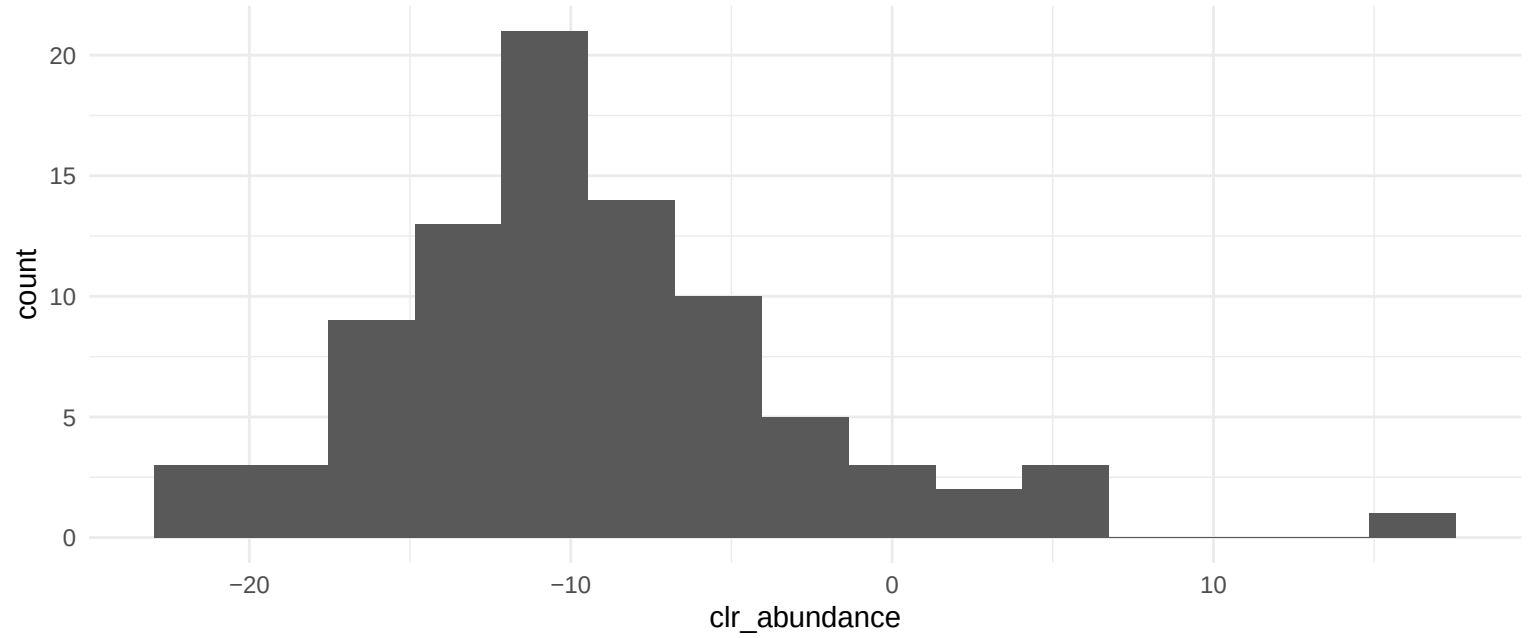
Distribution for 04540 Gap junction



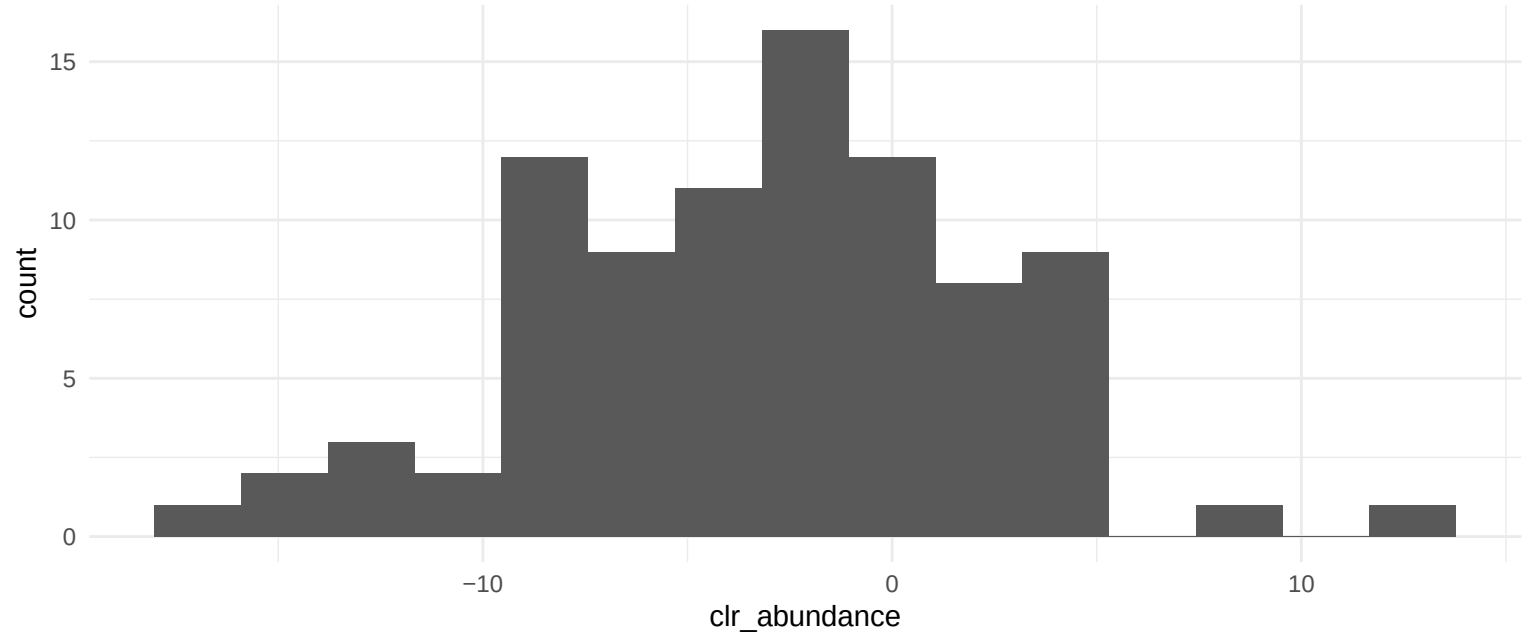
Distribution for 04550 Signaling pathways regulating pluripotency of stem cells



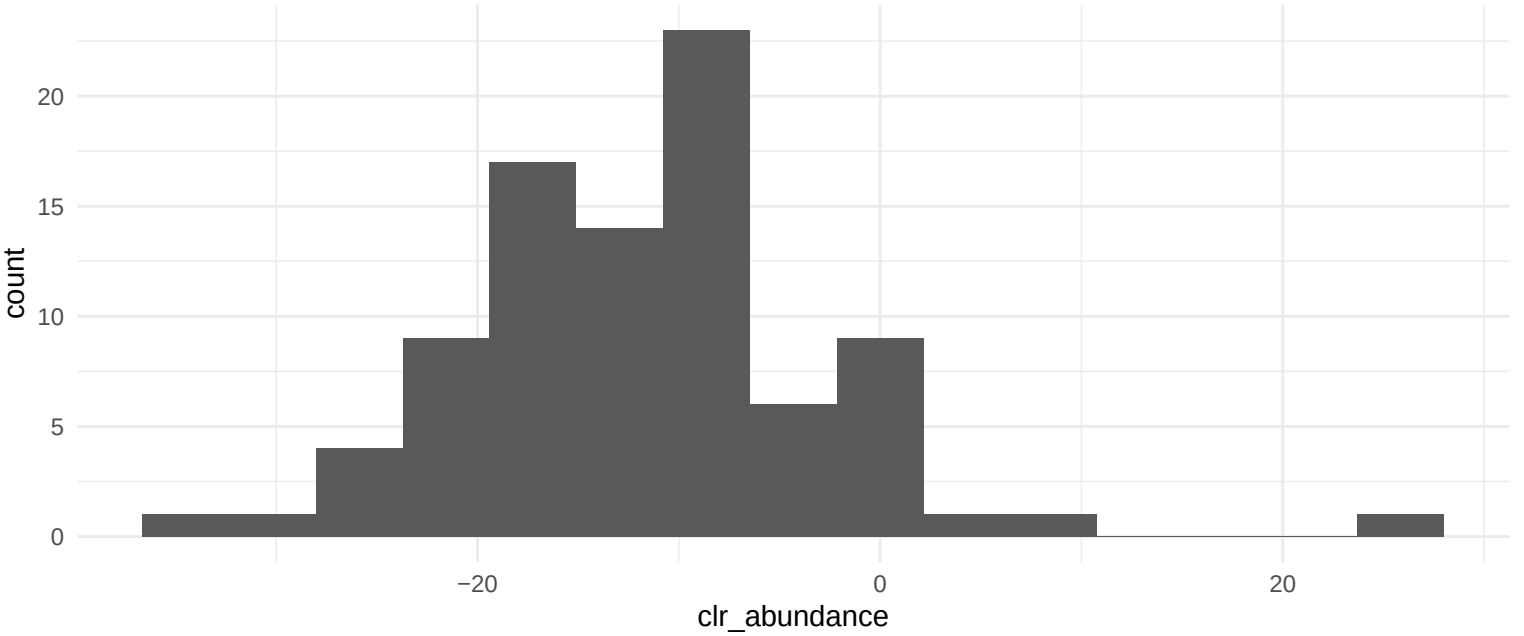
Distribution for 04611 Platelet activation



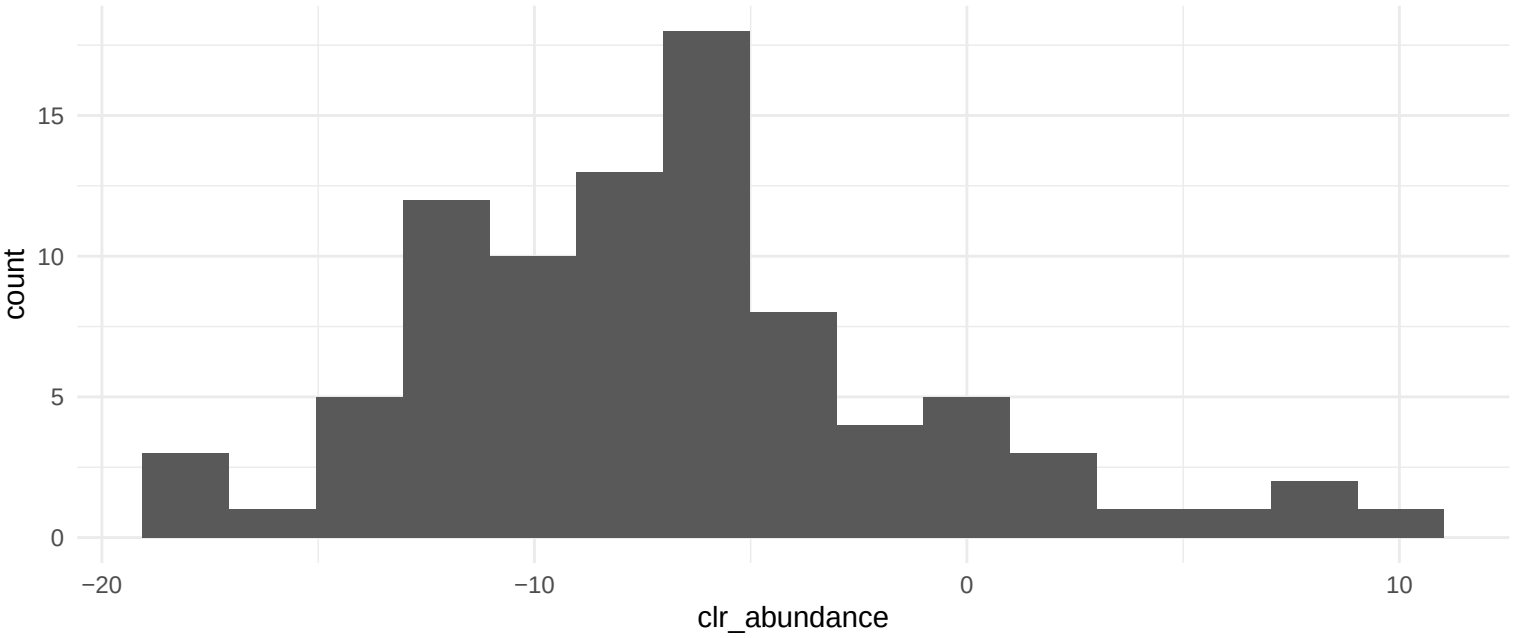
Distribution for 04612 Antigen processing and presentation



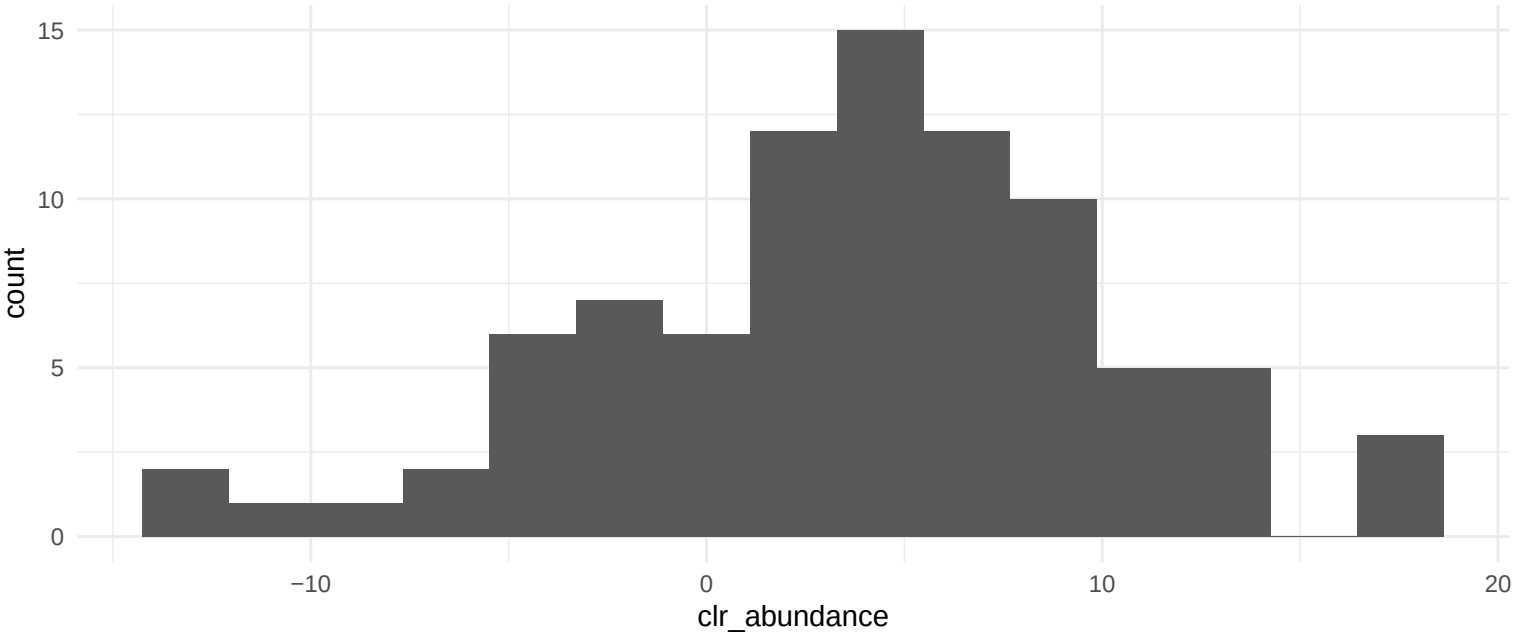
Distribution for 04613 Neutrophil extracellular trap formation

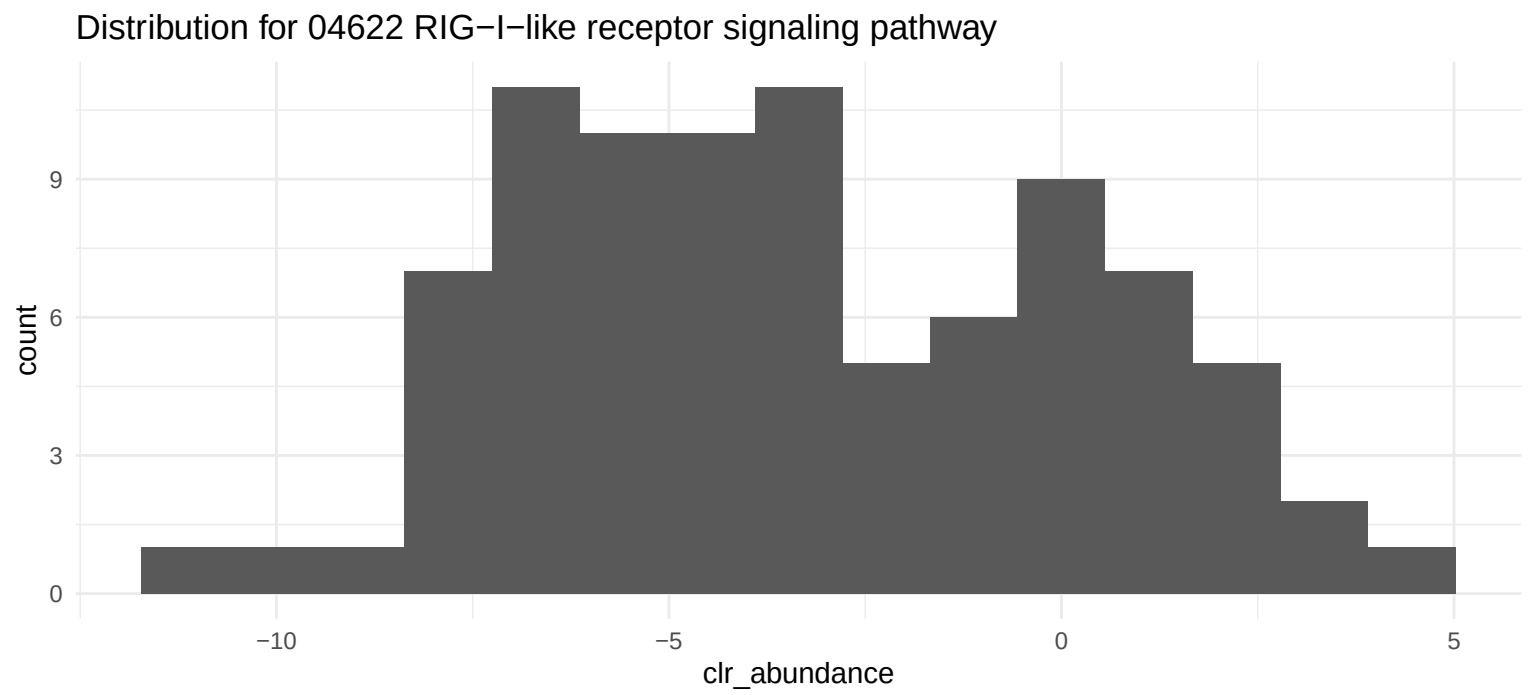


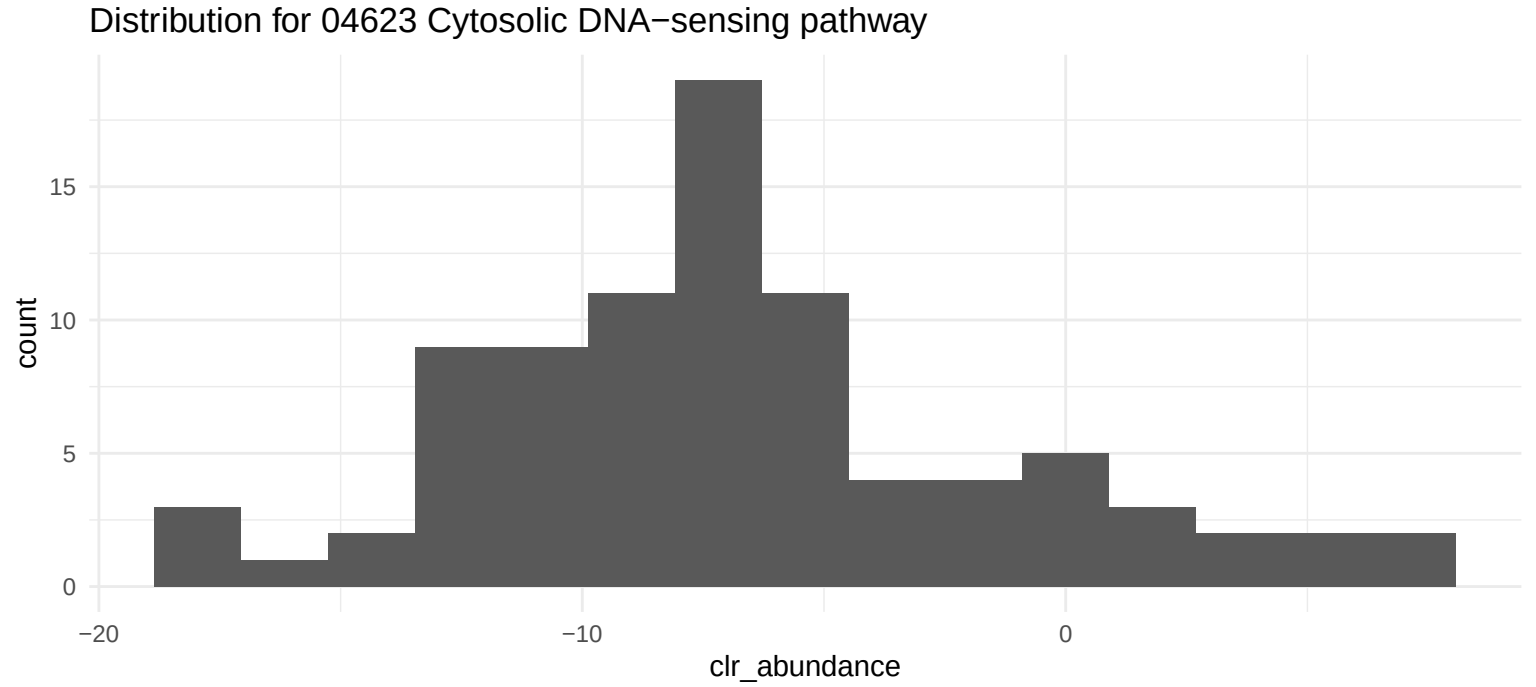
Distribution for 04620 Toll−like receptor signaling pathway

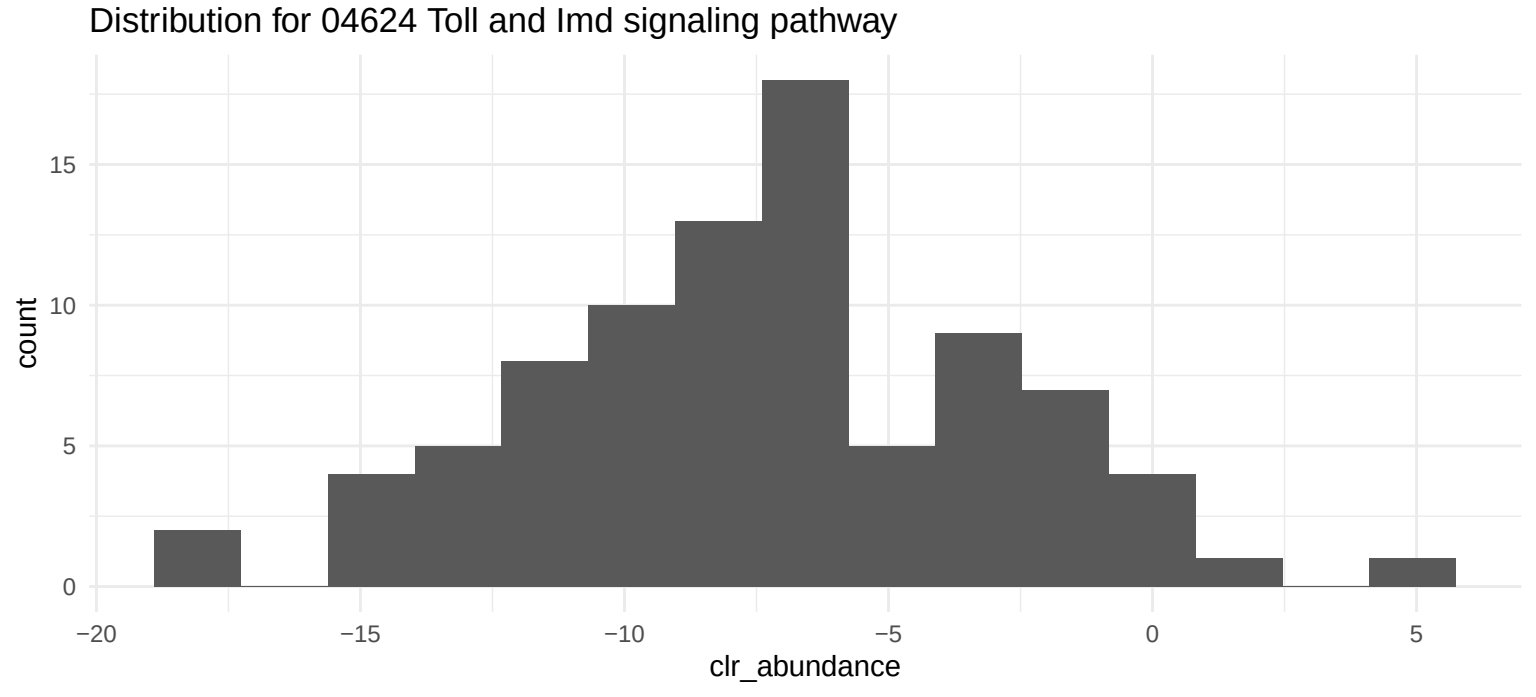


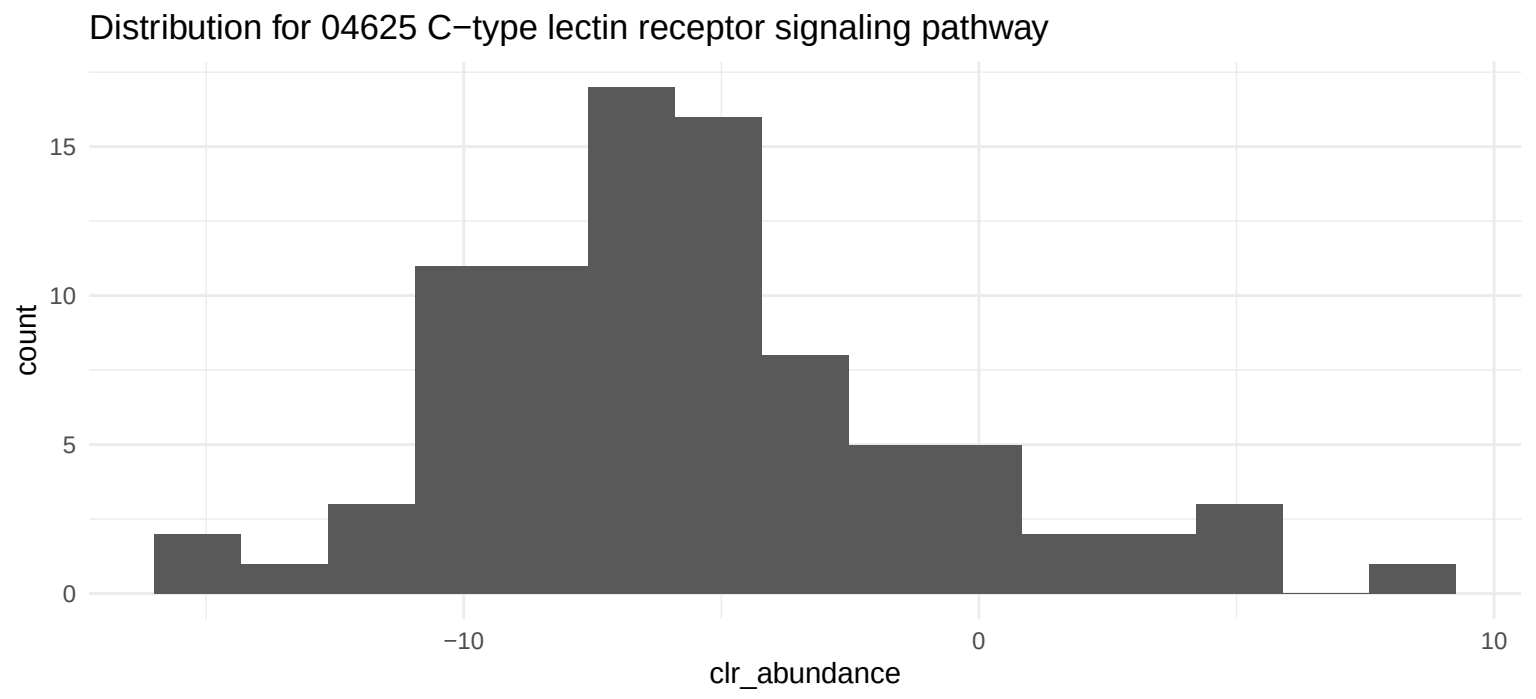
Distribution for 04621 NOD−like receptor signaling pathway

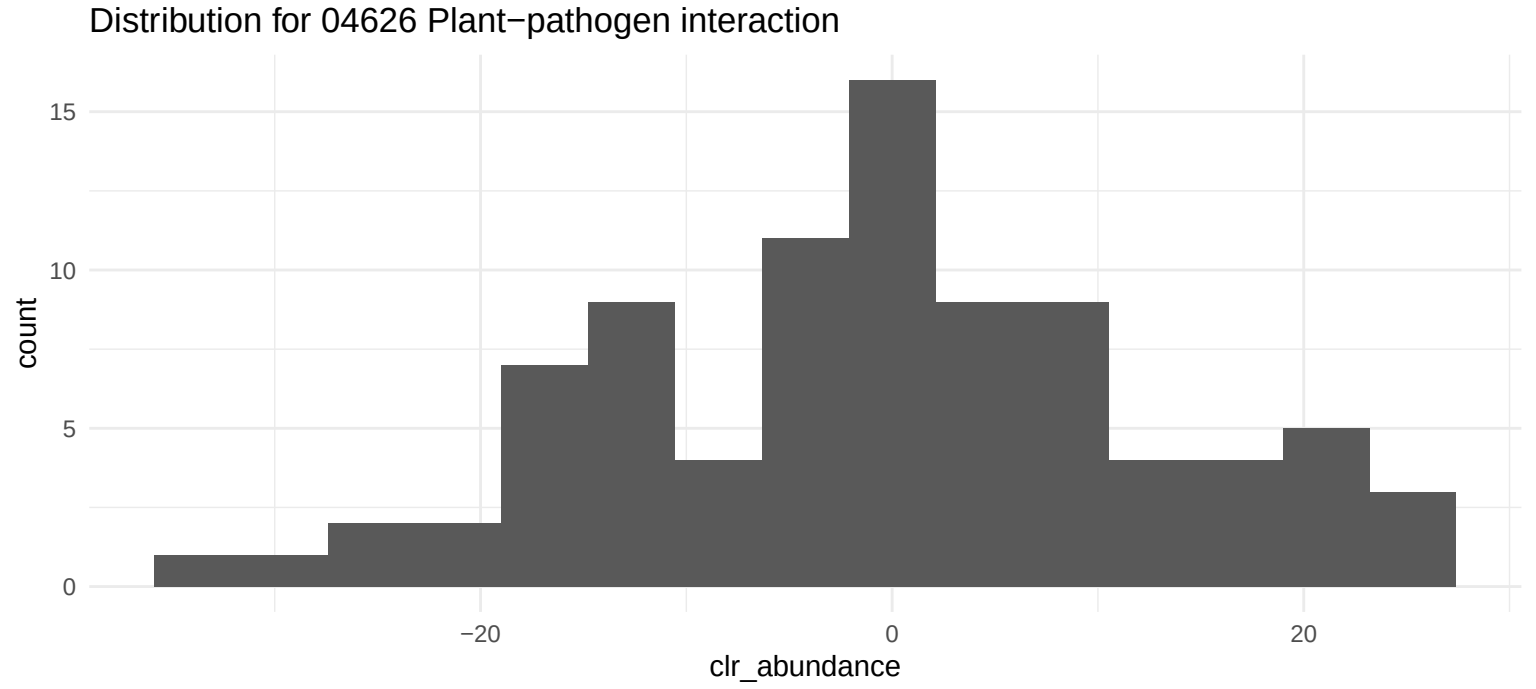


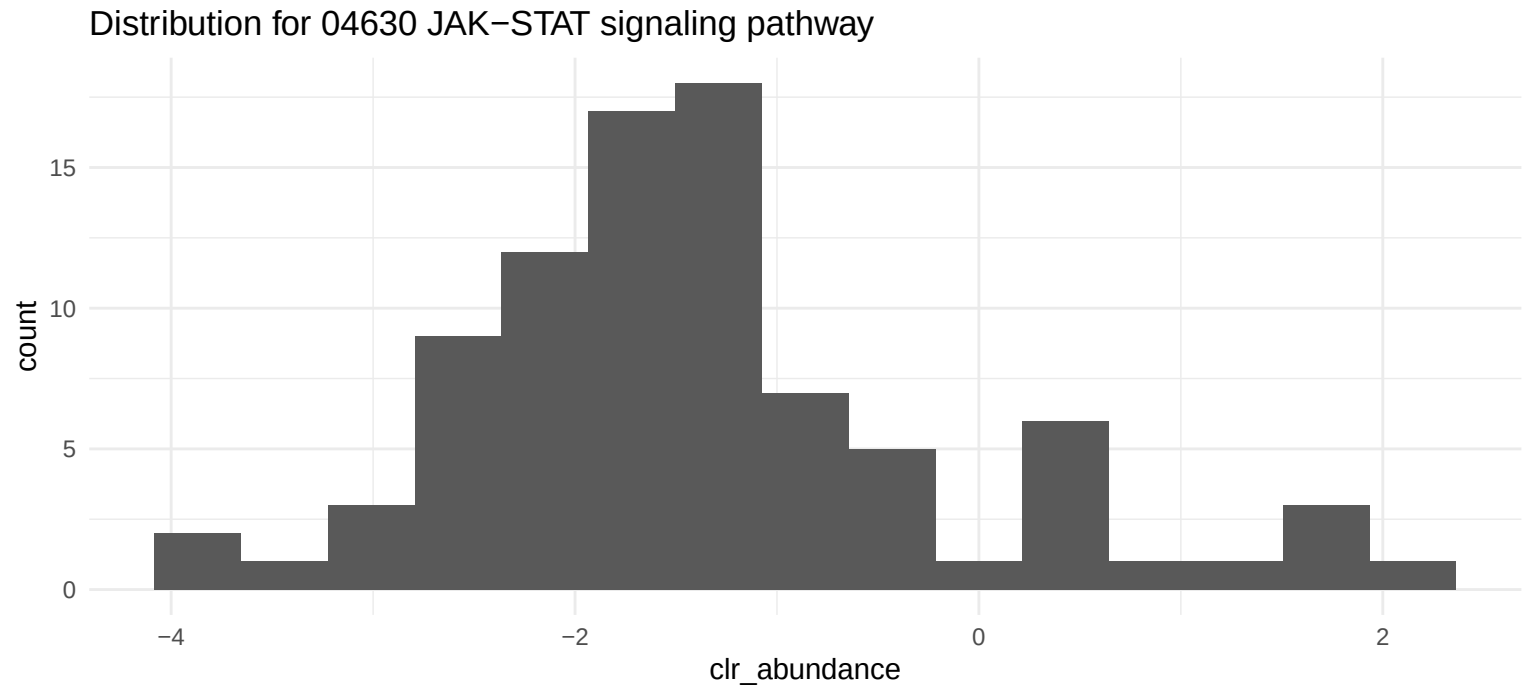

Distribution for 04622 RIG−I−like receptor signaling pathway
count
0
3
6
9
−10
−5
0
5
clr_abundance


Distribution for 04623 Cytosolic DNA−sensing pathway
count
0
5
10
15
−20
−10
0
clr_abundance

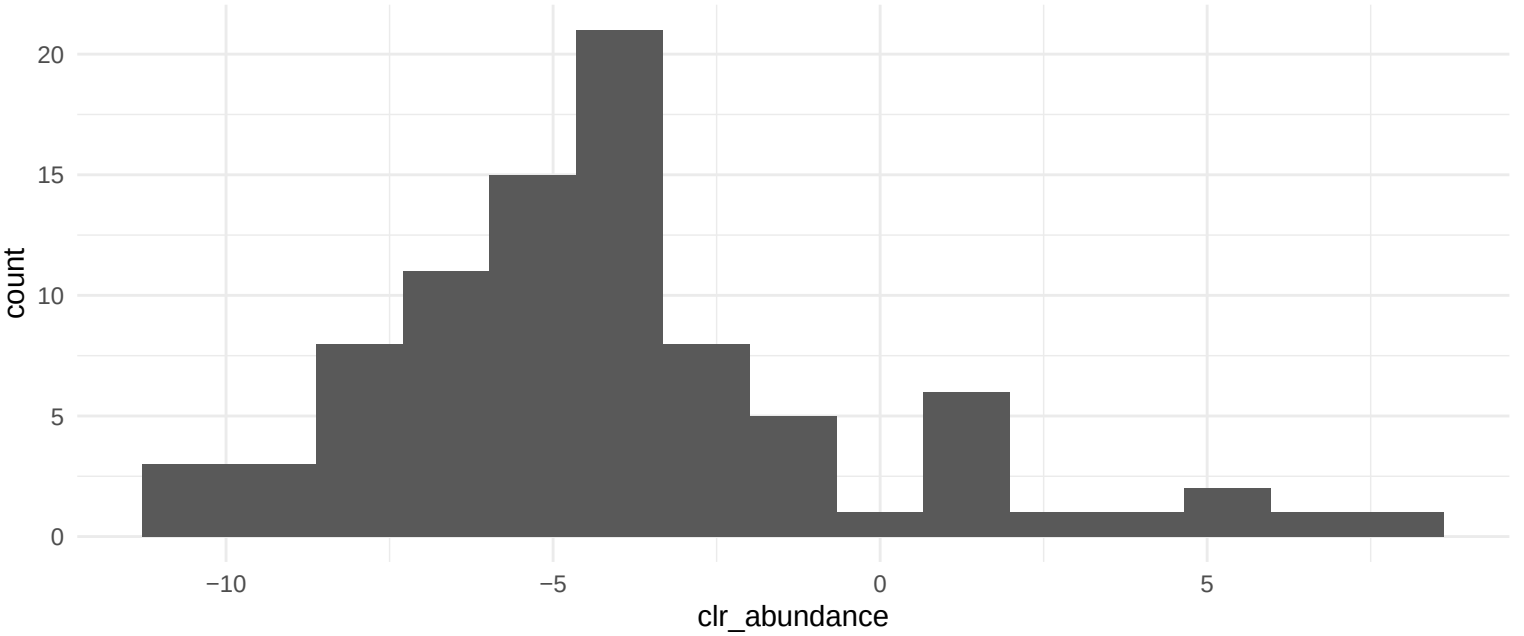

Distribution for 04624 Toll and Imd signaling pathway
count
0
5
10
15
−20
−15
−10
−5
0
5
clr_abundance


Distribution for 04625 C−type lectin receptor signaling pathway
count
15
10
5
0
−10
0
10
clr_abundance

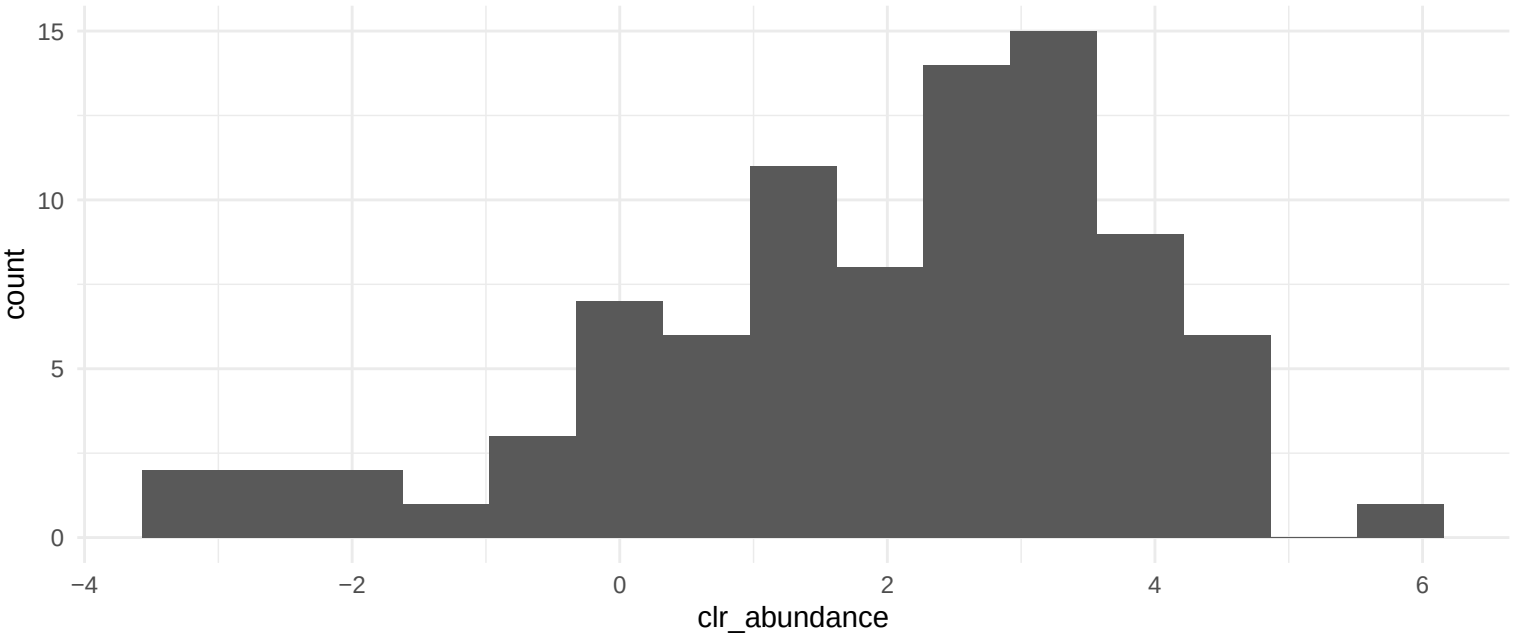

Distribution for 04626 Plant−pathogen interaction
count
15
10
5
0
−20
0
20
clr_abundance


Distribution for 04630 JAK−STAT signaling pathway
count
15
10
5
0
−4
−2
0
2
clr_abundance

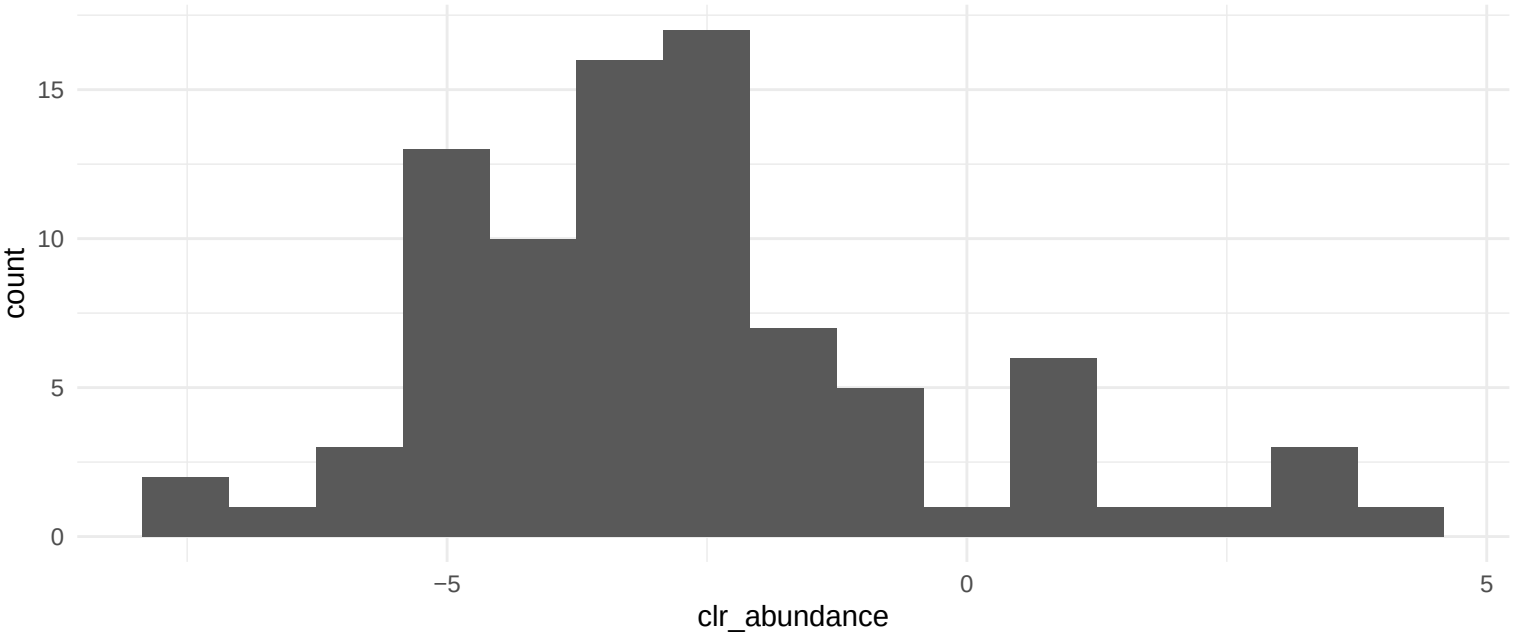
Distribution for 04650 Natural killer cell mediated cytotoxicity



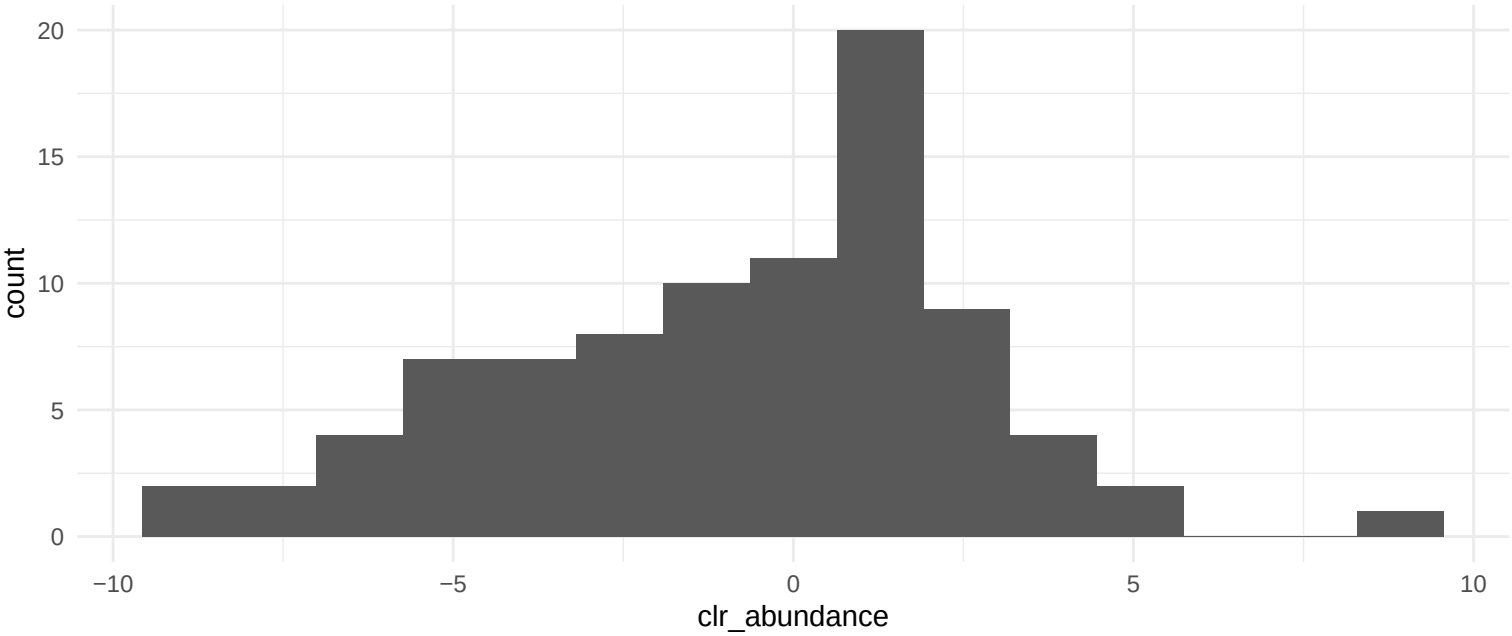
Distribution for 04657 IL−17 signaling pathway



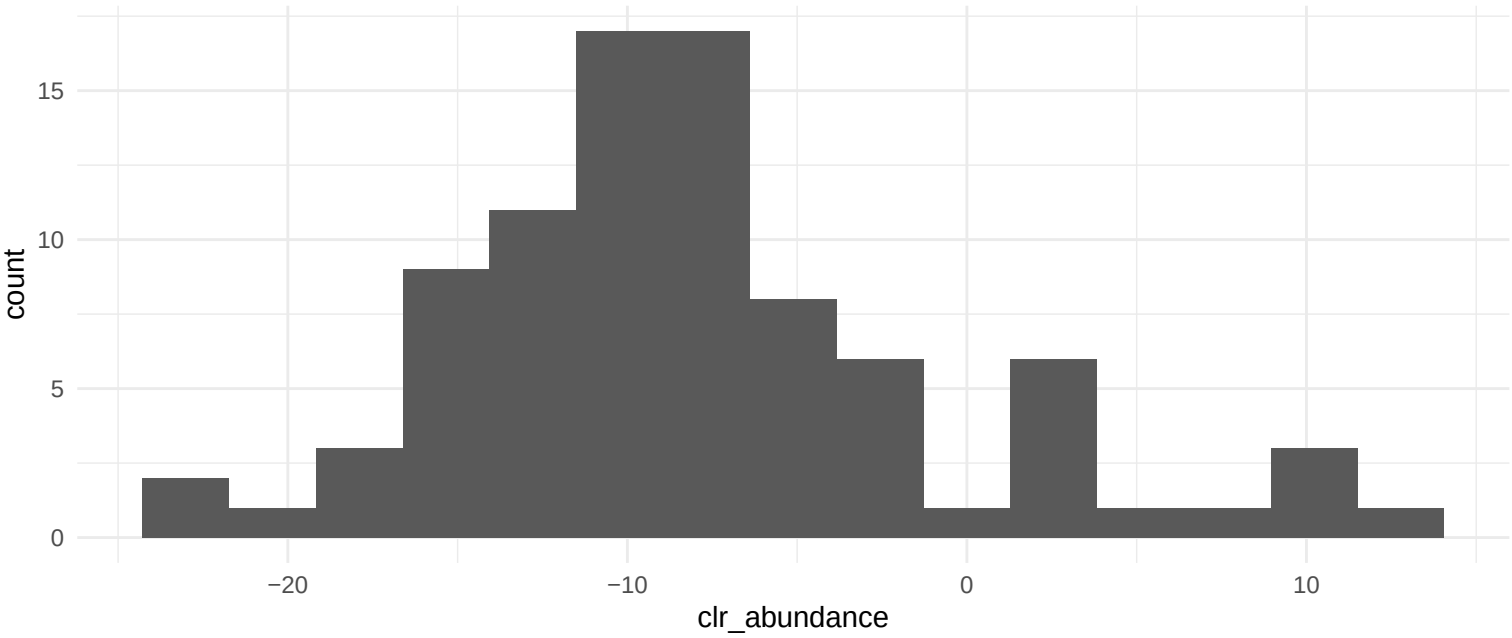
Distribution for 04658 Th1 and Th2 cell differentiation



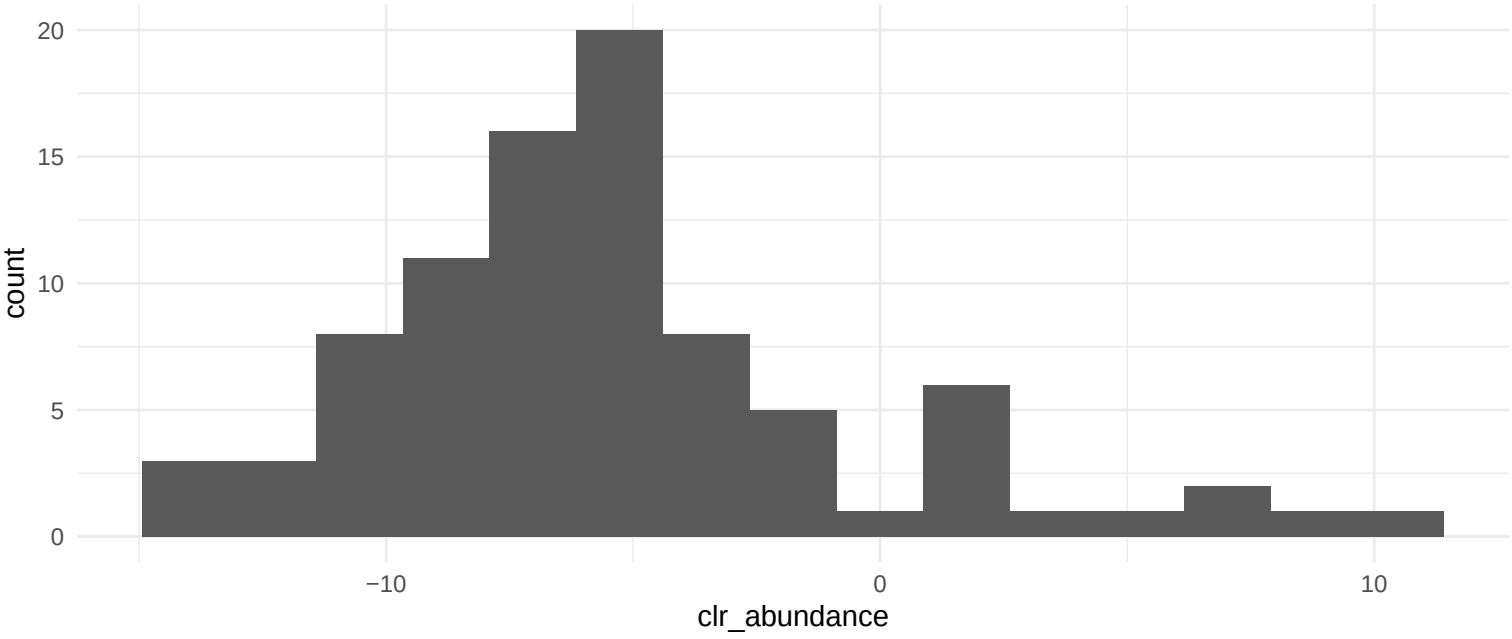
Distribution for 04659 Th17 cell differentiation



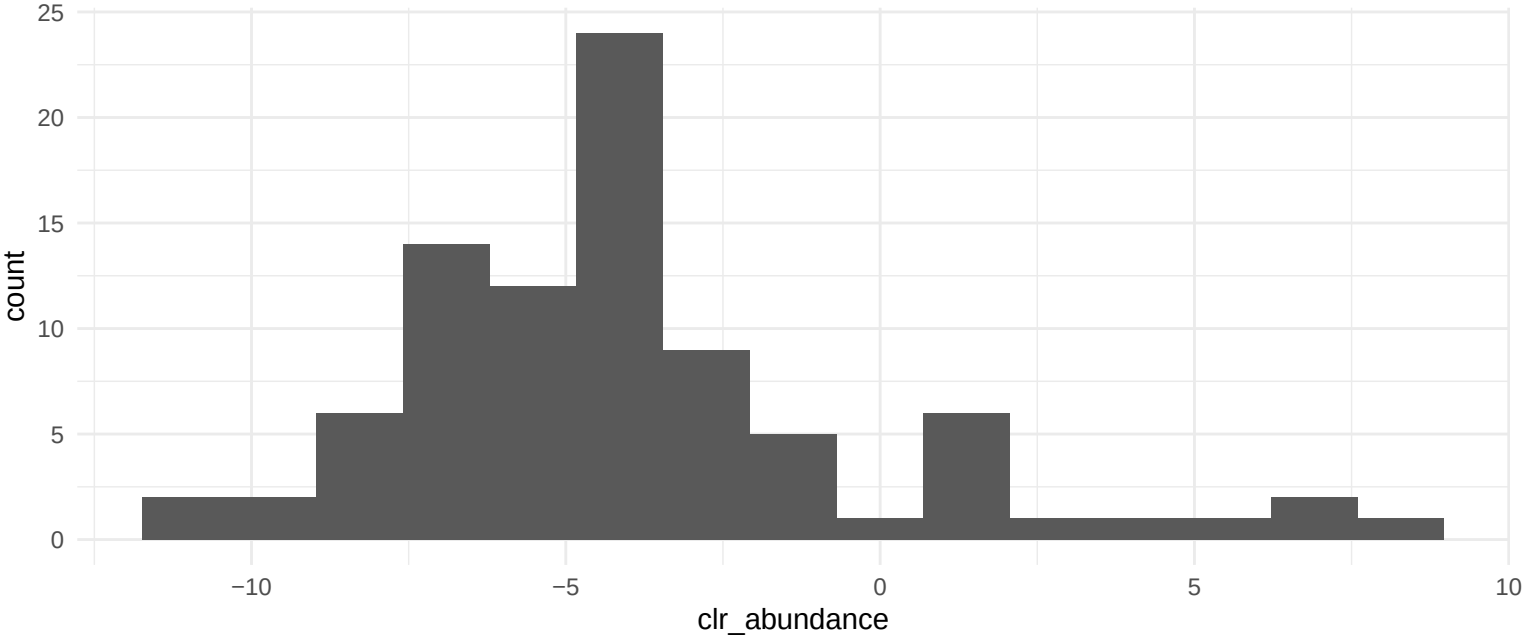
Distribution for 04660 T cell receptor signaling pathway



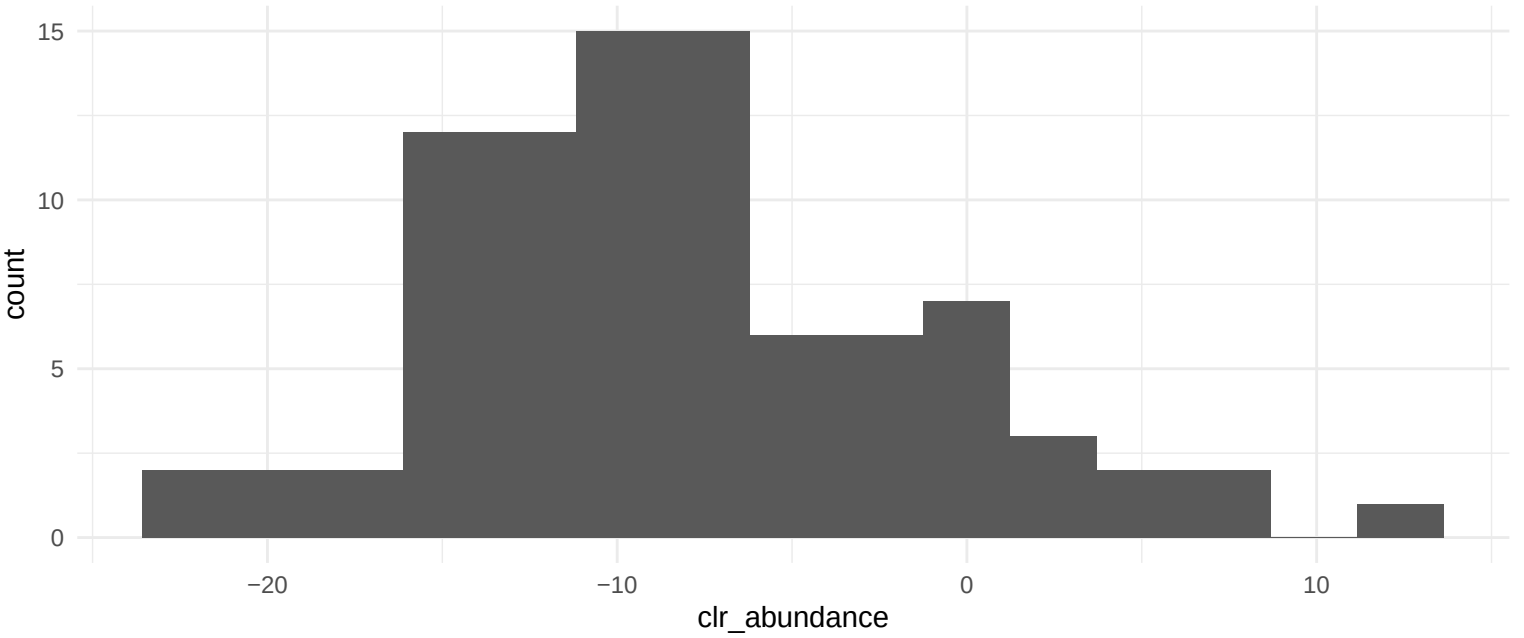
Distribution for 04662 B cell receptor signaling pathway



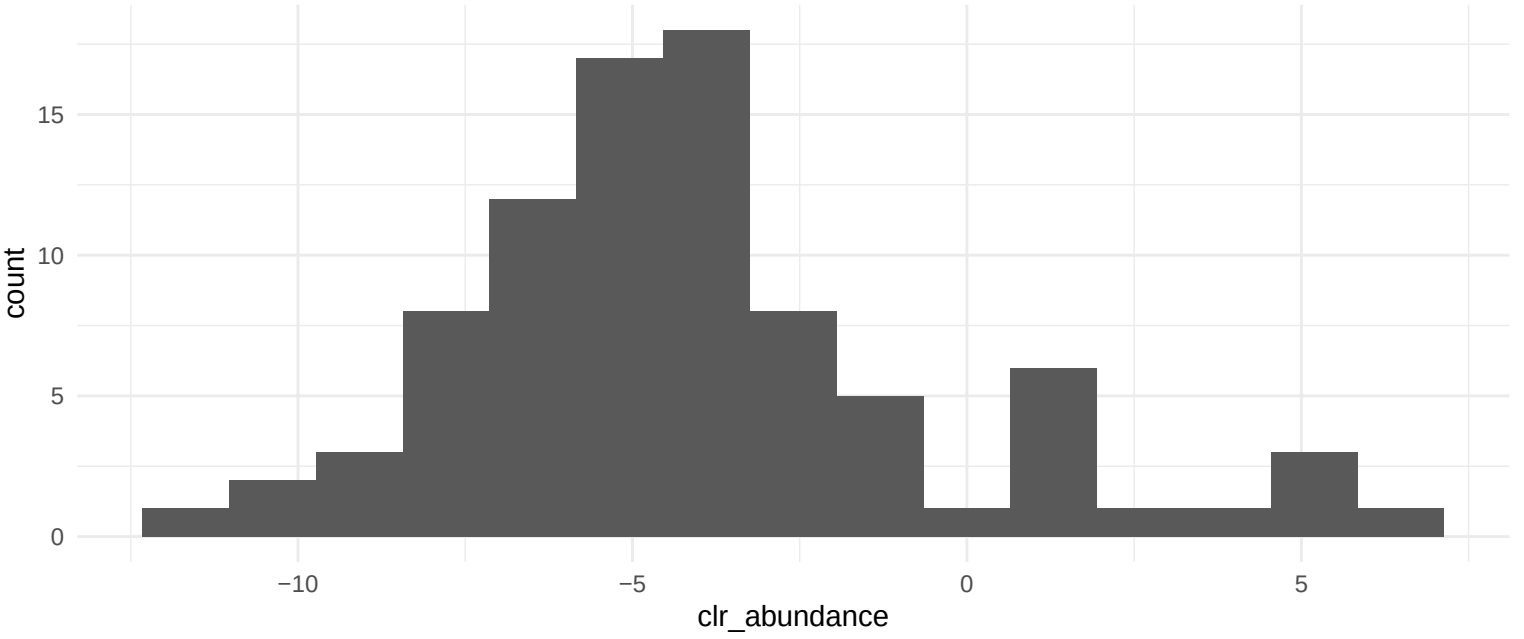
Distribution for 04664 Fc epsilon RI signaling pathway

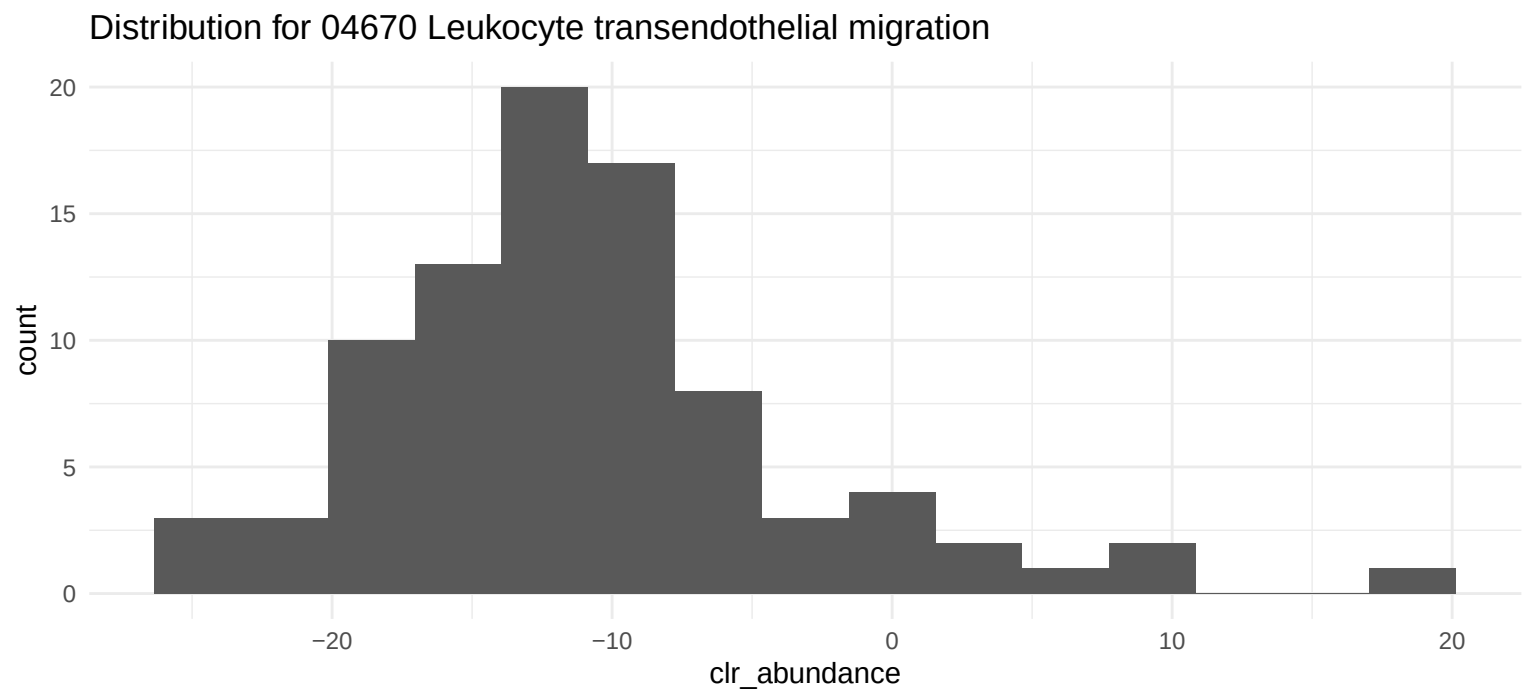


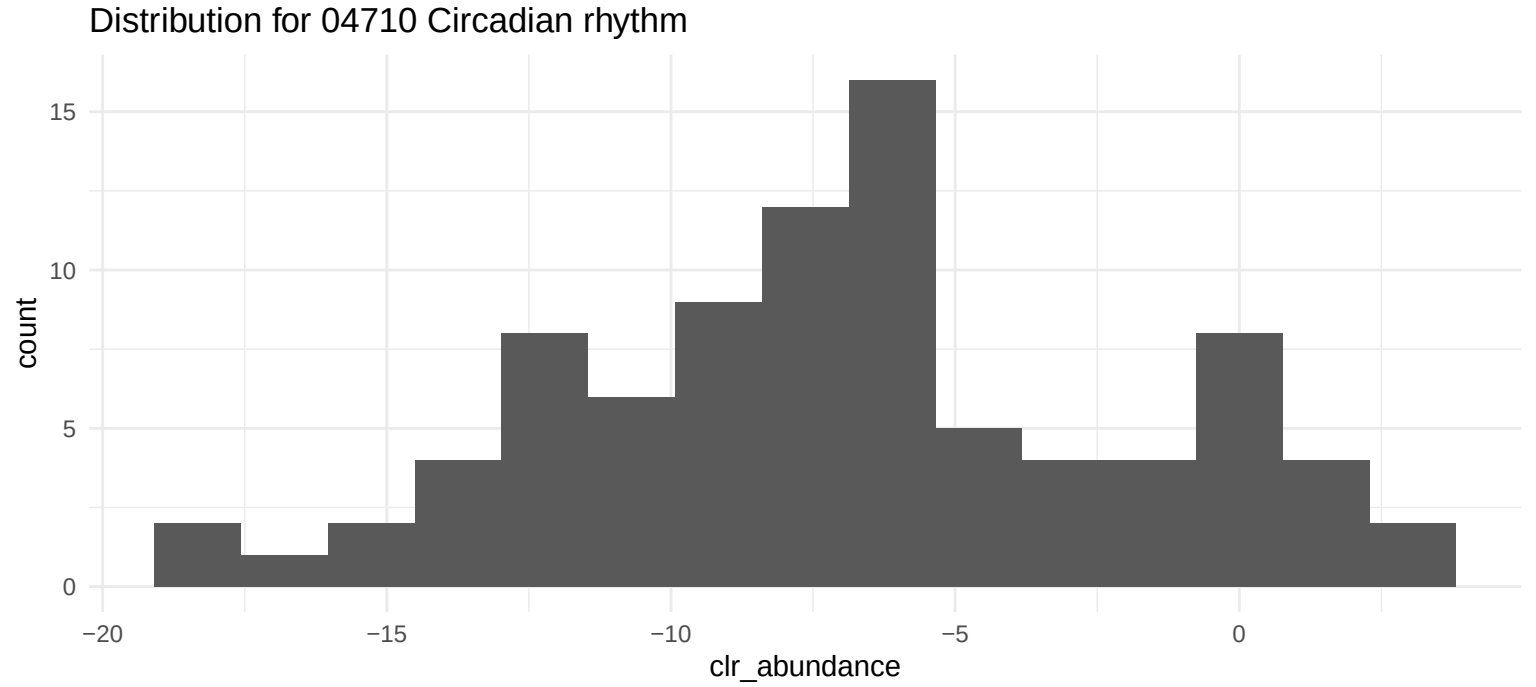
Distribution for 04666 Fc gamma R−mediated phagocytosis

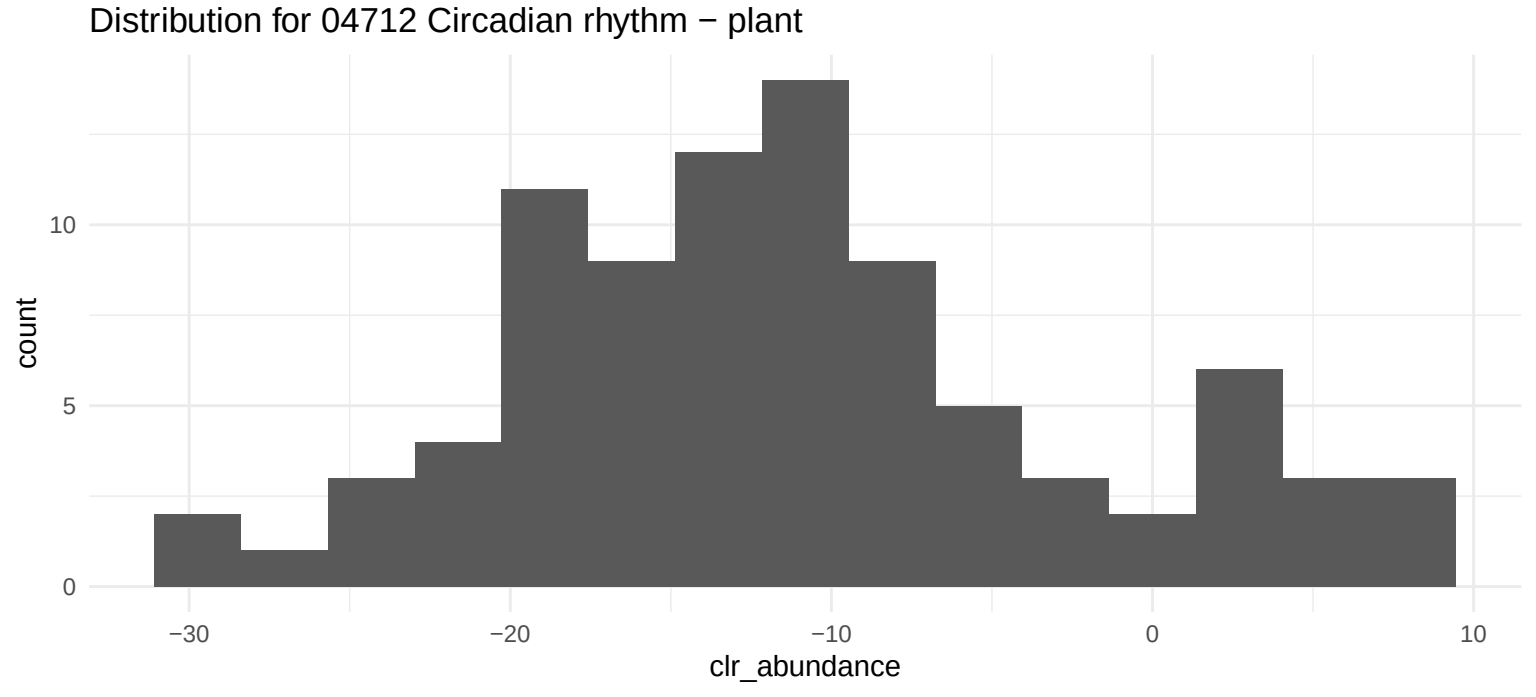


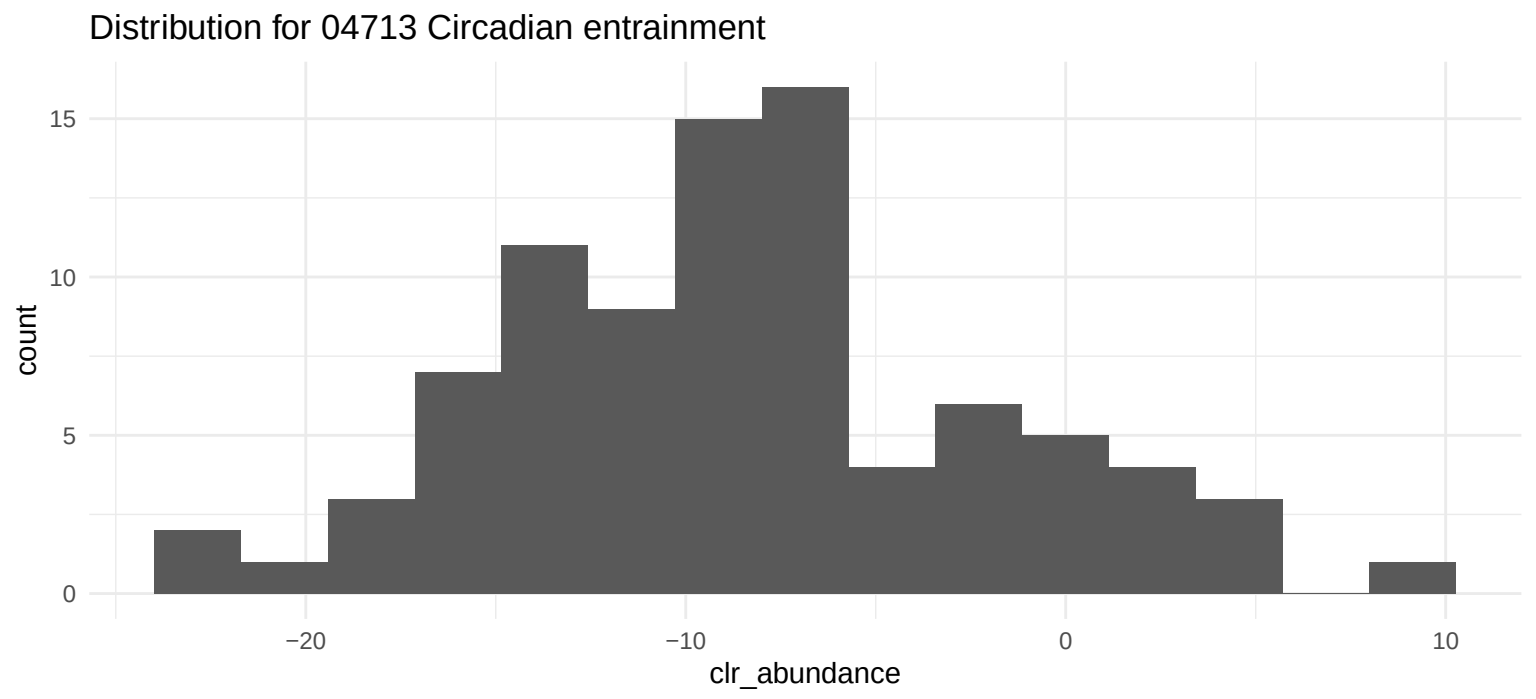
Distribution for 04668 TNF signaling pathway

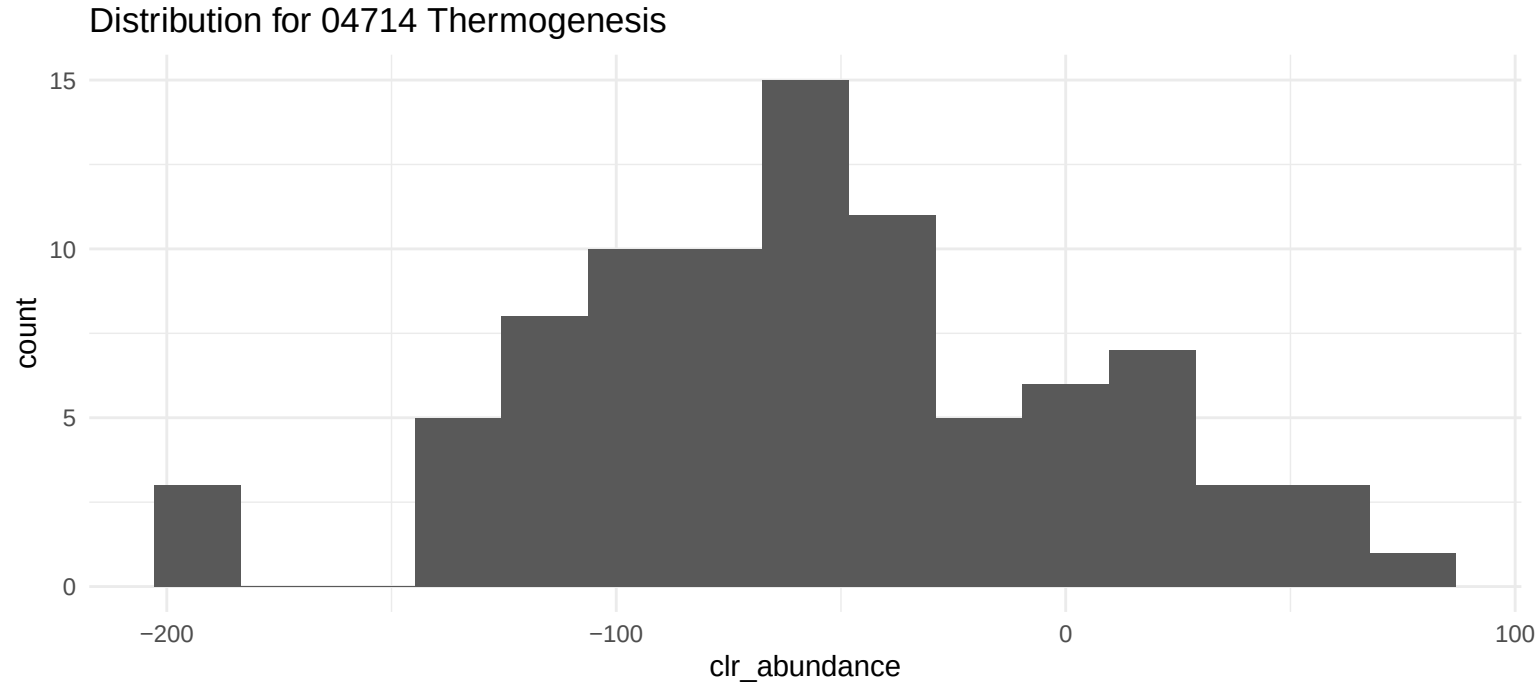


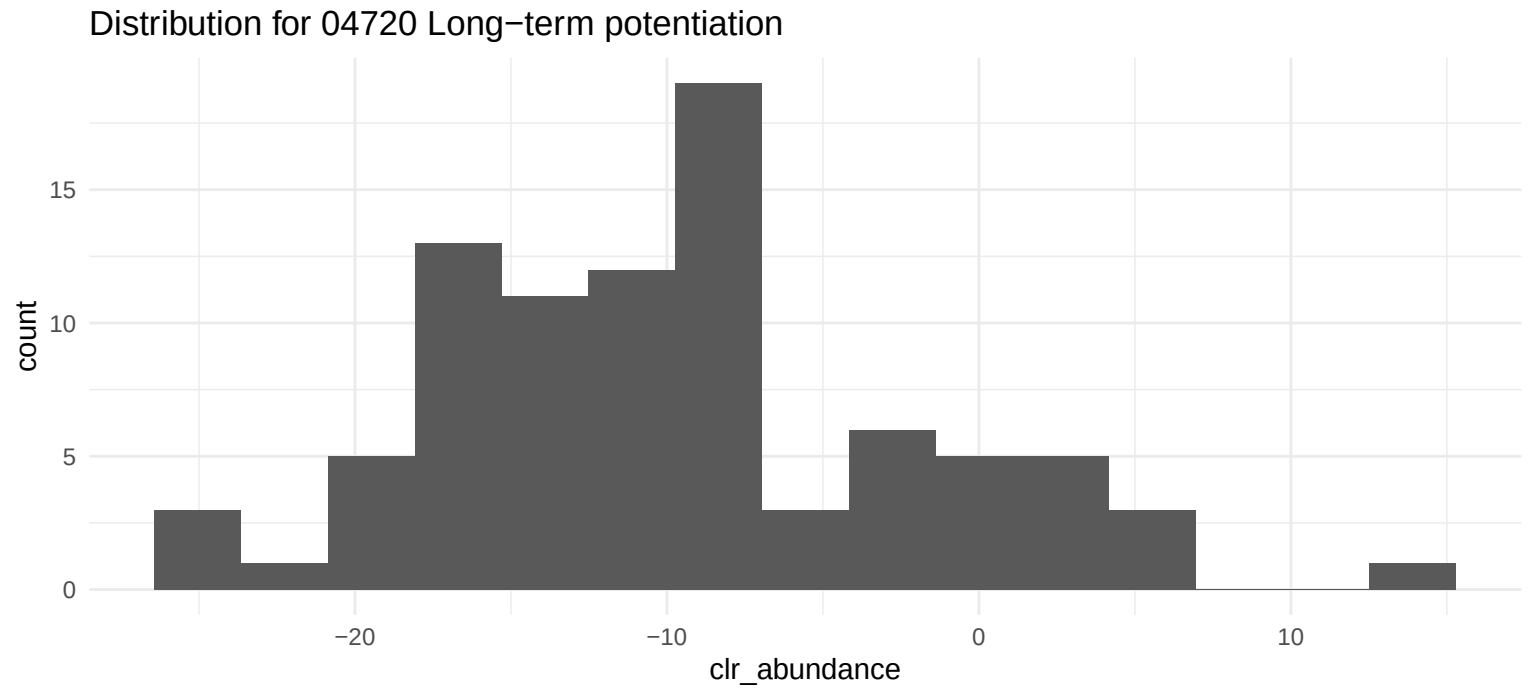

Distribution for 04670 Leukocyte transendothelial migration
count
clr_abundance
20
15
10
5
0
−20
−10
0
10
20


Distribution for 04710 Circadian rhythm
count
clr_abundance
15
10
5
0
−20
−15
−10
−5
0

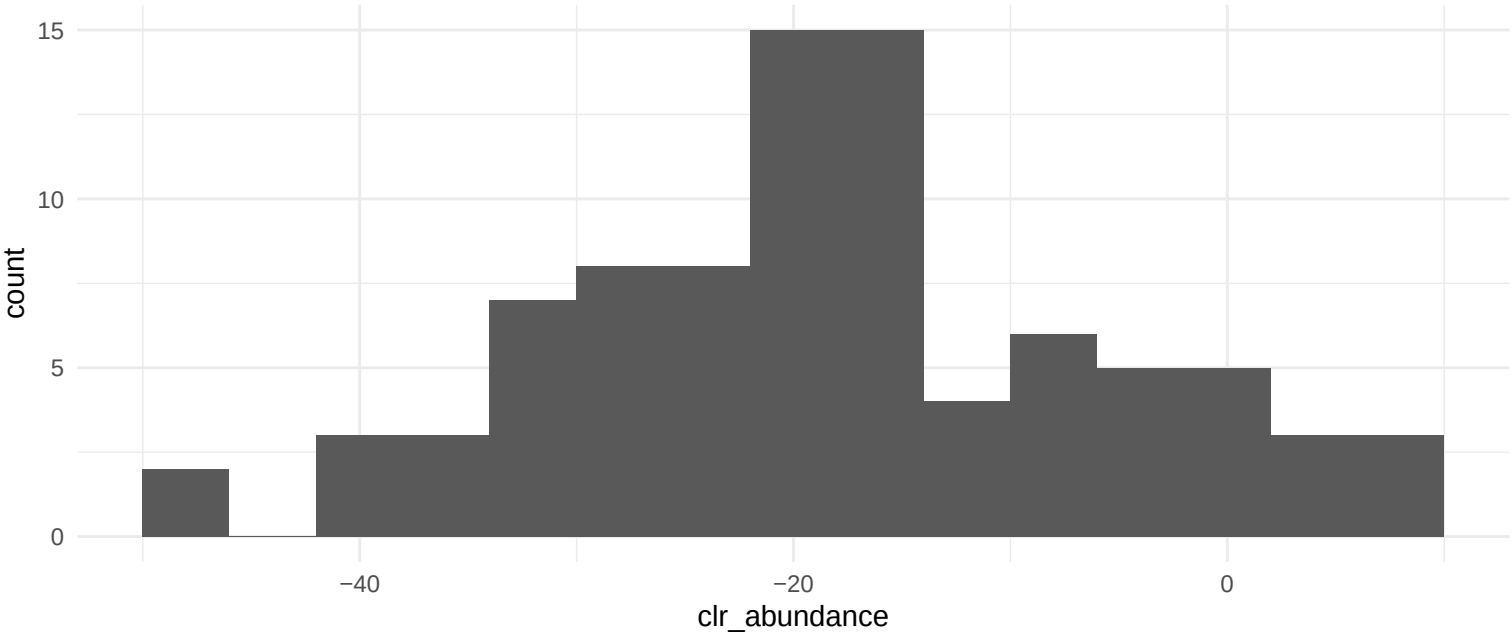

Distribution for 04712 Circadian rhythm − plant
count
clr_abundance
10
5
0
−30
−20
−10
0
10


Distribution for 04713 Circadian entrainment
count
clr_abundance
15
10
5
0
−20
−10
0
10

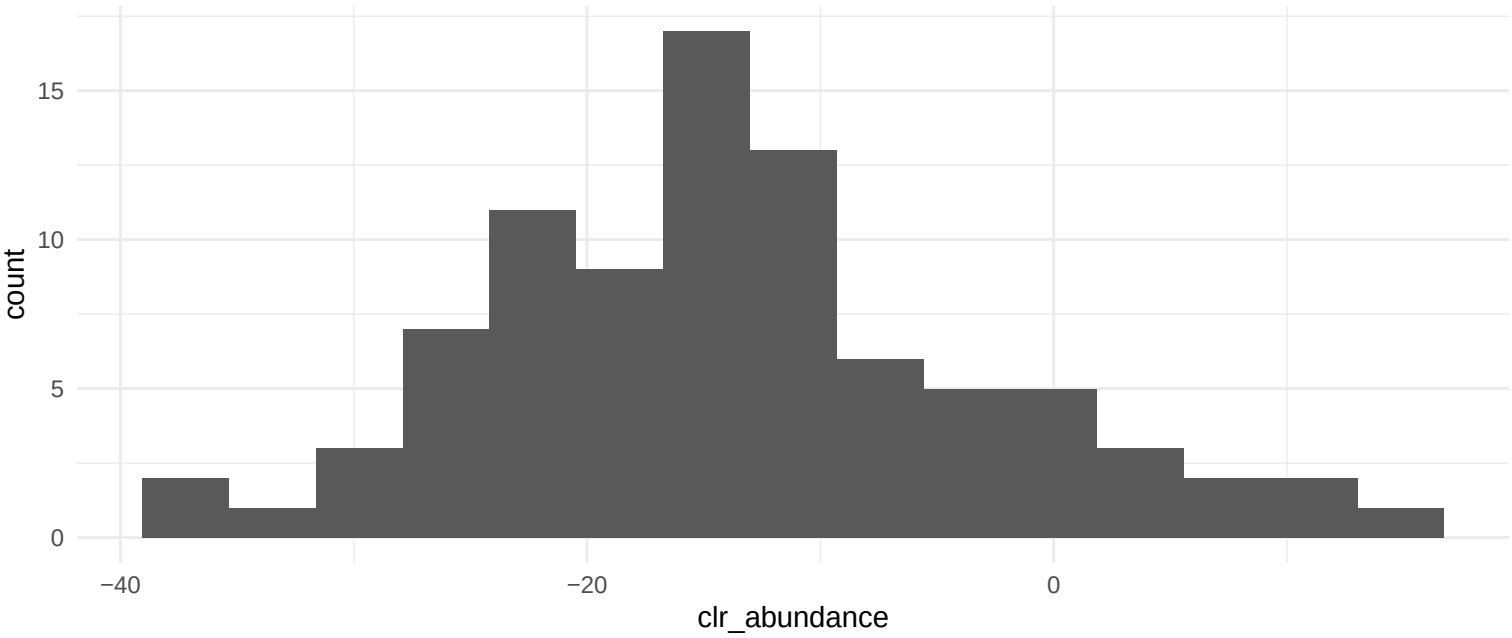

Distribution for 04714 Thermogenesis
count
clr_abundance
15
10
5
0
−200
−100
0
100


Distribution for 04720 Long−term potentiation
count
clr_abundance
15
10
5
0
−20
−10
0
10

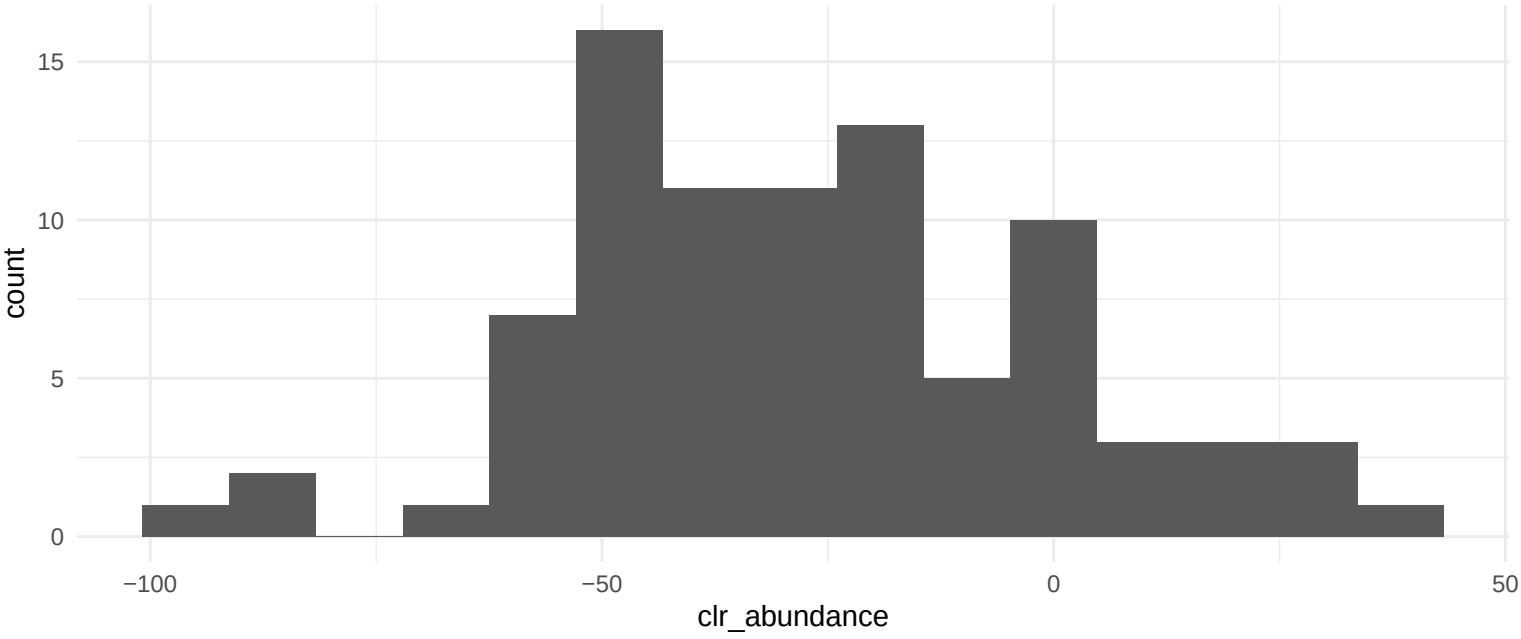
Distribution for 04721 Synaptic vesicle cycle



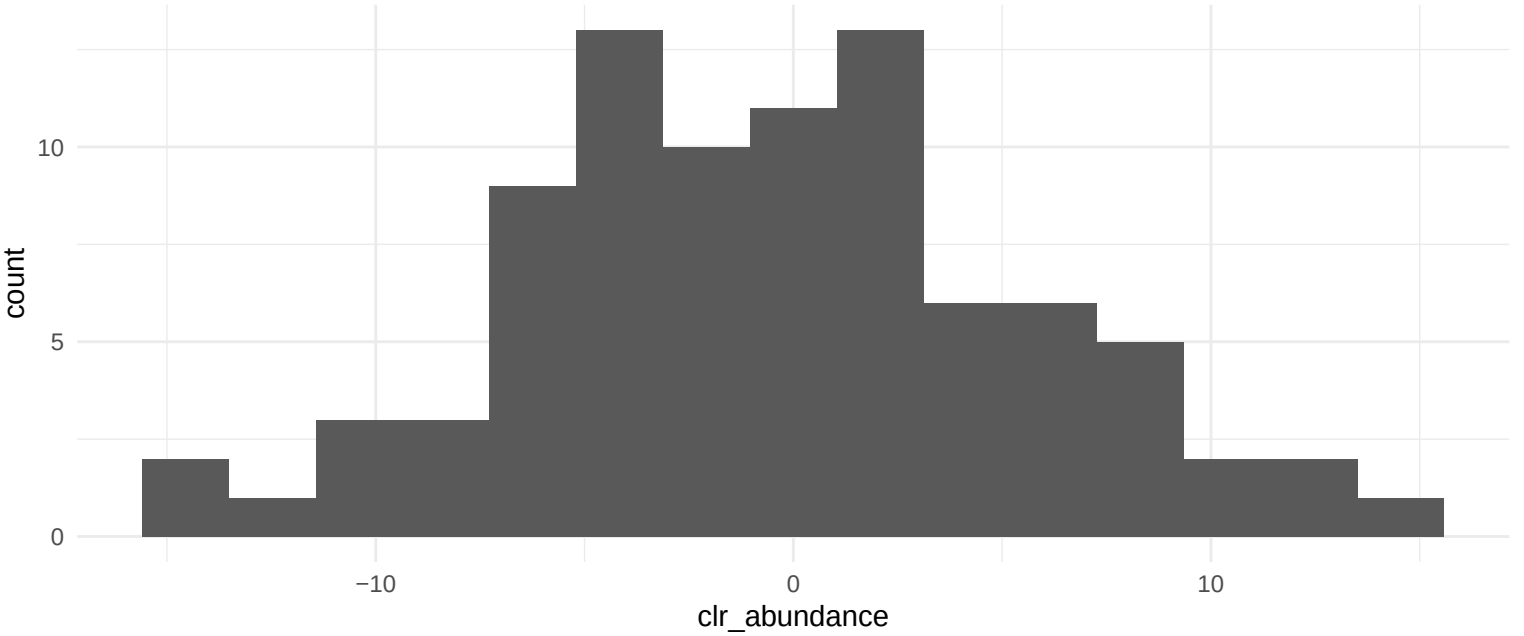
Distribution for 04722 Neurotrophin signaling pathway



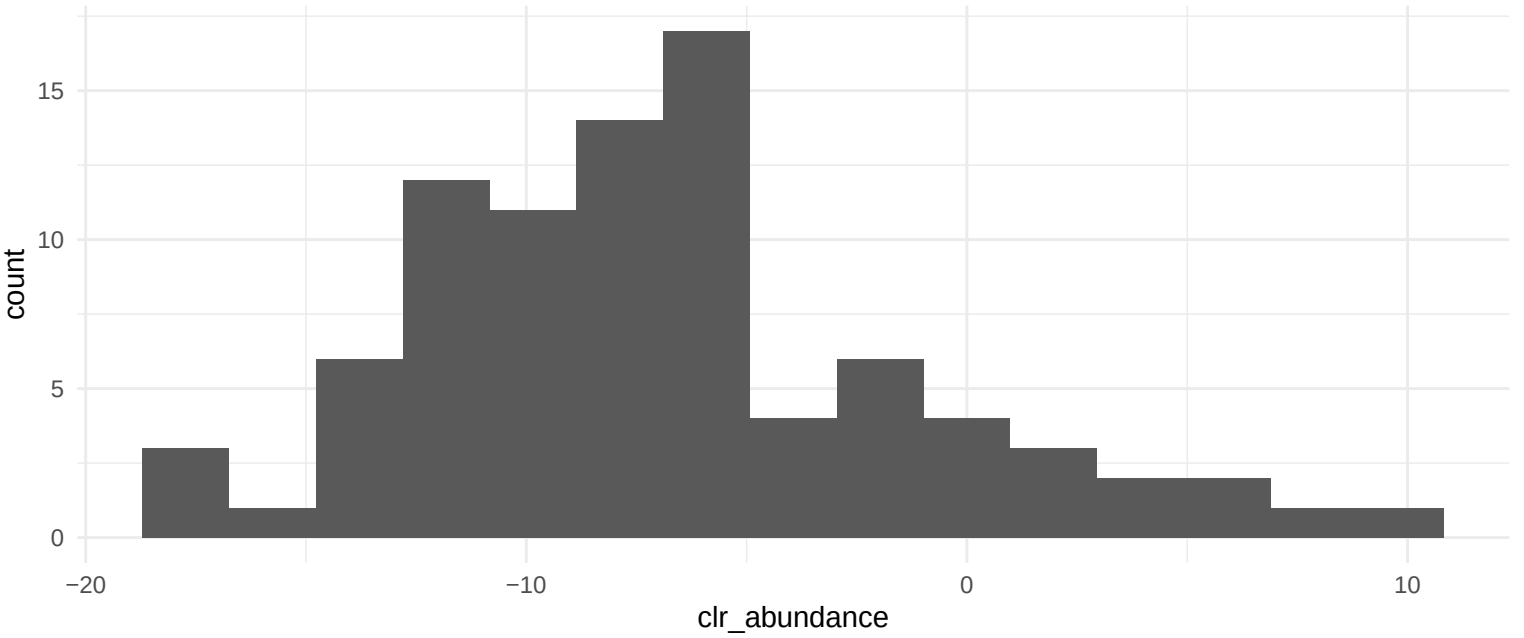
Distribution for 04723 Retrograde endocannabinoid signaling



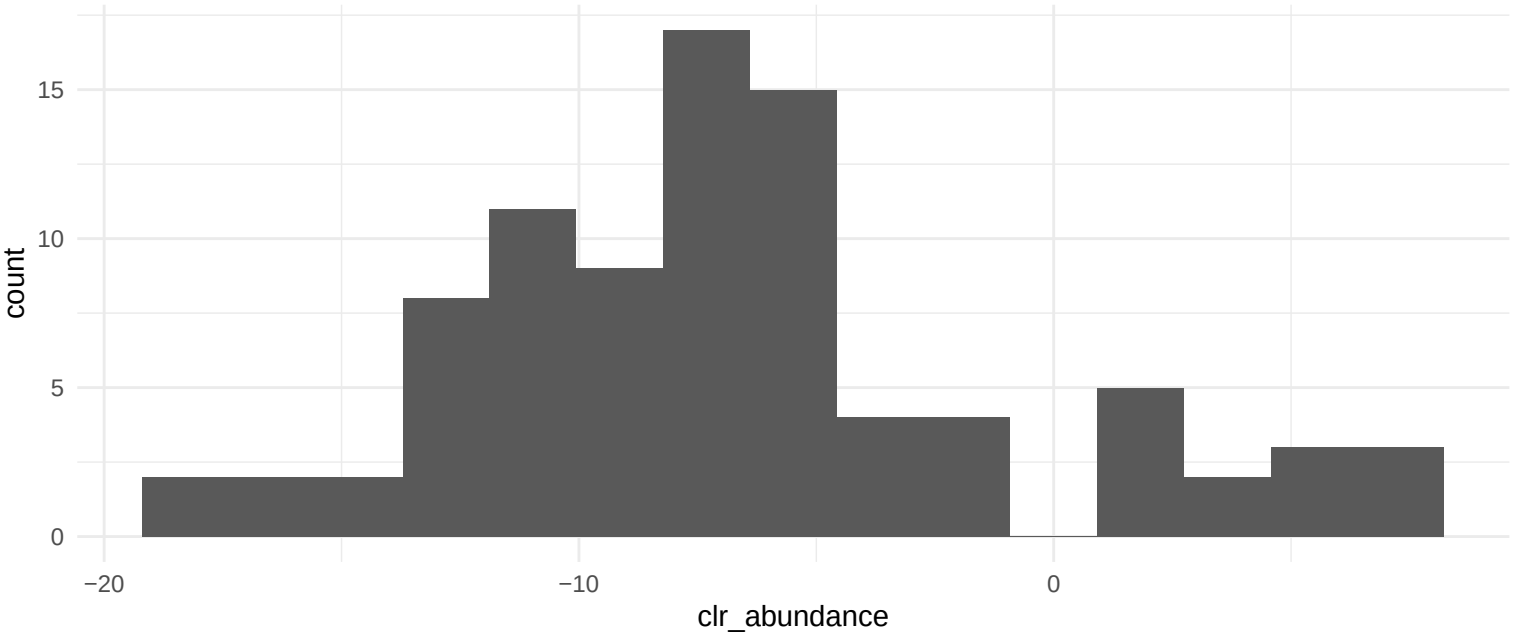
Distribution for 04724 Glutamatergic synapse

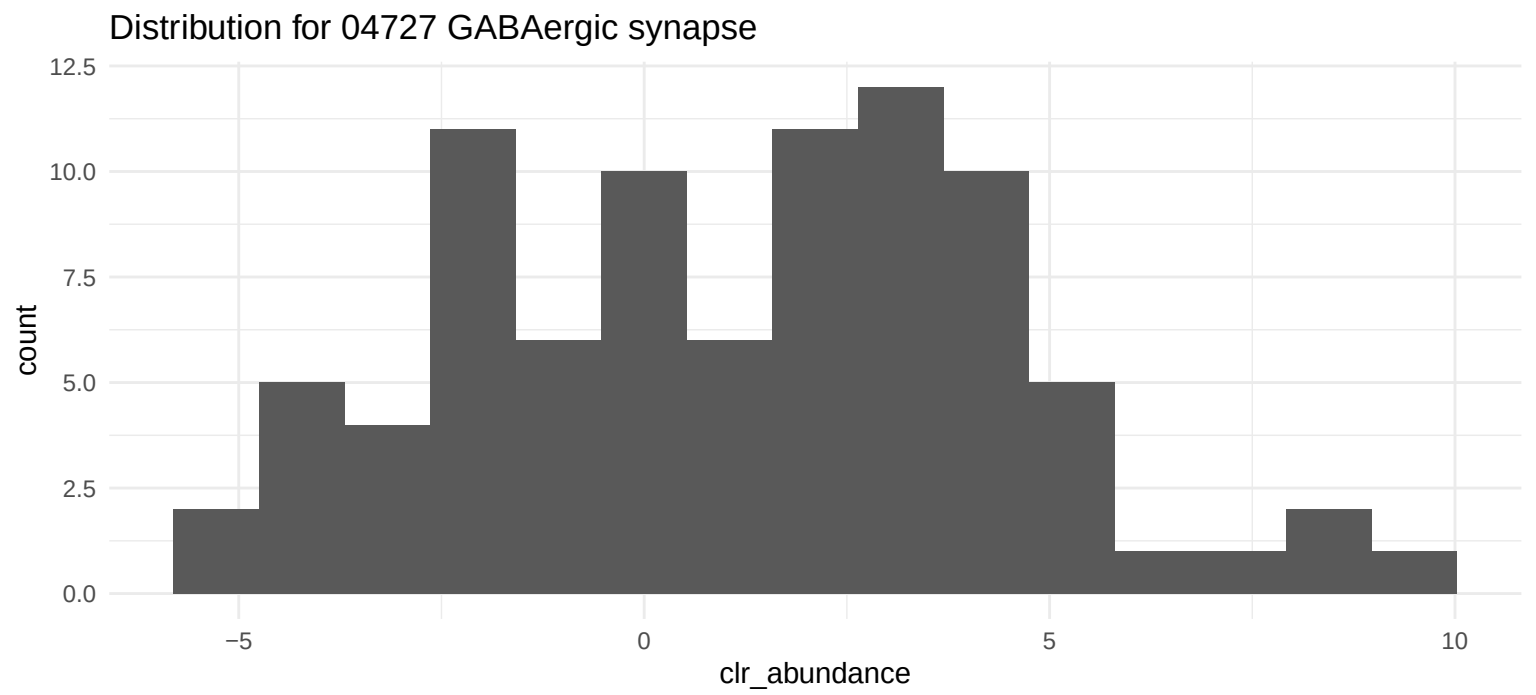


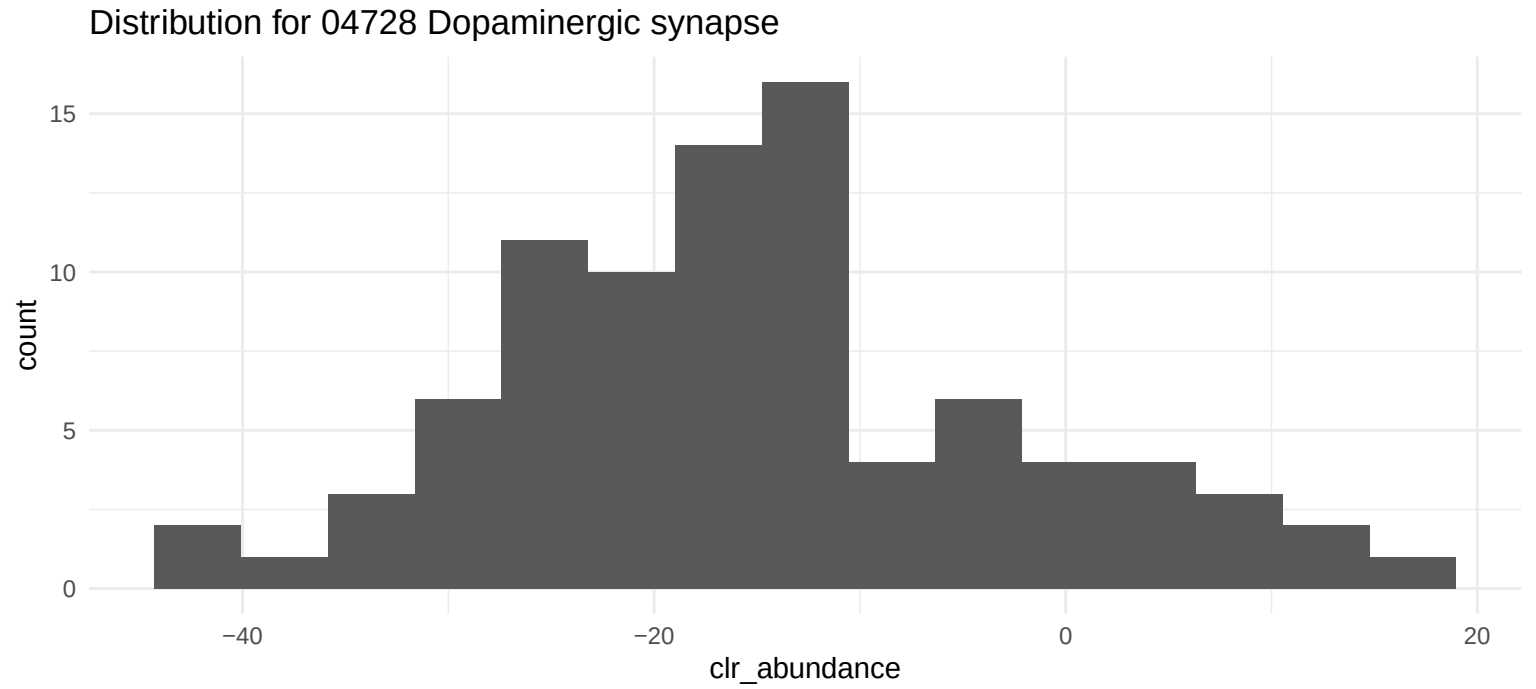
Distribution for 04725 Cholinergic synapse

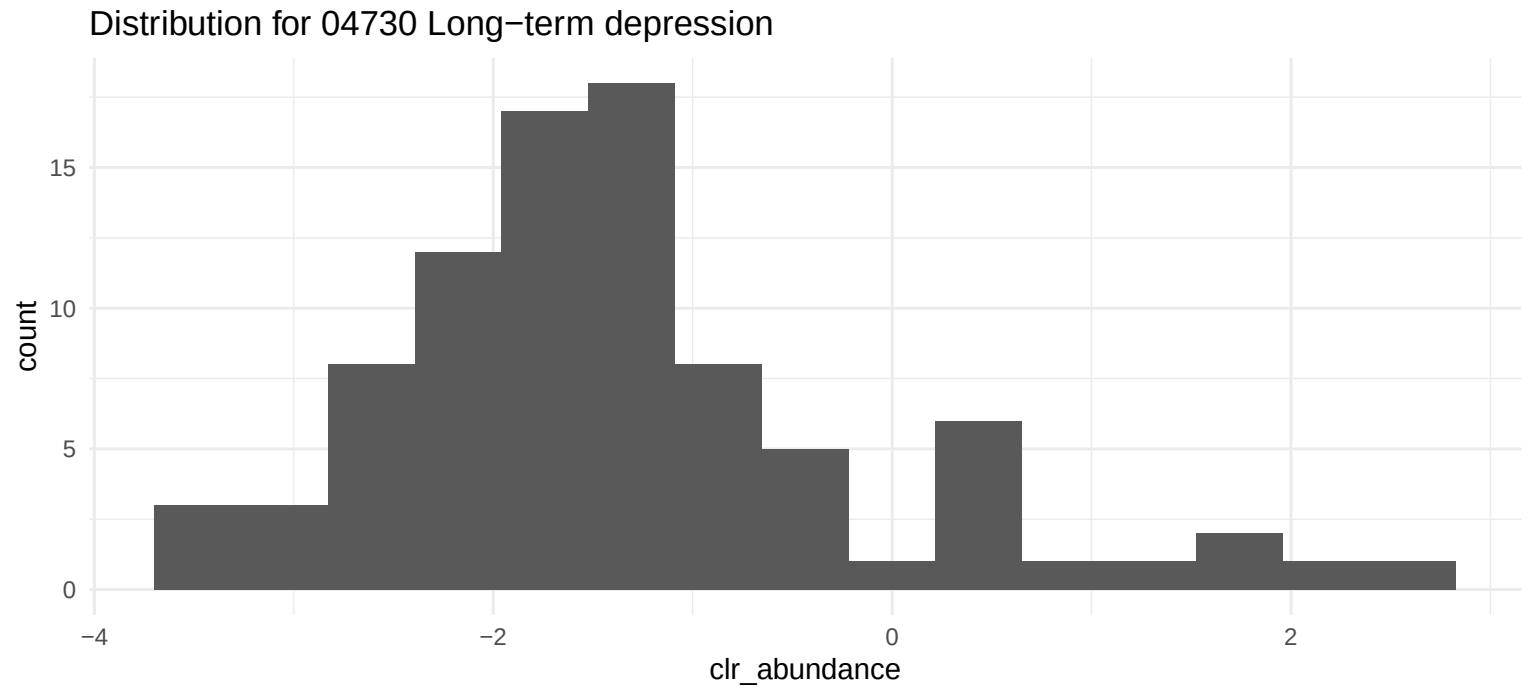


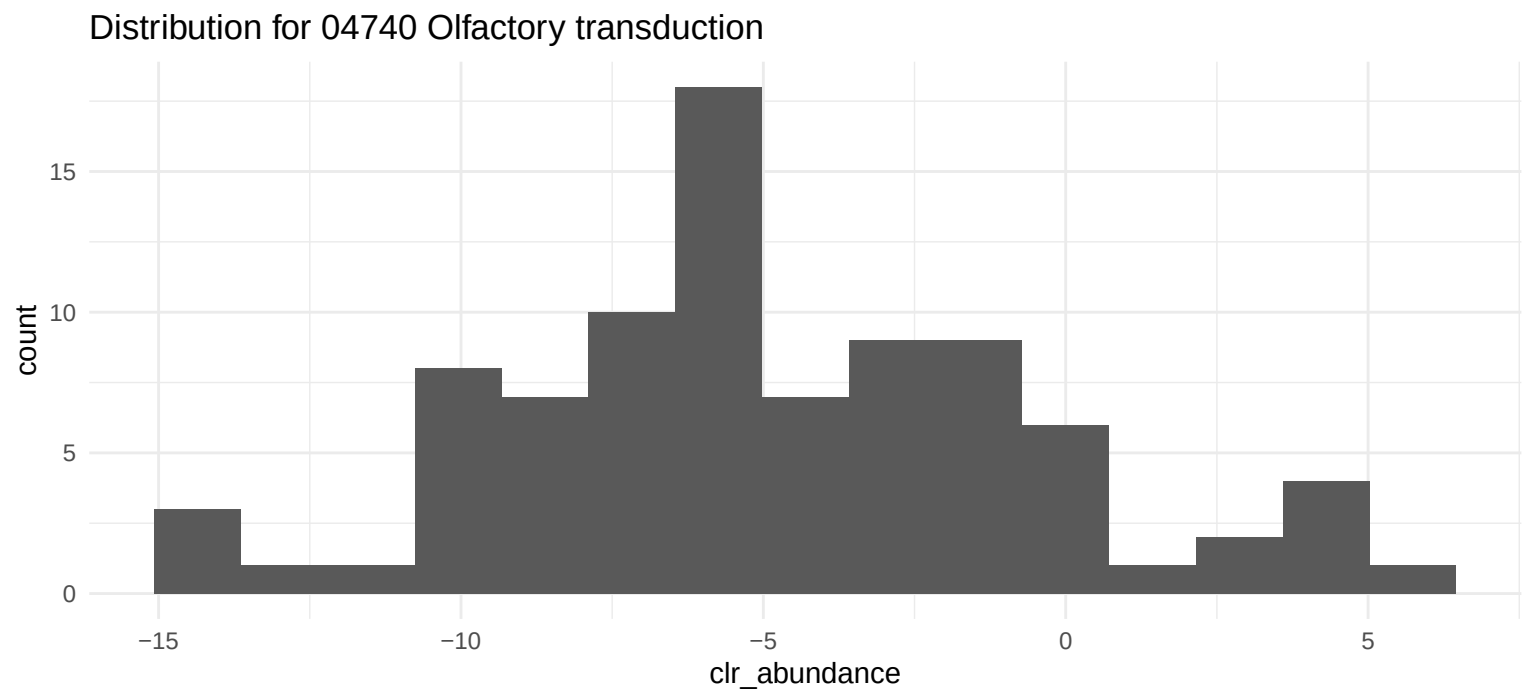
Distribution for 04726 Serotonergic synapse



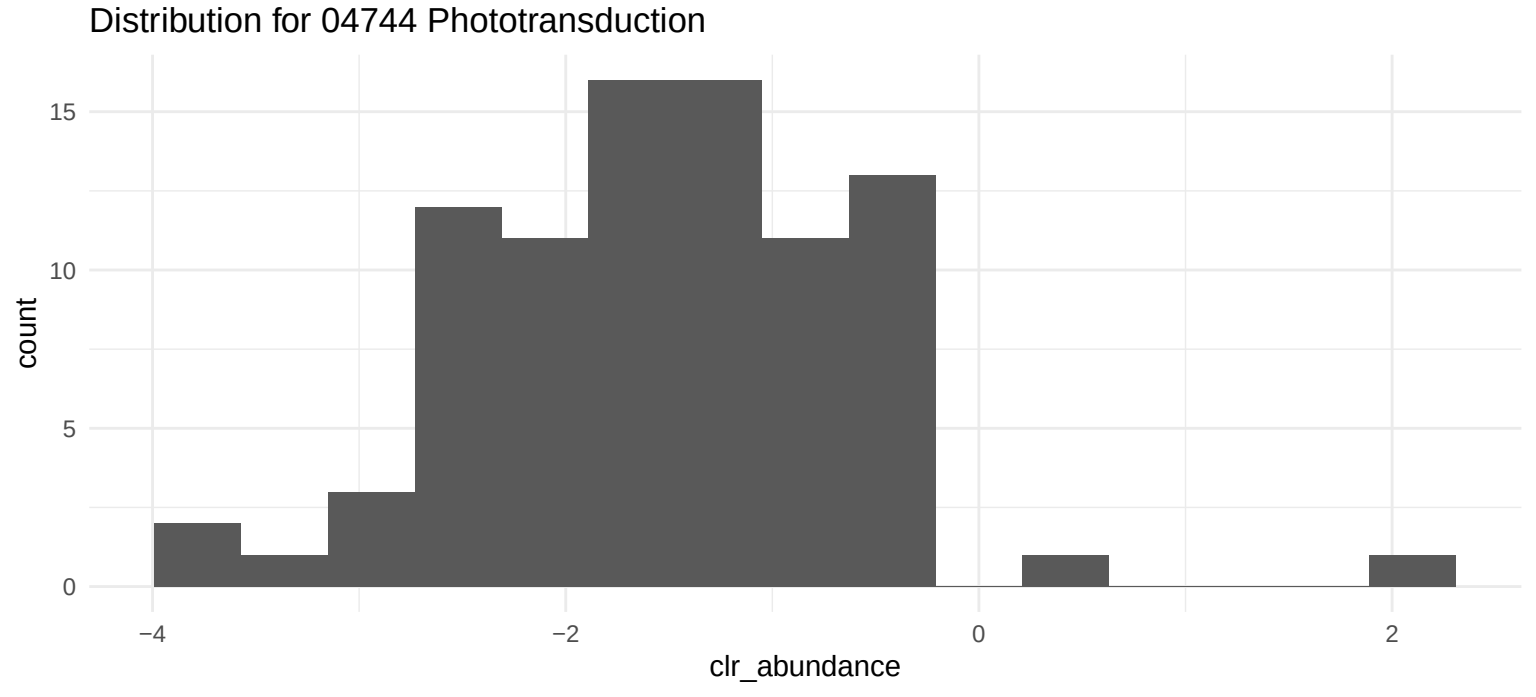

Distribution for 04727 GABAergic synapse
12.5
10.0
7.5
5.0
2.5
0.0
count
−5
0
5
10
clr_abundance


Distribution for 04728 Dopaminergic synapse
15
10
5
0
count
−40
−20
0
20
clr_abundance

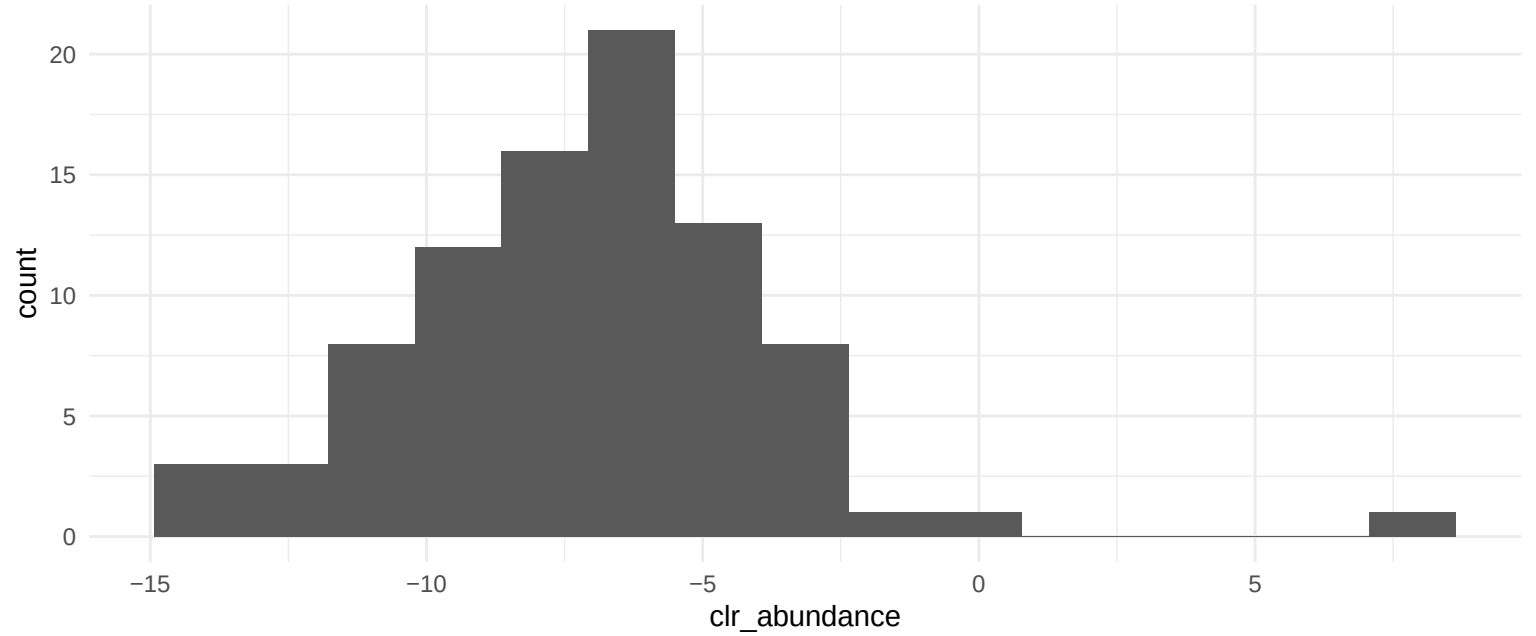

Distribution for 04730 Long−term depression
15
10
5
0
count
−4
−2
0
2
clr_abundance


Distribution for 04740 Olfactory transduction
count
15
10
5
0
−15
−10
−5
0
5
clr_abundance

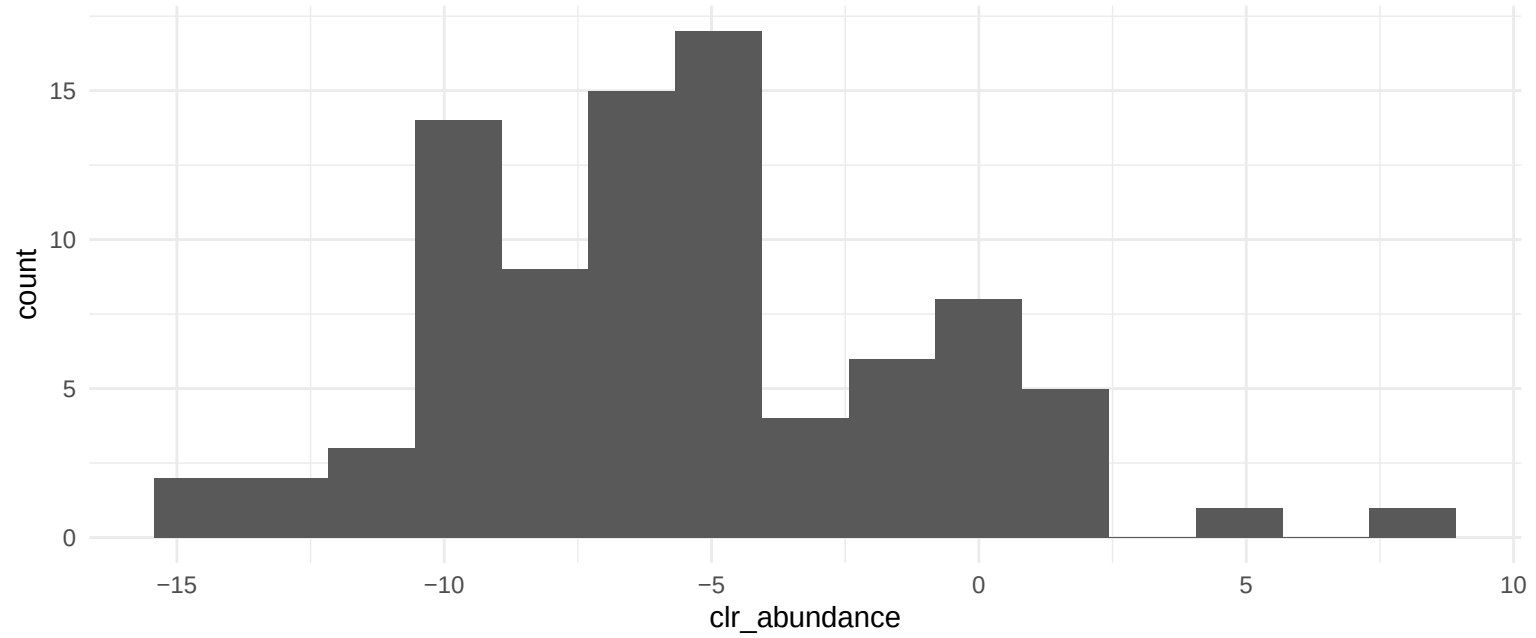
Distribution for 04742 Taste transduction
count
15
10
5
0
−4
−2
0
2
clr_abundance


Distribution for 04744 Phototransduction
count
15
10
5
0
−4
−2
0
2
clr_abundance

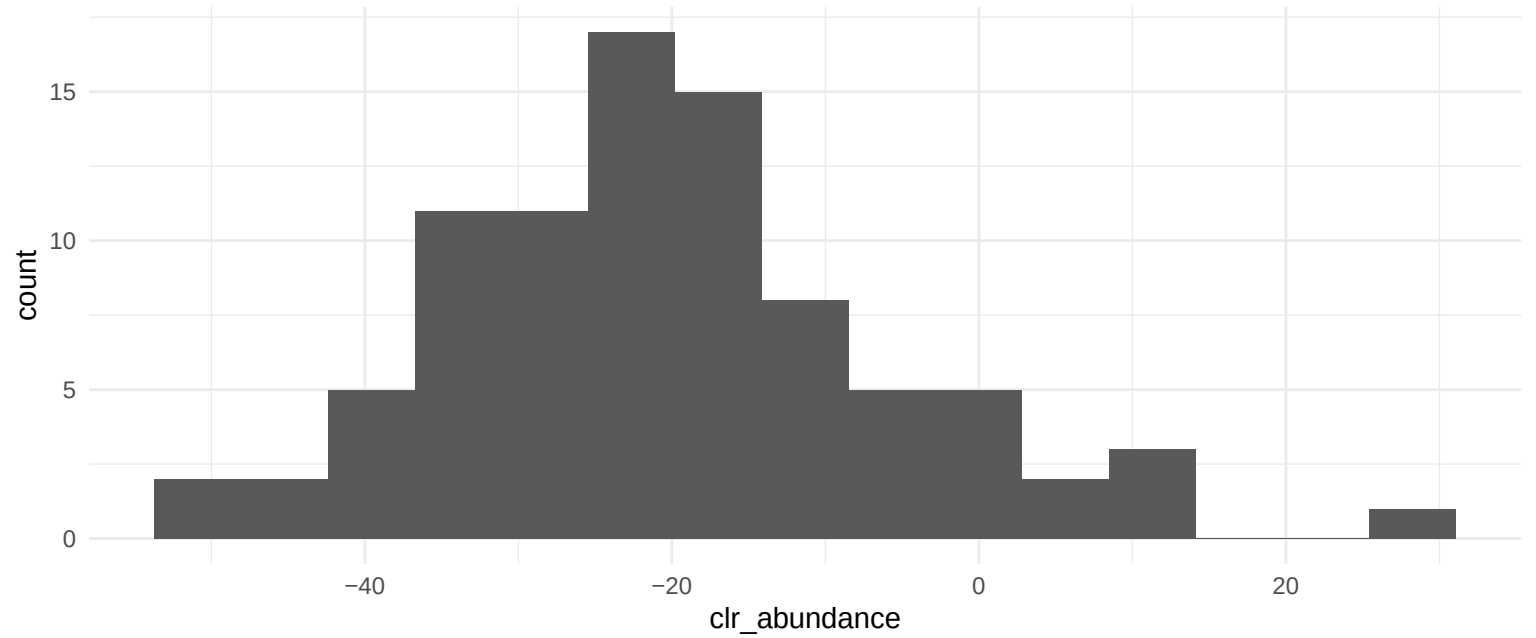
Distribution for 04745 Phototransduction − fly

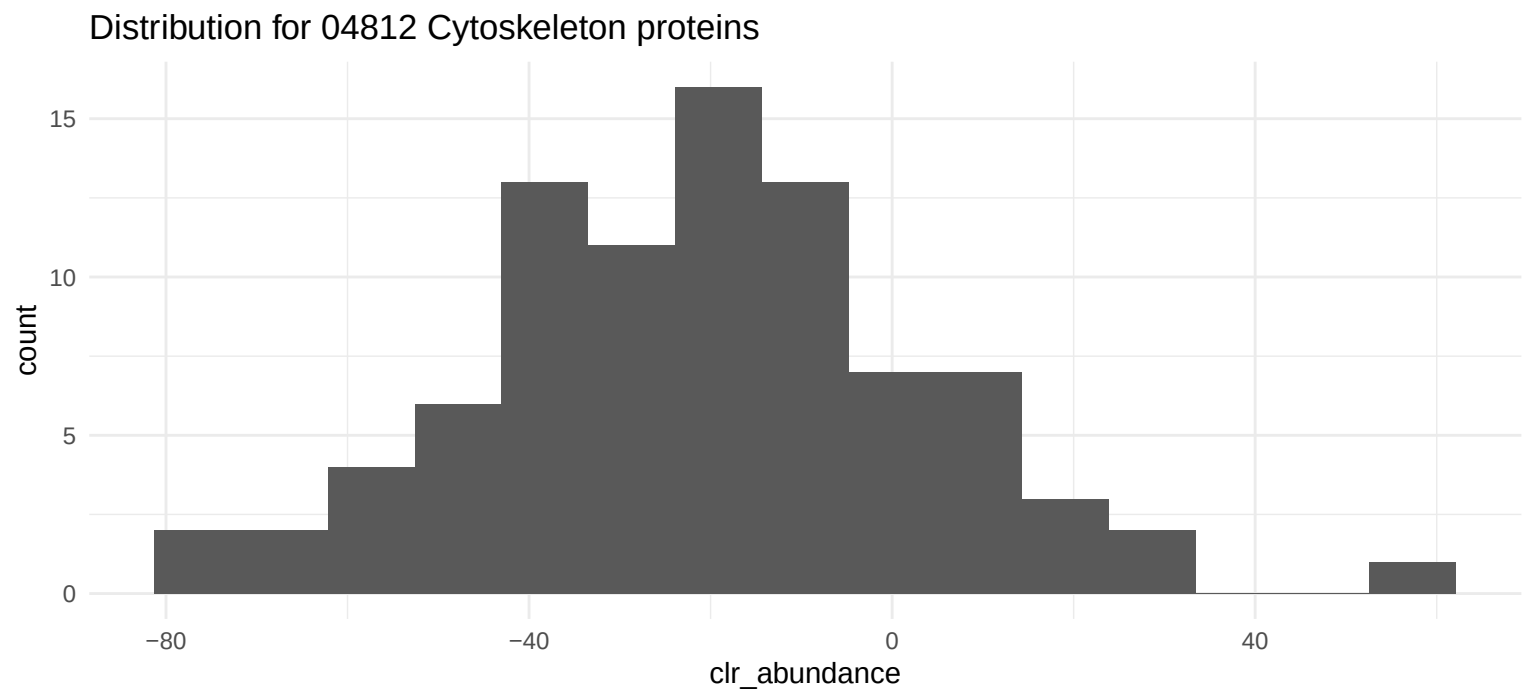


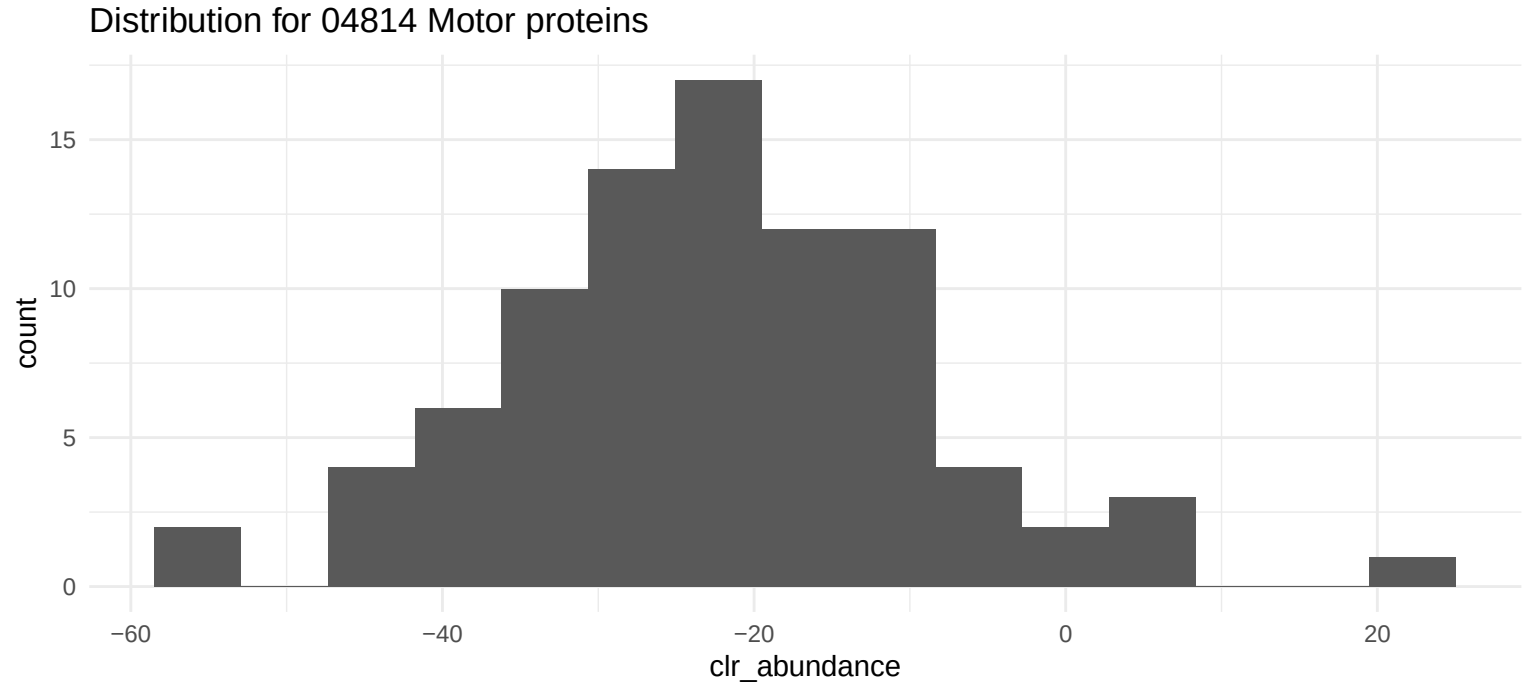
Distribution for 04750 Inflammatory mediator regulation of TRP channels

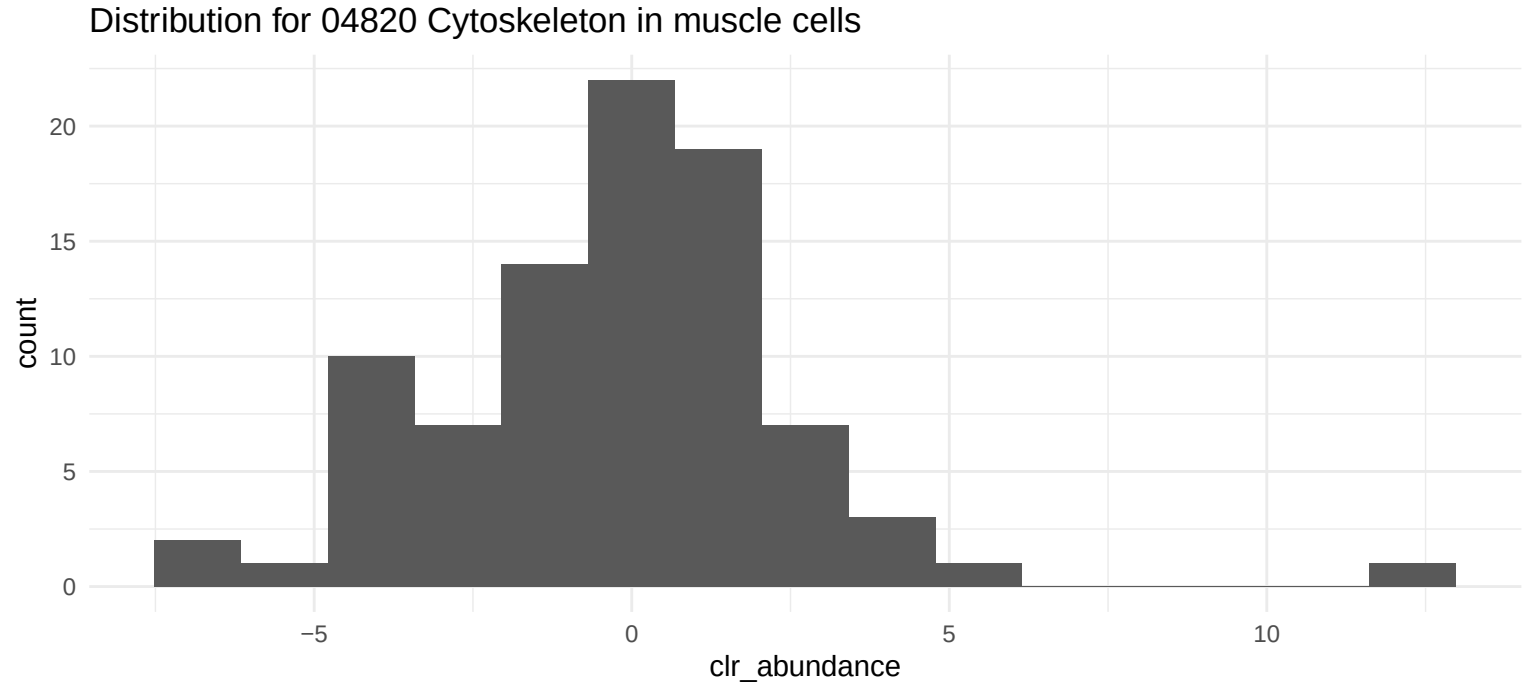


Distribution for 04810 Regulation of actin cytoskeleton

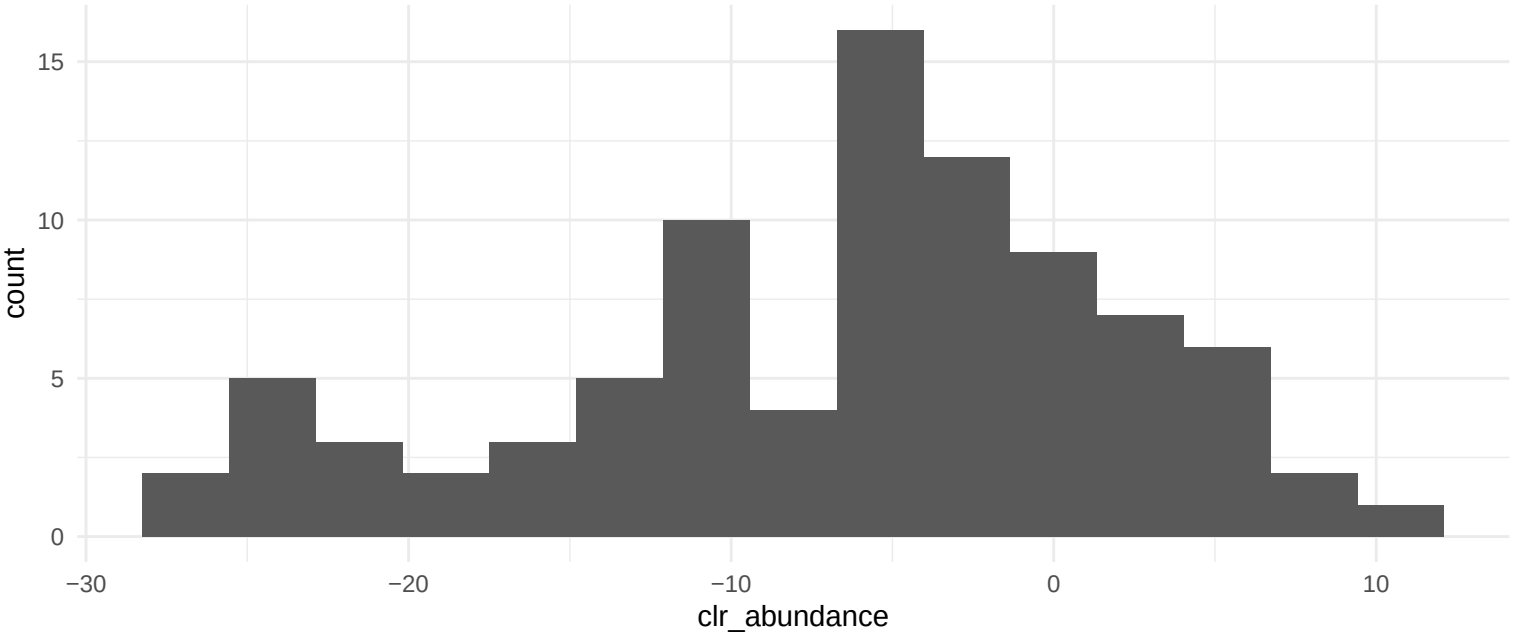



Distribution for 04812 Cytoskeleton proteins
count
clr_abundance

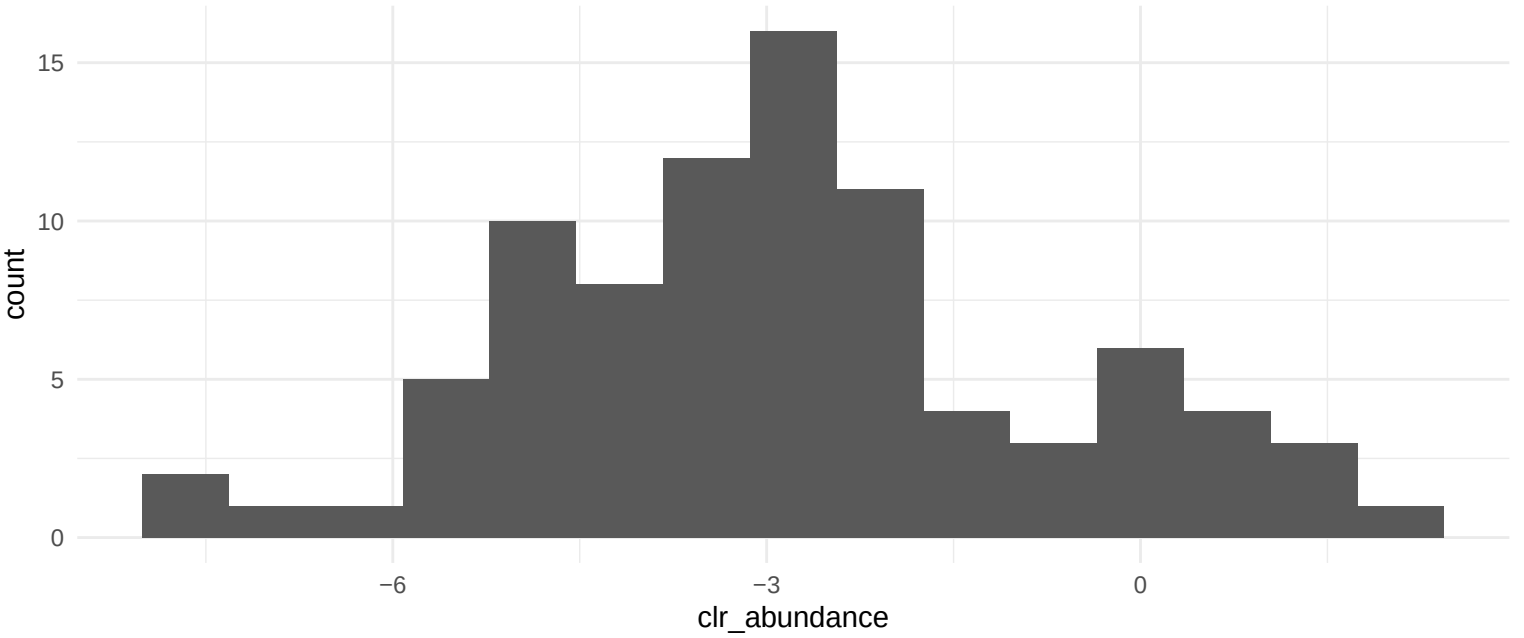

Distribution for 04814 Motor proteins
count
clr_abundance


Distribution for 04820 Cytoskeleton in muscle cells
count
clr_abundance

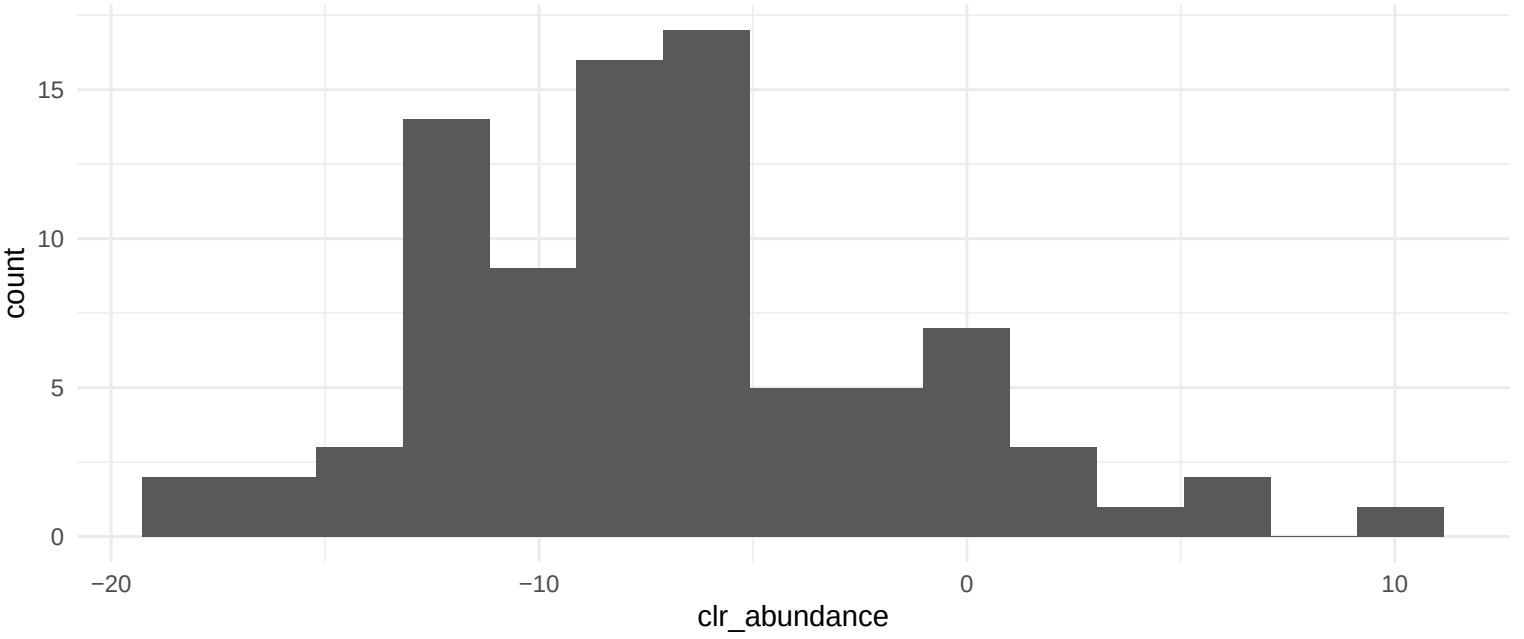
Distribution for 04910 Insulin signaling pathway



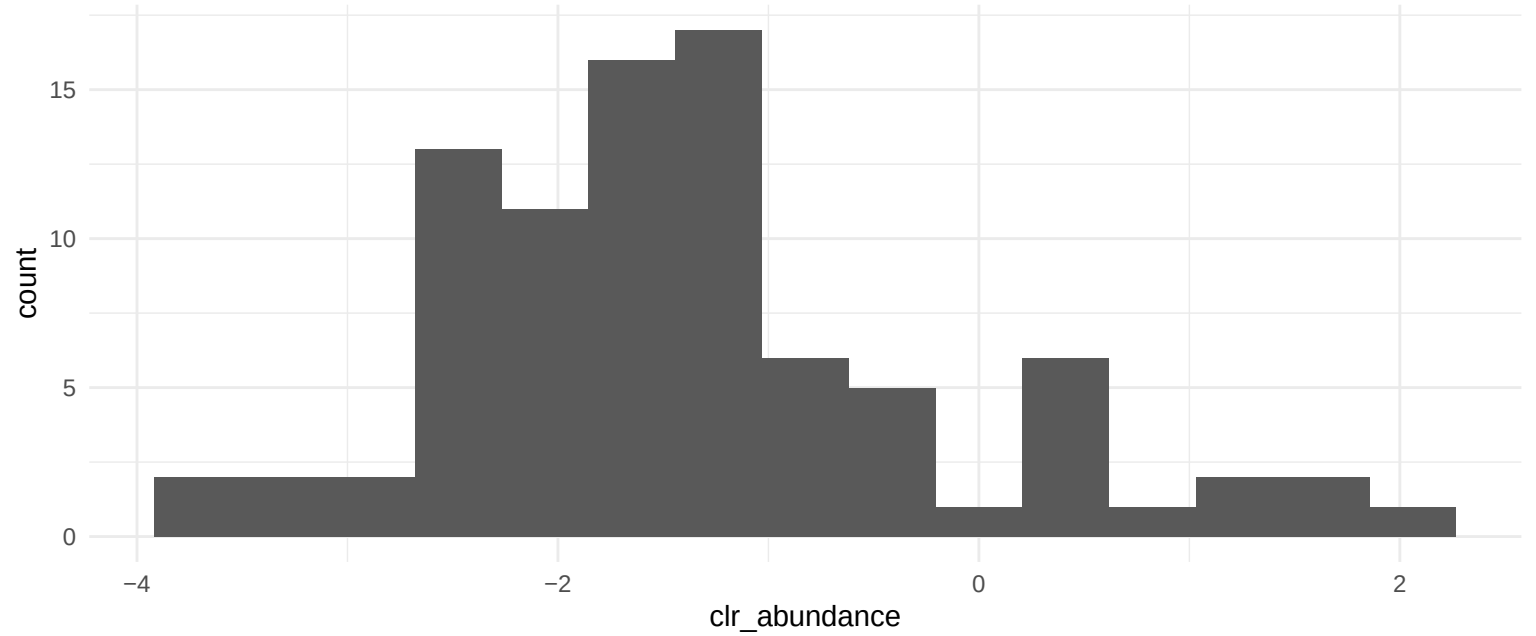
Distribution for 04911 Insulin secretion



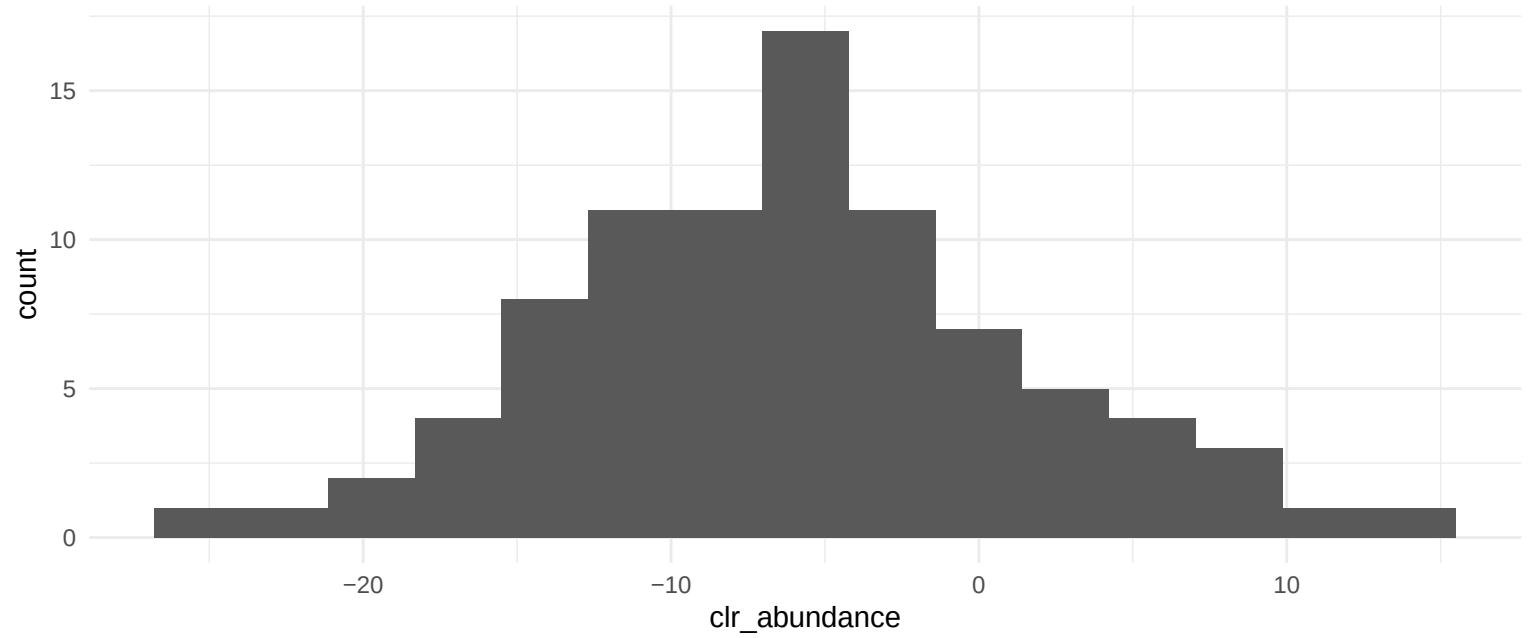
Distribution for 04912 GnRH signaling pathway



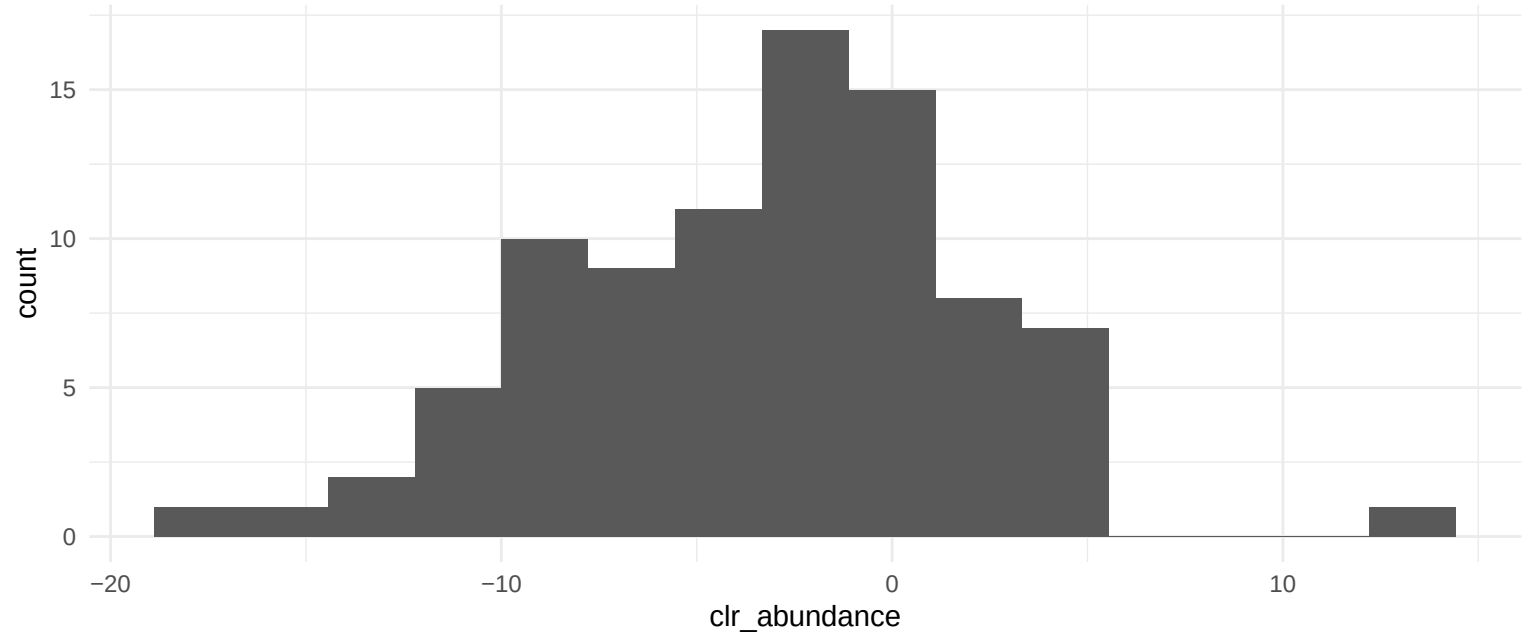
Distribution for 04913 Ovarian steroidogenesis



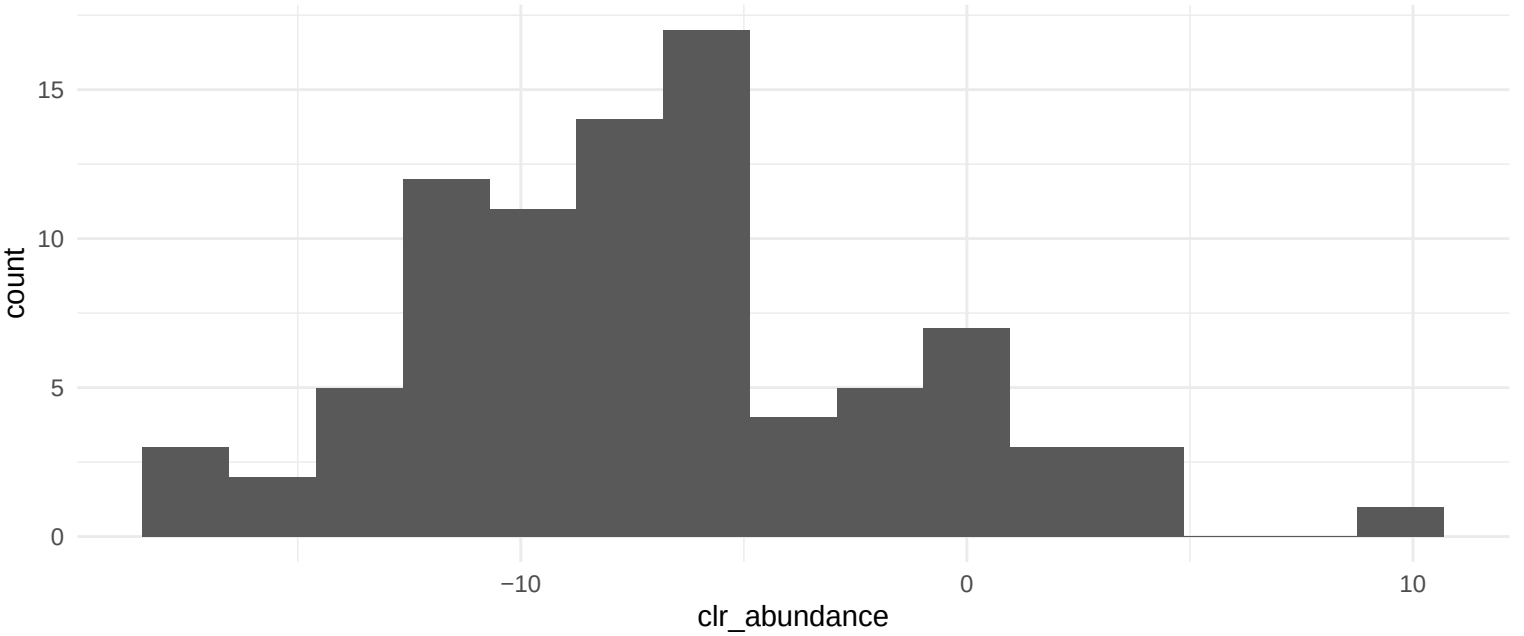
Distribution for 04914 Progesterone−mediated oocyte maturation



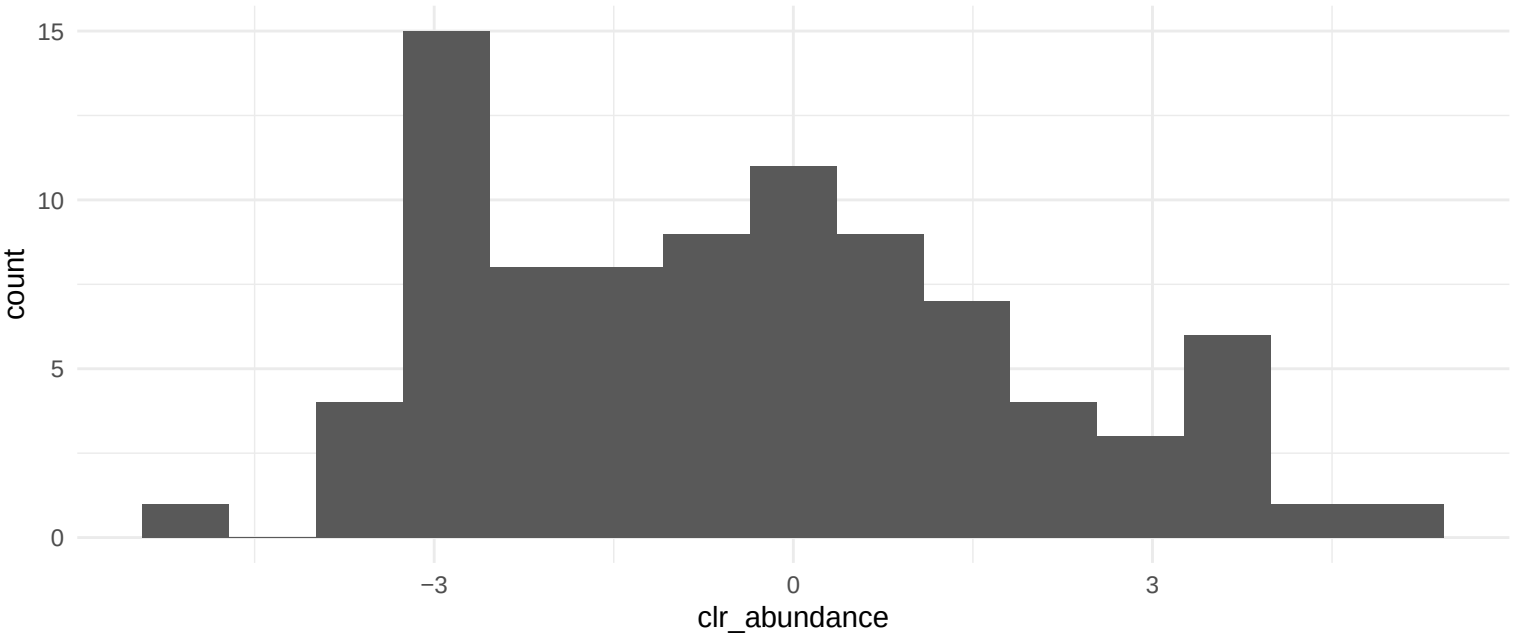
Distribution for 04915 Estrogen signaling pathway



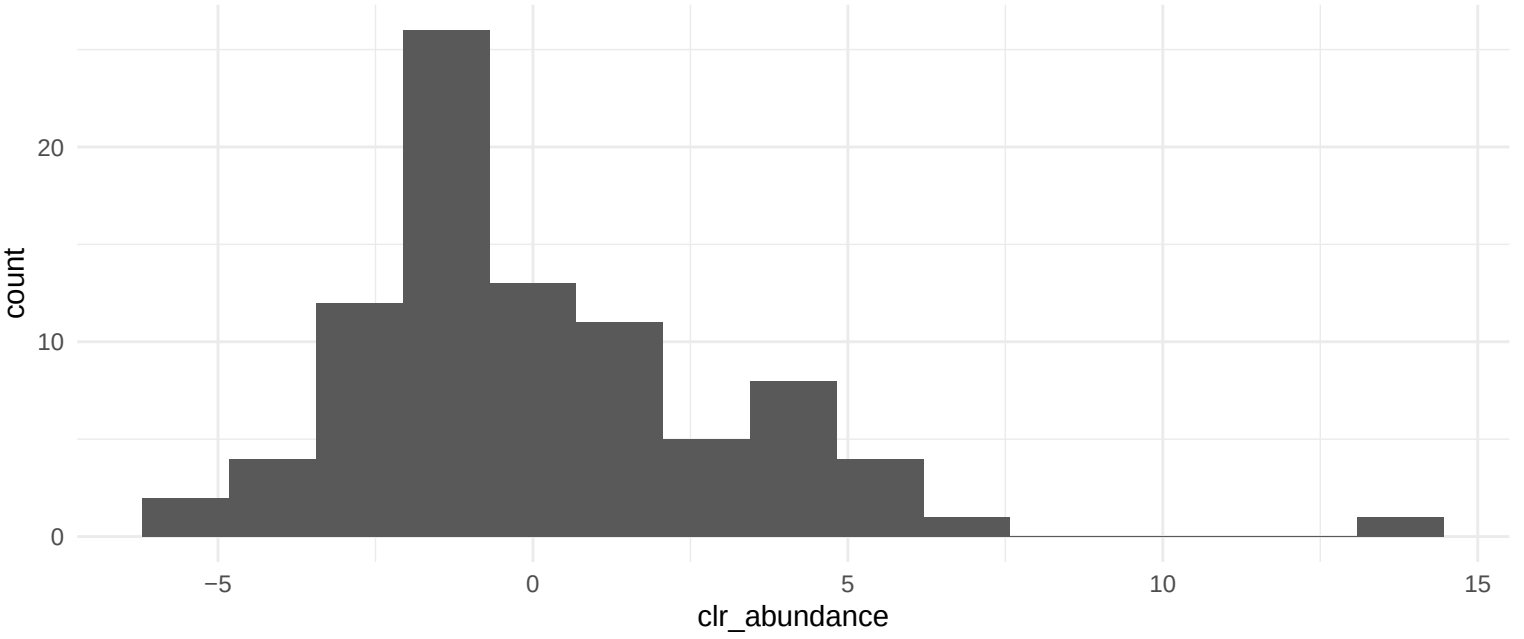
Distribution for 04916 Melanogenesis



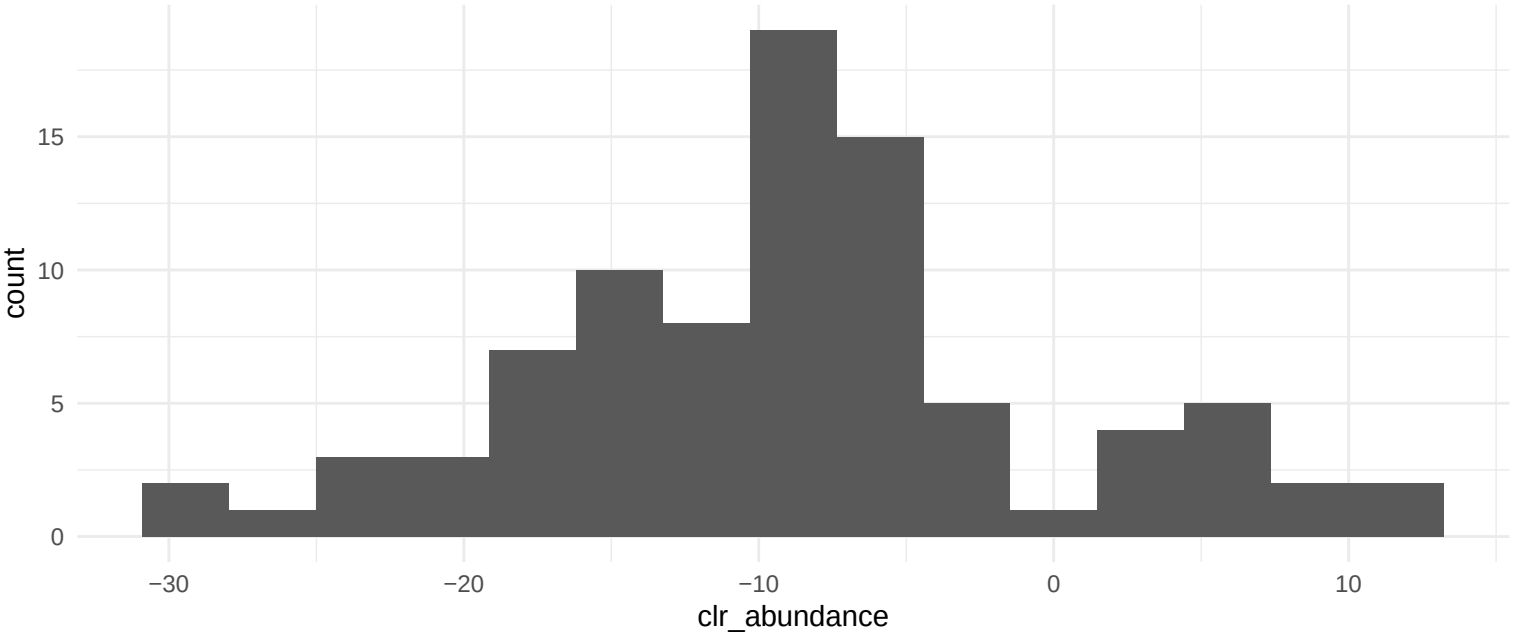
Distribution for 04917 Prolactin signaling pathway



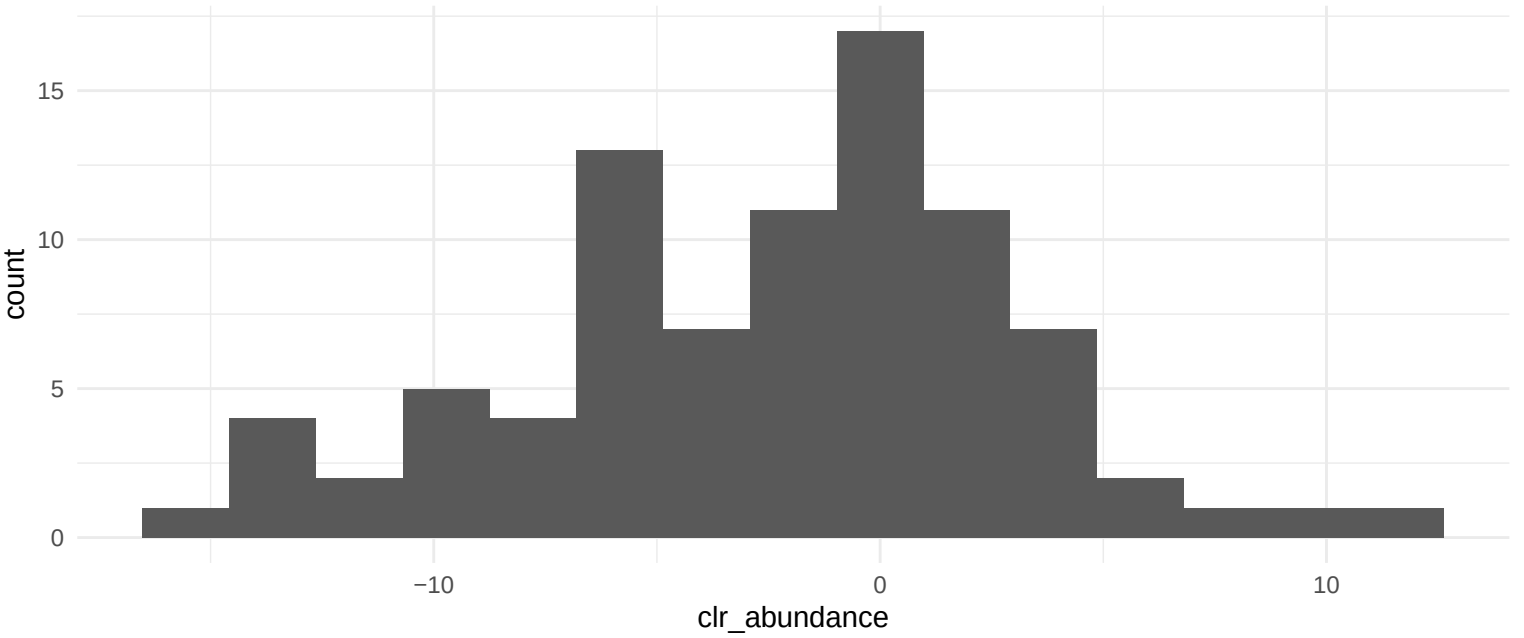
Distribution for 04918 Thyroid hormone synthesis



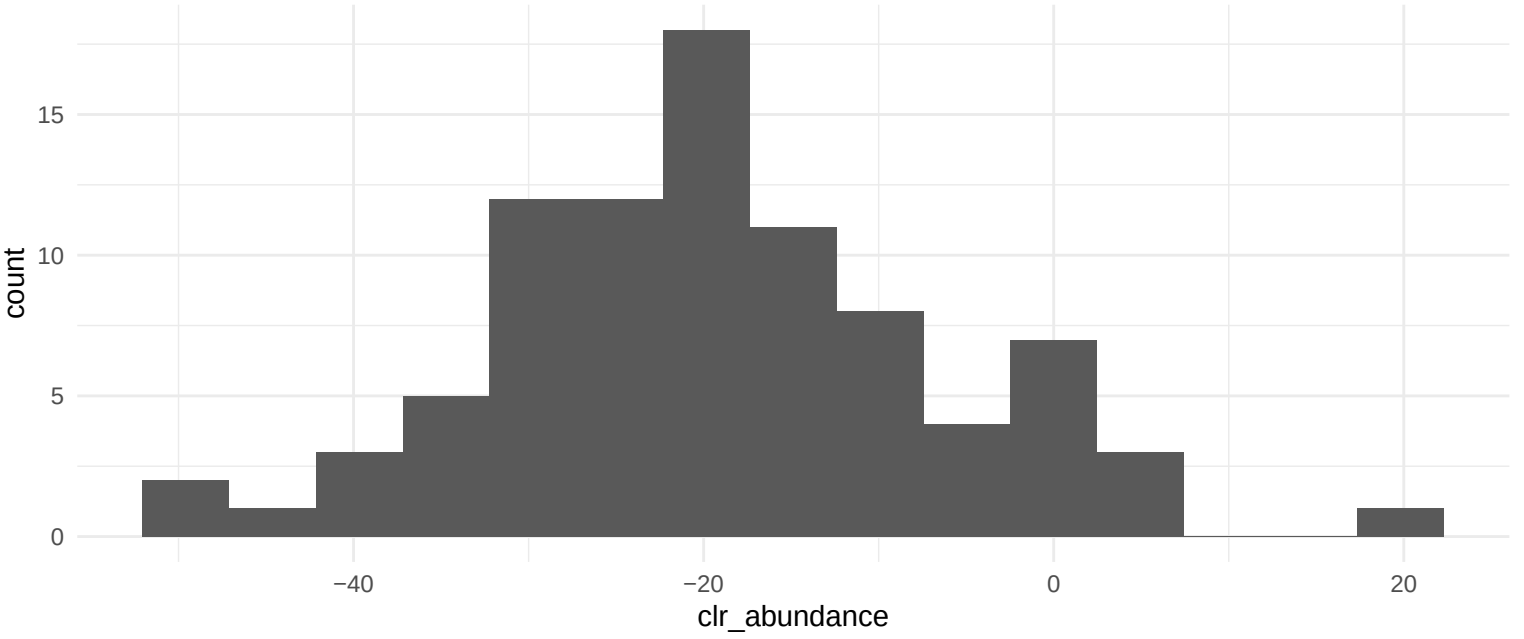
Distribution for 04919 Thyroid hormone signaling pathway



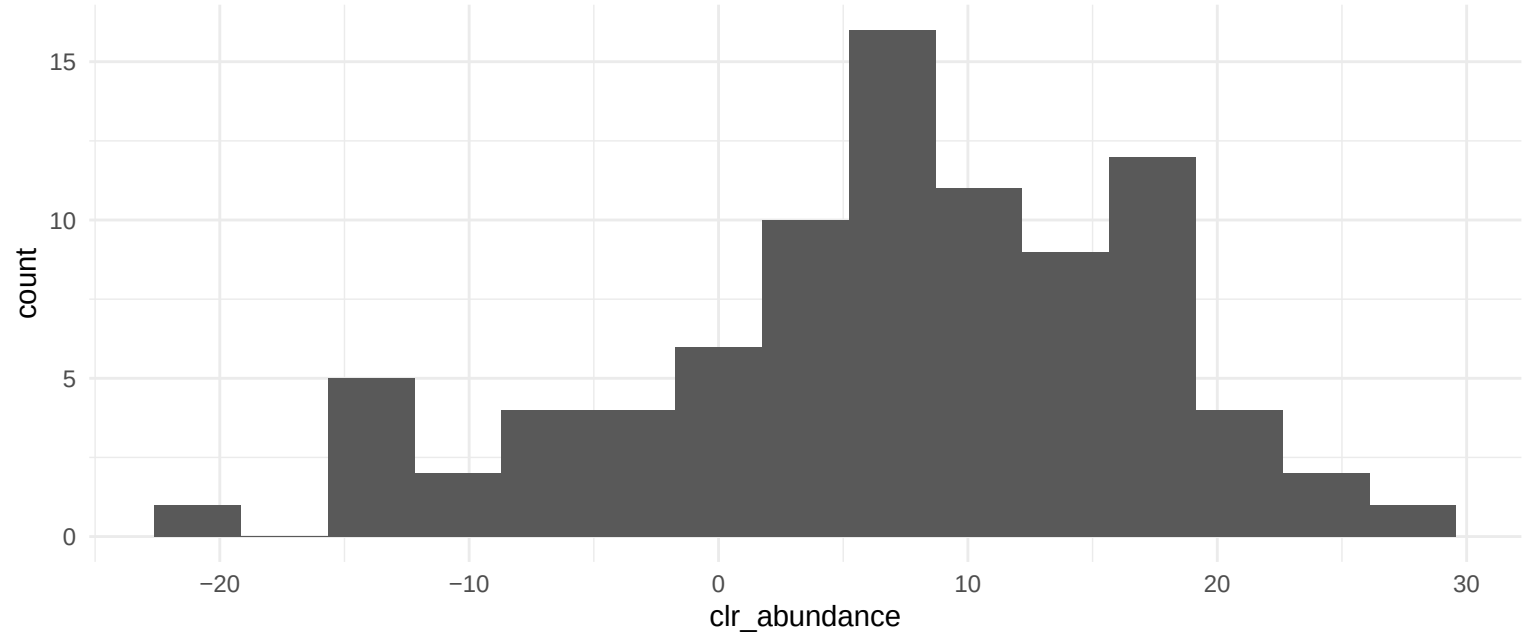
Distribution for 04920 Adipocytokine signaling pathway



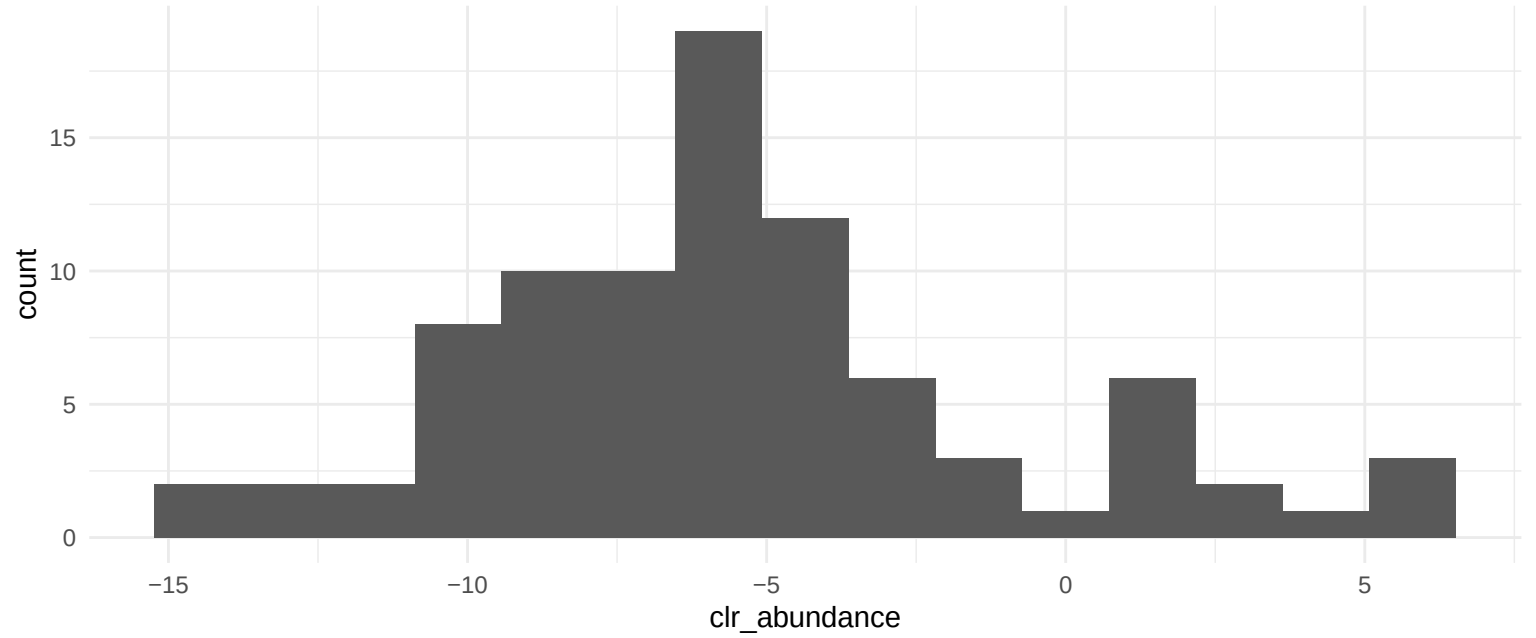
Distribution for 04921 Oxytocin signaling pathway



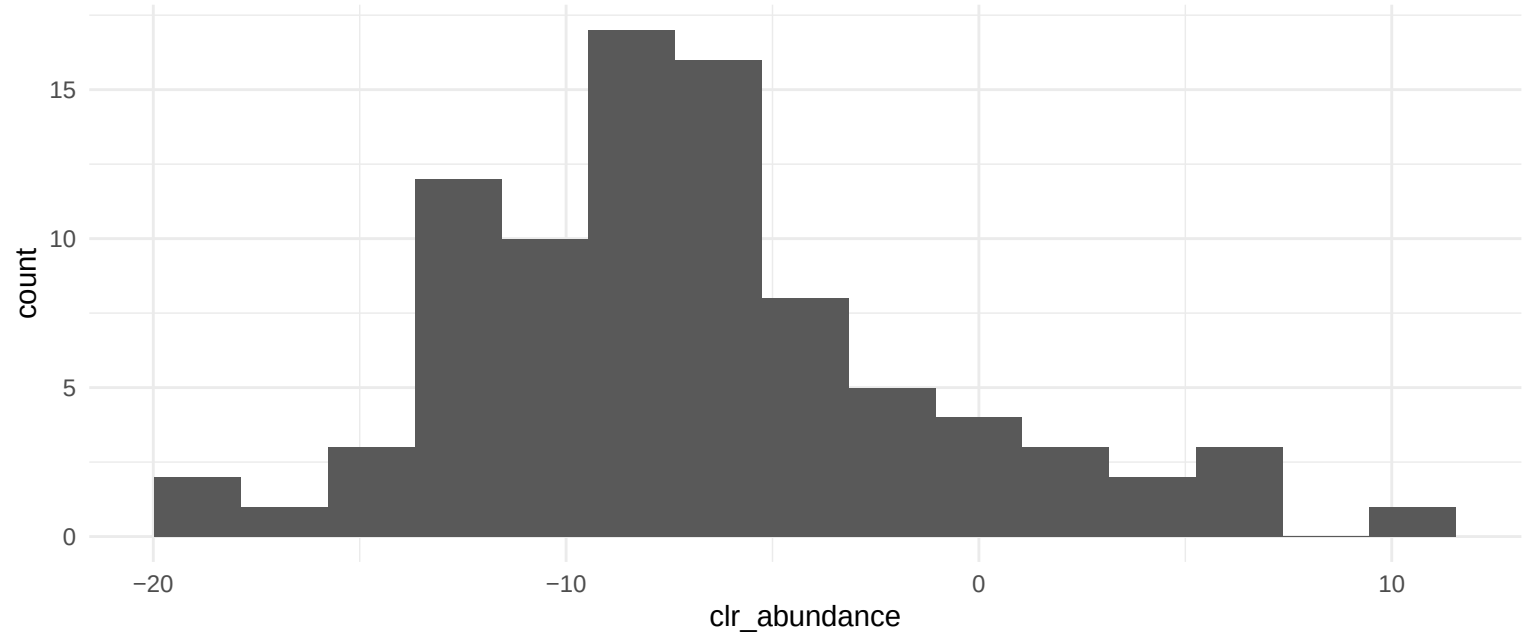
Distribution for 04922 Glucagon signaling pathway



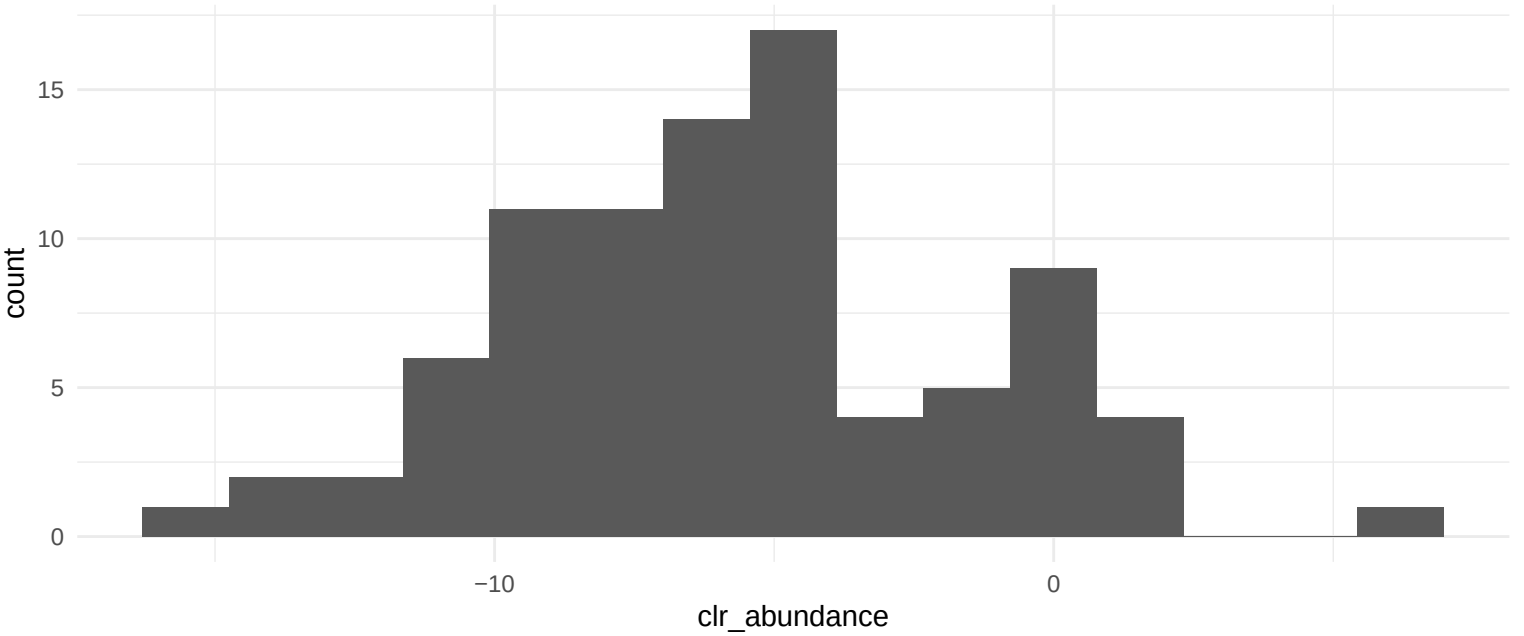
Distribution for 04923 Regulation of lipolysis in adipocytes



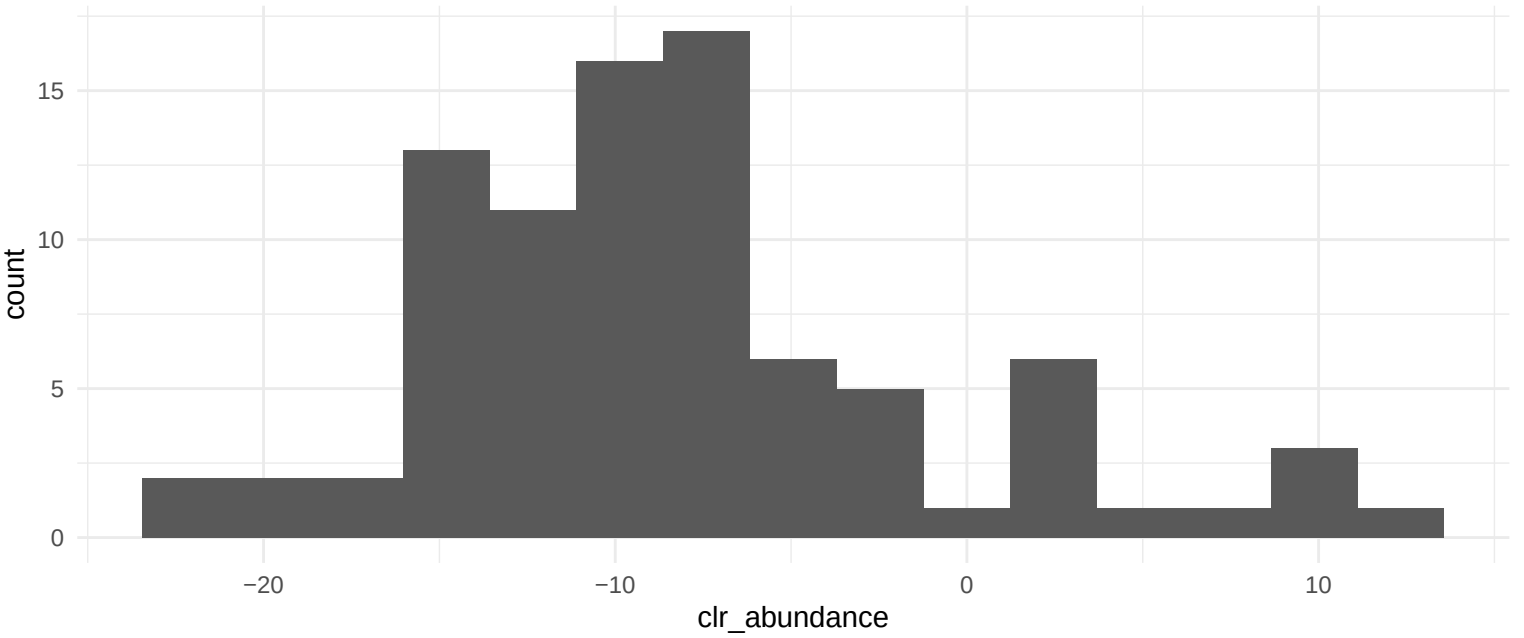
Distribution for 04924 Renin secretion



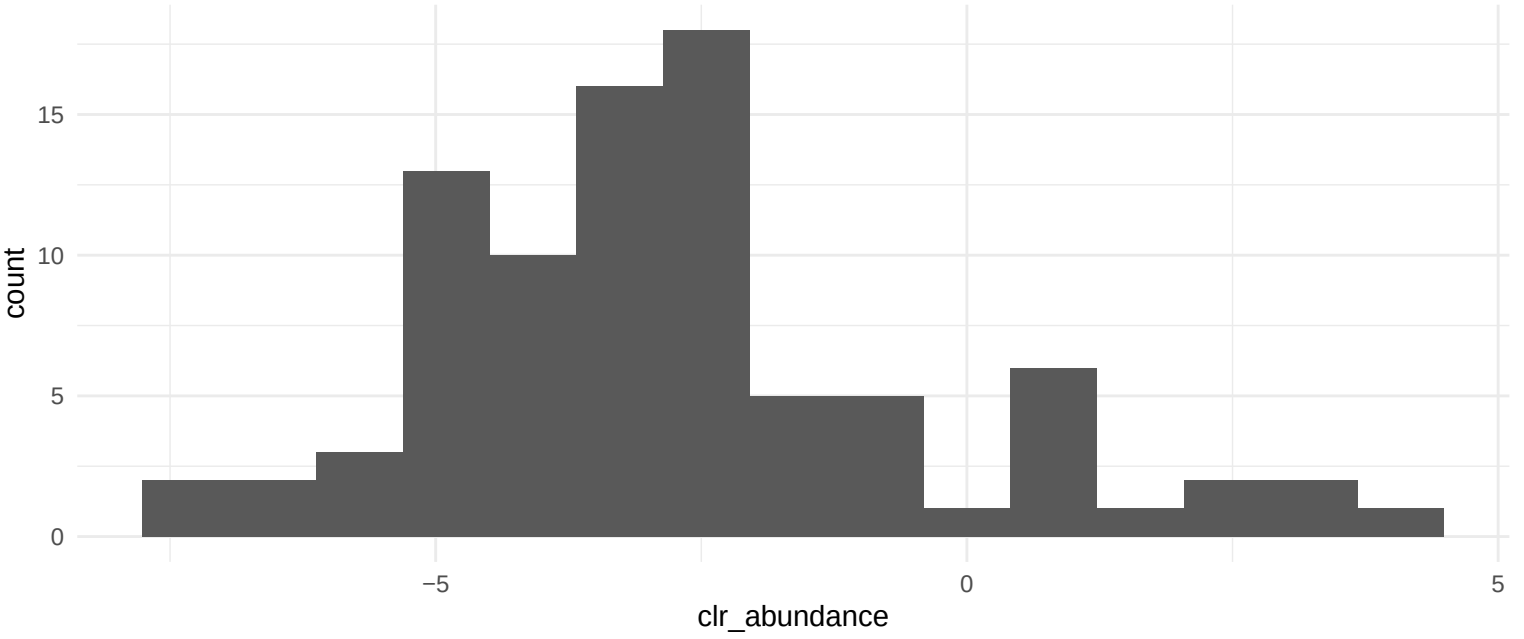
Distribution for 04925 Aldosterone synthesis and secretion



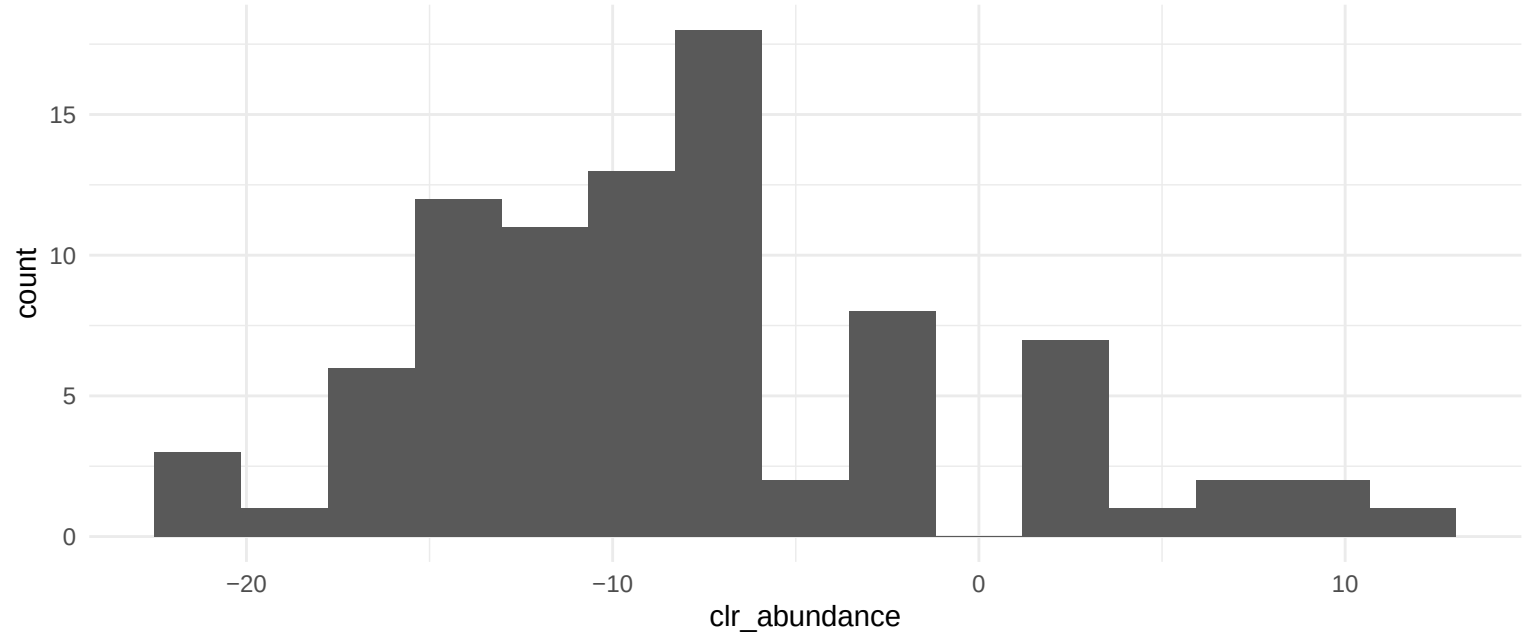
Distribution for 04926 Relaxin signaling pathway



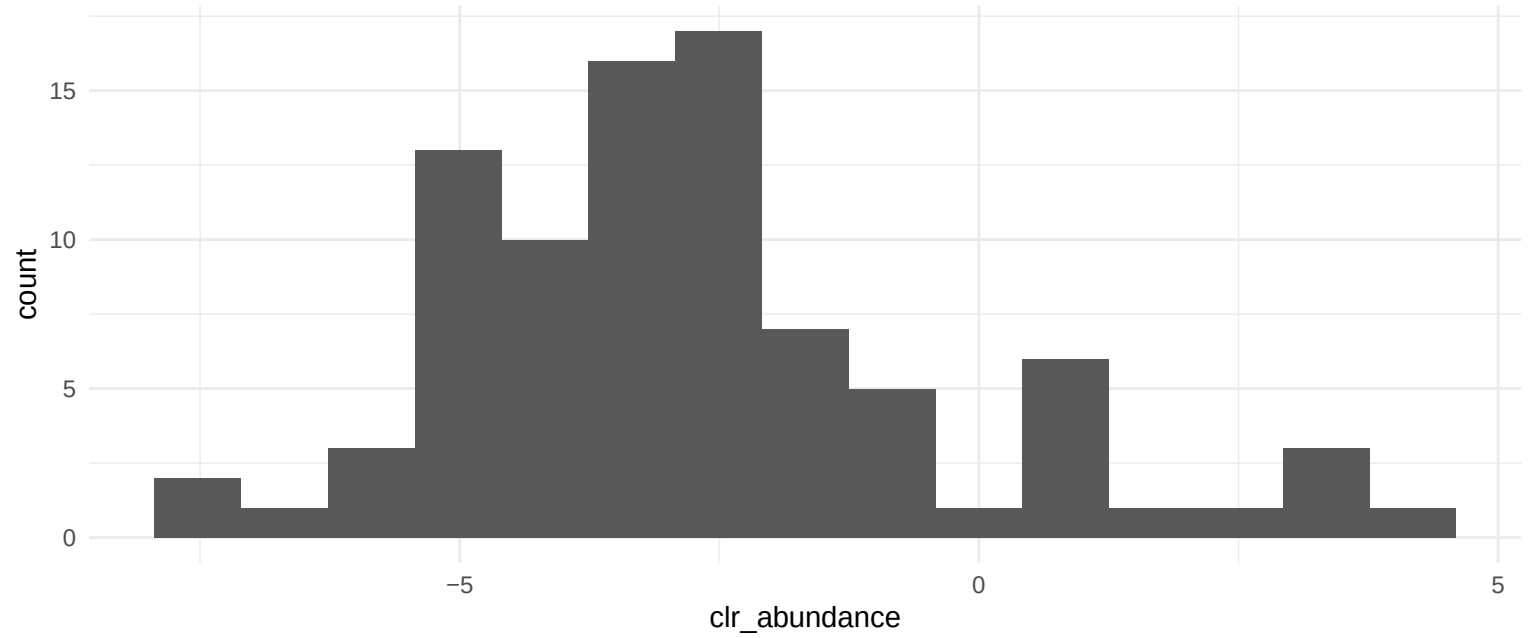
Distribution for 04927 Cortisol synthesis and secretion



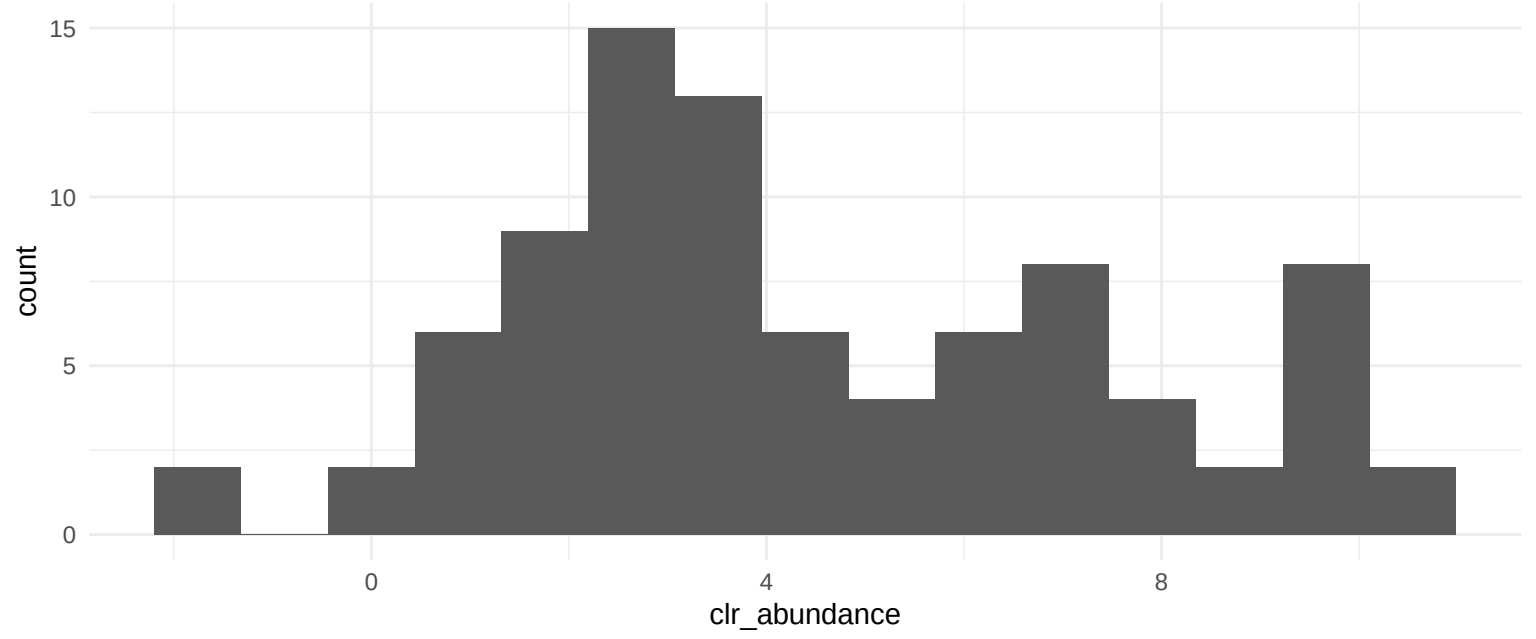
Distribution for 04928 Parathyroid hormone synthesis, secretion and action

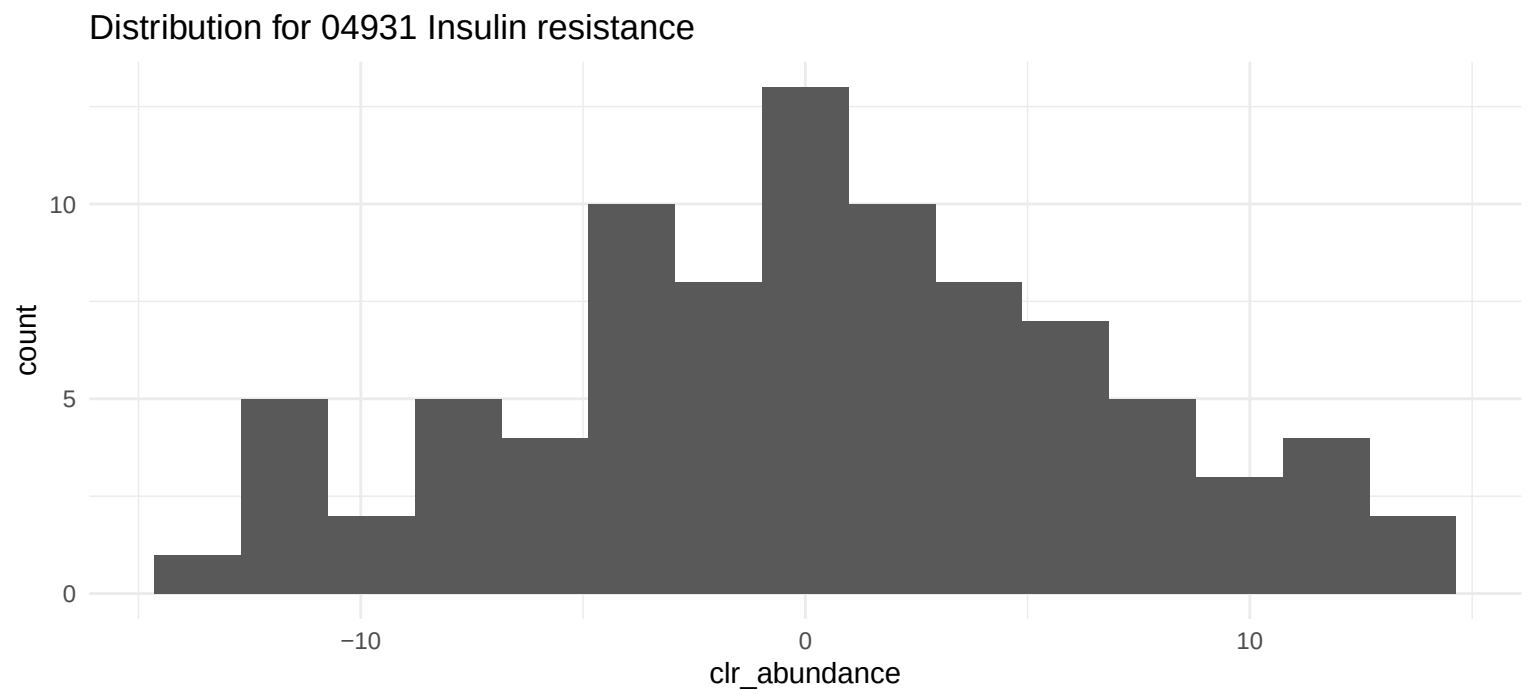


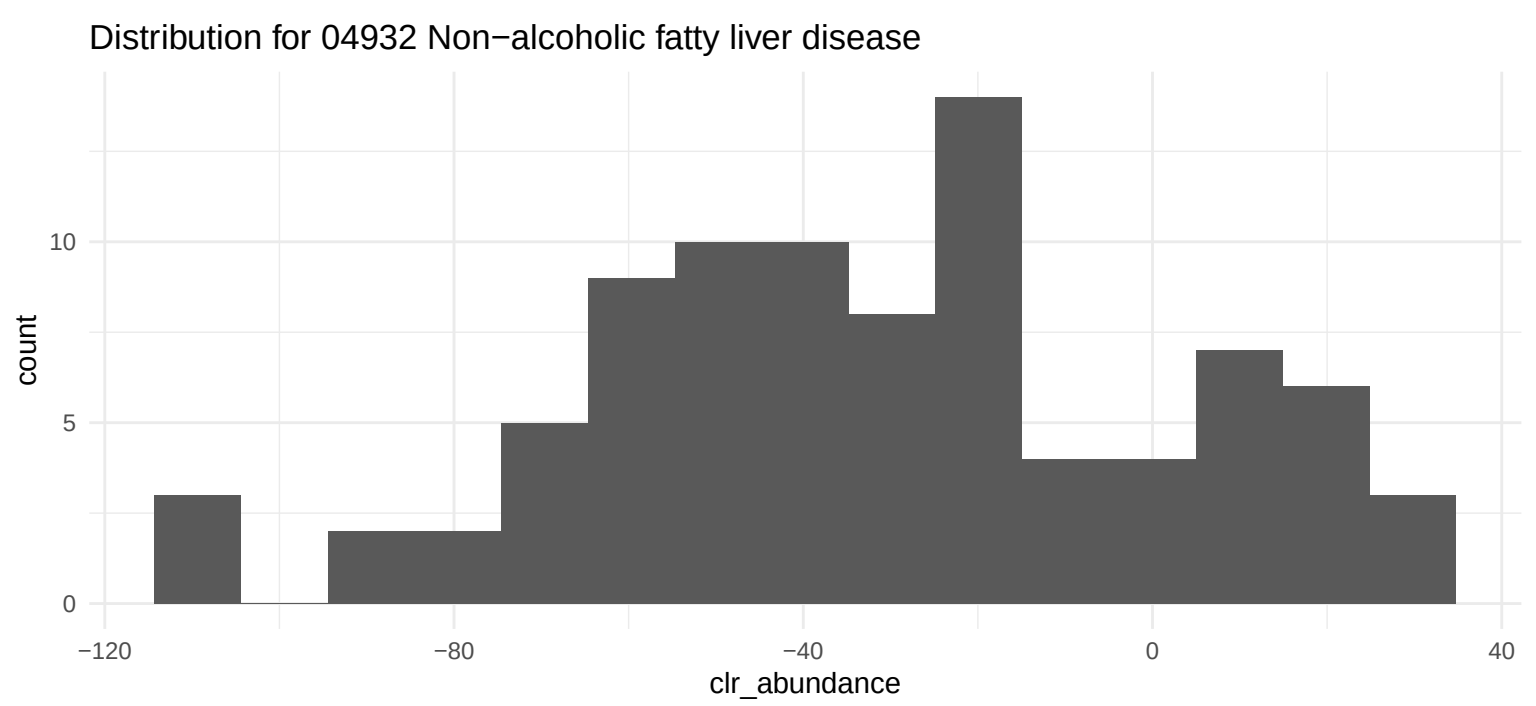
Distribution for 04929 GnRH secretion

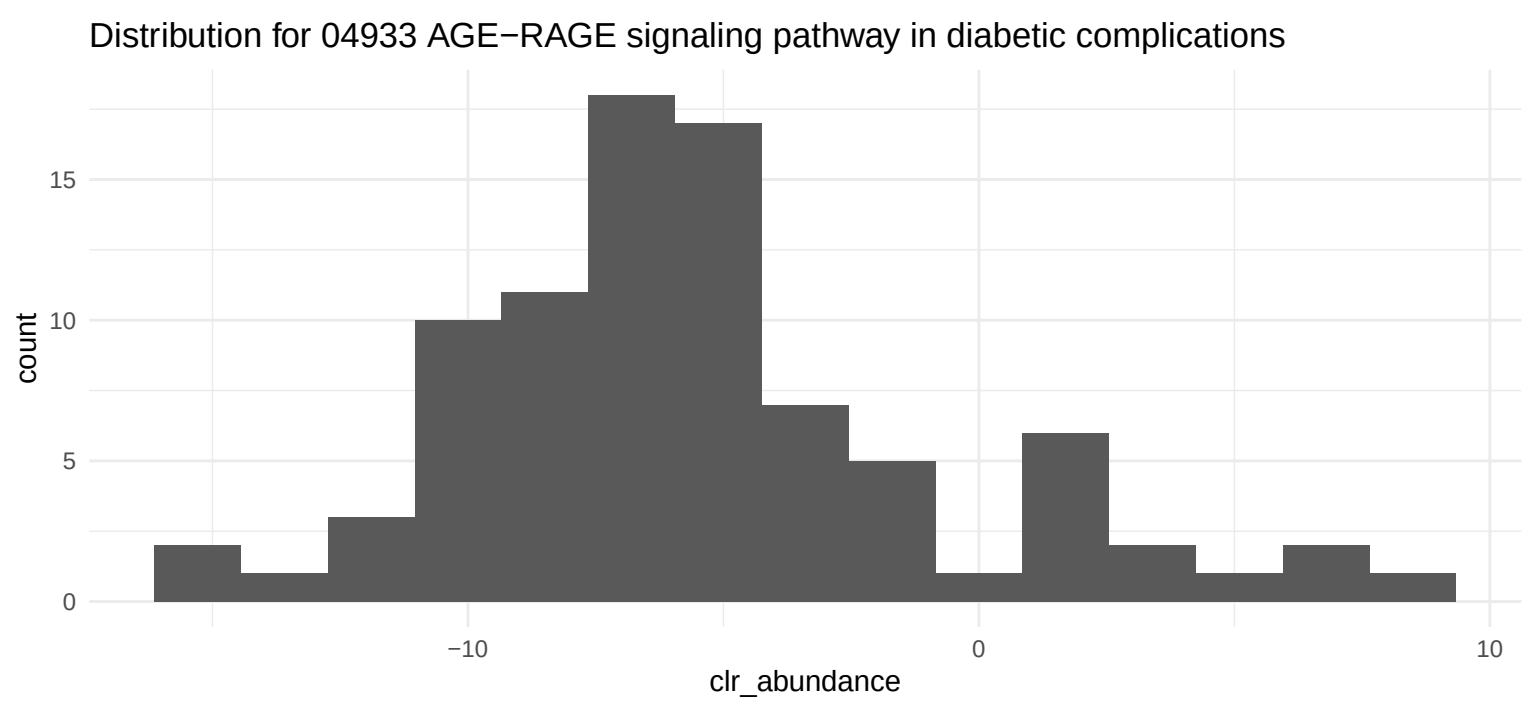


Distribution for 04930 Type II diabetes mellitus

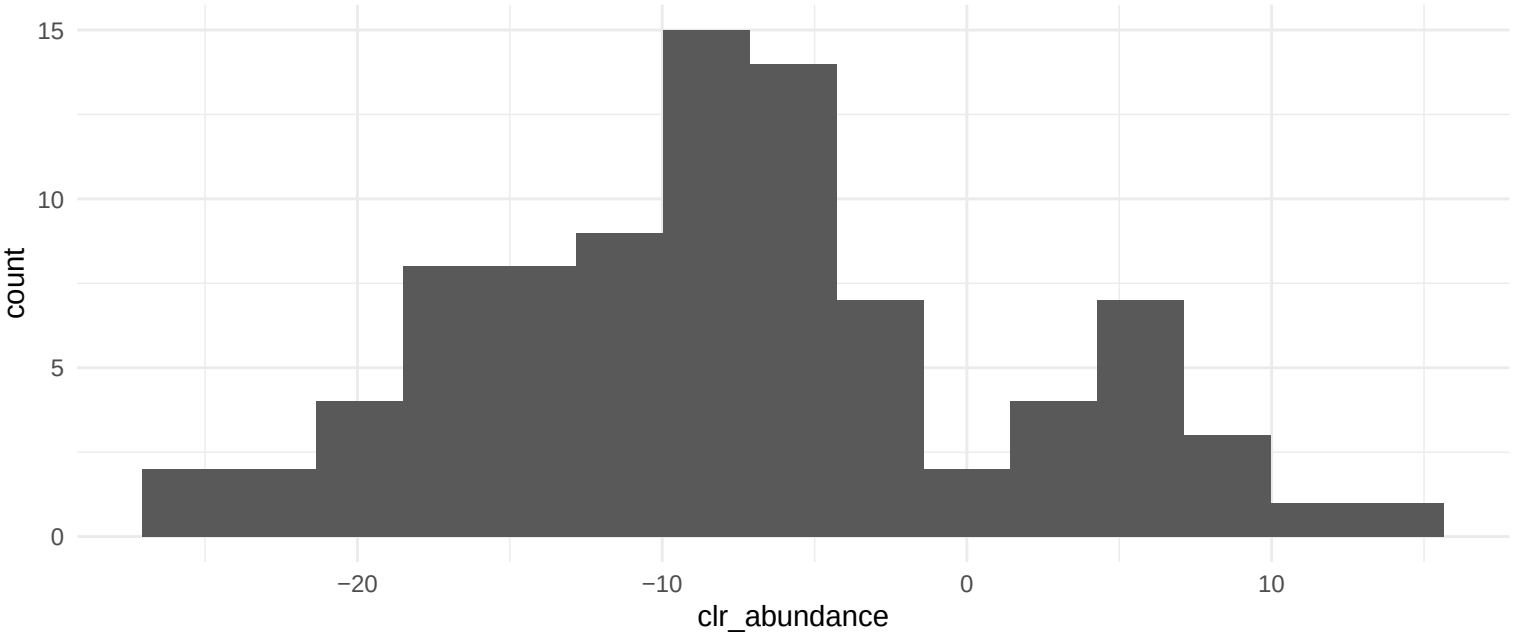



Distribution for 04931 Insulin resistance
count
0
5
10
−10
0
10
clr_abundance

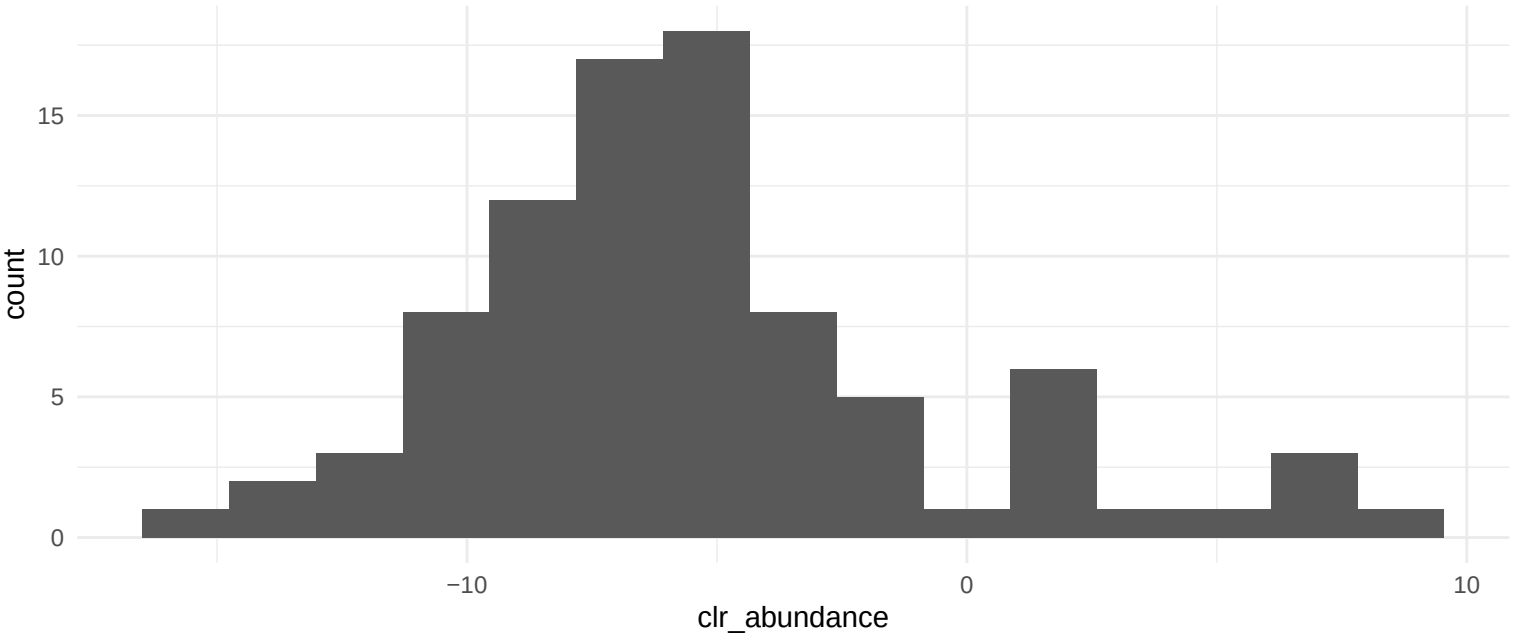

Distribution for 04932 Non−alcoholic fatty liver disease
count
0
5
10
−120
−80
−40
0
40
clr_abundance


Distribution for 04933 AGE−RAGE signaling pathway in diabetic complications
count
0
5
10
15
−10
0
10
clr_abundance

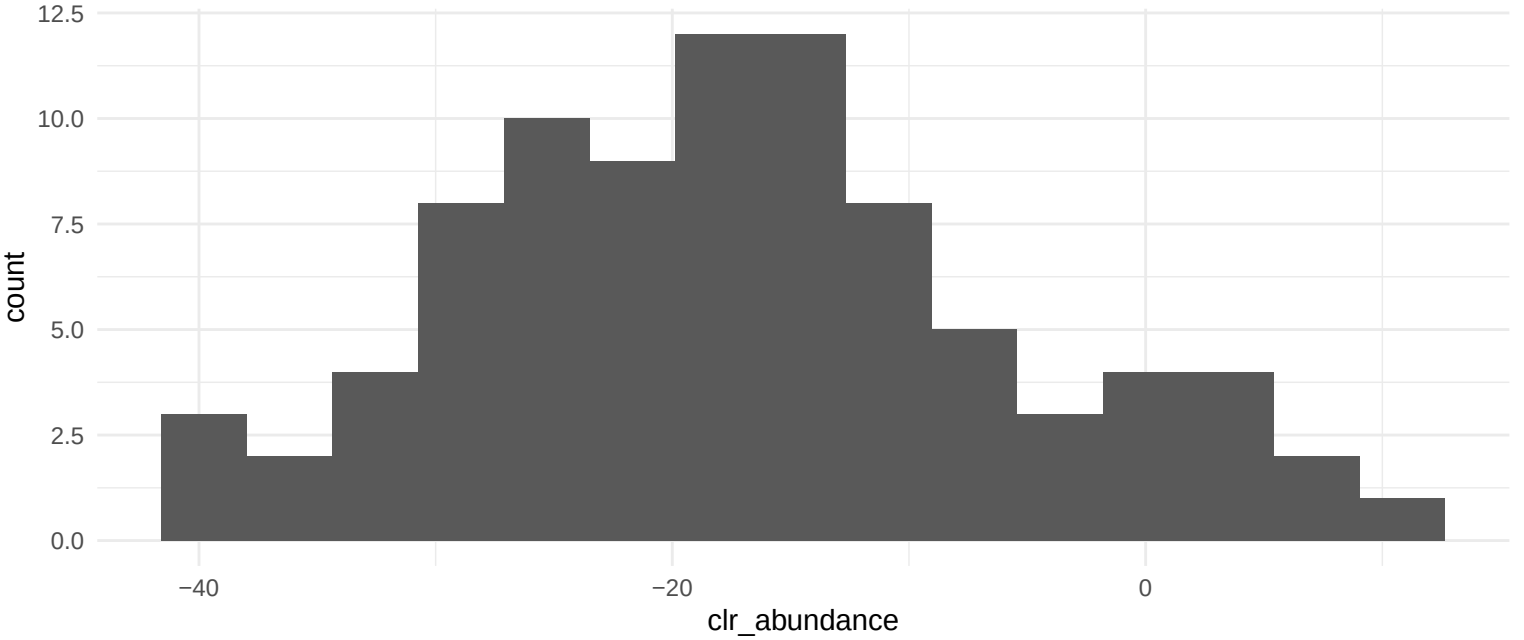
Distribution for 04934 Cushing syndrome

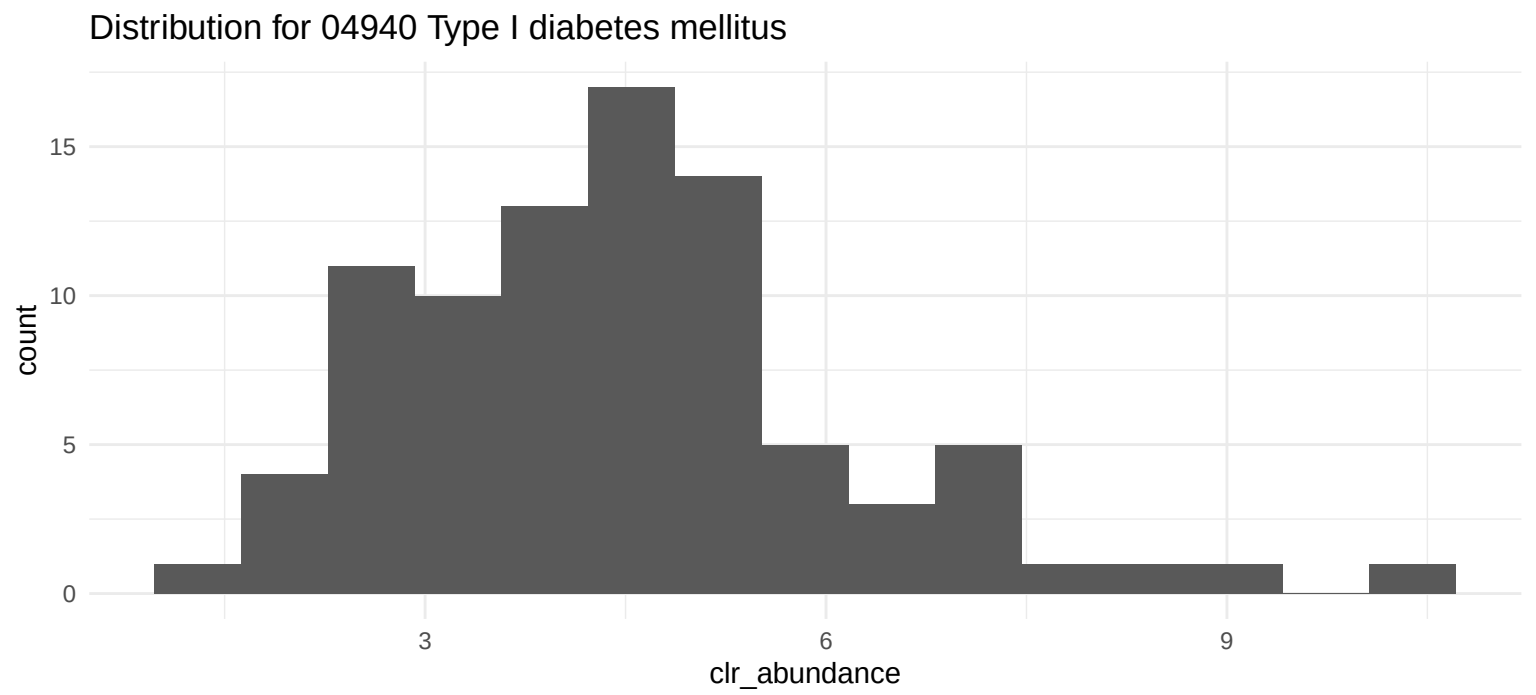


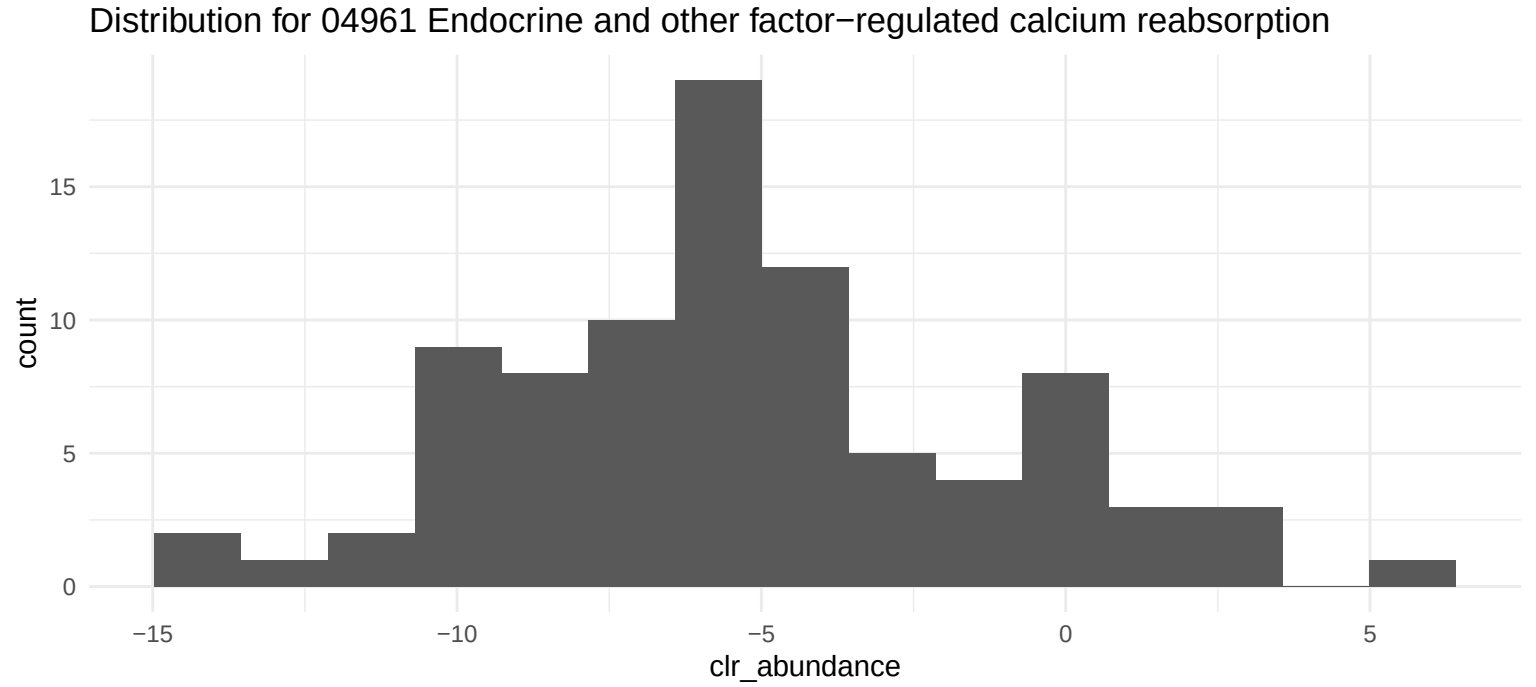
Distribution for 04935 Growth hormone synthesis, secretion and action

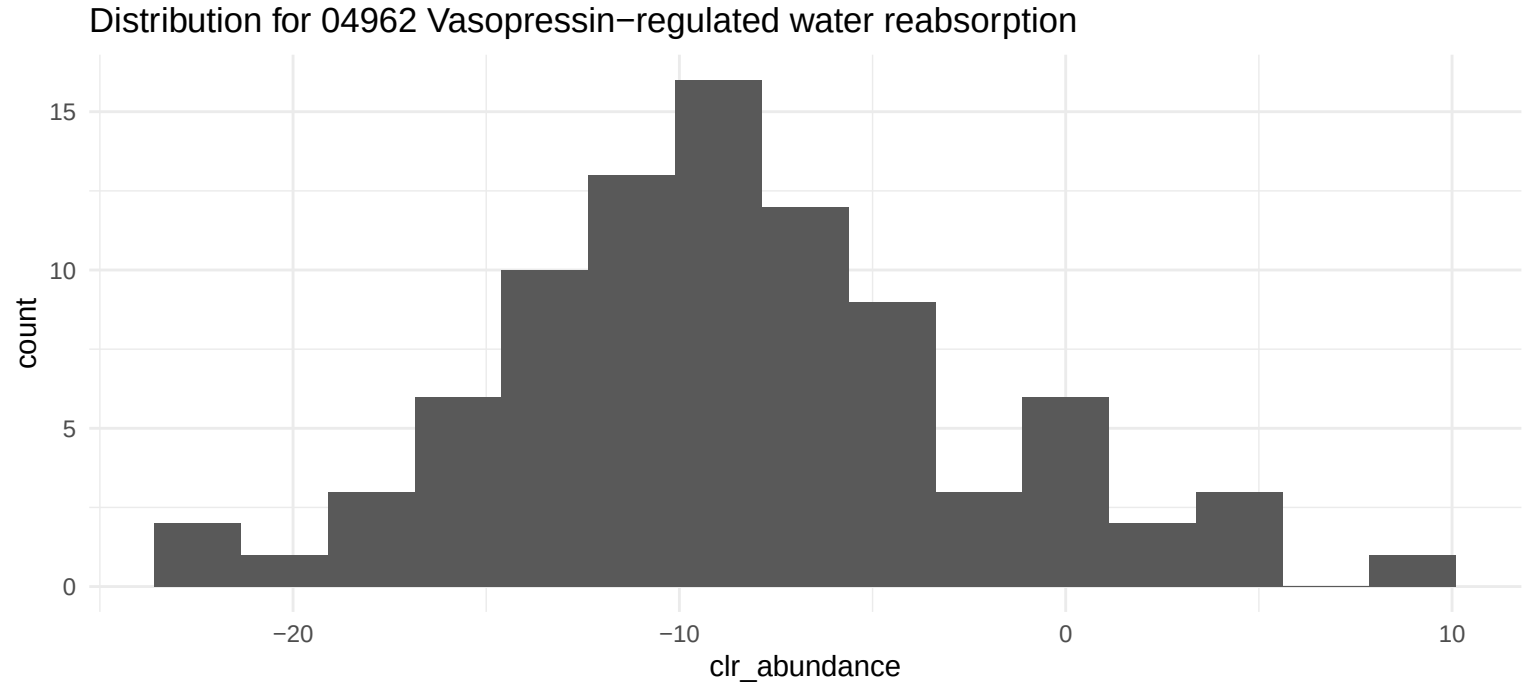


Distribution for 04936 Alcoholic liver disease

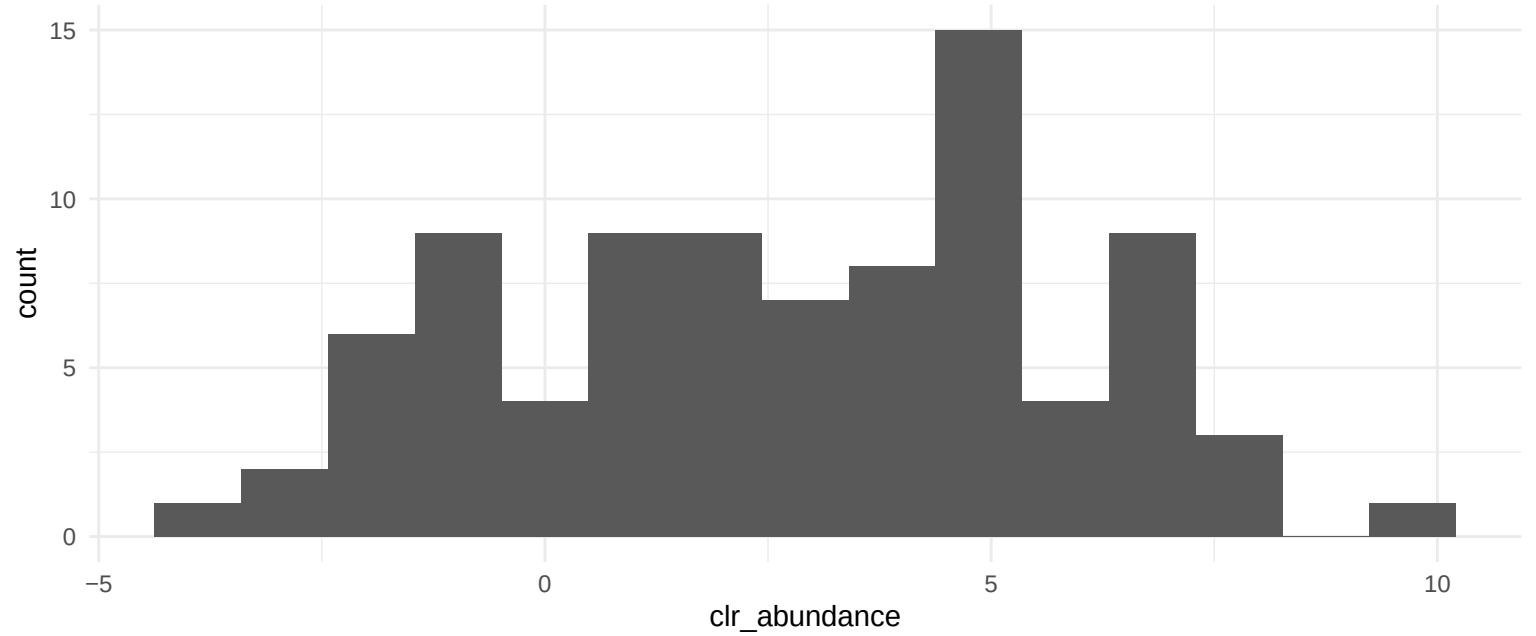



Distribution for 04940 Type I diabetes mellitus
count
15
10
5
0
3
6
9
clr_abundance

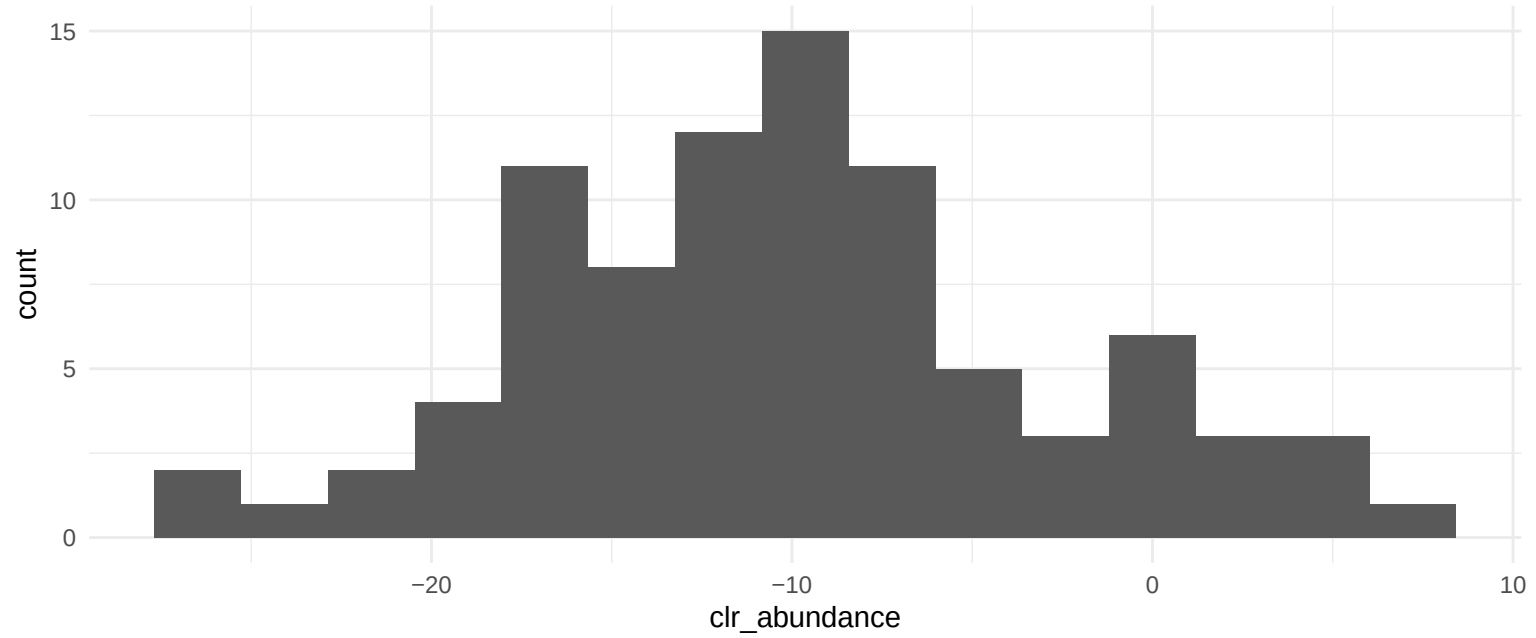

Distribution for 04961 Endocrine and other factor−regulated calcium reabsorption
count
15
10
5
0
−15
−10
−5
0
5
clr_abundance


Distribution for 04962 Vasopressin−regulated water reabsorption
count
15
10
5
0
−20
−10
0
10
clr_abundance

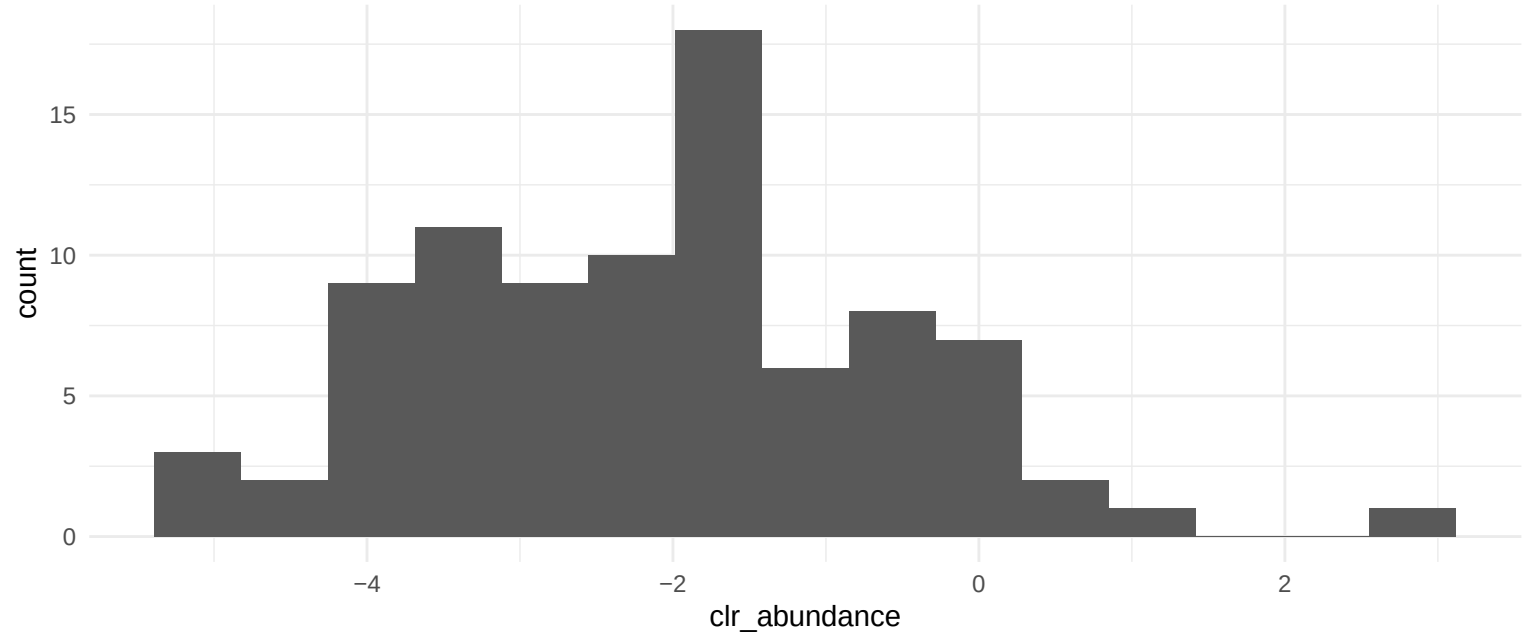
Distribution for 04964 Proximal tubule bicarbonate reclamation

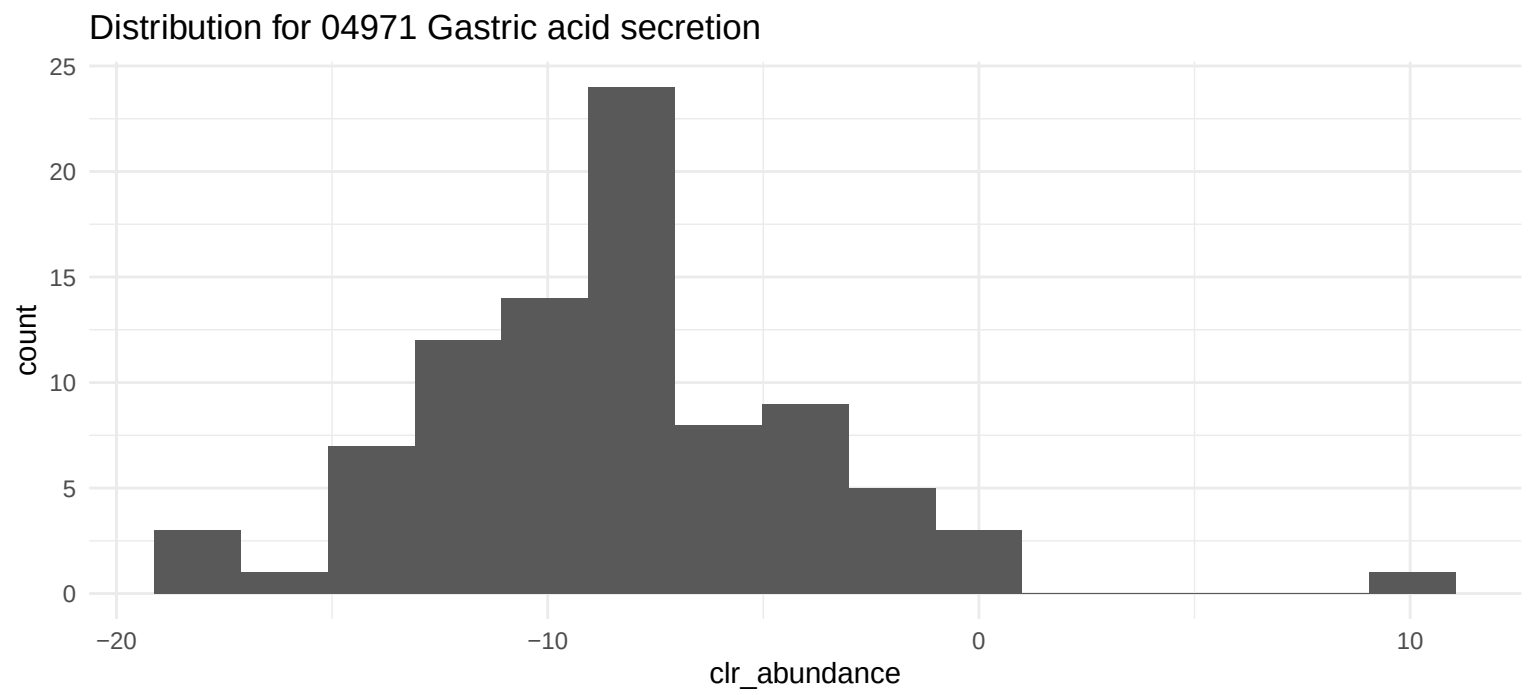


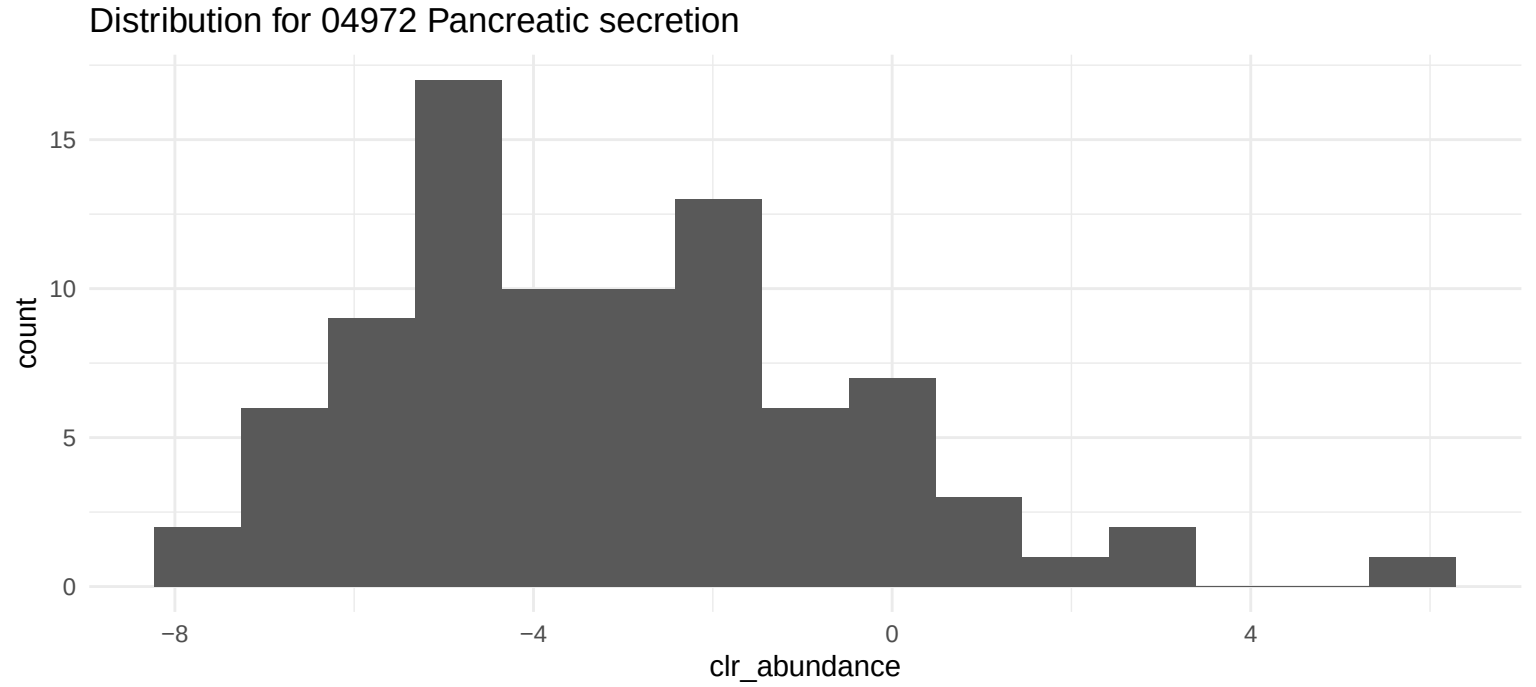
Distribution for 04966 Collecting duct acid secretion

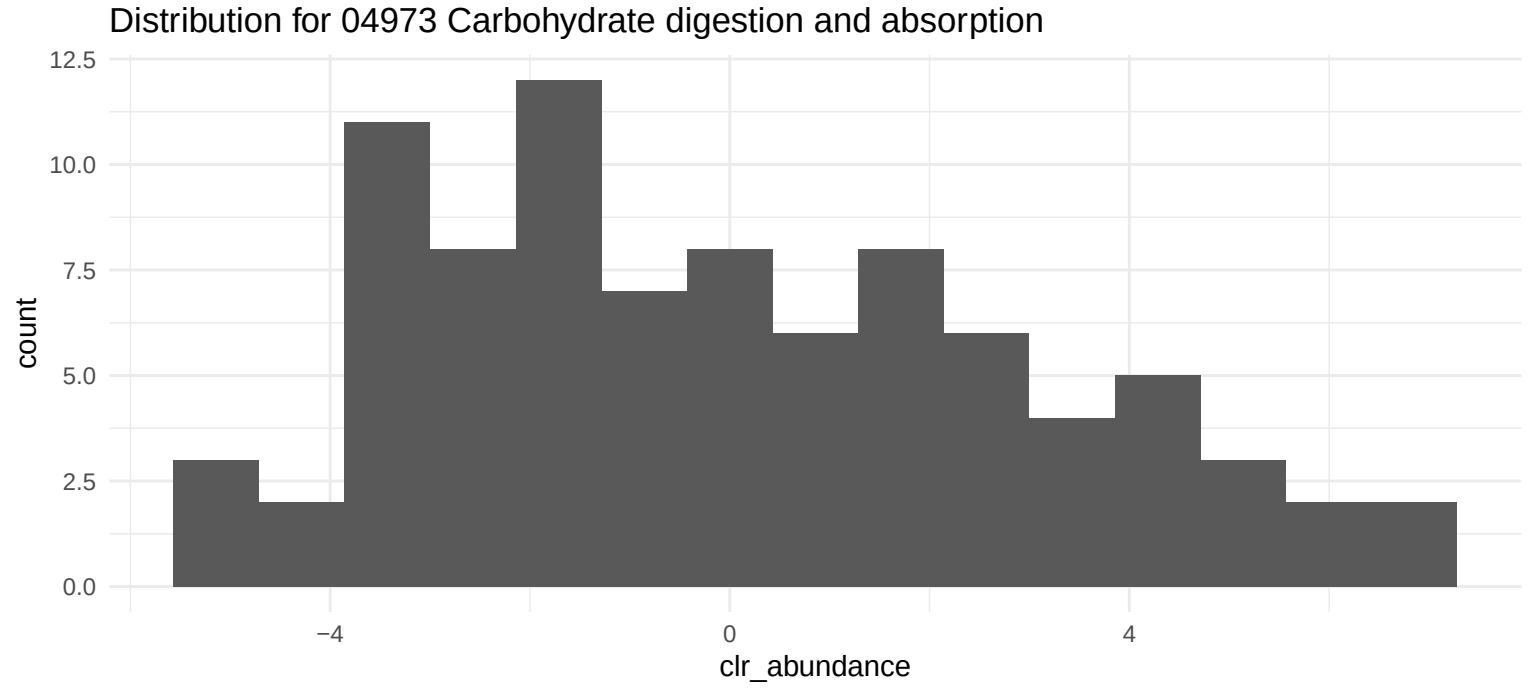


Distribution for 04970 Salivary secretion

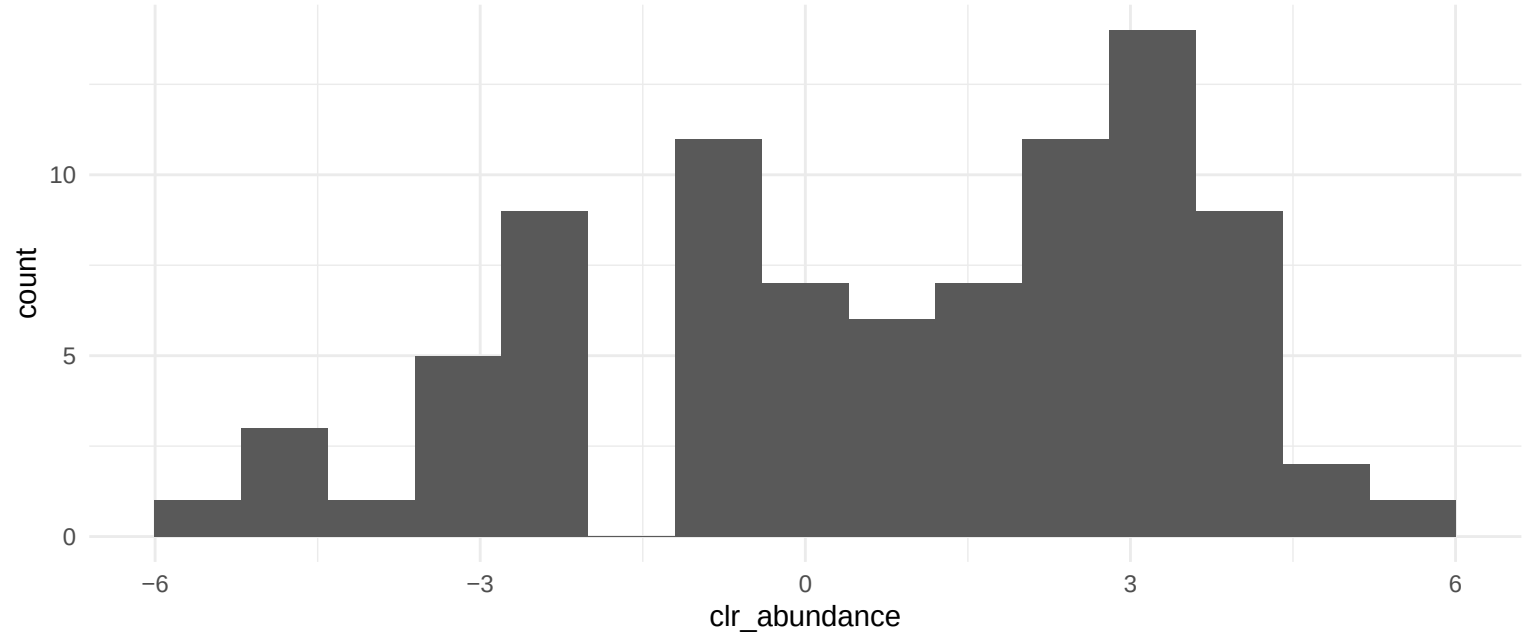



Distribution for 04971 Gastric acid secretion
count
25
20
15
10
5
0
−20
−10
0
10
clr_abundance

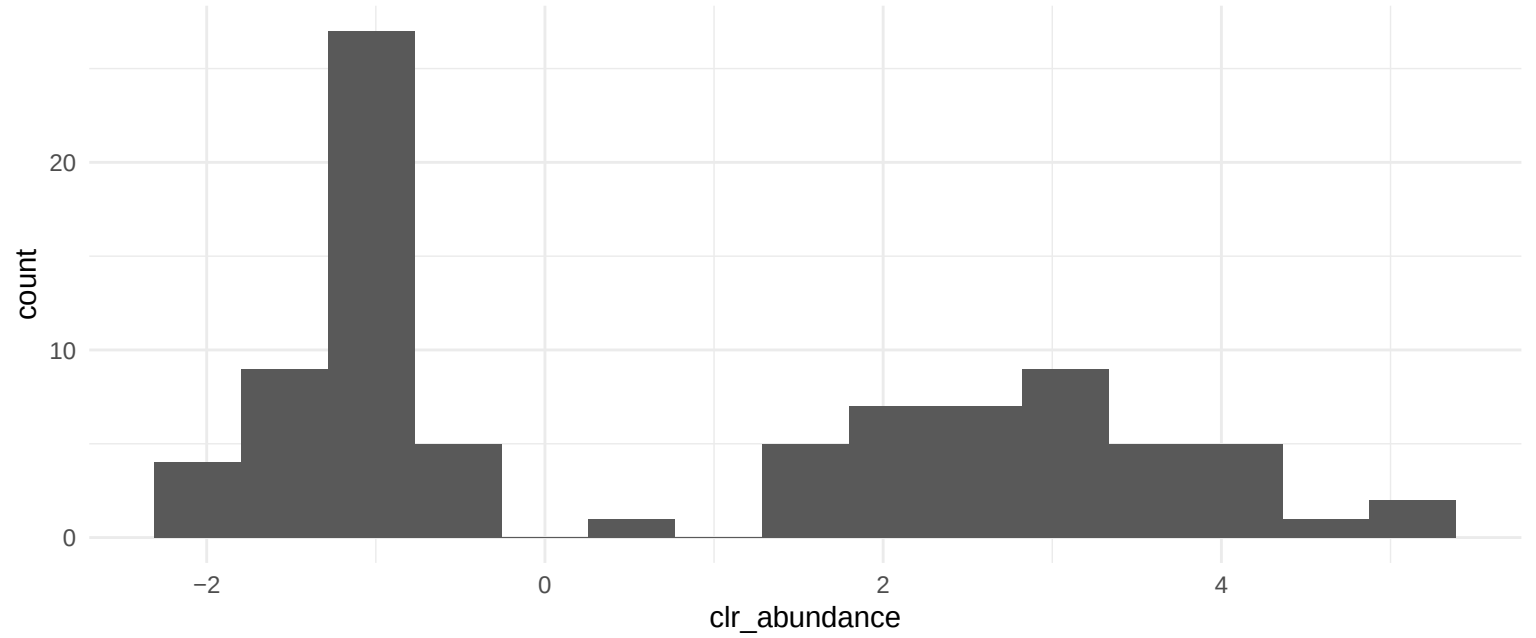

Distribution for 04972 Pancreatic secretion
count
15
10
5
0
−8
−4
0
4
clr_abundance


Distribution for 04973 Carbohydrate digestion and absorption
count
12.5
10.0
7.5
5.0
2.5
0.0
−4
0
4
clr_abundance

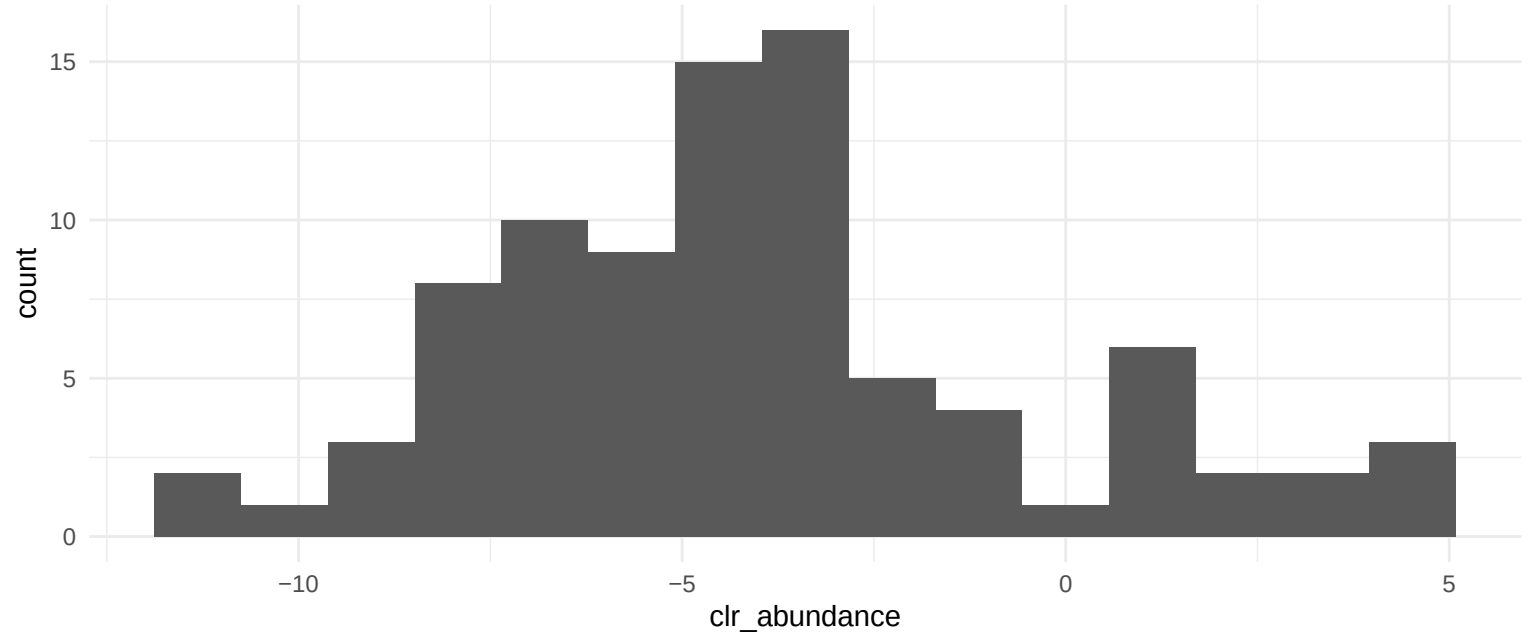
Distribution for 04974 Protein digestion and absorption

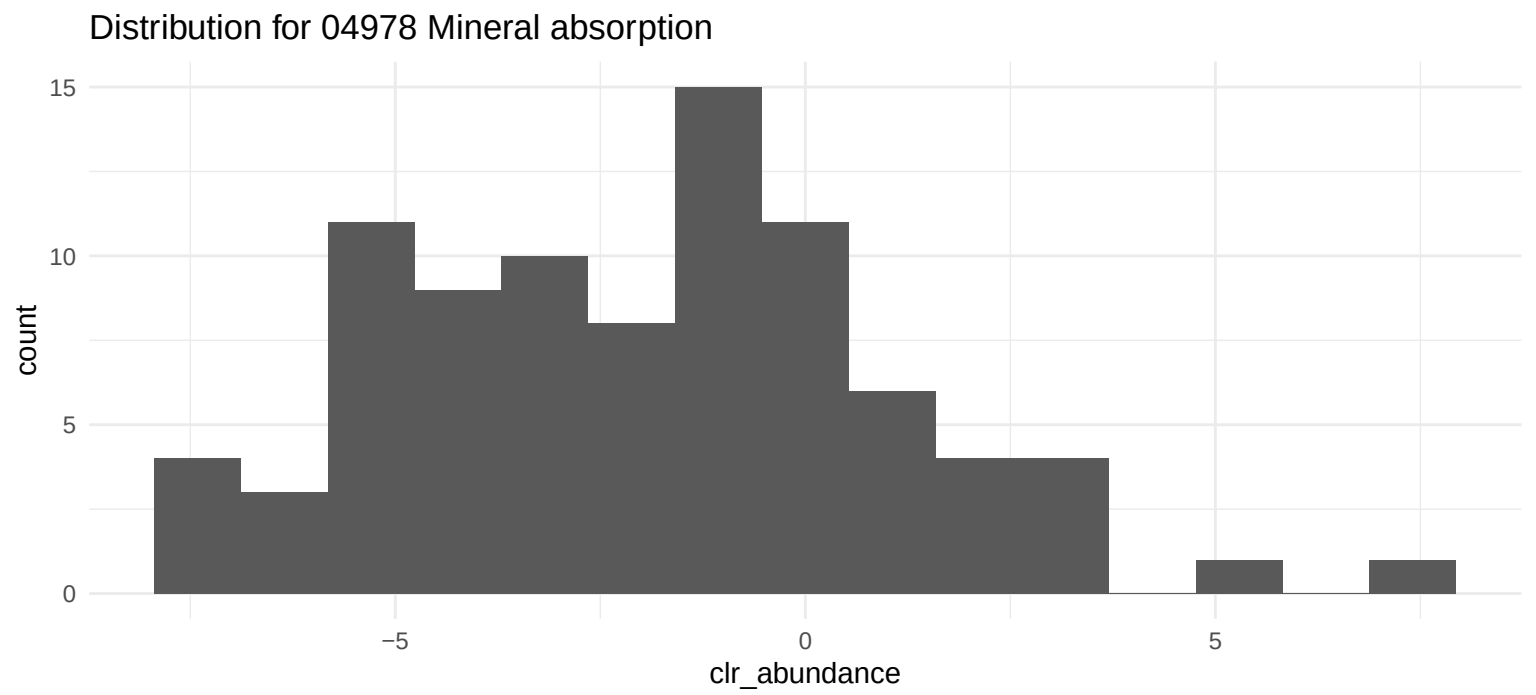


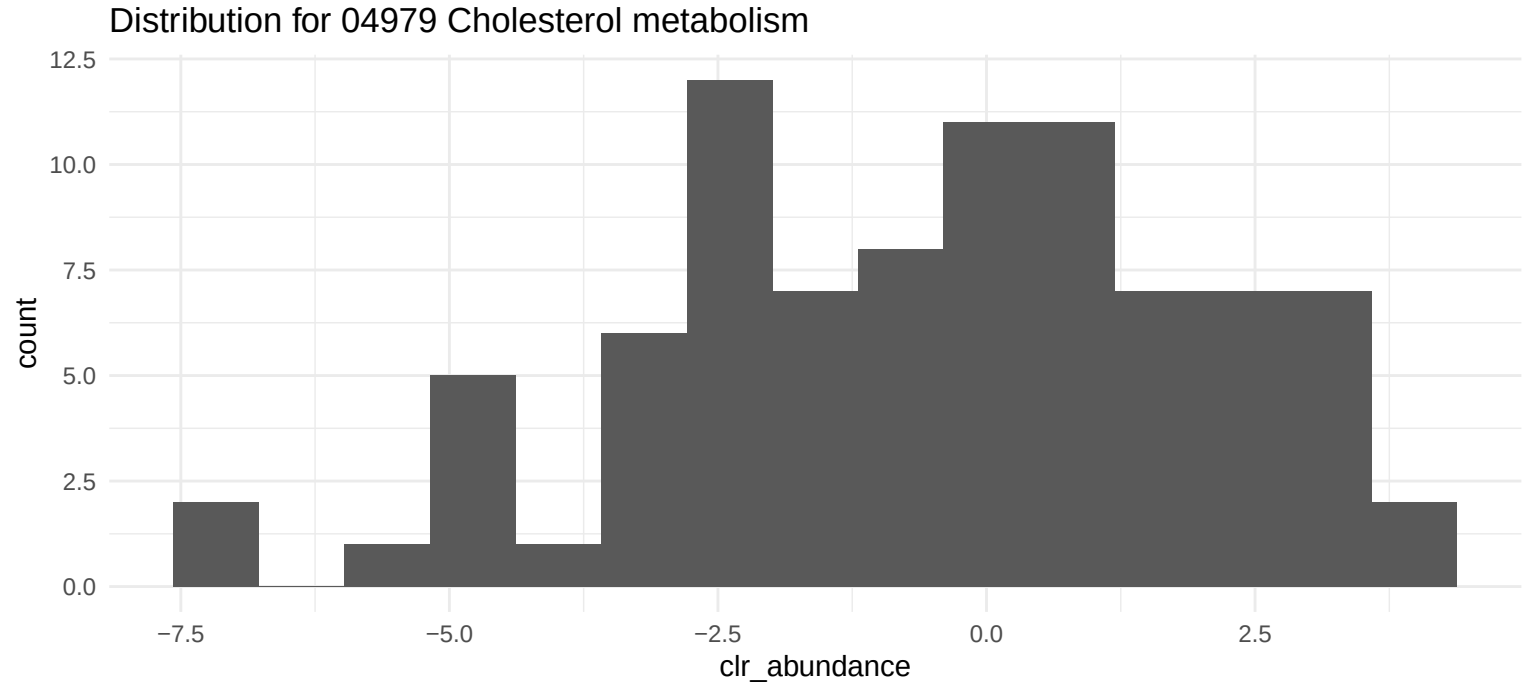
Distribution for 04975 Fat digestion and absorption

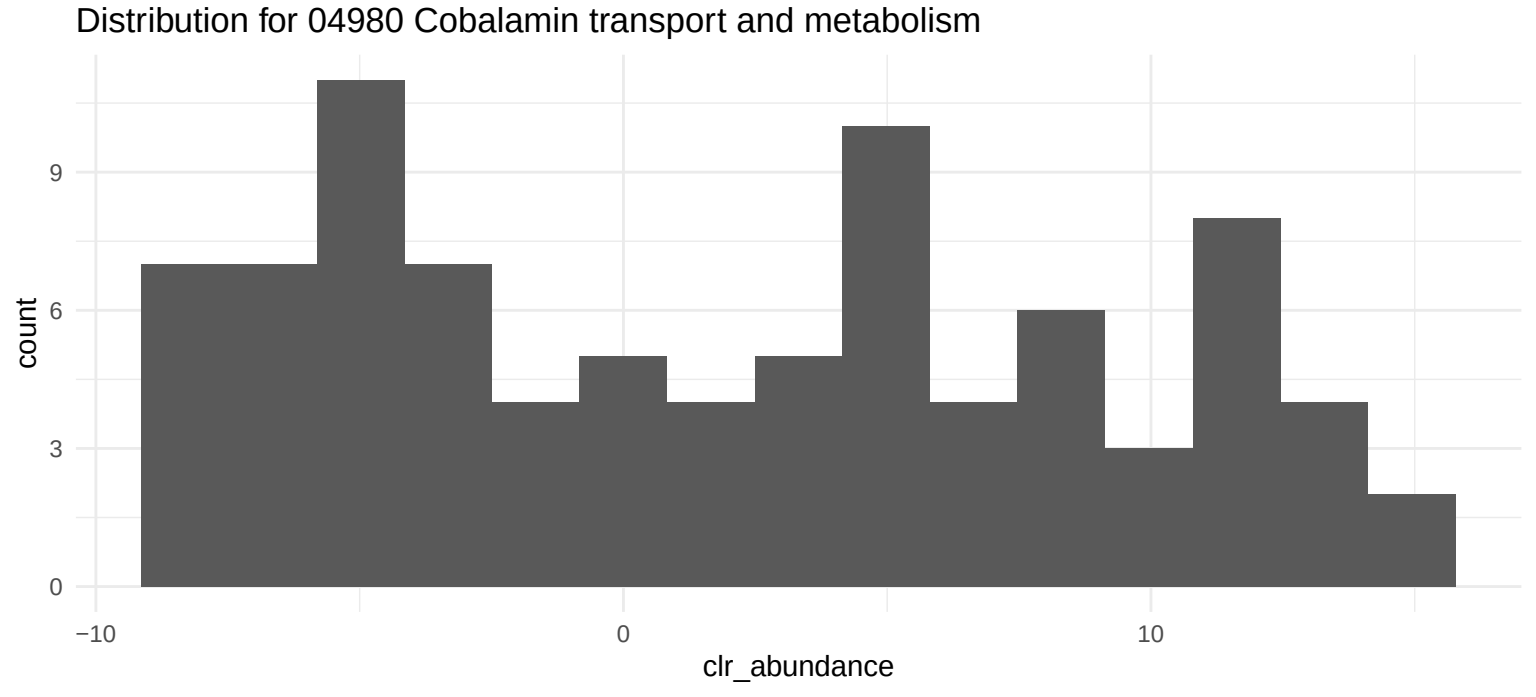


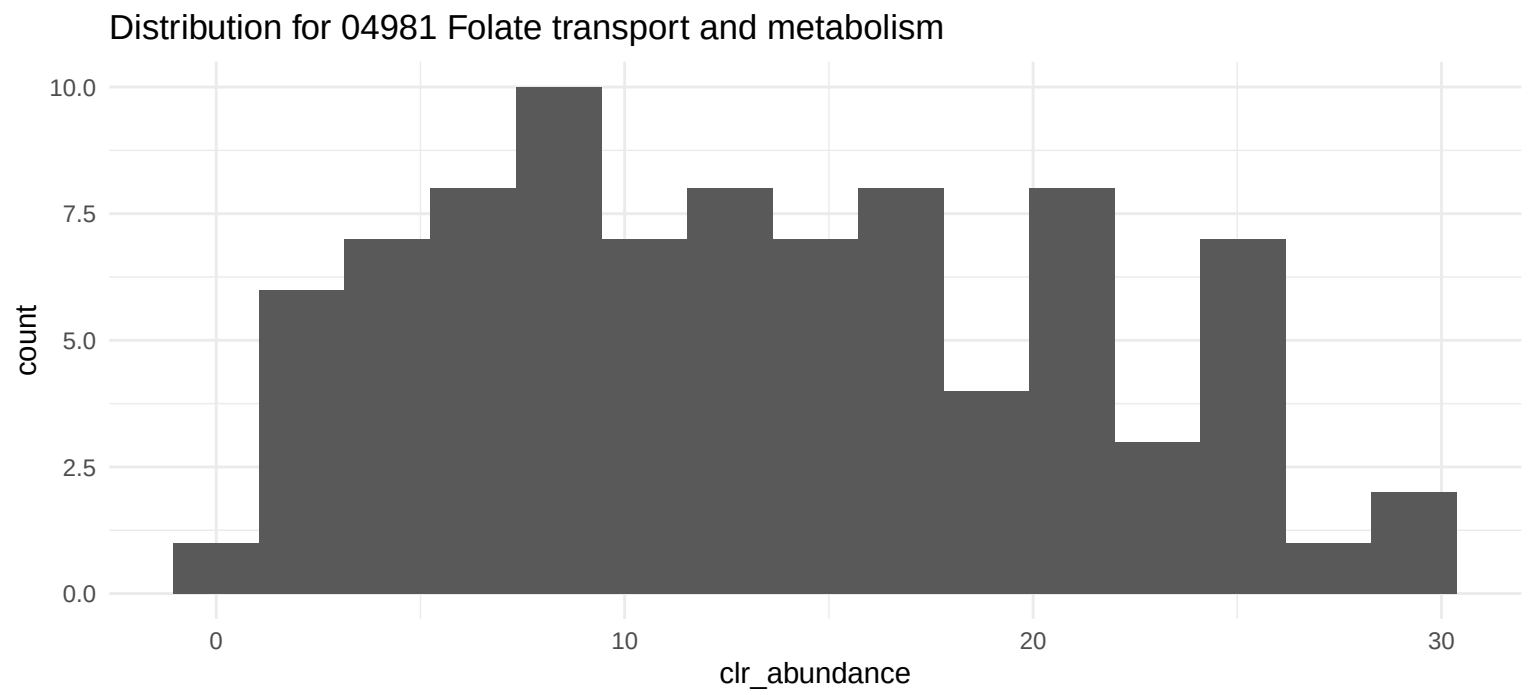
Distribution for 04976 Bile secretion

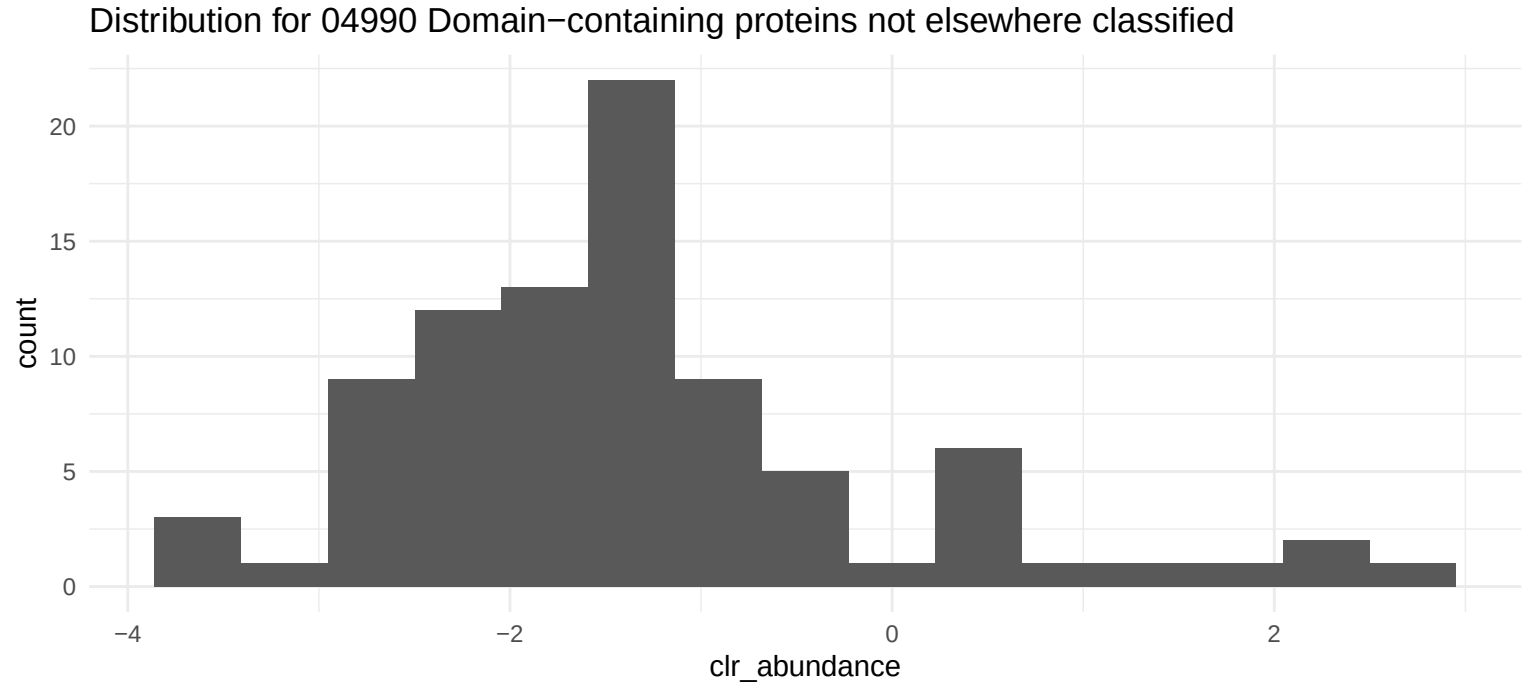


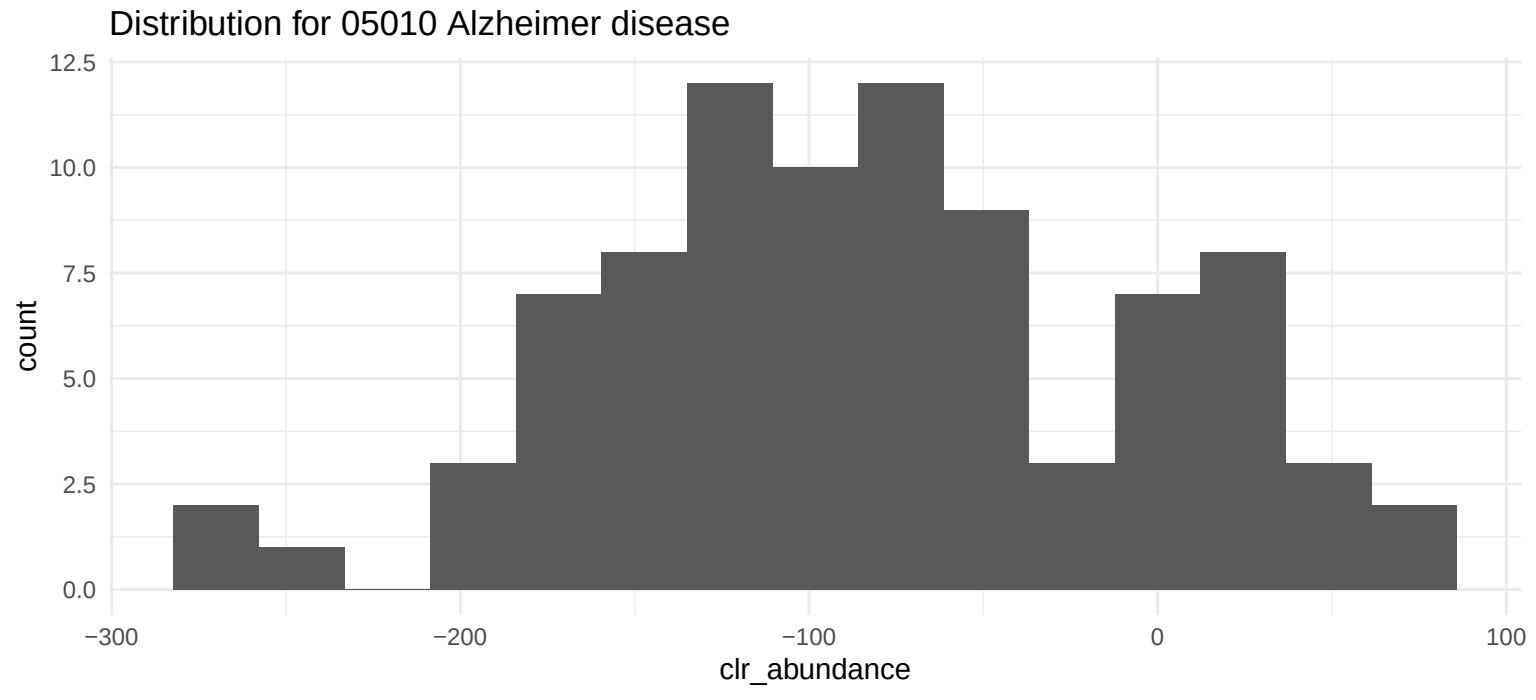

Distribution for 04978 Mineral absorption
15
10
5
0
count
−5
0
5
clr_abundance

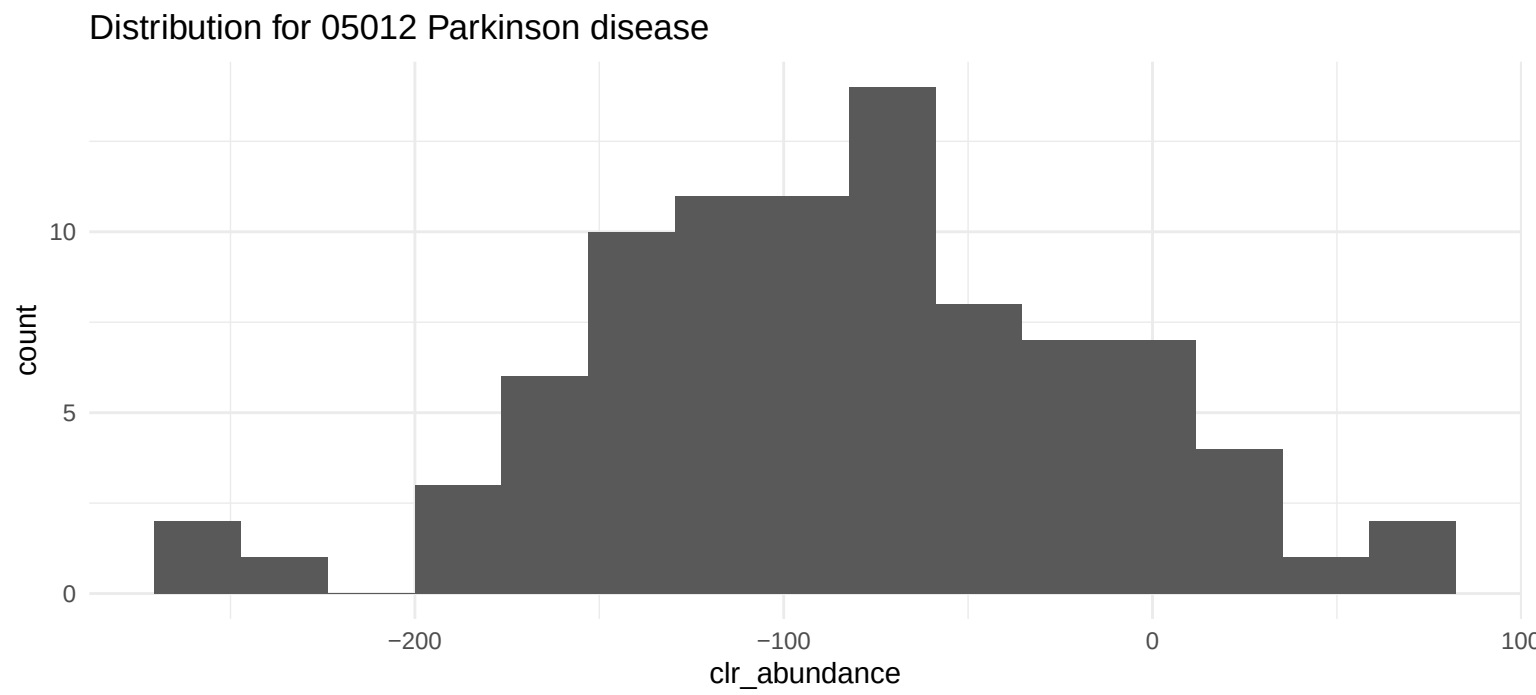

Distribution for 04979 Cholesterol metabolism
12.5
10.0
7.5
5.0
2.5
0.0
count
−7.5
−5.0
−2.5
0.0
2.5
clr_abundance

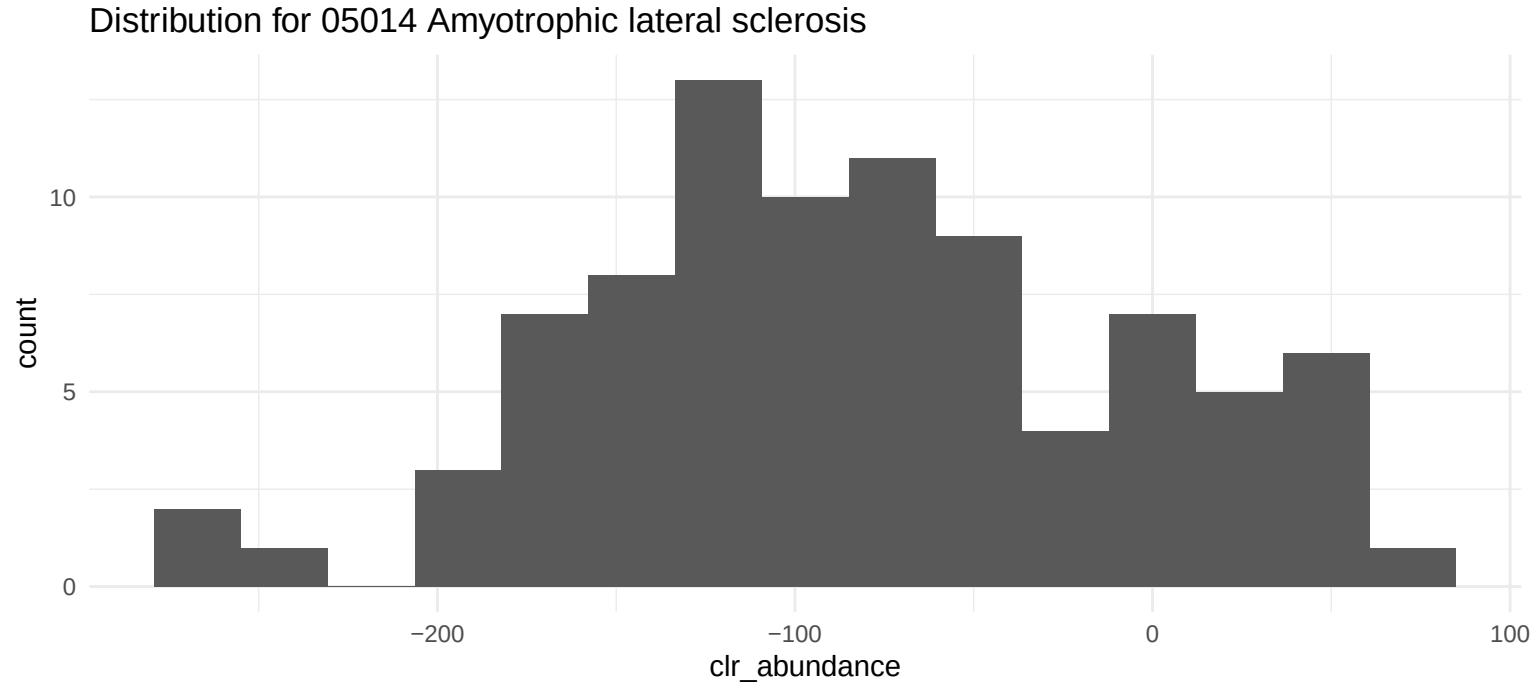

Distribution for 04980 Cobalamin transport and metabolism
9
6
3
0
count
−10
0
10
clr_abundance

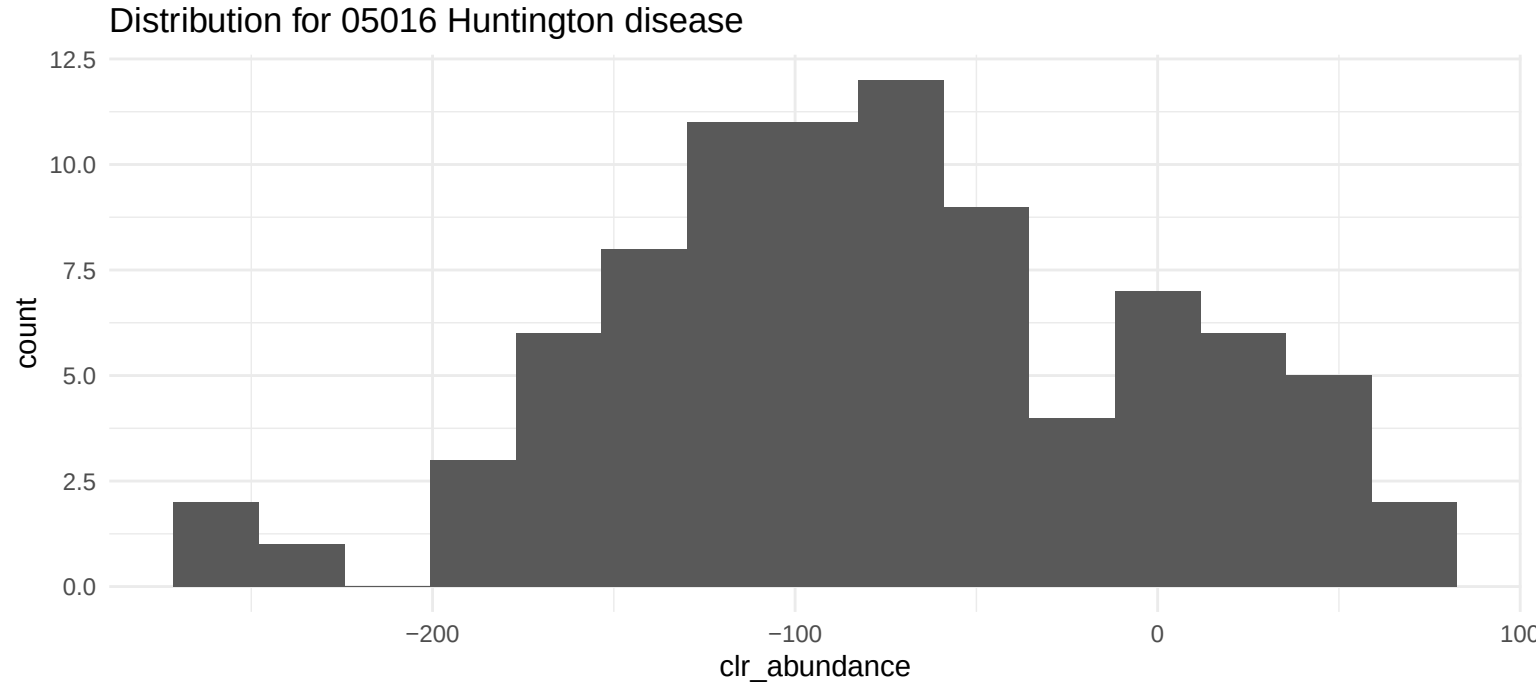

Distribution for 04981 Folate transport and metabolism
count
clr_abundance
10.0
7.5
5.0
2.5
0.0
0
10
20
30

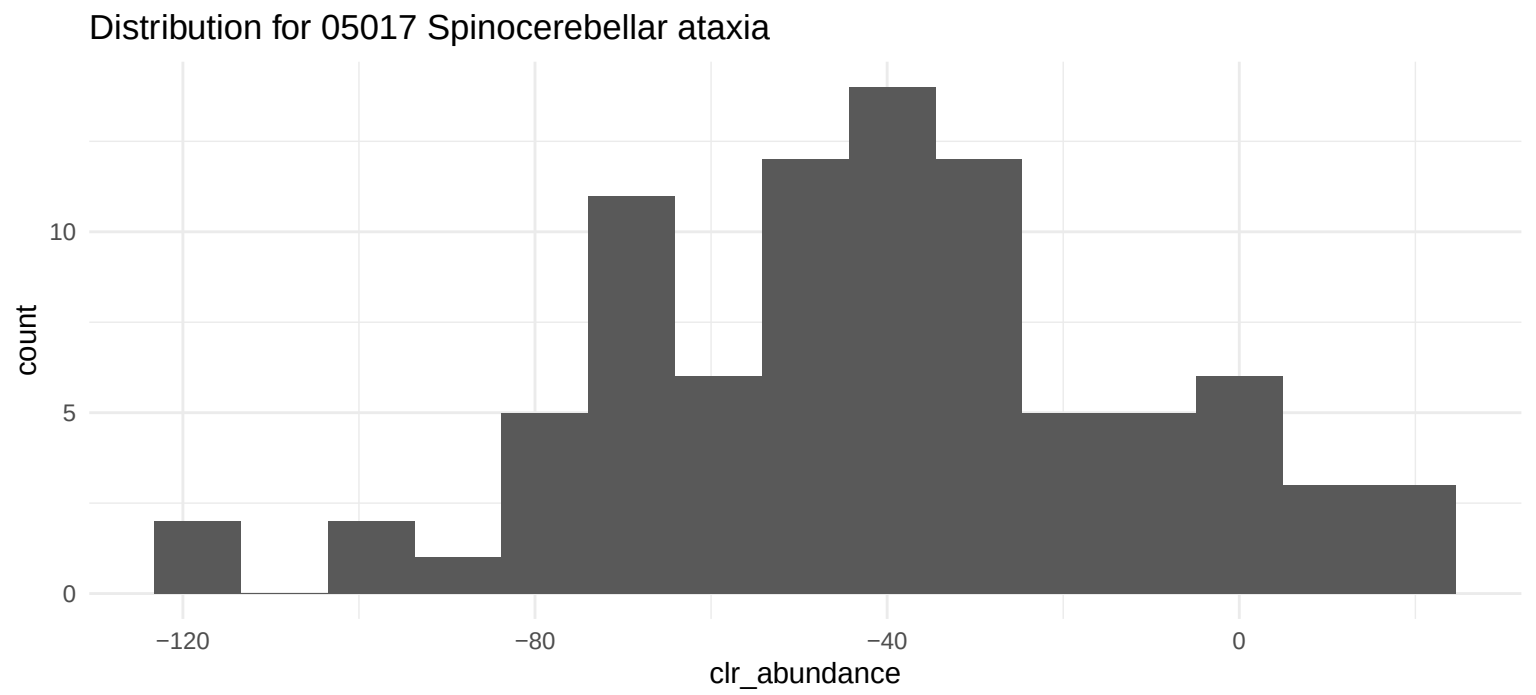

Distribution for 04990 Domain−containing proteins not elsewhere classified
count
clr_abundance
20
15
10
5
0
−4
−2
0
2

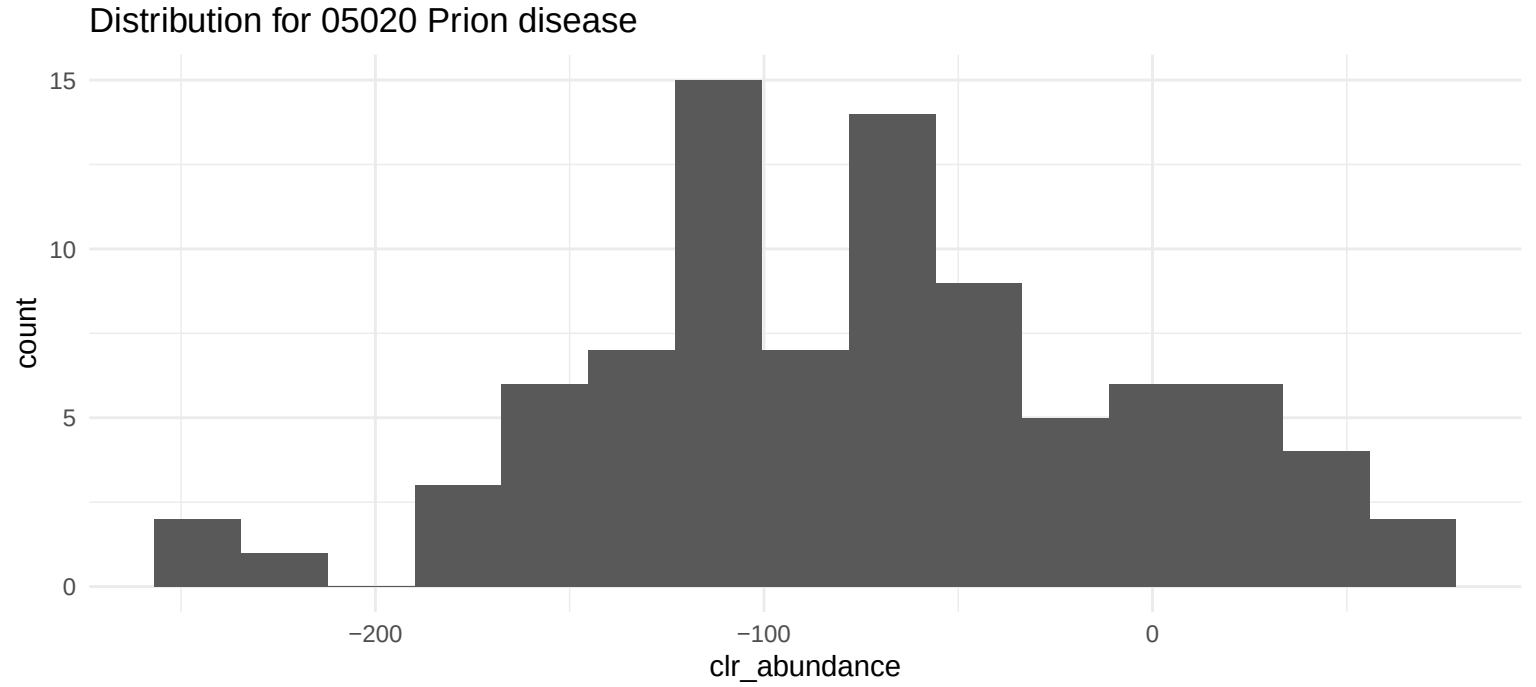

Distribution for 05010 Alzheimer disease
count
clr_abundance
12.5
10.0
7.5
5.0
2.5
0.0
−300
−200
−100
0
100

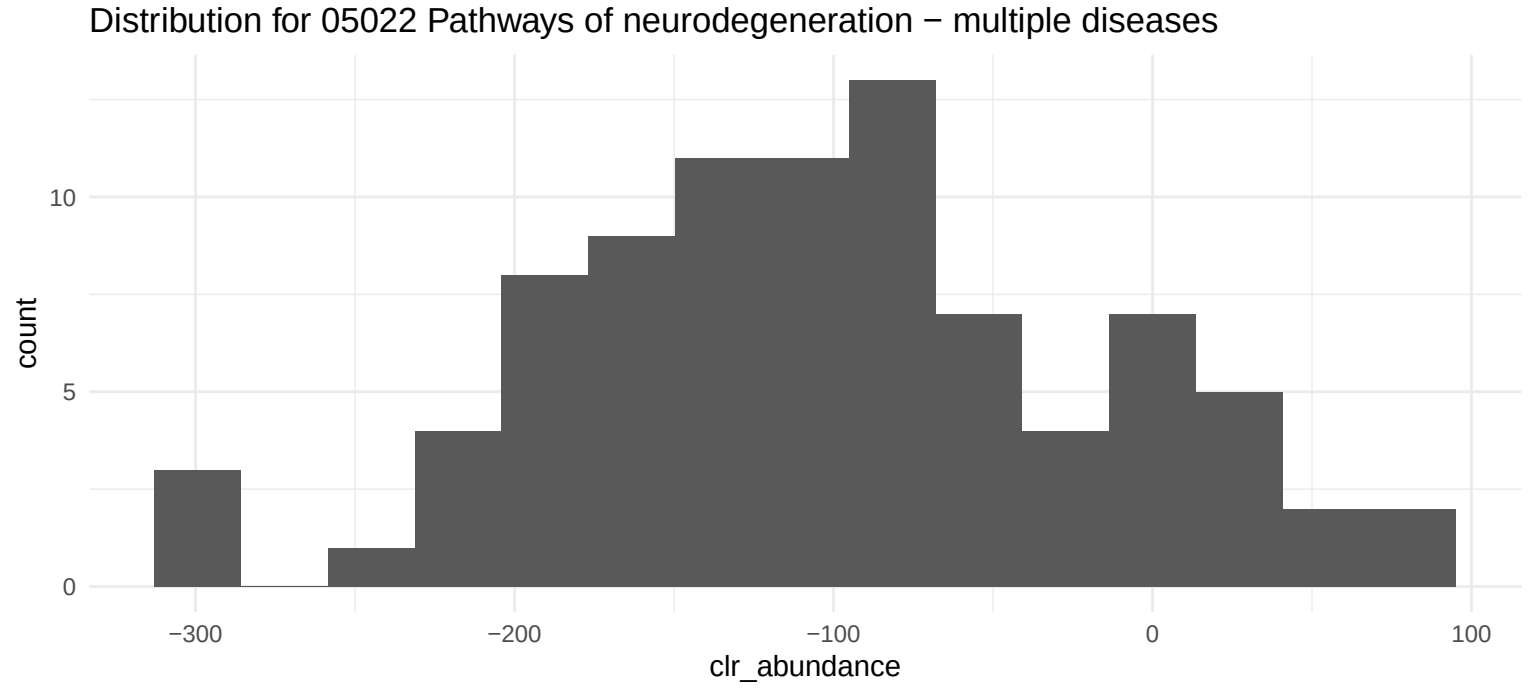

Distribution for 05012 Parkinson disease
count
clr_abundance
0
5
10
−200
−100
0
100


Distribution for 05014 Amyotrophic lateral sclerosis
count
clr_abundance
0
5
10
−200
−100
0
100

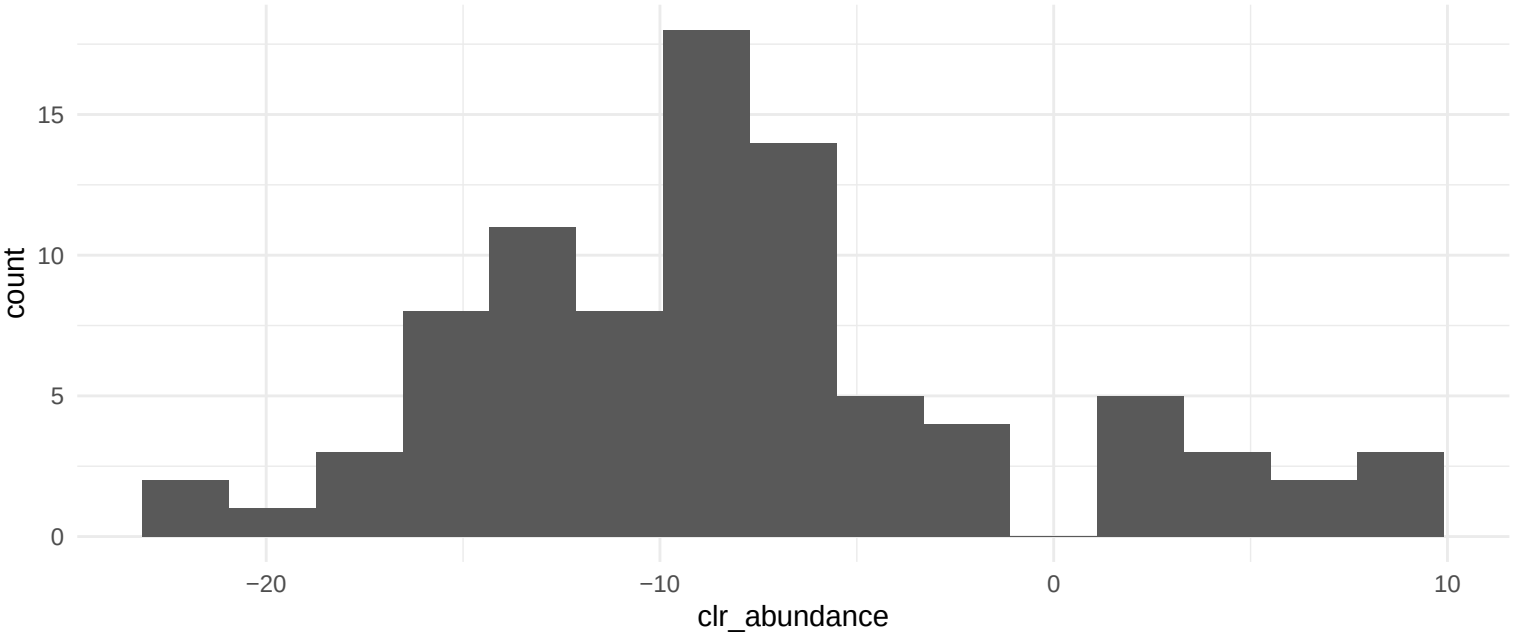

Distribution for 05016 Huntington disease
count
clr_abundance
0.0
2.5
5.0
7.5
10.0
12.5
−200
−100
0
100


Distribution for 05017 Spinocerebellar ataxia
count
clr_abundance
0
5
10
−120
−80
−40
0

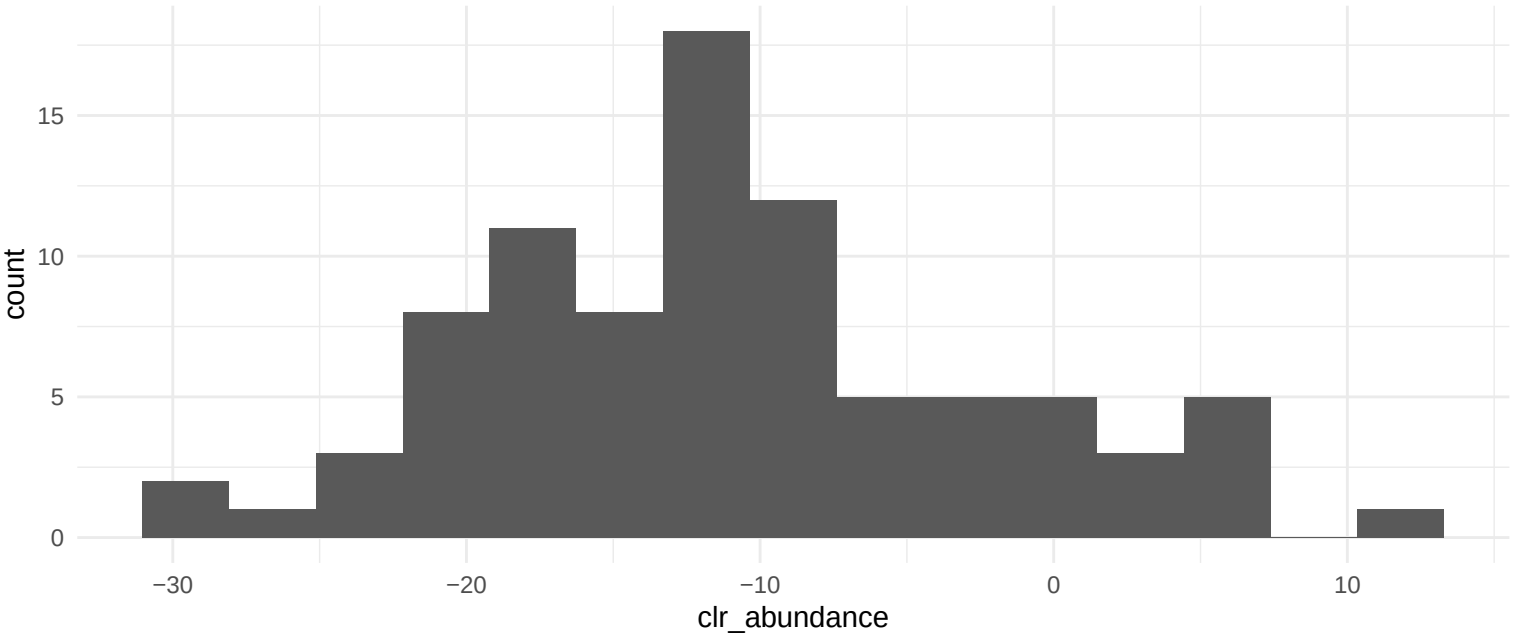

Distribution for 05020 Prion disease
count
clr_abundance
0
5
10
15
−200
−100
0


Distribution for 05022 Pathways of neurodegeneration − multiple diseases
count
clr_abundance
0
5
10
−300
−200
−100
0
100

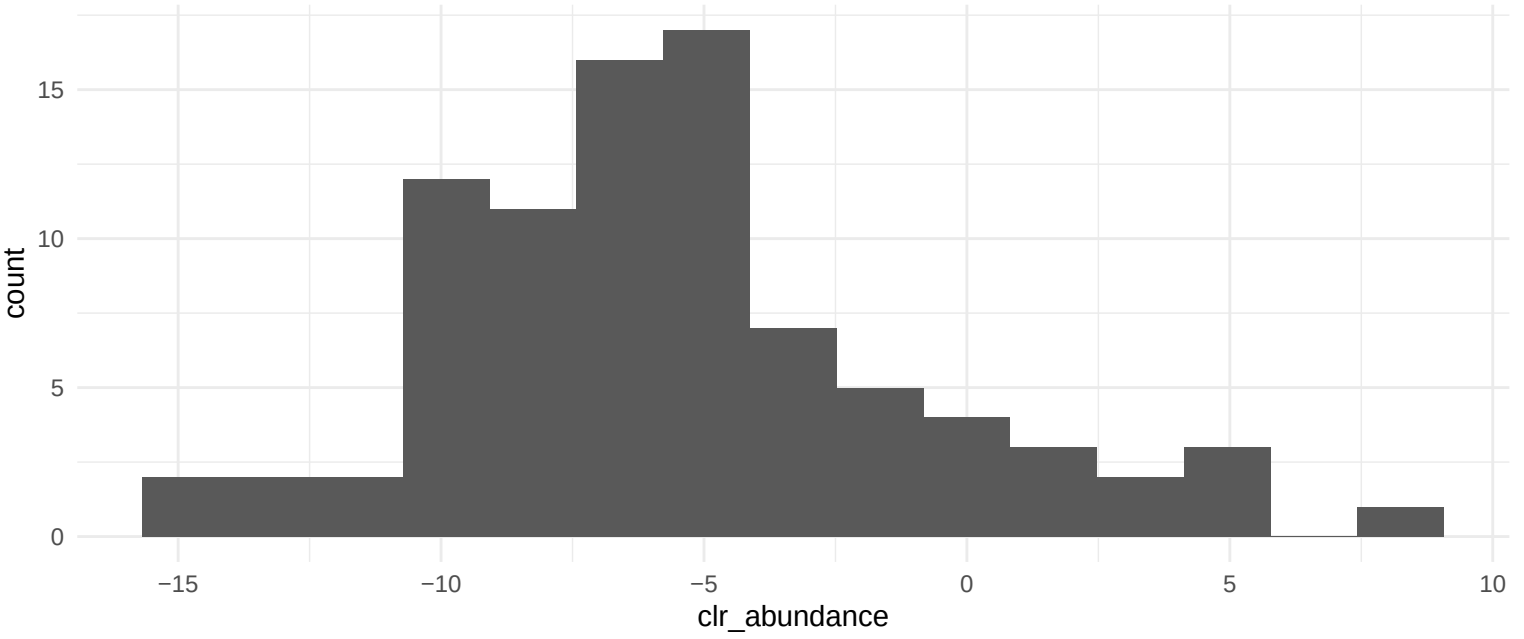
Distribution for 05030 Cocaine addiction

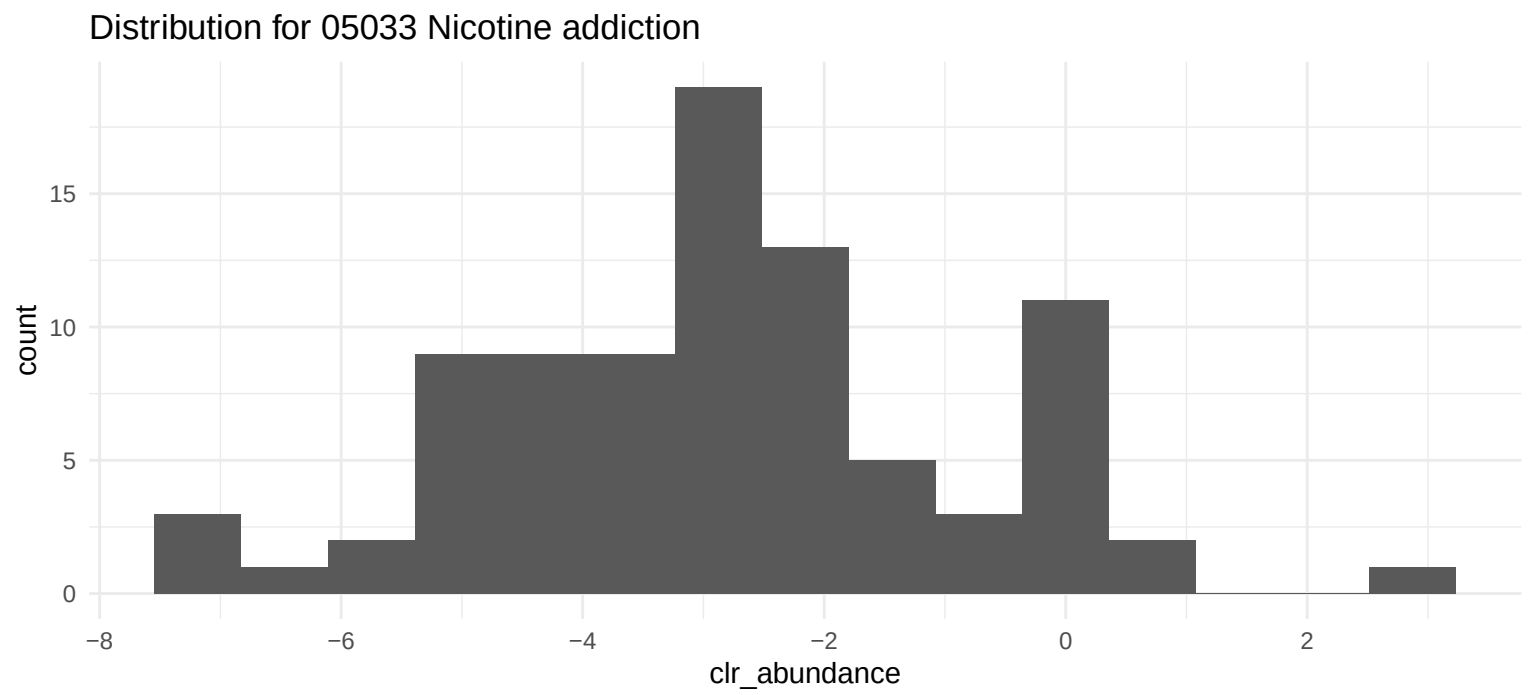


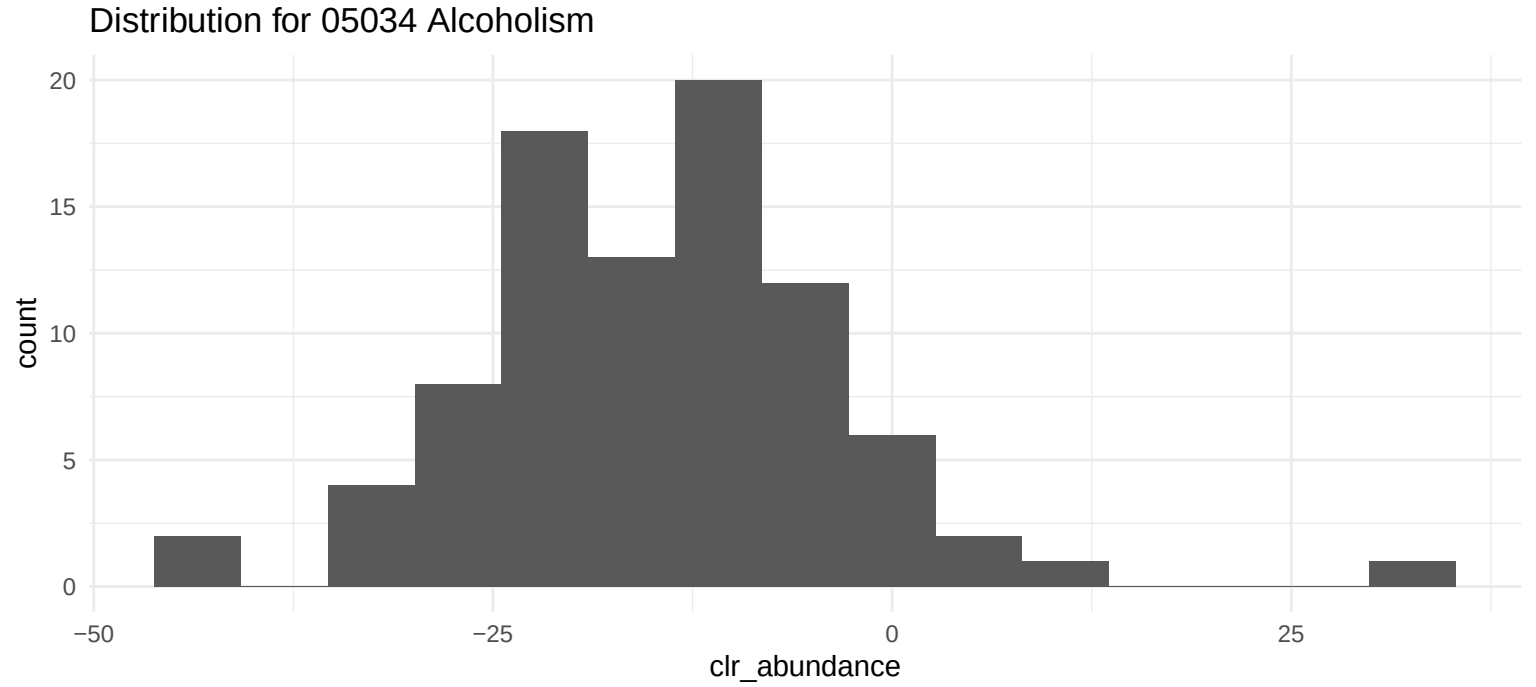
Distribution for 05031 Amphetamine addiction

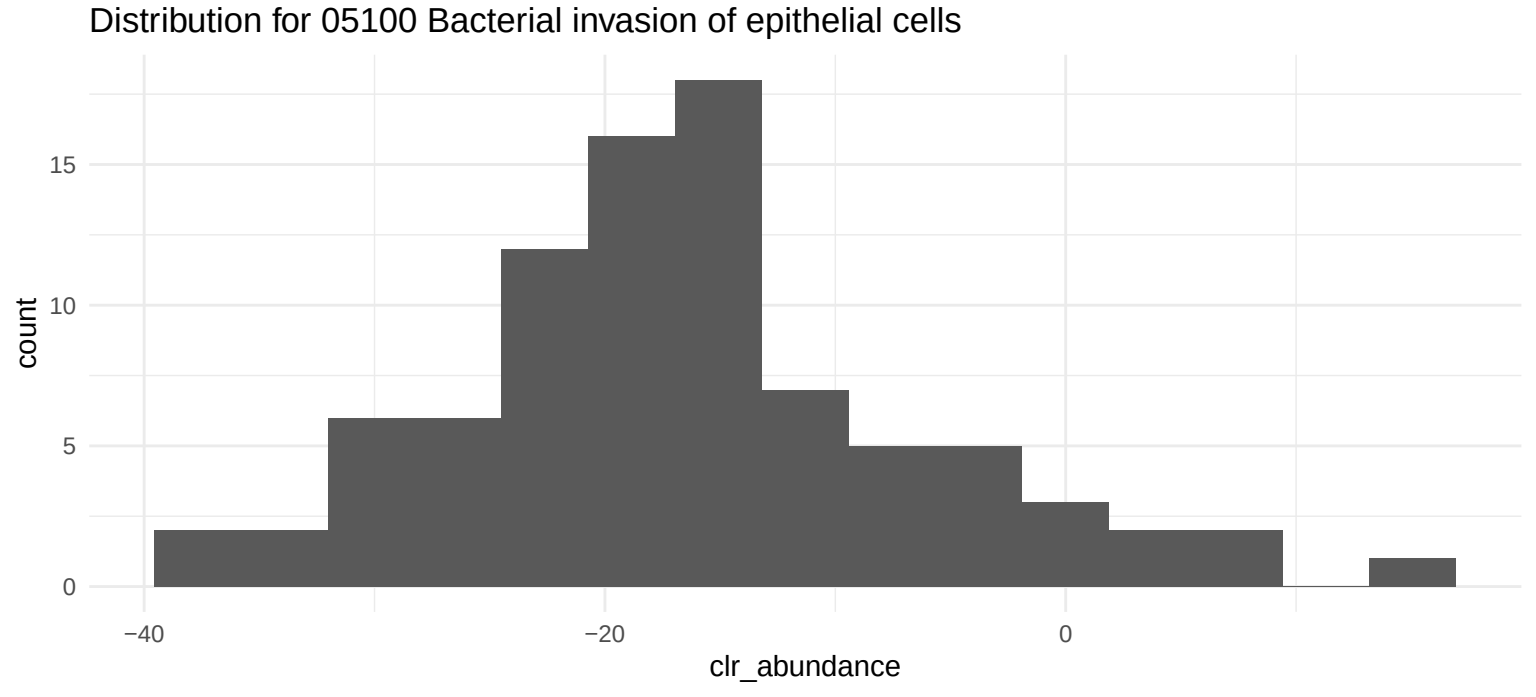


Distribution for 05032 Morphine addiction

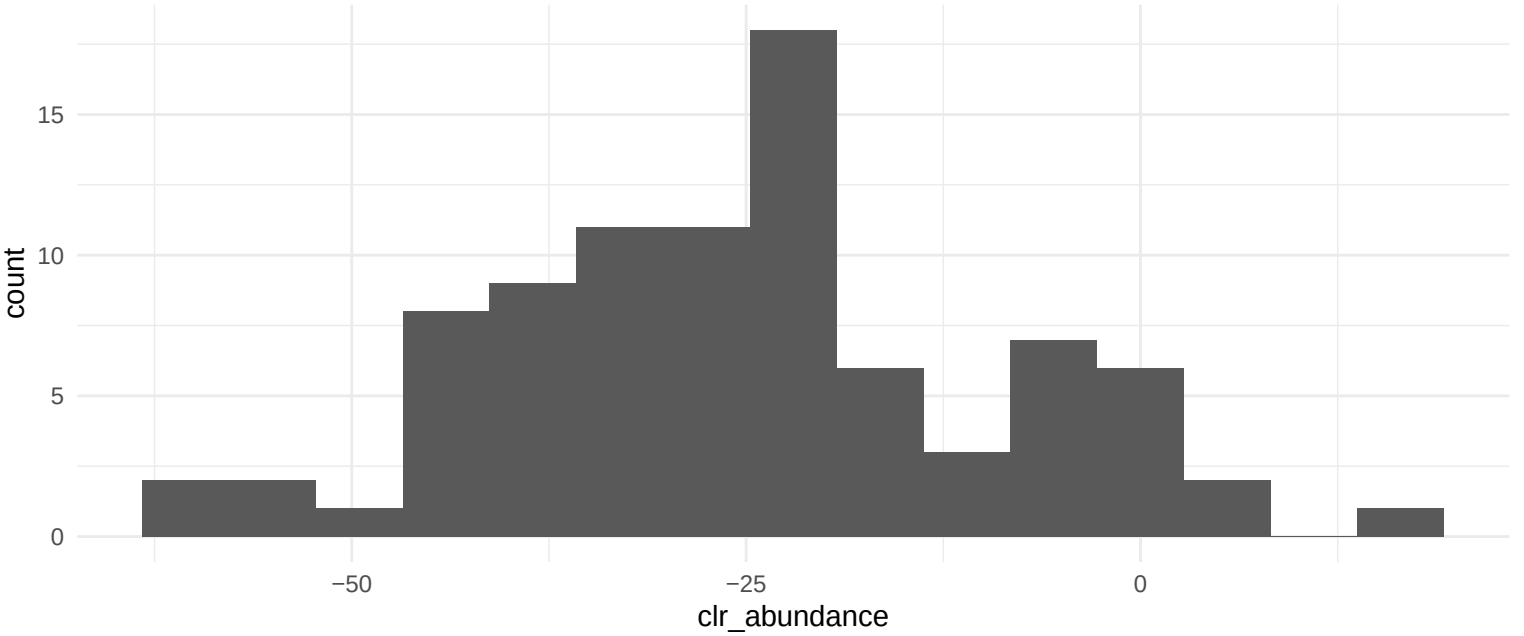



Distribution for 05033 Nicotine addiction
count
clr_abundance

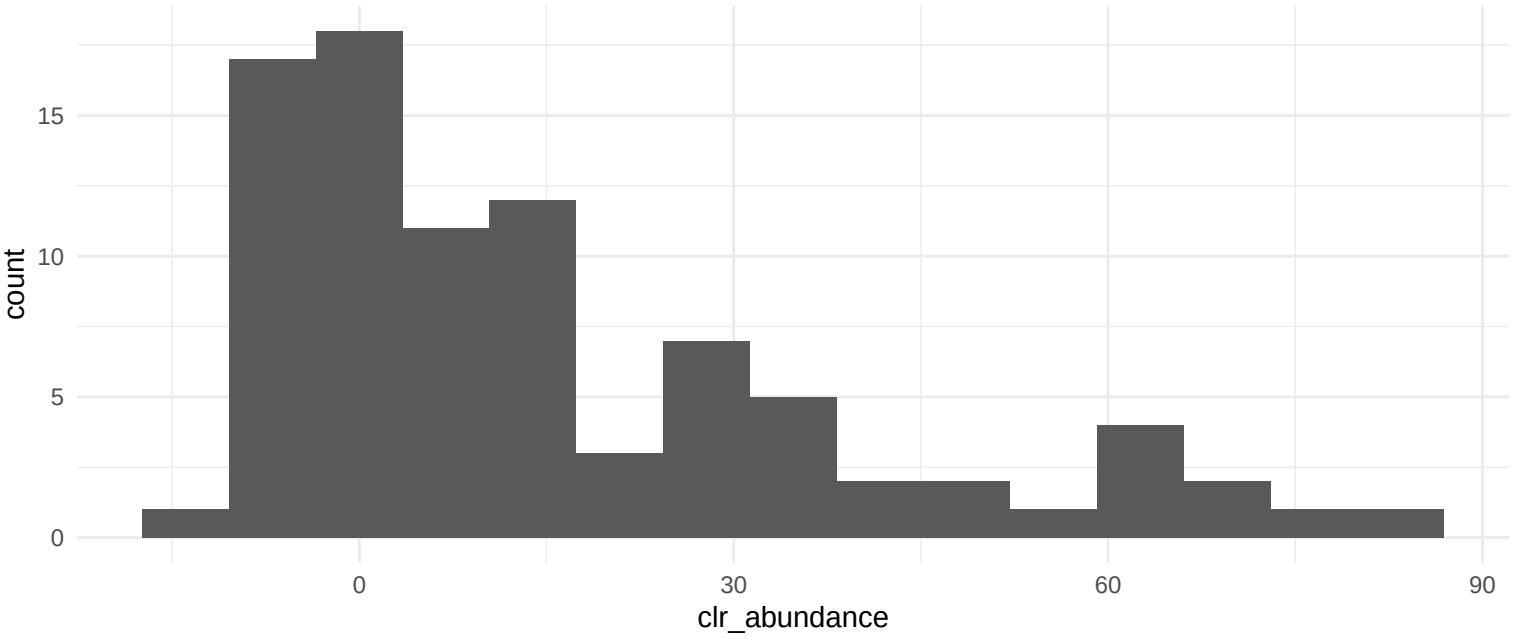

Distribution for 05034 Alcoholism
count
clr_abundance


Distribution for 05100 Bacterial invasion of epithelial cells
count
clr_abundance

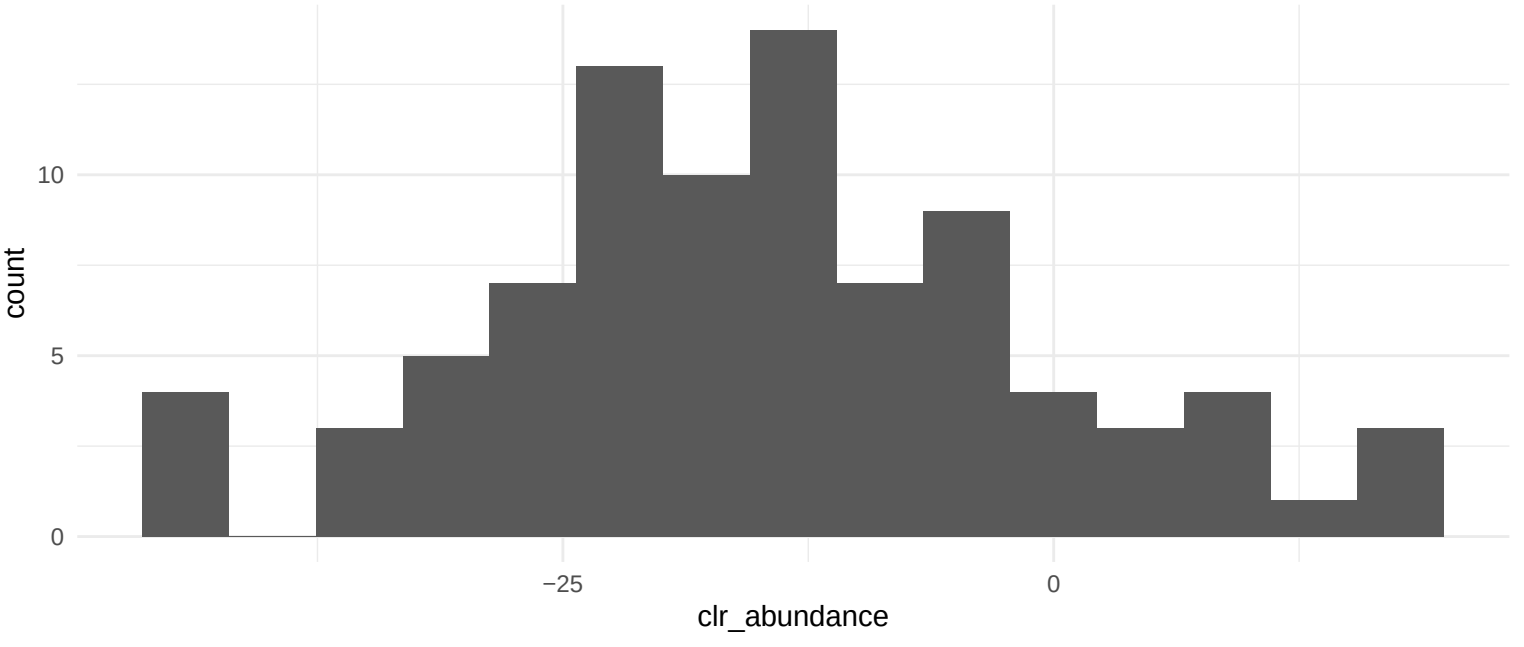
Distribution for 05110 Vibrio cholerae infection



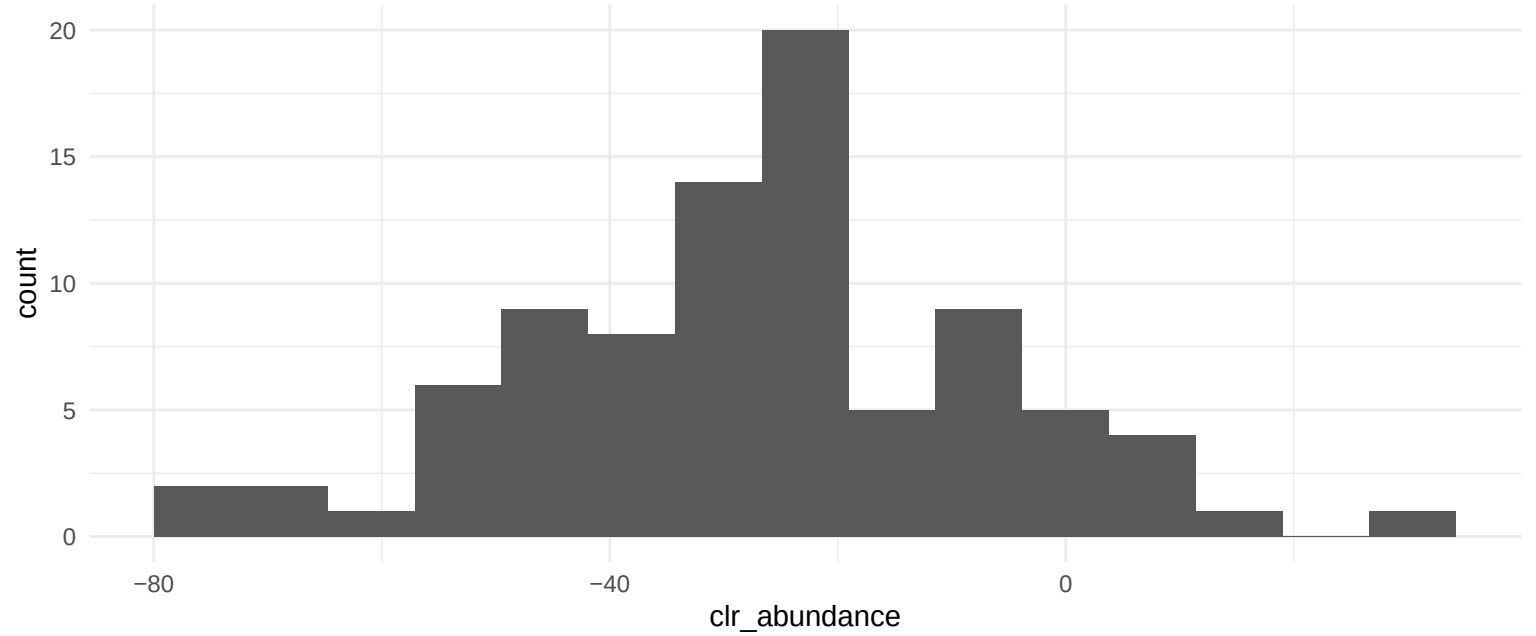
Distribution for 05111 Biofilm formation − Vibrio cholerae



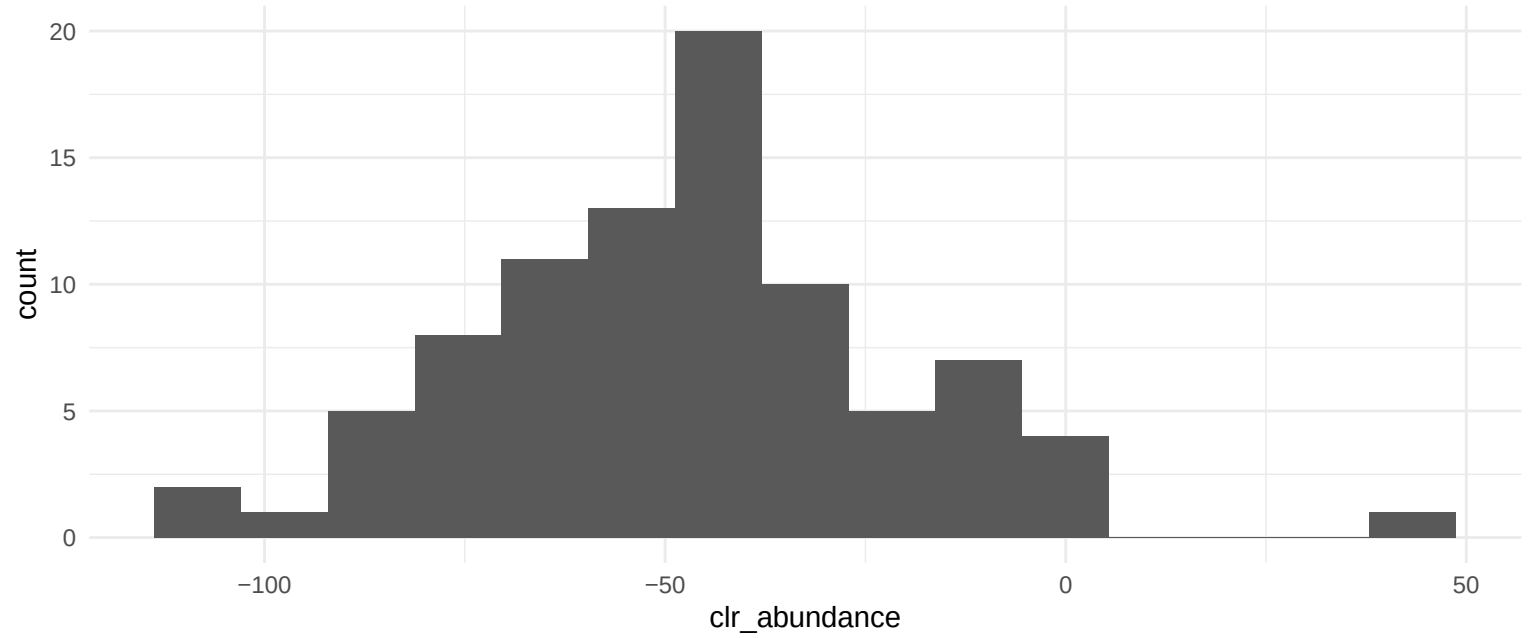
Distribution for 05120 Epithelial cell signaling in Helicobacter pylori infection



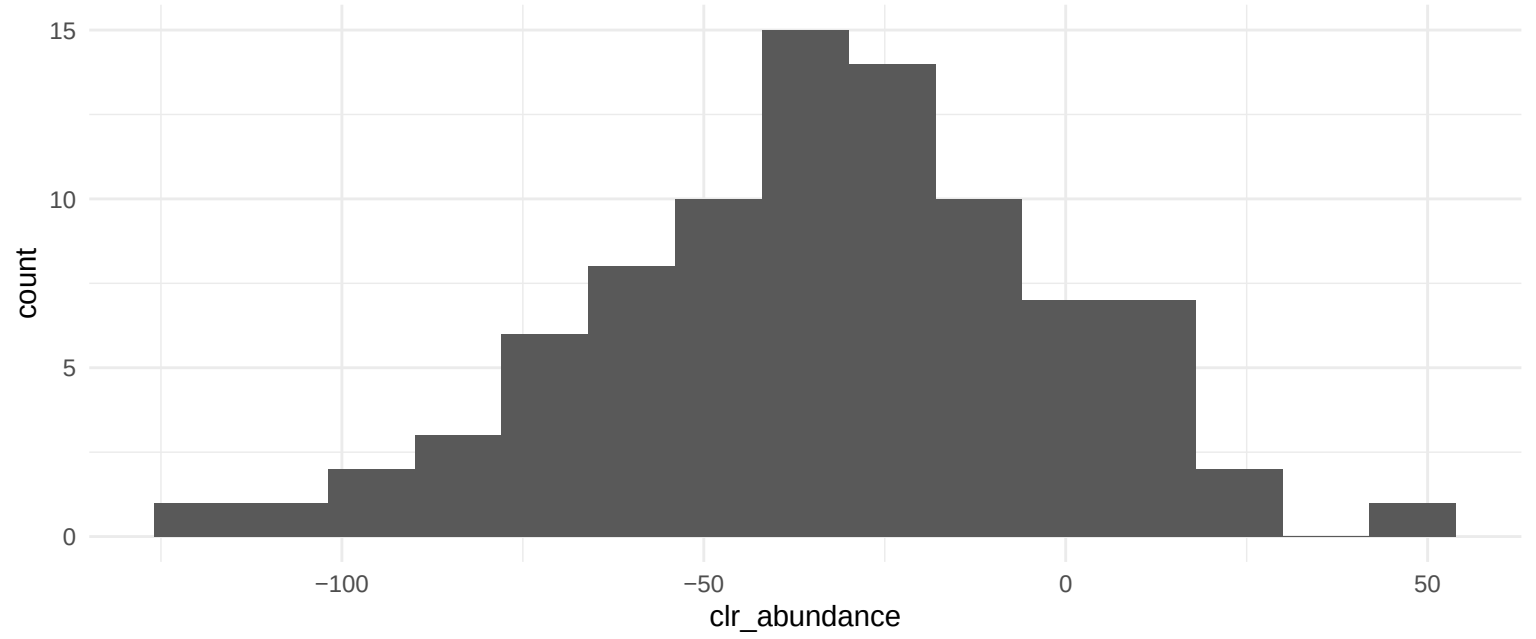
Distribution for 05130 Pathogenic Escherichia coli infection

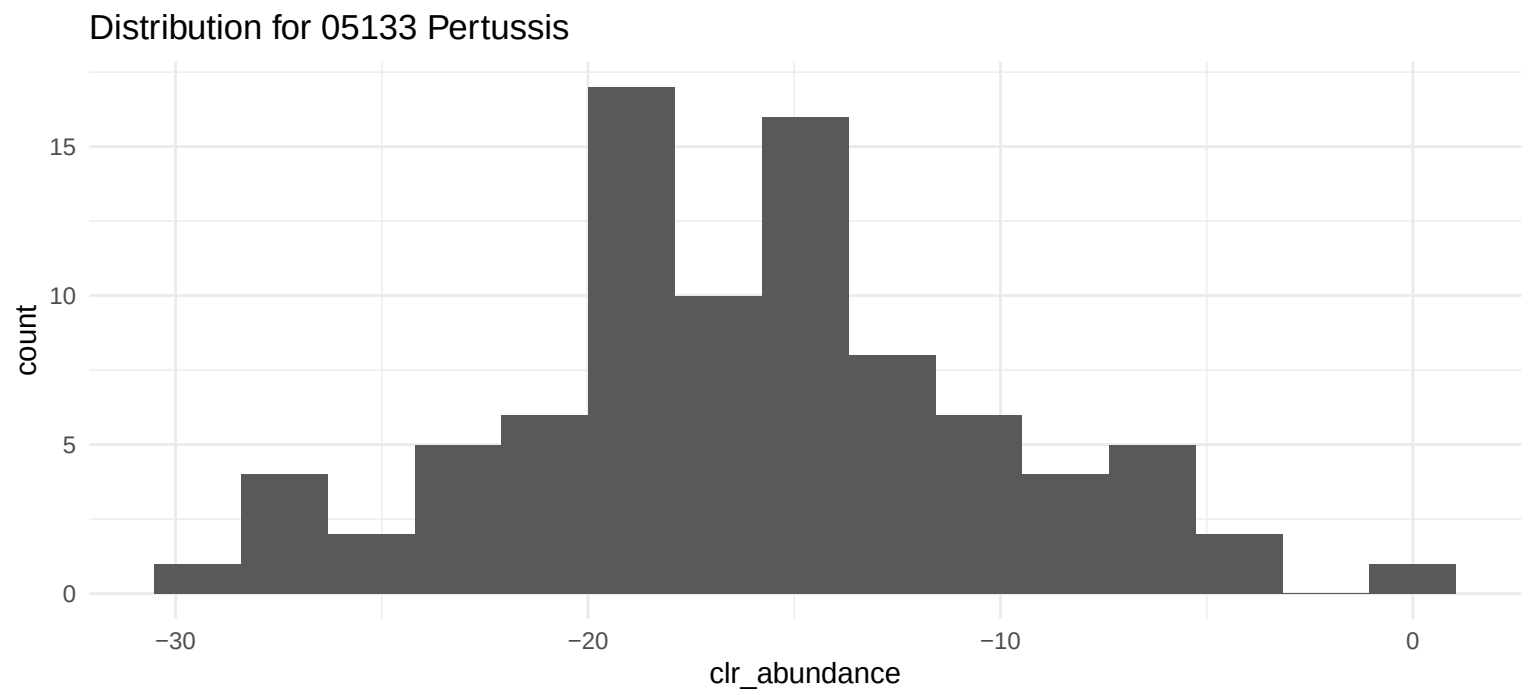


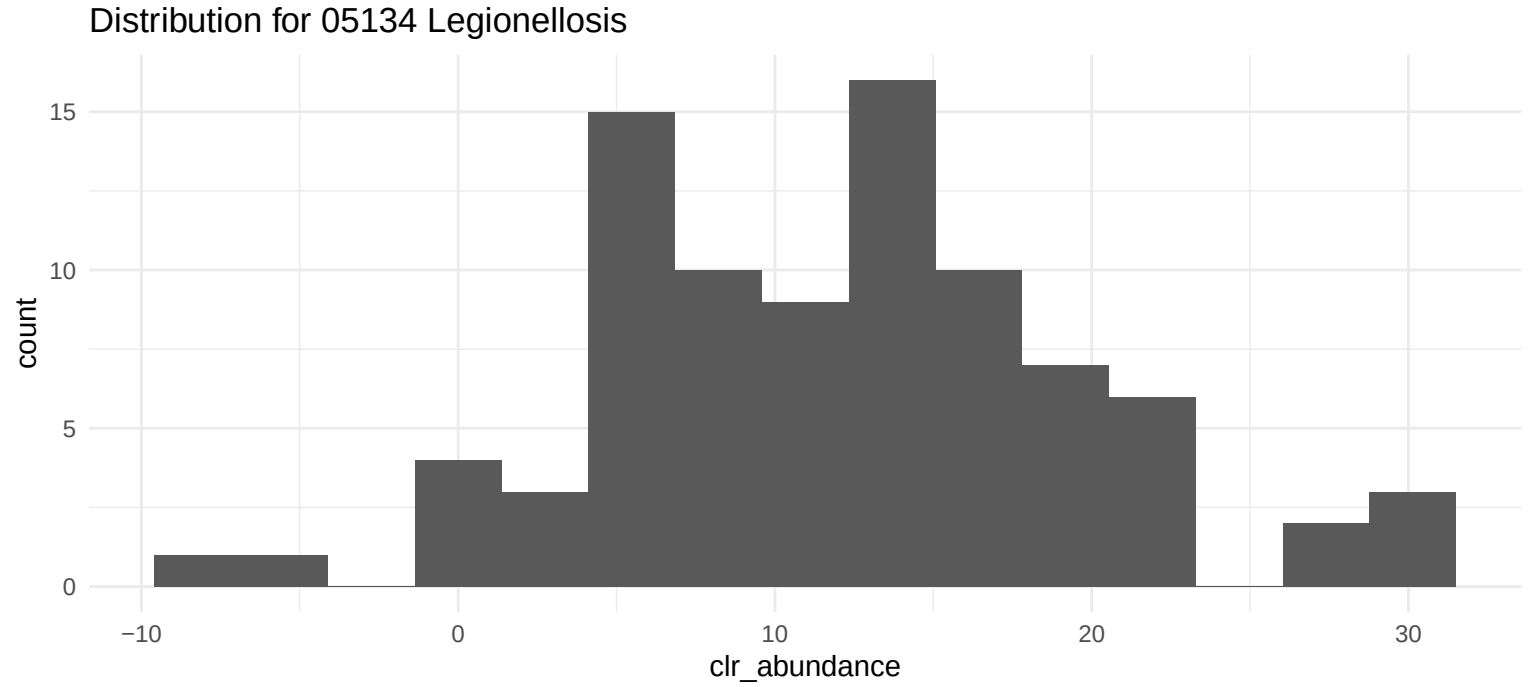
Distribution for 05131 Shigellosis

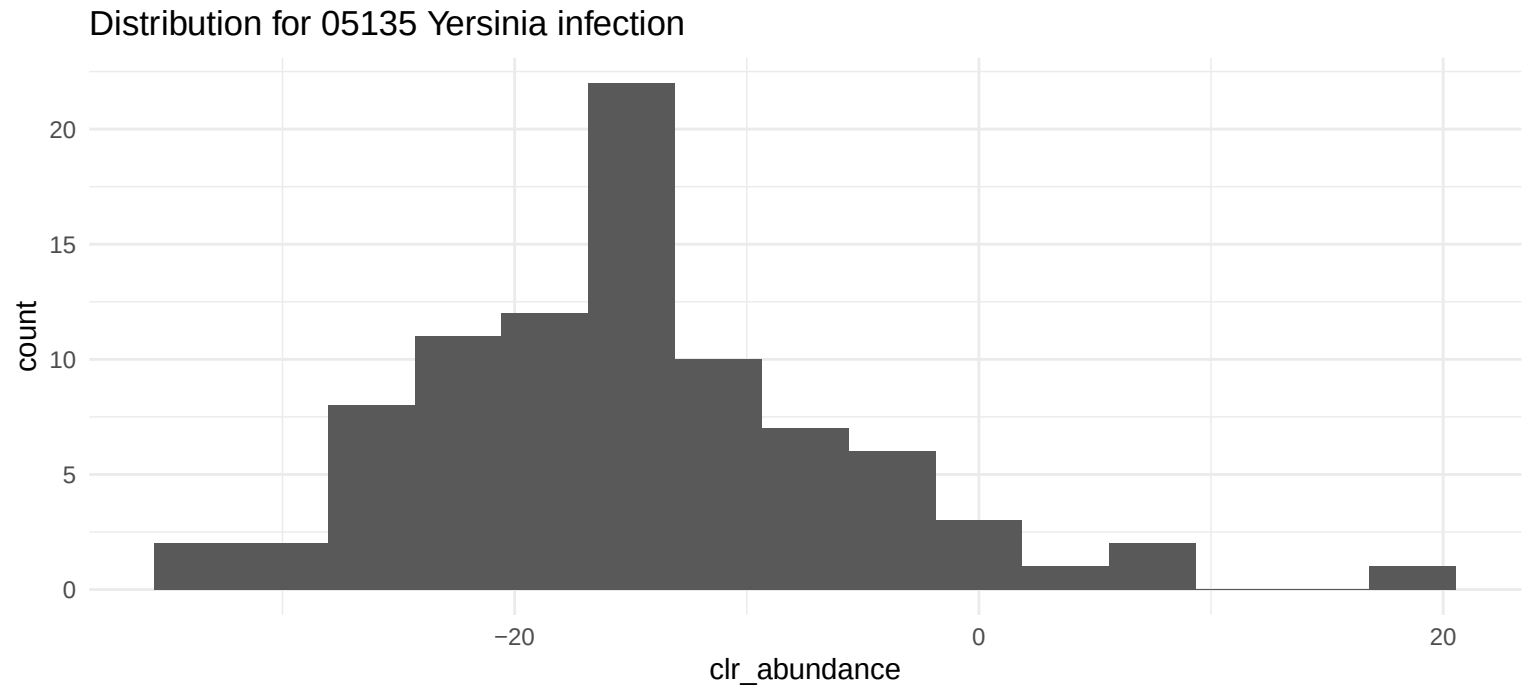


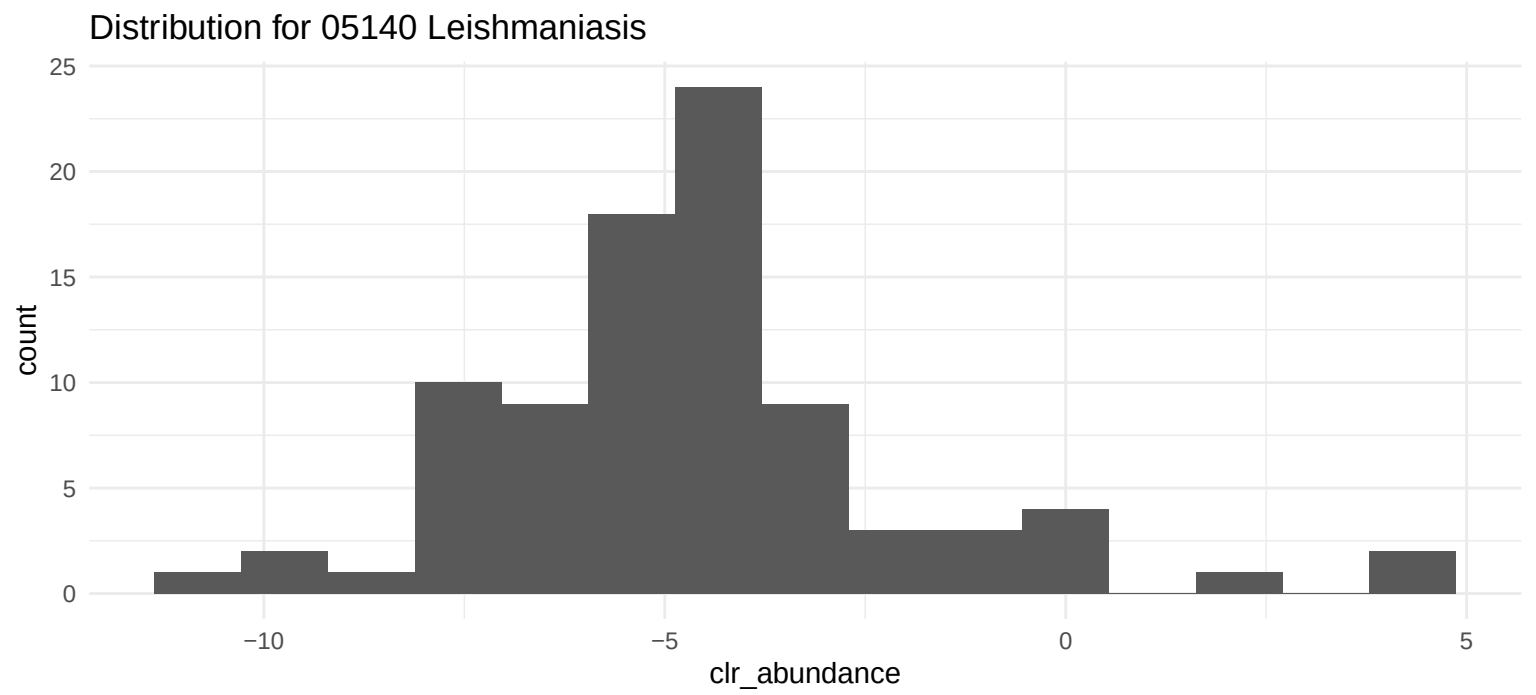
Distribution for 05132 Salmonella infection

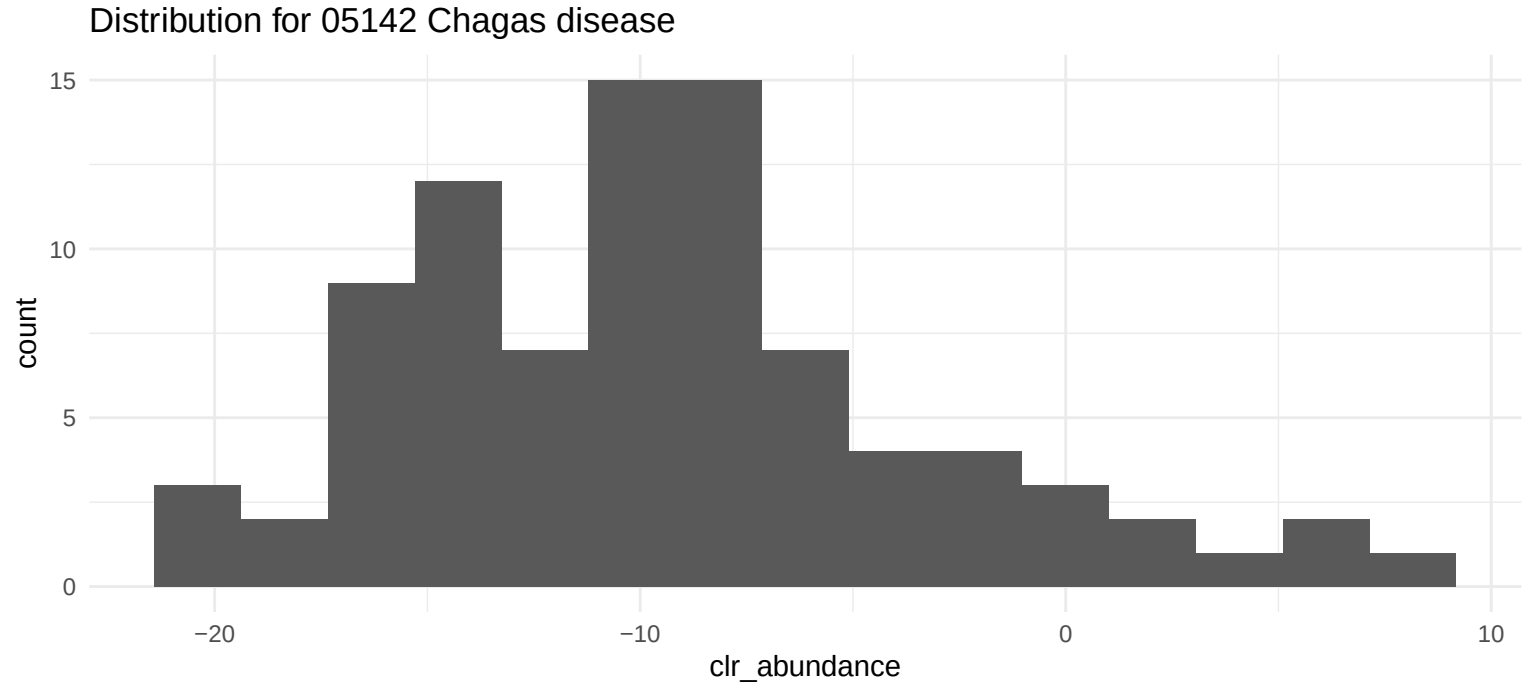


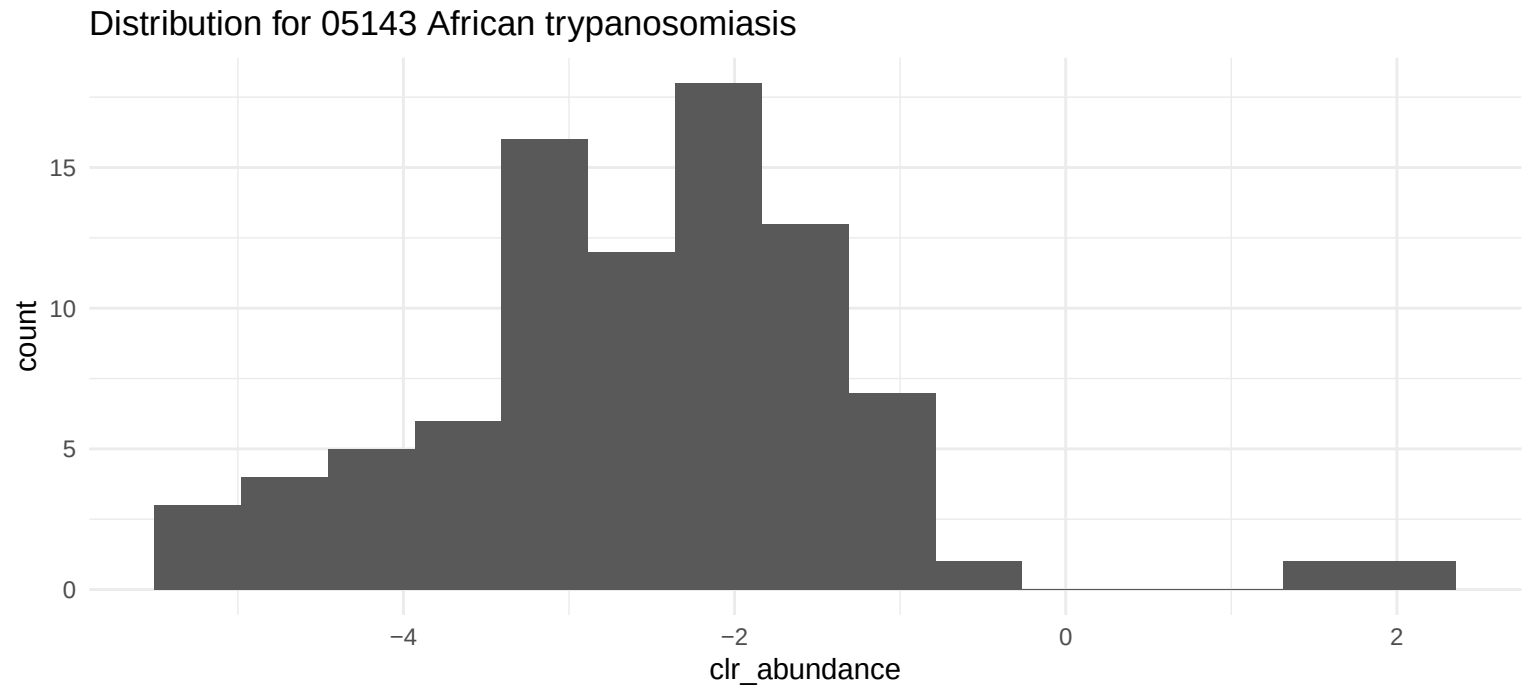

Distribution for 05133 Pertussis
count
clr_abundance
0
5
10
15
−30
−20
−10
0

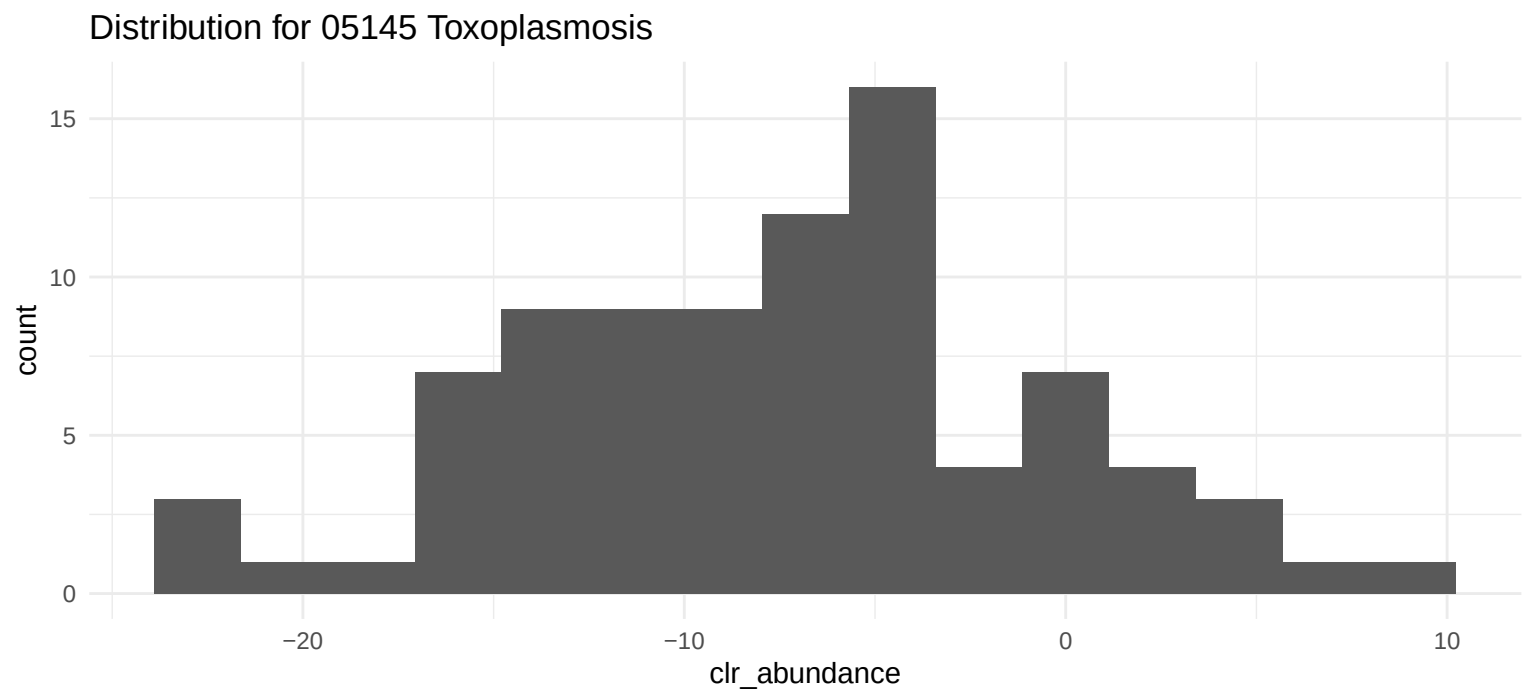

Distribution for 05134 Legionellosis
count
clr_abundance
0
5
10
15
−10
0
10
20
30

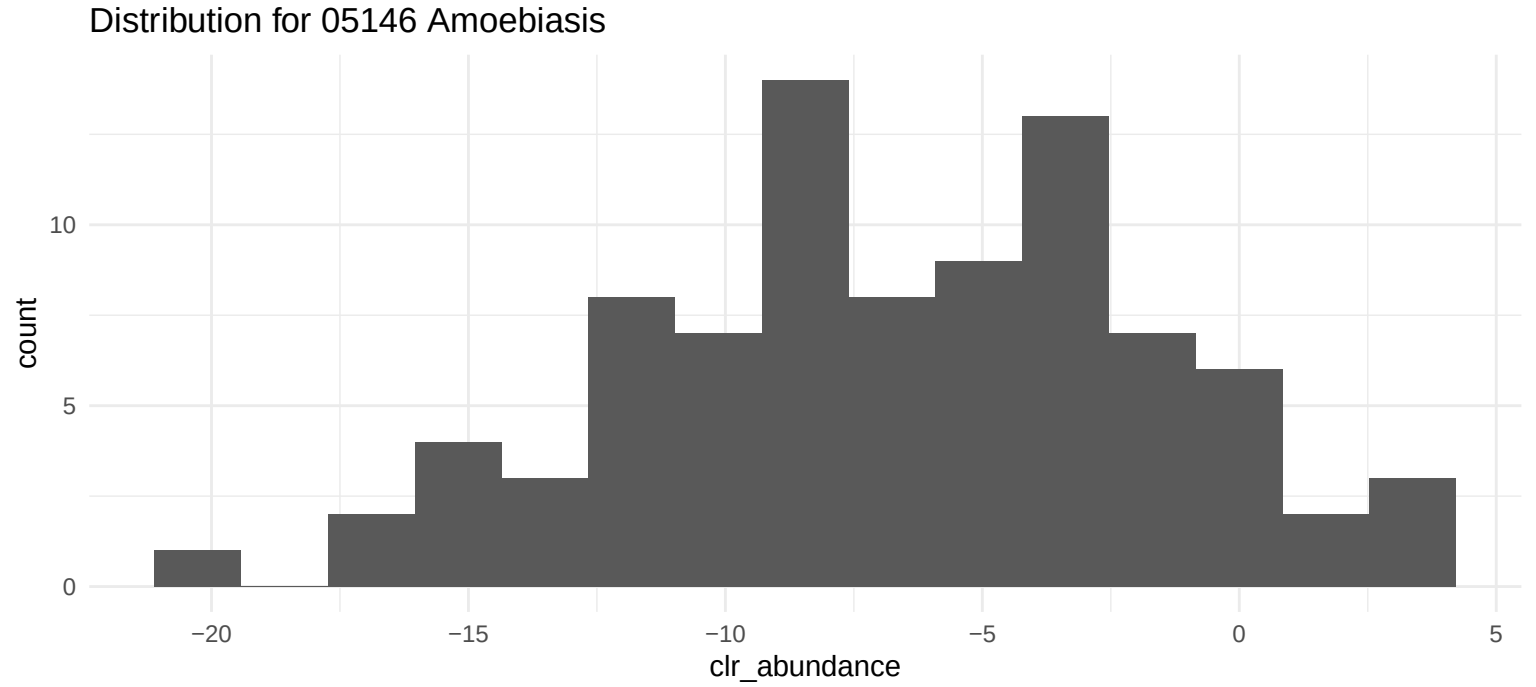

Distribution for 05135 Yersinia infection
count
clr_abundance
0
5
10
15
20
−20
0
20

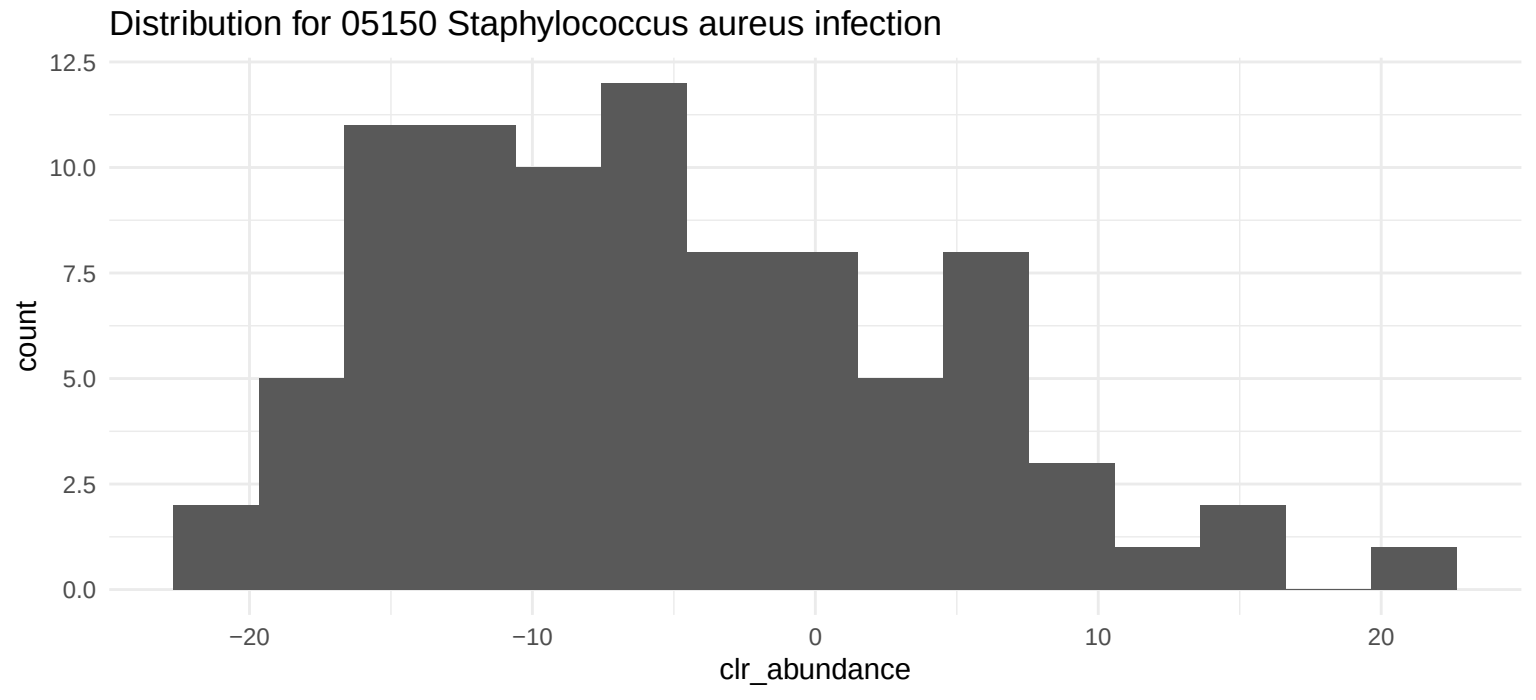

Distribution for 05140 Leishmaniasis
count
clr_abundance

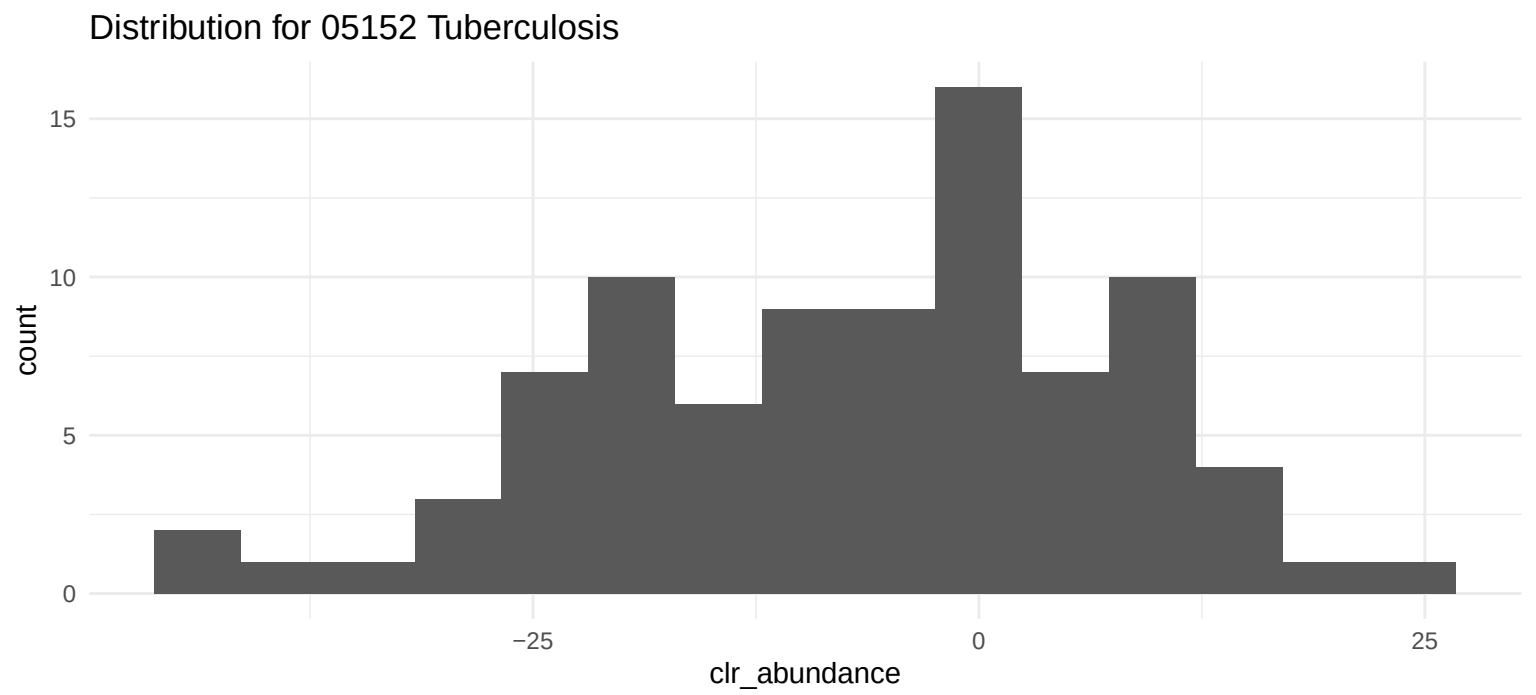

Distribution for 05142 Chagas disease
count
clr_abundance

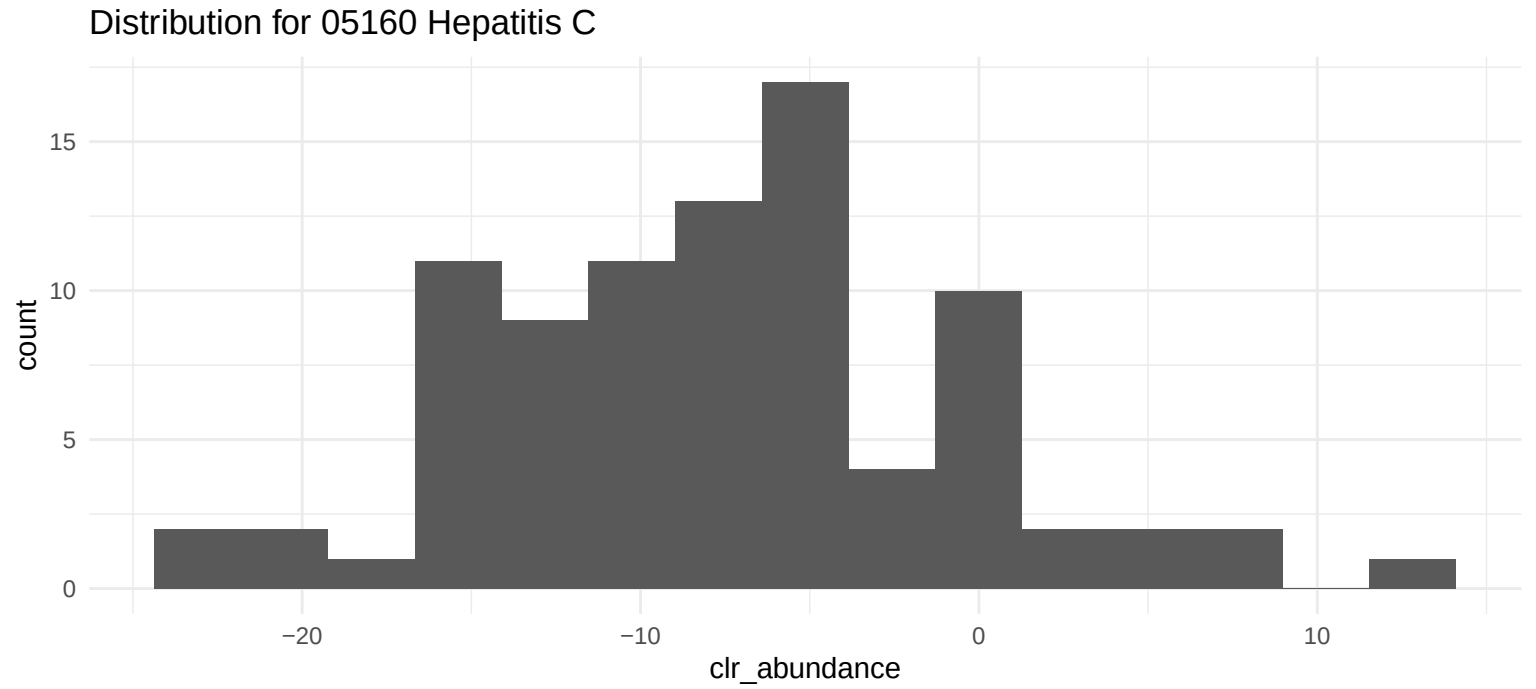

Distribution for 05143 African trypanosomiasis
count
clr_abundance

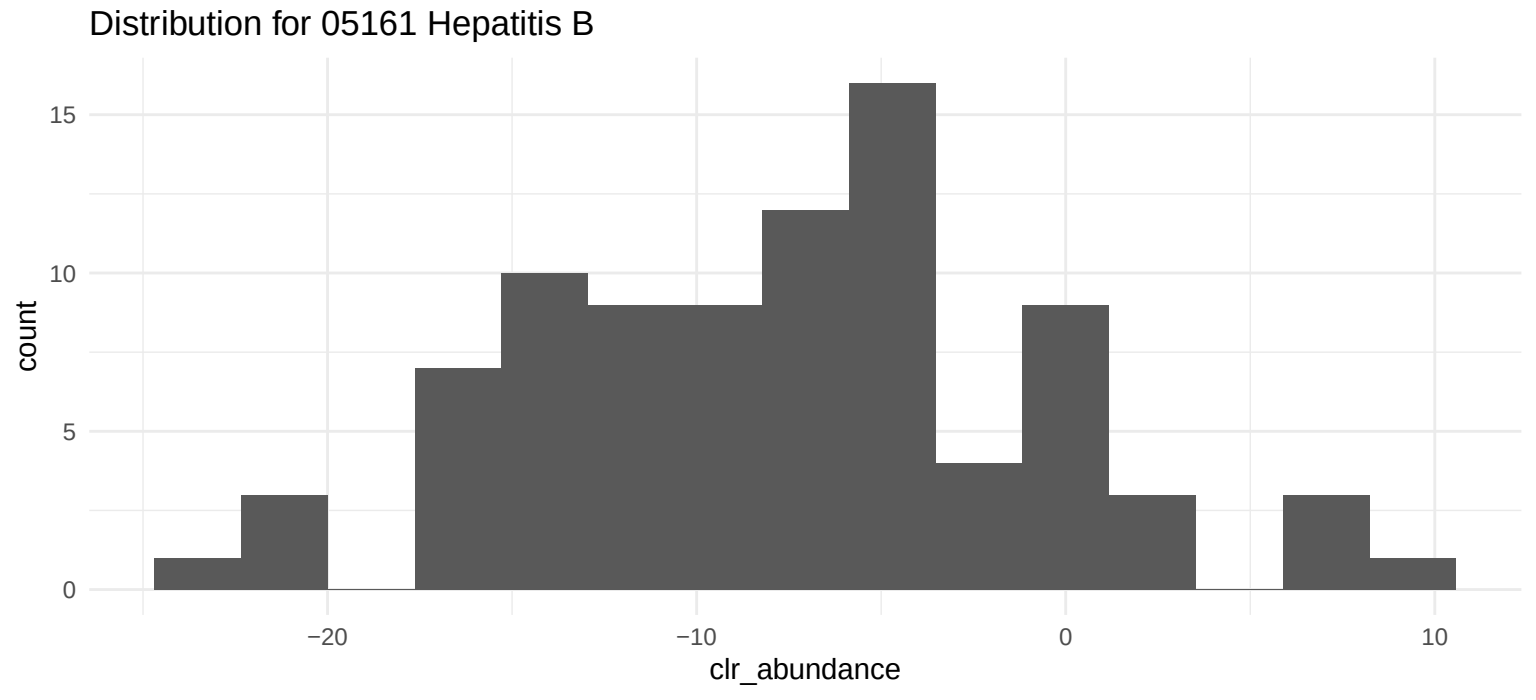

Distribution for 05145 Toxoplasmosis
count
clr_abundance
0
5
10
15
−20
−10
0
10


Distribution for 05146 Amoebiasis
count
clr_abundance
0
5
10
−20
−15
−10
−5
0
5

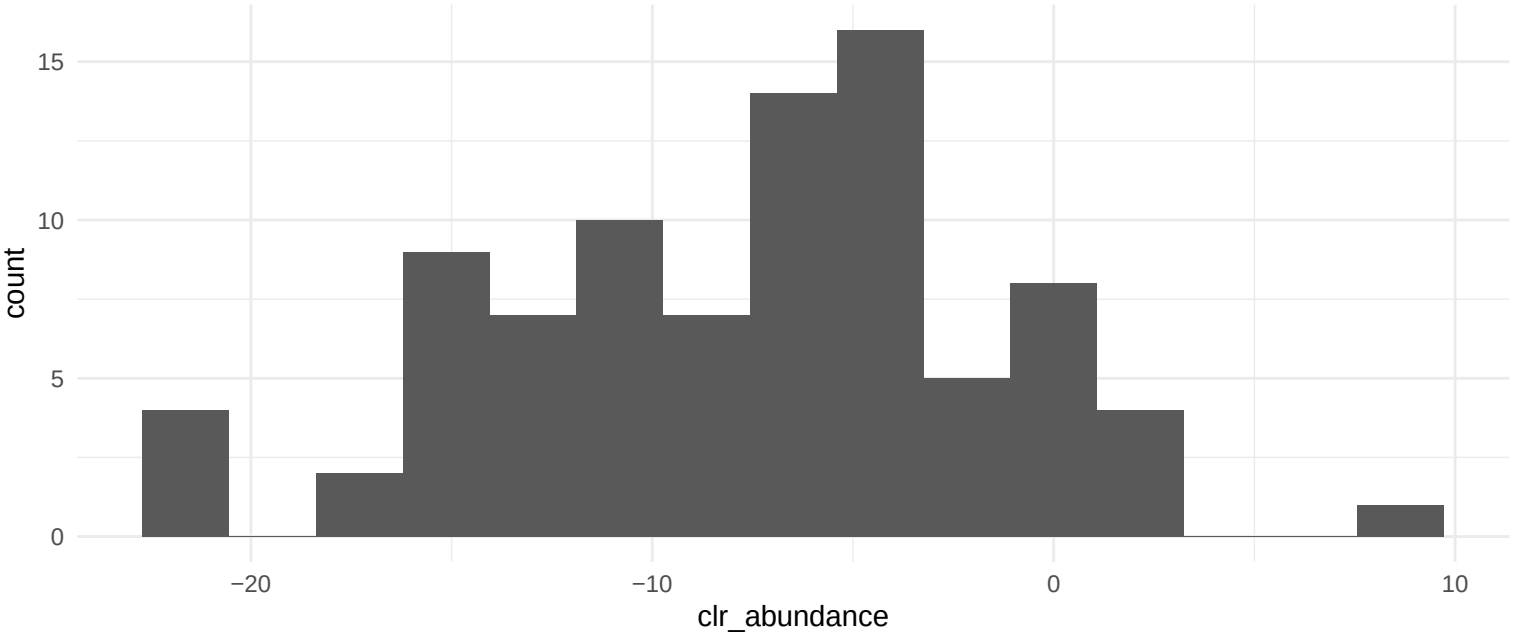

Distribution for 05150 Staphylococcus aureus infection
count
clr_abundance
0.0
2.5
5.0
7.5
10.0
12.5
−20
−10
0
10
20


Distribution for 05152 Tuberculosis
count
15
10
5
0
−25
0
25
clr_abundance

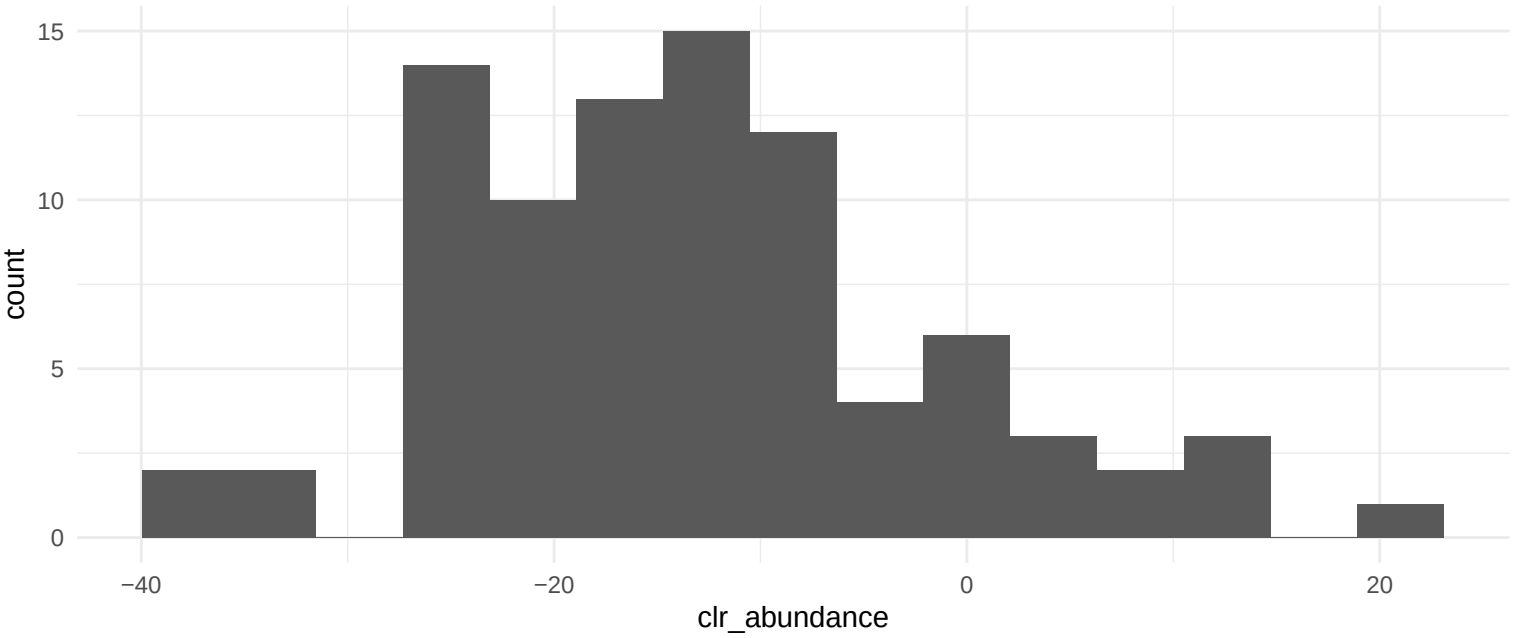

Distribution for 05160 Hepatitis C
count
15
10
5
0
−20
−10
0
10
clr_abundance


Distribution for 05161 Hepatitis B
count
15
10
5
0
−20
−10
0
10
clr_abundance

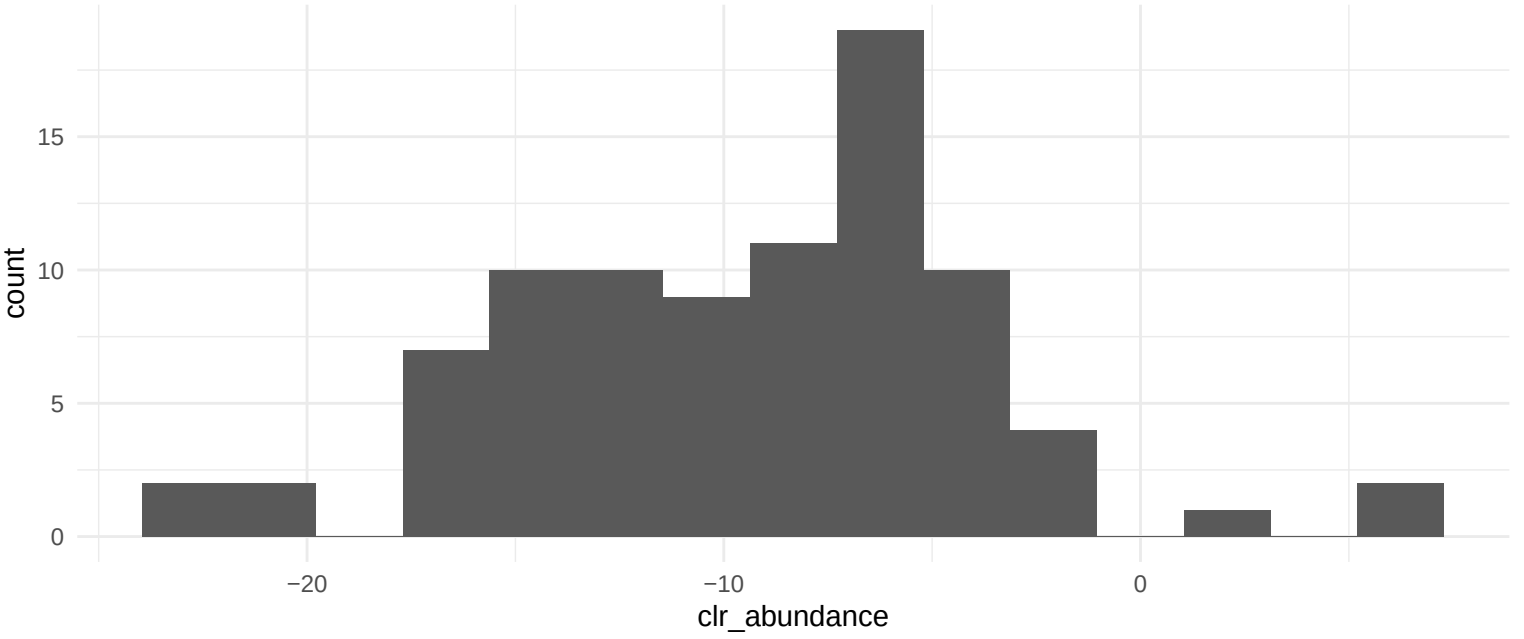
Distribution for 05162 Measles

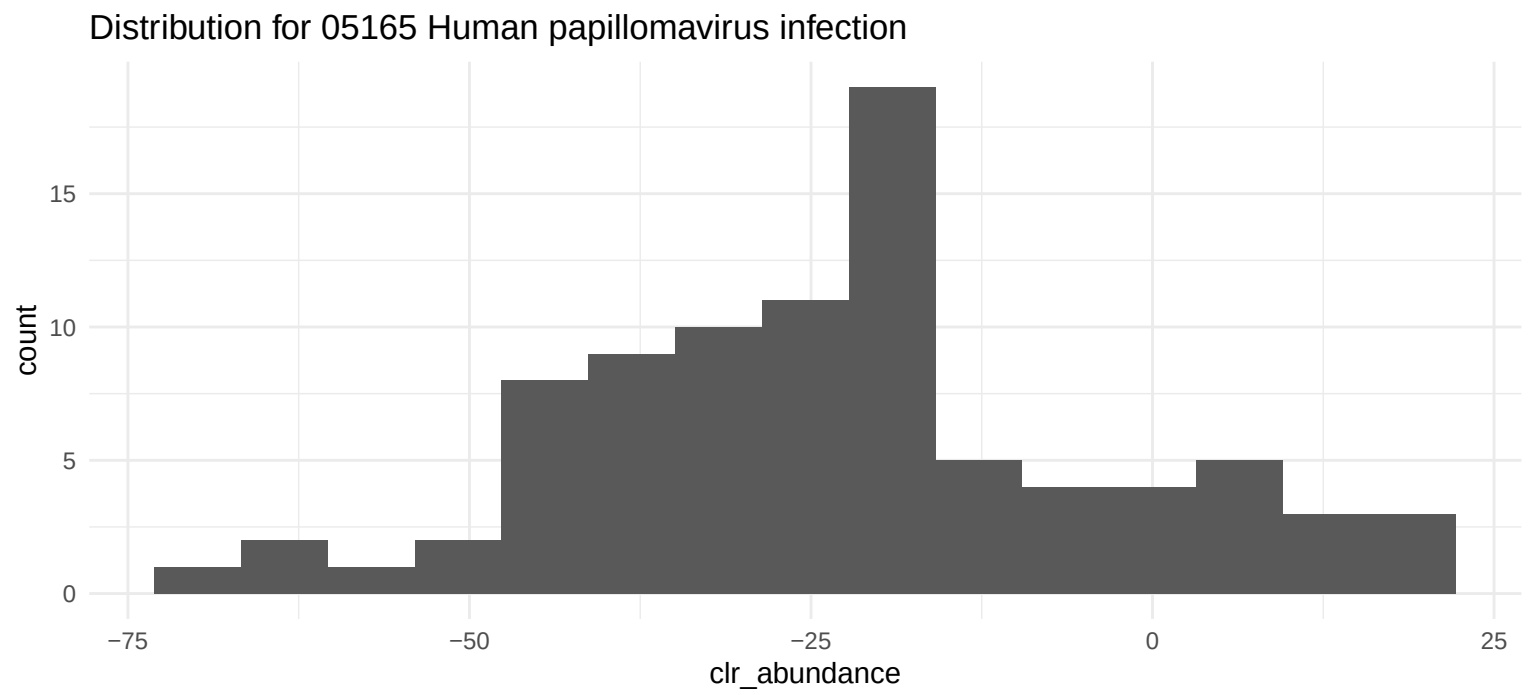


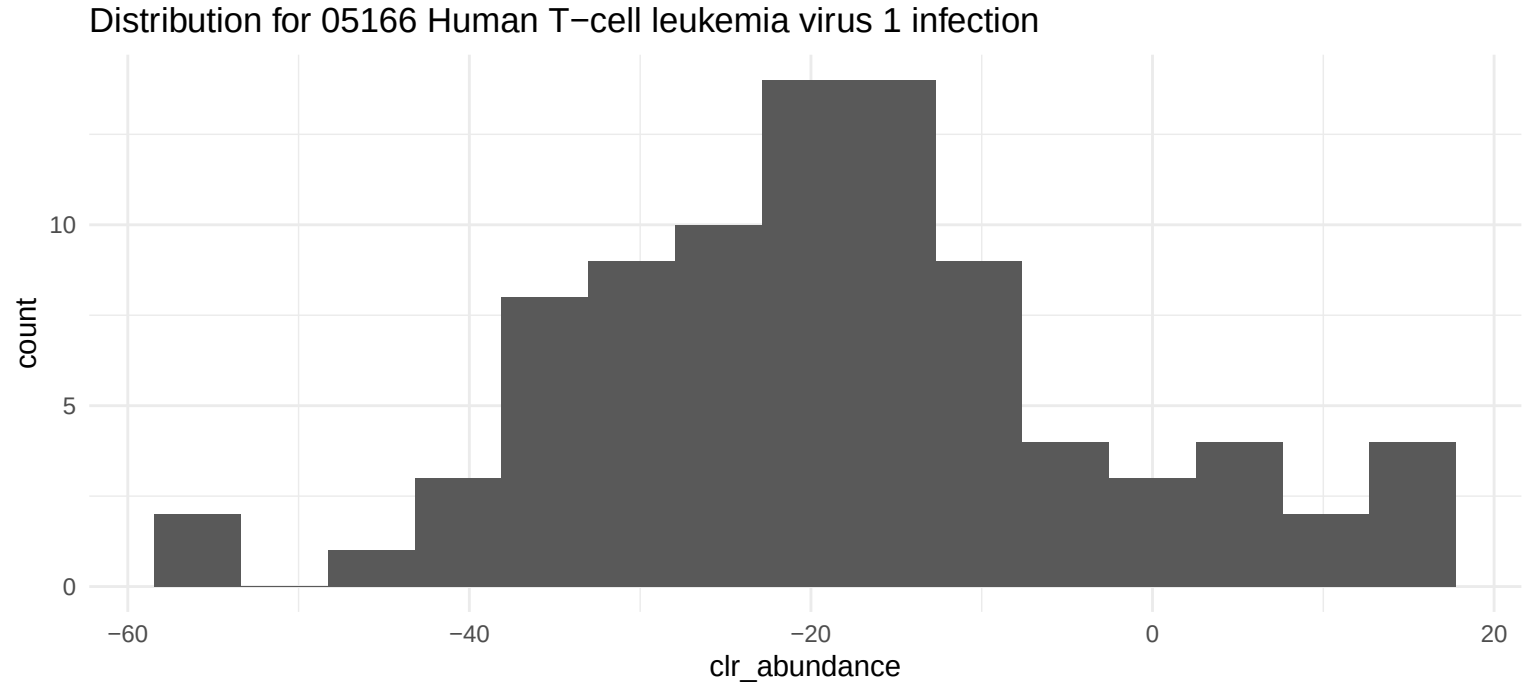
Distribution for 05163 Human cytomegalovirus infection

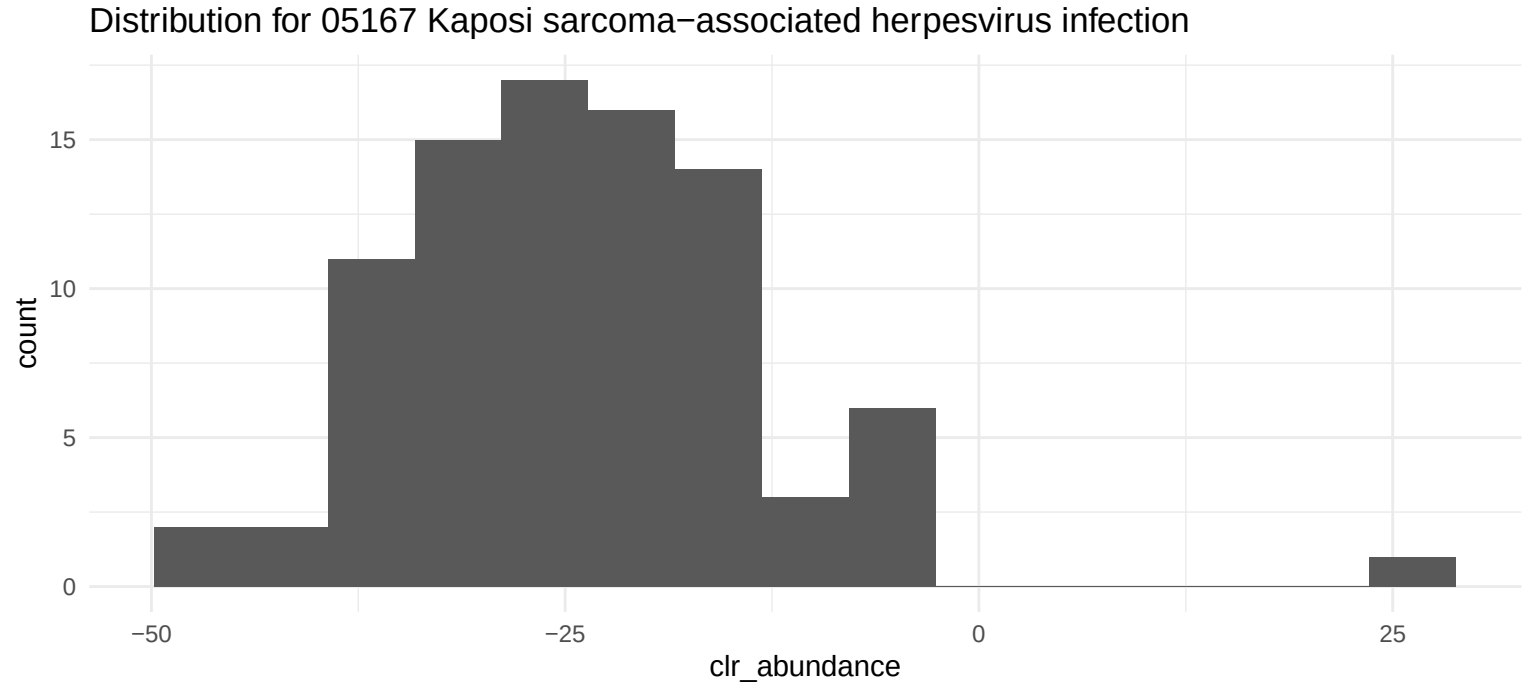


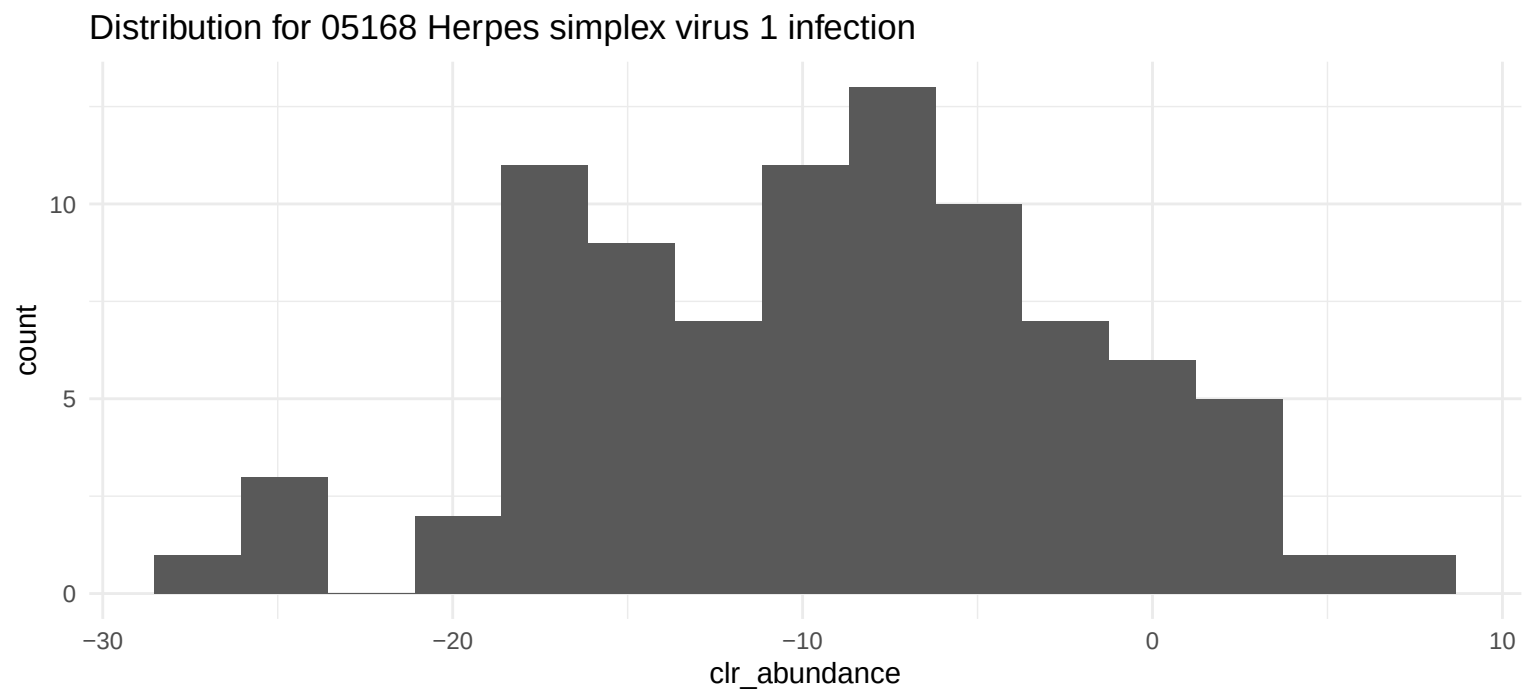
Distribution for 05164 Influenza A

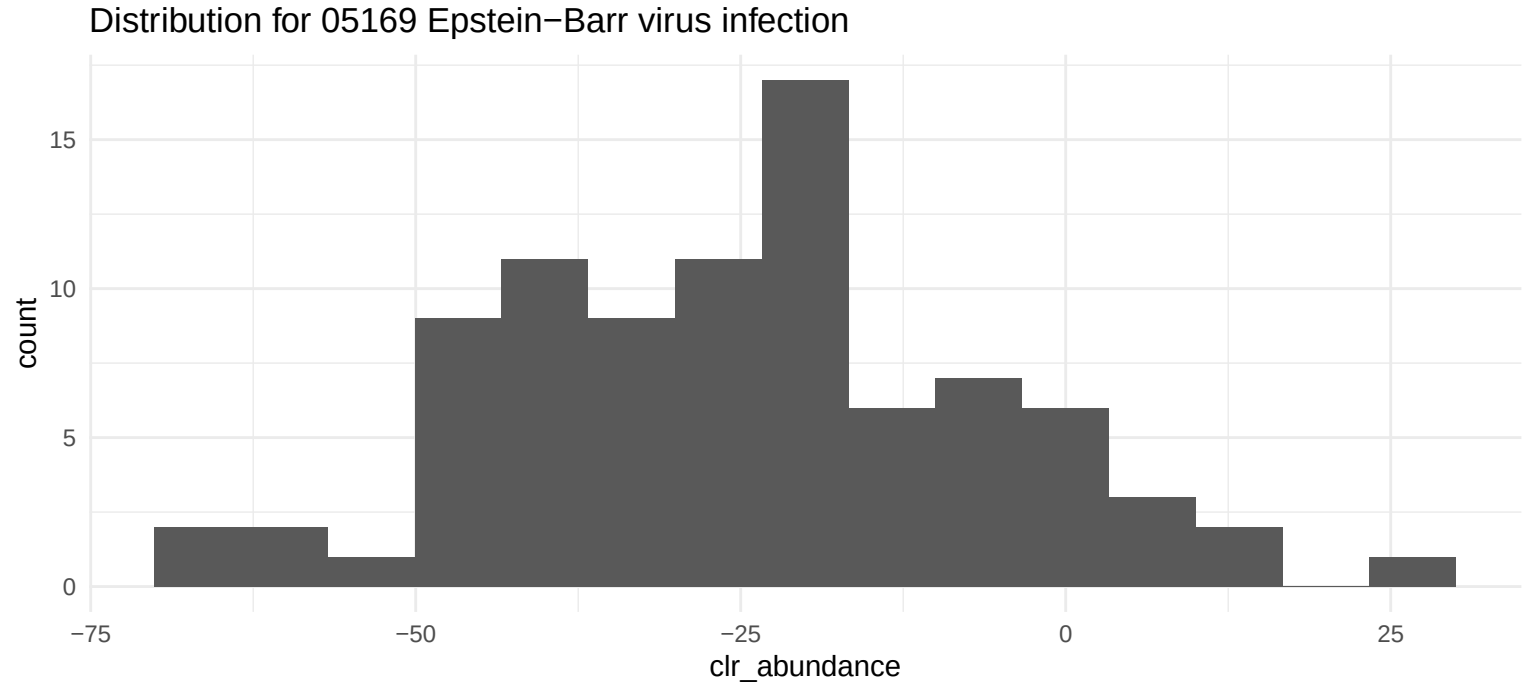


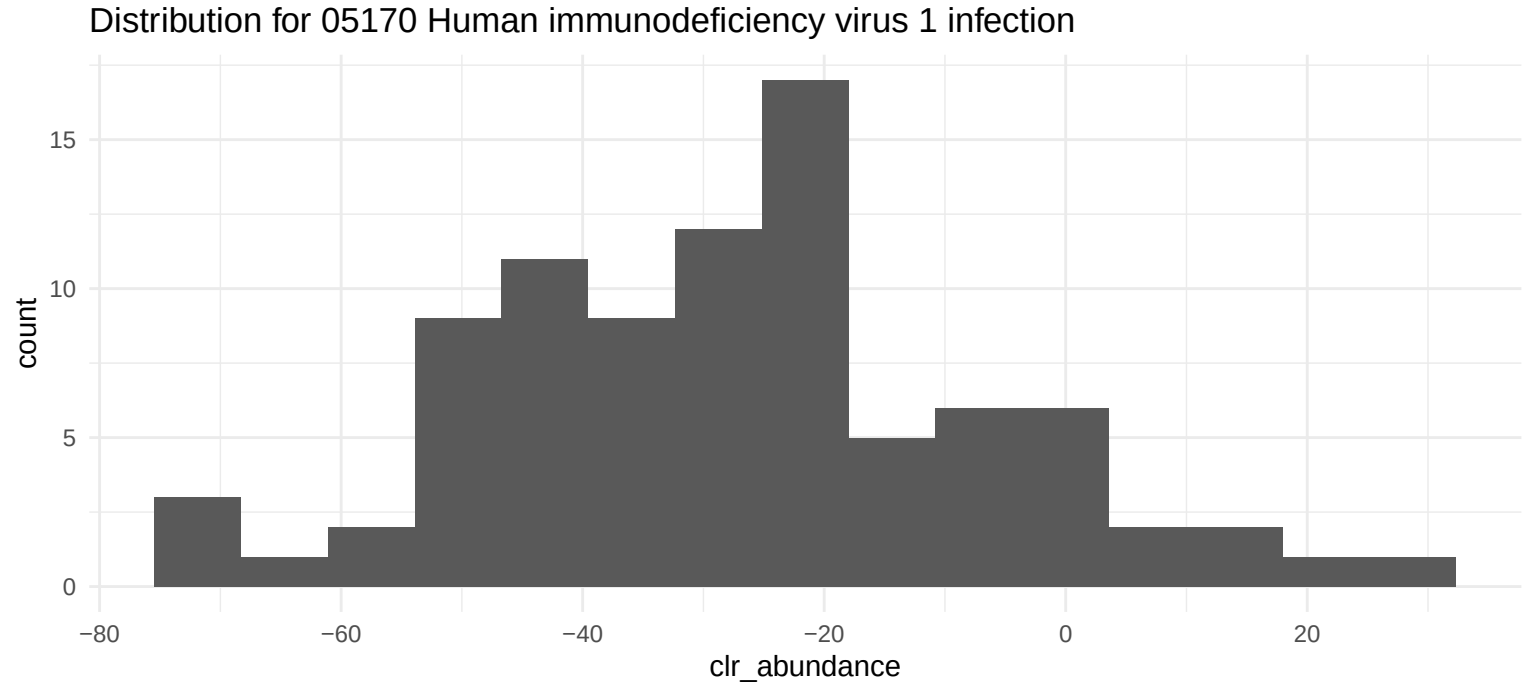

Distribution for 05165 Human papillomavirus infection
count
clr_abundance
0
5
10
15
−75
−50
−25
0
25

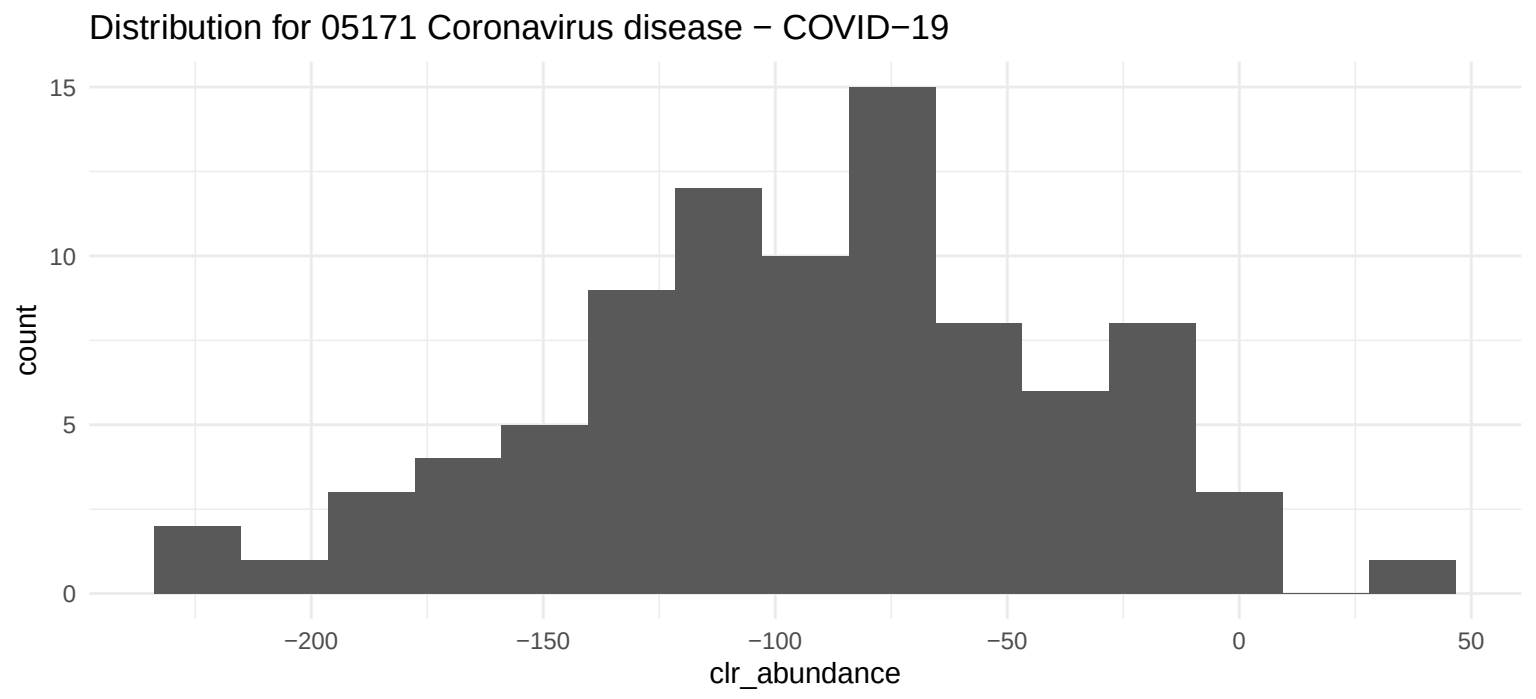

Distribution for 05166 Human T−cell leukemia virus 1 infection
count
clr_abundance
0
5
10
−60
−40
−20
0
20

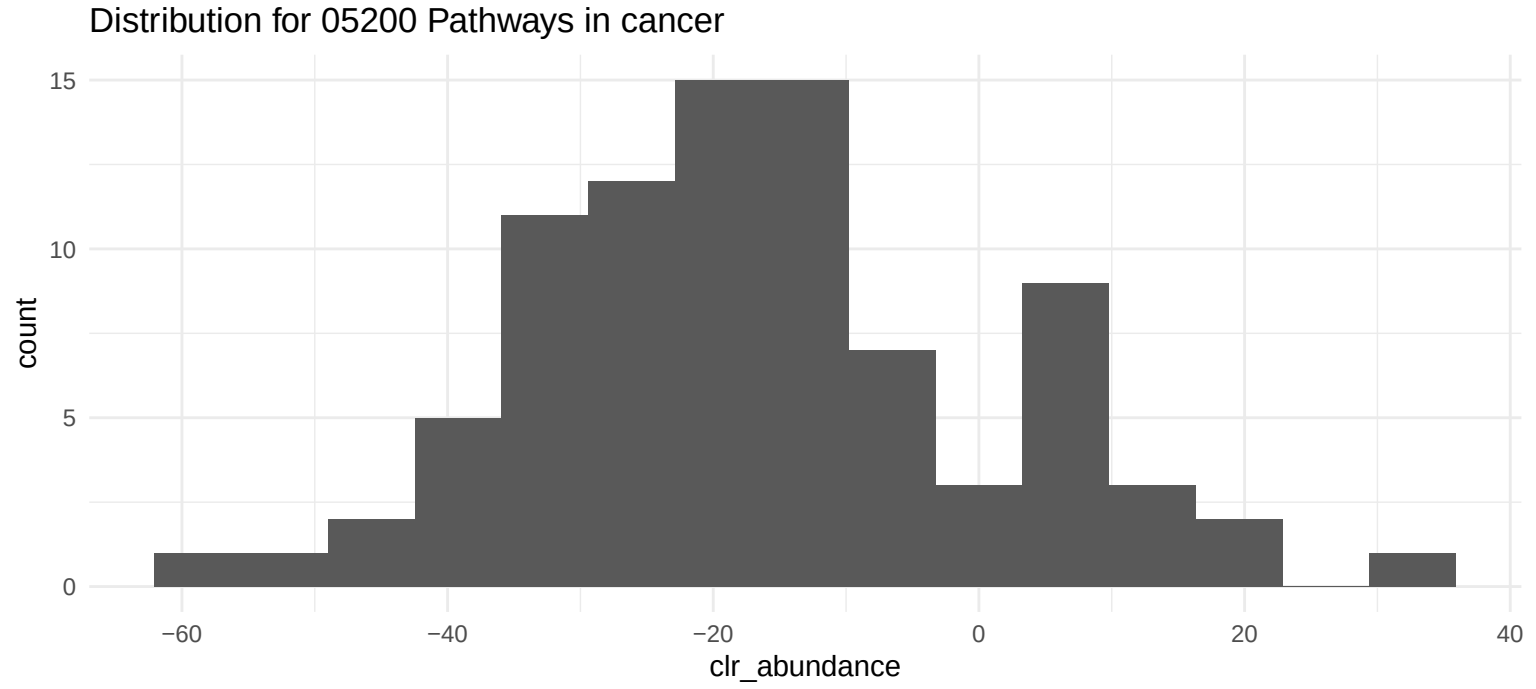

Distribution for 05167 Kaposi sarcoma−associated herpesvirus infection
count
clr_abundance
0
5
10
15
−50
−25
0
25

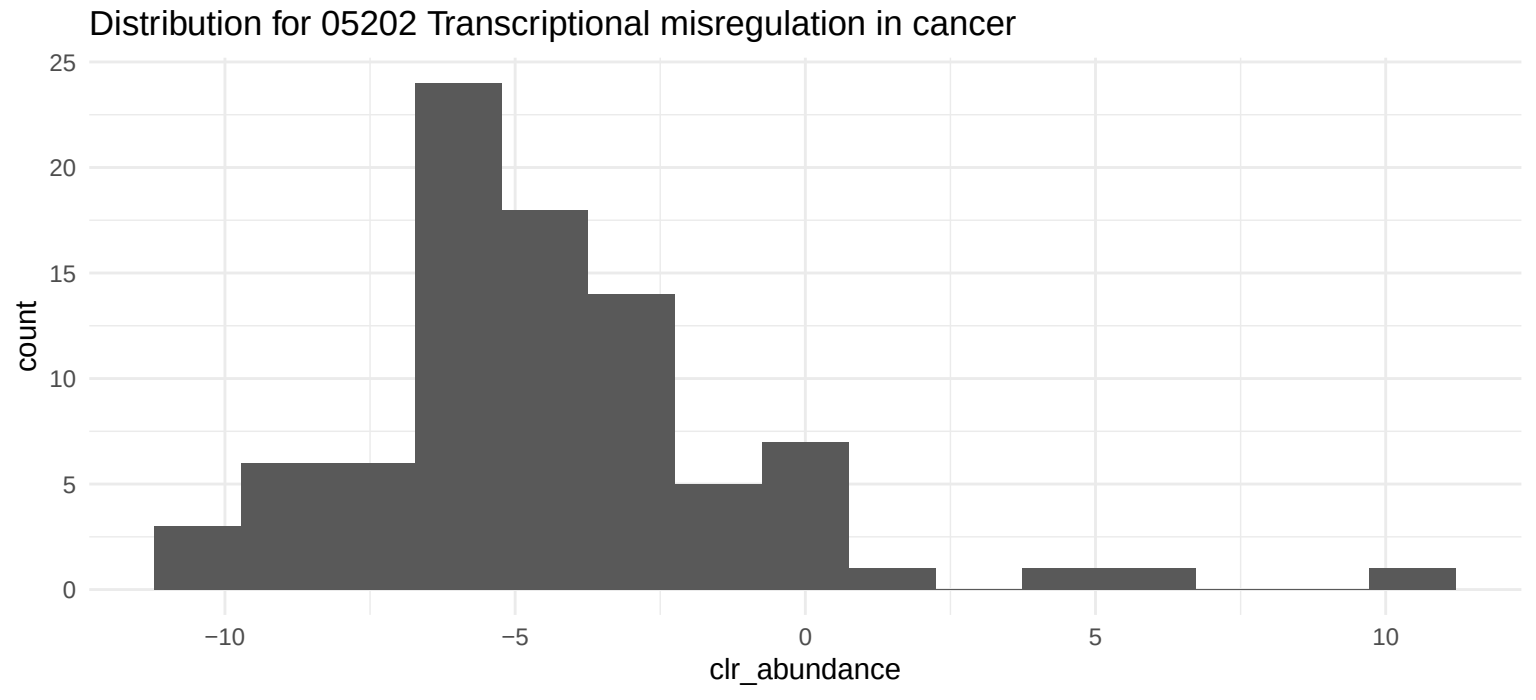

Distribution for 05168 Herpes simplex virus 1 infection
count
clr_abundance
0
5
10
−30
−20
−10
0
10

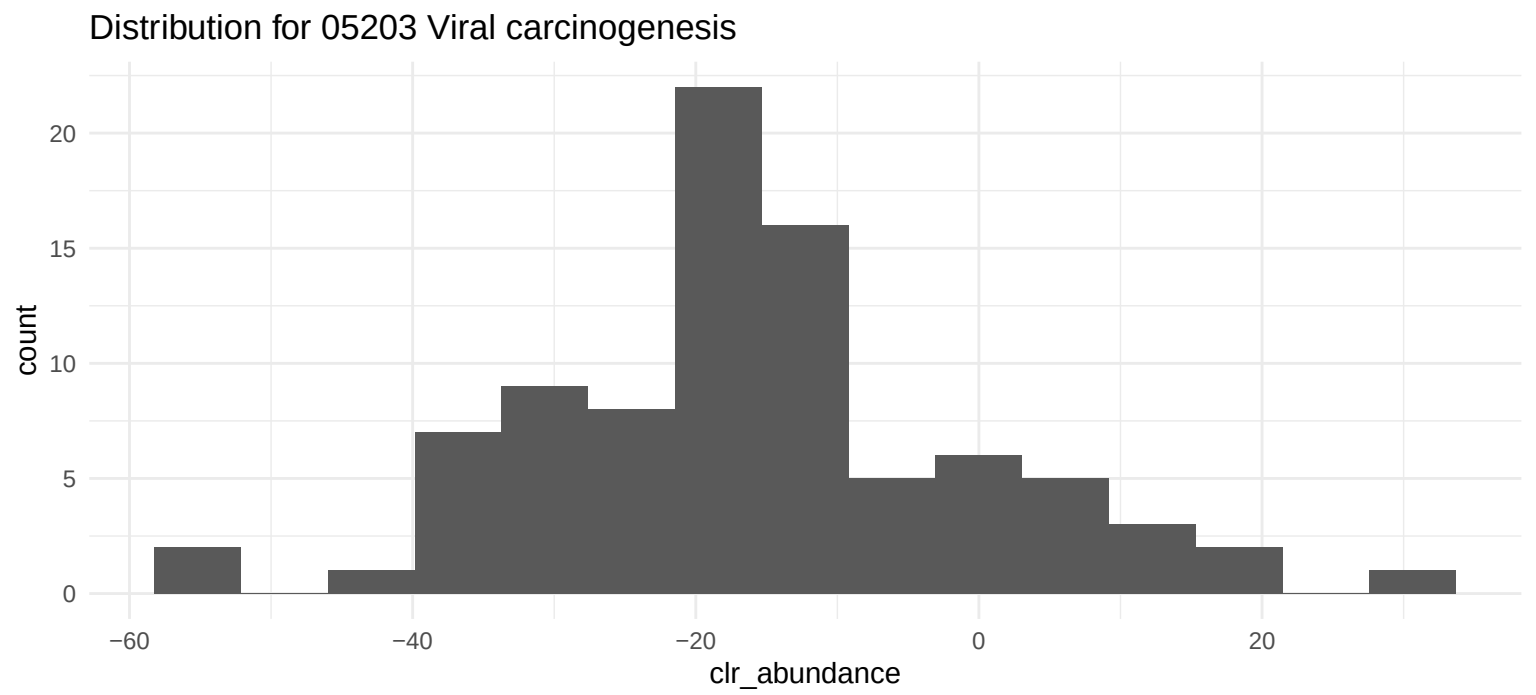

Distribution for 05169 Epstein−Barr virus infection
count
clr_abundance
0
5
10
15
−75
−50
−25
0
25

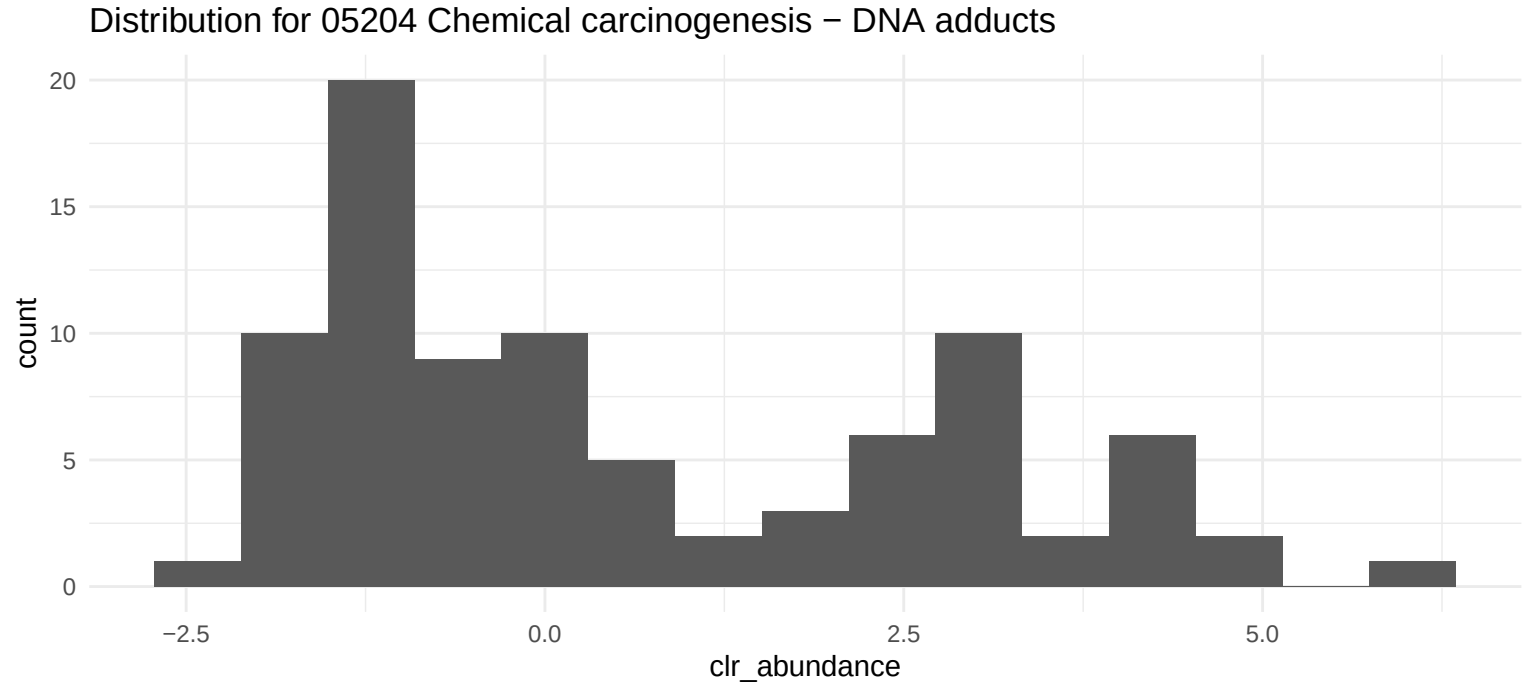

Distribution for 05170 Human immunodeficiency virus 1 infection
count
clr_abundance
0
5
10
15
−80
−60
−40
−20
0
20

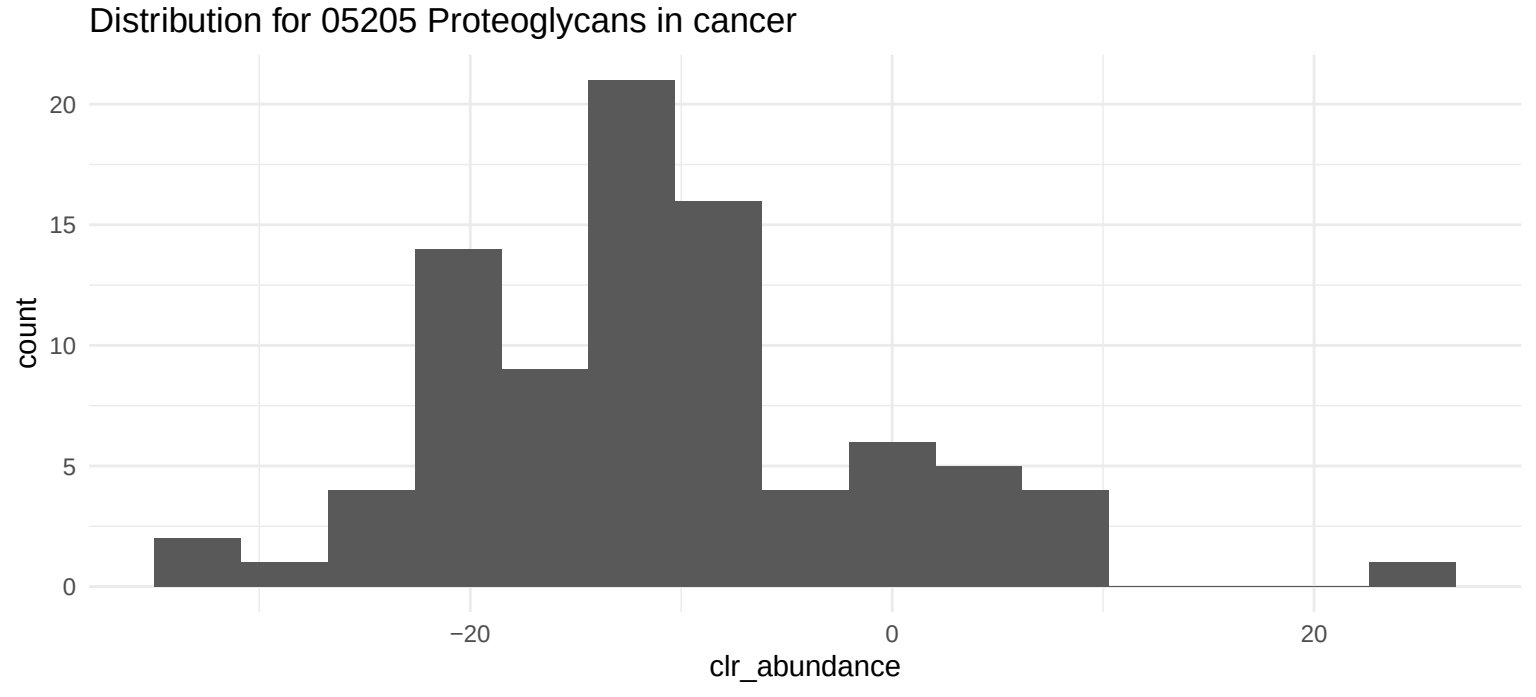

Distribution for 05171 Coronavirus disease − COVID−19
count
clr_abundance
15
10
5
0
−200
−150
−100
−50
0
50

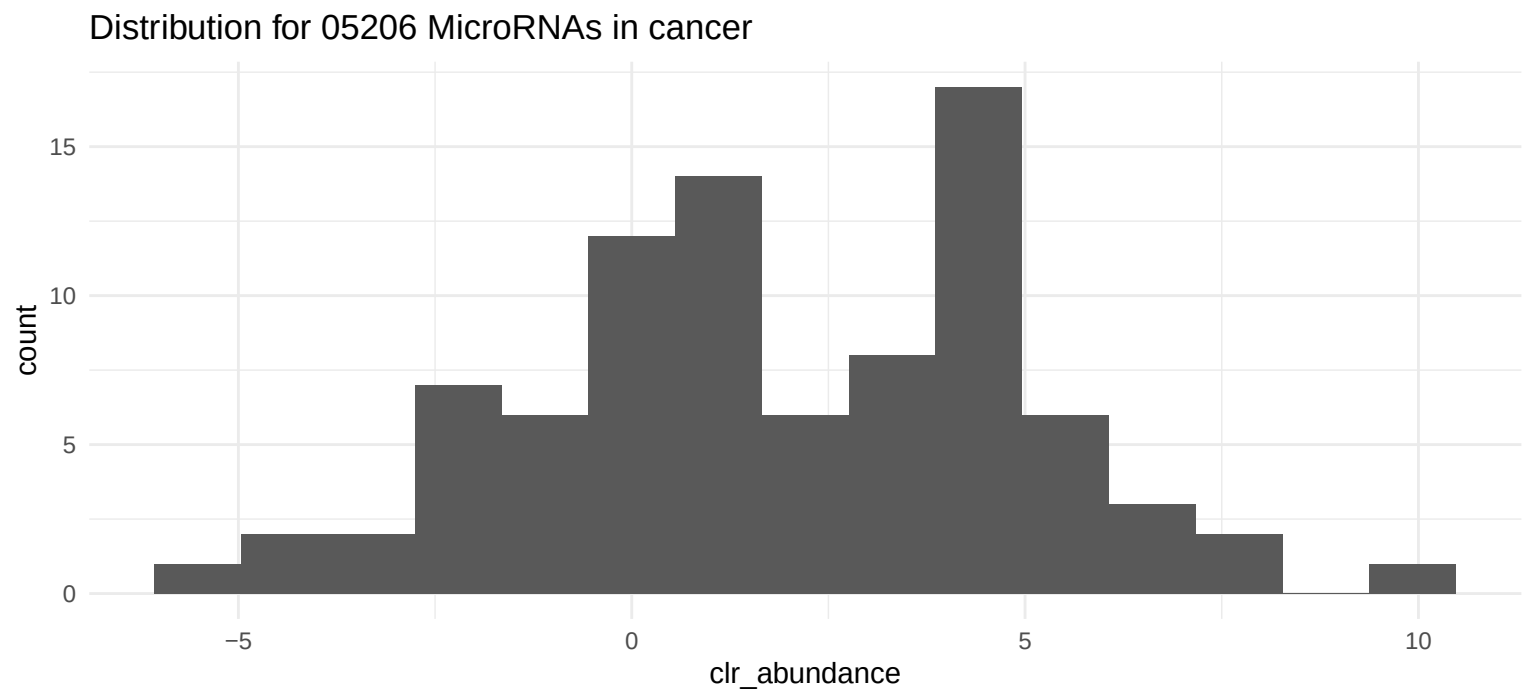

Distribution for 05200 Pathways in cancer
count
clr_abundance
15
10
5
0
−60
−40
−20
0
20
40

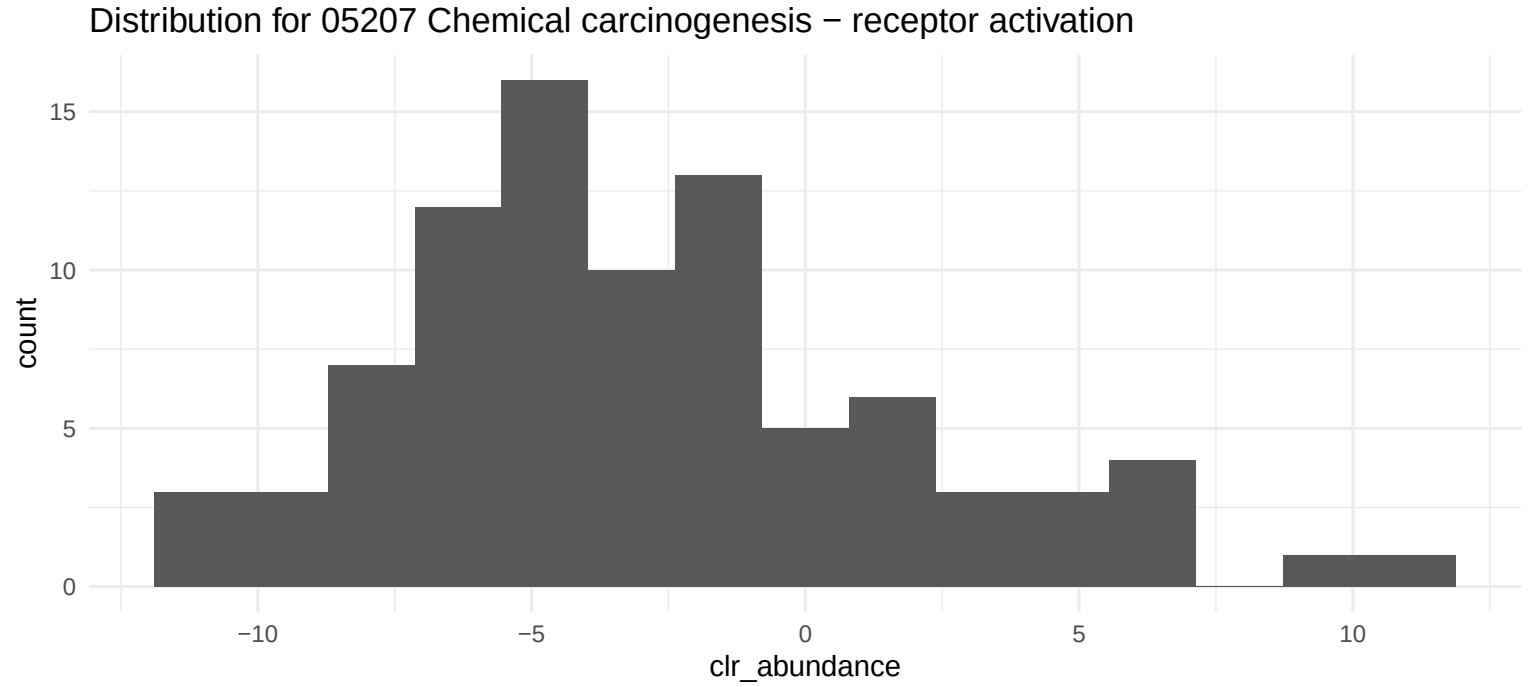

Distribution for 05202 Transcriptional misregulation in cancer
count
clr_abundance
25
20
15
10
5
0
−10
−5
0
5
10

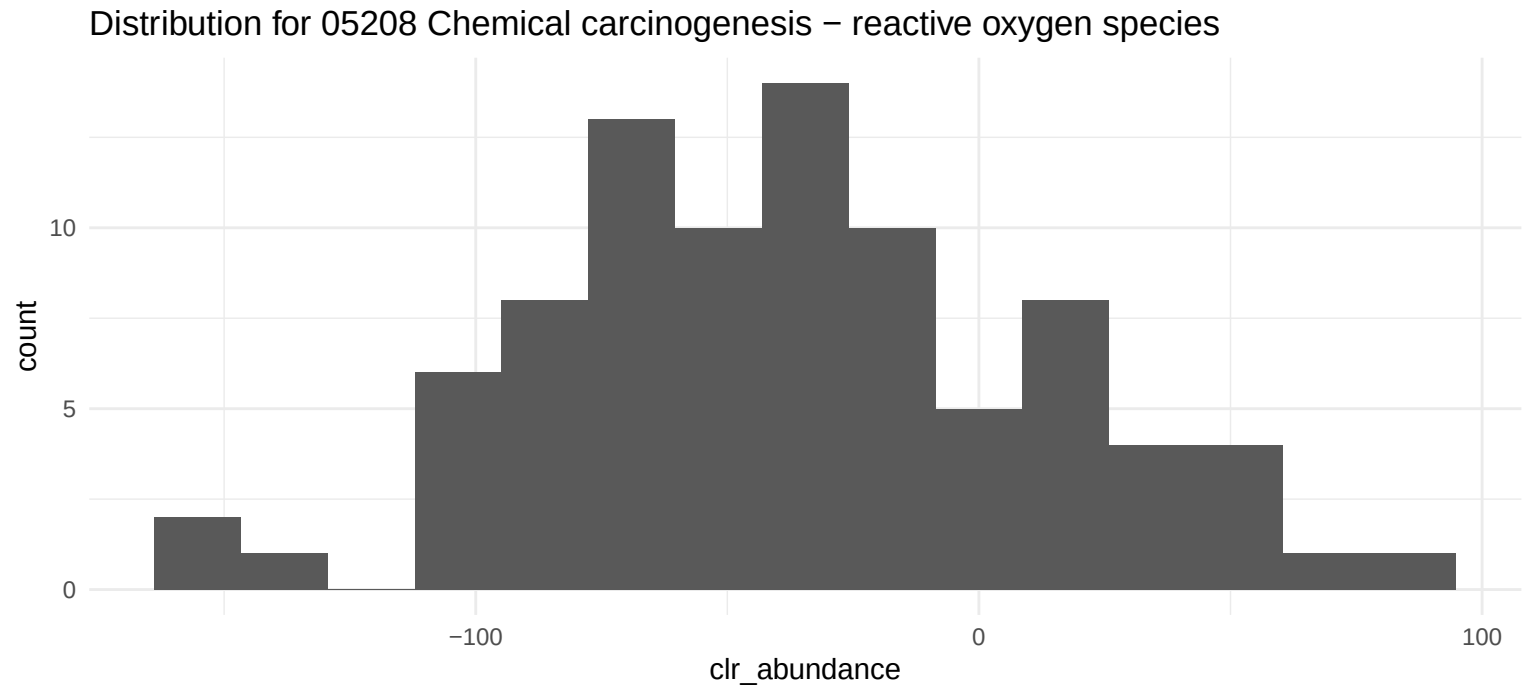

Distribution for 05203 Viral carcinogenesis
count
clr_abundance
0
5
10
15
20
−60
−40
−20
0
20

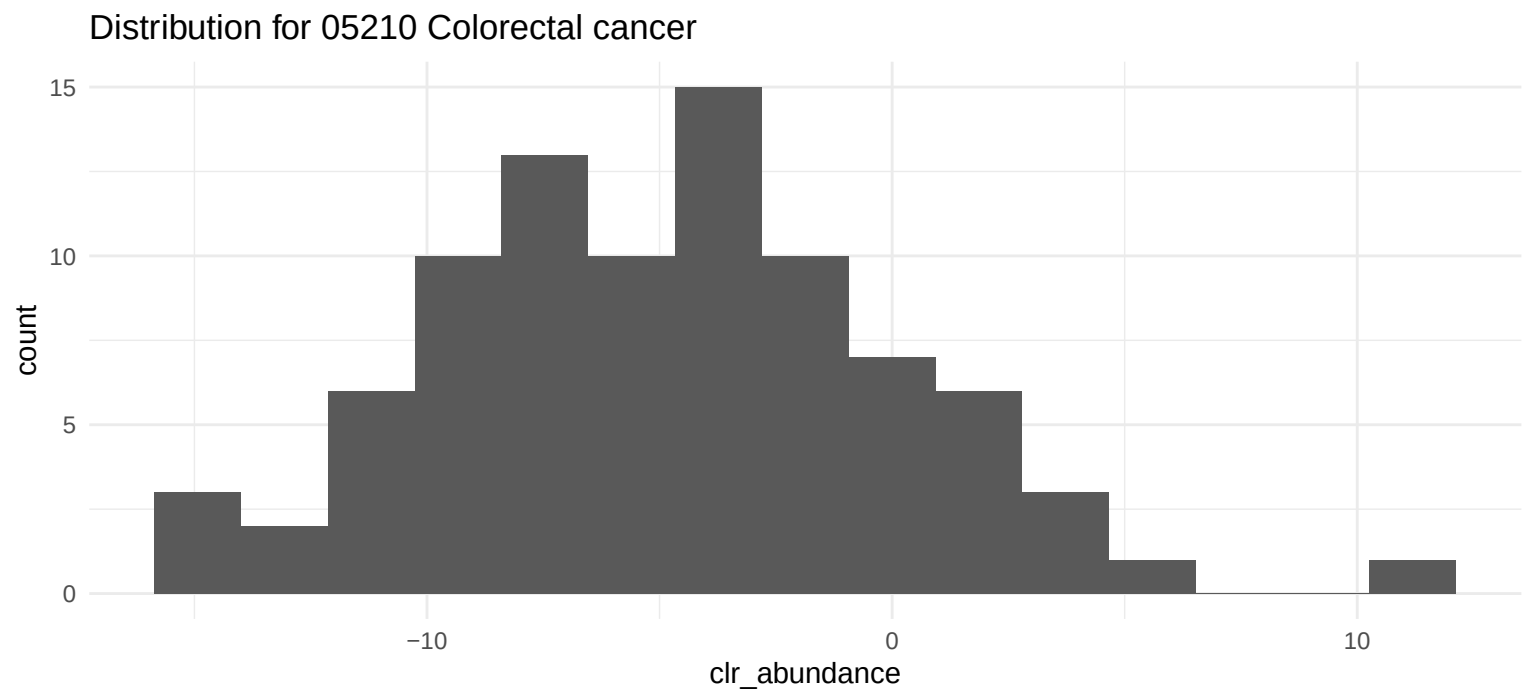

Distribution for 05204 Chemical carcinogenesis − DNA adducts
count
clr_abundance
0
5
10
15
20
−2.5
0.0
2.5
5.0

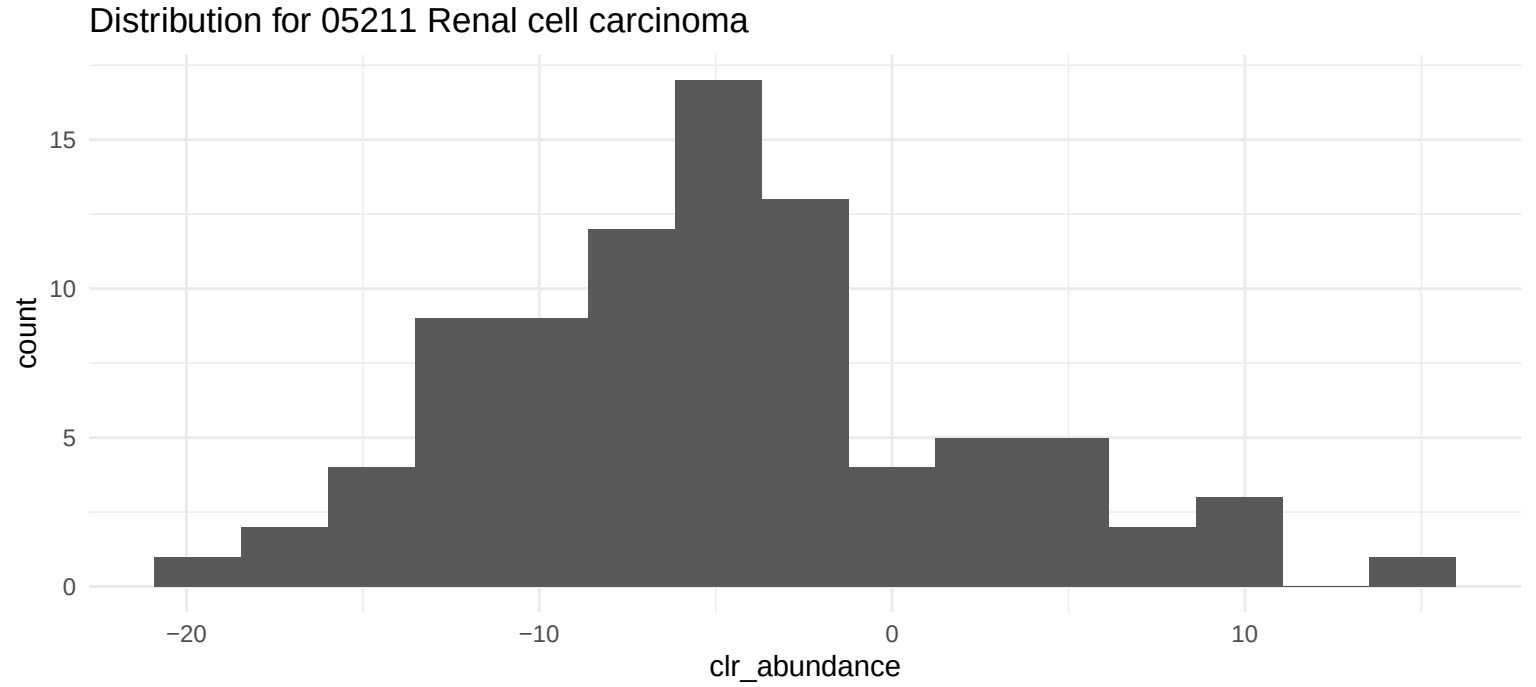

Distribution for 05205 Proteoglycans in cancer
count
clr_abundance
0
5
10
15
20
−20
0
20

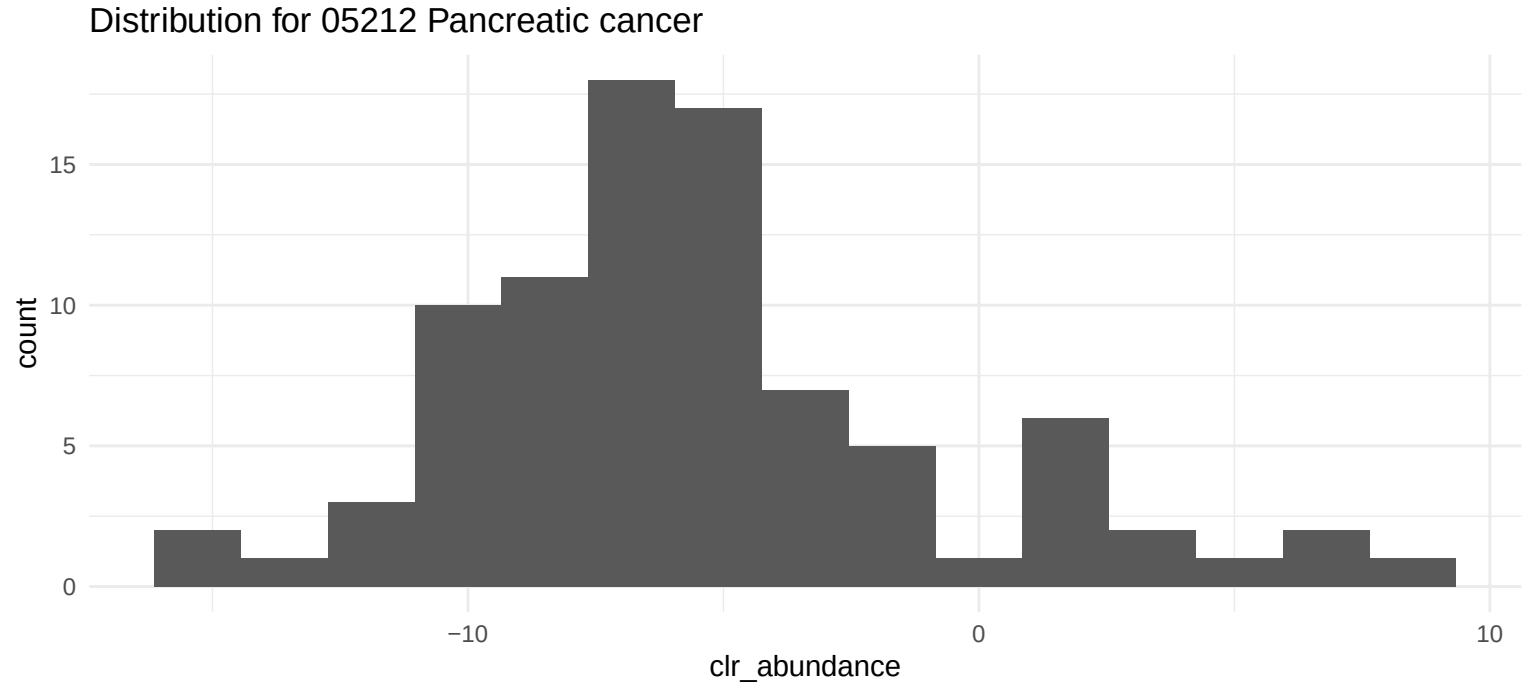

Distribution for 05206 MicroRNAs in cancer
count
15
10
5
0
−5
0
5
10
clr_abundance


Distribution for 05207 Chemical carcinogenesis − receptor activation
count
15
10
5
0
−10
−5
0
5
10
clr_abundance

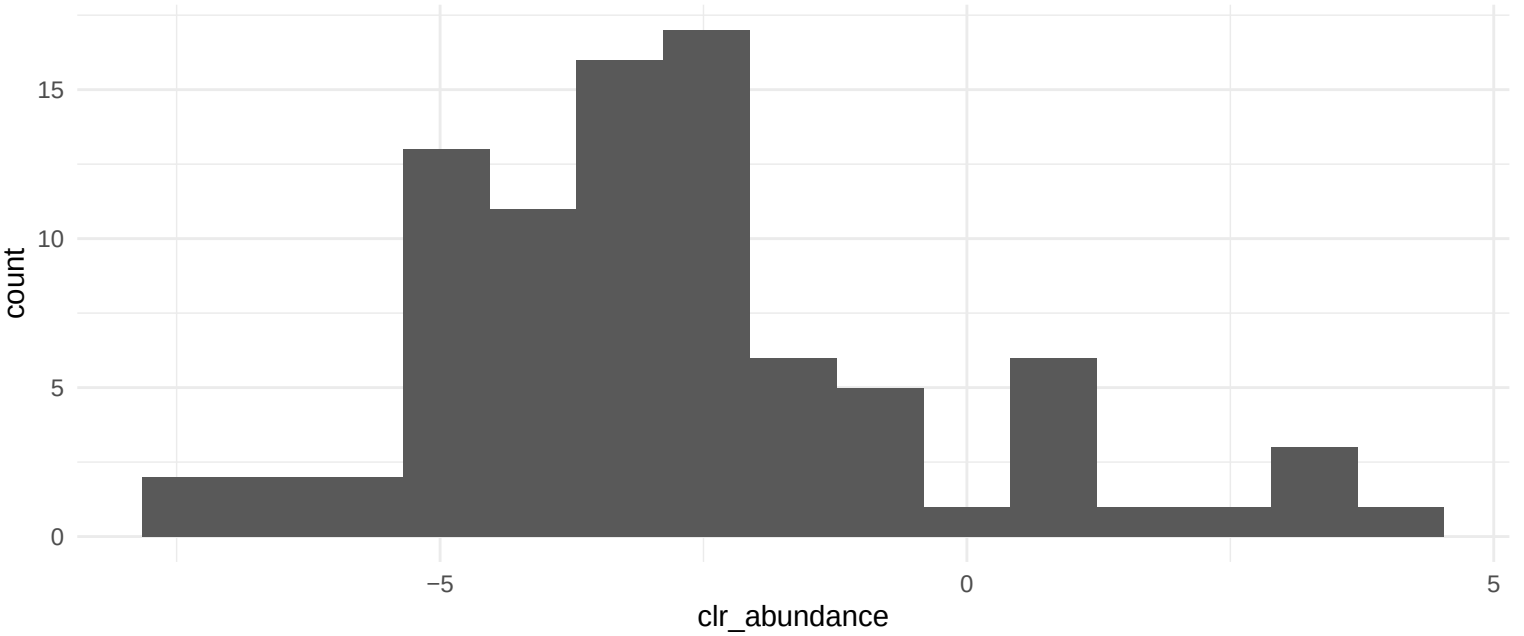

Distribution for 05208 Chemical carcinogenesis − reactive oxygen species
count
10
5
0
−100
0
100
clr_abundance


Distribution for 05210 Colorectal cancer
count
15
10
5
0
−10
0
10
clr_abundance

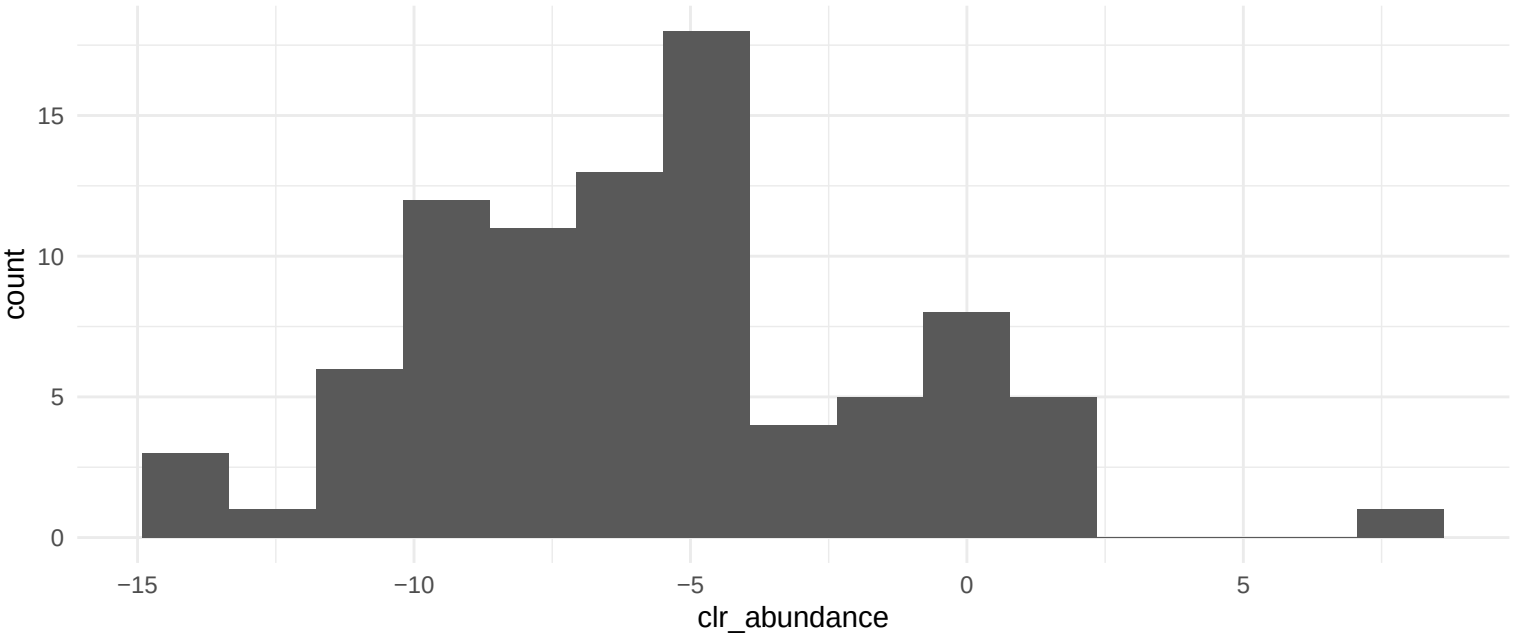

Distribution for 05211 Renal cell carcinoma
count
15
10
5
0
−20
−10
0
10
clr_abundance


Distribution for 05212 Pancreatic cancer
count
15
10
5
0
−10
0
10
clr_abundance

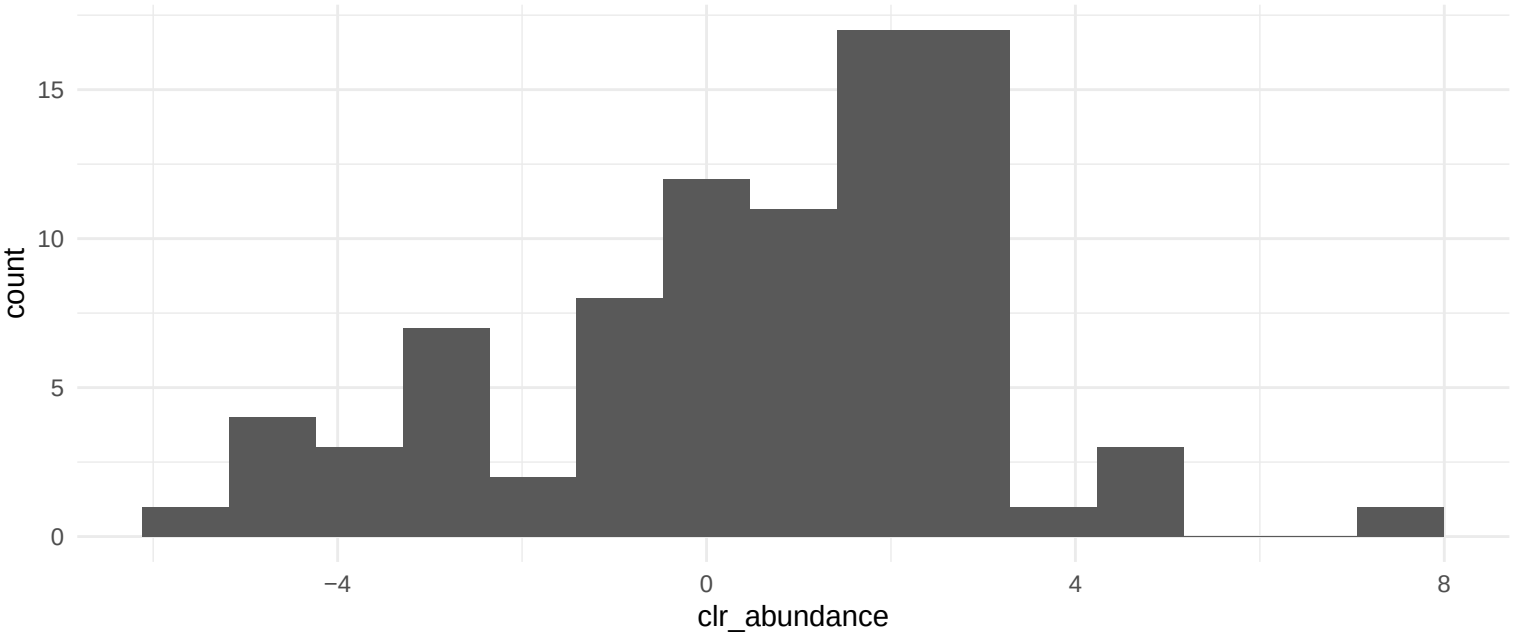
Distribution for 05213 Endometrial cancer

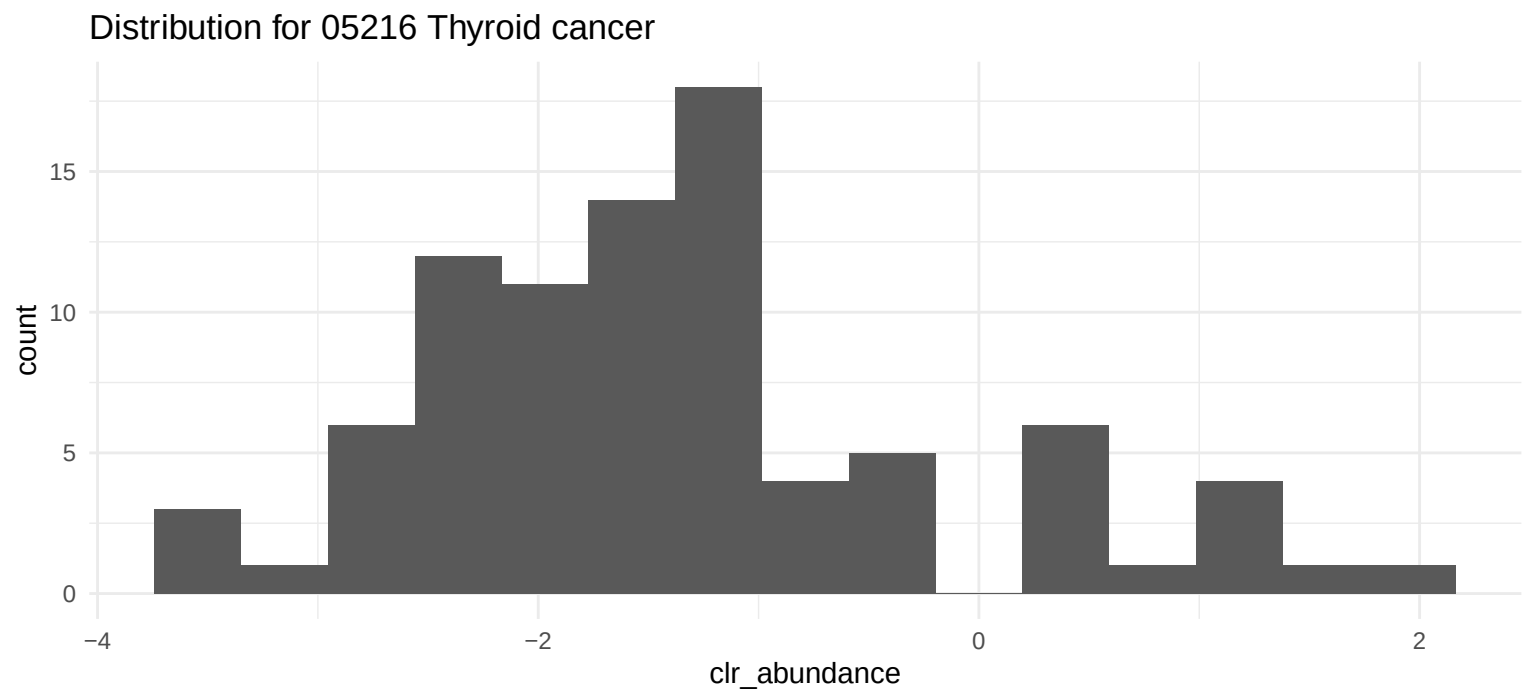


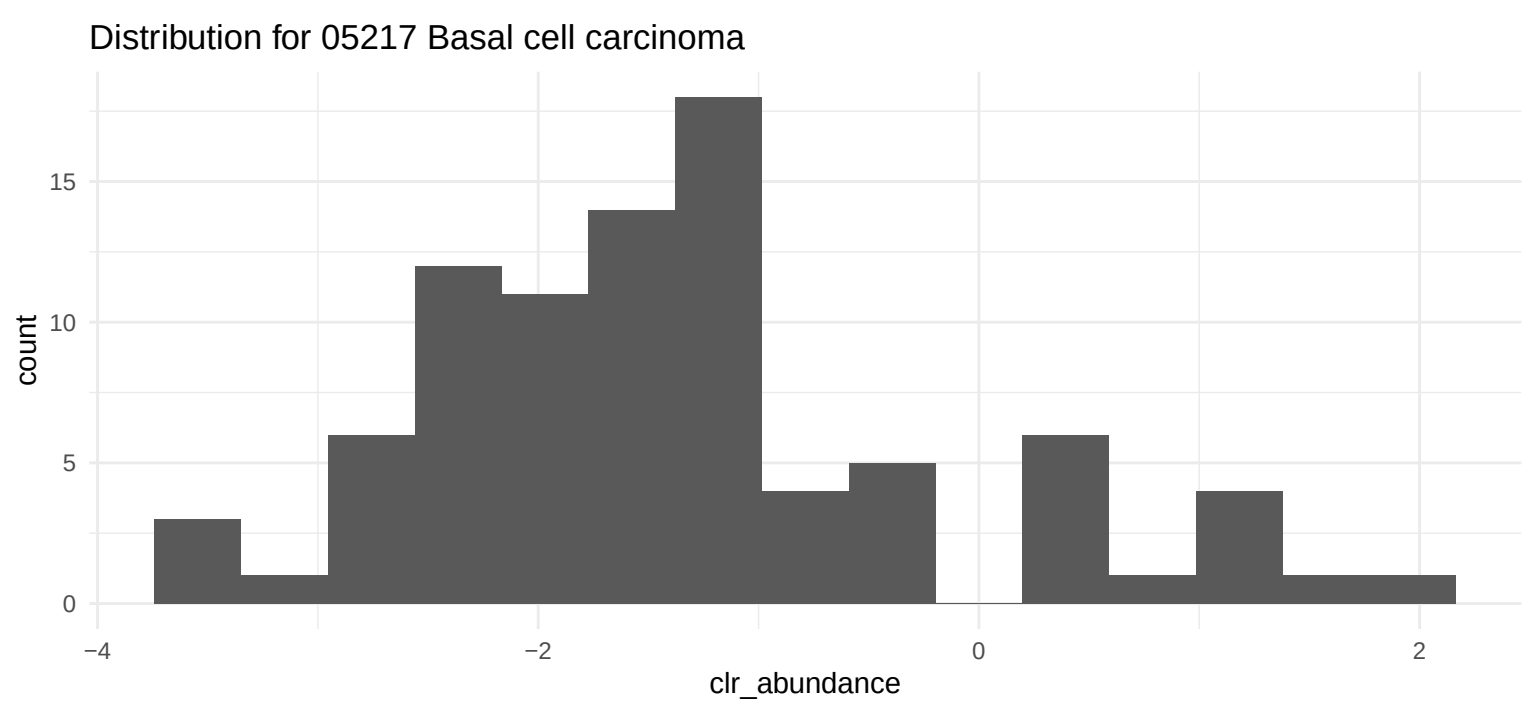
Distribution for 05214 Glioma

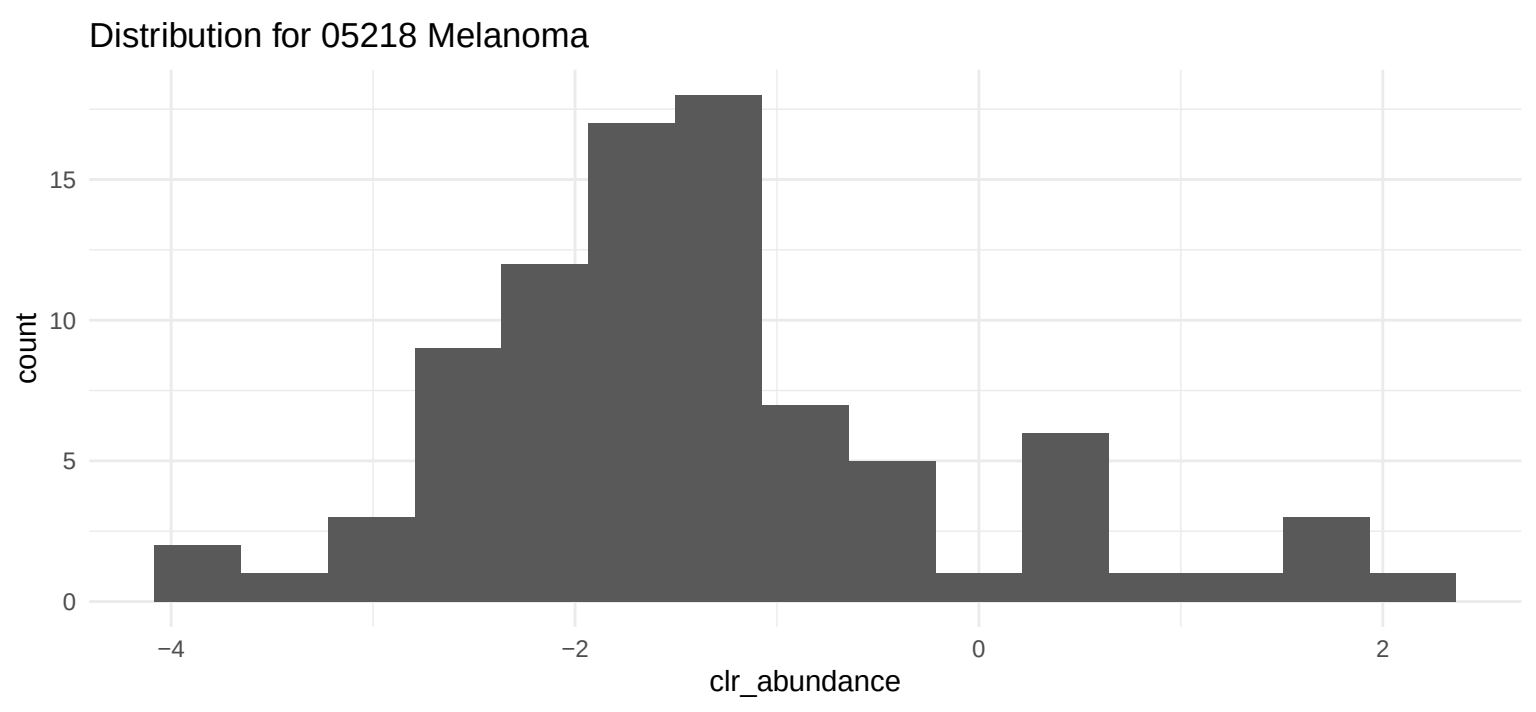


Distribution for 05215 Prostate cancer

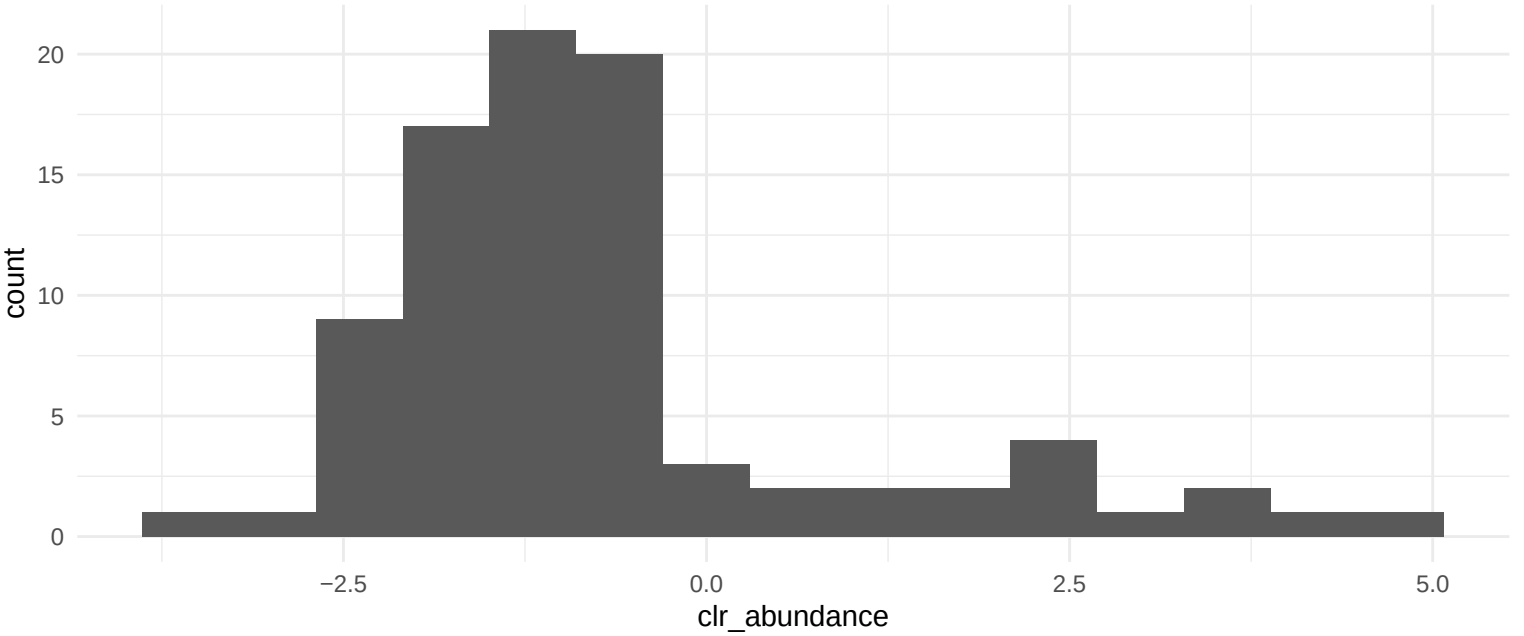



Distribution for 05216 Thyroid cancer
count
clr_abundance
0
5
10
15
−4
−2
0
2

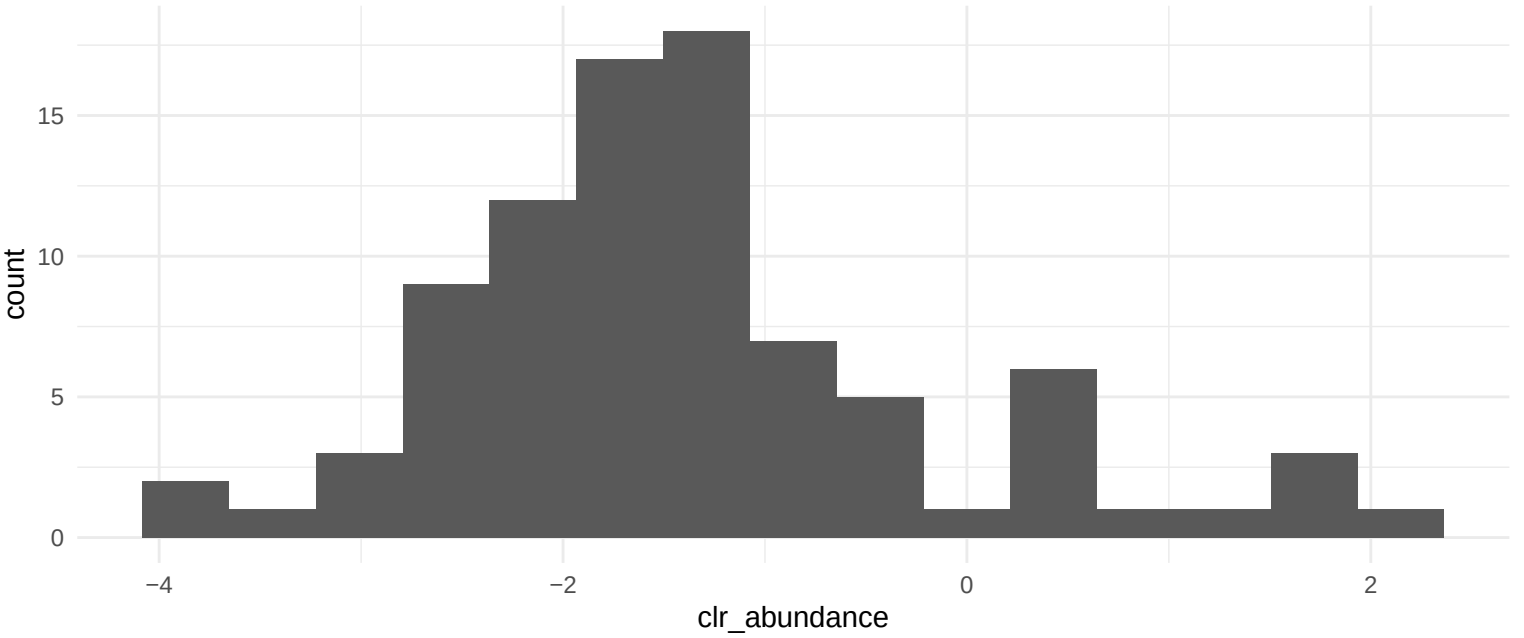

Distribution for 05217 Basal cell carcinoma
count
clr_abundance
0
5
10
15
−4
−2
0
2


Distribution for 05218 Melanoma
count
clr_abundance
0
5
10
15
−4
−2
0
2

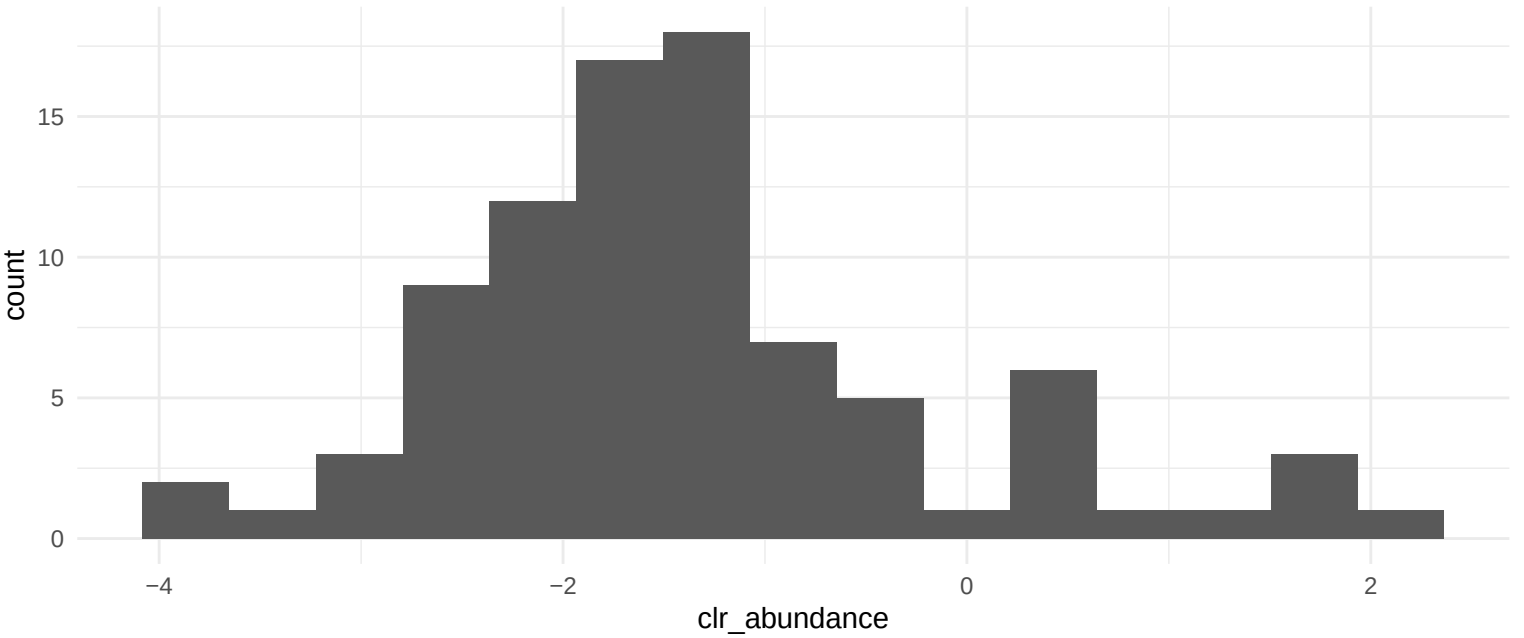
Distribution for 05219 Bladder cancer

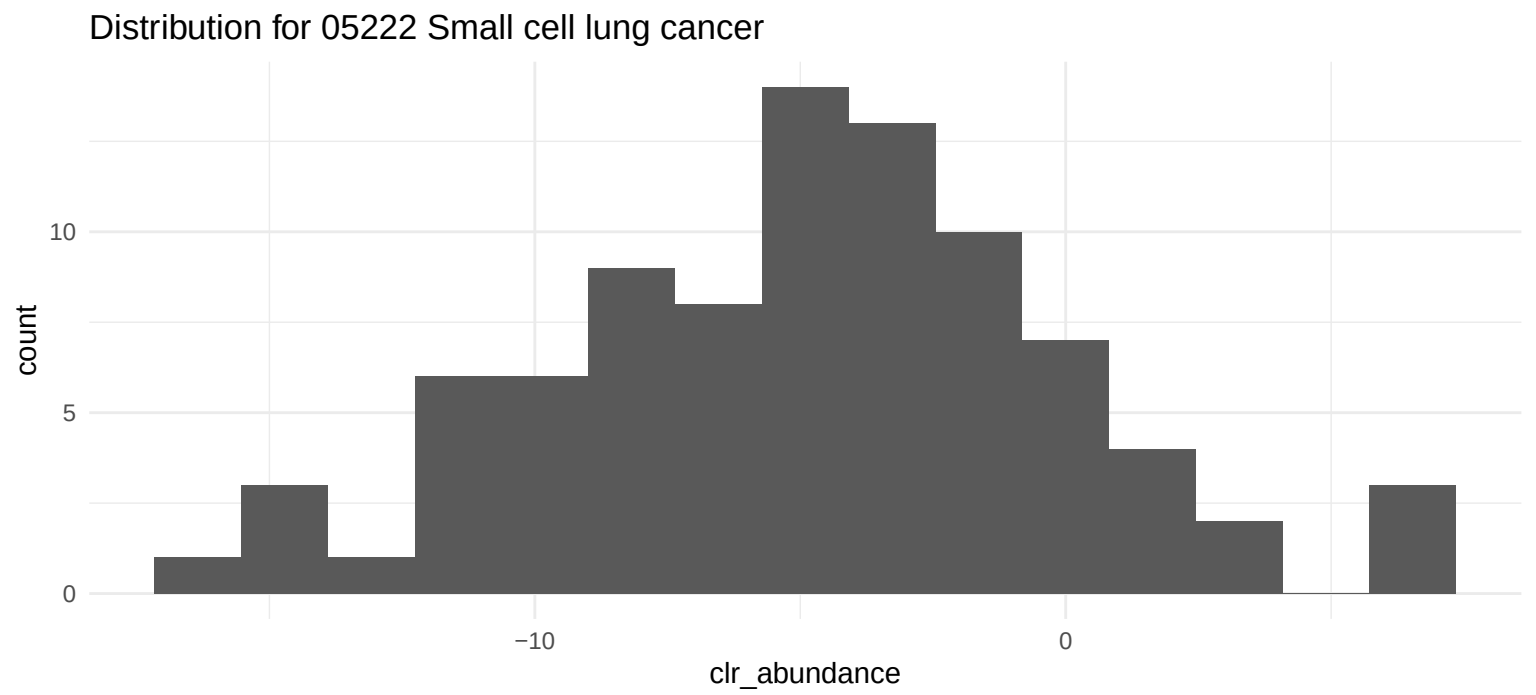


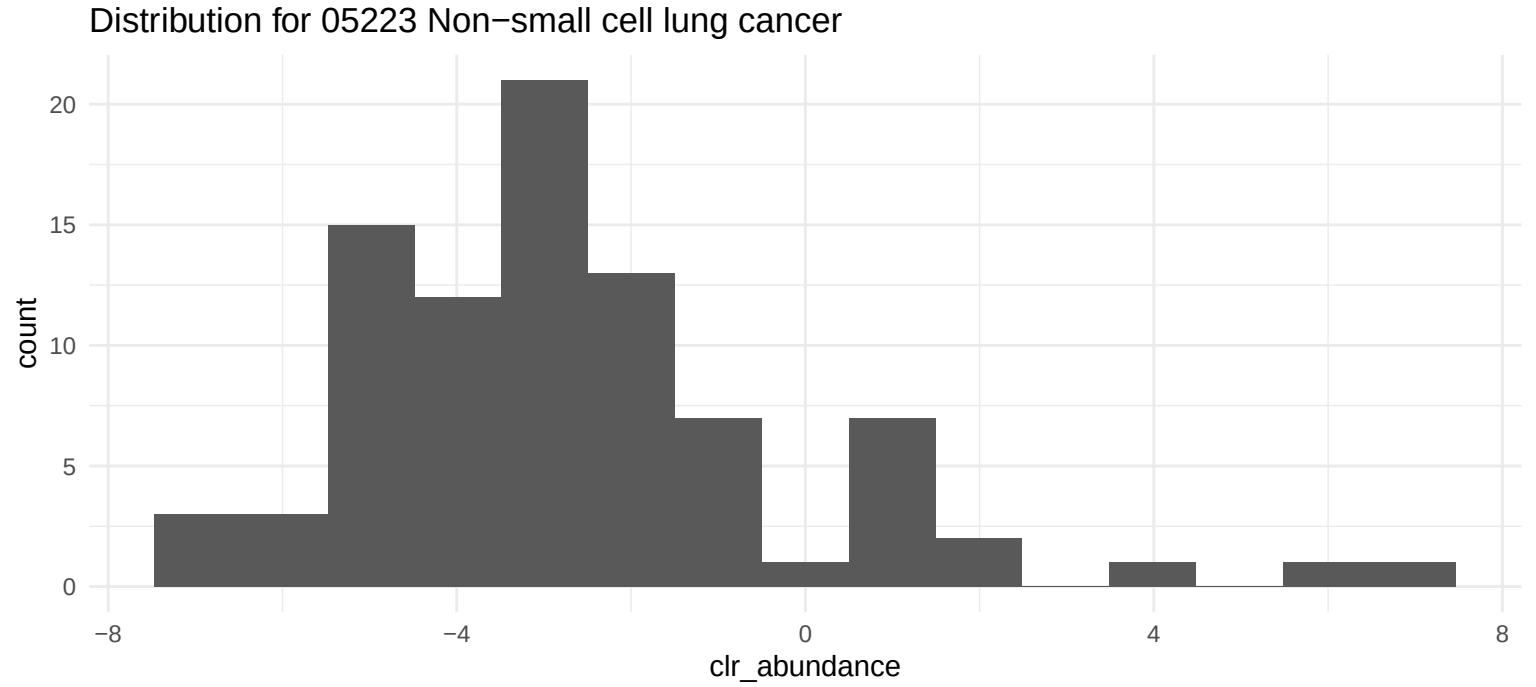
Distribution for 05220 Chronic myeloid leukemia

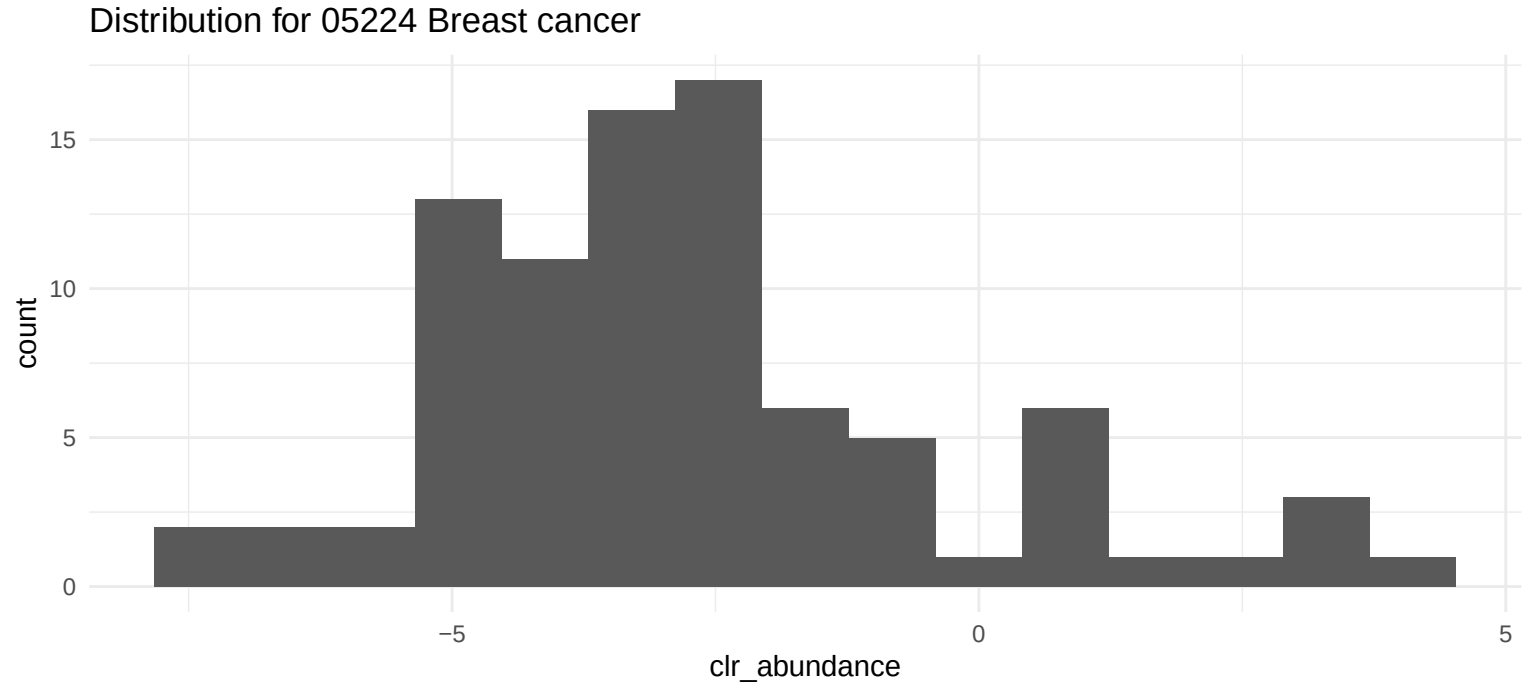


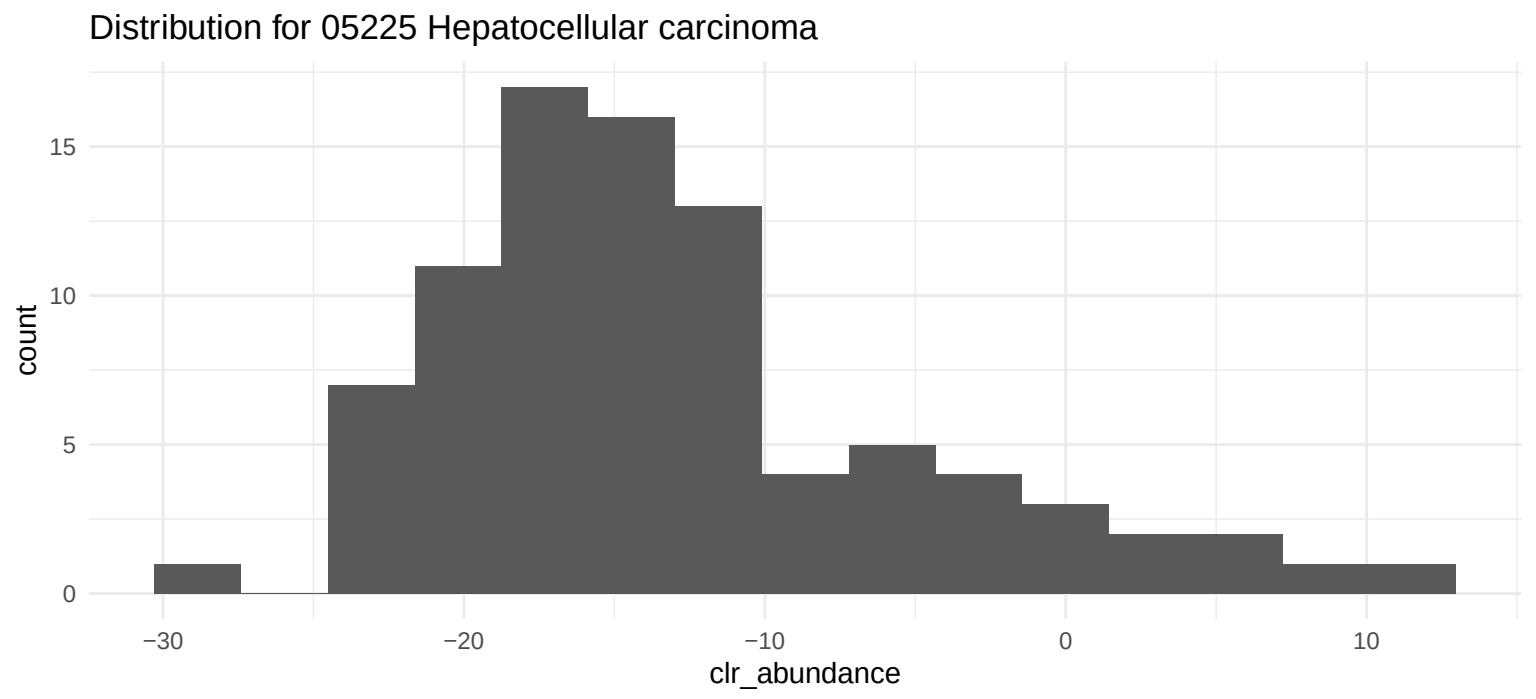
Distribution for 05221 Acute myeloid leukemia

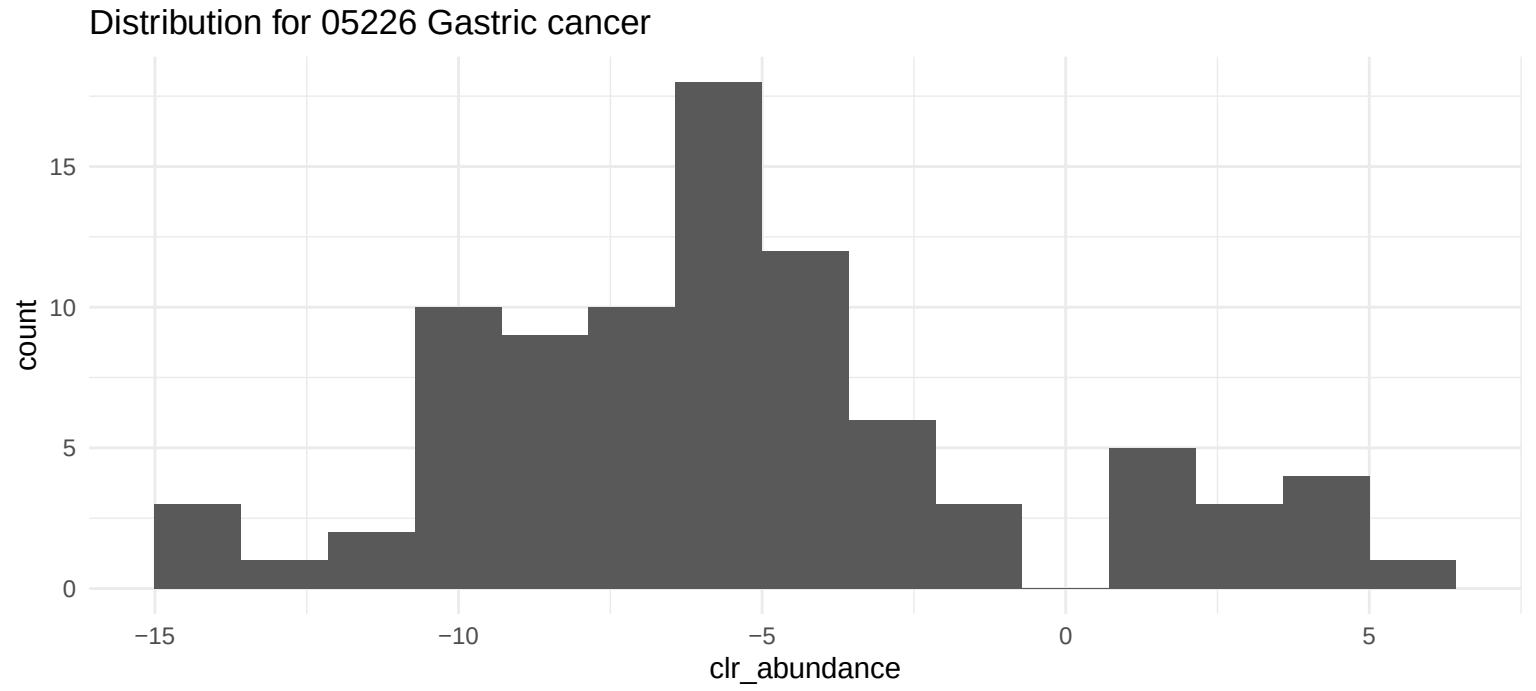


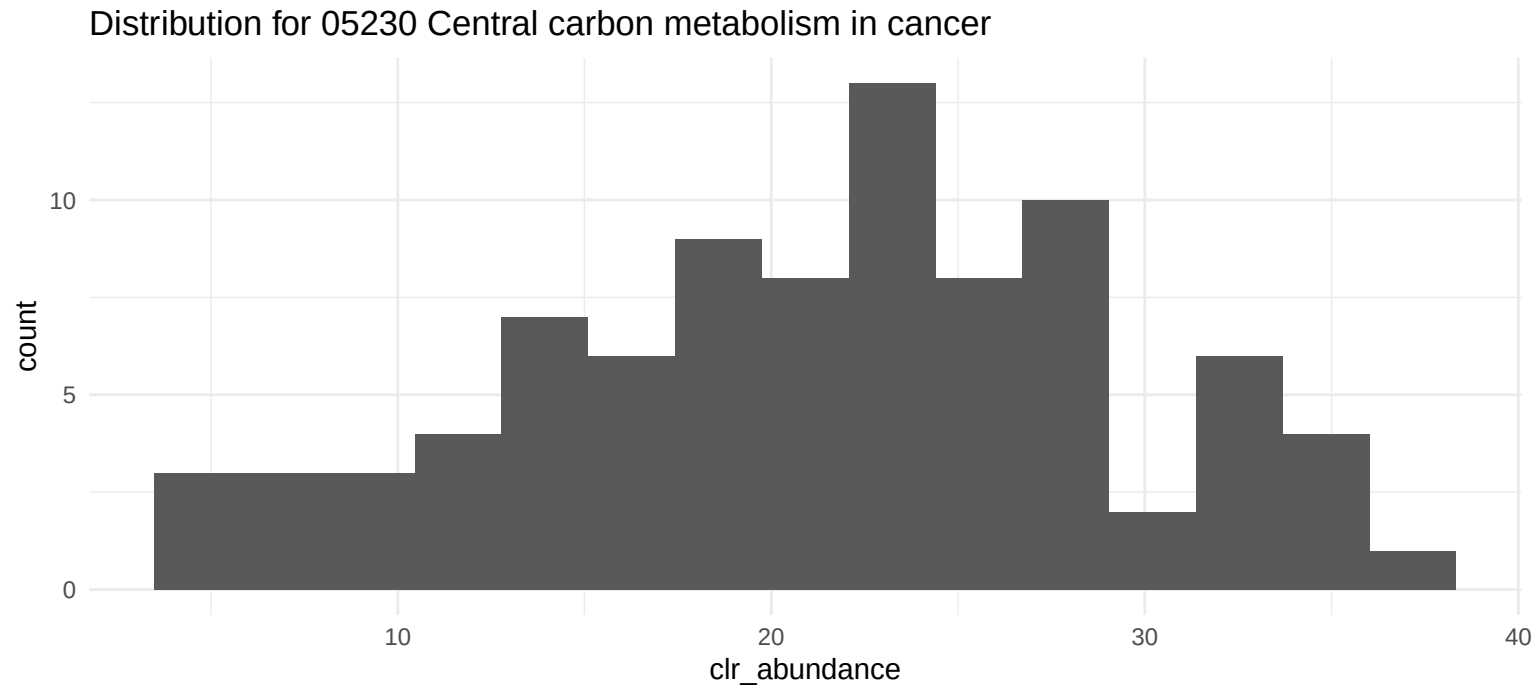

Distribution for 05222 Small cell lung cancer
count
10
5
0
−10
0
clr_abundance

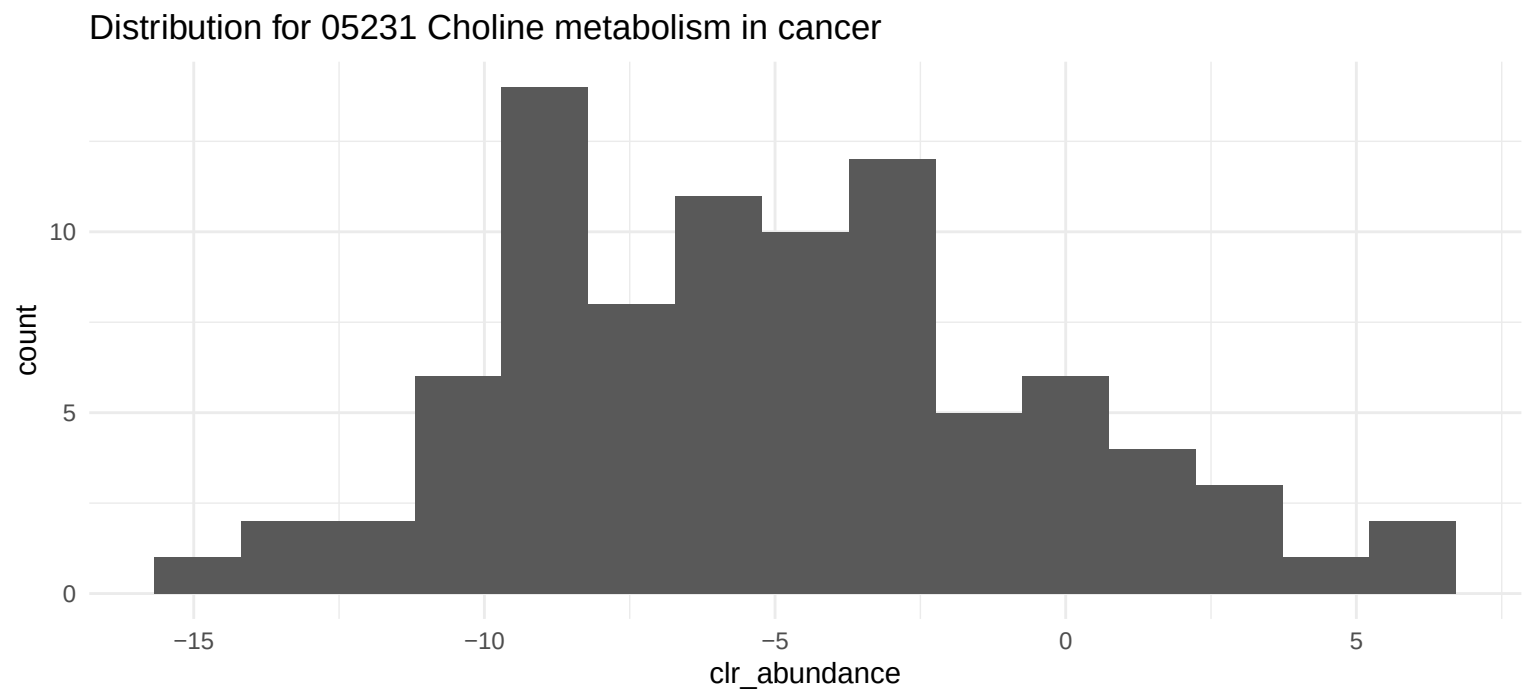

Distribution for 05223 Non−small cell lung cancer
count
20
15
10
5
0
−8
−4
0
4
8
clr_abundance

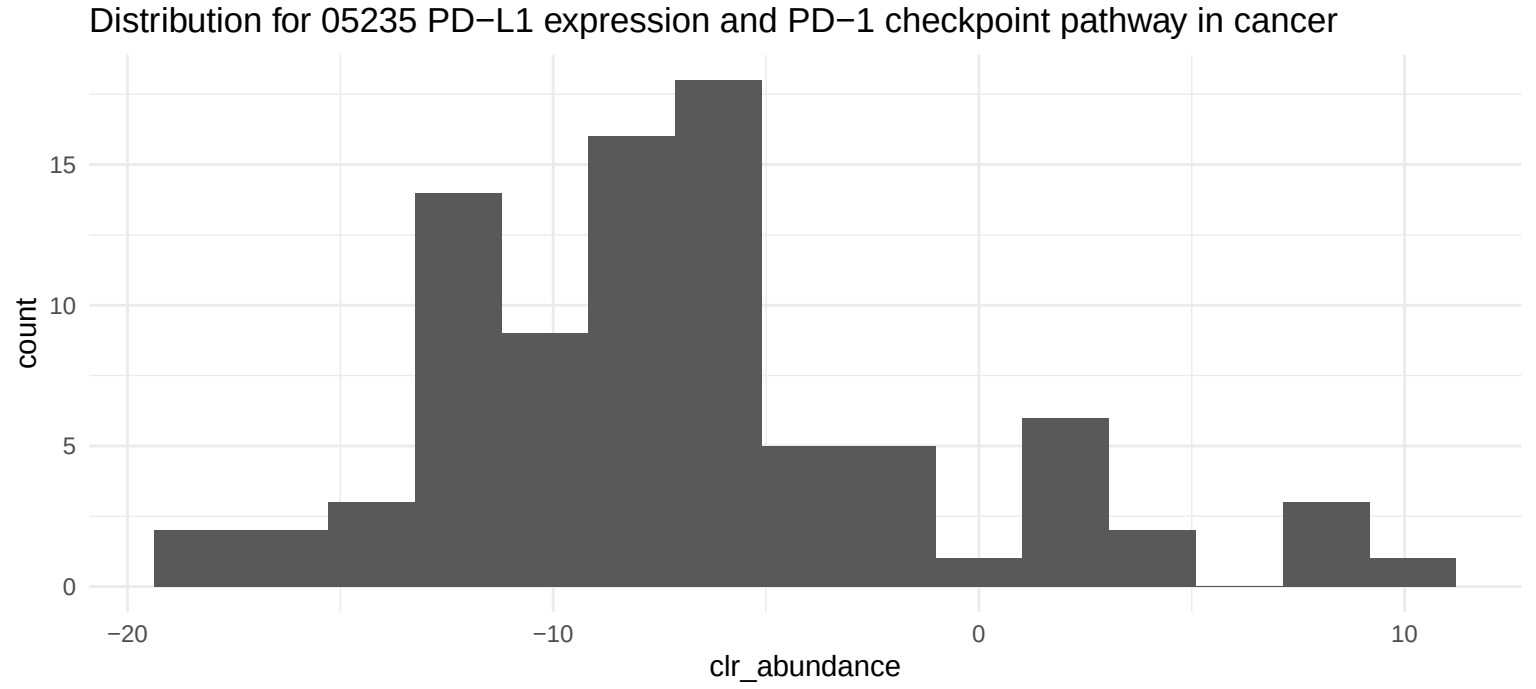

Distribution for 05224 Breast cancer
count
15
10
5
0
−5
0
5
clr_abundance

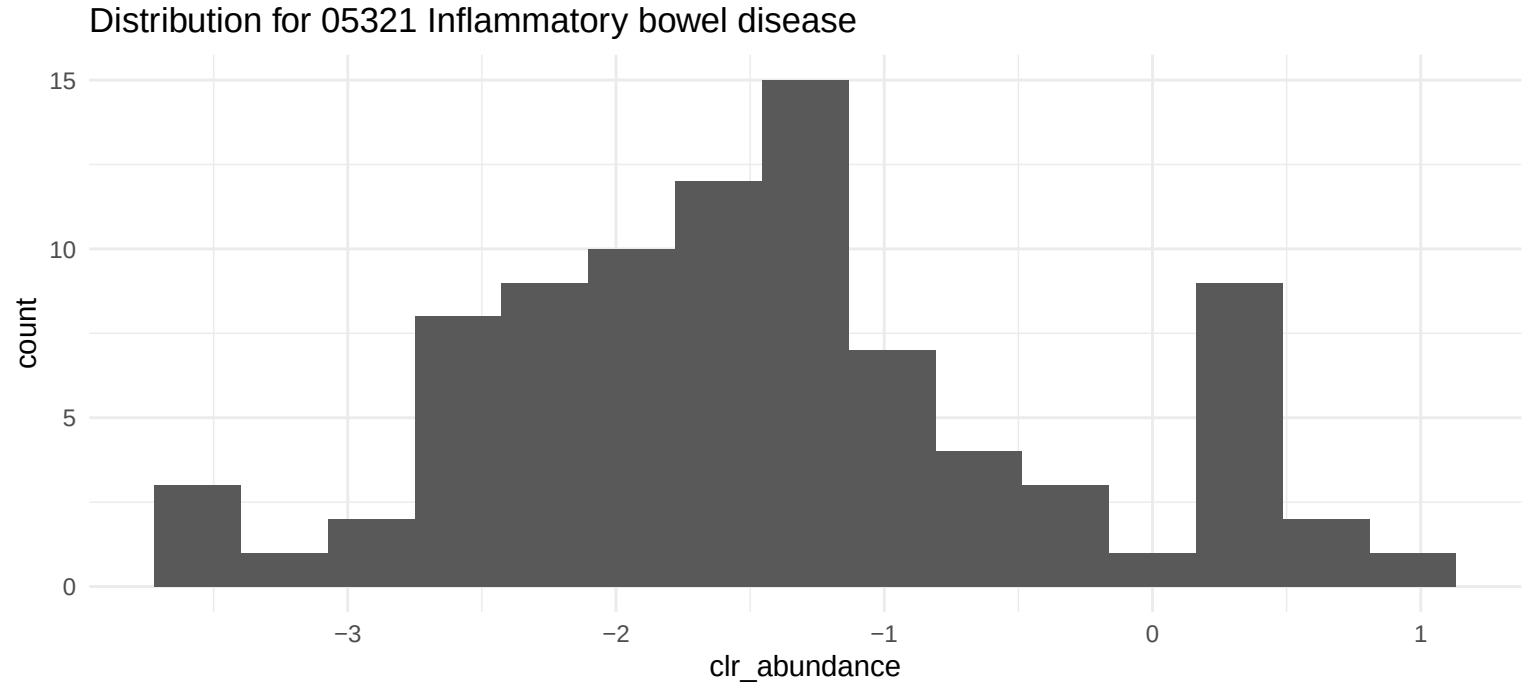

Distribution for 05225 Hepatocellular carcinoma
count
clr_abundance

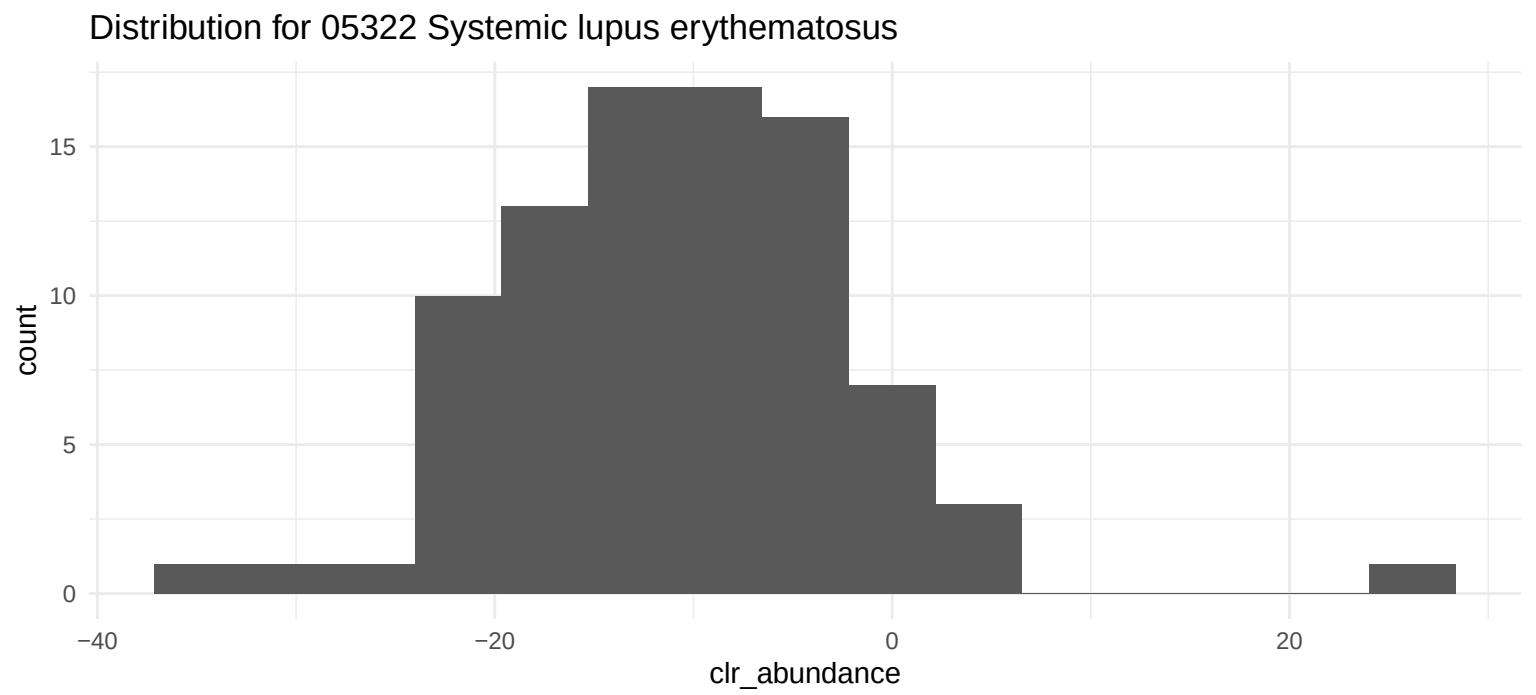

Distribution for 05226 Gastric cancer
count
clr_abundance

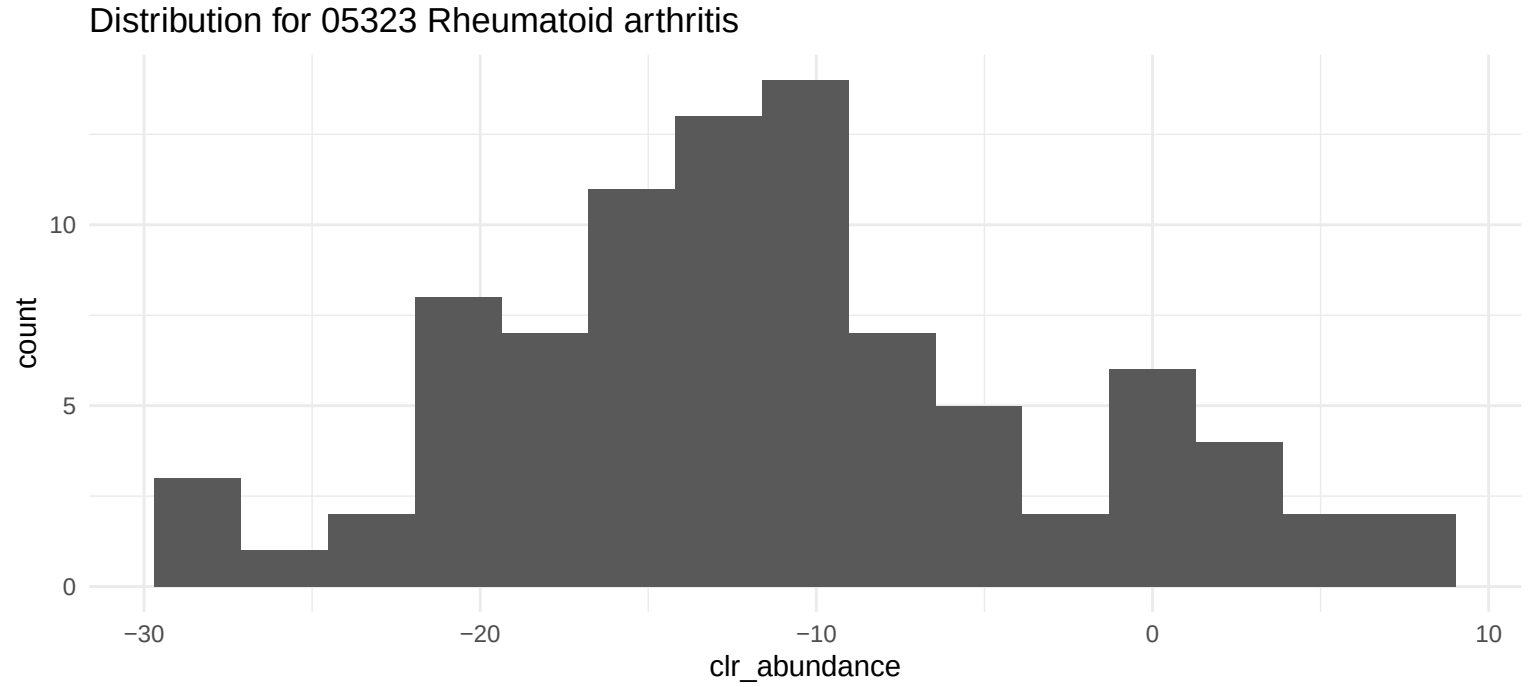

Distribution for 05230 Central carbon metabolism in cancer
count
clr_abundance

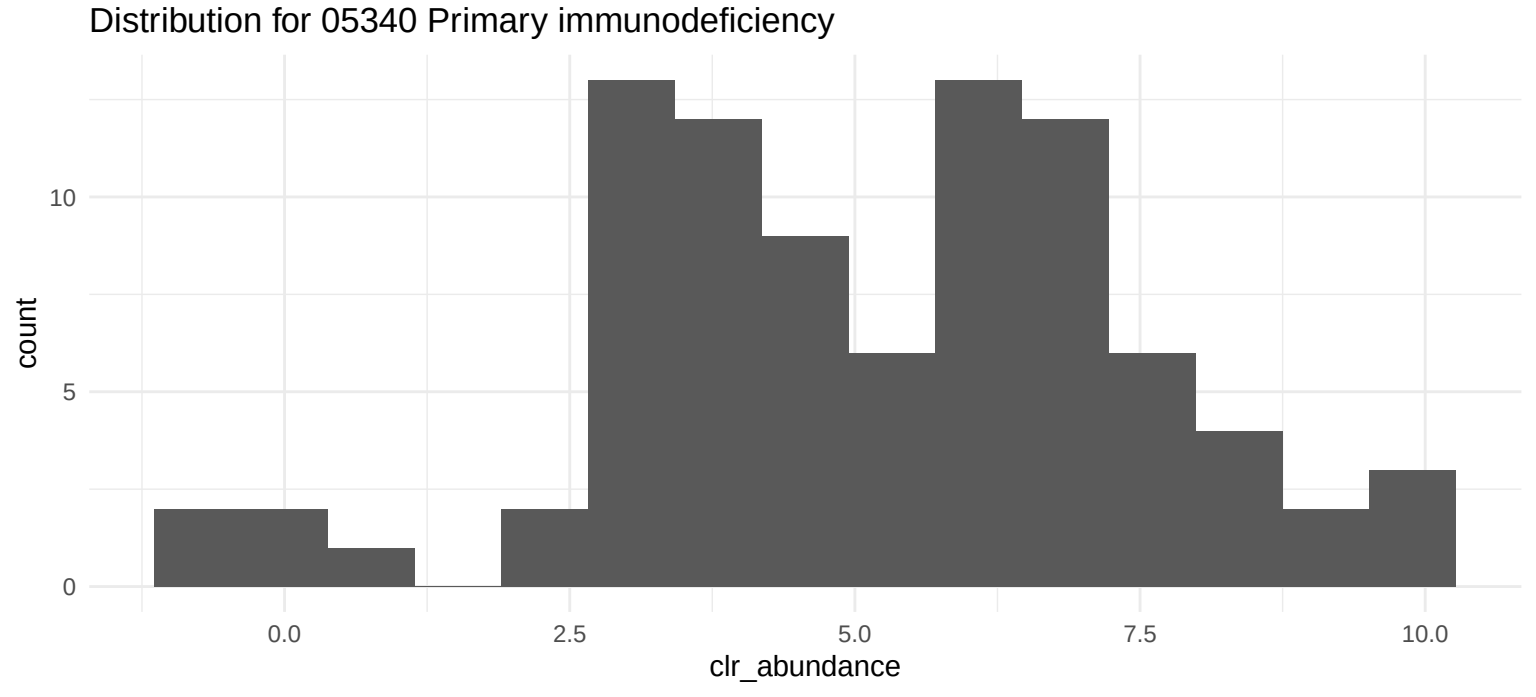

Distribution for 05231 Choline metabolism in cancer
count
clr_abundance

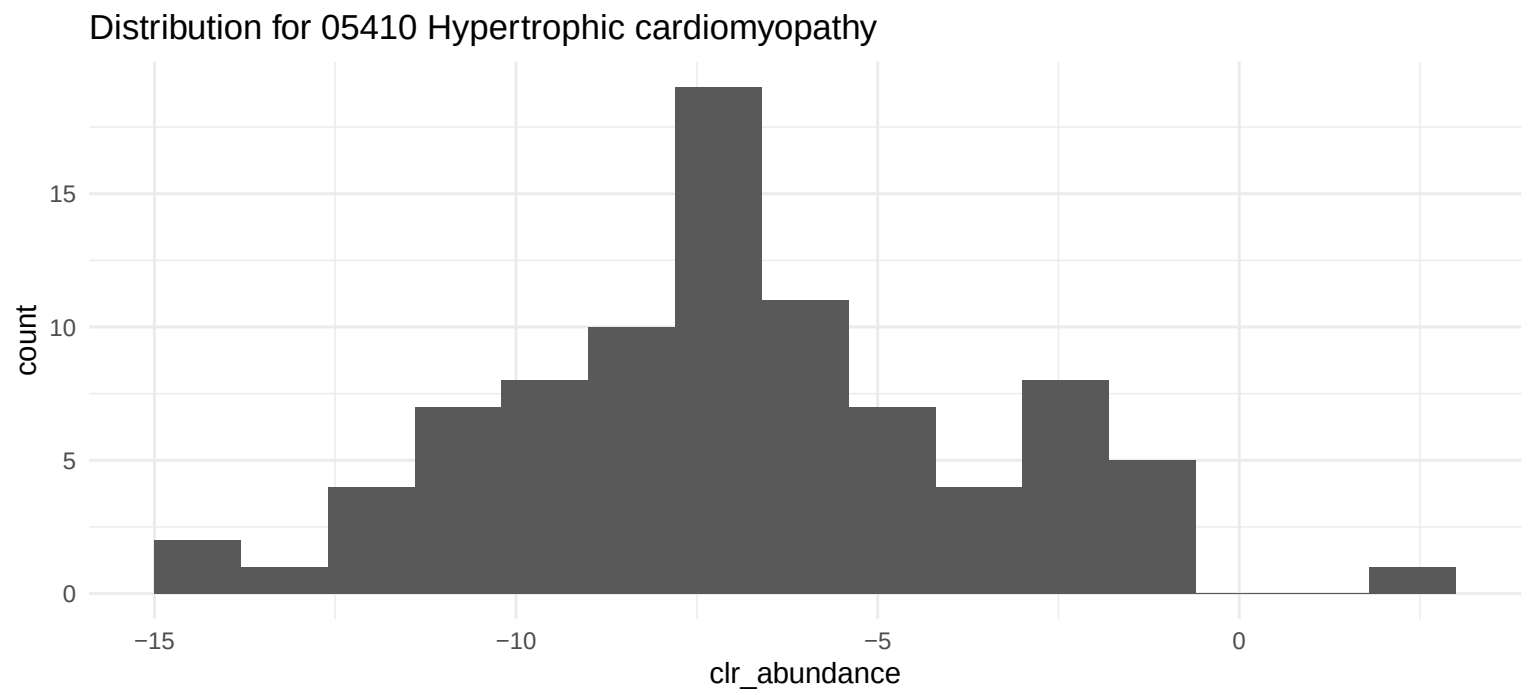

Distribution for 05235 PD−L1 expression and PD−1 checkpoint pathway in cancer
count
clr_abundance

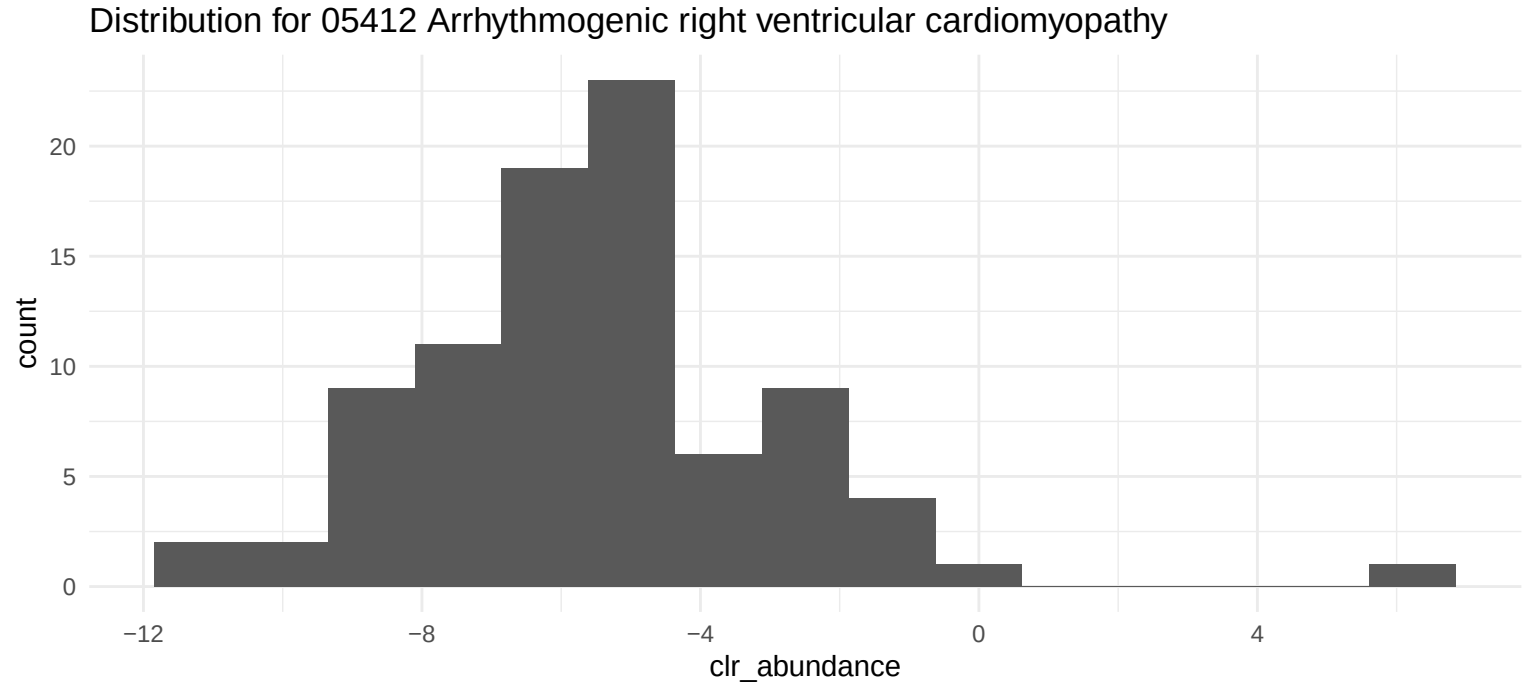

Distribution for 05321 Inflammatory bowel disease
count
clr_abundance

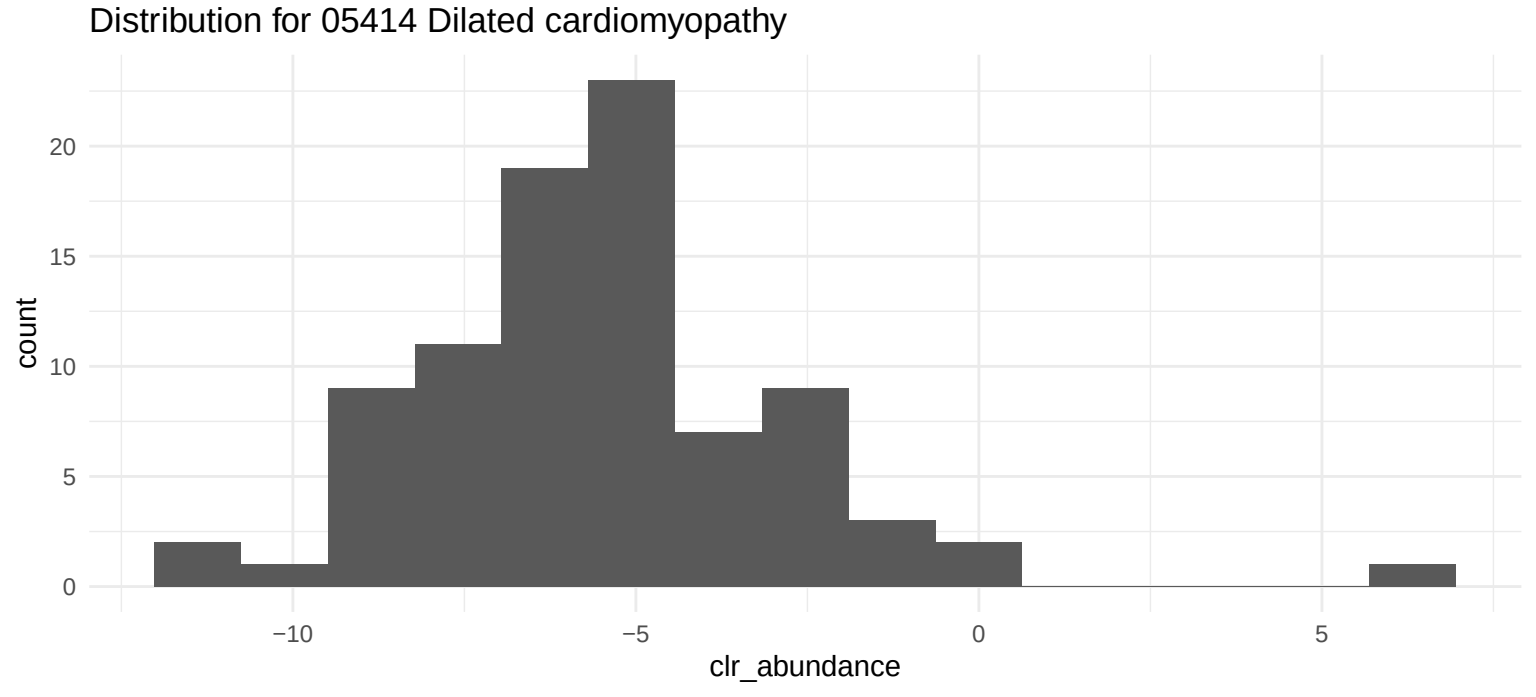

Distribution for 05322 Systemic lupus erythematosus
count
15
10
5
0
−40
−20
0
20
clr_abundance

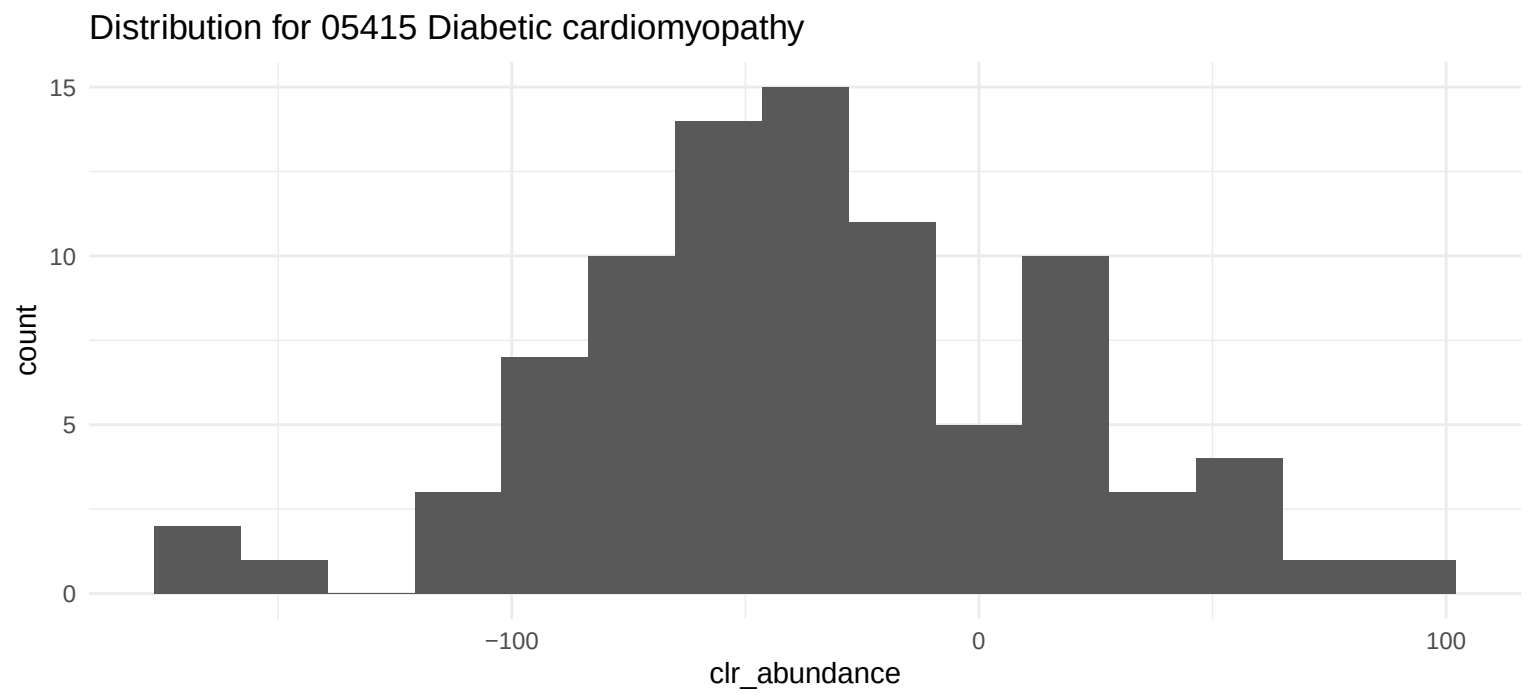

Distribution for 05323 Rheumatoid arthritis
count
10
5
0
−30
−20
−10
0
10
clr_abundance

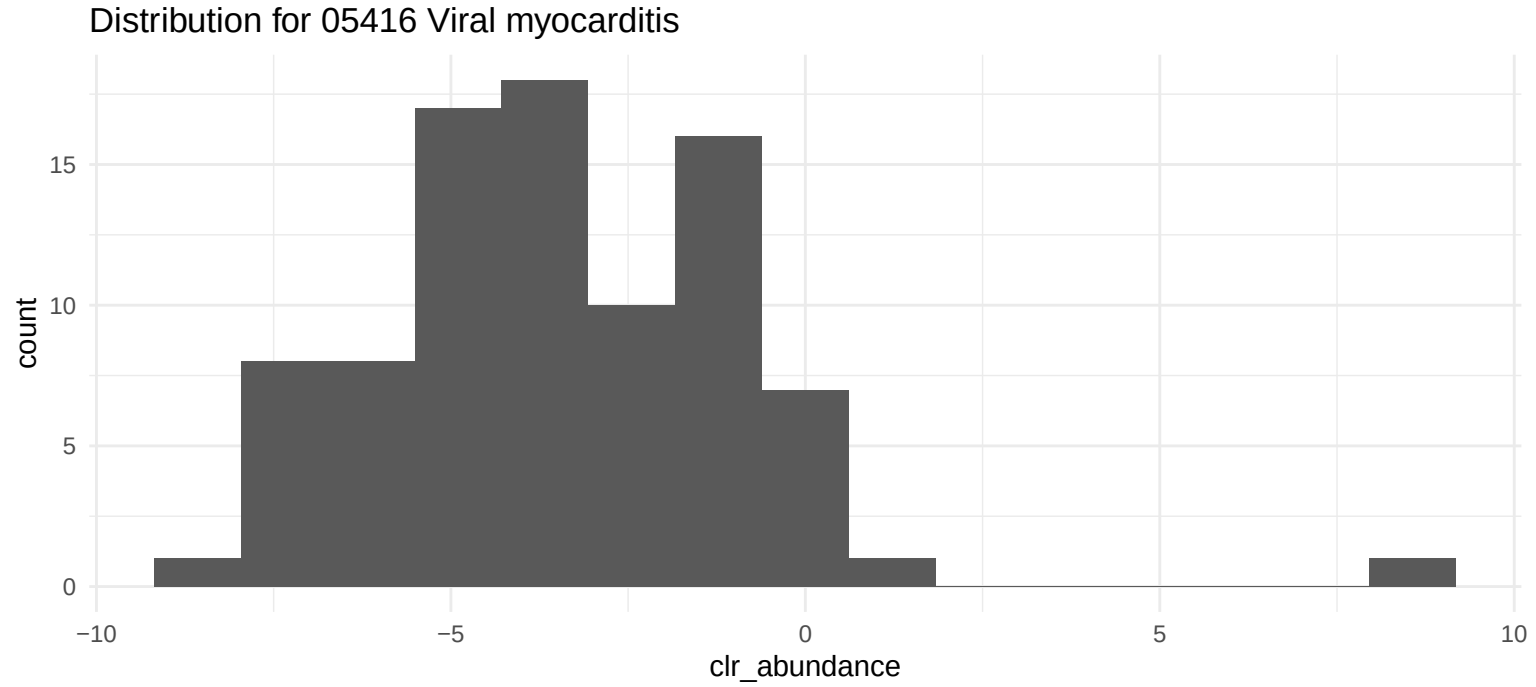

Distribution for 05340 Primary immunodeficiency
count
10
5
0
0.0
2.5
5.0
7.5
10.0
clr_abundance

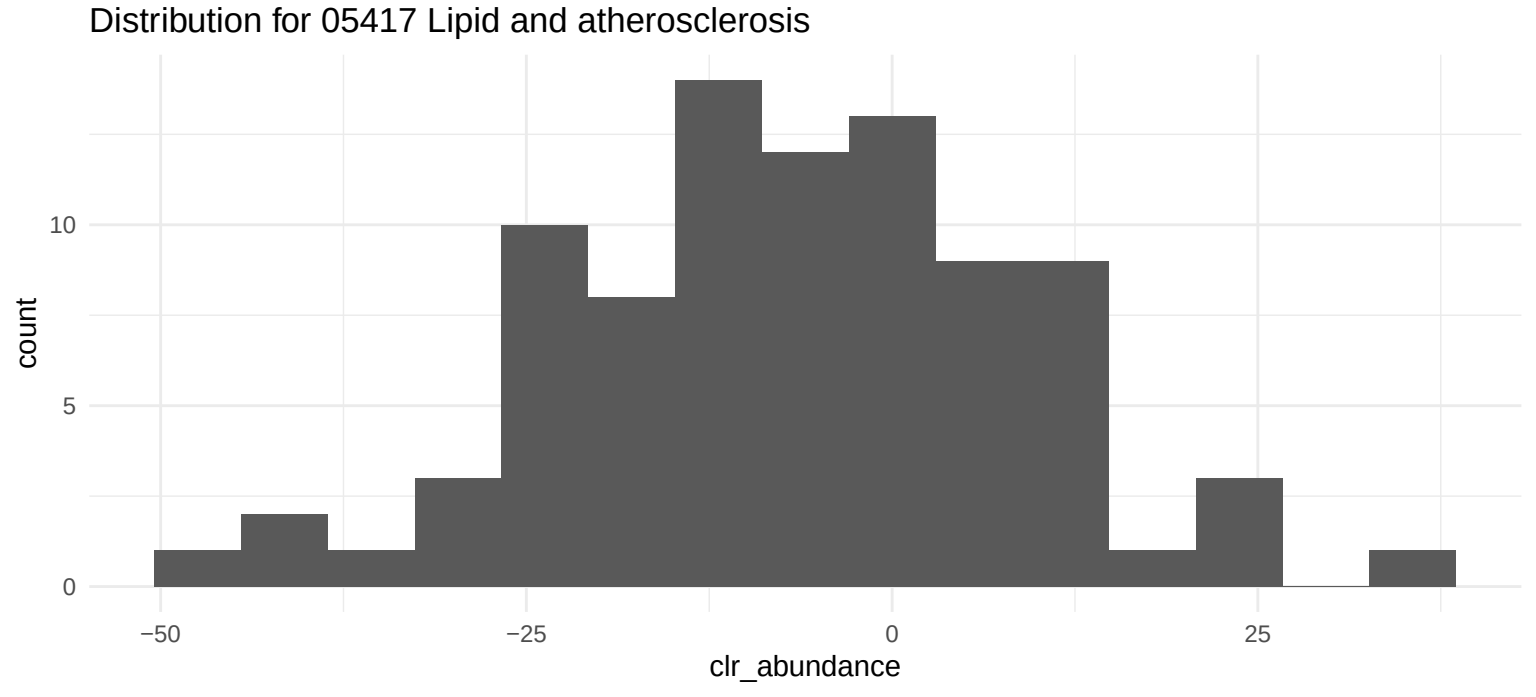

Distribution for 05410 Hypertrophic cardiomyopathy
count
clr_abundance
0
5
10
15
−15
−10
−5
0


Distribution for 05412 Arrhythmogenic right ventricular cardiomyopathy
count
clr_abundance
0
5
10
15
20
−12
−8
−4
0
4

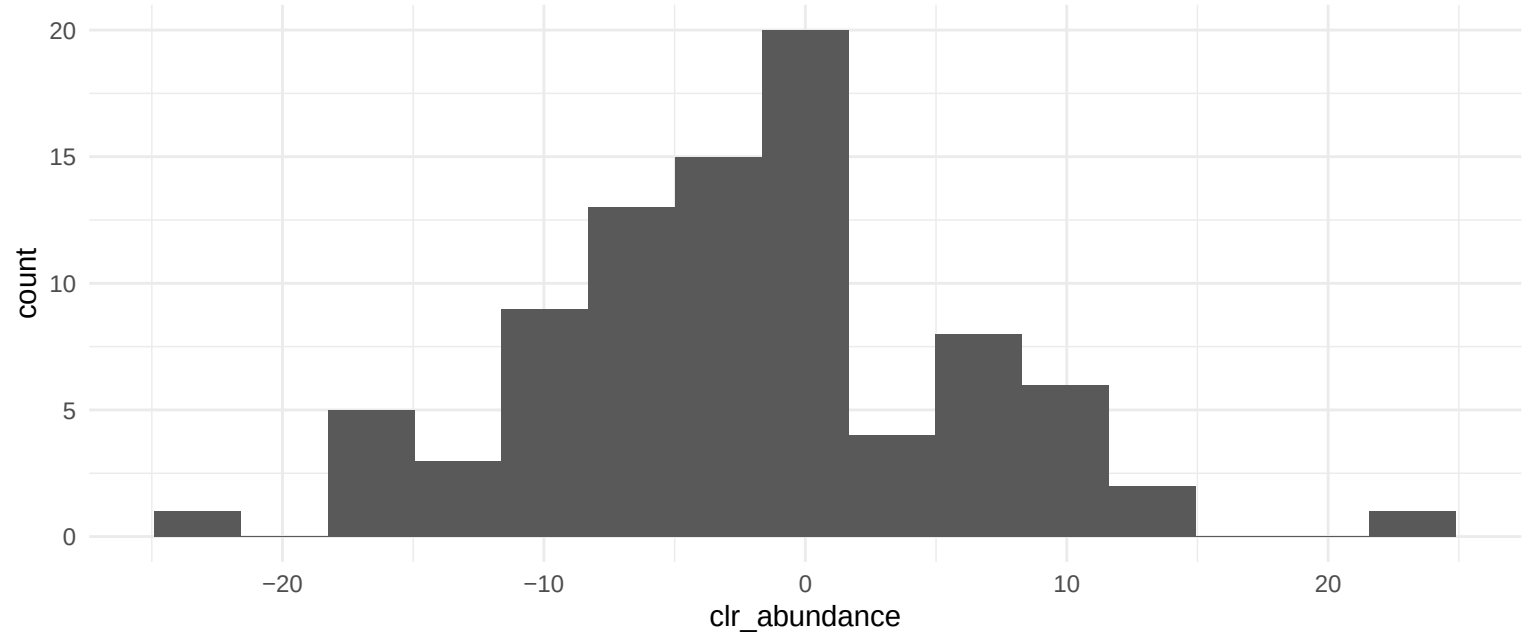

Distribution for 05414 Dilated cardiomyopathy
count
clr_abundance
0
5
10
15
20
−10
−5
0
5


Distribution for 05415 Diabetic cardiomyopathy
count
15
10
5
0
−100
0
100
clr_abundance

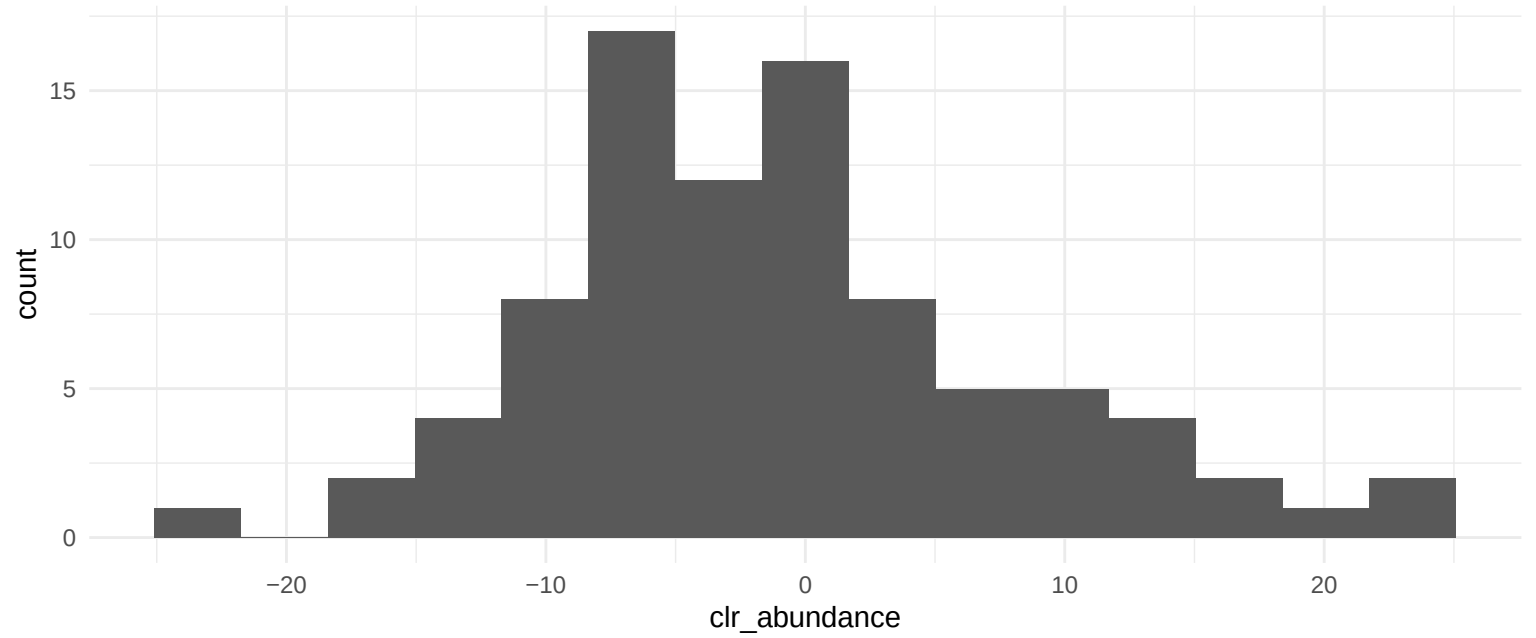

Distribution for 05416 Viral myocarditis
count
15
10
5
0
−10
−5
0
5
10
clr_abundance


Distribution for 05417 Lipid and atherosclerosis
count
10
5
0
−50
−25
0
25
clr_abundance

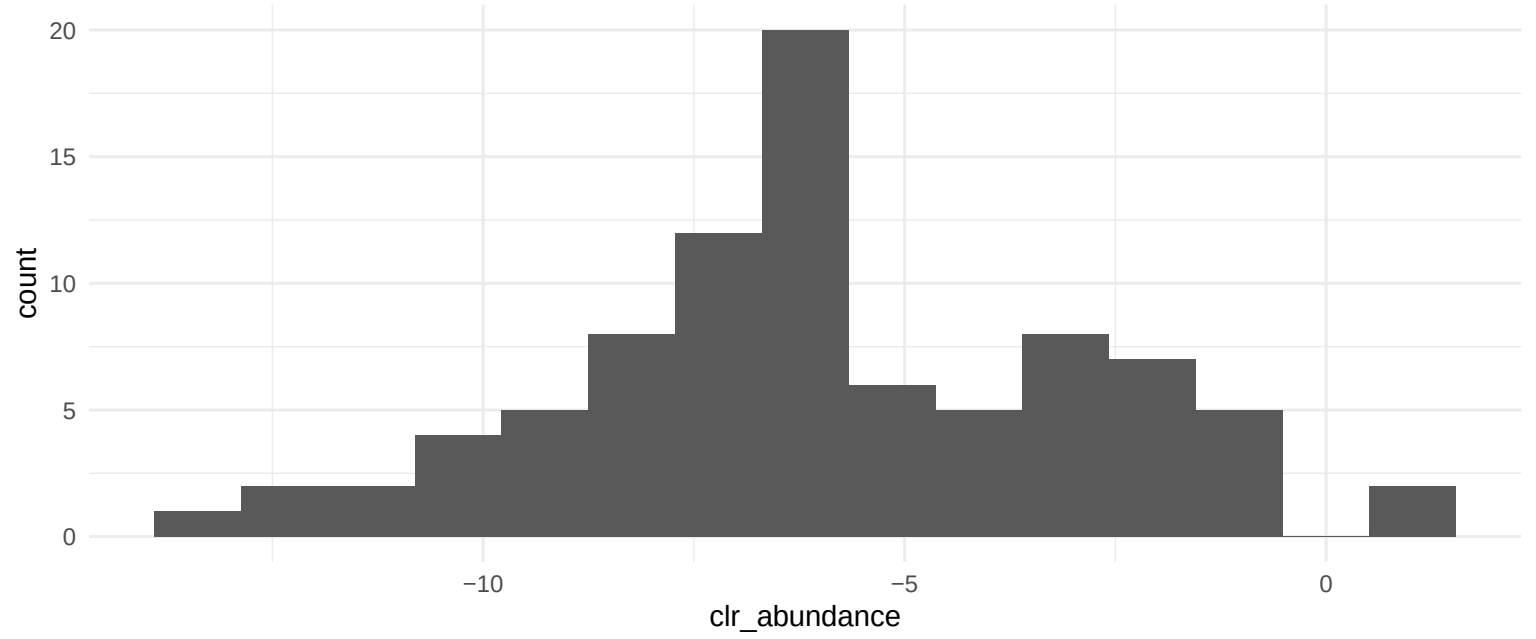
Distribution for 05418 Fluid shear stress and atherosclerosis

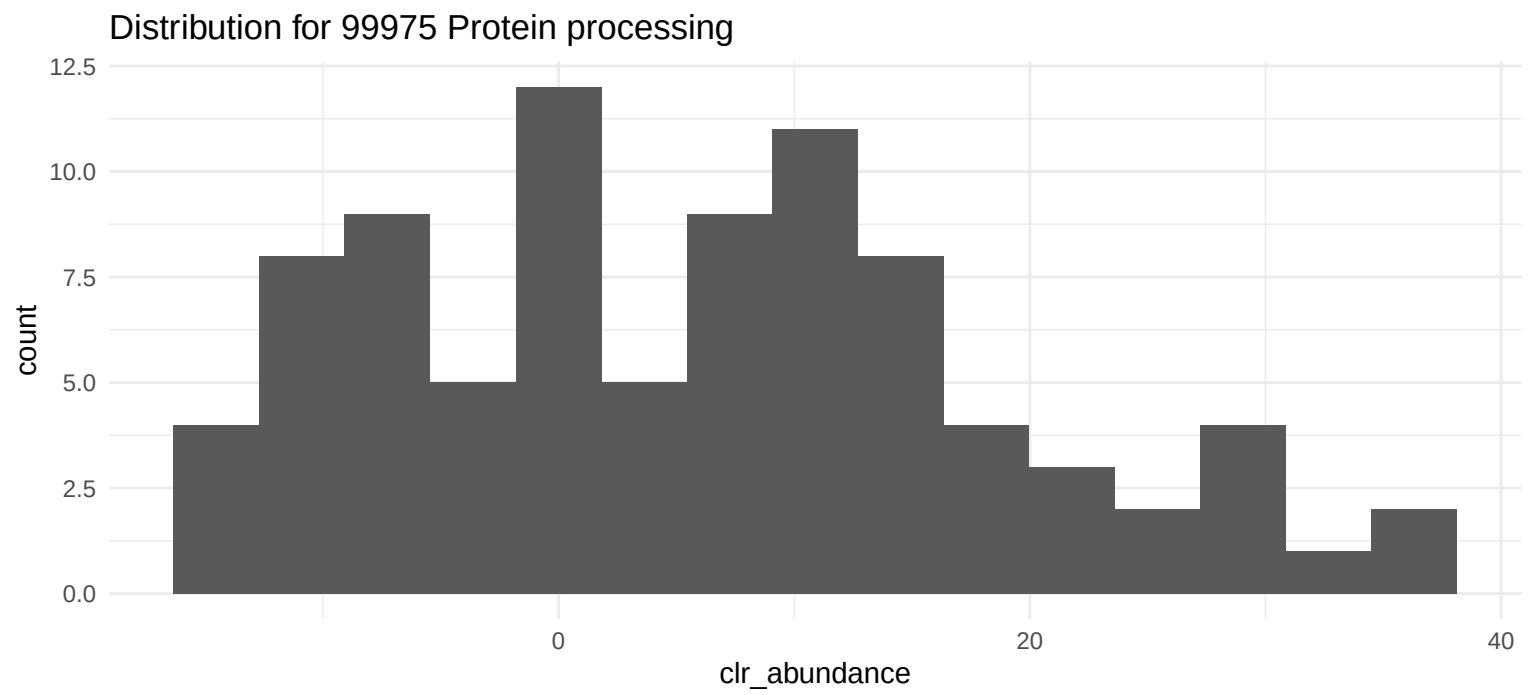


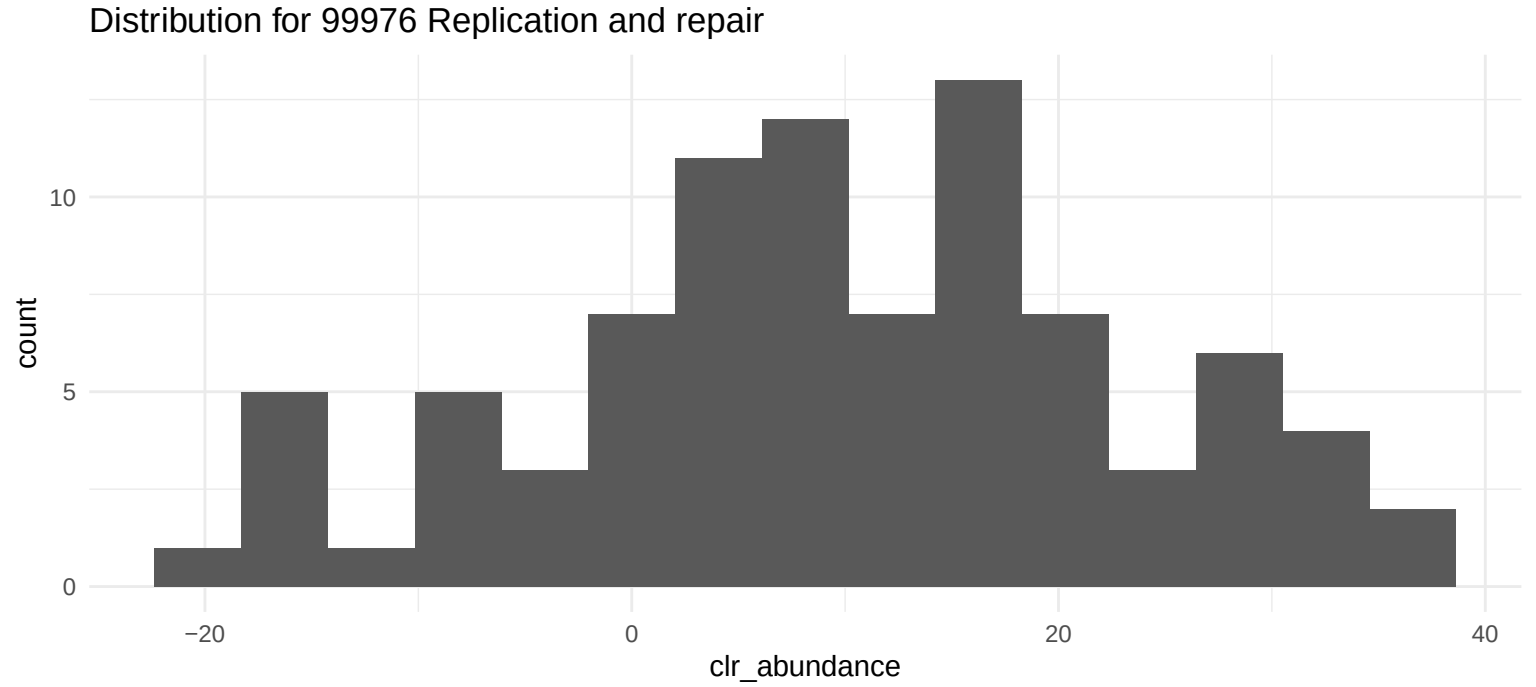
Distribution for 99973 Transcription

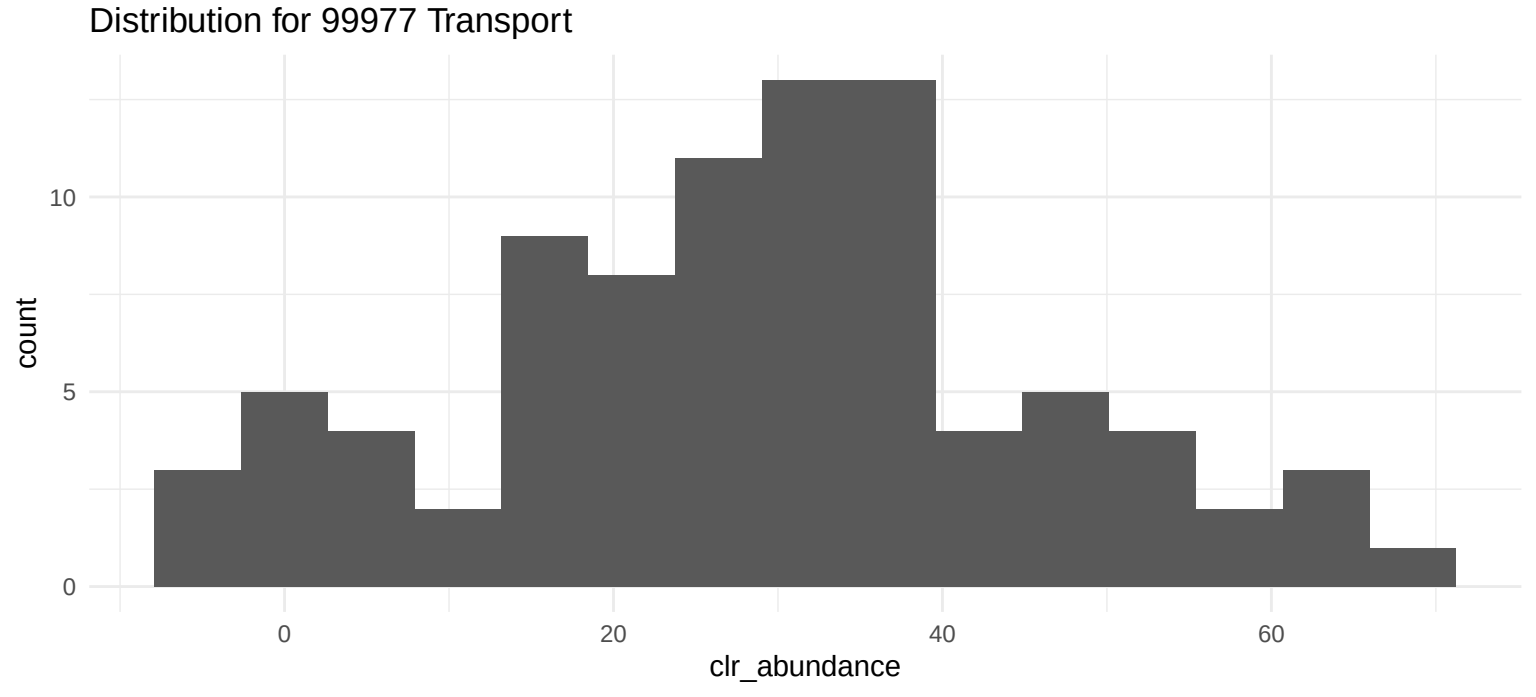


Distribution for 99974 Translation

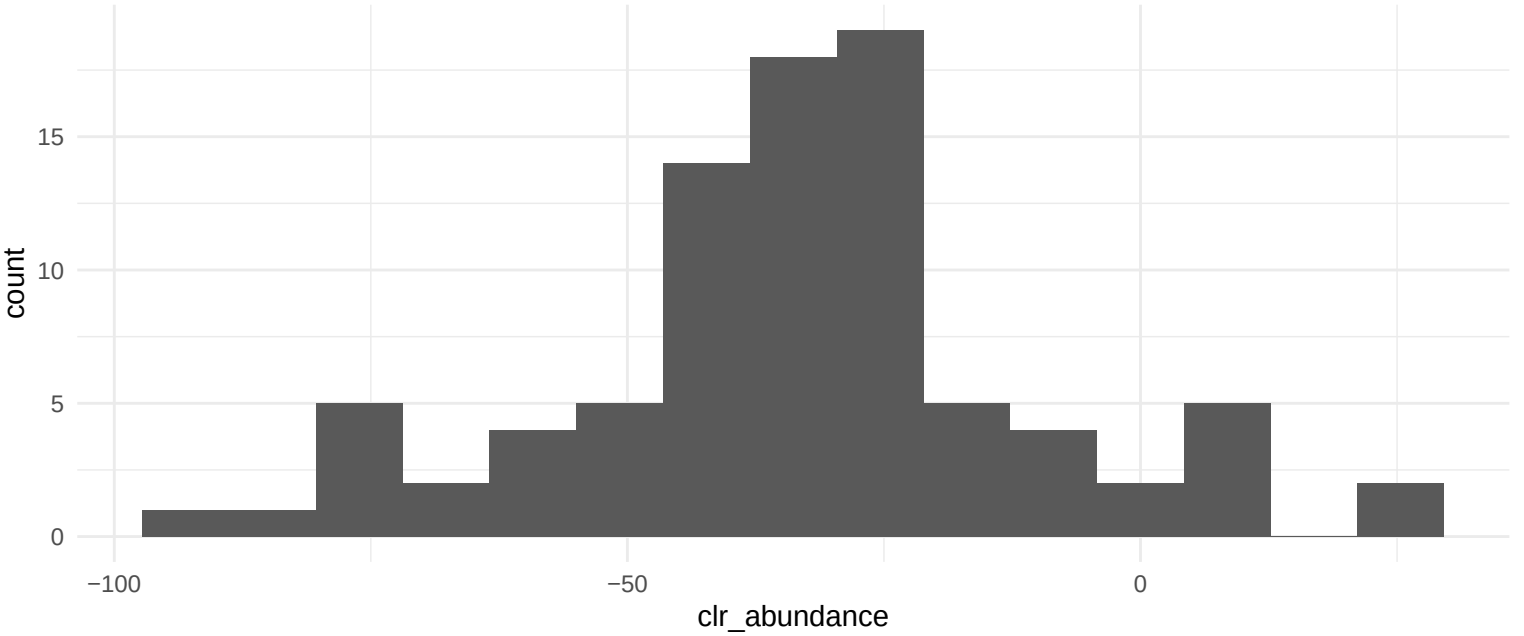



Distribution for 99975 Protein processing
12.5
10.0
7.5
5.0
2.5
0.0
count
0
20
40
clr_abundance

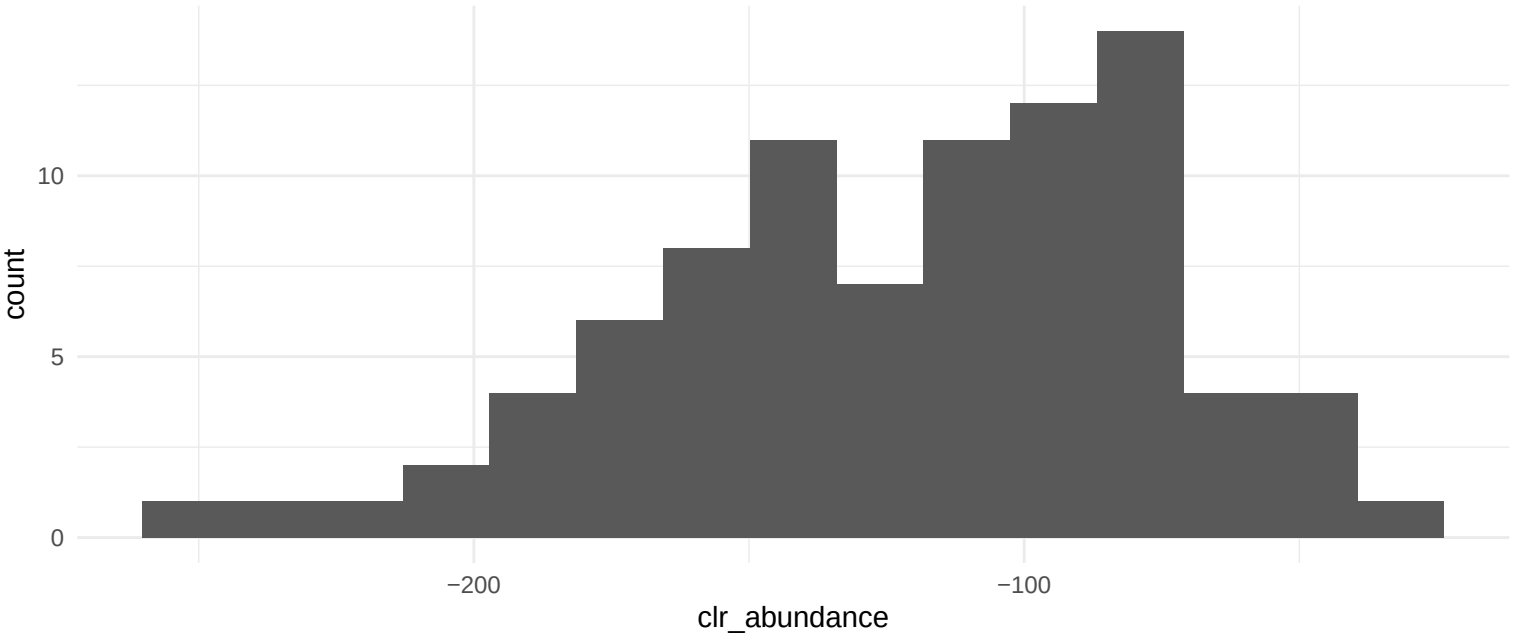

Distribution for 99976 Replication and repair
10
5
0
count
−20
0
20
40
clr_abundance


Distribution for 99977 Transport
10
5
0
count
0
20
40
60
clr_abundance

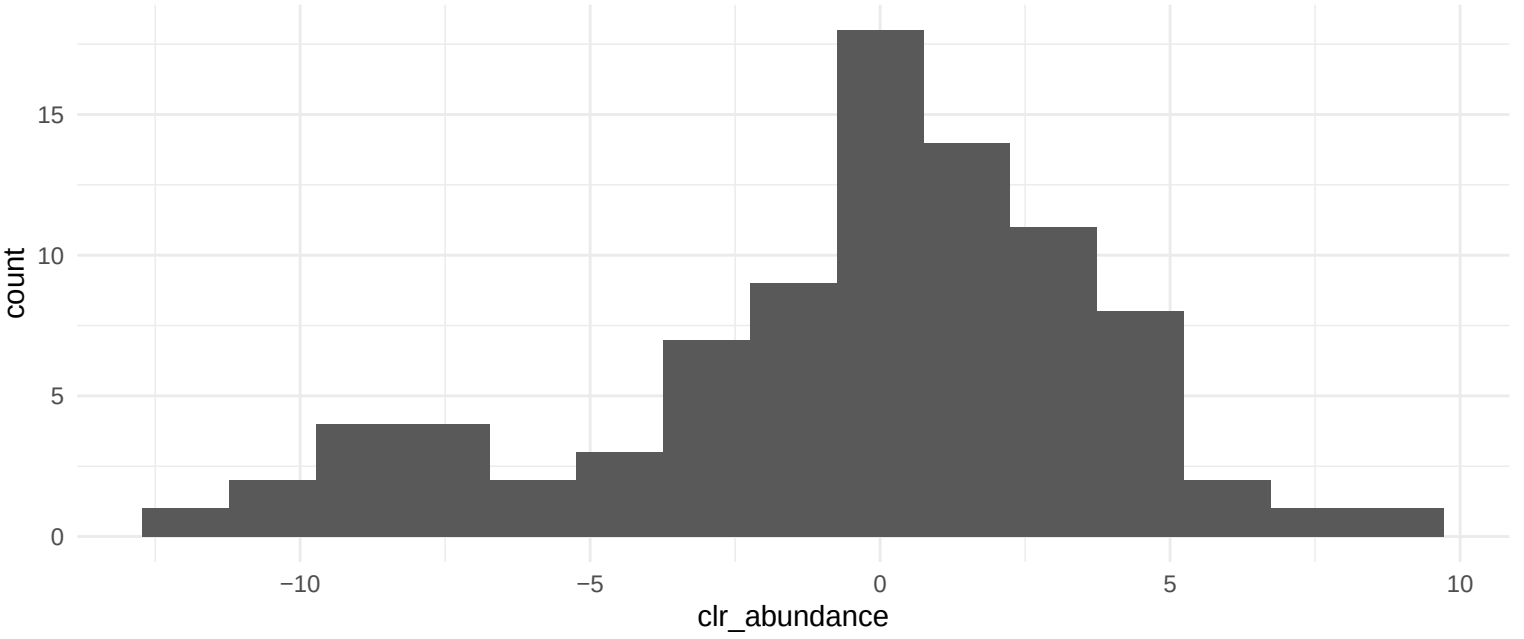
Distribution for 99978 Cell growth

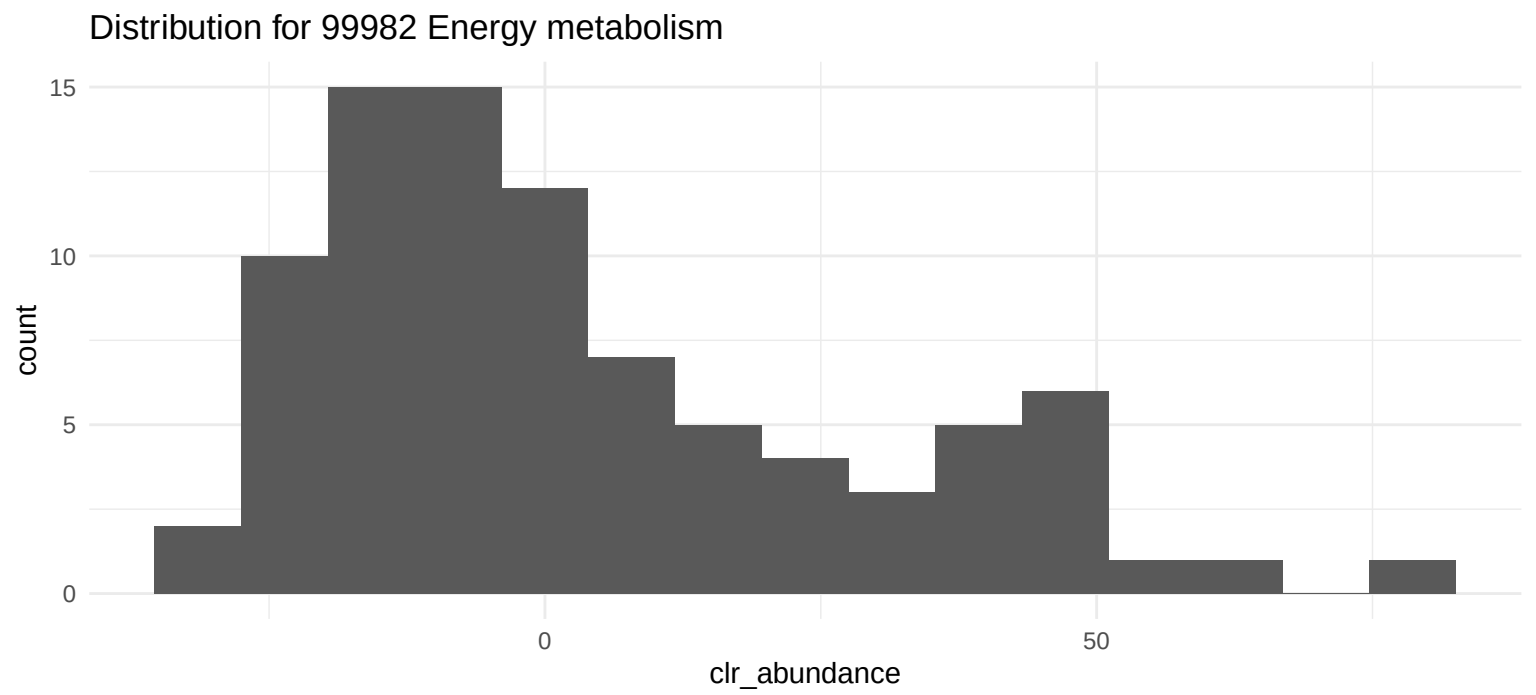


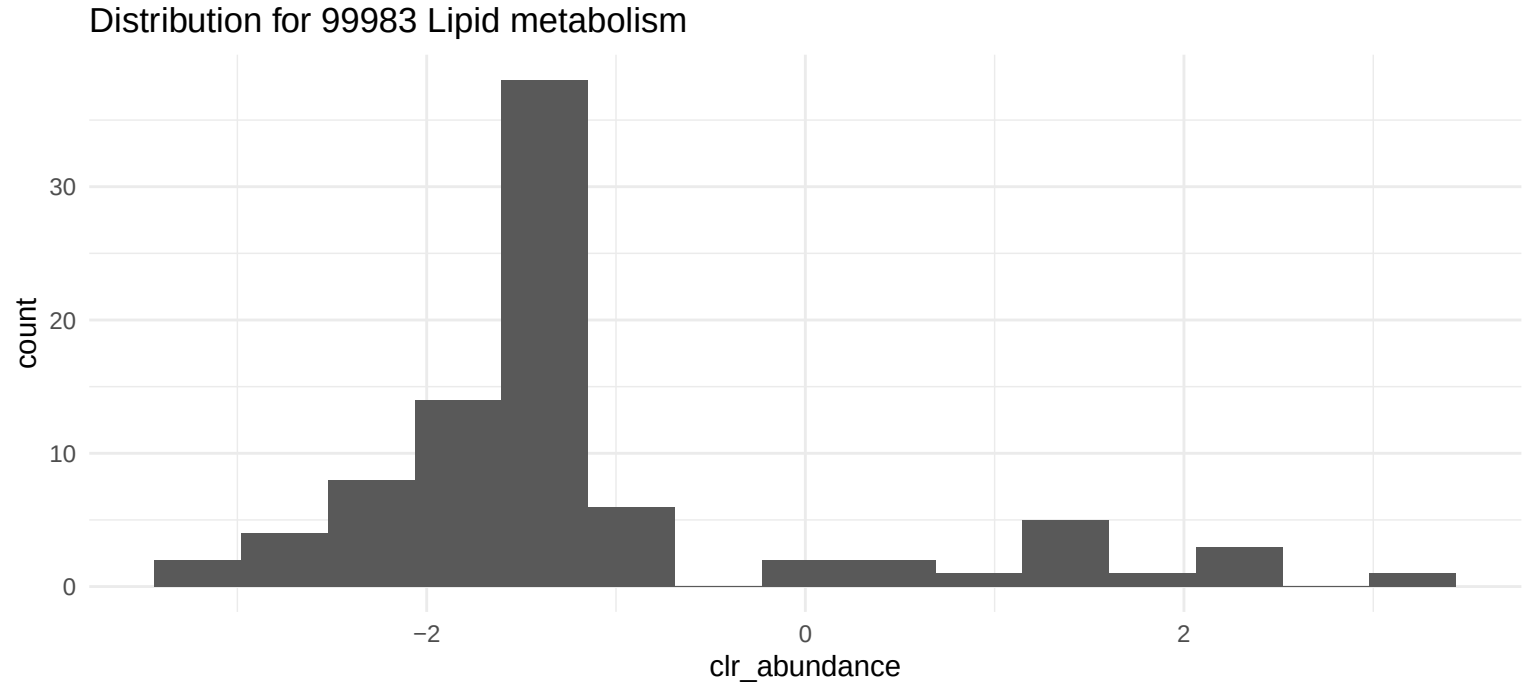
Distribution for 99980 Enzymes with EC numbers

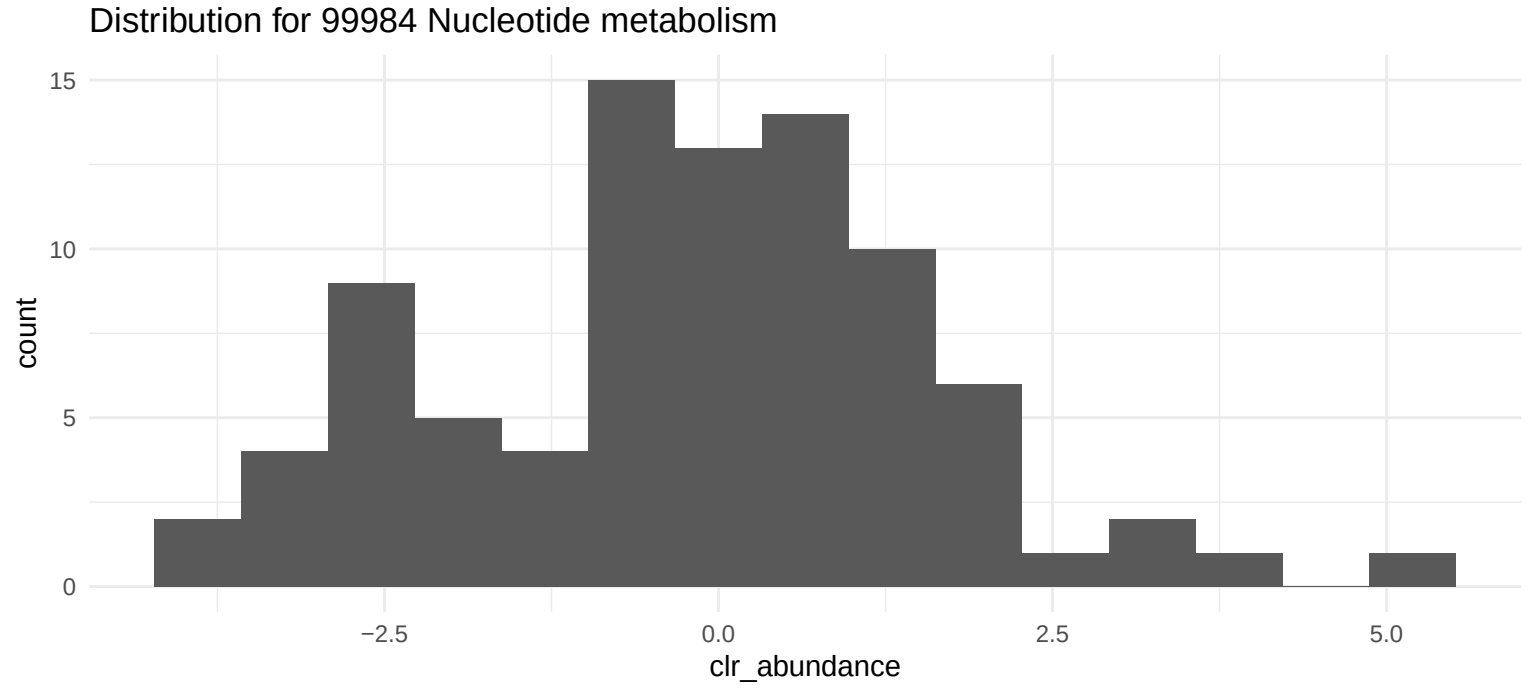


Distribution for 99981 Carbohydrate metabolism

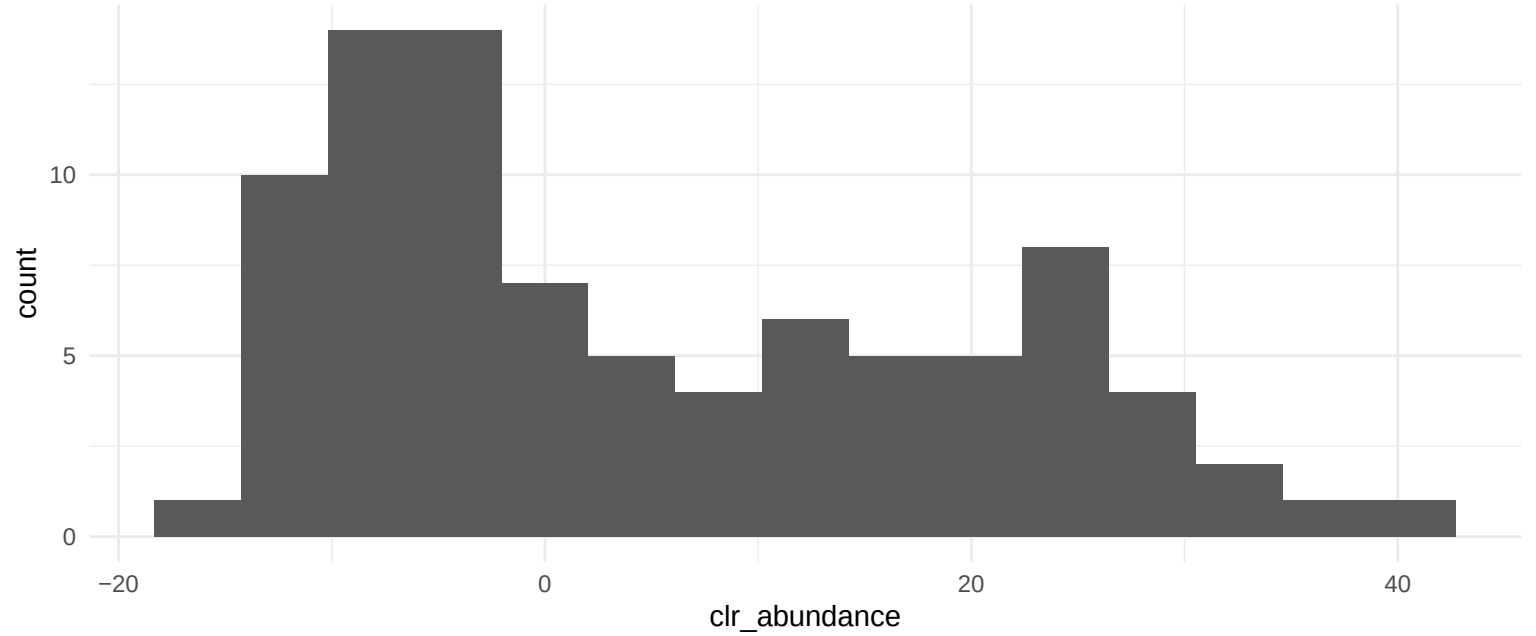



Distribution for 99982 Energy metabolism
count
15
10
5
0
0
50
clr_abundance

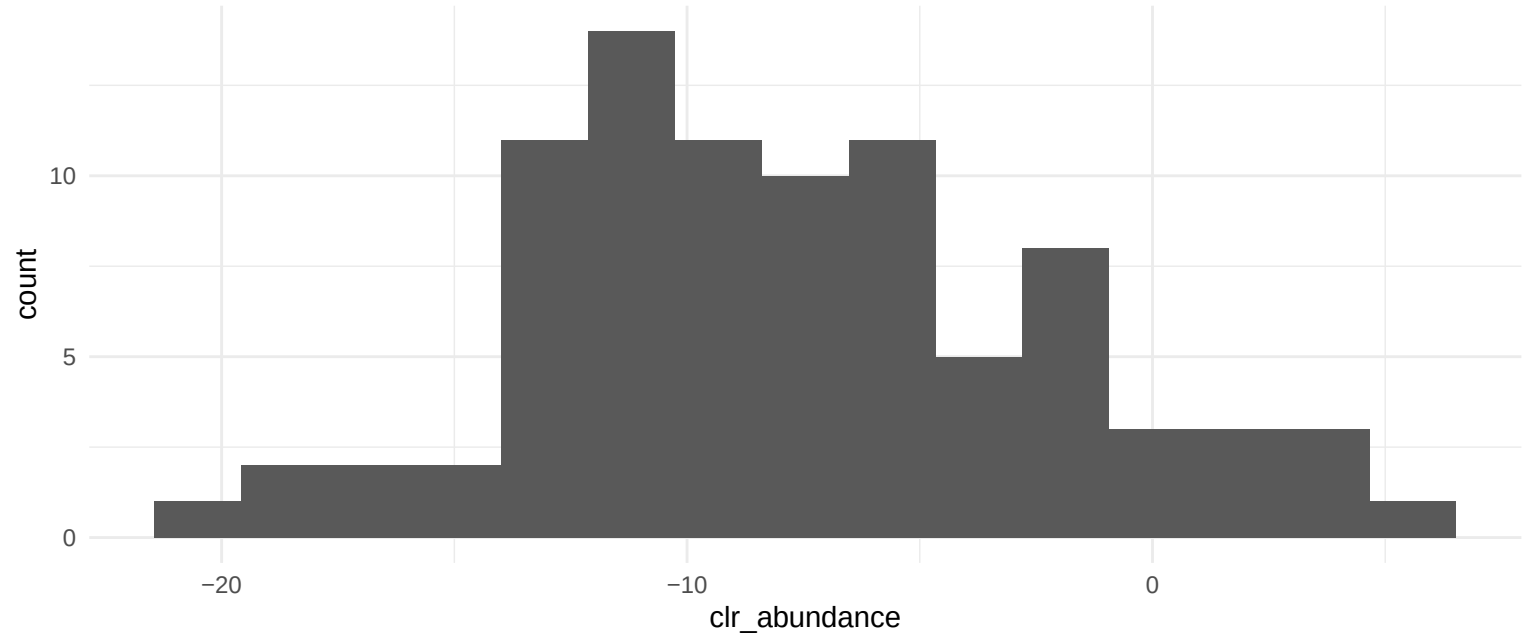

Distribution for 99983 Lipid metabolism
count
30
20
10
0
−2
0
2
clr_abundance


Distribution for 99984 Nucleotide metabolism
count
15
10
5
0
−2.5
0.0
2.5
5.0
clr_abundance

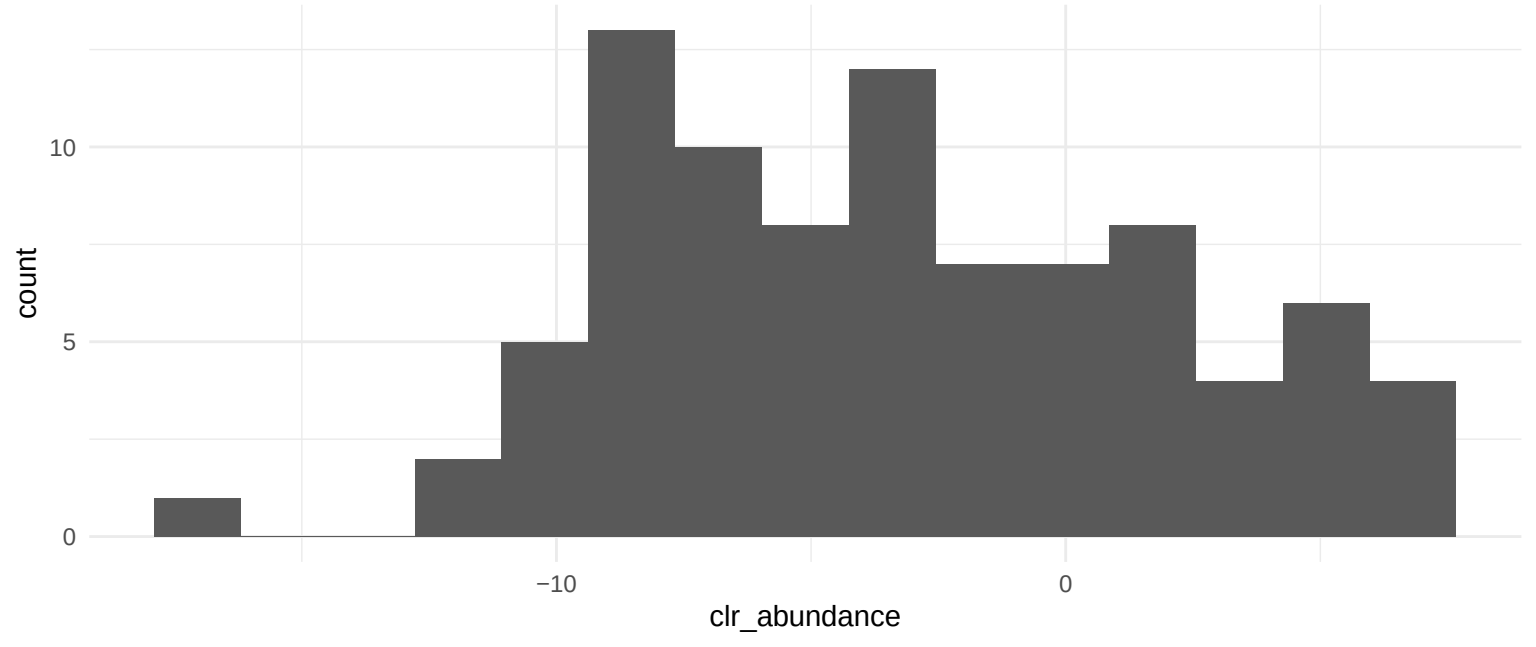
Distribution for 99985 Amino acid metabolism

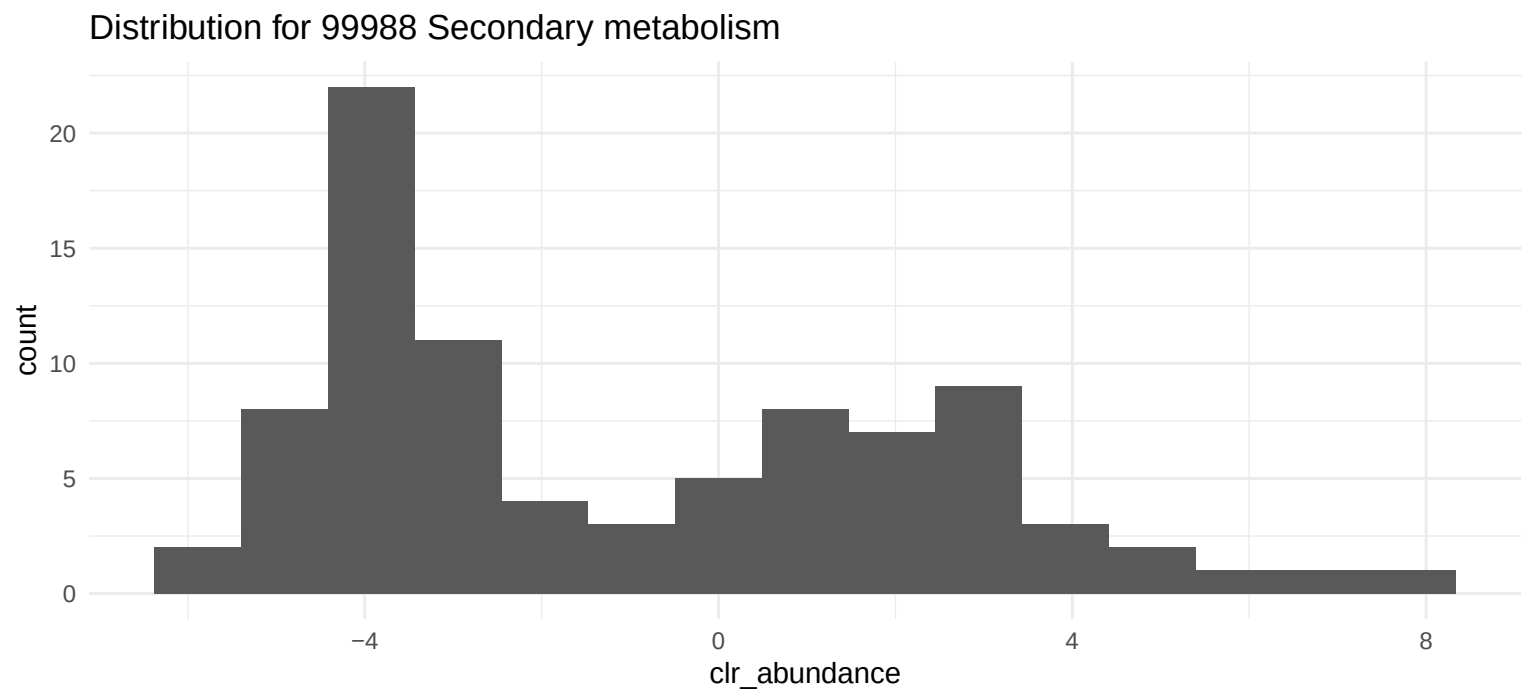


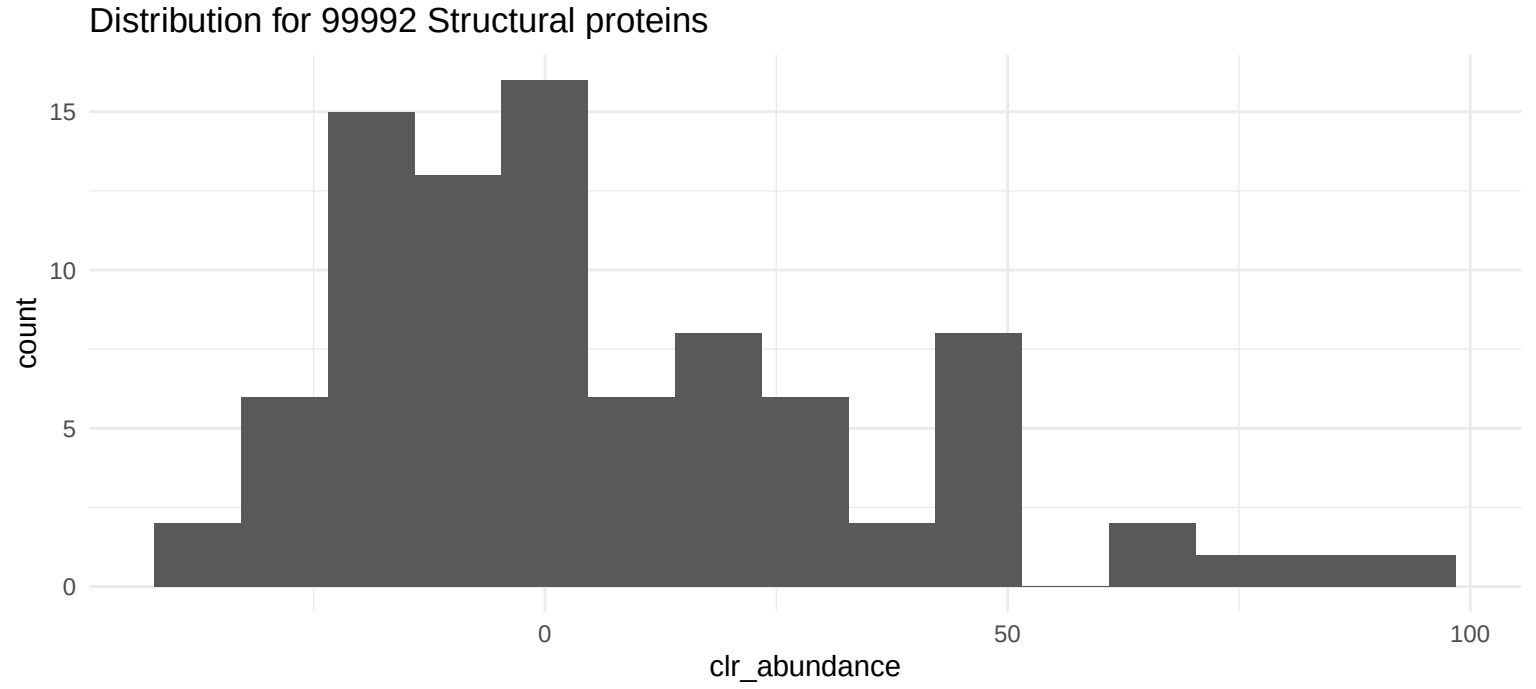
Distribution for 99986 Glycan metabolism

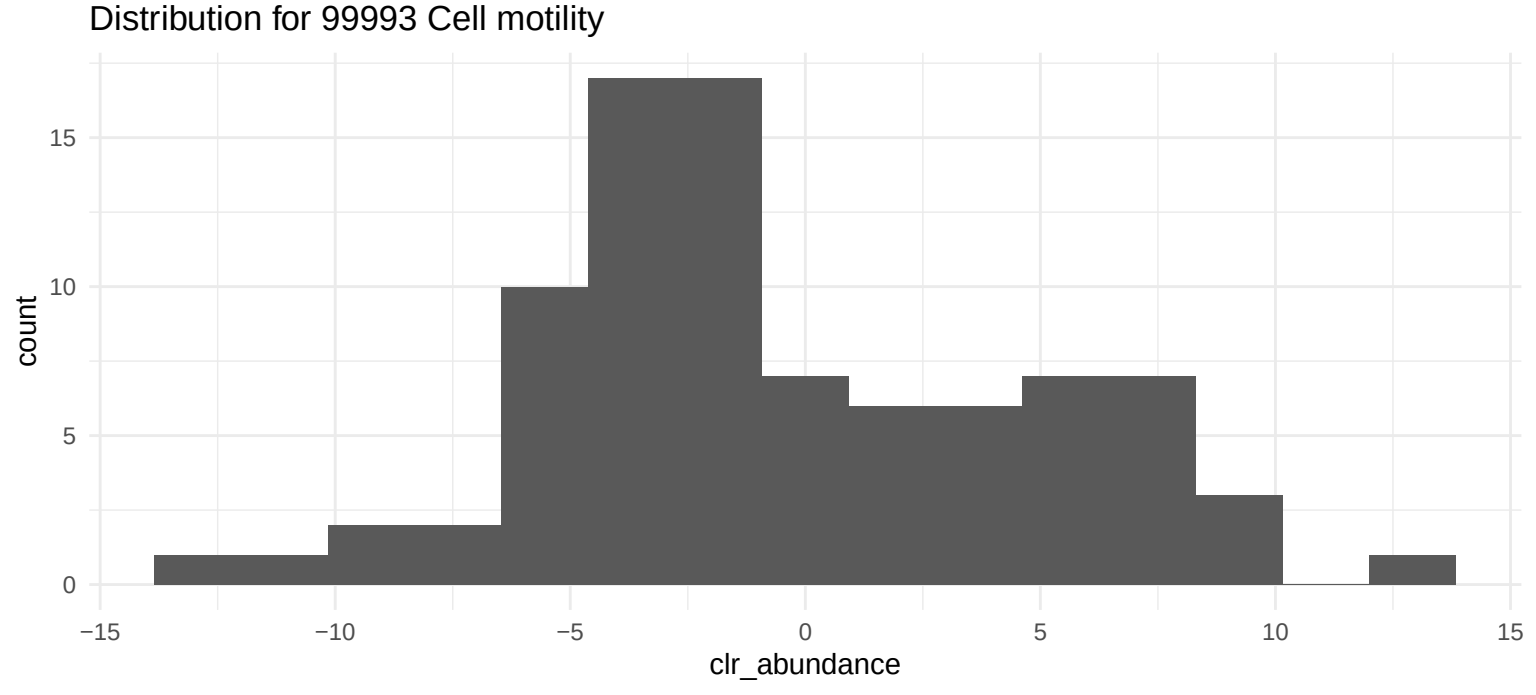


Distribution for 99987 Cofactor metabolism

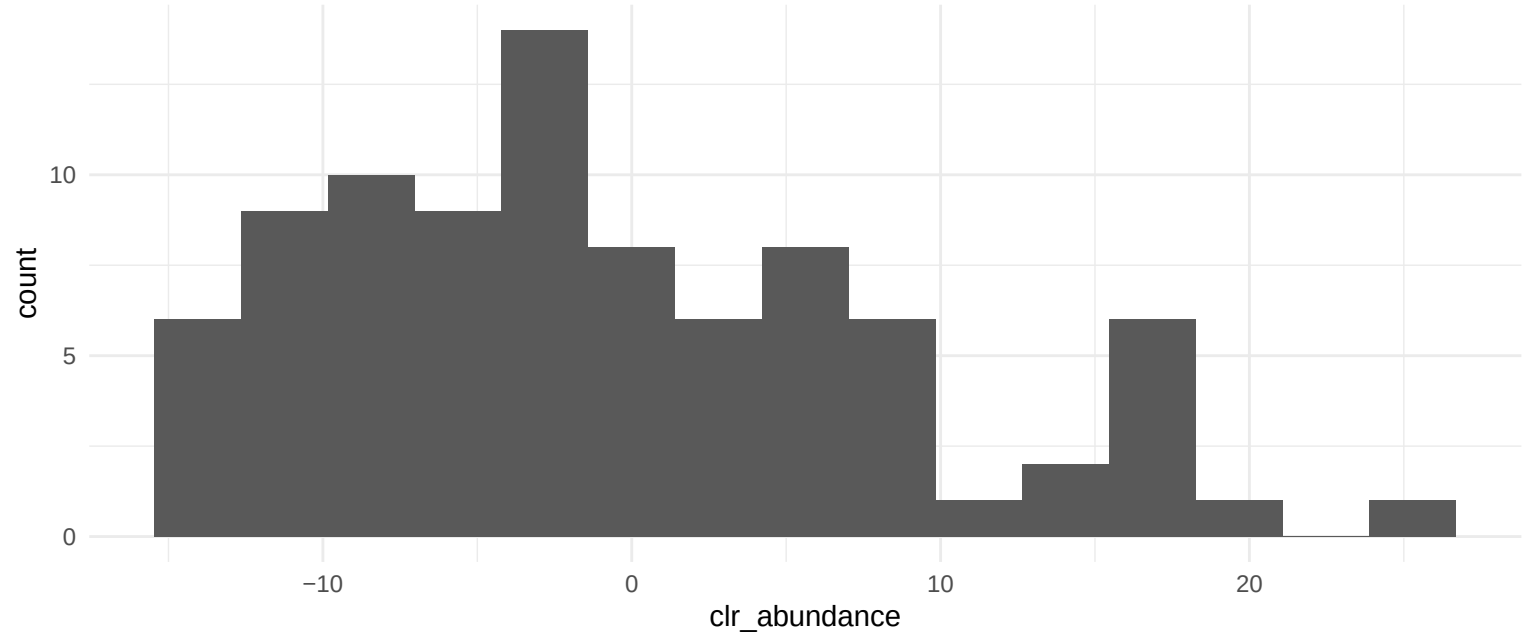



Distribution for 99988 Secondary metabolism
count
clr_abundance

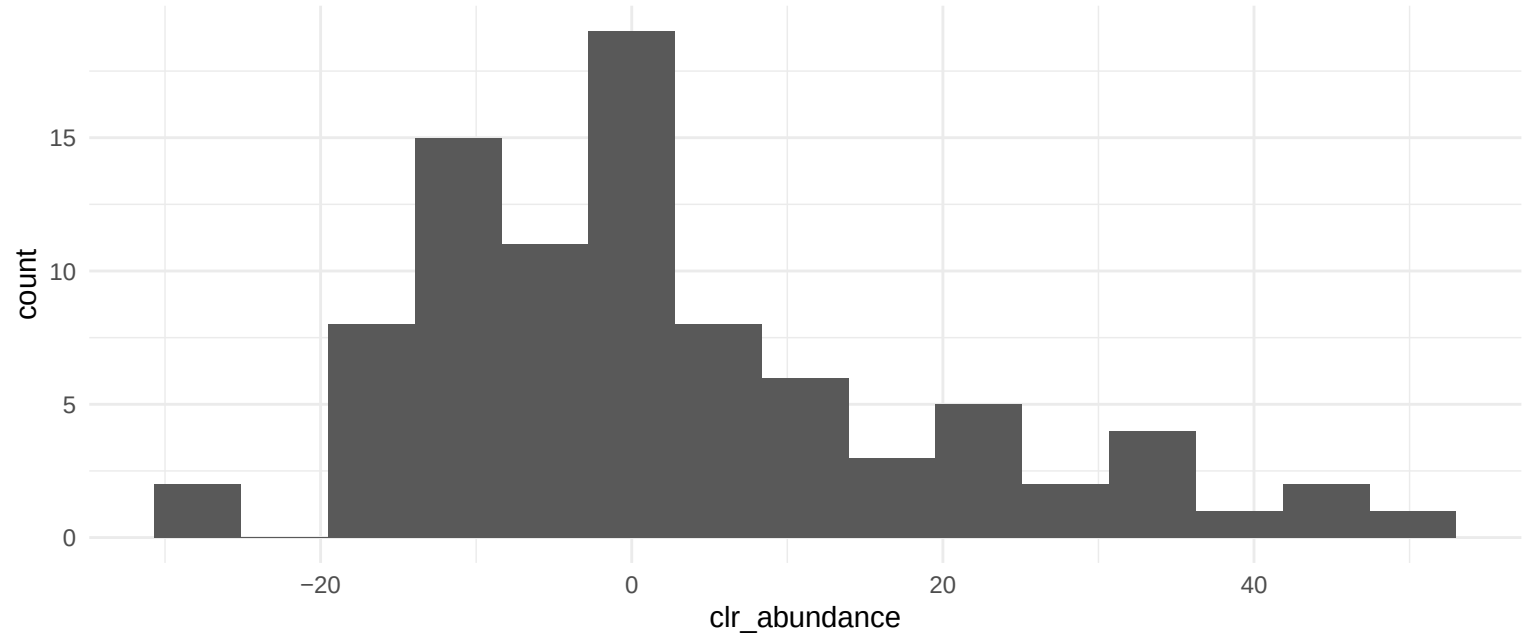

Distribution for 99992 Structural proteins
count
clr_abundance


Distribution for 99993 Cell motility
count
clr_abundance

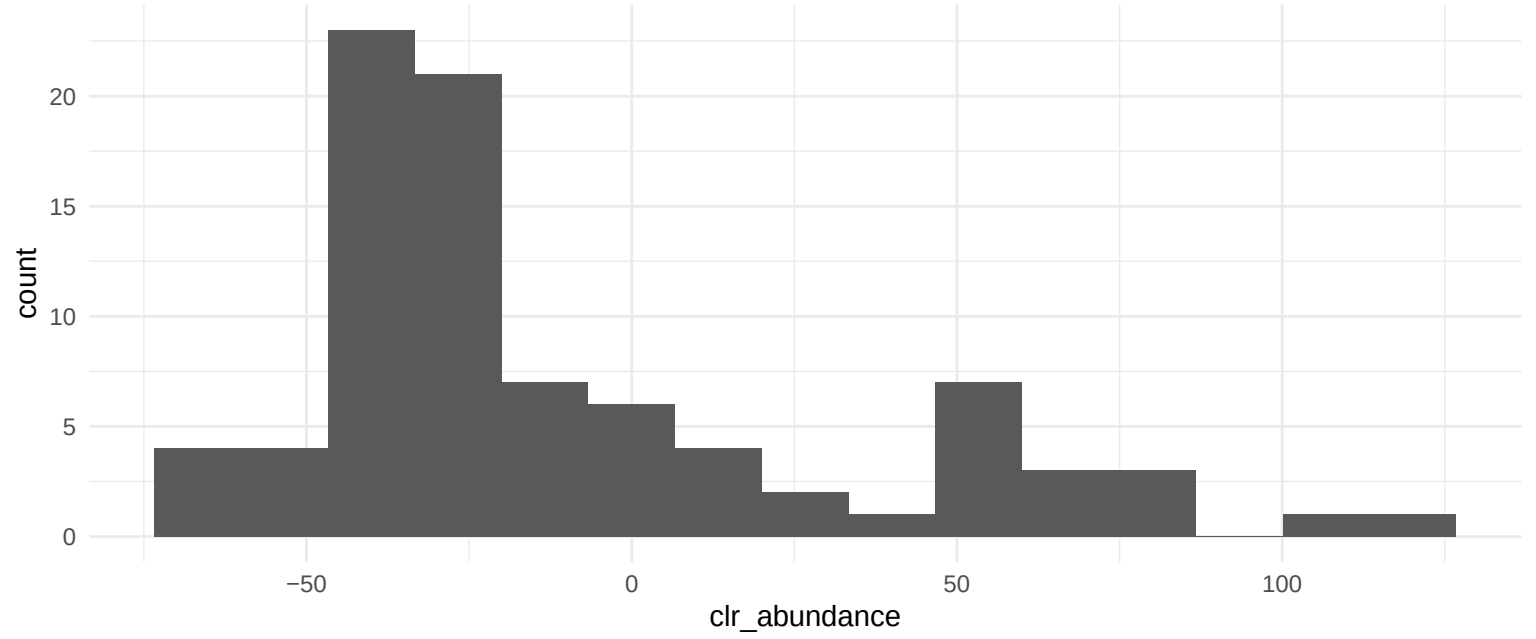
Distribution for 99994 Others

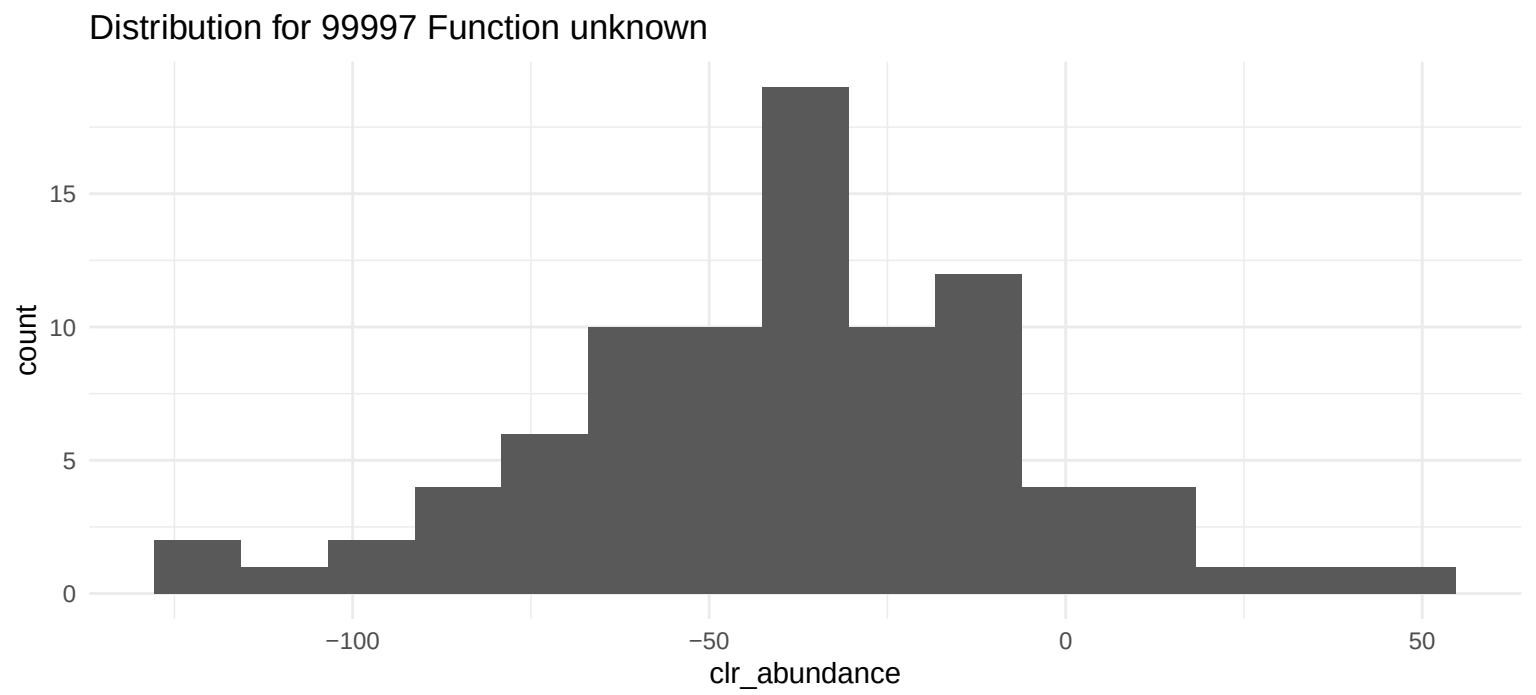


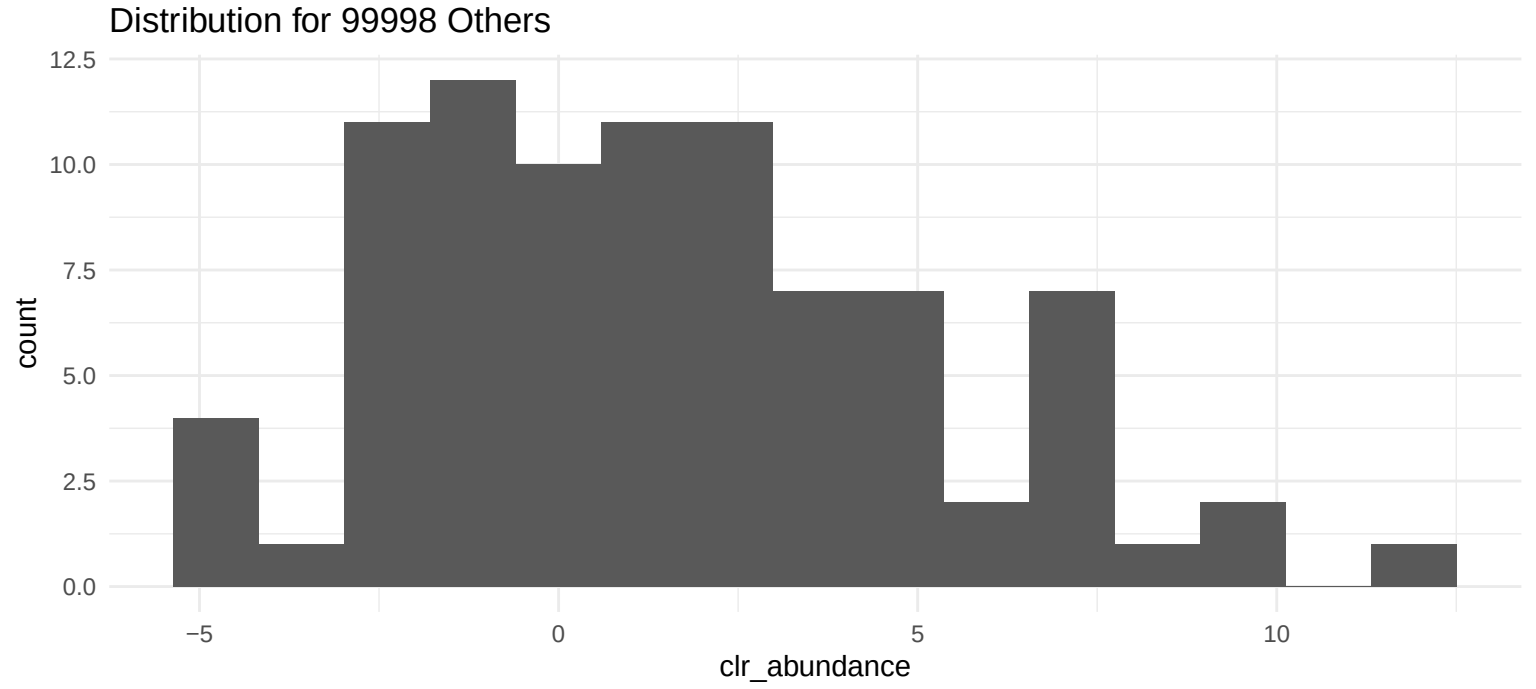
Distribution for 99995 Signaling proteins

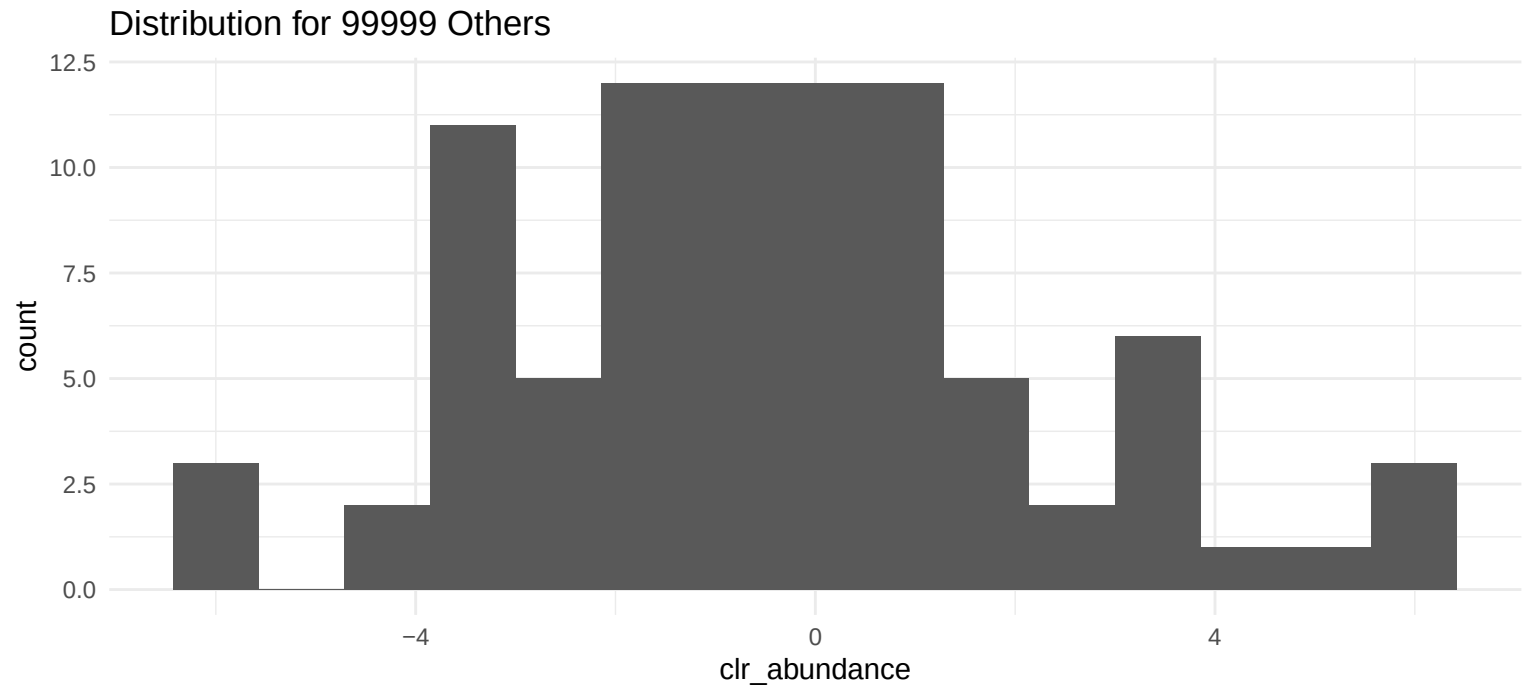


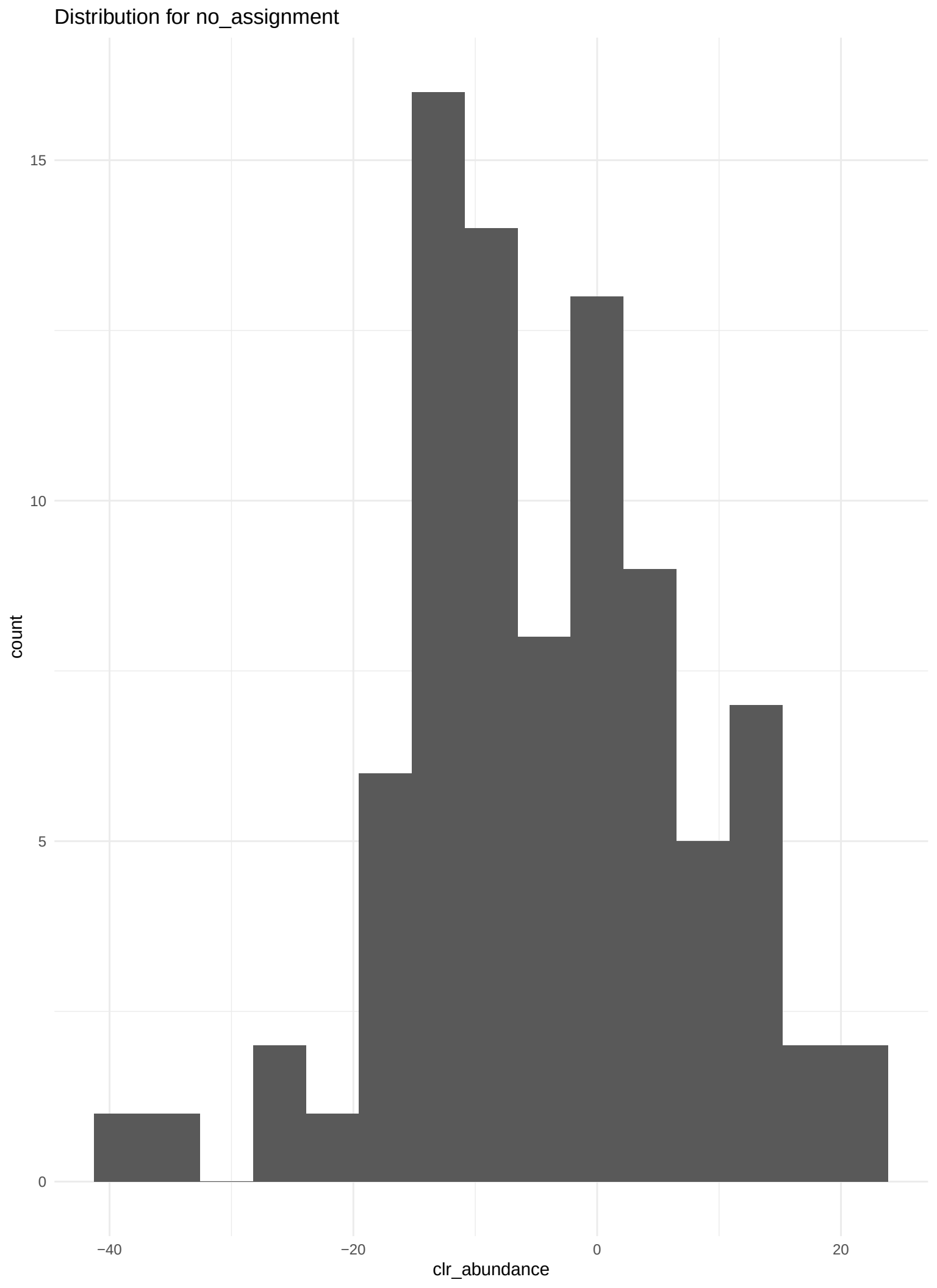
Distribution for 99996 General function prediction only




Distribution for 99997 Function unknown
count
15
10
5
0
−100
−50
0
50
clr_abundance


Distribution for 99998 Others
count
12.5
10.0
7.5
5.0
2.5
0.0
−5
0
5
10
clr_abundance


Distribution for 99999 Others
count
12.5
10.0
7.5
5.0
2.5
0.0
−4
0
4
clr_abundance


Distribution for no_assignment
15
10
5
0
count
−40
−20
0
20
clr_abundance